

PrimeSim™ Continuum Reference Manual: MOSFET Models

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About the PrimeSim Continuum MOSFET Models Manual

The *PrimeSim Continuum MOSFET Models Manual* describes available MOSFET models that you can use when simulating your circuit designs in the PrimeSim Continuum solution.

The material covered includes:

- Design model and simulation aspects of MOSFET models.
- Parameters of each model level, and associated equations.
- Parameters and equations for MOSFET diode and MOSFET capacitor models.

In addition, Synopsys has introduced LEVELs that are compatible with models developed by UC Berkeley, The University of Florida, Rensselaer Polytechnic Institute, and other institutions, developers, and foundries.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeSim Continuum simulator Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeSim Continuum solution, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Overview of MOSFET Models

Provides an overview of MOSFET model types; contains general information on using and selecting MOSFET models.

The following topics are discussed in these sections:

- [MOSFET Model Usage](#)
- [PrimeSim Continuum and MOSFET Libraries](#)
- [MOSFET Device Definition](#)
- [PrimeSim Continuum Automatic Model Selector](#)
- [General MOSFET Model Statement](#)
- [MOSFET Capacitors](#)
- [MOSFET Diodes](#)
- [MOSFET Control Options](#)
- [MOSFET Output Templates](#)
- [Vth Using the Maximum gm \(gmmax\) Method](#)
- [Saturation Voltage Correction — The G Function Method](#)
- [Safe Operating Area Voltage Warning](#)
- [Model Pre-Processing and Parameter Flattening](#)
- [Advanced MOSFET Self-Heating Simulation Solution](#)
- [Use of Example Syntax](#)
- [References](#)

MOSFET Model Usage

A circuit netlist describes the basic functionality of an electronic circuit that you are designing. In the PrimeSim Continuum simulators format, a netlist consists of a series of elements that define the individual components of the overall circuit. You can use the PrimeSim Continuum simulators -format netlist to help you verify, analyze, and debug your circuit design, before you turn that design into actual electronic circuitry.

Synopsys provides a series of standard models. Each model is like a template that defines various versions of each supported element type used in the PrimeSim Continuum simulators -format netlist. Individual elements in your netlist can refer to these standard models for their basic definitions. When you use these models, you can quickly and efficiently create a netlist and simulate your circuit design. Referring to standard models this way reduces the amount of time required to:

- Create the netlist
- Simulate and debug your circuit design
- Turn your circuit design into actual circuit hardware.

Within your netlist, each element that refers to a model is known as an *instance* of that model. When your netlist refers to predefined device models, you reduce both the time required to create and simulate a netlist, and the risk of errors, compared to fully defining each element within your netlist.

One type of model that you can use as a template to define an element in your netlist is a Metal Oxide Semiconductor Field Effect Transistor (MOSFET) device. This manual describes the MOSFET models supplied for use with the PrimeSim Continuum solution.

A MOSFET device is defined by the MOSFET model and element parameters, and two sub-models selected by the `CAPOP` and `ACM` model parameters.

- The `CAPOP` model parameter specifies the model for the MOSFET gate capacitances.
- The Area Calculation Method (`ACM`) parameter selects the type of diode model to use for the MOSFET bulk diodes.

Parameters in each submodel define the characteristics of the gate capacitances and bulk diodes.

MOSFET models are either p-channel or n-channel models; they are classified according to level, such as LEVEL 1 or LEVEL 50.

PrimeSim Continuum and MOSFET Libraries

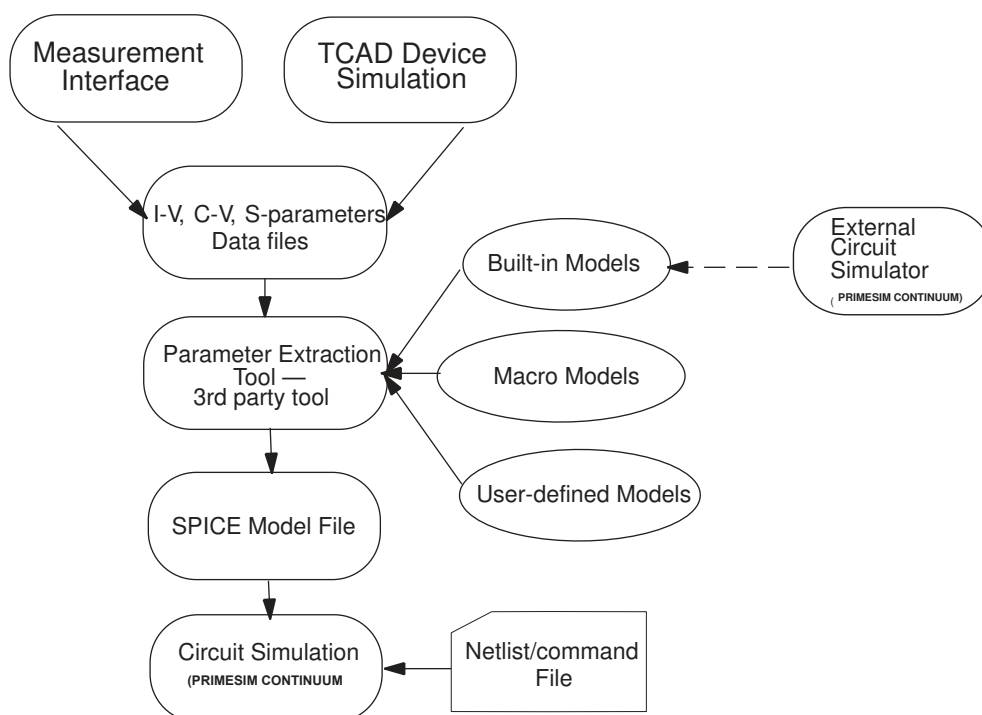
The following figure illustrates the generic flow of creating a MOSFET library-based circuit simulation. The `.LIB` call statement is used to call portions or all of a model file from a model library, see `.LIB` in the *PrimeSim Continuum Reference Manual: Commands and Control Options* and [Invoking MOSFET Library Files](#).

Support for Parameter Extraction:

The accuracy and a support for industry standard models makes the PrimeSim Continuum solution a strong option as an external circuit simulator in a parameter extraction flow. The PrimeSim Continuum built-in models are viable alternatives for any parameter extraction tool, such as the Agilent Technologies Integrated Circuits Characterization and Analysis Program (IC-CAP).

Note:

Run the PrimeSim Continuum simulators in client-server mode for best performance in a parameter extraction flow (See Using PrimeSim HSPICE in Client-Server Mode in the *PrimeSim™ HSPICE® User Guide: Basic Simulation and Analysis*).



MOSFET Device Definition

To define a MOSFET device in your netlist, use both an element statement and a `.MODEL` statement.

The element statement defines the connectivity of the transistor and references the `.MODEL` statement. The `.MODEL` statement specifies either an n- or p-channel device, the level of the model, and several user-selectable model parameters.

The following example specifies a PMOS MOSFET. PCH is the model reference name. The transistor is modeled using the [LEVEL 13: BSIM Model](#). Select the parameters from the MOSFET model parameter lists in this chapter.

```
M3 3 2 1 0 PCH <parameters>
.MODEL PCH PMOS LEVEL=13 <parameters>
```

Invoking MOSFET Library Files:

You can use the `.LIB` command to create and read from libraries of commonly used commands, device models, subcircuit analyses, and statements.

Use the following syntax for library calls:

```
.LIB '<filepath> filename' entryname
```

Use the following syntax to define library files:

```
.LIB entryname1
. $ ANY VALID SET OF HSPICE STATEMENTS
.ENDL entryname1
.LIB entryname2
.
. $ ANY VALID SET OF HSPICE STATEMENTS
.ENDL entryname2
.LIB entryname3
.
. $ ANY VALID ET OF HSPICE STATEMENTS
.ENDL entryname3
```

To build libraries (library file definition), use the `.LIB` statement in a library file. For each macro in a library, use a library definition statement (`.LIB entryname`) and an `.ENDL` statement. The `.LIB` statement begins the library macro and the `.ENDL` statement ends the library macro. The text after a library file entry name must consist of the PrimeSim Continuum statements. Library calls can call other libraries (nested library calls) if they are different files. You can nest library calls to any depth. Use nesting with the `.ALTER` statement to create a sequence of model runs. Each run can consist of similar components by using different model parameters without duplicating the entire input file.

The simulator uses the `.LIB` statement and the `.INCLUDE` statement to access the models and skew parameters. The library contains parameters that modify `.MODEL` statements.

This section covers the following topics:

- [Reliability Analysis for PrimeSim Continuum MOSFET Devices](#)
- [PrimeSim Continuum Custom Common Model Interface \(CMI\)](#)
- [TSMC Model Interface \(TMI\)](#)

Reliability Analysis for PrimeSim Continuum MOSFET Devices

As CMOS technology scales down, reliability requirements become more challenging and important in maintaining the long-term reliability of these devices. Two of the most critical reliability issues, the hot carrier injection (HCI) and the negative bias temperature instability (NBTI) effects have been demonstrated to change the characteristics of the MOS devices. The PrimeSim Continuum reliability analysis feature allows circuit designers to be able to predict the reliability of their designs such that there are enough margins for their circuits to function correctly over the entire lifetime. Refer to MOSFET Model Reliability Analysis (MOSRA) in the *PrimeSim HSPICE User Guide: Simulation and Analysis* for more information. In addition, a unified custom reliability modeling MOSRA API is available with an application note. Consult your Synopsys support team for full information.

PrimeSim Continuum Custom Common Model Interface (CMI)

The PrimeSim Continuum simulators can use a dynamically-linked shared library to integrate models with the Custom CMI with use of the `cmiflag` global option to load the dynamically linked Custom CMI library. Consult the PrimeSim Continuum technical support team for access to the PrimeSim Continuum CMI application note and source code.

TSMC Model Interface (TMI)

You can invoke the TMI flow using proprietary TSMC model files and compiled libraries. Jointly developed by Synopsys and TSMC, the TMI technology and API is a compact model with additional instance parameters and equations for an advanced modeling approach to support TSMC's extension of the standard BSIM4 model. Modeling API code is written in C and available in a compiled format for the PrimeSim Continuum simulators to link to during the simulation. TMI-required settings to invoke the flow and the location of a `.so` file are set by TSMC. The API also performs automatic platform selection on the `.so` file. Both the PrimeSim Continuum simulators provide the tool binaries and support the same `.so` file.

Use the existing PrimeSim Continuum simulator commands to run the simulation. For more information, see the *PrimeSim HSPICE Reference Manual: Commands and Control Options* for .OPTION TMIFLAG and .OPTION TMIPATH.

PrimeSim Continuum Automatic Model Selector

For libraries with multiple models of a specific element, you can use a built-in model selector in the PrimeSim Continuum simulators find the proper model for each transistor size automatically.

The automatic model selector uses the following criteria:

$$L_{MIN} + XL_{REF} \leq L + XL < L_{MAX} + XL_{REF}$$

$$W_{MIN} + XW_{REF} \leq W + XW < W_{MAX} + XW_{REF}$$

If you do not specify XL_{REF} , the simulation sets it to XL . If you do not specify XW_{REF} , the simulation sets it to XW .

The model selector syntax is based on a common model root name, with a unique extension for each model.

The following is an example of the PrimeSim Continuum syntax for MOSFET models:

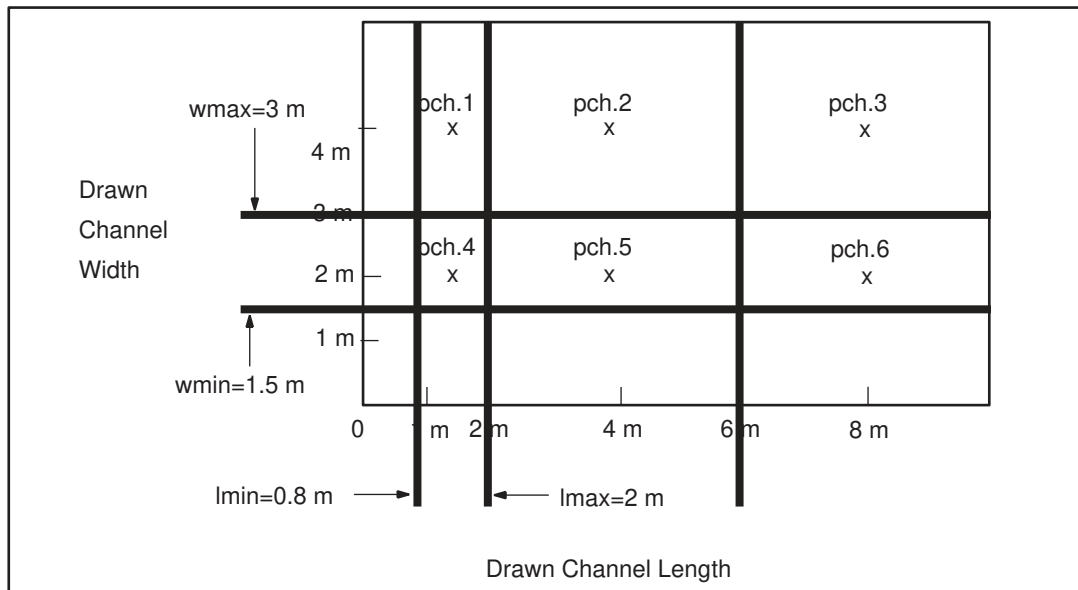
```
M1 drain gate source bulk NJ W=2u L=1u
.MODEL NJ4 NJF WMIN=1.5u WMAX=3u LMIN=.8u LMAX=2u
.MODEL NJ5 NJF WMIN=1.5u WMAX=3u LMIN=2u LMAX=6u
```

Figure 1 illustrates the MOSFET model selection method.

This example illustrates several pch.x models, with varying drawn channel lengths and widths, in the model library. The model root name is “pch”, and the extensions are 1, 2, ..., 6. The NJ4 instance of the NJ Element ($W=2\ \mu$, $L=1\ \mu$) requires a model for which $1.5\ \mu \leq$ channel width $\leq 3\ \mu$, and $0.8\ \mu \leq$ channel length $\leq 2\ \mu$.

The automatic model selector chooses the pch.4 model because that model satisfies these requirements. Similarly, the NJ5 transistor requires a model with $1.5\ \mu \leq$ channel width $\leq 3\ \mu$, and $2\ \mu \leq$ channel length $\leq 6\ \mu$. The pch.5 model satisfies these requirements. If a device size is out of range for all models, the automatic model selector issues an error message.

Figure 1 Automatic Model Selector Method



If the automatic model selector cannot find a model within a subcircuit, the automatic model selector searches the top level. If the automatic model selector fails to find a model, simulation terminates.

The following combination of conditions causes the automatic model selector to fail and terminates the simulation:

- The element statement uses a model name that contains a period (.).
- The model library was not designed for use with the automatic model selector.
- The simulation input includes either a multi-sweep specification or a `.TEMP` temperature analysis statement.

The following example illustrates how a period in a model name can cause problems in automatic model selection.

Example 1:

```
M1 d g s b N.CHN W=10u L=5u          * Element statement
.MODEL N.CHN LMIN=1u LMAX=4u WMIN=2u WMAX=100u
* .MODEL statement
```

Example 2:

```
.TEMP 25
.M1 d g s b N.CHN W=10u L=5u          * Element statement
.MODEL N.CHN LMIN=1u LMAX=4u WMIN=2u WMAX=100u
* .MODEL statement
```

Because Example 1 does not specify multi-sweep or temperature analysis, simulation does not invoke the model selector feature, so simulation uses the N.CHN model with no problems.

In Example 2, however, the `.TEMP` statement invokes the model selector feature. The model selector tries to find a model named “N.nnn” that fits within the length and width ranges specified in the element statement.

Because the length in the element statement (5 μm) is not within the 1 to 4 μm range specified in the `.MODEL` statement, the model selector cannot find a model that matches the element statement, and simulation issues a “device ‘N’ not found” error message.

General MOSFET Model Statement

You can use the `.MODEL` statement to include a MOSFET model in the PrimeSim Continuum simulators netlist. For more information, see `.MODEL` in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

The following syntax applies to all MOSFET model specifications. All related parameter levels are described in their respective sections.

Note:

The `ENCMODE` parameter can only be set in [LEVEL 54: BSIM4 MOSFET Model](#) to suppress warning messages. The `TMIMODEL` and `CMIMODEL` parameters avoid potential conflicts when the TMI or custom CMI model and other standard models are used together in a shared object file.

Syntax:

```
.MODEL mname [PMOS|NMOS] [ENCMODE=0|1]
+ ([LEVEL=val keyname1=val1 keyname2=val2...])
+ [VERSION=version_number]
+ [TMIMODEL=0|1] [CMIMODEL=0|1]
```

Parameter	Description
mname	Model name. Elements refer to the model by this name. See Model Name Identification Rule below.
PMOS	Identifies a p-channel MOSFET model.
NMOS	Identifies an n-channel MOSFET model.
ENCMODE	Applicable to BSIM4 (level 54 only); use to suppress warning messages originating in CMI code while inside encrypted code. Default is 0 (off). Set to off if this parameter is not present. This parameter cannot be overwritten through the instance line.

Parameter	Description
LEVEL	Use the LEVEL parameter to select from several MOSFET model types. Default=1.0.
VERSION	Specifies the version number of the model for LEVEL=13 BSIM and LEVEL=39 BSIM2 models only. See the .MODEL statement description for information about the effects of this parameter.
TMIMODEL	This parameter takes effect when .option tmiflag is on. If you set it to 1, the simulator seeks models from a shared object file to avoid potential conflicts. TMIMODEL default (1) with TMI model cards. Set it to 'zero' in non-TMI models when both TMI and non-TMI models are used in a design. When 0, the flag directs the simulator to ignore models in compiled TMI libraries.
CMIMODEL	This parameter takes effect when .option cmiflag is on. If you set it to 1, the simulator seeks models from a shared object file to avoid potential conflicts.

Example:

```
.MODEL MODP PMOS LEVEL=7 VTO=-3.25 GAMMA=1.0)
.MODEL MODN NMOS LEVEL=2 VTO=1.85 TOX=735e-10)
.MODEL MODN NMOS LEVEL=39 TOX=2.0e-02 TEMP=2.5e+01
+ VERSION=95.1
```

This section covers the following topics:

- [Model Name Identification Rule](#)
- [Measuring the Value of MOSFET Model Card Parameters](#)
- [Using a Model Card Defined with a Subckt Wrapper](#)

Model Name Identification Rule

MOSFETs can support up to 7 nodes. The PrimeSim Continuum model name identification uses the following rule: If the model names 'nch' and 'pch' appear at the same time, then 'nch' is regarded as a node name and 'pch' is considered the model. However, after the 4th node, the PrimeSim Continuum solution regards 'nch' as the model name instead of 'pch'. For example:

```
m1 n1 n2 n3 n4 nch pch p1 p2 p3
```

Measuring the Value of MOSFET Model Card Parameters

The keywords `val()` and `valm()` are supported by `.MEAS/.PRINT/.PROBE` commands (only).

The syntax for instance parameters is: `val(element.parameter)`

The syntax for model parameters is: `valm(elem_name.model_param)`

The parameters currently supported for `valm()` include: `vth0`, `lmin`, `lmax`, `wmin`, `wmax`, `lref`, `wref`, `xl`, `dl`, `dell`, `xw`, `dw`, `delw`, `scalm`, `lmlt`, `wmlt`, and levels 54, 57, and 70. If the modelcard in `valm()` is not a valid model parameter for the level, an error is reported.

For example:

```
.meas delvto1 param='val(m1.l)'  
.meas tran m_vth0 param='valm(m1.vth0)'
```

For cmi models, `valm()` only supports levels 54, 57, and 70 and model card parameters: `vth0`, `lmin`, `lmax`, `wmin`, `wmax`, `lref`, `wref`, `xl`, `dl`, `dll`, `xw`, `dw`, `delw`, `scalm`, `lmlt`, `wmlt`.

Using a Model Card Defined with a Subckt Wrapper

The increased complexity of MOSFET models sometimes renders the `.MODEL` construct inefficient to implement statistical, DFM, WPE modeling, and so on. As a countermeasure, foundries have used a macro model process in which they wrap `.MODEL` statements under a `.SUBCKT` structure to accommodate additional equations and parameter definitions. The arguments for MOSFET output templates when used with a macro model card differ slightly from a standard model card.

For example:

```
xm1 1 2 3 4 nmos_mac  
.subckt nmos_mac 1 2 3 4  
mn1 d g s b nmos ....  
.model nmos nmos ...  
.ends
```

In this case the format would be:

`template_name(circuit_element_name.subckt_element_name)`

For example: `LV9(xm1.mn1)`

In some cases, however, even though the model card is subcircuit-based, the netlist could still contain an `M`-element statement. Traditionally, the PrimeSim Continuum simulators require the `.SUBCKT` statement to be called by an `X`-element statement. But if you use `.OPTION MACMOD=3`, this requirement can be bypassed and the `M`-element statement can be directly mapped to a `.SUBCKT` statement in the absence of a `.MODEL` statement. For example, you can modify the example above as follows:

```
.option macmod=3  
m1 1 2 3 4 nmos_mac
```

In this case, the threshold voltage monitoring command would be:

```
lv9(m1.mn1)
```

For more information, see .SUBCKT and .OPTION MACMOD in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

MOSFET Capacitors

CAPOP is the MOSFET capacitance model parameter. This parameter determines which capacitor models to use when modeling the MOS gate capacitance; that is, the gate-to-drain capacitance, the gate-to-source capacitance, and the gate-to-bulk capacitance. Using the CAPOP parameter, you can select a specific version of the Meyer and charge conservation model.

Some capacitor models are tied to specific DC models; they are stated as such. Others are for general use by any DC model.

Parameter	Description
CAPOP=0	SPICE original Meyer gate-capacitance model (general)
CAPOP=1	Modified Meyer gate-capacitance model (general)
CAPOP=2	Modified Meyer gate-capacitance model with parameters (general default)
CAPOP=3	Modified Meyer gate-capacitance model with parameters and Simpson integration (general)
CAPOP=4	Charge conservation capacitance model (analytic), LEVELs 2, 3, 6, 7, 13, 28, and 39 only
CAPOP=5	No capacitor model
CAPOP=6	AMI capacitor model (LEVEL 5)
CAPOP=9	Charge conservation model (LEVEL 3)
CAPOP=11	Ward-Dutton model specialized (LEVEL 2)
CAPOP=12	Ward-Dutton model specialized (LEVEL 3)
CAPOP=13	Generic BSIM Charge-Conserving Gate Capacitance model (Default for Levels 13, 28, and 39)
CAPOP=39	BSIM2 Charge-Conserving Gate Capacitance Model (LEVEL 39)

CAPOP=4 selects the recommended charge-conserving model (from among CAPOP=11, 12, or 13) for the specified DC model.

MOS Level	Default CAPOP	CAPOP=4 selects:
2	2	11

MOS Level	Default CAPOP	CAPOP=4 selects:
3	2	12
13, 28, 39	13	13
others	2	11

LEVELs 49 and 53 use the Berkeley `CAPMOD` capacitance-model parameter. Proprietary models, and LEVELs 5, 17, 21, 22, 25, 27, 31, 33, 49, 53, 55, and 58, use built-in capacitance routines.

MOSFET Diodes

The Area Calculation Method (ACM) model parameter controls the geometry of the source and drain diffusions, and selects the modeling of the bulk-to-source and bulk-to-drain diodes of the MOSFET model. The diode model includes the diffusion resistance, capacitance, and DC currents to the substrate.

For more information on ACM, see [MOSFET Diode Model Parameters](#).

MOSFET Control Options

Specific control options (set in the `.OPTION` statement) used for MOSFET models include the following. For flag options, 0 is unset (off) and 1 is set (on).

Option	Description
ASPEC	This option uses ASPEC MOSFET model defaults and set units. Default=0.
BYPASS	This option avoids recomputing nonlinear functions that do not change with iterations. Default=1.
MBYPASS	BYPASS tolerance multiplier (BYTOL = MBYPASSxVNTOL). Default=1 if DVDT=0, 1, 2, or as 3. Default=2 if DVDT=4.
DEFAD	Default drain diode area. Default=0.
DEFAS	Default source diode area. Default=0.
DEFL	Default channel length. Default=1e-4m.
DEFNRD	Default number of squares for drain resistor. Default=0.
DEFNRS	Default number of squares for source resistor. Default=0.
DEFPD	Default drain diode perimeter. Default=0.

Option	Description
DEFPS	Default source diode perimeter. Default=0.
DEFW	Default channel width. Default=1e-4m.
GMIN	Pn junction parallel transient conductance. Default=1e-12mho.
GMINDC	Pn junction parallel DC conductance. Default=1e-12mho.
SCALE	Element scaling factor. Default=1.
SCALM	Model scaling factor. Default=1. Note: the SCALM parameter is only available in some models below Level 49. At Level 49 and higher, it is ignored.
WL	Reverses order in VSIZE MOS element from the default order (length-width) to width-length. Default=0.

- The `AD` element statement overrides the `DEFAD` default.
- The `AS` element statement overrides the `DEFAS` default.
- The `L` element statement overrides the `DEFL` default.
- The `NRD` element statement overrides the `DEFNRD` default.
- The `NRS` element statement overrides the `DEFNRS` default.
- The `PD` element statement overrides the `DEFPD` default.
- The `PS` element statement overrides the `DEFPS` default.
- The `W` element statement overrides the `DEFW` default.

The following sections discuss additional options:

- [Scale Units](#)
- [Bypass Option for Latent Devices](#)
- [Searching Models as Function of W, L](#)
- [Number of Fingers, WNFLAG Option](#)
- [Modify Element Temperature, DTEMP](#)

Scale Units

The `SCALE` and `SCALM` options control the units.

- `SCALE` scales element statement parameters.
- `SCALM` scales model statement parameters. It also affects the MOSFET gate capacitance and diode model parameters.

Note:

`SCALM` is ignored in Level 49 and higher.

In this chapter, scaling applies only to parameters that you specify as scaled. If you specify `SCALM` as a parameter in a `.MODEL` statement, it overrides the `SCALM` option. In this way, you can use models with different `SCALM` values in the same simulation. MOSFET parameter scaling follows the same rules as for other model parameters, for example:

Parameter Units	Parameter Value
meter	multiplied by <code>SCALM</code>
meter ²	multiplied by <code>SCALM</code> ²
meter ⁻¹	divided by <code>SCALM</code>
meter ⁻²	divided by <code>SCALM</code> ²

To override global model size scaling for individual MOSFET, diode, and BJT models that use the `.OPTION SCALM=val>` statement, include `SCALM= val` in the `.MODEL` statement. `.OPTION SCALM=<val>` applies globally for JFETs, resistors, transmission lines, and all models other than MOSFET, diode, and BJT models. You cannot override `SCALM` in the model.

Scaling for LEVEL 25 and 33

When using the proprietary LEVEL 25 (Rutherford CASMOS) or LEVEL 33 (National) models, the `SCALE` and `SCALM` options are automatically set to 1e-6. However, if you use these models with other scalable models, you must explicitly set the `SCALE=1e-6` and `SCALM=1e-6` options.

Bypass Option for Latent Devices

Use the `BYPASS` (latency) option to decrease simulation time in large designs. To speed simulation time, this option does not recalculate currents, capacitances, and conductances, if the voltages at the terminal device nodes have not changed. The `BYPASS` option applies to MOSFETs, MESFETs, JFETs, BJTs, and diodes. Use `.OPTION BYPASS` to set `BYPASS`.

`BYPASS` might reduce simulation accuracy for tightly-coupled circuits such as op-amps, high gain ring oscillators, and others. Use `.OPTION MBYPASS` to set `MBYPASS` to a smaller value for more accurate results.

Searching Models as Function of W, L

Model parameters are often the same for MOSFETs that have width and length dimensions within specific ranges. To take advantage of this, create a MOSFET model for a specific range of width and length. These model parameters help the simulator to select the appropriate model for the specified width and length.

The automatic model selection program searches a data file for a MOSFET model where the width and length are within the range specified in the MOSFET element statement. Simulation then uses this model statement.

To search a data file for MOSFET models within a specified range of width and length:

- Provide a root extension for the model reference name (in the `.MODEL` statement).
- Use the model geometric range parameters (`LMIN`, `LMAX`, `WMIN`, and `WMAX`). These model parameters define the range of physical length and width dimensions to which the MOSFET model applies.

Example 1:

If the model reference name in the element statement is `NCH`, the model selection program examines the models with the same root model reference name (`NCH`), such as `NCH.1`, `NCH.2`, or `NCH.A`.

The model selection program selects the first MOSFET model statement whose geometric range parameters include the width and length specified in the associated MOSFET element statement.

Example 2:

The following example shows how to call the MOSFET model selection program from a data file. The model selector program examines the `.MODEL` statements where the model reference names have the root extensions `NCHAN.2`, `NCHAN.3`, `NCHY.20`, and `NCHY.50`.

Note:

The `scalm` parameter is ignored in Level 49 and higher.

The following example is based on demonstration netlist `selector.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
file: selector.sp test of mos model selector
.option post list wl scale=1u scalm=1u nomod
.op
.probe i(m1)
```

```
v1 1 0 5
v2 2 0 4
v3 3 0 1
v4 4 0 -1
m1 1 2 3 4 nchan 10 2
m2 1 2 3 4 nchan 10 3
m3 1 2 3 4 nch 10 4
m4 1 2 3 4 nchx 10 5
m5 1 2 3 4 nchy 20 5
m6 1 2 3 4 nchy 50 5
$$$$$ for channel length selection
.model nchan.2 nmos level=2 vto=2.0 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=1 lmax=2.5 wmin=2 wmax=15
.model nchan.3 nmos level=2 vto=2.2 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=2.5 lmax=3.5 wmin=2 wmax=15
$$$$$$$ no selection for channel length and width
.model nch nmos level=2 vto=2.3 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=3.5 lmax=4.5 wmin=2 wmax=15
.model nchx nmos level=2 vto=2.4 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=4.5 lmax=100 wmin=2 wmax=15
$$$$$$$ for channel width selection
.model nchy.20 nmos level=2 vto=2.5 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=4.5 lmax=100 wmin=15 wmax=30
.model nchy.50 nmos level=2 vto=2.5 uo=800 tox=500 nsub=1e15
+ rd=10 rs=10 capop=5
+ lmin=4.5 lmax=100 wmin=30 wmax=500
.end
```

Number of Fingers, WNFLAG Option

This section discusses number of fingers, categorization bin, channel currents and NF vs. M parameter.

This section covers the following topics:

- [Number and Width of Fingers](#)
- [Categorizing Bins in Model File for MOSFETs](#)
- [Values for IDs with Different Values of NF](#)
- [How NF Differs from Multiplication Parameter M](#)

Number and Width of Fingers

Specify the number of fingers for fingered MOSFETs in the PrimeSim Continuum solution by using the instance parameter `NF`. The resultant width is explained below.

Example:

```
M1 out in vdd vdd pmos w=10u l=1u nf=5
```

M1 has a total drawn width of 10um with 5 fingers, each having a finger gate width of 2um. Both the PrimeSim Continuum solution and UCB BSIM4 models employ the same, consistent definition.

Categorizing Bins in Model File for MOSFETs

By default, the PrimeSim Continuum solution uses `W/nf` for model selection of the MOSFET. This is the default behavior because the parameter `wnflag` has been set to 1. If `wnflag = 1`, then the PrimeSim Continuum solution uses the ratio `W/nf` for model selection. If `wnflag = 0`, then the PrimeSim Continuum solution uses `W` for model selection. If you want to use the total width for binning model selection, you need to explicitly set `wnflag` to 0. Only BSIM4-based models can have `wnflag` as an instance parameter. But the netlist option, `.option wnflag`, is not strictly for BSIM4 model usage, it is for all levels of MOSFETs. For example,

For BSIM4 models only:

```
M1 out in vdd vdd pmos w=10u l=1u nf=5 wnflag=1
```

For all other levels:

```
.option wnflag  
M1 out in vdd vdd pmos w=10u l=1u nf=5
```

Values for IDs with Different Values of NF

While one might expect the MOSFET to have the same value of channel current even if you layout a device with a different number of fingers, you get different values of channel current (IDs).

Example:

```
M1 out in vdd vdd pmos w=10u l=1u nf=1  
M2 out in vdd vdd pmos w=10u l=1u nf=5
```

Ideally, M1 (one finger) and M2 (five fingers) should have same values of channel current since their `W` and `L` values are same, but these values are different because for the fingered MOSFETs, the channel current, effective width, and effective resistance of source and drain are functions of `nf`. Hence, the channel current value also differs slightly.

How NF Differs from Multiplication Parameter M

'M' stands for multiplicity factor. It means that 'M' identical transistors are in parallel. For example,

```
M1 out in vdd vdd pmos w=10u l=1u m=5
```

indicates that 5 identical transistors are connected in parallel with a total width of 50 μm .

Modify Element Temperature, DTEMP

To modify the temperature for a particular element, use the `DTEMP` parameter. The default circuit temperature is the value of `TNOM`. For additional information, see Temperature Variation Analysis in *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

MOSFET Output Templates

Many MOSFET models produce an output template, consisting of a set of parameters that specify the output of state variables, stored charges, capacitances, parasitic diode current, and capacitor currents. Different MOSFET model levels support different subsets of these output parameters.

For example, if your netlist contains four transistors `m0`, `m1`, `m2`, and `m3`, you can print the parameter values during the transient analysis using the following `.print` statement:

```
.print tran LX3(m1) LX4(m1) LX7(m1) LX8(m1)
```

This command prints the values of `Vds`, `Ids`, `Gm` and `Gds` to the `*.lis` file. For more information on MOSFET information, use the `.OPTION LIST` command as described in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

To enable the PrimeSim Continuum simulator to print PMOS template output voltage polarity as real bias, use the `.OPTION TMPLT_POL` as described in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

Wildcards are also supported, so you can also use the following, which prints the `Vds` of all transistors:

```
.print tran LX3(m*)
```

By default, the MOSFET output template is characterized based on the node bias associated with the node the MOSFET is physically connected to, also known as the physical notation. In some applications, the electrical notation is used which considers MOSFET source and drain node swapping at reversed bias conditions. In the MOSFET output template tables, the output templates characterized with the electrical notation are marked with a `**`.

The following output template tables are available. Refer to the appropriate table according to the MOSFET level.

- [Output Templates for General MOSFET \(Level<=71, Level 74\)](#)
- [Output Templates for BSIM-CMG \(Level 72\), BSIM6 \(Level 77\), and BSIM-IMG \(Level 78\)](#)
- [Output Templates for HiSIM_HV, Level=73](#)
- [Output Templates for UTSOI, Level=76](#)
- [MOSFET SUBCKT Output Templates](#)

Output Templates for General MOSFET (Level<=71, Level 74)

The following table lists parameters in the general MOSFET output templates.

Note:

** indicates electrical notation.

Name	Alias	Description	MOSFET Level <= 71, Level 74
L	LV1	Channel length (L) (also the effective channel length for all MOSFET models except Levels 54, 57, 69 and 70)	All
W	LV2	Channel width (W) (also the effective channel width for all MOSFET models except Levels 54, 57, 69 and 70)	All
AD	LV3	Area of the drain diode (AD)	All
AS	LV4	Area of the source diode (AS)	All
ICVDS	LV5	Initial condition for the drain-source voltage (VDS)	All
ICVGS	LV6	Initial condition for the gate-source voltage (VGS)	All
ICVBS	LV7	Initial condition for the bulk-source voltage (VBS)	All except 57, 58, 59, 70, 71
ICVES	LV7	Initial condition for the substrate-source voltage (VES)	57, 58, 59, 70, 71
–	LV8	Device polarity: • 1 = forward • -1 = reverse (not used after the PrimeSim Continuum solution release 95.3).	All
VTH	LV9	Threshold voltage (bias dependent)	All
VDSAT	LV10	Saturation voltage (VDSAT)	All
PD	LV11	Drain diode periphery (PD)	All

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
PS	LV12	Source diode periphery (PS)	All
RDS	LV13	Drain resistance (squares) (RDS) (equals the value of instance parameter nrd/nrs)	All
RSS	LV14	Source resistance (squares) (RSS) (equals the value of instance parameter nrd/nrs)	All
XQC	LV15	Charge-sharing coefficient (XQC).	All
GDEFF	LV16	Effective drain conductance (1/RDeff), rgeoMod is not 0	All
GSEFF	LV17	Effective source conductance (1/RSeff), rgeoMod is not 0	All
CDSAT	LV18	Drain-bulk saturation current, at -1 V bias.	All
CSSAT	LV19	Source-bulk saturation current, at -1 V bias.	All
VDBEFF	LV20	Effective drain bulk voltage.	All
BETAEFF	LV21	BETA effective	All
GAMMAEFF	LV22	GAMMA effective	All
DELTAL	LV23	ΔL (MOS6 amount of channel length modulation)	1, 2, 3, 6
UBEFF	LV24	UB effective	1, 2, 3, 6
VG	LV25	VG drive	1, 2, 3, 6
VFBEFF	LV26	VFB effective.	All
–	LV31	Drain current tolerance (not used in the PrimeSim Continuum solution releases after 95.3)	All
IDSTOL	LV32	Source-diode current tolerance	All
IDDTOL	LV33	Drain-diode current tolerance	All
COVLGS	LV36	Gate-source overlap and fringing capacitances	All
COVLGD	LV37	Gate-drain overlap and fringing capacitances	All
COVLGB	LV38	Gate-bulk overlap capacitances	All except 57, 59, 70, 71
COVLGE	LV38	Gate-substrate overlap capacitances	57, 59, 70, 71
VBD	LX0	Bulk-drain voltage	All
VBS	LX1	Bulk-source voltage (VBS)	All except 57, 59, 70, and 71

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
VES	LX1	Substrate-source voltage (VES)	57, 59, 70, 71
VGS	LX2	Gate-source voltage (VGS)	All
VDS	LX3	Drain-source voltage (VDS)	All
CDO**	LX4	Channel current (IDS)	All
CBSO	LX5	DC source-bulk diode current (CBSO)	All
CBDO	LX6	DC drain-bulk diode current (CBDO)	All
GMO**	LX7	DC MOSFET gate transconductance (GMO) • Current is I_{ds} , from drain-to-source, • Voltage is v_{gs}	All
GDSO**	LX8	DC drain-source conductance (GDSO)	All
GMBSO**	LX9	DC substrate transconductance (GMBSO)	All except 57, 58, 59, 70, 71
GMESO**	LX9	DC substrate transconductance (GMBSO/GMESO)	57, 58, 59, 70, 71
GBDO	LX10	Conductance of the drain diode (GBDO)	All
GBSO	LX11	Conductance of the source diode (GBSO)	All
QB	LX12	Total bulk (body) charge (QB)—Meyer and Charge Conservation	All
CQB	LX13	Bulk (body) charge current (CQB)—Meyer and Charge Conservation	All
QG	LX14	Total Gate charge (QG)—Meyer and Charge Conservation	All
CQG	LX15	Gate charge current (CQG)—Meyer and Charge Conservation	All
QD	LX16	Total Drain charge (QD)	49, 53
QD	LX16	Channel charge (QD)—Meyer and Charge Conservation	All except 49 and 53
CQD	LX17	Drain charge current (CQD)	49, 53
CQD	LX17	Channel charge current (CQD)—Meyer and Charge Conservation	All except 49 and 53
CGGBO	LX18	$CGGBO = dQ_g/dV_g = CGS + CGD + CGB$ - Meyer and Charge Conservation	All except 54, 57, 59, 60, 70, 71
CGGBO	LX18	Intrinsic gate capacitance	54, 57, 59, 60, 66, 70, 71

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
CGDBO	LX19	CGDBO = dQ_g/dV_d - Meyer and Charge Conservation; this cap is the total capacitance, including derivative of charge (dQ_g/dV_d) and overlap capacitance.	All except 54, 57, 59, 60, 66, 70, 71
CGDBO**	LX19	Intrinsic gate-to-drain capacitance	54, 57, 59, 60, 66, 70, 71
CGSBO	LX20	CGSBO = dQ_g/dV_s - Meyer and Charge Conservation	All except 54, 57, 59, 60, 66, 70, 71
CGSBO**	LX20	Intrinsic gate-to-source capacitance	54, 57, 59, 60, 66, 70, 71
CBGBO	LX21	CBGBO = dQ_b/dV_g - Meyer and Charge Conservation	All except 54, 57, 59, 60, 66, 70, 71
CBGBO	LX21	Intrinsic bulk-to-gate capacitance	54, 66
CBGBO	LX21	Intrinsic floating body-to-gate capacitance	57, 59, 60, 70, 71
CBDBO	LX22	CBDBO = dQ_b/dV_d - Meyer and Charge Conservation	All except 54, 57, 59, 60
CBDBO**	LX22	Intrinsic bulk-to-drain capacitance	54, 66
CBDBO**	LX22	Intrinsic floating body-to-drain capacitance	57, 59, 60, 70, 71
CBSBO	LX23	CBSBO = dQ_b/dV_s - Meyer and Charge Conservation	All except 54, 57, 59, 60, 66, 70, 71
CBSBO**	LX23	Intrinsic bulk-to-source capacitance	54, 66
CBSBO**	LX23	Intrinsic floating body-to-source capacitance	57, 59, 60, 70, 71
QBD	LX24	Drain-bulk charge (QBD)	All
–	LX25	Drain-bulk charge current (CQBD), (not used in the PrimeSim Continuum solution releases after 95.3).	All
QBS	LX26	Source-bulk charge (QBS)	All
–	LX27	Source-bulk charge current (CQBS), (not used after the PrimeSim Continuum solution release 95.3).	All
CAP_BS	LX28	Extrinsic drain to substrate Capacitances—Meyer and Charge Conservation. $CAP_BS = csbox + csesw$ - csbox is the substrate-to-source bottom capacitance - csesw is the substrate-to-source sidewall capacitance	57, 58, 70, 71
CAP_BS	LX28	Bias dependent bulk-source capacitance	All except 57, 58, 70, 71

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
CAP_BD	LX29	Extrinsic source to substrate Capacitances—Meyer and Charge Conservation. $CAP_BD = c_{dbox} + c_{desw}$ c_{dbox} is the substrate-to-drain bottom capacitance c_{desw} is the substrate-to-drain sidewall capacitance	57, 58, 70, 71
CAP_BD	LX29	Bias dependent bulk-drain capacitance	All except 57, 58, 70, 71
CQS	LX31	Channel-charge current (CQS).	All
CDGBO	LX32	CDGBO = dQ_d/dV_g - Meyer and Charge Conservation	All except 54, 57, 59, 60, 66, 70
CDGBO**	LX32	Intrinsic drain-to-gate capacitance	54, 57, 59, 60, 66, 70
CDDBO	LX33	CDDBO = dQ_d/dV_d - Meyer and Charge Conservation	All except 54, 57, 59, 60, 66, 70, 71
CDDBO**	LX33	Intrinsic drain capacitance	54, 57, 59, 60, 66, 70, 71
CDSBO	LX34	CDSBO = dQ_d/dV_s Drain-to-source capacitance - Meyer and Charge Conservation	All
GVDSAT (m*)	LX288 (m*)	vdsat according to G function method	54
QE	LX35	Substrate charge (QE)—Meyer and Charge Conservation	57, 58, 59, 70, 71
CQE	LX36	Substrate charge current (CQE)—Meyer and Charge Conservation	57, 58, 59, 70, 71
CDEBO**	LX37	CDEBO = dQ_d/dV_e intrinsic drain-to-substrate capacitance	57, 59, 70, 71
CBEBO	LX38	CBEBO = dQ_b/dV_e intrinsic floating body-to-substrate capacitance	59, 70, 71
igso	LX38	Gate-to-Source Current	49, 54, 69, 70
CEEBO	LX39	CEEBO = dQ_e/dV_e intrinsic substrate capacitance	59, 70, 71
igdo	LX39	Gate-to-Drain Current	49, 54, 69, 70
CEGBO	LX40	CEGBO = dQ_e/dV_g intrinsic substrate-to-gate capacitance	57, 59, 70, 71
CEDBO**	LX41	CEDBO = dQ_e/dV_d intrinsic substrate-to-drain capacitance	57, 59, 70, 71
CESBO**	LX42	CESBO = dQ_e/dV_s intrinsic substrate-to-source capacitance	57, 59, 70, 71

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
VBSI	LX43	Body-source voltage (VBS)—Meyer and Charge Conservation	57, 58, 59, 70, 71
ICH	LX44	Channel current—Meyer and Charge Conservation	57, 58, 59, 70, 71
IBJT	LX45	Parasitic BJT collector current—Meyer and Charge Conservation	57, 58, 59, 70, 71
III	LX46	Impact ionization current—Meyer and Charge Conservation	57, 58, 59, 70, 71
IGIDL	LX47	GIDL current—Meyer and Charge Conservation	57, 58, 59, 70, 71
ITUN	LX48	Tunneling current—Meyer and Charge Conservation	57, 58, 59, 70, 71
Qbacko	LX49	Back gate charge	57, 59, 70, 71
Ibp	LX50	Body contact current	57, 59, 70, 71
Sft	LX51	Value of the temperature node with shmod=1	57, 59, 70, 71
VBFLOAT	LX52	Internal body node voltage, if you do not specify the terminal	57, 59, 70, 71
Rbp	LX53	Combination of rbody and rhalo	57, 59, 70, 71
IGB	LX54	Gate tunneling current	57, 59, 70, 71
QSRCO	LX55	Total Source charge (Charge Conservation: $QS=-(QG+QD+QB)$)	49, 53
QSRCO	LX55	Total Source charge (Charge Conservation: $QS=-(QG+QD+QB+QE)$)	57, 59, 70, 71
CQs	LX56	Source charge current	57, 59, 70, 71
CGEBO	LX57	CGEBO = dQg/dV_e intrinsic gate-to-substrate capacitance	57, 59, 70, 71
CSSBO**	LX58	CSSBO = dQs/dV_s intrinsic source capacitance	57, 59, 70, 71
CSGBO**	LX59	CSGBO = dQs/dV_g intrinsic source-to-gate capacitance	57, 59, 70, 71
CSDBO**	LX60	CSDBO = dQs/dV_d intrinsic source-to-drain capacitance	57, 59, 70, 71
CSEBO**	LX61	CSEBO = dQs/dV_e intrinsic source-to-substrate capacitance	57, 59, 70, 71
weff	LX62	Effective channel width	54, 57, 66, 69, 70
leff	LX63	Effective channel length	54, 57, 66, 69, 70
weffcv	LX64	Effective channel width for CV	54, 66, 69

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
leffcv	LX65	Effective channel length for CV	54, 66, 69
igbo	LX66	Gate-to-Substrate Current ($I_{gb} = I_{gbacc} + I_{gbinv}$)	49, 54, 69
igcso	LX67	Source Partition of I_{gc}	49, 54, 69, 70
igcdo	LX68	Drain Partition of I_{gc}	49, 54, 69, 70
iimi**	LX69	Impact ionization current	54, 69
igidlo	LX70	Gate-induced drain leakage current	54, 69, 70
igdt	LX71	Gate Dielectric Tunneling Current ($I_g = I_{gs} + I_{gd} + I_{gcs} + I_{gcd} + I_{gb}$)	49, 54, 69
igc	LX72	Gate-to-Channel Current at zero V_{ds}	49, 54, 69
igbacc	LX73	Determined by ECB (Electron tunneling from the Conduction Band); significant in the accumulation	49, 54
igbinv	LX74	Determined by EVB (Electron tunneling from the Valence Band); significant in the inversion	49, 54
vfbsd	LX75	Flat-band Voltage between the Gate and S/D diffusions	54, 66
vgse	LX76	Effective Gate-to-Source Voltage	54, 66
vox	LX77	Voltage Across Oxide	54, 66
rdv	LX78	Asymmetric and Bias-Dependent Source Resistance, ($rdsMod = 1$)	54, 66
rsv	LX79	Asymmetric and Bias-Dependent Drain Resistance, ($rdsMod = 1$)	54, 66
cap_bsz	LX80	Zero voltage bias bulk-source capacitance	54, 66
cap_bdz	LX81	Zero voltage bias bulk-drain capacitance	54, 66
CGGBM	LX82	Total gate capacitance (including intrinsic), and all overlap and fringing components	54, 57, 59, 60, 66, 70, 69, 71
CGDBM	LX83	Total gate-to-drain capacitance (including intrinsic), and overlap and fringing components	54, 57, 59, 60, 66, 69, 70, 71
CGSBM	LX84	Total gate-to-source capacitance (including intrinsic), and overlap and fringing components	54, 57, 59, 60, 66, 69, 70, 71
CDDBM	LX85	Total drain capacitance (including intrinsic), overlap and fringing components, and junction capacitance	54, 57, 59, 60, 66, 69, 70, 71
CDSBM	LX86	Total drain-to-source capacitance	54, 57, 60, 66, 69, 70, 71

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
CDGBM	LX87	Total drain-to-gate capacitance (including intrinsic), and overlap and fringing components	54, 57, 59, 60, 66, 69, 70, 71
CBGBM	LX88	Total bulk-to-gate (floating body-to-gate) capacitance, including intrinsic and overlap components	54, 57, 59, 60, 66, 70, 71
CBDBM	LX89	Total bulk-to-drain capacitance (including intrinsic), and junction capacitance	54, 66
CBDBM	LX89	Total floating body-to-drain capacitance (including intrinsic), and junction capacitance.	57, 59, 60, 70, 71
CBSBM	LX90	Total bulk-to-source capacitance (including intrinsic), and junction capacitance	54, 66
CBSBM	LX90	Total floating body-to-source capacitance (including intrinsic), and junction capacitance.	57, 59, 60, 70, 71
CAPFG	LX91	Fringing capacitance	54, 66
CDEBM	LX92	Total drain-to-substrate capacitance (including intrinsic), and junction capacitance.	57, 59, 60, 70, 71
CSGBM	LX93	Total source-to-gate capacitance (including intrinsic), and overlap and fringing components.	57, 59, 60, 70, 71
CSSBM	LX94	Total source capacitance (including intrinsic), overlap and fringing components, and junction capacitance.	57, 59, 60, 70, 71
CSEBM	LX95	Total source-to-substrate capacitance (including intrinsic), and junction capacitance.	57, 59, 60, 70, 71
CEEBM	LX96	Total substrate capacitance (including intrinsic), overlap and fringing components, and junction capacitance.	57, 59, 60, 70, 71
QGI	LX97	Intrinsic Gate charge	49, 53
QSI	LX98	Intrinsic Source charge	49, 53
QDI	LX99	Intrinsic Drain charge	49, 53
QBI	LX100	Intrinsic Bulk charge (Charge Conservation: $QBI = -(QGI + QSI + QDI)$)	49, 53
CDDBI	LX101	Intrinsic drain capacitance; only includes derivative of charge	49, 53
CBDBI	LX102	Intrinsic bulk-to-drain capacitance; only includes derivative of charge	49, 53
CBSBI	LX103	Intrinsic bulk-to-source capacitance; only includes derivative of charge	49, 53

Name	Alias	Description	MOSFET Level ≤ 71, Level 74
VBDI	LX109	Body-drain voltage(VBD)—Meyer and Charge Conservation	57, 58, 59, 70, 71
IGISLO	LX110	Gate-induced source leakage current	54,70
GRIL	LX118	Intrinsic channel reflected gate conductance	54
GRGELTD	LX119	Gate electrode conductance	54
lbs1	LX120	Bulk to source diffusion current	57, 59, 60,70
lbd1	LX121	Bulk to drain diffusion current	57, 59, 60,70
lbs2	LX122	Bulk to source recombination/trap-assisted tunneling current	57, 59, 60,70
lbd2	LX123	Bulk to drain recombination/trap-assisted tunneling current	57, 59, 60,70
lbs3	LX124	Bulk to source recombination current in neutral body	57, 59, 60,70
lbd3	LX125	Bulk to drain recombination current in neutral body	57, 59, 60,70
lbs4	LX126	Bulk to source reversed bias tunneling leakage current	57, 59, 60,70
lbd4	LX127	Bulk to drain reversed bias tunneling leakage current	57, 59, 60,70
b4_sca	LX128	sca for WPE effect	54
b4_scb	LX129	scb for WPE effect	54
b4_scc	LX130	scc for WPE effect	54
b4_sc	LX131	sc for WPE effect	54
Ueff	LX132	Effective mobility at the specified analysis temperature.	66
VGB	LX133	Gate to bulk voltage	All
VDG	LX134	Drain to gate voltage	All
mult	LX135	Prints value of multiplier (M) for a specified MOSFET	All
b4_sa	LX136	sa for STI or LOD-induced mechanical stress-effects	54
b4_sb	LX137	sb for STI or LOD-induced mechanical stress-effects	54
b4_sd	LX138	sd for STI or LOD-induced mechanical stress-effects	54
b4_nf	LX139	nf for STI or LOD-induced mechanical stress-effects	54
b4_saeff	LX140	saeff for STI or LOD-induced mechanical stress-effects	54
b4_sbeff	LX141	sbeff for STI or LOD-induced mechanical stress-effects	54

Name	Alias	Description	MOSFET Level <= 71, Level 74
ivth (m*)	LX142 (m*)	Constant current-based Vth	49, 53, 54, 69, and 70
soiq0	LX143 (bqi)	Initial floating body charge at t=0 for BQI. If it is not given, 1e35 is printed. For tran analysis (t>0), the floating body charge is printed. This parameter only supports BQI.	57, 60, 70
LPOY	LX280	POLY Length	49,53,54,57, 69, 70
WPOLY	LX281	POLY Width	49,53,54,57, 69, 70
MOBEFF	LX282	Effective mobility at the specified analysis temperature. Note: For PSP, the effective mobility calculation considers geometry and temperature scaling, plus mechanical stress and electrical field-induced mobility change.	49,53,54,69,70
vdmargin (m*)	LX286 (m*)	Output based on .ivdmargin or .OPTION ivdmargin simulation	54
TOXE	LX287	Electrical gate equivalent oxide thickness	54
RDSW	LX289	Zero bias internal Rds per unit width	54
VSAT	LX290	Saturation velocity	54
liibjt	LX144	lii component induced by parasitic bjt	70
lgp	LX145	Gate tunneling component in the AGBCP2 region	70
Vbs0	LX146	Body-source built-in potential lowering	70
Vbseff	LX147	Effective body potential	70
GMVTH(m*)	LX148(m*)	Vth using the maximum gm method	54

Note:

** indicates electrical notation.

Additional Output Templates for PSP

The following output templates do not require the PrimeSim Continuum simulator alias and are all in support of the PSP model.

Name	Description	Model Level
VFB	Flat-band voltage at TR after geometry scaling	69
STVFB	Temperature dependence of VFB after geometry scaling	69
TOX	Gate oxide thickness after geometry scaling	69
EPSROX	Relative permittivity of gate dielectric after geometry scaling	69
NEFF	Substrate doping after geometry scaling	69
VNSUB	Effective doping bias-dependence parameter after geometry scaling	69
NSLP	Effective doping bias-dependence parameter after geometry scaling	69
DNSUB	Effective doping bias-dependence parameter after geometry scaling	69
DPHIB	Offset of ϕ_B after geometry scaling	69
NP	NP Gate poly-silicon doping after geometry scaling	69
CT	Interface states factor after geometry scaling	69
TOXOV	Overlap oxide thickness after geometry scaling	69
NOV	Effective doping of overlap region after geometry scaling	69
CF	DIBL parameter after geometry scaling	69
CFB	Back-bias dependence of CF after geometry scaling	69
BETN	Product of channel aspect ratio and zero field mobility at TR after geometry scaling	69
STBET	Temperature dependence of BETN after geometry scaling	69
MUE	Mobility reduction coefficient at TR after geometry scaling	69
STMUE	Temperature dependence of MUE after geometry scaling	69
THEMU	Mobility reduction exponent at TR after geometry scaling	69
STTHEMU	Temperature dependence of THEMU after geometry scaling	69
CS	Coulomb scattering parameter at TR after geometry scaling	69
STCS	Temperature dependence of CS after geometry scaling	69
XCOR	Non-universality parameter after geometry scaling	69

Name	Description	Model Level
STXCOR	Temperature dependence of XCOR after geometry scaling	69
FETA	Effective field parameter after geometry scaling	69
RS	Source/drain series resistance at TR after geometry scaling	69
STRS	Temperature dependence of RS after geometry scaling	69
RSB	Back-bias dependence of RS after geometry scaling	69
RSG	Gate-bias dependence of RS) after geometry scaling	69
THESAT	Velocity saturation parameter at TR after geometry scaling	69
STTHESAT	Temperature dependence of THESAT after geometry scaling	69
THESATB	Back-bias dependence of velocity saturation after geometry scaling	69
THESATG	Gate-bias dependence of velocity saturation) after geometry scaling	69
AX	Linear/saturation transition factor after geometry scaling	69
ALP	CLM pre-factor after geometry scaling	69
ALP1	CLM enhancement factor above threshold) after geometry scaling	69
ALP2	CLM enhancement factor below threshold) after geometry scaling	69
VPO	CLM logarithmic dependence parameter) after geometry scaling	69
A1	Impact-ionization pre-factor after geometry scaling	69
A2	Impact-ionization exponent at TR after geometry scaling	69
STA2	Temperature dependence of A2 after geometry scaling	69
A3	Saturation-voltage dependence of II after geometry scaling	69
A4	Back-bias dependence of II after geometry scaling	69
GCO	Gate tunneling energy adjustment after geometry scaling	69
IGINV	Gate channel current pre-factor after geometry scaling	69
IGOV	Gate overlap current pre-factor after geometry scaling	69

Name	Description	Model Level
STIG	Temperature dependence of gate current after geometry scaling	69
GC2	Gate current slope factor after geometry scaling	69
GC3	Gate current curvature factor after geometry scaling	69
CHIB	Tunneling barrier height) after geometry scaling	69
AGIDL	GIDL pre-factor after geometry scaling	69
BGIDL	GIDL probability factor at TR after geometry scaling	69
STBGIDL	Temperature dependence of BGIDL after geometry scaling	69
CGIDL	Back-bias dependence of GIDL after geometry scaling	69
COX	Oxide capacitance for intrinsic channel after geometry scaling	69
CGOV	Oxide capacitance for gate-drain/source overlap after geometry scaling	69
CGBOV	Oxide capacitance for gate-bulk overlap after geometry scaling	69
CFR	Outer fringe capacitance after geometry scaling	69
NFA	First coefficient of flicker noise after geometry scaling	69
NFB	Second coefficient of flicker noise) after geometry scaling	69
NFC	Third coefficient of flicker noise after geometry scaling	69
KVTHOWE	Threshold shift parameter after geometry scaling	69
KUOWE	Mobility degradation factor after geometry scaling	69
TOXOVD	Overlap oxide thickness for drain side after geometry scaling	69
NOVD	Effective doping of overlap region for drain side after geometry scaling	69
IGOVD	Gate overlap current pre-factor for drain side after geometry scaling	69
AGIDLD	GIDL pre-factor for drain side after geometry scaling	69
BGIDLD	GIDL probability factor at TR for drain side after geometry scaling	69
STBGIDLD	Temperature dependence of BGIDL for drain side after geometry scaling	69

Name	Description	Model Level
CGIDLD	Back-bias dependence of GIDL for drain side after geometry scaling	69
CGOVD	Oxide capacitance for gate-drain/source overlap for drain side after geometry scaling	69
CFRD	Outer fringe capacitance for drain side after geometry scaling	69
EF	Flicker noise frequency exponent after geometry scaling	69
FACNEFFAC	Pre-factor for effective substrate doping in separate charge calculation when SWDELVTAC = 1 after geometry scaling	69
GFACNUD	Body factor change due to NUD-effect after geometry scaling	69
VSBNUD	Lower VSB-value for NUD-effect after geometry scaling	69
DVSBNUD	VSB-range for NUD-effect after geometry scaling	69
DELVTAC	Offset of ϕB in separate charge calculation when SWDELVTAC=1 after geometry scaling	69
CGGI	Intrinsic gate capacitance	69
CGDI	Intrinsic gate to drain capacitance	69
CGSI	Intrinsic gate to source capacitance	69

Output Templates for BSIM-CMG (Level 72), BSIM6 (Level 77), and BSIM-IMG (Level 78)

The following table lists parameters in the MOSFET BSIM-CMG, BSIM6, and BSIM-IMG output templates. The last column indicates which model levels support each parameter.

Name	Alias	Description	MOSFET Level
ICVDS	LV5	Initial condition for the drain-source voltage (VDS)	72, 77
ICVGS	LV6	Initial condition for the gate-source voltage (VGS)	72, 77
ICVBS	LV7	Initial condition for the bulk-source voltage (VBS)	72, 77
IVTH	LX142/LX592	Constant current-based VTH	72, 78
VTH	LV9	Threshold voltage (bias dependent)	72, 77
BETAEFF	LV21	BETA effective	72, 77, 78

Name	Alias	Description	MOSFET Level
VBD	LX0	Bulk-drain voltage	72, 77, 78
VBS	LX1	Bulk-source voltage (VBS)	72, 77, 78
VGS	LX2	Gate-source voltage (VGS)	72, 77, 78
VDS	LX3	Drain-source voltage (VDS)	72, 77, 78
CDO**	LX4	Channel current (IDS)	72, 77, 78
CBSO	LX5	DC source-bulk diode current (CBSO)	72, 77
CBDO	LX6	DC drain-bulk diode current (CBDO)	72, 77
GMO**	LX7	DC MOSFET gate transconductance (GMO)	72, 77, 78
GDSO**	LX8	DC drain-source conductance (GDSO)	72, 77, 78
GMBSO**	LX9	DC substrate transconductance (GMBSO)	72, 77, 78
GVDSAT	LX288	vdsat according to G function method	72, 77, 78
VGB	LX133	Gate to bulk voltage	72, 77, 78
VDG	LX134	Drain to gate voltage	72, 77, 78
IVTH	LX142	Constant current based VTH	72, 77, 78
VDMARGIN	LX286	Output based on .ivdmargin or .OPTION ivdmargin simulation	72, 77, 78
LEFF	LX501	Effective channel length	72, 77, 78
WEFF	LX502	Effective channel width per fin	72
WEFF	LX502	Effective channel width	77, 78
LEFFCV	LX503	Effective channel length for CV	72, 77, 78
WEFFCV	LX504	Effective channel width per fin for CV	72
WEFFCV	LX504	Effective channel width for CV	77, 78
AS	LX505	Source area	77, 78
AS	LX505	Source area (version <=104.1)	72
AD	LX506	Drain area	77, 78
AD	LX506	Drain area (version <=104.1)	72

Name	Alias	Description	MOSFET Level
PS	LX507	Source perimeter	77, 78
PS1	LX507	1st part of the source perimeter (version <=104.1)	72
PD	LX508	Drain perimeter	77, 78
PD1	LX508	1st part of the drain perimeter (version <=104.1)	72
PS2	LX509	2nd part of the source perimeter (version <=104.1)	72
PD2	LX510	2nd part of the drain perimeter (version <=104.1)	72
D	LX511	Diameter of cylinder	72
NFIN	LX512	Number of fins per finger	72
NFINTOTAL	LX513	Total fins (NFIN*NF)	72
TFIN	LX514	Fin thickness	72
CGSP	LX515	Constant gate to source fringe capacitance	72
CGDP	LX516	Constant gate to drain fringe capacitance	72
CDSP	LX517	Constant drain to source fringe capacitance	72
NBODY	LX518	Channel doping concentration, effective	72, 78
PHIG	LX519	Work function of gate, effective	72
NGATE	LX520	Poly Gate doping, effective	72, 77
U0	LX521	Low field mobility, effective	72, 77, 78
MEXP	LX522	Smoothing function factor for Vdsat, effective	72, 78
VSAT	LX523	Saturation velocity, effective	72, 77, 78
UA	LX524	Phonon / surface roughness scattering, effective	72, 77, 78
UD	LX525	Columbic scattering, effective	72, 77, 78
CS	LX526	Coulombic scattering parameter, effective	77, 78
THETAMU	LX527	Mobility reduction exponent, effective	72
THETASAT	LX528	Velocity saturation parameter, effective	72
PTWG	LX529	Correction factor for velocity saturation, effective	72, 77, 78
MUE	LX530	Mobility reduction co-efficient, effective	72

Name	Alias	Description	MOSFET Level
NSD	LX531	S/D doping concentration, effective	72, 77, 78
RSOURCEGO	LX532	Bias independent source resistance	72
RDRAIN GEO	LX533	Bias independent drain resistance	72
GTH	LX534	Thermal conductance	72, 77, 78
CTH	LX535	Thermal capacitance	72, 77, 78
VDSAT	LV10/LX536	Saturation voltage	72, 77, 78
UEFF	LX537	Effective mobility	72, 77, 78
RSOURCE	LX538	Total source resistance	72, 77, 78
RDRAIN	LX539	Total drain resistance	72, 77, 78
RGELTD	LX540	Gate electrode resistance	72, 77, 78
III**	LX541	Impact ionization current	72, 77, 78
IGBINV	LX542	Gate to body current	72, 77, 78
IGC	LX543	Gate-to-Channel current at zero Vds	72
IGIDL	LX544	Gate-induced drain leakage current	72, 77, 78
IGISL	LX545	Gate-induced source leakage current	72, 77, 78
IGSO	LX546	Total gate-to-source current (Igs+Igcs)	72, 77, 78
IGDO	LX547	Total gate-to-drain current (Igd+Igcd)	72, 77, 78
QD	LX548	Intrinsic drain charge	72, 77, 78
QG	LX549	Intrinsic gate charge	72, 77, 78
IDS	LX550	Channel current	72, 77, 78
QG_TOTAL	LX551	Total Gate charge (intrinsic+extrinsic)	72, 77, 78
QD_TOTAL	LX552	Total Drain charge (intrinsic+extrinsic)	72, 77, 78
QS_TOTAL	LX553	Total Source charge (intrinsic+extrinsic)	72, 77, 78
QB_TOTAL	LX554	Total Bulk charge (intrinsic+extrinsic)	72, 77, 78
IGS	LX555	Tunneling current due to gate-to-source overlap (Igs).	72, 77, 78
IGD	LX556	Tunneling current due to gate-to-drain overlap (Igd).	72, 77, 78

Name	Alias	Description	MOSFET Level
IGBS	LX557	Source part of gate to Body Tunneling current for BULKMOD=0	72
IGBD	LX558	Drain part of gate to Body Tunneling current for BULKMOD=0	72
IGCS	LX559	Source Partition of IGC	72, 77, 78
IGCD	LX560	Drain Partition of IGC	72, 77, 78
RSGEO	LX561	Bias independent Source Resistance (Alias to RSOURCEGO)	72
RDGEO	LX562	Bias independent Drain Resistance (Alias to RDRAINGO)	72
CFGEO	LX563	Geometric Parasitic Cap for CGEOMOD=2	72
DIDSDVG	LX564	Gate transconductance (dI_{ds}/dV_g)	72, 77, 78
DIDSDVS	LX565	Source conductance (dI_{ds}/dV_s)	72, 77, 78
DIDSDVD	LX566	Drain conductance (dI_{ds}/dV_d)	72, 77, 78
DIDSDVTH	LX567	Temperature dependence of channel current (dI_{ds}/dT)	72, 77, 78
CGG	LX568	Total gate capacitance ($CGG = dQ_g/dV_g$)	72, 77, 78
CGS	LX569	Total gate-to-source capacitance ($CGS = -dQ_g/dV_s$)	72, 77, 78
CGD	LX570	Total gate-to-drain capacitance ($CGD = -dQ_g/dV_d$)	72, 77, 78
CSG	LX571	Total source-to-gate capacitance ($CSG = -dQ_s/dV_g$)	72, 77, 78
CSD	LX572	Total source-to-drain capacitance ($CSD = -dQ_s/dV_d$)	72, 77, 78
CSS	LX573	Total Source capacitance ($CSS = dQ_s/dV_s$)	72, 77, 78
CDG	LX574	Total drain-to-gate capacitance ($CDG = -dQ_d/dV_g$)	72, 77, 78
CDD	LX575	Total drain capacitance ($CDD = dQ_d/dV_d$)	72, 77, 78
CDS	LX576	Total drain-to-source capacitance ($CDS = -dQ_d/dV_s$)	72, 77, 78
CGT	LX577	Temperature dependence of gate charge (dQ_g/dT)	72, 77, 78
CST	LX578	Temperature dependence of source charge (dQ_s/dT)	72, 77, 78
CDT	LX579	Temperature dependence of drain charge (dQ_d/dT)	72, 77, 78
ITH	LX580	Channel power ($I_{ds} \cdot V_{ds}$)	72, 77, 78

Name	Alias	Description	MOSFET Level
CJDB	LX581	Bias dependent bulk-drain capacitance	72, 77
CJSB	LX582	Bias dependent bulk-source capacitance	72, 77
COVLGS	LX583	Gate-source overlap and fringing capacitances	72, 77, 78
COVLGD	LX584	Gate-drain overlap and fringing capacitances	72, 77, 78
COVLGB	LX585	Gate-bulk overlap capacitances	72, 77, 78
CGGBO	LX586	Intrinsic gate capacitance	72, 77, 78
CGDBO	LX587	Intrinsic gate-to-drain capacitance	72, 77, 78
CGSBO	LX588	Intrinsic gate-to-source capacitance	72, 77, 78
CAP_BS	LX589	Bias dependent bulk-source capacitance (alias to CJSB)	72, 77, 78
CAP_BD	LX590	Bias dependent bulk-drain capacitance (alias to CJDB)	72, 77, 78
CGGBM	LX591	Total gate capacitance (alias to CGG)	72, 77, 78
IVTH	LX142/LX592	Constant current-based VTH	72
DIDSDVB	LX593	Bulk transconductance (dI_{ds}/dV_b)	72, 77, 78
SFT	LX594	Self-heating temperature	72, 77, 78
CGB	LX595	Total gate-to-bulk capacitance ($CGB = -dQ_g/dV_b$)	72, 77, 78
CDB	LX596	Total drain-to-bulk capacitance ($CDB = -dQ_d/dV_b$)	72, 77, 78
CSB	LX597	Total source-to-bulk capacitance ($CSB = -dQ_s/dV_b$)	72, 77, 78
CBG	LX598	Total bulk-to-gate capacitance ($CBG = -dQ_b/dV_g$)	72, 77, 78
CBD	LX599	Total bulk-top-drain capacitance ($CBD = -dQ_b/dV_d$)	72, 77, 78
CBS	LX600	Total bulk-to-source capacitance ($CBS = -dQ_b/dV_s$)	72, 77, 78
CBB	LX601	Total bulk capacitance ($CBB = dQ_b/dV_b$)	72, 77, 78
CGBBO	LX602	Intrinsic gate-to-bulk capacitance	72, 77, 78
CSGBO	LX603	Intrinsic source-to-gate capacitance	72, 77, 78
CSDBO	LX604	Intrinsic source-to-drain capacitance	72, 77, 78
CSSBO	LX605	Intrinsic source capacitance	72, 77, 78
CSBBO	LX606	Intrinsic source-to-bulk capacitance	72, 77, 78

Name	Alias	Description	MOSFET Level
CDGBO	LX607	Intrinsic drain-to-gate capacitance	72, 77, 78
CDDBO	LX608	Intrinsic drain capacitance	72, 77, 78
CDSBO	LX609	Intrinsic drain-to-source capacitance	72, 77, 78
CDBBO	LX610	Intrinsic drain-to-bulk capacitance	72, 77, 78
CBGBO	LX611	Intrinsic bulk-to-gate capacitance	72, 77, 78
CBDBO	LX612	Intrinsic bulk-to-drain capacitance	72, 77, 78
CBSBO	LX613	Intrinsic bulk-to-source capacitance	72, 77, 78
CBBBO	LX614	Intrinsic bulk capacitance	72, 77, 78
CGSBM	Lx615	Total gate-to-source capacitance (alias to CGS)	77, 78
CGDBM	Lx616	Total gate-to-drain capacitance (alias to CGD)	77, 78
CAP_BSZ	Lx617	Zero voltage bias bulk-source capacitance	77
CAP_BDZ	Lx618	Zero voltage bias bulk-drain capacitance	77
CAPFG	Lx619	Fringing capacitance	77, 78
PHIS**	Lx620	Surface potential at source end	72
PHID**	Lx621	Surface potential at drain end	72
IGB	LX622	Sum of gate to body currents ($I_{gb} = I_{gbacc} + I_{gbinv}$)	72
IGBACC	LX623	Gate to body current under accumulation	72
GMVTH	LX148	Vth using the maximum gm method	72, 77, 78

Note:

** indicates electrical notation.

Output Templates for HiSIM_HV, Level=73

Note:

** indicates electrical notation.

Name	Alias	Description
L	LX291	Channel Length (L)
W	LX292	Channel Width (W)
AD	LX293	Area of the drain diode (AD)
AS	LX294	Area of the source diode (AS)
ICVDS	LV5/LX295	Initial condition for the drain-source voltage (VDS)
ICVGS	LV6/LX296	Initial condition for the gate-source voltage (VGS)
ICVBS	LV7/LX297	Initial condition for the bulk-source voltage (VBS)
PD	LX298	Drain diode periphery (PD)
PS	LX299	Source diode periphery (PS)
RDS	LX300	Drain resistance (squares) (RDS) (equals the value of instance parameter rd)
RSS	LX301	Drain resistance (squares) (RSS) (equals the value of instance parameter rs)
WEFF	LX302	Effective channel width
LEFF	LX303	Effective channel length
CAPFG	LX304	Fringing capacitance
VTH	LV9/LX305	Threshold voltage (bias dependent)
VDSAT	LV10/LX306	Saturation voltage (VDSAT)
BETAEFF	LX310	BETA effective
GAMMAEFF	LX311	GAMMA effective
VFBEFF	LX312	VFB effective
COVLGS	LX313	Gate-source overlap and fringing capacitances
COVLGD	LX314	Gate-drain overlap and fringing capacitances

Name	Alias	Description
COVLGB	LX315	Gate-bulk overlap capacitances
VBD	LX0/LX316	Bulk-drain voltage
VBS	LX1/LX317	Bulk-source voltage (VBS)
VGS	LX2/LX318	Gate-source voltage (VGS)
VDS	LX3/LX319	Drain-source voltage (VDS)
CDO	LX4/LX320	Channel current (IDS)
CBSO	LX5/LX321	DC source-bulk diode current (CBSO)
CBDO	LX6/LX322	DC drain-bulk diode current (CBDO)
GMO	LX7/LX323	DC MOSFET gate transconductance (GMO)
GDSO	LX8/LX324	DC drain-source conductance (GDSO)
GMBSO	LX9/LX325	DC substrate transconductance (GMBSO)
GBDO	LX326	Conductance of the drain diode (GBDO)
GBSO	LX327	Conductance of the source diode (GBSO)
QB	LX328	Total bulk (body) charge (QB)
CQB	LX329	Bulk (body) charge current (CQB)
QG	LX330	Total Gate charge (QG)
CQG	LX331	Gate charge current (CQG)
QD	LX332	Total internal drain charge (QD)
CQD	LX333	Drain charge current (CQD)
CGGBO	LX334	Intrinsic gate capacitance
CGDBO	LX335	Intrinsic gate-to-drain capacitance
CGSBO	LX336	Intrinsic gate-to-source capacitance
CBGBO	LX337	Intrinsic bulk-to-gate capacitance
CBDBO	LX338	Intrinsic bulk-to-drain capacitance

Name	Alias	Description
CBSBO	LX339	CBSBO = $-dQ_b/dV_s$
QBD	LX340	Drain-bulk charge (QBD)
QBS	LX341	Source-bulk charge (QBS)
CAP_BS	LX342	Bias dependent bulk-source capacitance
CAP_BD	LX343	Bias dependent bulk-drain capacitance
CDGBO	LX344	Intrinsic drain-to-gate capacitance
CDDBO	LX345	Intrinsic drain capacitance
CDSBO	LX346	CDSBO = $-dQ_d/dV_s$, Drain-to-source capacitance
SFT	LX347	Value of the temperature node
weffcv	LX348	Effective channel width for CV
leffcv	LX349	Effective channel length for CV
igso	LX350	Gate-to-Source Current (including $igcso$)
igdo	LX351	Gate-to-Drain Current (including $igcdo$)
igbo	LX352	Gate-to-Substrate Current
igcso	LX353	Source Partition of I_{gc}
igcdo	LX354	Drain Partition of I_{gc}
iimi**	LX355	Impact ionization current
igidlo	LX356	Gate-induced drain leakage current
igislo	LX357	Gate-induced source leakage current
igdt	LX358	Gate Dielectric Tunneling Current ($I_{gs} + I_{gd} + I_{gb}$)
rdv	LX360	Asymmetric and Bias-Dependent Source Resistance
rsv	LX361	Asymmetric and Bias-Dependent Drain Resistance

Name	Alias	Description
cap_bsz	LX362	Zero voltage bias bulk-source capacitance
cap_bdz	LX363	Zero voltage bias bulk-drain capacitance
CGGBM	LX364	Total gate capacitance (including intrinsic), and overlap and fringing components
CGDBM	LX365	Total gate-to-drain capacitance (including intrinsic), and overlap and fringing components
CGSBM	LX366	Total gate-to-source capacitance (including intrinsic), and overlap and fringing components
CDDBM	LX367	Total drain capacitance (including intrinsic), overlap and fringing components, and junction capacitance
CDSBM	LX368	Total drain-to-source capacitance
CDGBM	LX369	Total drain-to-gate capacitance (including intrinsic), and overlap and fringing components
CBGBM	LX370	Total bulk-to-gate (floating body-to-gate) capacitance, including intrinsic and overlap components
CBDBM	LX371	Total bulk-to-drain capacitance (including intrinsic) and junction capacitance
CBSBM	LX372	Total bulk-to-source capacitance (including intrinsic), and junction capacitance
Ueff	LX373	Effective mobility at the specified analysis temperature
VGB	LX374	Gate to bulk voltage
VDG	LX375	Drain to gate voltage
mult	LX376	Prints value of multiplier (M) for a specified MOSFET

Name	Alias	Description
sa	LX377	sa for STI or LOD-induced mechanical stress-effects
sb	LX378	sb for STI or LOD-induced mechanical stress-effects
sd	LX379	sd for STI or LOD-induced mechanical stress-effects
hsmhv_nf	LX380	nf for STI or LOD-induced mechanical stress-effects
IVTH	LX142	Constant current-based Vth

Output Templates for UTSOI, Level=76

Name	Alias	Description
L	LV1	Channel length (L)
W	LV2	Channel width (W)
VTH	LV9	Threshold voltage
VDSAT	LV10	Saturation voltage
VBD	LX0	Bulk-drain voltage
VBS	LX1	Bulk-source voltage
VGS	LX2	Gate-source voltage
VDS	LX3	Drain-source voltage
CDO**	LX4	Channel current (IDS)
GMO**	LX7	DC MOSFET gate transconductance
GDSO**	LX8	DC drain-source conductance
GMBSO**	LX9	DC substrate transconductance
GVDSAT	LX288	vdsat according to G function method
QB	LX12	Total bulk charge
CQB	LX13	Bulk charge current

Name	Alias	Description
QG	LX14	Total gate charge
CQG	LX15	Gate charge current
QD	LX16	Total drain charge
CQD	LX17	Drain charge current
CGGBO	LX18	Total gate capacitance (Including intrinsic/overlap/fringing)
CGDBO	LX19	Gate to drain capacitance (Including intrinsic/overlap/fringing)
CGSBO	LX20	Gate to source capacitance (Including intrinsic/overlap/fringing)
CBGBO	LX21	Substrate to gate capacitance (Including intrinsic/overlap/fringing)
CBDBO	LX22	Substrate to drain capacitance (Including intrinsic/overlap/fringing)
CBSBO	LX23	Substrate to source capacitance (Including intrinsic/overlap/fringing)
CDGBO	LX32	Drain to gate capacitance (Including intrinsic/overlap/fringing)
CDDBO	LX33	Total drain capacitance (Including intrinsic/overlap/fringing)
CDSBO	LX34	Drain to source capacitance (Including intrinsic/overlap/fringing)
IGSO	LX501	Gate to source current
IGDO	LX502	Gate to drain current
IGIDL	LX503	Gate induced drain leakage current
IGISL	LX504	Gate induced source leakage current
CGSF	LX505	Fringing capacitance at source side
CGDF	LX506	Fringing capacitance at drain side
CGSOV	LX507	Overlap capacitance at source side
CGDOV	LX508	Overlap capacitance at drain side
CGBOV	LX509	Oxide capacitance for gate to substrate overlap
CBDE	LX510	Drain to substrate extrinsic capacitance
CBSE	LX511	Source to substrate extrinsic capacitance

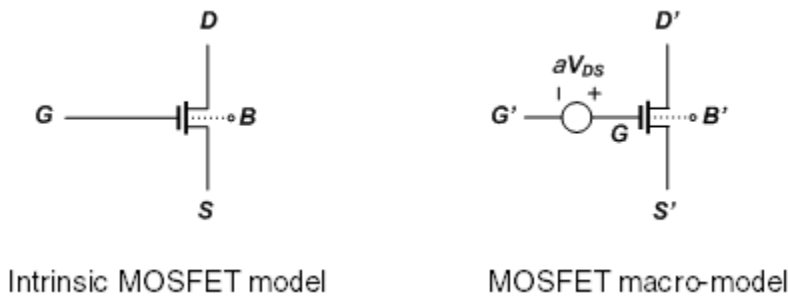
Name	Alias	Description
CDSE	LX512	Drain to source coupling capacitance
SFT	LX513	Value of the temperature node
WEFF	LX514	Effective channel width
LEFF	LX515	Effective channel length
GMVTH	LX148	Vth using the maximum gm method

Note:

** indicates electrical notation.

MOSFET SUBCKT Output Templates

Given modeling limitations of intrinsic MOSFET models such as BSIM4 across process corners, it is necessary to define a MOSFET model as a SUBCKT macro-model comprising not only the intrinsic MOSFET model but also additional circuit elements, so that the resulting SUBCKT describes expected silicon behavior more accurately. The following is an illustration of a simple SUBCKT macro-model where an intrinsic BSIM4.6 MOSFET model has been modified by a voltage-dependent voltage source in series with the gate input to modify DIBL behavior

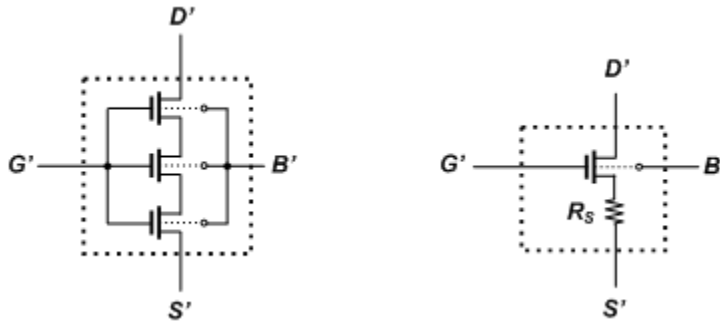


The conventional PrimeSim Continuum simulator output template infrastructure provides MOSFET biasing information such as V_{GS} , V_{GD} , V_{DS} , V_T , I_D , I_G , I_B , g_m , g_{ds} , and V_{Dsat} for only the intrinsic MOSFET model inside the SUBCKT, whereas the output template of the overall SUBCKT macro-model is really desired. To overcome this limitation, the output template infrastructure has been extended to report critical device characteristics of a SUBCKT macro-model.

Conventional output template = $f(V_D, V_G, V_S, V_B)$

Macro-model output template = $f(V_{D'}, V_{G'}, V_{S'}, V_{B'})$

The output template capability has been generalized to facilitate macro-model hierarchy for a generalized 4-terminal MOSFET structure. The following illustration shows a macro-model defined as a composite stack of MOSFETs (each MOSFET potentially a macro-model itself) with a common gate connection or as a MOSFET with resistive source degeneration. These are typical approaches in nanoscale analog design to overcome channel length and output resistance limitations.



In these situations, the designer is primarily interested in the collective behavior of the stacked MOSFETs, for example, as opposed to the individual behavior of each MOSFET in the stack.

Macro-model output templates are calculated based on probing the SUBCKT ports or terminals and observing the subsequent “black-box” behavior. Such terminal-based measurements facilitate easier correlation of simulations to silicon measurements. Parameters such as V_{dsat} that are based on model-specific equations are no longer available as output templates for the SUBCKT macro-model.

Analysis Type: DC/AC/OP and user specified TRAN point (normal TRAN analysis is not supported as runtime impact would be too significant).

Example of user-specified TRAN point: `.op 1n 2.5n 10n` (The SUBCKT output templates are probed and available only at the three specified time instances and are not available at other time instances in the TRAN analysis).

For example:

```
.print dc LX7(X1)
.print ac LX7(X1)
.print tran LX7(X1)
```

Models: No restrictions except for U, W, T, S, B, P and VEC elements.

Date Input Requirement: SUBCKT consists of exactly 4 terminals with order fixed as Drain (d), Gate (g), Source (s), and Bulk (b) which can constrain number of levels of hierarchy.

For example:

```
.subckt nch_mac    d g s b
+ w=0 l=0
rd d d1 r=20
rg g g1 r=20
rs s s1 r=20
main d1 g1 s1 b nch w='w' l='l'
.ends
.subckt nmos      d g s b
+ w=0 l=0
rd d d1 r=100
rg g g1 r=100
rs s s1 r=100
XMN d1 g1 s1 b nch_mac w='w' l='l'
.ends
X1 d g s b nmos  w='w' l='l'
```

Note:

LX7(X1) and LX7(X1.XMN) are all supported.

Note:

** indicates electrical notation.

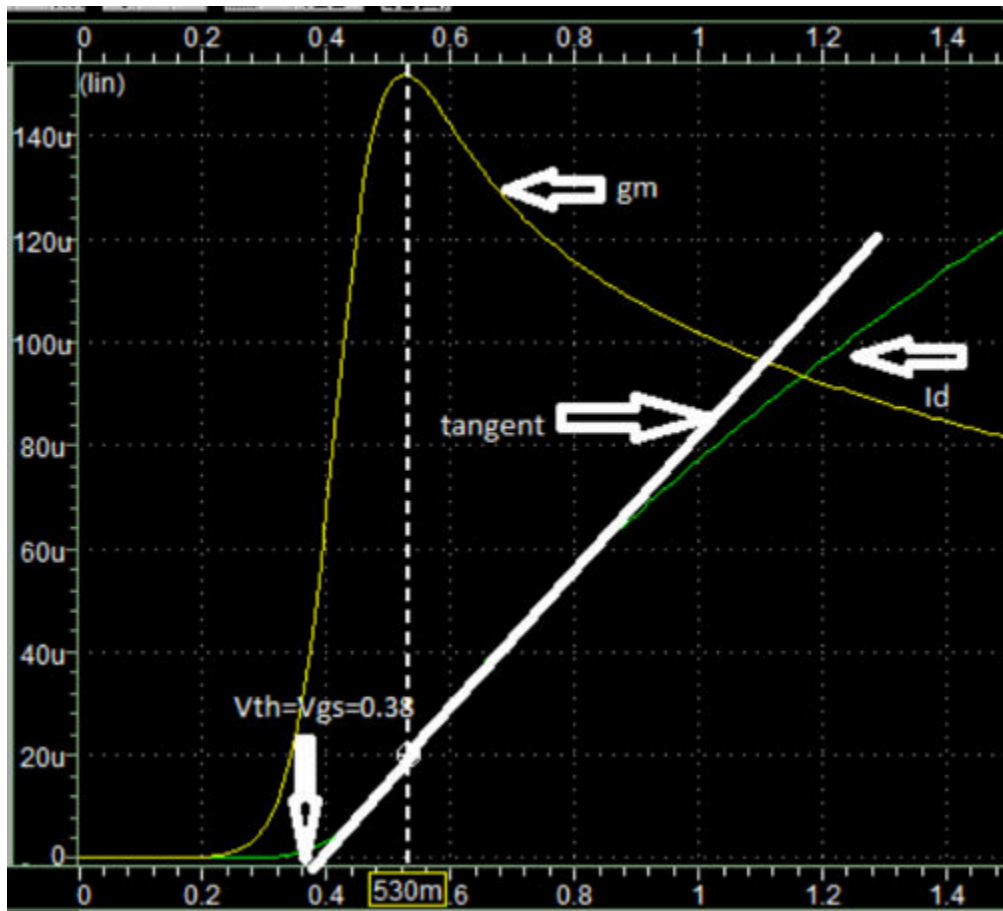
Name	Alias	Description
VBD	LX0	SUBCKT body to drain voltage ($V_{B'} - V_{D'}$).
VBS	LX1	SUBCKT body to source voltage ($V_{B'} - V_{S'}$).
VGS	LX2	SUBCKT gate to source voltage ($V_{G'} - V_{S'}$).
VDS	LX3	SUBCKT drain to source voltage ($V_{D'} - V_{S'}$).
VDG	LX134	SUBCKT drain to gate voltage ($V_{D'} - V_{G'}$).
ID		Current flow into/out of SUBCKT D', flow into SUBCKT is "+".
IG		Current flow into/out of SUBCKT G', flow into SUBCKT is "+".
IS		Current flow into/out of SUBCKT S', flow into SUBCKT is "+".
IB		Current flow into/out of SUBCKT B', flow into SUBCKT is "+".
GM**	LX7	SUBCKT gate transconductance ($d i(V_{D'}) / d V_{G'}$).
GDS**	LX8	SUBCKT drain conductance ($d i(V_{D'}) / d V_{D'}$).
GMBS**	LX9	SUBCKT body transconductance ($d i(V_{D'}) / d V_{B'}$).
GVDSAT	LX288	vdsat according to G function method

Name	Alias	Description
IVTH	LX142	Constant-current based threshold voltage. Device W and L must be explicitly declared to compute LX142.
VDMARGIN	LX286	MOSFET drain voltage margin available before MOSFET drain conductance degrades (increases) by user-specified target with respect to existing bias point.
CGG	LX82	Subckt gate capacitance dQ_g/dV_g .
CGD	LX83	Subckt gate-drain capacitance dQ_g/dV_d .
CGS	LX84	Subckt gate-source capacitance dQ_g/dV_s .
CDD	LX85	Subckt drain capacitance dQ_d/dV_d .
CDS	LX86	Subckt drain-source capacitance dQ_d/dV_s .
CDG	LX87	Subckt drain-gate capacitance dQ_d/dV_g .
CBG	LX88	Subckt body-gate capacitance dQ_b/dV_g .
CBD	LX89	Subckt body-drain capacitance dQ_b/dV_d .
CBS	LX90	Subckt body-source capacitance dQ_b/dV_s .
GMVTH	LX148	Vth using the maximum gm method.

Vth Using the Maximum gm (gmmax) Method

The gmmax-based Vth is an approximated $I_{ds_vs_V_{gs}}$ equation in the linear region. The $I_{ds_vs_V_{gs}}$ at the linear region is approximated as $I_{ds} \sim gm * (V_{gs} - V_{th} - 1/2 * V_{ds})$.

To perform the calculation, keep the transistor biased in the linear region and sweep V_{gs} . Set the tangent at the maximum gm point on the $I_{ds_vs_V_{gs}}$ curve. The intersection of the tangent line to the x-axis (V_{gs}) is the $V_{th} + 1/2 * V_{ds}$ point and then derive the Vth.



Vth calculation is given as follows:

$$V_{th} = V_{gs1} - I_{d1} / g_{mmax} - 1/2 V_{ds}$$

V_{gs1} is V_{gs} at the maximum gm point, I_{d1} is the I_d at V_{gs1} , g_{mmax} is maximum gm.

Note:

When gm_{vth} is in use, the operation region of specified MOSFET is determined by gm_{vth} .

Examples

Example 1: The `mn01` instance uses gm_{vth} .

```
mn00 d g s s nch l=1u w=150e-9
.print dc gm_vth(mn01)
```

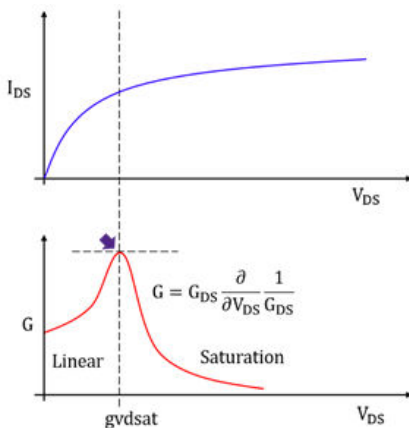
Saturation Voltage Correction — The G Function Method

Owing to channel pinch off and drift velocity saturation phenomenon, the drain current of the MOSFET biased above the threshold voltage is separated into two regions, namely the linear region and saturation region. The saturation voltage is the critical parameter to determine the boundary between the two regions. However, in advance technology nodes, the saturation voltage calculated by the SPICE compact model may not always be consistent with the measurement result. To resolve this issue, the PrimeSim Continuum simulators can invoke a saturation voltage correction method to determine the saturation voltage of a small-geometry MOSFET, to provide a more accurate result. The G function method is based on general device theory; so, is more versatile and can be applied to small-geometry MOSFET from any technology.

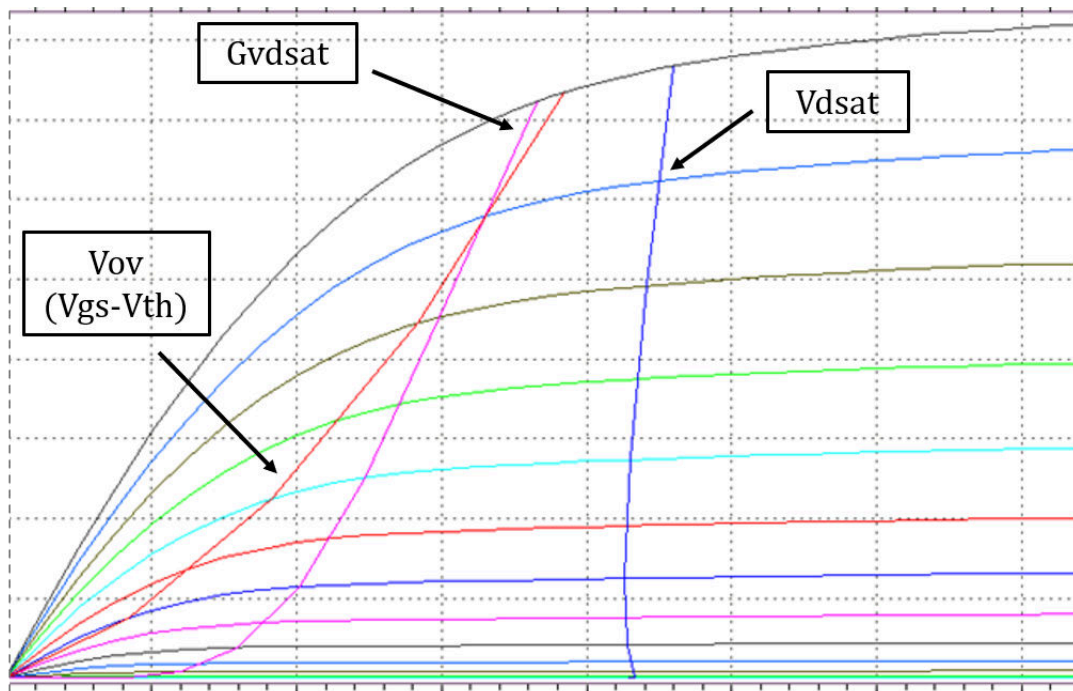
The G function is defined as follows:

$$G = G_{DS} \frac{\partial}{\partial V_{DS}} \frac{1}{G_{DS}}$$

According to the general device physics analysis in [1], it is found that the G function increases in the linear region to a peak value then decreases in the saturation region. Therefore, the newly defined saturation voltage (Gvdsat) is identified as the voltage of the peak point in the plot of G versus the drain source voltage (V_{DS}), as shown in the following figure:



A comparison of Gvdsat, Vdsat, and Vov ($V_{GS} - V_{th}$), in a state-of-the-art NMOS Id-Vd curve (see the following figure), shows that the Vdsat which is calculated by the SPICE compact model is largely overestimated while Gvdsat provides a more reasonable result.



Use the `gvdsat_vgsmin` correction parameter to generate more reasonable result at V_{gs} approaches to or lower than V_{th} . The default value of `gvdsat_vgsmin` is $V_{th} + 0.1$ V.

When V_{gs} is smaller than `gvdsat_vgsmin`.

$$Gvdsat = Gvdsat(gvdsat_vgsmin) - [Vdsat(gvdsat_vgsmin) - Vdsat(V_{gs})];$$

For more information , see `.OPTION GVDSAT_VGSMIN`.

Note:

When `Gvdsat` is in use, the operation region of specified MOSFET is determined by `Gvdsat` instead of `Vov`.

Examples

Example 1 The `mn01` instance uses `Gvdsat`

```
mn00 d g s s nch l=1u w=150e-9
mn01 d g s s nch l=1u w=150e-9
.print dc gvdsat(mn01)
```

Example 2 Both `mn00` and `mn01` use `Gvdsat`

```
mn00 d g s s nch l=1u w=150e-9
mn01 d g s s nch l=1u w=150e-9
.print dc gvdsat(mn00) gvdsat(mn01)
```

Example 3 Stack devices *xmn* and *xmn.mn01* use *Gvdsat*

```
xmn d g s s nseries
.subckt nseries d2 g2 s2 b2
mn01 d2 g2 s1i b2 nch l=1u w=150e-9
mn02 s1i g2 s2 b2 nch l=1u w=150e-9
.ends
.print dc gvdsat(xmn)
.print dc gvdsat(xmn.mn01)
```

Safe Operating Area Voltage Warning

The following warning message is issued when terminal voltages of a device (MOSFET, BJT, Diode, Resistor, Capacitor, etc.) exceed their safe operating area (SOA):

```
**warning**(filename:line number) resulted during SOA check
<node voltage name> (=val) of <device/element name> has exceeded <node
voltage name>_max (=val) To turn it off use .option WARN=0
```

Example Warnings

```
**Warning**(res1 : r1): Vr = 1.00      has exceeded Bv_max =      0.900
**Warning**(cap1 : c1): Vr = 1.00      has exceeded Bv_max =      0.900
**Warning**(dio1 : d1): Vr = 1.00      has exceeded Bv_max =      0.900
**Warning**(bjt1: q_gp1):Vbe = 1.00 has exceeded Vbe_max =      0.800
**Warning**(mos49 : m1): Vgs =  1.00 has exceeded Vgs_max =      0.900
**Warning**(mos49 : m1): Vgb =  1.00 has exceeded Vgs_max =      0.900
**Warning**(mos49 : m1): Vds =  1.00 has exceeded Vds_max =      0.900
**Warning**(mos49 : m1): Vbd = -1.00 has exceeded Vbd_max =      0.900
```

See the following control options for details:

- .OPTION WARN
- .OPTION MAXWARNS

For details on the warnings issued, see Safe Operating Area (SOA) Warnings in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

Model Pre-Processing and Parameter Flattening

You can invoke model pre-processing and parameter flattening by using `.OPTION MODPRT=1`. For details, see `.OPTION MODPRT` in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

Advanced MOSFET Self-Heating Simulation Solution

The PrimeSim Continuum solution enables the advanced device self-heating simulation technique for designs that need to account for MOSFET self-heating effect. With this new technique, the runtime and convergence concerns associated with the conventional compact self-heating model are eliminated and with no accuracy loss.

This new solution works directly with standard foundry SPICE models. No additional command syntax is required. BSIM-CMG, BSIM-BULK, BSIM-IMG, and UTSOI support this new technique.

Use of Example Syntax

To copy and paste proven syntax use the demonstration files shipped with your installation of the PrimeSim Continuum solution (see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists*). Attempting to copy and paste from the book or help documentation may present unexpected results, as text used in formatting may include hidden characters, white space, and so on. for visual clarity.

References

- [1] W. Jang, C. Wu and H. Wu, (1988). A new experimental method to determine the saturation voltage of a small-geometry MOSFET. *Solid-State Electronics*, 31(9), pp.1421-1431.

2

MOSFET Models

Lists and describes the standard MOSFET models.

This section covers the following MOSFET models:

Synopsys Level: Model	Model Parameters	Comments
	Common MOSFET Model Parameters	
LEVEL 82: MVSG Model	See the MVSG manual released by Compact Model Coalition.	A GaN high electron mobility transistor model.
LEVEL 81: ASM-HEMT Model	See the ASM-HEMT manual released by Compact Model Coalition.	An advanced physics-based GaN high electron mobility transistor model.
LEVEL 80: HISIM_SOI Model	HISIM_SOI Model Parameters and Equations	For all possible SOI-MOSFET structures, developed by Hiroshima University.
LEVEL 79: EKV3 Model	EKV3 Model—Parameters	An analytical compact MOSFET model that relies on charge sheet theory to describe its behavior.
Level 78: BSIM-IMG Model	BSIM-IMG Model Parameters	Independent multi-gate model.
LEVEL 77: BSIM6 MOSFET Model	BSIM6 MOSFET Model Parameters	-
LEVEL 76: LETI-UTSOI MOSFET Model	UTSOI MOSFET Model — Global Model Flags and Parameters UTSOI MOSFET Model — Local Level Parameters (SWSCALE=0) UTSOI MOSFET Model — Global Level Parameters (SWSCALE=1)	-

Synopsys Level: Model	Model Parameters	Comments
LEVEL 74: MOS Model 20 Model	MOS Model 20 Model — Instances and Parameters	-
LEVEL 73: HiSIM_HV Model	HiSIM_HV Model Parameters	CMC Open Standards
LEVEL 72: BSIM-CMG MOSFET Model	BSIM-CMG Model Parameters (Version >= 105) BSIM-CMG Model Parameters (Version < 105) BSIM-CMG Model (Version=105.03) — WPE Model Support	CMC Open Standards. Popularly known as FinFET.
LEVEL 71: TFT Model	TFT Model Parameters	-
	BSIMSOI4 Model Parameters	CMC Open Standards
LEVEL 69: PSP Model	PSP 103.1 Model Parameters PSP 103.1 Instance Parameters	CMC Open Standards
LEVEL 68: STARC HiSIM2 MOSFET Model	STARC HiSIM2 Model Parameters STARC HiSIM2 Model Instance Parameters	CMC Open Standards
LEVEL 66: PrimeSim Continuum HVMOS Model	PrimeSim Continuum Solution HVMOS Model Parameters	-
LEVEL 65: SSIMSOI Model	SSIMSOI Model Parameters	-
LEVEL 64: STARC HiSIM MOSFET Model	STARC HiSIM Model Parameters	-
LEVEL 63: Philips MOS11 Model	Philips MOS11 Model Parameters	-
LEVEL 62: RPI Poli-Si TFT Model	RPI Poli-Si TFT Model Parameters	-
LEVEL 61: RPI a-Si TFT Model	RPI a-Si TFT Model Parameters	-
LEVEL 60: BSIM3 SOI DD Model	BSIM3 SOI DD Model Parameters	-
LEVEL 59: BSIM3 SOI FD Model	BSIM3 SOI FD Model Parameters	-
LEVEL 58: UFSOI MOSFET Model	UFSOI Model — FDSOI Parameters UFSOI Model — NFDSOI Parameters	-
LEVEL 57: BSIM3 SOI Model	BSIM3 SOI Model Parameters	CMC Open Standards
LEVEL 55: EPFL-EKV MOSFET Model	-	-
LEVEL 54: BSIM4 MOSFET Model	BSIM4 MOSFET Model Parameters	CMC Open Standards

Synopsys Level: Model	Model Parameters	Comments
LEVEL 50: Philips MOS9 Model	Philips MOS9 Model Parameters Philips MOS9 Model — JUNCAP Parameters	-
LEVEL 49 and 53: BSIM3 Version 3 MOSFET Model	BSIM3 Version 3 MOSFET Model Parameters	CMC Open Standards
LEVEL 47: BSIM3 Version 2 MOSFET Model	BSIM3 Version 2 MOSFET Model Parameters	-
LEVEL 40: HP a-Si TFT Model	-	-
LEVEL 39: BSIM2 Model	BSIM2 Model Parameters	-
LEVEL 38: Cypress Depletion Model	Cypress Depletion Model Parameters	Used to model depletion MOS devices.
LEVEL 28: BSIM Derived Model	BSIM Derived Model Parameters	-
LEVEL 27: SOSFET Model	SOSFET Model Parameters	Used to model Silicon-On-Sapphire MOS devices. You can include photocurrent effects at this level.
LEVEL 13: BSIM Model	BSIM Model Parameters	-
LEVEL 8: Advanced LEVEL 2: MOS2 Grove-Frohman (SPICE 2G) Model	Advanced LEVEL 2: MOS2 Grove-Frohman (SPICE 2G) Model Parameters	-
LEVEL 7: Lattin-Jenkins-Grove (SPICE) Model	-	-
LEVEL 6: Lattin-Jenkins-Grove (ASPEC) Model	Lattin-Jenkins-Grove (ASPEC) Model Parameters	Compatible with models originally developed using ASPEC. Used to model ion-implanted devices.
LEVEL 5: AMI-ASPEC Depletion and Enhancement (Taylor-Huang) Model	AMI-ASPEC Model Parameters	Used to model depletion MOS devices.
LEVEL 4: Grove-Frohman-LEVEL 2 Model Derived from SPICE 2E.3	-	-

Synopsys Level: Model	Model Parameters	Comments
LEVEL 3: MOS3 Empirical Model (SPICE 2G)	MOS3 Empirical Model Parameters	Requires less simulation time, provides as much accuracy as LEVEL 2, and has a greater tendency to converge.
LEVEL 2: MOS2 Grove-Frohman (SPICE 2G) Model	Grove-Frohman (SPICE 2G) Model Parameters	Considers bulk charge effects on current.
LEVEL 1: Schichman-Hodges Model	Schichman-Hodges Model Parameters	-

Common MOSFET Model Parameters

This section covers the following topics:

- [Basic Model Parameters](#)
- [Effective Width and Length Parameters](#)
- [Threshold Voltage Parameters](#)
- [Mobility Parameters](#)

Basic Model Parameters

Name (Alias)	Units	Default	Description	Level
LEVEL		1.0	DC model selector: • LEVEL=1 (default) is the Schichman-Hodges model. • LEVEL=2 is the Grove-Frohmman model. • LEVEL=3 is an empirical model. • LEVEL=4 is a modified version of Level 2. • LEVEL=5 is the IDS model with enhancement and depletion modes. • LEVEL=6 is the Lattin-Jenkins-Grove model using ASPEC-style parasitics. • LEVEL=7 is the Lattin-Jenkins-Grove model using SPICE-style parasitics. • LEVEL=8 is an advanced model using finite differences. • LEVEL=13 is the University of California (UC) Berkeley BSIM1 model. • LEVEL=27 is the SOSFET model. • LEVEL=28 is a Synopsys proprietary model, based on the UC Berkeley BSIM1 model, Level 13. • LEVEL=38 is the Cypress Depletion model. • LEVEL=39 is the UC Berkeley BSIM2 model. • LEVEL=40 is the Hewlett-Packard amorphous-silicon Thin-Film Transistor (a-Si TFT) model. • LEVEL=47 is the UC Berkeley BSIM3 version 2 model • LEVEL=49 is a Synopsys proprietary model, based on the UC Berkeley BSIM3 version 3 model, Level 53.	All

Name (Alias)	Units	Default	Description	Level
LEVEL (continued)			• LEVEL=50 is the Philips MOS9 model. • LEVEL=53 is the original UC Berkeley BSIM3 version 3 model, not modified as Level 49 is. • LEVEL=54 is the UC Berkeley BSIM4 model • LEVEL=55 is the EPFL-EKV model. • LEVEL=57 is the UC Berkeley BSIM3-SOI Partially-Depleted (PD) model. • LEVEL=58 is the University of Florida SOI model • LEVEL=59 is the UC Berkeley BSIM3-SOI Fully-Depleted (FD) model • LEVEL=60 is the UC Berkeley BSIM3-SOI Dynamically-Depleted (DD) model • LEVEL=61 is the Rensselaer Polytechnic Institute (RPI) a-Si TFT model • LEVEL=62 is the Rensselaer Polytechnic Institute (RPI) poly-silicon Thin-Film Transistor (Poli-Si TFT) model • LEVEL=63 is the Philips MOS11 model • LEVEL=64 is the Hiroshima University Semiconductor Technology Academic Research Center (STARC) IGFET (HiSIM) model. • LEVEL=69 is Pennsylvania State University and Philips Research PSP100 model	All
ACM	-	0	Selects MOS S/D parasitics. ACM=0 is SPICE style. Use ACM=2 or 3 for LDD.	39
ALPHA	V ⁻¹	0	Impact ionization coefficient. This parameter includes geometry-sensitivity parameters. Choose between BSIM2 (A10>0 and the PrimeSim Continuum simulators (ALPHA>0) impact ionization modeling. <i>Do not use both.</i>	39
CAPOP	-	*	MOS gate cap model selector: CAPOP=39 for BSIM2 4, 13 39 or CAPOP=13 for BSIM1. CAPOP=4 is the same as CAPOP=13. • If SPICE3=0, default CAPOP=13 • If SPICE3=1, default CAPOP=39	4, 13 39
CGBO	F/m	-	Gate-to-bulk overlap capacitance. If you specify WD and TOX, but you do not specify CGBO, then simulation calculates CGBO.	39
CGDO	F	1.0p	TFT gate-to-drain overlap capacitance.	40
	F/m	-	Gate-to-drain overlap capacitance. If you specify TOX, and you specify either LD or METO, but you do not specify CGDO, then simulation calculates CGDO.	39
CGSO	F	1.0p	TFT gate-to-source overlap capacitance.	40

Name (Alias)	Units	Default	Description	Level
	F/m	-	Gate-to-source overlap capacitance. If you specify TOX, but you do not specify either LD or METO, and you do not specify CGSO, then simulation calculates CGSO.	39
CHI		0.5	Temperature exponential part.	40
CJ	F/m ²	0	Source/drain bulk zero-bias junction capacitance.	39
CJSW	F/m	0	Sidewall junction capacitance.	39
CLM (GDS)		0.0	Selects a channel length modulation equation.	6, 7, 8
COX	F/m ²	3.453e-4	Oxide capacitance per unit gate area. If you do not specify COX, simulation calculates it from TOX.	1, 2, 3, 8
CSC	F/m ²	10μ	Space charge capacitance.	40
DEFF		2.0	Drain voltage effect for the TFT leakage current.	40
DERIV		1	Derivative method selector: • DERIV=0: analytic • DERIV=1: finite difference	3, 39
DP	μm	1.0	Implant depth (depletion model only).	5, 38
ECRIT (ESAT)	V/cm	0.0	Critical electric field for the carrier velocity saturation. From Grove: • electrons 6e4 • holes 2.4e4. Zero indicates an infinite value. The ECRIT equation is more stable than VMAX. Simulation estimates ECRIT as: ECRIT=100 * (VMAX / UO)	2, 8
	V/cm	0.0	Drain-source critical field. Zero indicates an infinite value, typically 40,000 V/cm.	6, 7
ECV	V/μm	1000	Critical field.	5, 38
FEFF		0.5	Frequency effect constant.	40
FREQ	Hz	400	Frequency of the device.	40
GO	ohm ⁻¹	10e-15	Conductance of the TFT leakage current.	40
IIRAT	-	0	Impact ionization source bulk current partitioning factor. One corresponds to 100% source. Zero corresponds to 100% bulk.	39
JS	A/m ²	0	Source/drain bulk diode reverse saturation current density.	39

Name (Alias)	Units	Default	Description	Level
K2		2.0	Temperature exponential part.	40
KAPPA	V^{-1}	0.2	Saturation field factor. The channel length modulation equation uses this parameter.	3
KCS		2.77	Implant capacitance integration constant.	38
KP (BET, BETA)	A/V^2		Intrinsic transconductance parameter. If you specify U0 and TOX, but you do not specify KP, simulation computes the parameter from: $KP = UO * COX$ - Level 1 default=2.0718e-5 (NMOS), 8.632e-6 (PMOS). - Level 2, 3 default=2.0e-5	1, 2, 3
LAMBDA (LAM, LA)	V^{-1}	0.0	Channel length modulation.	2, 8
MJ	-	0.5	Source/drain bulk junction grading coefficient.	39
MJSW		0.33	Sidewall junction grading coefficient.	39
NEFF		1.0	Total channel charge (fixed and mobile) coefficient.	2
NI	cm^{-2}	2e11	Implant doping (depletion model only).	5, 38
NU		0.0	First order temperature gradient.	40
PB	V	0.8	Source/drain bulk junction potential.	39
PBSW	V	PB	Sidewall junction potential.	39
PSI		1e-20	Temperature exponential part.	40
RD	ohm	1.0K	(External) drain resistance.	40
RS	ohm	1.0K	(External) source resistance.	40
RSH	ohm/sq	0	Source/drain sheet resistance.	39
SNVB	$1/(V \cdot cm^3)$	0.0	Slope of the doping concentration versus vsb (element parameter). (Multiplied by 1e6).	8
SPICE3	-	0	Selects SPICE3 model compatibility. For accurate SPICE3 BSIM2, set SPICE3=1.	39
TAU	s	10n	Relaxation time constant.	40

Name (Alias)	Units	Default	Description	Level
TCV	V/C	0	Zero-bias threshold voltage temperature coefficient. The sign of TCV adjusts automatically for NMOS and PMOS to decrease the magnitude of the threshold with rising temperature.	39
TOX	m	1e-7	Gate oxide thickness.	1, 2, 3, 8
	Å	0.0	Oxide thickness.	5, 38
	m	7.0e-8	Oxide thickness.	27
TRD	1/K	0.0	Temperature coefficient for the Rd drain diffusion and contact resistances.	54
TREF		1.5	Temperature gradient of UO.	40
TRS	1/K	0.0	Temperature coefficient for the Rs source diffusion and contact resistances.	54
TUH		1.5	Implant channel mobility temperature exponent (depletion model only).	5
UO	cm ² /(V·s)		Carrier mobility. Default for LEVEL 40 is 1.0.	1, 40
UO (UB, UBO)	cm ² /(V·s)	600 (N) 250 (P)	Low-field bulk mobility. Simulation calculates this parameter from the KP value that you specify.	2
VB0 (VB)	V	0.0	Reference voltage for the GAMMA switch. • If vsb < VB0, the equation uses GAMMA. • If vsb > VB0, the equation uses LGAMMA.	6, 7
VCR	V	0	Impact ionization critical voltage. This parameter includes geometry-sensitivity parameters.	39
VMAX (VMX, VSAT)	m/s	0.0	Maximum drift velocity of the carriers. Zero indicates an infinite value. Default VMAX value for Level 40 is 1e6.	2, 3, 8, 40
VMAX (VMX)	cm/s	0.0	Maximum drift velocity of the carriers. Selects a calculation scheme to use for vdsat. Zero indicates an infinite value. Typical values: • electrons 8.4e6 cm/s • holes 4.3e6 cm/s	6, 7
VTIME	s	10m	Voltage stress.	40
ZENH		1.0	Mode flag (enhancement). Set ZENH=0.0 for the depletion mode.	5

Effective Width and Length Parameters

Name (Alias)	Units	Default	Description	Level
DEL	m	0.0	Channel length reduction on each side: $DEL_{scaled}=DEL \cdot SCALMMOSFET$ Level 13 does not support DEL.	1, 2, 3, 6, 7, 8, 38
DEL (WDEL)	μm	0.0	Channel length reduction on each side	5
DELVTO	V	0	Threshold voltage shift. This parameter is type sensitive. For example, $DELVTO>0$ increases the magnitude of the n-channel threshold, decreases the magnitude of the p-channel threshold, and adds to the element-line DELVTO parameter.	39, 40, 47, 49, 53, 54, 70,
LATD (LD)	μm	$1.7 \cdot XJ$	Lateral diffusion on each side	5, 38
LDAC	m		This parameter is the same as LD, but if you specify LDAC in the .MODEL statement, it replaces LD in the L_{eff} calculation for the AC gate capacitance.	1, 2, 3, 6, 7, 8, 13, 28, 38, 39
LMLT		1.0	Gate length shrink factor.	1, 2, 3, 5, 6, 7, 8, 13, 28, 38, 39
			Scale MOSFET drawn length	54
LD (DLAT, LATD)	m		Lateral diffusion into the channel from the source and the drain diffusion. If you do not specify LD and XJ: LD Default=0.0 If you specify XJ, but you do not specify LD, simulation calculates LD as: LD default= $0.75 \cdot XJ$ $LD_{scaled}=LD \cdot SCALM$	1, 2, 3, 6, 7, 8, 13, 28
LD	m	0	Lateral diffusion under the gate (per side) of the S/D junction. Use this parameter to calculate L_{eff} only if $DL=0$: $LD_{scaled}=LD \cdot SCALM$	39
LREF	m	0.0	Channel length reference: $LREF_{scaled}=LREF \cdot SCALM$ If the Level 13 model does not define LREF and WREF, their value is infinity.	2, 3, 6, 7, 8, 13, 28
	m	0 (A)	Reference channel length to adjust the length of the BSIM model parameters. For Berkeley compatibility ($LREF \rightarrow \lambda$), use: $LREF=0$. $LREF_{scaled}=LREF \cdot SCALM$	39
OXETCH	μm	0.0	Oxide etch	5, 38

Name (Alias)	Units	Default	Description	Level
Px	[x] $\mu\mu^2$	0	Px is a Synopsys proprietary, WL-product sensitivity parameter, where x is a model parameter with length and width sensitivity.	39
WD	m	0.0	Lateral diffusion into the channel from the bulk along the width: $WD_{scaled}=WD \cdot SCALM$	1, 2, 3, 6, 7, 8, 13
	m	0	Channel stop lateral diffusion under the gate (per side). Use this parameter to calculate W_{eff} only if $DW=0$. $WD_{scaled}=WD \cdot SCALM$	39
WDAC	m		This parameter is the same as WD, but if you specify WDAC in the .MODEL statement, it replaces WD in the W_{eff} calculation for the AC gate capacitance.	1, 2, 3, 6, 7, 8, 13, 28, 39
WMLT		1.0	Diffusion layer and width shrink factor.	1, 2, 3, 5, 6, 7, 8, 13, 28, 38, 54
	-	1.0	Diffusion and gate width shrink factor.	39
			Scale MOSFET drawn width	54
WREF	m	0.0	Channel width reference: $WREF_{scaled}=WREF \cdot SCALM$ If the Level 13 model does not define LREF and WREF, their value is infinity.	2, 3, 6, 7, 8, 13, 28
	m	0 (∞)	Reference device width to adjust the width of the BSIM model parameters. For Berkeley compatibility ($WREF \rightarrow \infty$), use $WREF=0$: $WREF_{scaled}=WREF \cdot SCALM$	39
XJ	m	0.0	Metallurgical junction depth: $XJ_{scaled}=XJ \cdot SCALM$	1, 2, 3, 6, 7, 8
	μm	1.5	Junction depth	5, 38
XL (DL, LDEL)	m	0.0	Length bias accounts for the masking and etching effects (length): $XL_{scaled}=XL \cdot SCALM$	1, 2, 3, 6, 7, 8, 13, 28, 39
XL	m	0	Difference between the physical (on the wafer) and the drawn reference channel length. Use this parameter to calculate L_{eff} only if $DL=0$: $XL_{scaled}=XL \cdot SCALM$ Note: SCALM is ignored for Level 54.	39, 54
XLREF	m	0.0	Difference between the physical (on the wafer) and the drawn reference channel length: $XLREF_{scaled}=XLREF \cdot SCALM$	28, 39

Name (Alias)	Units	Default	Description	Level
XW (DW, WDEL)	m	0.0	Width bias accounts for the masking and etching effects (width): $XW_{scaled}=XW \cdot SCALM$	1, 2, 3, 6, 7, 8, 13, 28
XW	m	0	Difference between the physical (on the wafer) and the drawn S/D active width. Use this parameter to calculate W_{eff} only if DW=0: $XW_{scaled}=XW \cdot SCALM$ Note: SCALM is ignored for Level 54.	39, 54
XWREF	m	0.0	Difference between the physical (on the wafer) and the drawn reference channel width: $XWREF_{scaled}=XWREF \cdot SCALM$	28, 39

Threshold Voltage Parameters

Name (Alias)	Units	Default	Description	Level
BetaGam		1.0	Body effect transition ratio.	38
CAV		0.0	Thermal voltage multiplier for the weak inversion equation.	8
DELTA		0.0	Narrow width factor for adjusting the threshold.	2, 3, 8
DNB (NSUB)	cm^{-3}	0.0	Surface doping density.	5, 38
	$1/cm^3$	1.0e15	Substrate doping.	6, 7
DNS (NI)	$1/cm^3$	0.0	Surface substrate doping.	6, 7
DVIN	V	0.0	Adjusts the empirical surface inversion voltage.	38
DVSBC	V	0.0	Adjusts the empirical body effect transition voltage.	38
E1		3.9	Dielectric constant of first film.	40
E2		0.0	Dielectric constant of second film.	40
ETA	V^{-1} (Level 40)	0.0	Static feedback factor for adjusting the threshold voltage (difficulty of band bending).	3, 40
		0.0	Drain-induced barrier lowering (DIBL) effect coefficient for the threshold voltage.	8
		0.0	Channel-length independent drain-induced barrier lowering.	38

Name (Alias)	Units	Default	Description	Level
FDS		0.0	Field, drain to source. Controls the threshold reduction due to the source-drain electric field.	6, 7
FSS (NFS)	$\text{cm}^{-2} \cdot \text{V}^{-1}$	0.0	Number of fast surface states	5, 38
GAMMA	$\text{V}^{1/2}$	0.5276	Body effect factor. If you do not specify GAMMA, simulation calculates it from NSUB.	1, 2, 3, 8
	$\text{V}^{1/2}$		Body effect factor. - If you do not specify GAMMA, simulation calculates it from DNB. - GAMMA is the body effect, if $v_{sb} < V_{B0}$. - If $v_{sb} > V_{B0}$, simulation uses LGAMMA. GAMMA, LGAMMA, and V_{B0} perform a two-step approximation of a non-homogeneous substrate.	6, 7
LBetaGam.	μm	0.0	BetaGam dependence on the channel length.	38
LDVSBC	$\text{V} \cdot \mu\text{m}$	0.0	Adjusts the L-dependent body effect transition voltage.	38
LETA(DIBL)	μm	0.0	Channel-length dependent drain-induced barrier lowering.	38
LGAMMA	$\text{V}^{1/2}$	0.0	This parameter is the body effect factor if $v_{sb} > V_{B0}$. If you use the Poon-Yau GAMMA expression, LGAMMA is the junction depth in microns. Simulation multiplies LGAMMA by SCALM.	6, 7
LND	$\mu\text{m}/\text{V}$	0.0	ND length sensitivity.	2, 3, 6, 7, 8
LN0	μm	0.0	N0 length sensitivity.	2, 3, 6, 7, 8
LVT (LVTO)	$\text{V} \cdot \mu\text{m}$	0.0	VT dependence on the channel length.	38
ND	$\text{V}^{-1} (1/\text{V})$	0.0	Drain subthreshold factor. Typical value=1.	2, 3, 6, 7, 8
N0		0.0	Gate subthreshold factor. Typical value=1.	2, 3, 6, 7, 8
NFS (DFS, NF, DNF)	$\text{cm}^{-2} \cdot \text{V}^{-1}$	0.0	Fast surface state density.	1, 2, 3, 6, 7, 8
NFS	cm^2	0.0	Fast surface state density.	40
NSS	cm^2	0.0	Surface state density.	40

Name (Alias)	Units	Default	Description	Level
NSUB (DNB, NB)	cm ⁻³	1e15	Bulk surface doping. If you do not specify NSUB, simulation calculates it from GAMMA.	1, 2, 3, 8
NWE	m	0.0	Narrow width effect, direct compensation of VTO: $NWE_{scaled} = NWE \cdot SCALM$	6, 7
NWM		0.0	Narrow width modifier.	5, 38
		0.0	Narrow width modulation of GAMMA.	6, 7
PHI	V	0.576	Surface inversion potential. If you do not specify PHI, the PrimeSim Continuum simulators calculate it from NSUB.	1, 2, 3, 8
	V	0.8	Built-in potential.	5, 38
	V	0.0	Surface potential.	40
SCM		0.0	Short-channel drain source voltage multiplier	5, 38
		0.0	Short-channel modulation of GAMMA.	6, 7
T1	m	280n	First thin film thickness.	40
T2	m	0.0	Second thin film thickness.	40
TDVSBC	V/K	0.0	Body effect transition voltage shift due to the temperature.	38
UFDS		0.0	High field FDS.	6, 7
UPDATE		0.0	Selects different versions of the LEVEL 6 model. For the UPDATE=1 or 2 alternate saturation voltage, simulation modifies the MOB=3 mobility equation and the RS and RD series resistances so they are compatible with ASPEC. UPDATE=1 is a continuous Multi-Level GAMMA model.	6, 7
VFDS	V	0.0	Reference voltage for selecting FDS or UFDS: Uses FDS if $v_{ds} \leq VFDS$. Uses UFDS if $v_{ds} > VFDS$.	6, 7
VSH	V	0.0	Threshold voltage shifter for reducing the zero-bias threshold voltage (VTO) as a function of the ratio of LD to Leff.	6, 7
VT (VTO)	V	0.0	Extrapolated threshold voltage	5, 38
VTO (VT)	V	0.0	Zero-bias threshold voltage. If you do not specify VTO, simulation calculates it.	1, 2, 3, 6, 7, 8, 40

Name (Alias)	Units	Default	Description	Level
WBetaGam	μm	0.0	BetaGam dependence on the channel width.	38
WDVSBC	$\text{V}\cdot\mu\text{m}$	0.0	Adjusts the W-dependent body effect transition voltage.	38
WETA	μm	0.0	Channel-width dependent drain-induced barrier lowering.	38
WEX			Weak inversion exponent.	6, 7
WIC		0.0	Subthreshold model selector.	2, 3, 6, 7, 8
WND	$\mu\text{m}/\text{V}$	0.0	ND width sensitivity.	2, 3, 6, 7, 8
WN0	μm	0.0	N0 width sensitivity.	2, 3, 6, 7
WVT (WVTO)	$\text{V}\cdot\mu\text{m}$	0.0	VT dependence on the channel width.	38

Mobility Parameters

Use curve fitting to determine the mobility parameters. Generally, you should set U_{TRA} between 0.0 and 0.5. Nonzero values for U_{TRA} can result in negative resistance regions at the onset of saturation.

Name (Alias)	Units	Default	Description	Level
BEX	-	-1.5	Surface channel mobility temperature exponent.	38, 39
BFRC	$\text{\AA}\cdot\text{s}/(\text{cm}^2\cdot\text{V})$	0.0	Field reduction coefficient variation due to the substrate bias.	38
FACTOR			Mobility degradation factor. Default=1.0.	6, 7
FEX	-	0	Temperature exponent for velocity saturation.	39
FRC	$\text{\AA}\cdot\text{s}/\text{cm}^2$	0.0	Field reduction coefficient.	5, 38
FRCEX (F1EX)		0.0	Temperature coefficient for FRC	38
FSB	$\text{V}^{1/2}\cdot\text{s}/\text{cm}^2$	0.0	Lateral mobility coefficient.	5, 38
HEX(TUH)		-1.5	Implant channel mobility temperature exponent.	38

Name (Alias)	Units	Default	Description	Level
KBeta1		1.0	Effective implant-channel mobility modifier.	38
KIO(KIO)		1.0	Residue current coefficient.	38
LBFRC	$10^{-4} \text{ Å} \cdot \text{s}/(\text{cm} \cdot \text{V})$	0.0	BFRC sensitivity to the effective channel length.	38
LFRC	$10^{-4} \text{ Å} \cdot \text{s}/\text{cm}$	0.0	FRC sensitivity to the effective channel length.	38
LKBeta1	μm	0.0	Length-dependent implant-channel mobility modifier.	38
LKIO(LKIO)	μm	0.0	Length-dependent residue current coefficient.	38
LUO(LUB)	$\text{cm}^2 \cdot \mu\text{m}/(\text{V} \cdot \text{s})$	0.0	UO sensitivity to the effective channel length.	38
LVFRC	$10^{-4} \text{ Å} \cdot \text{s}/(\text{cm} \cdot \text{V})$	0.0	VFRC sensitivity to the effective channel length.	38
LFSB	$10^{-4} \text{ V}^{1/2} \cdot \text{s}/\text{cm}$	0.0	FSB sensitivity to the effective channel length.	38
MOB		0.0	Selects a mobility equation. You can set this parameter to MOB=0 or MOB=7. MOB=7 changes both the model and the channel length calculation. The MOB=7 flag invokes the channel length modulation and mobility equations in MOSFET LEVEL 3. In MOSFET Level 8, you can set MOB to 2, 3, 6, or 7.	2, 6, 7, 8
THETA	V^{-1}	0.0	Mobility modulation. MOSFET models use THETA only if MOB=7. A typical value is THETA=5e-2.	2, 40
	V^{-1}	0.0	Mobility degradation factor.	3
UB (UO)	$\text{cm}^2/(\text{V} \cdot \text{s})$	0.0	Low field bulk mobility	5
UCRIT	V/cm	1.0e4	Critical field for mobility degradation, UCRIT. This parameter is the limit where UO surface mobility begins to decrease as specified in the empirical relation.	2

Name (Alias)	Units	Default	Description	Level
	V/cm	1e4	MOB=6, UEXP>0 Critical field for the mobility degradation. UEXP operates as a switch. MOB=6, UEXP≤0 Critical field for mobility degradation. Typical value is 0.01 V ⁻¹ .	8
UEFF			Effective mobility at the specified analysis temperature.	6, 7
UEXP (F2)		0.0	Critical field exponent in the empirical formula that characterizes the surface mobility degradation. Typical value in MOSFET Level 8 with MOB=6 is 0.01 V ⁻¹ .	2, 8
UH	cm ² /(V·s)	900 (N) 300 (P)*	Implant - channel mobility * (For depletion model only)	5, 38
UHSAT	μm/V	0.0	Implant-channel mobility saturation factor.	38
UO (UB, UBO)	cm ² /(V·s)	600 (N) 250 (P)	Low-field bulk mobility. Simulation calculates this parameter from the KP value that you specify.	2, 3, 6, 7, 8, 38
UTRA		0.0	Transverse field coefficient (mobility). Traditional SPICE does not use UTRA. The PrimeSim Continuum simulators can use UTRA, but simulation issues a warning, because UTRA can hinder convergence.	2, 8
VFRC	Å·s/(cm ² ·V)	0.0	Field reduction coefficient variation due to the drain bias.	38
VST	cm/s	0.0	Saturation velocity.	5, 38
WBFRC	10 ⁻⁴ Å·s/(cm·V)	0.0	BFRC sensitivity to the effective channel width.	38
WFRC	10 ⁻⁴ Å·s/cm	0.0	FRC sensitivity to the effective channel width.	38
WFSB	10 ⁻⁴ V ^{1/2} · s/cm	0.0	FSB sensitivity to the effective channel width.	38
WKBeta1	μm	0.0	Width-dependent implant-channel mobility modifier.	38

Name (Alias)	Units	Default	Description	Level
WKIO (WKIO)	μm	0.0	Width-dependent residue current coefficient.	38
WUO(WUB)	$\text{cm}^2 \cdot \mu\text{m}/(\text{V} \cdot \text{s})$	0.0	UO sensitivity to the effective channel width.	38
WVFRC	$10^{-4} \text{A} \cdot \text{s}/(\text{cm} \cdot \text{V})$	0.0	VFRC sensitivity to the effective channel width.	38

LEVEL 1: Schichman-Hodges Model

Use this model if accuracy is less important to you than simulation turn-around time. For digital switching circuits, especially if you need only a “qualitative” simulation of the timing and the function, LEVEL 1 run-time can be about half that of a simulation using the LEVEL 2 model. The agreement in timing is approximately 10%. The LEVEL 1 model, however, results in severe inaccuracies in DC transfer functions of any TTL-compatible input buffers in the circuit.

The `LAMBDA` channel-length modulation parameter is equivalent to the inverse of the Early voltage for the bipolar transistor. `LAMBDA` measures the output conductance in the saturation. If you specify this parameter, the MOSFET has a finite but constant output conductance in saturation. If you do not specify `LAMBDA`, the LEVEL 1 model assumes zero output conductance.

This section covers the following topics:

- [Schichman-Hodges Model Parameters](#)
- [Schichman-Hodges Model Equations](#)

Schichman-Hodges Model Parameters

MOSFET Level 1 uses only the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section.

Schichman-Hodges Model Equations

This section covers the following topics:

- [Schichman-Hodges Model — IDS Equations](#)
- [Schichman-Hodges Model — Effective Channel Length and Width](#)

Schichman-Hodges Model — IDS Equations

The LEVEL 1 model does not include the carrier mobility degradation, the carrier saturation effect, or the weak inversion model. This model determines the DC current:

Cutoff Region:

$$v_{gs} \leq v_{th}, I_{ds} = 0.0$$

Linear Region, $v_{ds} < v_{gs} - v_{th}$

$$I_{ds} = KP \cdot \frac{W_{eff}}{L_{eff}} \cdot (1 + LAMBDA \cdot v_{ds}) \cdot \left(v_{gs} - v_{th} - \frac{v_{ds}}{2} \right) \cdot v_{ds}$$

Saturation Region, -

$$I_{ds} = \frac{KP}{2} \cdot \frac{W_{eff}}{L_{eff}} \cdot (1 + LAMBDA \cdot v_{ds}) \cdot (v_{gs} - v_{th})^2$$

Schichman-Hodges Model — Effective Channel Length and Width

The Level 1 model calculates the effective channel length and width from the drawn length and width:

$$L_{eff} = L_{scaled} \cdot LMLT + XL_{scaled}^{-2} (LD_{scaled} + DEL_{scaled})$$

$$W_{eff} = M \cdot (W_{scaled} WMLT + XW_{scaled}^{-2} WD_{scaled})$$

Threshold Voltage, v_{th}

$$v_{sb} \geq 0, v_{th} = v_{bi} + GAMMA \cdot (PHI + V_{sb})^{1/2}$$

$$v_{sb} < 0, v_{th} = v_{bi} + GAMMA \cdot \left(PHI^{1/2} + 0.5 \frac{v_{sb}}{PHI^{1/2}} \right)$$

The preceding equations define the built-in voltage (v_{bi}) as:

$$v_{bi} = v_{fb} + PHI, v_{bi} = VTO - GAMMA PHI^{1/2}$$

See [Common Threshold Voltage Equations](#) for calculation of VTO , $GAMMA$, and PHI if you do not specify them.

Saturation Voltage, v_{sat}

The saturation voltage for the LEVEL 1 model is due to the channel pinch-off at the drain side. The following equation computes this voltage:

$$v_{sat} = v_{gs} - v_{th}$$

The LEVEL 1 model does not include the carrier velocity saturation effect.

LEVEL 2: MOS2 Grove-Frohmman (SPICE 2G) Model

This section describes the parameters and equations for the LEVEL 2: MOS2 Grove-Frohmman model:

- [Grove-Frohmman \(SPICE 2G\) Model Parameters](#)
- [Grove-Frohmman \(SPICE 2G\) Model Equations](#)

Grove-Frohmman (SPICE 2G) Model Parameters

This model uses only the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section.

Grove-Frohmman (SPICE 2G) Model Equations

This section covers the following topics:

- [Grove-Frohmman \(SPICE 2G\) Model — IDS Equations](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Effective Channel Length and Width](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Threshold Voltage, vth](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Saturation Voltage, vdsat](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Mobility Reduction, ueff](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Channel Length Modulation](#)
- [Grove-Frohmman \(SPICE 2G\) Model — Subthreshold Current, Ids](#)

Grove-Frohmman (SPICE 2G) Model — IDS Equations

This section describes how this model calculates the drain current of n-channel and p-channel MOSFETs.

Cutoff Region, $v_{gs} \leq v_{th}$, $I_{ds} = 0$ (see subthreshold current)

On Region, $v_{gs} > v_{th}$

$$i_{ds} = \beta \cdot \left\{ \left(v_{gs} - v_{th} - \frac{\eta \cdot v_{de}}{2} \right) \cdot v_{de} - \frac{2}{3} \gamma \left[(phi + v_{de} + v_{sb})^{\frac{3}{2}} - (phi + v_{sb})^{\frac{3}{2}} \right] \right\}$$

The following equations calculate values used in the preceding equation:

$$v_{de} = \min(v_{ds}, v_{dsat})$$

$$\eta = 1 + DELTA \cdot \frac{\pi \cdot \epsilon_{Si}}{4 \cdot COX \cdot W_{eff}}, \beta = KP \cdot \frac{W_{eff}}{L_{eff}}$$

Grove-Frohmman (SPICE 2G) Model — Effective Channel Length and Width

The Level 2 model calculates effective channel length and width from the drawn length and width:

$$L_{eff} = L_{scaled} \cdot LMLT + XL_{scaled} - 2(LD_{scaled} + DEL_{scaled})$$

$$W_{eff} = M \cdot (W_{scaled} \cdot WMLT + XW_{scaled} \cdot 2WD_{scaled})$$

$$LREF_{eff} = LREF_{scaled} \cdot LMLT + XL_{scaled} \cdot 2(LD_{scaled} + DEL_{scaled})$$

$$WREF_{eff} = M \cdot (WREF_{scaled} \cdot WMLT + XW_{scaled} \cdot 2WD_{scaled})$$

Grove-Frohmman (SPICE 2G) Model — Threshold Voltage, v_{th}

The V_{TO} model parameter is an extrapolated zero-bias threshold voltage for a large device. The following equation calculates the effective threshold voltage, including the device size effects and the terminal voltages:

$$v_{th} = v_{bi} + \gamma \cdot (PHI + v_{sb})^{\frac{1}{2}}$$

The following equation calculates the v_{bi} value used in the preceding equation:

$$v_{bi} = VTO - GAMMA(PHI)^{\frac{1}{2}} + (\eta - 1) \cdot (PHI + v_{sb})$$

To include the narrow width effect, use v_{bi} and η . To include the narrow width effect, specify the Δ model parameter. The effective γ specifies the short-channel effect. To include short-channel effects, the XJ model parameter must be greater than zero.

$$\gamma = GAMMA \cdot \left\{ 1 - \frac{XJ_{scaled}}{2 \cdot L_{eff}} \left[\left(1 + \frac{2 \cdot W_s}{XJ_{scaled}} \right)^{\frac{1}{2}} + \left(1 + \frac{2 \cdot W_d}{XJ_{scaled}} \right)^{\frac{1}{2}} - 2 \right] \right\}$$

The following equations determine the W_s and W_d depletion widths:

$$W_s = \left[\frac{2 \cdot E_{si}}{q \cdot NSUB} \cdot (PHI + v_{sb}) \right]^{\frac{1}{2}}$$

$$W_d = \left[\frac{2 \cdot E_{si}}{q \cdot NSUB} \cdot (PHI + v_{ds} + v_{sb}) \right]^{\frac{1}{2}}$$

If you do not specify parameters such as V_{TO} , $GAMMA$, and PHI , simulation calculates them automatically. The Level 2 model uses these parameters to calculate the threshold voltage. (See [Common Threshold Voltage Equations](#)).

Grove-Frohmman (SPICE 2G) Model — Saturation Voltage, v_{dsat}

If you do not specify the V_{MAX} model parameter, the program computes the saturation voltage due to channel pinch off at the drain side. If you specify the corrections for small-size effects, then:

$$v_{sat} = \frac{v_{gs} - v_{bi}}{\eta} + \frac{1}{2} \left(\frac{\gamma}{\eta} \right)^2 \cdot \left\{ 1 - \left[1 + 4 \cdot \left(\frac{\eta}{\gamma} \right)^2 \cdot \left(\frac{v_{gs} - v_{bi}}{\eta} + PHI + v_{sb} \right) \right]^{\frac{1}{2}} \right\}$$

$$v_{dsat} = v_{sat}$$

If you specify `ECRIT`, the program modifies v_{sat} to include carrier velocity saturation effect:

$$v_{dsat} = v_{sat} + v_c \cdot (v_{sat}^2 + v_c^2)^{-\frac{1}{2}}$$

The following equation calculates the v_c value used in the preceding equation:

$$v_c = ECRIT \cdot L_{eff}$$

Note:

If you specify `vMAX`, simulation calculates a different v_{dsat} value. Refer to the Vladimirescu document [1] for details.

Grove-Frohmman (SPICE 2G) Model — Mobility Reduction, u_{eff}

The mobility of carriers in the channel decreases as speeds of the carriers approach their scattering limited velocity. The mobility degradation for the LEVEL 2 MOS model uses two different equations, depending on the `MOB` mobility equation selector value.

If `MOB=0`, (default):

$$u_{eff} = UO \cdot \left[\frac{UCRIT \cdot E_{si}}{COX \cdot (v_{gs} - v_{th} - UTRAP \cdot v_{ds})} \right]^{UEXP}$$

Because u_{eff} is less than UO , the program uses the above equation if the bracket term is less than one; otherwise the program uses $u_{eff}=UO$.

If `MOB=7`, `THETA` $\neq 0$

$$u_{eff} = \frac{UO}{1 + THETA \cdot (v_{gs} - v_{th})}$$

$v_{gs} < v_{th}$, $u_{eff}=UO$

If `MOB=7`, `THETA=0`

$$u_{eff} = UO \cdot \left[\frac{UCRIT \cdot E_{si}}{COX \cdot (v_{gs} - v_{th})} \right]^{UEXP}$$

If `MOB=7`, `vMAX>0`

$$u_{eff} = \frac{u_{eff}}{1 + u_{eff} \cdot \frac{v_{de}}{vMAX \cdot L_{eff}}}$$

Grove-Frohmman (SPICE 2G) Model — Channel Length Modulation

To include the channel length modulation effect, the LEVEL 2 MOS model modifies the I_{ds} current:

$$I_{ds} = \frac{I_{ds}}{1 - \lambda v_{ds}}$$

If you do not specify the `LAMBDA` model parameter, the model calculates the λ value.

LAMBDA>0:

$$\lambda = \text{LAMBDA}$$

`VMAX`>0, `NSUB`>0, and `LAMBDA`<0

$$\lambda = \frac{X_d}{n_{eff}^{\frac{1}{2}} \cdot l_{eff} \cdot v_{ds}} \cdot \left\{ \left[\left(\frac{v_{max} \cdot X_d}{2 \cdot n_{eff}^{\frac{1}{2}} \cdot u_{eff}} \right)^2 + v_{ds} - v_{dsat} \right]^{\frac{1}{2}} - \frac{v_{max} \cdot X_d}{2 \cdot n_{eff}^{\frac{1}{2}} \cdot u_{eff}} \right\}$$

`VMAX`=0, `NSUB`>0, and `LAMBDA`<0

If `MOB`=0

$$\lambda = \frac{X_d}{L_{eff} \cdot v_{ds}} \cdot \left\{ \frac{v_{ds} - v_{dsat}}{4} + \left[1 + \left(\frac{v_{ds} - v_{dsat}}{4} \right)^2 \right]^{\frac{1}{2}} \right\}^{\frac{1}{2}}$$

This equation does not include the effect of the field between the gate and the drain. It also tends to overestimate the output conductance in the saturation region.

If `MOB`=7

$$\lambda = \frac{X_d}{L_{eff} \cdot v_{ds}} \cdot \left\{ \left[\frac{v_{ds} - v_{dsat}}{4} + \left(1 + \left(\frac{v_{ds} - v_{dsat}}{4} \right)^2 \right)^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

This equation does not include the effect of the field between the gate and the pinch-off point. It also tends to overestimate the output conductance in the saturation region.

The following equation calculates the X_d value used in the two preceding equations:

$$X_d = \left(\frac{2 \cdot E_{si}}{q \cdot \text{NSUB}} \right)^{\frac{1}{2}}$$

Modifying I_{ds} by a factor of $(1 - \lambda \cdot v_{ds})$ is equivalent to replacing L_{eff} with:

$$L_e = L_{eff} - \lambda_{ds} \cdot L_{eff}$$

To prevent the channel length (L_e) from becoming negative, the value of L_e is limited.

If $L_e < x_{wb}$, then simulation replaces L_e with:

$$\frac{x_{wb}}{1 + \frac{x_{wb} - L_e}{x_{wb}}}$$

The following equation calculates the x_{wb} value used in the preceding equation:

$$x_{wb} = X_d \cdot PB^{\frac{1}{2}}$$

Grove-Frohmman (SPICE 2G) Model — Subthreshold Current, I_{ds}

The fast surface states model parameter (NFS) characterizes this region of operation. For $NFS > 0$ the model determines the modified threshold voltage (v_{on}):

$$v_{on} = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = v_t \cdot \left[\eta + (PHI + v_{sb})^{\frac{1}{2}} \cdot \frac{\partial \gamma}{\partial v_{sb}} + \frac{\gamma}{2 \cdot (PHI + v_{sb})^{\frac{1}{2}}} + \frac{q \cdot NFS}{COX} \right]$$

In the preceding equations, v_t is the thermal voltage.

The following equation calculates the I_{ds} current for $v_{gs} < v_{on}$:

$$I_{ds} = I_{ds}(v_{on}, v_{de}, v_{sb}) \cdot e^{\frac{v_{gs} - v_{on}}{fast}}$$

$v_{gs} > v_{on}$:

$$I_{ds} = I_{ds}(v_{gs}, v_{de}, v_{sb})$$

The following equation calculates the v_{de} value used in the preceding equation:

$$v_{de} = \min(v_{ds}, v_{dsat})$$

Note:

The modified threshold voltage (v_{on}), due to NFS specification, is also used in strong inversion instead of v_{th} , mostly in the mobility equations.

If $WIC=3$, the Level 2 model calculates the subthreshold current differently. In this case the I_{ds} current is:

$$I_{ds} = I_{ds}(v_{gs}, v_{de}, v_{sb}) + isub(N0_{eff}, ND_{eff}, v_{gs}, v_{ds})$$

$N0_{eff}$ and ND_{eff} are functions of effective device width and length.

LEVEL 3: MOS3 Empirical Model (SPICE 2G)

This section describes the LEVEL 3 IDS: MOS3 Empirical model parameters and equations.

- [MOS3 Empirical Model Parameters](#)
- [MOS3 Empirical Model Equations](#)
- [MOS3 Empirical Model — Compatibility Notes](#)
- [MOS3 Empirical Model – Temperature Compensation Example](#)

MOS3 Empirical Model Parameters

This model uses only the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section.

MOS3 Empirical Model Equations

This section covers the following topics:

- [MOS3 Empirical Model — IDS Equations](#)
- [MOS3 Empirical Model — Effective Channel Length and Width](#)
- [MOS3 Empirical Model — Threshold Voltage, \$v_{th}\$](#)
- [MOS3 Empirical Model — Saturation Voltage, \$v_{dsat}\$](#)
- [MOS3 Empirical Model — Effective Mobility, \$\mu_{eff}\$](#)
- [MOS3 Empirical Model — Channel Length Modulation](#)
- [MOS3 Empirical Model — Subthreshold Current, \$I_{ds}\$](#)

MOS3 Empirical Model — IDS Equations

The following equations describe how this model calculates the I_{ds} drain current.

Cutoff Region, $v_{gs} < v_{th}$

$$I_{ds} = 0 \text{ (See subthreshold current)}$$

On Region, $v_{gs} > v_{th}$

$$I_{ds} = \beta \cdot \left(V_{gs} - V_{th} - \frac{1 + fb}{2} \cdot v_{de} \right) \cdot v_{de}$$

The following equations calculate values used in the preceding equation:

$$\beta = K \cdot P \cdot \frac{W_{eff}}{L_{eff}} = u_{eff} \cdot COX \cdot \frac{W_{eff}}{L_{eff}}$$

Since

$$K \cdot P = u_{eff} \cdot COX$$

$$vde = \min(v_{ds}, v_{dsat})$$

$$fb = f_n + \frac{GAMMA \cdot f_s}{4 \cdot (PHI + v_{sb})^{\frac{1}{2}}}$$

Note:

In the above equation, the factor 4 should be 2, but because SPICE uses a factor of 4, this model uses a factor of 4 as well.

The f_n parameter specifies the narrow width effect:

$$f_n = \frac{DELTA}{W_{eff}} \cdot \frac{1}{4} \cdot \frac{2\pi \cdot E_{si}}{COX}$$

The f_s term expresses the effect of the short channel:

$$f_s = 1 - \frac{XJ_{scaled}}{L_{eff}} \left\{ \frac{LD_{scaled} + W_c}{XJ_{scaled}} \cdot \left[1 - \left(\frac{W_p}{XJ_{scaled} + W_p} \right)^2 \right]^{\frac{1}{2}} - \frac{LD_{scaled}}{XJ_{scaled}} \right\}$$

$$W_p = X_d \cdot (PHI + v_{sb})^{\frac{1}{2}}$$

$$X_d = \left(\frac{2 \cdot E_{si}}{q \cdot NSUB} \right)^{\frac{1}{2}}$$

$$W_c = XJ_{scaled} \cdot \left[0.0631353 + 0.8013292 \cdot \left(\frac{W_p}{XJ_{scaled}} \right) - 0.01110777 \left(\frac{W_p}{XJ_{scaled}} \right)^2 \right]$$

MOS3 Empirical Model — Effective Channel Length and Width

The following equations determine the effective channel length and width in the LEVEL 3 model:

$$L_{eff} = L_{scaled} \cdot LMLT + XL_{scaled} - 2(LD_{scaled} + DEL_{scaled})$$

$$w_{eff} = m \cdot (w_{scaled} \cdot wmlt + xw_{scaled} - 2wd_{scaled})$$

$$LREFF = LREF_{scaled} \cdot LMLT + XL_{scaled} \cdot 2(LD_{scaled} + DEL_{scaled})$$

$$WREF_{eff} = M \cdot (WREF_{scaled} \cdot WMLT + XW_{scaled} \cdot 2WD_{scaled})$$

MOS3 Empirical Model — Threshold Voltage, v_{th}

The following equation calculates the effective threshold voltage, including the device size and terminal voltage effects:

$$v_{th} = v_{bi} - \frac{8.14e-22 \eta}{COX \cdot I_{eff}^3} = v_{ds} + \gamma \cdot f_s \cdot (\phi + v_{sb})^{\frac{1}{2}} + f_n \cdot (\phi + v_{sb})$$

The following equation calculates the v_{bi} value used in the preceding equation:

$$v_{bi} = v_{fb} + PHI \text{ or } v_{bi} = VTO - GAMMA \cdot PHI^{\frac{1}{2}}$$

VTO is the extrapolated zero-bias threshold voltage of a large device. If you do not specify VTO , $GAMMA$, or PHI , simulation computes these values (see [Common Threshold Voltage Equations](#)).

MOS3 Empirical Model — Saturation Voltage, v_{dsat}

The LEVEL 3 model determines the saturation voltage due to the channel pinch-off at the drain side. The V_{MAX} parameter specifies the reduction of the saturation voltage due to the carrier velocity saturation effect.

$$v_{sat} = \frac{v_{gs} - v_{th}}{1 + f_b}$$

$$v_{dsat} = v_{sat} + v_c \cdot (v_{sat}^2 + v_c^2)^{\frac{1}{2}}$$

The following equation calculates the v_c value used in the preceding equations:

$$v_c = \frac{V_{MAX} \cdot L_{eff}}{u_s}$$

The next section defines the u_s surface mobility parameter. If you do not specify the V_{MAX} model parameter, then:

$$v_{dsat} = v_{sat}$$

MOS3 Empirical Model — Effective Mobility, u_{eff}

The Level 3 model defines the carrier mobility reduction due to the normal field as the effective surface mobility (u_s).

$$v_{gs} > v_{th}$$

$$u_s = \frac{UO}{1 + THETA \cdot (v_{gs} - v_{th})}$$

The v_{MAX} model parameter model determines the degradation of mobility due to the lateral field and the carrier velocity saturation.

$v_{MAX} > 0$:

$$u_{eff} = \frac{u_s}{1 + \frac{v_{de}}{v_c}}$$

Otherwise: $u_{eff} = u_s$

MOS3 Empirical Model — Channel Length Modulation

For $v_{ds} > v_{dsat}$, this model computes the channel length modulation factor. The v_{MAX} model parameter value determines the amount of channel length reduction (ΔL).

$v_{MAX} = 0$

$$\Delta L = X_d \cdot [KAPPA \cdot (v_{ds} - v_{dsat})]^{\frac{1}{2}}$$

$v_{MAX} > 0$

$$\Delta L = -\frac{E_p \cdot X_d^2}{2} + \left[\left(\frac{E_p \cdot X_d^2}{2} \right)^2 + KAPPA \cdot X_d^2 \cdot (v_{ds} - v_{dsat}) \right]^{\frac{1}{2}}$$

In the preceding equation, E_p is the lateral electric field at the pinch off point. The following equation approximates its value:

$$E_p = \frac{v_c \cdot (v_c + v_{dsat})}{L_{eff} \cdot v_{dsat}}$$

The LEVEL 3 model modifies the I_{ds} current to include the channel length modulation effect:

$$I_{ds} = \frac{I_{ds}}{1 - \frac{\Delta L}{L_{eff}}}$$

To prevent a zero denominator, the ΔL value is limited:

$$\text{If } \Delta L > \frac{L_{eff}}{2}, \text{ then } \Delta L = L_{eff} - \frac{\left(\frac{L_{eff}}{2} \right)^2}{\Delta L}$$

MOS3 Empirical Model — Subthreshold Current, I_{ds}

This region of operation is characterized by the model parameter for the fast surface state (NFS). The following equation determines the modified threshold voltage (v_{on}):

$$NFS > 0 \quad v_{on} = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = v_{tm} \cdot \left[1 + \frac{q \cdot NFS}{COX} + \frac{GAMMA \cdot f_s \cdot (PHI + v_{sb})^{\frac{1}{2}} + f_n \cdot (PHI + v_{sb})}{2 \cdot (PHI + v_{sb})} \right]$$

The following equations calculate the I_{ds} current:

$v_{gs} < v_{on}$:

$$I_{ds} = I_{ds}(v_{on}, v_{de}, v_{sb}) \cdot e^{\frac{v_{gs} - v_{on}}{fast}}$$

$v_{gs} > v_{on}$:

$$I_{ds} = I_{ds}(v_{gs}, v_{de}, v_{sb})$$

Note:

Strong inversion does not use the modified threshold voltage.

If $WIC=3$, the model calculates subthreshold current differently. In this case, the I_{ds} current is:

$$I_{ds} = I_{ds}(v_{gs}, v_{de}, v_{sb}) + isub(N0_{eff}, ND_{eff}, v_{gs}, v_{ds})$$

The $isub$ subthreshold current for $LEVEL=3$ is the same as for $LEVEL=13$ (see [BSIM Model — \$i_{ds}\$ Subthreshold Current](#)).

$N0_{eff}$ and ND_{eff} are functions of the effective device width and length.

MOS3 Empirical Model — Compatibility Notes

Differences between the Synopsys Level 3 MOSFET device model and Berkeley $SPICE3$ can arise in the following situations:

This section covers the following topics:

- [MOS3 Empirical Model — Compatibility Notes: Small XJ](#)
- [MOS3 Empirical Model — Compatibility Notes: ETA](#)
- [MOS3 Empirical Model — Compatibility Notes: NSUB Missing](#)

- [MOS3 Empirical Model — Compatibility Notes: LD Missing](#)
- [MOS3 Empirical Model — Compatibility Notes: KP Given and U0 Not Given](#)

MOS3 Empirical Model — Compatibility Notes: Small XJ

LEVEL 3 and SPICE3 differ for small XJ values, typically >0.05 microns. Do not use such small values for XJ; they are physically unreasonable. XJ calculates the short-channel reduction of the GAMMA effect:

$$GAMMA \rightarrow f_s \cdot GAMMA$$

f_s is normally less than or equal to 1. For very small values of XJ, f_s can be greater than one. The Synopsys Level 3 model imposes the limit $f_s \leq 1.0$, but SPICE3 allows $f_s > 1.0$.

MOS3 Empirical Model — Compatibility Notes: ETA

In this model, 8.14 is the constant in the ETA equation, which varies the v_{ds} threshold. Berkeley SPICE3 uses 8.15.

Solution: To convert a SPICE3 model to the Synopsys Level 3 MOSFET device model, multiply ETA by 815/814.

MOS3 Empirical Model — Compatibility Notes: NSUB Missing

If you do not specify NSUB in SPICE3, the KAPPA equation becomes inactive. The Synopsys Level 3 MOSFET model generates a default NSUB from GAMMA, and the KAPPA equation is active.

Solution: If you do not specify NSUB in the SPICE3 model, set KAPPA=0 in the Synopsys Level 3 MOSFET model.

MOS3 Empirical Model — Compatibility Notes: LD Missing

If you do not specify LD, simulation uses the default (0.75XJ). The SPICE3 default for LD is zero.

Solution: If you do not specify LD in the SPICE3 model, set LD=0 in the Level 3 MOSFET model.

MOS3 Empirical Model — Compatibility Notes: KP Given and U0 Not Given

If KP is specified but U0 is not specified, the PrimeSim Continuumsimulators calculate U0 from KP. SPICE3 uses the default value (600cm²/Vs) for U0.

In summary, MOS3 in the PrimeSim Continuum simulators is not compatible with that of SPICE3. However, the same I-V characteristics can still be obtained from both SPICE

and the PrimeSim Continuum simulators, when proper model parameters are given in the model cards.

Some of the key model parameters include:

1. TOX, U0, VT0, NSUB, XJ, LD: Key technology scaling parameters.
2. VMAX, KAPPA, ETA, THETA: Velocity saturation, DIBL, mobility degradation parameters.

These parameters are used to address the major features of MOS3, and they should be extracted, and explicitly given in the model cards to simulate MOS3 model based circuit design.

Name	Symbol	Value
Boltzmann constant	k	=1.3806226e-23J·K ⁻¹
Electron charge	e	=1.6021918e-19C
Permittivity of silicon dioxide	ε _{ox}	=3.45314379969e-11F/m
Permittivity of silicon	ε _{si}	=1.035943139907e-10F/m

MOS3 Empirical Model – Temperature Compensation Example

This example is based on demonstration netlist `tempdep.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
$ test of temp dependence for LEVEL=3 Tlevc=0 Tlev=1
.option ingold=2 numdgt=6 post=2
.temp 25 100
vd d 0 5
vg g 0 2
m1 d g 0 0 nch w=10u L=1u
.op
.print id=LX4(m1) vdsat=lv10(m1)
.model nch nmos LEVEL=3 tlev=1 tlevc=0 acm=3
+ uo=600 tox=172.6572
+ vto=0.8 gamma=0.8 phi=0.64
+ kappa=0 xj=0
+ nsub=1e16 rsh=0
+ tcv=1.5e-3 bex=-1.5
.end
```

This simple model, with `XJ=0` and `KAPPA=0`, has a saturation current:

$$I_{ds} = \frac{\beta \cdot 0.5 \cdot (v_{gs} - v_{tm})^2}{1 + fb}$$

$$beta = COX \cdot \left(\frac{W}{L} \right) \cdot UO(t), fb = \frac{GAMMA}{(4 \cdot \sqrt{phi(t)})}$$

Using the model parameters in the input file, and the preceding equations, produces these results:

$$beta = (1.2e-3) \cdot \left(\frac{t}{tref} \right)^{BEX}$$

$$v_{tm} = 0.8 \cdot TCV \cdot (t - tref)$$

$$phi(t) = 0.64 \cdot \left(\frac{t}{tref} \right)^{-vtherm} \cdot (egarg + 3 \cdot \log \left(\frac{t}{tref} \right))$$

At room temperature:

$$beta = (1.2e-3)$$

$$v_{tm} = 0.8$$

$$phi(t) = 0.64$$

$$I_{ds} = (1.2e-3) \cdot 0.5 \cdot \frac{(2-0.8)^2}{1 + \frac{0.2}{\sqrt{0.64}}} = 6.912e-4$$

At T=100:

$$beta = 1.2e-3 \cdot (1.251551)^{-1.5} = 0.570545e-4$$

$$v_{tm} = 0.8 - (1.5e-3) \cdot 75 = 0.6875$$

$$egarg = 9.399920, vtherm = 3.215466e-2$$

$$phi(t) = 0.64 \cdot 1.251551 - 0.3238962 = 0.4770964$$

$$I_{ds} = beta \cdot 0.5 \cdot \frac{(2-vt)^2}{1 + \frac{0.2}{\sqrt{phi(t)}}} = 5.724507e-4$$

Simulation results:

T=25, id=6.91200e-04

T=100, id=5.72451e-04

These results agree with the hand calculations.

LEVEL 4: Grove-Frohmman-LEVEL 2 Model Derived from SPICE 2E.3

This model is the same as [LEVEL 2: MOS2 Grove-Frohmman \(SPICE 2G\) Model](#) with the following exceptions:

- No narrow width effects: $h = 1$
- No short-channel effects: $\gamma = GAMMA$
- For lateral diffusion, $LD_{scaled} = LD \cdot XJ \cdot SCALM$. If you specify XJ , the LD default=0.75. If you do not specify XJ , the default is 0.
- TPG , the model parameter for type of gate materials, defaults to zero (AL gate). The default is 1 for other levels. If you do not specify V_{TO} , this parameter computes V_{TO} (see [Common Threshold Voltage Equations](#)).
- Starting in 2001.4.2, MOSFET LEVEL 4 and LEVEL 9 support both M and AREA scaling.

LEVEL 5: AMI-ASPEC Depletion and Enhancement (Taylor-Huang) Model

This model uses micrometer units rather than the typical meter units. Units and defaults are often unique in LEVEL 5. Level 5 does not use the `SCALM` option:

- [AMI-ASPEC Model Parameters](#)
- [AMI-ASPEC Model Equations — Cutoff Region \(\$V_{gs} < V_{th}\$ \)](#)
- [AMI-ASPEC Model Equations — Depletion \(\$V_{gs} - V_{fb} < 0\$ \)](#)
- [AMI-ASPEC Model — Depletion Mode DC Model, \$ZENH=0\$](#)

AMI-ASPEC Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, it also uses the following parameters:

Name (Alias)	Units	Default	Description
AFC		1.0	Area factor for MOSFET capacitance
CAPOP		6	Gate capacitance selector
METO	μm	0.0	Metal overlap on gate

Use the `ZENH` flag mode parameter to select one of two modes: enhancement or depletion.

Parameter	Description
<code>ZENH=1</code>	This enhancement model (default mode) is a portion of the Synopsys MOS5 device model, and is identical to AMI SPICE MOS LEVEL 4.
<code>ZENH=0</code>	This depletion model is revised in the Synopsys MOS5 device model (from previous depletion mode) and is identical to AMI SPICE MOS LEVEL 5.

The Synopsys enhancement and depletion modes are basically identical to the AMI models. However, the Synopsys enhancement and depletion modes let you choose either SPICE or ASPEC temperature compensation.

- `TLEV=1` (default) uses ASPEC-style temperature compensation.
- `TLEV=0` uses SPICE-style temperature compensation.

`CAPOP=6` represents AMI Gate Capacitance in the Synopsys device models. `CAPOP=6` is the default setting for LEVEL 5 only. LEVEL 5 models can also use `CAPOP=1`, `2`, and `3`.

The `ACM` parameter defaults to 0 in LEVEL 5, invoking SPICE-style parasitics. You can also set `ACM` to 1 (`ASPEC`) or to 2 (Synopsys device models). All MOSFET models follow this convention.

You can use `.OPTION SCALE` with the LEVEL 5 model; however, you cannot use the `SCALM` option, due to the difference in units.

You *must* specify the following parameters for MOS LEVEL 5: `VTO` (`VT`), `TOX`, `UO` (`UB`), `FRC`, and `NSUB` (`DNB`).

AMI-ASPEC Model Equations — Cutoff Region ($V_{gs} < V_{th}$)

This section covers the following topics:

- [AMI-ASPEC Model — Cutoff Region, \$v_{gs} < v_{th}\$](#)
- [AMI-ASPEC Model — Effective Channel Length and Width](#)
- [AMI-ASPEC Model — Threshold Voltage, \$v_{th}\$](#)
- [AMI-ASPEC Model — Saturation Voltage, \$v_{dsat}\$](#)
- [AMI-ASPEC Model — Mobility Reduction, \$UB_{eff}\$](#)
- [AMI-ASPEC Model — Channel Length Modulation](#)
- [AMI-ASPEC Model — Subthreshold Current, \$I_{ds}\$](#)

AMI-ASPEC Model — Cutoff Region, $v_{gs} < v_{th}$

$I_{ds} = 0$ (See [AMI-ASPEC Model — Subthreshold Current, \$I_{ds}\$](#))

On Region, $v_{gs} > v_{th}$

$$i_{ds} = \beta \cdot \left\{ \left(v_{gs} - v_{bi} - \frac{v_{de}}{2} \right) \cdot v_{de} - \frac{2}{3} b \gamma b \left[(\phi_f + v_{de} + v_{sb})^{\frac{3}{2}} - (\phi_f + v_{sb})^{\frac{3}{2}} \right] \right\}$$

The following equations calculate values used in the preceding equation:

$$v_{de} = \min(v_{ds}, v_{dsat})$$

$$\beta = UB_{eff} \cdot COX \cdot \frac{W_{eff}}{L_{eff}}$$

$$\Phi_f = 2 \cdot v_{tm} \cdot \ln\left(\frac{DNB}{ni}\right)$$

The following equation calculates the gate oxide capacitances per unit area:

$$COX = \frac{E_{ox}}{TOX \cdot 1E-10} \text{ F/m}$$

AMI-ASPEC Model — Effective Channel Length and Width

The following equations determine the effective channel length and width in the LEVEL 5 model:

$$W_{eff} = W_{scaled} \cdot WMLT + OXETCH$$

$$L_{eff} = L_{scaled} \cdot LMLT - 2P(LATD + DEL)$$

AMI-ASPEC Model — Threshold Voltage, v_{th}

The V_{TO} model parameter is an extrapolated zero-bias threshold voltage for a large device. The following equation calculates the effective threshold voltage, including the device size effects and the terminal voltages:

$$v_{th} = v_{bi} + \gamma \cdot (\Phi_f + v_{sb})^{\frac{1}{2}}$$

The following equations calculate values used in the preceding equation:

$$v_{bi} = v_{fb} + \Phi_f = VTO - \gamma_0 P \Phi_f^{\frac{1}{2}}$$

$$\gamma_0 = \frac{(2 \cdot E_{si} \cdot q \cdot DNB)^{\frac{1}{2}}}{COX}$$

Note:

You must specify DNB and V_{TO} parameters for the LEVEL 5 model. The Synopsys device model uses DNB to compute γ_0 , and ignores the $GAMMA$ model parameter.

The following equation computes the γ effective body effect, including the device size effects:

$$\gamma = \gamma_0 \cdot (1 - scf) \cdot (1 + ncf)$$

If $SCM < 0$, then $scf=0$.

$$\text{Otherwise: } scf = \frac{XJ}{L_{eff}} \cdot \left\{ \left[1 + \frac{2Xd}{XJ} \cdot (SCM \cdot v_{ds} + v_{sb} + \Phi_f)^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

If $NWM < 0$, then $ncf=0$.

$$\text{Otherwise: } ncf = \frac{NWM \cdot X_d \cdot (\Phi_f)^{\frac{1}{2}}}{W_{eff}}$$

The following equation calculates the x_d value used in the preceding equations:

$$X_d = \left(\frac{2 \cdot E_{si}}{q \cdot DNB} \right)^{\frac{1}{2}}$$

AMI-ASPEC Model — Saturation Voltage, v_{dsat}

The following equation computes the saturation voltage due to the channel pinch-off at the drain side:

$$v_{sat} = v_{gs} - v_{bi} + \frac{v^2}{2} \cdot \left\{ 1 - \left[1 + \frac{4}{v^2} \cdot (v_{gs} - v_{bi} + \Phi_f + v_{sb}) \right]^{\frac{1}{2}} \right\}$$

$$v_{dsat} = v_{sat}$$

If ECV does not equal 1000, the program modifies v_{sat} to include the carrier velocity saturation effect:

$$v_{dsat} = v_{sat} + v_c \cdot (v_{sat}^2 + v_c^2)^{\frac{1}{2}}$$

where:

$$v_c = ECV \cdot L_{eff}$$

AMI-ASPEC Model — Mobility Reduction, UB_{eff}

The following equation computes the mobility degradation effect in the LEVEL 5 MOSFET model:

$$UB_{eff} = \frac{1}{\frac{1}{UB} + \frac{FRC(v_{gs} - v_{th})}{TOX} + \frac{v_{de}}{VST \cdot L_e} + FSB \cdot v_{sb}^{1/2}}$$

The following equations calculate the L_e value used in the preceding equation:

$$L_e = L_{eff} \text{ linear region}$$

$$L_e = L_{eff} \cdot \Delta L \text{ saturation region}$$

The next section describes the ΔL channel length modulation effect.

AMI-ASPEC Model — Channel Length Modulation

The LEVEL 5 model modifies the I_{ds} current to include the channel length modulation effect:

$$I_{ds} = \frac{I_{ds}}{1 - \frac{\Delta L}{L_{eff}}}$$

The following equation calculates the ΔL value used in the preceding equation:

$$\Delta L = 1e4 \cdot \left[\frac{2.73e3 \cdot XJ}{DNB \cdot \ln\left(\frac{1e20}{DNB}\right)} \right]^{\frac{1}{3}} \cdot \left[(v_{ds} - v_{dsat} + PHI)^{\frac{1}{3}} - PHI^{\frac{1}{3}} \right]$$

ΔL is in microns, if XJ is in microns and DNB is in cm^{-3} .

AMI-ASPEC Model — Subthreshold Current, I_{ds}

The Fast Surface State (FSS) characterizes this region of operation if it is greater than $1e10$. The following equation then calculates the effective threshold voltage, separating the strong inversion region from the weak inversion region:

$$v_{on} = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = v_{tm} \cdot \left[1 + \frac{q \cdot FSS}{COX} + \frac{\gamma}{2 \cdot (\Phi_f + v_{sb})^{\frac{1}{2}}} \right]$$

In the preceding equations, v_t is the thermal voltage.

The following equations calculate I_{ds} .

Weak Inversion Region, $v_{gs} < v_{th}$

$$I_{ds} = (v_{on}, v_{de}, v_{sb}) \cdot e^{\frac{v_{gs} - v_{on}}{fast}}$$

Strong Inversion Region, $v_{gs} > v_{th}$

$$I_{ds} = I_{ds}(v_{gs}, v_{de}, v_{sb})$$

Note:

Strong inversion also use the modified threshold voltage (v_{on}) that FSS produces; that is, the mobility equations use v_{on} instead of v_{th} .

AMI-ASPEC Model Equations — Depletion ($V_{gs}-V_{fb}<0$)

$$I_{ds} = \beta_1 \cdot \left\{ q \cdot NI \cdot v_{de} + cav \cdot \left[(v_{gs}-v_{fb}) \cdot v_{de} - \frac{v_{de}^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \gamma \cdot \left[(v_{de} + v_{sb} + \Phi_d)^{\frac{3}{2}} - (v_{sb} + \Phi_d)^{\frac{3}{2}} \right] \right\}$$

Enhancement, $v_{gs}-v_{fb} > v_{de} > 0$

$$I_{ds} = \beta_1 \cdot \left\{ q \cdot NI \cdot v_{de} - \frac{2}{3} \beta cav \gamma \left[(v_{de} + v_{sb} + \Phi_d)^{\frac{3}{2}} - (v_{sb} + \Phi_d)^{\frac{3}{2}} \right] \right\} \beta \cdot \left[(v_{gs}-v_{fb}) \cdot v_{de} - \frac{v_{de}^2}{2} \right]$$

Partial Enhancement, $v_{gs}-v_{fb} < v_{de}$

$$I_{ds} = \beta_1 \cdot \left\{ q \cdot NI \cdot v_{de} + cav \cdot \left[(v_{gs}-v_{fb}) \cdot v_{de} - \frac{v_{de}^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \gamma \cdot \left[(v_{de} + v_{sb} + \Phi_d)^{\frac{3}{2}} - (v_{sb} + \Phi_d)^{\frac{3}{2}} \right] \right\} \left(\frac{1}{2} \beta - \beta_1 \beta cav \right) \cdot (v_{gs}-v_{fb})^2$$

The following equations calculate values used in the preceding equations:

$$\beta_1 = UH \cdot \frac{W_{eff}}{L_{eff}}$$

$$\beta = UB_{eff} \cdot cox \cdot \frac{W_{eff}}{L_{eff}}$$

$$cav = \frac{cox \cdot cs}{cox + cs}$$

$$cs = \frac{2.77 E_{si}}{DP \cdot 1e-4}$$

$$\Phi_d = v_{tm} \cdot \ln \left(\frac{DNB \cdot nd}{ni^2} \right)$$

$$nd = \frac{NI \cdot 1e4}{DP}$$

$$v_{de} = \min(v_{ds}, v_{dsat})$$

The following sections describe the saturation voltage, threshold voltage, and effective γ .

This section covers the following topics:

- [AMI-ASPEC Model — Depletion: Threshold Voltage, \$v_{th}\$](#)
- [AMI-ASPEC Model — Depletion: Saturation Voltage, \$v_{dsat}\$](#)
- [AMI-ASPEC Model — Depletion: Mobility Reduction, \$U_{Beff}\$](#)
- [AMI-ASPEC Model — Depletion: Channel Length Modulation](#)
- [AMI-ASPEC Model — Depletion: Subthreshold Current, \$I_{ds}\$](#)

AMI-ASPEC Model — Depletion: Threshold Voltage, v_{th}

The v_{TO} model parameter is an extrapolated zero-bias threshold voltage for a large device. The following equations calculate the effective threshold voltage, including the device size effects and the terminal voltages:

$$v_{th} = v_{fb} - \beta d \mathbb{P} \left[v_{ch} - \gamma \mathbb{P}(\Phi_d + v_{sb})^{\frac{1}{2}} \right]$$

The following equations calculate values used in the preceding equation:

$$v_{fb} = V_{TO} + \beta d \cdot (v_{ch} - \gamma_0 \mathbb{P} \Phi_d^{1/2})$$

$$\beta d = \frac{U_H \cdot c_{av}}{U_B \cdot c_{ox}}$$

$$v_{ch} = \frac{q \cdot N I}{c_{av}}$$

$$\gamma_0 = \frac{(2 \cdot E_{si} \cdot q \cdot na1)^{\frac{1}{2}}}{c_{av}}$$

$$na1 = \frac{nd \cdot DNB}{nd + DNB}$$

$$nd = \frac{N I}{D P \cdot 1e-4}$$

The following equation computes the effective γ , including the small device size effects:

$$\gamma = \gamma_0 \cdot (1 - scf) \cdot (1 + ncf)$$

The following equations calculate values used in the preceding equation:

If $SCM \leq 0$, then $scf=0$. Otherwise:

$$scf = \frac{XJ}{L_{eff}} \cdot \left\{ \left[1 + \frac{2xd}{XJ} \cdot (SCM \cdot v_{ds} + v_b + \Phi_d)^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

If $N_{WM} \leq 0$, then $ncf=0$. Otherwise:

$$ncf = \frac{N_{WM} \cdot X_d \cdot \Phi_d^{\frac{1}{2}}}{W_{eff}}$$

The following equation calculates the x_d value used in the preceding equation:

$$X_d = \left(\frac{2 \cdot E_{si}}{q \cdot DNB} \right)^{\frac{1}{2}}$$

Note:

If $v_{gs} \leq v_{th}$, the surface is inverted and a residual DC current exists. If v_{sb} is large enough to make $v_{th} > v_{in_{th}}$, then v_{th} is the inversion threshold voltage. To determine the residual current, this model inserts $v_{in_{th}}$ into the I_{ds} , v_{sat} , and mobility equation in place of v_{gs} (except for v_{gs} in the exponential term of the subthreshold current).

The inversion threshold voltage at a specified v_{sb} is $v_{in_{th}}$, which the following equation computes:

$$v_{inth} = v_{fb} - \frac{q \cdot NI}{COX} - v_{sb}$$

AMI-ASPEC Model — Depletion: Saturation Voltage, v_{dsat}

The following equation computes the saturation voltage (v_{sat}):

$$v_{sat} = v_{gs} - v_{fb} + v_{ch} + \frac{v^2}{2} \cdot \left\{ 1 - \left[1 + \frac{4}{v^2} \cdot (v_{gs} - v_{fb} + v_{ch} + v_{sb} + \Phi_d) \right]^{\frac{1}{2}} \right\}$$

$$v_{dsat} = v_{sat}$$

IF ECV is not equal to 1000 (V/ μ m), the Synopsys device models modify v_{sat} to include the carrier velocity saturation effect:

$$v_{dsat} = v_{sat} + v_c - (v_{sat} + v_c^2)^{\frac{1}{2}}$$

The following equation calculates the v_c value used in the preceding equation:

$$v_c = ECV \cdot L_{eff}$$

AMI-ASPEC Model — Depletion: Mobility Reduction, UB_{eff}

The surface mobility (UB) depends on terminal voltages as follows:

$$UB_{eff} = \frac{1}{\frac{1}{UB} + \frac{FRC(v_{gs}-v_{th})}{TOX} + \frac{v_{de}}{VST \cdot le} + FSB \cdot v_{sb}^{1/2}}$$

The following equations calculate values used in the preceding equation:

$$L_e = L_{eff} \text{ Linear region}$$

$$L_e = L_{eff} - \Delta L \text{ Saturation region}$$

The next section describes the ΔL channel length modulation effect.

AMI-ASPEC Model — Depletion: Channel Length Modulation

Modify the I_{ds} current to model the channel length modulation effect:

$$I_{ds} = \frac{I_{ds}}{1 - \frac{\Delta L}{L_{eff}}}$$

The following equation calculates the ΔL value used in the preceding equation:

$$\Delta L = 1e4 \cdot \left[\frac{2.73e3 \cdot XJ}{na1 \cdot \ln\left(\frac{1e20}{na1}\right)} \right]^{\frac{1}{3}} \cdot \left[(v_{ds} - v_{dsat} + PHI)^{\frac{1}{3}} - PHI^{\frac{1}{3}} \right]$$

The ΔL parameter is in microns, if XJ is in microns and $na1$ is in cm^{-3} .

AMI-ASPEC Model — Depletion: Subthreshold Current, I_{ds}

If device leakage currents become important for operation near or below the normal threshold voltage, then this model considers the subthreshold characteristics. The Level 5 MOSFET model uses the subthreshold model only if the number of fast surface states (FSS) is greater than $1e10$. The following equation determines the effective threshold voltage (von):

$$von = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = v_{tm} \cdot \left[1 + \frac{q \cdot FSS}{COX} + \frac{\gamma}{2 \cdot (\Phi_d + v_{sb})^{\frac{1}{2}}} \right]$$

If $von < vin_{th}$, then simulation substitutes vin_{th} for von .

Note:

The Level 5 MOSFET device model uses the following subthreshold model only if $v_{gs} < v_{on}$, and if the device is either in partial or full enhancement mode. Otherwise, it uses the model in enhancement mode ($Z_{ENH}=1$). The subthreshold current calculated below includes the residual DC current.

If $v_{gs} < v_{on}$, then:

Partial Enhancement, $v_{gs}-v_{fb} < v_{de}$

$$I_{ds} = \beta_1 \cdot \left\{ q \cdot NI \cdot v_{de} + c_{av} \cdot \left[(v_{on} - v_{fb}) \cdot v_{de} - \frac{v_{de}^2}{2} \right] - \frac{2}{3} \cdot c_{av} \cdot \gamma \cdot \left[(v_{de} + v_{sb} + \Phi_d)^{\frac{3}{2}} - (v_{sb} + \Phi_d)^{\frac{3}{2}} \right] \right\} \cdot \frac{1}{2} \cdot \left(\beta \cdot e^{\frac{v_{gs}-v_{on}}{fast}} - \beta_1 \cdot c_{av} \right) \cdot (v_{on} - v_{fb})^2$$

Full enhancement: $v_{gs}-v_{fb} \geq v_{de} > 0$

$$I_{ds} = \beta_1 \cdot \left\{ q \cdot NI \cdot v_{de} - \frac{2}{3} \cdot c_{av} \cdot \gamma \cdot \left[(v_{de} + v_{sb} + \Phi_d)^{\frac{3}{2}} - (v_{sb} + \Phi_d)^{\frac{3}{2}} \right] \right\} \cdot \beta \cdot \left[(v_{on} - v_{fb}) \cdot v_{de} - \frac{v_{de}^2}{2} \right] \cdot e^{\frac{v_{gs}-v_{on}}{fast}}$$

This example is based on demonstration netlist `m15iv.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
FILE ML5IV.SP HSPICE LEVEL 5 MODEL EXAMPLES
*OUTPUT CHARACTERISTICS FOR ENHANCEMENT & DEPLETION MODE
.OPT ACCT LIST CO=132 POST=2
.OP
VDS 3 0 .1
VGS 2 0
M1 1 2 0 0 MODEN L=20U W=20U
.MODEL MODEN NMOS LEVEL=5
+ VT=.7 TOX=292 FRC=2.739E-2 DNB=2.423E16 UB=642.8
+ OXETCH=-.98 XJ=.29 LATD=.34 ECV=4 VST=5.595E7
+ FSB=7.095E-5 SCM=.4 FSS=2.2E11 NWM=.93 PHI=.61
+ TCV=1.45E-3 PTC=9E-5 BEX=1.8
*
VIDS 3 1
.DC VGS 0 5 0.2
.PRINT DC I(VIDS) V(2)
.PRINT DC I(VIDS)
$$$$$
.ALTER
$$$$$
M1 1 2 0 0 MODDP L=20U W=20U
.MODEL MODDP NMOS LEVEL=5 ZENH=0.
```

```
+ VT=-4.0 FRC=.03 TOX=800 DNB=6E14 XJ=0.8 LATD=0.7
+ DEL=0.4 CJ=0.1E-3 PHI=0.6 EXA=0.5 EXP=0.5 FSB=3E-5
+ ECV=5 VST=4E7 UB=850 SCM=0.5 NI=5.5E11 DP=0.7 UH=1200
*
.END
```

AMI-ASPEC Model — Depletion Mode DC Model, ZENH=0

This model uses depletion mode devices as the load element in contemporary standard n-channel technologies [2]. This model assumes a silicon gate construction with an ion implant used to obtain the depletion characteristics. A special model is required for depletion devices, because the implant used to create the negative threshold also results in a complicated impurity concentration profile in the substrate. The implant profile changes the basis for the traditional calculation of the QB bulk charge. The additional charge from the implant, QBI, must be calculated.

This implanted layer also forms an additional channel, offering a conductive pathway through the bulk silicon as well as through the surface channel. This second pathway can cause difficulties when trying to model a depletion device using existing MOS models.

The surface channel partially shields the bulk channel from the oxide interface, and the mobility of the bulk silicon can be substantially higher. Yet with all of these differences, a depletion model still can share the same theoretical basis as the Iantola and Moll gradual channel model.

The depletion model differs from the Iantola and Moll model:

- Implant charge accounted for.
- Finite implant thickness (DP).
- Assumes two channels: a surface channel and a bulk channel.
- Bulk channel has a bulk mobility (UH).
- Assumes that the bulk gain is different from the surface gain.

In the depletion model, the gain is lower at low gate voltages and higher at high gate voltages. Due to this variation in gain, the enhancement models cannot accurately represent a depletion device. The physical model for a depletion device is basically the same as an enhancement model, except that a one-step profile with DP depth approximates the depletion implant.

Due to the implant profile, simulation calculates the drain current equation by region. The MOSFET Level 5 model has three regions: depletion, enhancement, and partial enhancement.

Depletion Region, $v_{gs} - v_{fb} < 0$

The bulk channel dominates the low gate voltage region.

Enhancement region: $v_{gs} - v_{fb} > 0$, $v_{ds} < v_{gs} - v_{fb}$

High gate voltage and low drain voltage define the enhancement region. In this region, both channels are fully turned on.

Partial enhancement region: $v_{gs} - v_{fb} > 0$, $v_{ds} > v_{gs} - v_{fb}$

The region has high gate and drain voltages so the surface region is partially turned on and the bulk region is fully turned on.

LEVEL 6: Lattin-Jenkins-Grove (ASPEC) Model

These models represent ASPEC, MSINC, and ISPICE MOSFET model equations. The only difference between LEVEL 6: Lattin-Jenkins-Grove (ASPEC) Model and [LEVEL 7: Lattin-Jenkins-Grove \(SPICE\) Model](#) equations is the handling of the parasitic elements and the method of temperature compensation. See [Grove-Frohman \(SPICE 2G\) Model — Channel Length Modulation](#) for those model parameters.

This section covers the following topics:

- [Lattin-Jenkins-Grove \(ASPEC\) Model Parameters](#)
- [Lattin-Jenkins-Grove \(ASPEC\) Model — UPDATE Parameter](#)
- [Lattin-Jenkins-Grove \(ASPEC\) Model Equations, \(UPDATE=0 or 2\)](#)
- [Lattin-Jenkins-Grove \(ASPEC\) Model Equations, UPDATE=1](#)
- [Lattin-Jenkins-Grove \(ASPEC\) Model — Compatibility Notes](#)

Lattin-Jenkins-Grove (ASPEC) Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this model also uses the following parameters:

Alternate Saturation Model Parameters

Name (Alias)	Units	Default	Description
KA		1.0	Alternate saturation model: coefficient for the short-channel vds scaling factor.
KU		0.0	Lateral field mobility parameter.
MAL		0.5	Alternate saturation model: exponent of the short-channel vds scaling factor.
MBL		1.0	Exponent for mobility reduction due to the source-drain electric field.
NU		1.0	Mobility reduction due to the source-drain electric field.

Lattin-Jenkins-Grove (ASPEC) Model — UPDATE Parameter

The general form of the I_{ds} equation for this model 6 is the same as [LEVEL 2: MOS2 Grove-Frohman \(SPICE 2G\) Model](#). However, the small size effects, mobility reduction,

and channel length modulation are included differently. Also, you can use the multi-level GAMMA capability to model MOS transistors with ion-implanted channels.

This model can represent the ASPEC, MSINC, or ISPICE MOSFET model. Use the UPDATE model parameter to invoke different versions of this model.

Statement	Description
UPDATE=0	This is the original Synopsys Level 6 MOSFET device model, which is not quite compatible with the ASPEC model. It has some discontinuities in the weak inversion, mobility equations ($MOB=3$), and multi-level GAMMA equations.
UPDATE=1	This enhanced version of the LEVEL 6 model contains improved multi-level GAMMA equations. The saturation voltage, drain-source current, and conductances are continuous.
UPDATE=2	This version of the LEVEL 6 model is compatible with the ASPEC model. The multi-level GAMMA model is not continuous as it is in the ASPEC program. See Lattin-Jenkins-Grove (ASPEC) Model — Compatibility Notes. Set UPDATE to 1.0 to use changes to the device equations. Set UPDATE to 1.0 or 2 to use the default RS and RD handling. These values and changes provide a more accurate ASPEC model.
UPDATE=1 or 2	TOX=690 UO (UB)=750 cm2/(V · s) (N-ch) UTRA (F3)=0.0

Statement	Description
UPDATE=0	TOX=1000 $UO (UB)=750 \text{ cm}^2/(V \cdot s) \text{ (N-ch)}$ $UTRA (F3)=0.0$ If you do not specify LDIF, then the RD and RS values change in the MOSFET:
UPDATE=1 or 2 and LDIF=0	$RS = \frac{(RS + NRS \cdot RL)}{M}$ $RD = \frac{(RD + NRD \cdot RL)}{M}$ Note: The ASPEC program does not use the M multiplier. LDIF $\neq$ 0: $RS = \frac{LATDscaled + LDIFscaled}{Weff} \cdot RS + NRS \cdot \frac{RL}{M}$ $RD = \frac{LATDscaled + LDIFscaled}{Weff} \cdot RD + NRD \cdot \frac{RL}{M}$ The vde value in the mobility equations change for the alternate saturation model: $vde = \min\left(\frac{vds}{vfa}, v_{sat}\right), \text{ UPDATE}=1 \text{ or } 2$ $vde = \min(vds, vfa \cdot v_{sat}), \text{ UPDATE}=0$ The impact ionization equation calculates the saturation voltage: $vdsat = vfa \cdot v_{sat}, \text{ UPDATE}=1 \text{ or } 2$ $vdsat = v_{sat}, \text{ UPDATE}=0$ The MOB=3 mobility equation changes: $ueff = \frac{UB}{F4 + (F1-F3) \cdot VF1 + F3 \cdot (vgs-vth)^{F2}}$
UPDATE=1 or 2 and (vgs-vth) ^{F2} > VF1	
UPDATE=0 and (vgs-vth) ^{F2} > VF1:	$ueff = \frac{UB}{F4 + F3 \cdot (vgs-vth)^{F2}}$

Lattin-Jenkins-Grove (ASPEC) Model Equations, (UPDATE=0 or 2)

This section covers the following topics:

- [ASPEC Model, \(UPDATE=0 or 2\) — IDS Equation](#)
- [ASPEC Model, \(UPDATE=0 or 2\) — Effective Channel Length and Width](#)
- [ASPEC Model, \(UPDATE=0 or 2\) — Threshold Voltage, vth](#)

- ASPEC Model, (UPDATE=0 or 2) — Single-Gamma, VBO=0
- ASPEC Model, (UPDATE=0 or 2) — Effective Built-in Voltage, vbi
- ASPEC Model, (UPDATE=0 or 2) — Multi-Level Gamma, VBO>0
- ASPEC Model, (UPDATE=0 or 2) — Effective Built-in Voltage, vbi for VBO>0
- ASPEC Model, (UPDATE=0 or 2) — Saturation Voltage, vdsat
- ASPEC Model, (UPDATE=0 or 2) — Improved Multi-Level Gamma, UPDATE=1
- ASPEC Model, (UPDATE=0 or 2) — Saturation Voltage, vsat

ASPEC Model, (UPDATE=0 or 2) — IDS Equation

$$ids = \beta \cdot \left\{ \left(vgs - vbi - \frac{\eta \cdot vde}{2} \right) \cdot vde - \frac{2}{3} \gamma \left[(PHI + vde + vsb)^{\frac{3}{2}} - (PHI + vsb)^{\frac{3}{2}} \right] \right\}$$

The following equations calculate values used in the preceding equation:

$$vde = \min(vds, vdsat)$$

$$\eta = 1 + \frac{NWEscaled}{weff}$$

$$\beta = ueff \cdot COX \cdot \frac{weff}{Leff}$$

- The η , vbi, and γ values define the narrow-width effect.
- The NWE or NWM model parameters also specify the narrow-width effect.
- The vbi and γ parameters specify the short-channel effect.

ASPEC Model, (UPDATE=0 or 2) — Effective Channel Length and Width

The following equations calculate the effective channel length and width from the drawn length and width:

$$leff = Lscaled \cdot LMLT + XLscaled - 2 \cdot (LDscaled + DELscaled)$$

$$weff = M \cdot (Wscaled \cdot WMLT + XWscaled - 2 \cdot WDscaled)$$

$$LREFeff = LREFscaled \cdot LMLT + XLscaled - 2 \cdot (LDscaled + DELscaled)$$

$$WREFeff = M \cdot (WREFscaled \cdot WMLT + XWscaled - 2 \cdot WDscaled)$$

ASPEC Model, (UPDATE=0 or 2) — Threshold Voltage, v_{th}

The following equation determines the effective threshold voltage:

$$v_{th} = v_{bi} + \gamma \cdot (PHI + v_{sb})^{\frac{1}{2}}$$

The v_{bi} and γ built-in voltage value depends on the specified model parameters.

ASPEC Model, (UPDATE=0 or 2) — Single-Gamma, $V_{BO}=0$

If you set the V_{BO} model parameter to zero, simulation uses the single-gamma model, which treats the $LGAMMA$ parameter as a junction depth. To modify the $GAMMA$ parameter for the short-channel effect, this model then uses the scf factor, which the Poon and Yau formulation computes. In this case, simulation multiplies $LGAMMA$ by the $SCALM$ option.

$$scf = 1 - \frac{LGAMMA}{l_{eff}} \cdot P \left\{ \left[1 + \frac{2 \cdot LAMBDA}{LGAMMA} \cdot (PHI + v_{sb})^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

The XJ model parameter modifies the $GAMMA$ model parameter by the short-channel factor (gl):

$$gl = 1 - \frac{XJ_{scaled}}{l_{eff}} \cdot P \left\{ \left[1 + \frac{2 \cdot LAMBDA}{XJ_{scaled}} \cdot (PHI + v_{sb} + SCM \cdot v_{ds})^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

The gl factor generally replaces the scf factor for the multi-level $GAMMA$ model.

The gw factor modifies $GAMMA$ to compute the narrow-width effect:

$$gw = \frac{1 + NWM \cdot x_d}{w_{eff}}$$

The following equation calculates the x_d value used in the preceding equation:

$$x_d = \left(\frac{2 \cdot \epsilon_{si}}{q \cdot DNB} \right)^{\frac{1}{2}}$$

Finally, the effective γ , including short-channel and narrow-width effects, is

$$\gamma = GAMMA \cdot gw \cdot gl \cdot scf$$

ASPEC Model, (UPDATE=0 or 2) — Effective Built-in Voltage, v_{bi}

The Level 6 model includes the narrow-width effect. This effect is the increase in threshold voltage due to the extra bulk charge at the edge of the channel. To use this effect with the NWE model parameter, modify v_{bi} .

Modify v_{bi} to use the short-channel effect, which decreases threshold voltage due to the induced potential barrier-lowering effect. To include this effect, you must specify either the FDS parameter, or the $UFDS$ and $VFDS$ model parameters.

The following equations calculate v_{bi} , which sums up the preceding features.

$v_{ds} < V_{FDS}$, or $V_{FDS} = 0$

$$v_{bi} = v_{to} - \gamma \sqrt{\phi_i^2 + (\eta - 1) \cdot (\phi_i + v_{sb})} - \frac{I_{d,scaled}}{I_{eff}} \ln v_{sh} - \frac{\epsilon_{si}}{C_{ox} \cdot I_{eff}} \ln f_{ds} \ln v_{ds}$$

$v_{ds} > V_{FDS}$

$$v_{bi} = v_{to} - \gamma \sqrt{\phi_i^2 + (\eta - 1) \cdot (\phi_i + v_{sb})} - \frac{I_{d,scaled}}{I_{eff}} \ln v_{sh} - \frac{\epsilon_{si}}{C_{ox} \cdot I_{eff}}$$

$$[(FDS - UFDS) \cdot V_{FDS} + UFDS \cdot v_{ds}]$$

The preceding equations describe piecewise linear variations of v_{bi} as a function of v_{ds} . If you do not specify V_{FDS} , this model uses the first equation for v_{bi} .

Note:

The Level 6 MOSFET device model calculates model parameters such as V_{TO} , ϕ_i , and γ , if you did not specify them (see [Common Threshold Voltage Equations](#)).

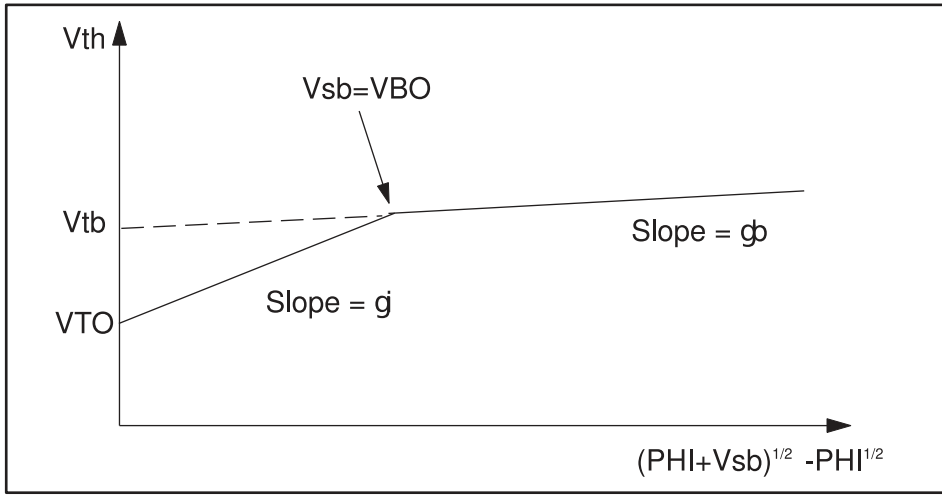
ASPEC Model, (UPDATE=0 or 2) — Multi-Level Gamma, $V_{BO} > 0$

Use Multi-Level Gamma to model MOS transistors with Ion-Implanted channels. The doping concentration under the gate is approximated as step functions.

- γ represents the corresponding body effects coefficient for the implant layer.
- γ_{sub} represents the corresponding body effects coefficient for the substrate.

[Figure 2](#) shows the variation of v_{th} as a function of v_{sb} for Multi-Level Gamma.

Figure 2 Threshold Voltage Variation



The following equations calculate the threshold voltage for different regions:

Channel Depletion Region is in the Implant Layer, $v_{sb} < V_{BO}$

$$\gamma = \gamma_i$$

$$v_{th} = v_{bi} + \gamma_i \cdot (v_{sb} + \phi)^{\frac{1}{2}}$$

$$v_{bi} = V_{TO} - \gamma_i \phi^{\frac{1}{2}}$$

Channel Depletion Region Expands into the Bulk, $v_{sb} > V_{BO}$

$$\gamma = \gamma_b$$

$$v_{th} = v_{bi} + \gamma_b \cdot (v_{sb} + \phi)^{\frac{1}{2}}$$

$$v_{bi} = v_{tb} - \gamma_b \phi^{\frac{1}{2}}$$

For the threshold voltage to be continuous at $v_{sb} = V_{BO}$, v_{tb} must be:

$$v_{tb} = V_{TO} + (\gamma_i - \gamma_b) \cdot \left[(V_{BO} + \phi)^{\frac{1}{2}} - (\phi)^{\frac{1}{2}} \right]$$

- γ_i is the effective value of GAMMA .
- γ_b is the effective value of LGAMMA .

The model computes them as γ in single-gamma models, except the scf factor is 1.0.

$$\gamma_i = \text{GAMMA} \cdot g_w \cdot g_l$$

$$\gamma b = LGAMMA \cdot gw \cdot gl$$

ASPEC Model, (UPDATE=0 or 2) — Effective Built-in Voltage, vbi for VBO>0

For vds < VFDS

if vsb ≤ VBO:

$$vbi = VTO - \gamma \sqrt{PHI} + (\eta - 1) \cdot (PHI + vsb) - \frac{LDscaled}{Leff} \cdot VSH - \frac{\epsilon si}{COX \cdot Leff} \cdot FDS \cdot vds$$

if vsb > VBO:

$$vbi = VTO - \gamma \sqrt{PHI} + (\gamma - 1) \cdot \left[(VBO + PHI)^{\frac{1}{2}} - (PHI)^{\frac{1}{2}} \right] + (\eta - 1) \cdot (vbi + vsb) - \frac{LDscaled}{Leff} \cdot VSH - \frac{\epsilon si}{COX \cdot Leff} \cdot FDS \cdot vds$$

For vds > VFDS

if vsb ≤ VBO:

$$vbi = VTO - \gamma \sqrt{PHI} + (\eta - 1) \cdot (PHI + vsb) - \frac{LDscaled}{Leff} \cdot VSH - \frac{\epsilon si}{COX \cdot Leff} \cdot \left[(FDS - UFDS) \cdot VFDS + UFDS \cdot vds \right]$$

if vsb > VBO:

$$vbi = VTO - \gamma \sqrt{PHI} + (\gamma - 1) \cdot \left[(VBO + PHI)^{\frac{1}{2}} - (PHI)^{\frac{1}{2}} \right] + (\eta - 1) \cdot (vbi + vsb) - \frac{LDscaled}{Leff} \cdot VSH - \frac{\epsilon si}{COX \cdot Leff} \cdot \left[(FDS - UFDS) \cdot VFDS + UFDS \cdot vds \right]$$

ASPEC Model, (UPDATE=0 or 2) — Saturation Voltage, vdsat

The following formula determines the saturation voltage due to channel pinch-off at the drain side:

$$vdsat = \frac{vgs - vbi}{\eta} + \frac{1}{2} \left(\frac{\gamma}{\eta} \right)^2 \cdot \left\{ 1 - \left[1 + \left(\frac{2 \cdot \eta}{\gamma} \right)^2 \cdot \left(\frac{vgs - vbi}{\eta} + PHI + vsb \right) \right]^{\frac{1}{2}} \right\}$$

The following equation calculates the reduction of the saturation voltage due to the carrier velocity saturation effect:

$$vdsat = vdsat + vc \cdot (vdsat^2 + vc^2)^{\frac{1}{2}}$$

In the preceding equation, determines v_c if the $ECRIT$ model parameter >0 , or $V_{MAX} >0$, and $KU \leq 1$. If you specify both $ECRIT$ and V_{MAX} , then simulation uses only the V_{MAX} equation. However, this model does not use the V_{MAX} equation if $MOB=4$ or $MOB=5$, because these mobility equations already contain a velocity saturation term.

$$v_c = ECRIT \cdot Leff \text{ or } v_c = \frac{V_{MAX} \cdot Leff}{ueff}$$

Because $v_{sb} > V_{BO}$, γ switches from γ_I to γ_β , and the i_{ds} , v_{sat} , and conductance values are not continuous as in the following example. To correct this discontinuity problem, specify the $UPDATE=1$ model parameter. The next section discusses this improvement.

This example is based on demonstration netlist `tgam2.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
$ tgam2.sp---multi-level gamma model
* this data is for the comparison of multi-level gamma
* update=0 or 2 and the improved multi-level gamma update=1.
*
.options post aspec nomod vntol=.1u reli=.001 relv=.0001
*
.model nch nmos bulk=99 update=2
+ fds=0.9 ku=1.6 mal=0.5 mob=1 clm=1
+ latd=0.2 phi=0.3 vt=0.9 gamma=0.72 lgamma=0.14
+ vb0=1.2 fl=0.08 esat=8.6e+4 kl=0.05
+ lambda=3.2u ub=638 f3=0.22
+ ka=0.97 mbl=0.76 nfs=1.0e+12 wic=0
+ ldel=0.084 wdel=0.037 tox=365 vsh=0.7
*
vd 1 0 5
vb 0 99 0
vg 2 0 1
ma 1 2 0 99 nch 26.0 1.4
.dc vb 1.0 1.3 .01
.probe ids=par('i(ma)') vth=par('lv9(ma)') vdsat=par('lv10(ma)')
.probe gm=par('LX7(ma)') ds=par('LX8(ma)') gmbs=par('LX9(ma)')
.end
```

Figure 3 Variation of I_{DS} , V_{TH} , and V_{DSAT} for $UPDATE=0$

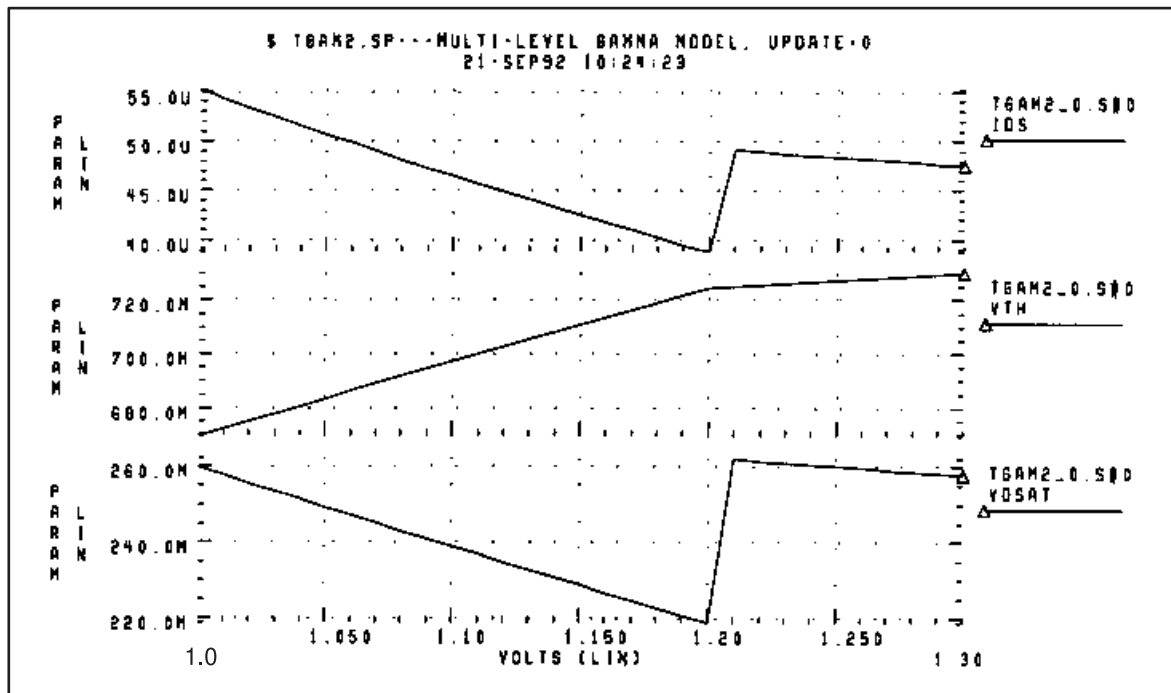
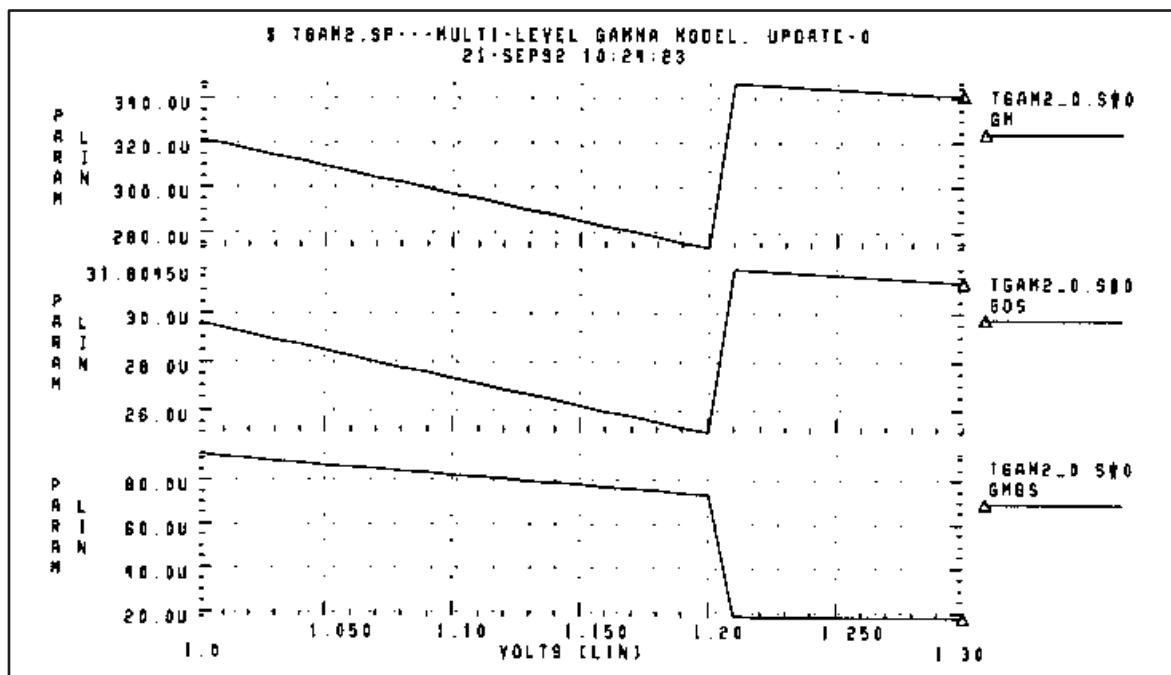


Figure 4 Variation of G_M , G_{DS} , and G_{MBS} for $UPDATE=0$



Each plot compares IDS, VTH, VDSAT, GM, GDS, and GMBS as a function of vsb for UPDATE=0.

ASPEC Model, (UPDATE=0 or 2) – Improved Multi-Level Gamma, UPDATE=1

As demonstrated in previous sections, the regular Multi-Level Gamma displays some discontinuities in saturation voltage and drain current. This occurs because when v_{sb} is less than VBO, simulation sets γ to γ_i and uses it to calculate i_{ds} and v_{sat} . This is not correct; if $(v_{ds} + v_{sb})$ exceeds VBO, then the depletion regions at the drain side expands into the substrate region, and the v_{sat} computation must use γ_b instead of γ_i . Because $v_{sat} = v_{gs} - v_{th}(\text{drain})$, this model uses γ_i to compute the threshold voltage at the drain for $v_{sb} < VBO$. As a result, the existing model overestimates the threshold voltage ($\gamma_i > \gamma_b$), and underestimates the saturation voltage and the drain current in the saturation region.

This causes a discontinuous increase in the saturation drain current, crossing from the $v_{sb} < VBO$ region to the $v_{sb} > VBO$ region.

The improved Multi-Level model upgrades the saturation voltage and drain current equations, compared to the regular Multi-Level model. To use the improved model, set the model parameter to UPDATE=1.

You can see an example of a multi-level gamma model with UPDATE=2 using a netlist from a previous example. Change UPDATE=0 to UPDATE=2 in the netlist located in directory \$installdir/demo/hspice/mos/tgam2.sp:

```
$ tgam2.sp---multi-level gamma model
* this data is for the comparison of multi-level gamma
* update=0 or 2 and the improved multi-level gamma update=1.
*
.options post aspect nomod vntol=.1u reli=.001 relv=.0001
*
.model nch nmos bulk=99 update=2
+ fds=0.9 ku=1.6 mal=0.5 mob=1 clm=1
+ latd=0.2 phi=0.3 vt=0.9 gamma=0.72 lgamma=0.14
+ vb0=1.2 fl=0.08 esat=8.6e+4 kl=0.05
+ lambda=3.2u ub=638 f3=0.22
+ ka=0.97 mbl=0.76 nfs=1.0e+12 wic=0
+ ldel=0.084 wdel=0.037 tox=365 vsh=0.7
*
vd 1 0 5
vb 0 99 0
vg 2 0 1
ma 1 2 0 99 nch 26.0 1.4
.dc vb 1.0 1.3 .01
.probe ids=par('i(ma)') vth=par('lv9(ma)') vdsat=par('lv10(ma)')
.probe gm=par('LX7(ma)') ds=par('LX8(ma)') gmbs=par('LX9(ma)')
.end
```

Figure 5 Variation of I_{DS} , V_{TH} , and V_{DSAT} for $UPDATE=2$

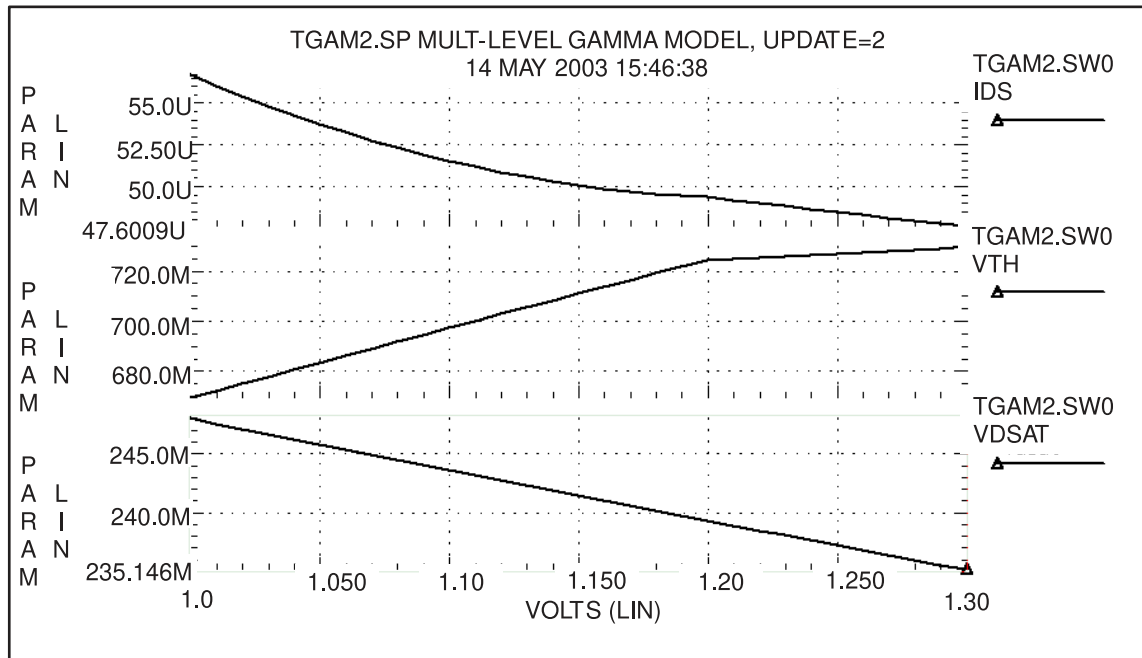
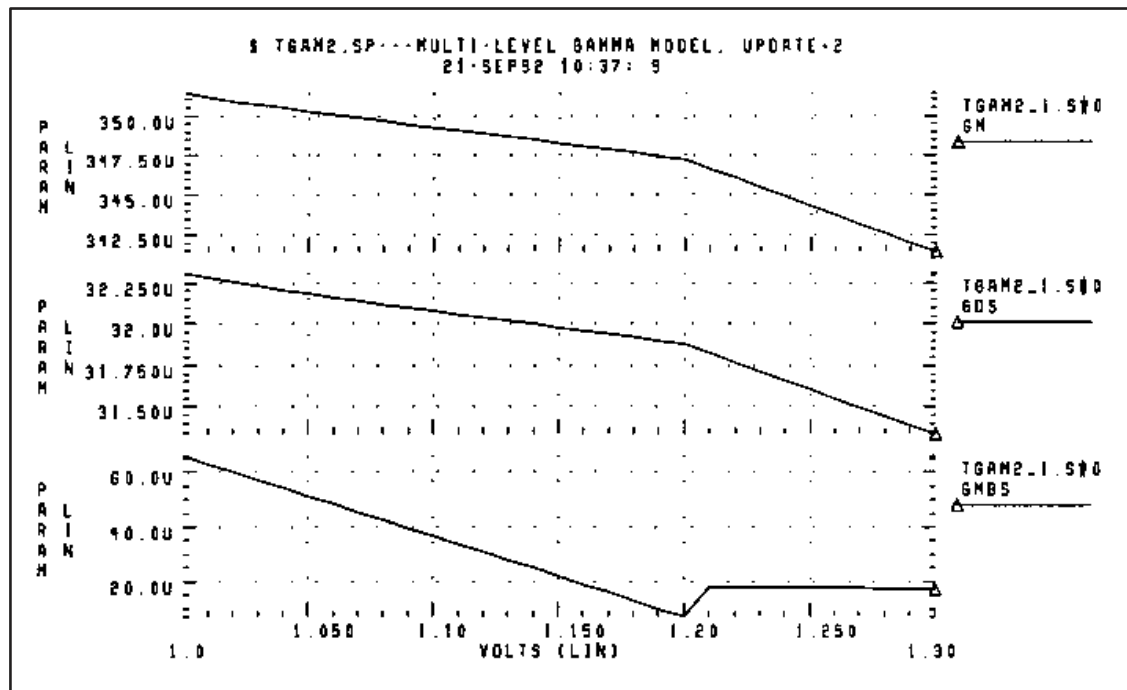


Figure 6 Variation of GM , GDS , and $GMBS$ for $UPDATE=2$



Each plot compares IDS, VTH, VDSAT, GM, GDS, and GMBS as a function of vsb for UPDATE=1.

ASPEC Model, (UPDATE=0 or 2) — Saturation Voltage, vsat

To obtain the right value for vsat, the following equations calculate two trial values of vsat corresponding to γ_i and γ_b :

$$v_{sat1} = \frac{v_{gs}-v_{bi1}}{\eta} + \frac{1}{2} \left(\frac{\gamma_i}{\eta} \right)^2 \cdot \left\{ 1 - \left[1 + \left(\frac{2 \cdot \eta}{\gamma_i} \right)^2 \cdot \left(\frac{v_{gs}-v_{bi1}}{\eta} + PHI + v_{sb} \right) \right]^{\frac{1}{2}} \right\}$$

$$v_{sat2} = \frac{v_{gs}-v_{bi2}}{\eta} + \frac{1}{2} \left(\frac{\gamma_b}{\eta} \right)^2 \cdot \left\{ 1 - \left[1 + \left(\frac{2 \cdot \eta}{\gamma_b} \right)^2 \cdot \left(\frac{v_{gs}-v_{bi2}}{\eta} + PHI + v_{sb} \right) \right]^{\frac{1}{2}} \right\}$$

- vbi1 is the built-in potential corresponding to γ_i .
- vbi2 is the built-in potential corresponding to γ_b .
- If $(v_{dsat1} + v_{sb}) \leq VBO$, then $v_{dsat}=v_{dsat1}$
- If $(v_{dsat2} + v_{sb}) > VBO$, then $v_{dsat}=v_{dsat2}$

To obtain vdsat, vc modifies vsat for the carrier velocity saturation.

Lattin-Jenkins-Grove (ASPEC) Model Equations, UPDATE=1

You can use one of three equations for ids, depending on the region of operation. To derive these equations, this model integrates the bulk charge ($v_{gs} - v_{th}(v) - v$) from the source to the drain.

For $v_{sb} < VBO - v_{de}$, the model forms an entire gate depletion region in the implant layer.

$$ids = \beta \cdot \left\{ \left(v_{gs}-v_{bi1}-\frac{\eta \cdot v_{de}}{2} \right) \cdot v_{de} - \frac{2}{3} \cdot \gamma_i \cdot \left[(PHI + v_{de} + v_{sb})^{\frac{3}{2}} - (PHI + v_{sb})^{\frac{3}{2}} \right] \right\}$$

In the preceding equation, vbi1 is the same as vbi for $v_{sb} \leq VBO$.

For $v_{sb} \geq VBO$, the entire gate depletion region expands into the bulk area.

$$ids = \beta \cdot \left\{ \left(v_{gs}-v_{bi2}-\frac{\eta \cdot v_{de}}{2} \right) \cdot v_{de} - \frac{2}{3} \cdot \gamma_b \cdot \left[(PHI + v_{de} + v_{sb})^{\frac{3}{2}} - (PHI + v_{sb})^{\frac{3}{2}} \right] \right\}$$

In the preceding equation, vbi2 is the same as vbi for $v_{sb} > VBO$.

$$ids = \beta \cdot \left\{ \left(v_{gs} - v_{bi1} - \frac{\eta \cdot v_{de}}{2} \right) \cdot v_{de} - \frac{2}{3} \cdot \gamma_i \cdot \left[(PHI + v_{de} + v_{sb})^{3/2} - (PHI + v_{sb})^{3/2} \right] \right\}$$

For $V_{BO} - v_{de} < v_{sb} < V_{BO}$, the source side gate depletion region is in the implant layer, but the drain side gate depletion region expands into the bulk area.

This section covers the following topics:

- [ASPEC Model, \(UPDATE=1\) — Alternate DC Model \(ISPICE model\)](#)
- [ASPEC Model, \(UPDATE=1\) — Subthreshold Current, \$i_{ds}\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$WIC=0\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$WIC=1\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$WIC=2\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$WIC=3\$](#)
- [ASPEC Model, \(UPDATE=1\) — Effective Mobility, \$\mu_{eff}\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=0\$ Default, No Mobility](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=1\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=2\$, Frohman-Bentchkowski Equation](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=3\$, Normal Field Equation](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=4\$ or \$5\$, Universal Field Mobility Reduction](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=5\$](#)
- [ASPEC Model, \(UPDATE=1\) — \$MOB=6\$ or \$7\$, Modified \$MOB=3\$:](#)
- [ASPEC Model, \(UPDATE=1\) — Channel Length Modulation](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=0\$, No Channel Modulation—Default](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=1\$, Step Depletion Equation](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=2\$, Electrostatic Fringing Field](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=3\$, Carrier Velocity Saturation Equation](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=4\$, Wang's Equation](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=5\$, Channel Length Modulation](#)
- [ASPEC Model, \(UPDATE=1\) — \$CLM=6\$, \$\Delta L\$ Equation](#)

ASPEC Model, (UPDATE=1) — Alternate DC Model (ISPICE model)

To invoke this model, set the $KU > 1$ model parameter. Then, the model computes v_{fu} and v_{fa} scale factors to scale both the v_{ds} voltage and the i_{ds} current. These scale factors are

functions of $ECRIT$ and the vgs voltage. The following equations compute the vfa and vf_u factors:

$$vf_u = 1 - \frac{KU}{(\alpha^2 + KU^2)^{\frac{1}{2}} + \alpha(KU - 1)}$$

$$vfa = KA \cdot vf_u^{(2MAL)}$$

The following equation calculates the α value used in the preceding equations:

$$\alpha = \frac{ECRIT \cdot Leff}{vgs - vth}$$

Note:

The vf_u factor is always less than one.

The following equation modifies the ids current:

$$NU = 1$$

$$ids = vf_u^{(2MBL)} \cdot ids$$

For $NU = 0$, the

$$vf_u^{(2MBL)}$$

factor is set to one.

The ids current is a function of the effective drain to source voltage (vde):

$$vde = \min(vds/vfa, vsat)$$

$$vdsat = vfa \cdot vsat$$

This alternate model is generally coupled with the mobility normal field equations ($MOB = 3$) and the channel length modulation drain field equation ($CLM = 3$).

The mobility equations use the following vde and vds values:

$$vde = \min(vds, vfa \cdot vsat), \text{UPDATE} = 0$$

$$vds = \min(vds/vfa, vsat), \text{UPDATE} = 1, 2$$

ASPEC Model, (UPDATE=1) — Subthreshold Current, ids

is the choice of two different equations, selected through The WIC (Weak Inversion Choice) model parameter characterizes this region of operation.

Parameter	Description
WIC=0	No weak inversion (default)
WIC=1	ASPEC-style weak inversion
WIC=2	Enhanced the PrimeSim Continuum simulators -style weak inversion

In addition to `WIC`, set the `NFS` parameter. `NFS` represents the number of fast states per centimeter squared. Reasonable values for `NFS` range from 1e10 to 1e12.

ASPEC Model, (UPDATE=1) — WIC=0

No weak inversion.

ASPEC Model, (UPDATE=1) — WIC=1

The v_{th} threshold voltage increases by the fast term.

$$v_{on} = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = v_t \cdot \left[1 + \frac{q \cdot NFS}{COX} + \frac{\gamma}{2 \cdot (v_{sb} + PHI)^{\frac{1}{2}}} \right]$$

In the preceding equations, v_t is the thermal voltage. The following equation specifies the i_{ds} current for $v_{gs} < v_{on}$:

$$i_{ds} = i_{ds}(v_{on}, v_{de}, v_{sb}) \cdot e^{\frac{v_{gs} - v_{on}}{fast}}$$

if $v_{gs} < v_{on}$, then

$$i_{ds} = i_{ds}(v_{ge}, v_{de}, v_{sb})$$

Note:

Strong inversion conditions do not use the modified threshold voltage (v_{on}).

ASPEC Model, (UPDATE=1) — WIC=2

The subthreshold region is limited between the cutoff region and the strong inversion region. If the gate voltage is less than v_{th-PHI} , this model cannot include any weak inversion conduction. However, this model can still include diffusion conduction from the drain-to-bulk rather than from the drain-to-source.

$$v_{on} = v_{th} + fast$$

The following equation calculates the fast value used in the preceding equation:

$$fast = vt \cdot \left[1 + \frac{q \cdot NFS}{COX} + \frac{\gamma}{2 \cdot (vsb + PHI)^{\frac{1}{2}}} \right]$$

Cutoff Region, $vgs \leq vth - PHI$

$$ids = 0$$

Weak Inversion, $vth - PHI < vgs \leq von$

$$ids = ids(von, vde, vsb) \cdot \left(1 - \frac{von - vgs}{fast + PHI} \right)^{WEX}$$

Strong Inversion, $vgs > von$

$$ids = ids(vgs, vde, vsb)$$

Note:

Strong inversion conditions do not use the modified threshold voltage (von).

ASPEC Model, (UPDATE=1) — WIC=3

If $WIC=3$, simulation calculates the subthreshold current differently. In this case, the ids current is:

$$ids = ids(vgs, vde, vsb) + isub(N0eff, NDeff, vgs, vds)$$

N0eff and NDeff are functions of the effective device width and length.

ASPEC Model, (UPDATE=1) — Effective Mobility, ueff

All mobility equations have the following general form:

$$ueff = UO \cdot factor$$

Parameter	Description
UEFF	Effective mobility at the specified a specified analysis temperature.
FACTOR	Mobility degradation factor. Default=1.0

Use the MOB model parameter to select the mobility modulation equation used in the Level 6 MOSFET model.

Parameter	Description
$MOB=0$	No mobility reduction (default)

Parameter	Description
MOB=1	Gm equation
MOB=2	Frohmman-Bentchkowski equation
MOB=3	Normal field equation
MOB=4	Universal field mobility reduction
MOB=5	Universal field mobility reduction with an independent drain field
MOB=6	Modified MOB=3 equations (lateral field effect included)
MOB=7	Modified MOB=3 equations (lateral field effect not included)

The following sections describe these equations.

ASPEC Model, (UPDATE=1) — MOB=0 Default, No Mobility

FACTOR=1.0 No mobility reduction

ASPEC Model, (UPDATE=1) — MOB=1

Name (Alias)	Units	Default	Description
F1	1/V	0.0	Gate field mobility reduction
UTRA (F3)	factor	0.0	Source-drain mobility reduction factor

Use the MOB=1 equation for transistors with constant source-to-bulk voltage, because the factor does not contain a vsb term. This equation sometimes over-estimates mobility for small gate voltages and large back-bias, such as depletion pull-ups.

$$factor = \frac{1}{1 + F1 \cdot (vgs - vb1 - F3 \cdot vde)}$$

$$vde = \min(vds, vdsat)$$

Note:

In the alternate saturation model, vde is different if UPDATE=0 than if UPDATE=1. See [ASPEC Model, \(UPDATE=1\) — Alternate DC Model \(ISPICE model\)](#). Also, if VMAX>0, then vde=min (vds, vsat). If you do not specify VMAX, then vde=min (vds, vdsat).

ASPEC Model, (UPDATE=1) — MOB=2, Frohman-Bentchkowski Equation

Name (Alias)	Units	Default	Description
F1	V/cm	0.0	Critical gate-bulk electric field at which mobility reduction becomes significant.
UEXP (F2)		0.0	Mobility exponent. Use a factor of 0.36 for n-channel and 0.15 for p-channel.
UTRA (F3)	factor	0.0	Source-drain mobility reduction factor.
VMAX (VMX)	cm/s	0.0	Maximum drift velocity of carriers. The VMAX setting determines which calculation scheme vdsat uses. Zero indicates an infinite value.

The mobility reduction equation ($MOB=2$) [2] produces good results for high gate voltages and drain fields with constant back-bias. Typically, you can use this equation for p-channel pull-ups and n-channel pull-downs. The V_{MAX} value selects the proper vdsat calculation scheme. $MOB=2$ (SPICE default) corresponds to MSINC UN=2.

$$factor = \left[\frac{F1 \cdot \epsilon_{Si}}{COX \cdot (v_{gs} - v_{bi} - F3 \cdot v_{de})} \right]^{F2}$$

vde is the same in this equation as in the $MOB=1$ equation.

ASPEC Model, (UPDATE=1) — MOB=3, Normal Field Equation

Name (Alias)	Units	Default	Description
F1	1/V	0.0	Low-field mobility multiplier
F4		1.0	Mobility summing constant
UEXP (F2)		0.0	Mobility exponent
UTRA (F3)	1/V	0.0	High-field mobility multiplier
VF1	V	0.0	Low to high field mobility (voltage switch)

This equation is the same as MSINC UN=1:

$$(v_{gs} - v_{th})^{F2}$$

$\leq VF1$:

$$factor = \frac{1}{F4 + F1 \cdot (vgs - vth)^{F2}}$$

If $UPDATE=0$, and $(vgs - vth)^{F2} > VF1$:

$$factor = \frac{1}{F4 + F3 \cdot (vgs - vth)^{F2}}$$

If $UPDATE=1, 2$ and $(vgs - vth)^{F2} > VF1$:

$$factor = \frac{1}{F4 + (F1 - F3) \cdot VF1 + F3 \cdot (vgs - vth)^{F2}}$$

ASPEC Model, (UPDATE=1) — MOB=4 or 5, Universal Field Mobility Reduction

Name (Alias)	Units	Default	Description
ECRIT	V/cm	0.0	Critical electric drain field for mobility reduction. Zero indicates an infinite value.
F1	V/cm	0.0	Source-drain mobility reduction field (typical values are 1e4 to 5e8).
MOB		0.0	Selects a mobility equation: - Set MOB=4 for the critical field equation. - Set MOB=5 for the critical field equation with an independent drain field.
UEXP (F2)	$1/V^{1/2}$	0.0	Bulk mobility reduction factor (typical values are 0 to 0.5).
UTRA (F3)	V/cm	0.0	Critical electric drain field for mobility reduction.

The $MOB=4$ equation is the same as the $MSINC_{UN=3}$ equation.

ASPEC Model, (UPDATE=1) — MOB=5

The $MOB=5$ equation is the same as $MOB=4$, except that $F3$ substitutes for $ECRIT$ in the vc expression.

The $MOB=5$ equation provides a better fit for CMOS devices in the saturation region. Do not specify a $VMAX$ value, because the mobility equation calculates the velocity saturation.

$$factor = \frac{1}{1 + \frac{COX}{F1 \epsilon_{OX}} \cdot (vgs - cth) + \frac{vde}{vc} + F2 \cdot (vsb + PHI)^{\frac{1}{2}}}$$

- If $MOB=4$, then

$$v_c = ECRIT \cdot Leff$$

- If $MOB=5$, then

$$v_c = F3 \cdot Leff$$

Note:

If you use the alternate saturation model, v_{de} is different for $UPDATE=0$ than it is for $UPDATE=1, 2$.

ASPEC Model, (UPDATE=1) — MOB=6 or 7, Modified MOB=3:

This mobility equation is the same as $MOB=3$, except that the equation uses v_{TO} instead of v_{th} . If you specify $MOB=6$, the following equation modifies the i_{ds} current:

$$i_{ds} = \frac{i_{ds}}{1 + F1 \cdot \left(v_{gs} - v_{th} - \frac{v_{de}}{2} \right) + \frac{UTRA}{Leff} \cdot v_{de}}$$

ASPEC Model, (UPDATE=1) — Channel Length Modulation

The basic MOSFET current equation for i_{ds} describes a parabola, where the peak corresponds to the drain-to-source saturation voltage (v_{dsat}). Long-channel MOSFETs generally demonstrate ideal behavior. For v_{ds} voltages greater than v_{dsat} , i_{ds} current does not increase. As channel length decreases, current in the saturation region continues to increase.

The simulator models this increase in current as a decrease in the effective channel length. Except for $CLM=5$ and 6 , this model calculates the channel length modulation equations only when the device is in the saturation region.

The Level 6 MOSFET model provides several channel length modulation equations; all (except $CLM=5$) modify the i_{ds} equation:

$$i_{ds} = \frac{i_{ds}}{1 - \frac{\Delta L}{Leff}}$$

ΔL is the change in channel length due to MOSFET electric fields.

The CLM model parameter designates the channel length modulation equation for the Level 6 MOSFET device model:

Parameter	Description
$CLM=0$	No channel length modulation (default)
$CLM=1$	One-sided step depletion layer drain field equation

Parameter	Description
CLM=2	Frohmman's electrostatic fringing field equation
CLM=3	One-sided step depletion layer drain field equation with carrier velocity saturation
CLM=4	Wang's equation: linearly graded depletion layer
CLM=5	Synopsys channel length modulation
CLM=6	Synopsys D_L equations

The following sections describe these equations and the associated model parameters.

ASPEC Model, (UPDATE=1) — CLM=0, No Channel Modulation—Default

$$\Delta L = 0$$

This is the default channel length equation, representing no channel length modulation; it corresponds to $MSINC_{GDS}=0.0$.

ASPEC Model, (UPDATE=1) — CLM=1, Step Depletion Equation

Name (Alias)	Units	Default	Description
KL		0.0	Empirical constant (saturation voltage)
LAMBDA (LAM, LA)	cm/V ^{1/2}	1.137e-4	Channel length modulation. If you do not specify s, simulation calculates it from NSUB. The default LAMBDA corresponds to the default NSUB value.

$$\Delta L = LAMBDA \cdot (vds - vdsat)^{\frac{1}{2}} \cdot \left(\frac{vdsat}{vsat} \right)^{KL}$$

If you do not specify LAMBDA, simulation calculates it as:

$$LAMBDA = \left(\frac{2 \cdot \epsilon si}{q \cdot DNB} \right)^{\frac{1}{2}}$$

This is a one-sided step depletion region formulation by Grove: ΔL varies with the depletion layer width, which is a function of the difference between the effective saturation voltage ($vdsat$) and the drain-to-source channel voltage (vds). Typically, you can use this equation for long channels and high dopant concentrations. This corresponds to $GDS=1$ in $MSINC$.

ASPEC Model, (UPDATE=1) — CLM=2, Electrostatic Fringing Field

Name (Alias)	Units	Default	Description
A1		0.2	First fringing field factor, gate-drain
A2		0.6	Second fringing field factor, gate-vdsat

$$\Delta L = \frac{\varepsilon_{si}}{COX} \cdot \frac{v_{ds} - v_{dsat}}{A1 \cdot (v_{ds} - v_{gs} + v_{bi}) + A2 \cdot (v_{gs} - v_{bi} - v_{dsat})}$$

You can use the fringing field equation or electrostatic channel length reduction (developed by Frohman-Bentchkowski) to model short-channel enhancement transistors. In MSINC, the equivalent equation is $GDS=2$.

ASPEC Model, (UPDATE=1) — CLM=3, Carrier Velocity Saturation Equation

Name (Alias)	Units	Default	Description
KA		1.0	vds scaling factor for velocity saturation.
KCL		1.0	Exponent for vsb scaling factor.
KU		0.0	Velocity saturation switch. If KU ≤ 1, simulation uses the standard velocity saturation equation.
LAMBDA (LAM, LA)	cm/V ^{1/2}	1.137e-4	Channel length modulation. If you do not specify LAMBDA, simulation calculates it from NSUB. The default LAMBDA corresponds to the default NSUB value.
MAL		0.5	vds exponent for velocity saturation.
MCL		1.0	Short channel exponent.

$$\Delta L = vfu^{(2MCL)} \cdot LAMBDA$$

$$\left[(v_{ds} - v_{fa} \cdot v_{sat} + kcl \cdot v_{sb} + phi)^{\frac{1}{2}} - (kcl \cdot v_{sb} + phi)^{\frac{1}{2}} \right]$$

This equation is an extension of the first depletion layer equation, $CLM=1$. It includes effects of carrier velocity saturation, and source-to-bulk voltage (vsb) depletion layer width. It represents the basic ISPIECE equation. See [ASPEC Model, \(UPDATE=1\) — Alternate DC Model \(ISPIECE model\)](#) for definitions of vfa and vfu .

ASPEC Model, (UPDATE=1) — CLM=4, Wang's Equation

Name (Alias)	Units	Default	Description
A1	m	0.2	Junction depth: $A1_{scaled} = A1 \cdot SCALM$
DND	cm ⁻³	1e20	Drain diffusion concentration

Linearly Graded Depletion Layer:

$$\Delta L = \left[\frac{2.73e5 \cdot A1_{scaled}}{DND \cdot \ln\left(\frac{DND}{DNB}\right)} \right]^{\frac{1}{3}} \cdot \left[(vds - vdsat + PHI)^{\frac{1}{3}} - PHI^{\frac{1}{3}} \right]$$

Wang's equation can include junction characteristics to calculate the channel length modulation. The equation assumes that the junction approximates a linearly-graded junction, and provides a value of 0.33 for the exponent. This equation is similar to MSINC GDS=3.

ASPEC Model, (UPDATE=1) — CLM=5, Channel Length Modulation

Name (Alias)	Units	Default	Description
LAMBDA	amp/V ²	0	Constant coefficient
VGLAM	1/V	0	Constant coefficient

If CLM=5, the ids current increases by idssat:

$$idssat = \frac{w_{eff}}{L_{eff}} \cdot LAMBDA \cdot vds \cdot (vgs - vth) \cdot [1 + VGLAM \cdot (vgs - vth)]$$

$$ids = ids + idssat$$

Note:

The equation adds the idssat term to ids in all regions of operation. Also, LAMBDA is a function of the temperature.

ASPEC Model, (UPDATE=1) — CLM=6, ΔL Equation

Name (Alias)	Units	Default	Description
LAMBDA	1/V ^{KL}	0	vds coefficient

Name (Alias)	Units	Default	Description
LAM1	1/m	0	Channel length coefficient
KL		0	vds exponent
VGLAM	1/V	0	Gate drive coefficient

Unlike the other `CLM` values, this equation calculates the channel length modulation (ΔL) in all regions of operations, and uses it to modify the `ids` current.

$$\Delta L = \frac{Leff \cdot LAMBDA \cdot vds^{KL} \cdot [1 + VGLAM \cdot (vgs - vth)]}{1 + LAM1 \cdot Leff}$$

$$ids = \frac{ids}{1 - \frac{\Delta L}{Leff}}$$

Note:

`LAMBDA` is a function of the temperature.

Lattin-Jenkins-Grove (ASPEC) Model — Compatibility Notes

To make MOSFET models compatible with ASPEC, specify `ASPEC=1` in the `.OPTION` statement and `LEVEL=6` in the associated MOSFET model statement.

If you assign the element parameters without keynames, specify the parameters in the same sequence as in the general format. The Level 6 MOSFET model assigns parameters in the order that you list them in the element statement. If parameter names are also element keynames, simulation reports errors.

If you use the `ASPEC` option, several program variations occur. The `LEVEL` model parameter is set to 6.

Note:

Setting `LEVEL=6` in the model does not invoke ASPEC.

MOSFET control option	WL=1
General control options	SCALE=1e-6SCALM=1e-6

ASPEC sets the `SCALE` and `SCALM` options so it effectively changes the default units in parameters that these options affect. Parameter values must be consistent with these scaling factors.

LEVEL	=	6
ACM	=	1
CJ	=	0.0
IS	=	0.0
NSUB	=	1e15
PHI	=	$1 \cdot F_f$ (the Fermi potential)
TLEV	=	1
TLEVC	=	1

Note:

Do not calculate `NSUB` from `GAMMA`, if `UPDATE=1` or `2`.

`TLEV` (`TLEVC`) selects the ASPEC method of updating temperatures for the `CJ`, `CJSW`, `PB`, `PHP`, `VTO`, and `PHI` parameters.

Note:

If you explicitly enter `PHI`, this model does not update it for temperature. `SCALM` does not affect how simulation scales parameters for the ASPEC mode. If you specify `SCALM` when you use ASPEC, the Level 7 MOSFET model generates an error stating that it ignores `SCALM`.

LEVEL 7: Lattin-Jenkins-Grove (SPICE) Model

The LEVEL 7: Lattin-Jenkins-Grove (SPICE) Model is the same as [LEVEL 6: Lattin-Jenkins-Grove \(ASPEC\) Model](#) except for the ϕ_s value. The only difference between the two is the equations that handle the parasitic elements and the method of temperature compensation.

If you specify ϕ_s , then:

For LEVEL=6

$$\phi_s = \frac{\phi_s}{2}, \text{ where } \phi_s$$

is the surface potential.

For LEVEL=7

$$\phi_s = \phi_s$$

To transform a LEVEL 7 equation to LEVEL 6, make the following substitution:

$$\phi_s \rightarrow 2 \cdot \phi_s$$

To transform a LEVEL 6 model into a LEVEL 7 model, make the following substitution:

$$\phi_s(\text{Level7}) = \phi_s(\text{Level6}) / 2$$

LEVEL 8: Advanced LEVEL 2: MOS2 Grove-Frohmman Model (SPICE 2G)

This model is derived from research at Intersil and General Electric, is an enhanced version of the [LEVEL 2: MOS2 Grove-Frohmman \(SPICE 2G\) Model](#) ids equations. LEVEL 2 differs from LEVEL 8 in the following areas:

- Effective substrate doping
- Threshold voltage
- Effective mobility
- Channel length modulation
- Subthreshold current.

This section covers the following topics:

- [Advanced LEVEL 2: MOS2 Grove-Frohmman \(SPICE 2G\) Model Parameters](#)
- [Advanced LEVEL 2: MOS2 Grove-Frohmman \(SPICE 2G\) — Model Equations](#)

Advanced LEVEL 2: MOS2 Grove-Frohmman (SPICE 2G) Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this level also uses the following parameters:

Name (Alias)	Units	Default	Description
A1		0.2	Channel length modulation exponent (CLM=8)
CLM		7	Channel length modulation equation selector
LAM1	1/m	0.0	Channel length modulation length correction
LAMBDA (LAM, LA)		0.0	Channel length modulation coefficient

Advanced LEVEL 2: MOS2 Grove-Frohmman (SPICE 2G) — Model Equations

This section covers the following topics:

- [Advanced LEVEL 2 Model — IDS Equations](#)
- [Advanced LEVEL 2 Model — Effective Channel Length and Width](#)
- [Advanced LEVEL 2 Model — Effective Substrate Doping, \$n_{sub}\$](#)
- [Advanced LEVEL 2 Model — Threshold Voltage, \$v_{th}\$](#)
- [Advanced LEVEL 2 Model — Saturation Voltage \$v_{dsat}\$](#)
- [Advanced LEVEL 2 Model — \$ECRIT > 0\$:](#)
- [Advanced LEVEL 2 Model — \$ECRIT \leq 0\$ or \$MOB=7\$:](#)
- [Advanced LEVEL 2 Model — Effective Mobility, \$\mu_{eff}\$](#)
- [Advanced LEVEL 2 Model — \$MOB=2\$ Mobility Reduction](#)
- [Advanced LEVEL 2 Model — \$MOB=3\$ Mobility Reduction](#)
- [Advanced LEVEL 2 Model — \$MOB=6\$ Mobility Reduction](#)
- [Advanced LEVEL 2 Model — \$MOB=7\$ Mobility Reduction](#)
- [Advanced LEVEL 2 Model — Channel Length Modulation](#)
- [Advanced LEVEL 2 Model — \$CLM=6\$, SPICE Channel Length Modulation](#)
- [Advanced LEVEL 2 Model — \$CLM=7\$, Intersil Channel Length Modulation](#)
- [Advanced LEVEL 2 Model — \$CLM=8\$](#)
- [Advanced LEVEL 2 Model — Subthreshold Current \$I_{ds}\$](#)

Advanced LEVEL 2 Model — IDS Equations

The i_{ds} equations are the same as that of [LEVEL 2: MOS2 Grove-Frohmman \(SPICE 2G\) Model](#).

Advanced LEVEL 2 Model — Effective Channel Length and Width

This model calculates the effective channel length and width from the drawn length and width. See [Grove-Frohmman \(SPICE 2G\) Model Equations](#).

Advanced LEVEL 2 Model — Effective Substrate Doping, n_{sub}

The $SNVB$ model parameter varies the substrate doping concentration linearly as a function of v_{sb} :

$$n_{sub} = NSUB + SNVB \cdot v_{sb}$$

The preceding equation computes γ , ϕ , and x_d parameters for n_{sub} :

$$\gamma = \frac{\sqrt{2 \cdot \epsilon_{si} \cdot q \cdot n_{sub}}}{COX}$$

$$\Phi = 2 \cdot v_t \cdot \ln\left(\frac{n_{sub}}{n_i}\right)$$

$$x_d = \sqrt{\frac{2 \cdot \epsilon_{si}}{q \cdot n_{sub}}}$$

If $SNVB$ is zero, then $\gamma = \gamma_{GAMMA}$. You can adjust the γ value for the short-channel effect the same way as in the LEVEL 2 model. $NSUB$ calculates the ϕ value.

Advanced LEVEL 2 Model — Threshold Voltage, v_{th}

ETA specifies the threshold voltage reduction due to the potential barrier lowering effect.

$$v_{bi} = VTO - g \sqrt{\Phi} - \frac{8.14e-22 \cdot ETA}{COX \cdot L_{eff}^3} p_{vds} + (\eta - 1) \cdot (v_{sb} + \Phi)$$

$$v_{th} = v_{bi} + g \cdot \sqrt{v_{sb} + \Phi}$$

Modify γ for the short-channel effect, the same as in the LEVEL 2 model to obtain the effective γ .

Advanced LEVEL 2 Model — Saturation Voltage v_{dsat}

Level 8 computes the v_{sat} saturation voltage the same way as in the LEVEL 2 model. This model includes the carrier velocity effect only if $ECRIT$ is greater than zero.

Advanced LEVEL 2 Model — $ECRIT > 0$:

$$v_{dsat} = v_{sat} + v_c \sqrt{v_{sat}^2 + v_c^2}$$

The following equation calculates the v_c value used in the preceding equation:

$$v_c = ECRIT \cdot L_{eff}$$

Advanced LEVEL 2 Model — ECRIT ≤ 0 or MOB=7:

$$vdsat = vsat$$

This model computes $vsat$ as in the `LEVEL=2` model (see [Grove-Frohman \(SPICE 2G\) Model — Saturation Voltage, \$vdsat\$](#)).

Advanced LEVEL 2 Model — Effective Mobility, $ueff$

The `MOB` mobility equation selector controls the mobility reduction equations. In the LEVEL 8 model, set `MOB` to 2, 3, 6, or 7. Default=6.

Advanced LEVEL 2 Model — MOB=2 Mobility Reduction

$$u_{eff} = UO \cdot \left[\frac{\epsilon_{se} \cdot UCRIT}{COX \cdot (vgs - vth - UTRA \cdot vde)} \right]^{UEXP}$$

Advanced LEVEL 2 Model — MOB=3 Mobility Reduction

$$u_{eff} = \frac{UO}{1 + \frac{2.1e-8(vgs + vth + egfet - \Phi)}{6TOX}}$$

In the preceding equation, $egfet$ is the silicon energy gap at the analysis temperature:

$$egfet = 1.16 \cdot \frac{7.02e-4 \cdot t^2}{t + 1108}$$

In the preceding equation, t is the temperature in degrees Kelvin.

If `VMAX`>1:

$$u_{eff} = \frac{u_{eff}}{1 + \frac{u_{eff}}{VMAX \cdot Leff} \cdot vde}$$

Advanced LEVEL 2 Model — MOB=6 Mobility Reduction

For `UEXP`>0:

$$\text{If } (vgs - vth) > \frac{\epsilon_{si} \cdot UCRIT}{COX}, \text{ then } u_{eff} = \frac{UO \cdot \left[\frac{\epsilon_{si} \cdot UCRIT}{COX \cdot (vgs - vth)} \right]^{UEXP}}{1 + \frac{UTRA}{Leff} \cdot vde}$$

$$\text{Otherwise: } u_{eff} = \frac{UO}{1 + \frac{UTRA}{Leff} \cdot vde}$$

For $U_{EXP}=0$:

$$u_{eff} = \frac{UO}{[1 + UCRIT \cdot (v_{gs} - v_{th})] \cdot \left(1 + \frac{UTRA}{L_{eff}} \cdot v_{de}\right)}$$

$UCRIT$ for $U_{EXP}=0$ has a dimension of $(1/V)$.

Advanced LEVEL 2 Model — MOB=7 Mobility Reduction

$$u_{eff} = \frac{UO}{1 + UTRA \cdot \left(v_{gs} - v_{bi} - \eta \sqrt{\frac{v_{de}}{2} + \frac{body}{v_{de}}}\right)}$$

The following equation calculates the body value used in the preceding equation:

$$body = \frac{2}{3} \cdot \gamma \cdot \left[(v_{de} + v_{sb} + \Phi)^{\frac{3}{2}} - (v_{sb} + \Phi)^{\frac{3}{2}} \right]$$

Advanced LEVEL 2 Model — Channel Length Modulation

The CLM equation selector controls the channel length modulation equations. In the LEVEL 8 model, set CLM to 6, 7, or 8. Default=7.

Advanced LEVEL 2 Model — CLM=6, SPICE Channel Length Modulation

If $LAMBDA=0$:

$$\lambda = \frac{\chi d}{l_{eff} \cdot v_{ds}} \cdot \sqrt{\frac{v_{ds} - v_{dsat}}{4}} + \sqrt{1 + \left(\frac{v_{ds} - v_{dsat}}{4}\right)^2}$$

Otherwise, $\lambda = LAMBDA$. Then: $\Delta L = \frac{\lambda \cdot L_{eff} \cdot v_{ds}}{1 + LAM1 \cdot L_{eff}}$

Note:

The LEVEL 2 model has no $LAM1$ term.

This model modifies the current for the channel length modulation effect in the entire regions:

$$ids = \frac{id_s}{1 - \frac{\Delta L}{L_{eff}}}$$

Advanced LEVEL 2 Model — CLM=7, Intersil Channel Length Modulation

If $CLM=7$, this model computes ΔL only for the saturation region.

$$v_{ds} > v_{dsat}$$

$$\Delta L = \frac{LAMBDA \cdot L_{eff}}{1 + LAM1 \cdot L_{eff}} \cdot (v_{ds} - v_{dsat})$$

$$i_{ds} = \frac{i_{ds}}{L - \frac{\Delta L}{L_{eff}}}$$

Advanced LEVEL 2 Model — CLM=8

If $CLM=8$, this model computes ΔL only for the saturation region.

$$v_{ds} > v_{dsat}$$

$$\Delta L = \frac{L_{eff}}{1 + \frac{(1 + LAM1 \cdot L_{eff})(1 + v_{de})^{A1}}{LAMBDA(v_{ds} - v_{de})}}$$

$$i_{ds} = \frac{i_{ds}}{1 - \frac{\Delta L}{L_{eff}}}$$

Advanced LEVEL 2 Model — Subthreshold Current i_{ds}

The LEVEL 8 model has different subthreshold current equations, depending on the value of the CAV model parameter.

$$\text{Define: } fast = vt \cdot \left[\eta + \frac{q \cdot NFS}{COX} + \frac{\gamma}{2 \cdot (v_{sb} + \Phi)^{\frac{1}{2}}} + \frac{\epsilon_{si} \cdot q \cdot SNVB \cdot \sqrt{v_{sb} + \Phi}}{\gamma \cdot COX^2} \right]$$

For $CAV \neq 0$

$$v_{on} = v_{th} + CAV \cdot fast$$

Subthreshold Region, $v_{gs} < v_{on}$

If $v_{gs} > v_{th}$:

$$i_{ds} = i_{ds}(v_{on}, v_{de}, v_{sb}) \cdot e^{\left(1 - \frac{CAV}{2}\right)} \cdot e^{\left[\frac{1}{fast} - \frac{(CAV - 2)(v_{gs} - v_{th})}{2CAV^2 fast^2}\right](v_{gs} - v_{th})}$$

If $v_{gs} \leq v_{th}$

$$i_{ds} = i_{ds}(v_{on}, v_{de}, v_{sb}) \cdot e^{\left(1 - \frac{CAV}{2}\right)} \cdot e^{\left(\frac{v_{gs} - v_{th}}{fast}\right)}$$

For $CAV=0$

$$\text{If } CLM=8: v_{on} = v_{th} + 3 \cdot fast$$

$$\text{Otherwise, } v_{on} = v_{th} + 2 \cdot fast$$

Subthreshold Region, $v_{gs} < v_{on}$

$$ids = ids(v_{on}, v_{de}, v_{sb}) \cdot e^{\left(\frac{v_{gs} - v_{on}}{f_{ast}}\right)}$$

If $WIC=3$, the next equation calculates the ids subthreshold current:

$$ids = ids(v_{gs}, v_{de}, v_{sb}) + isub(N0eff, NDeff, v_{gs}, v_{ds})$$

$N0eff$ and $NDeff$ are functions of effective device width and length.

LEVEL 13: BSIM Model

This model is based on the SPICE 2G.6 BSIM model, which models the device physics of small-geometry MOS transistors. To invoke the subthreshold region, set the `NO` model parameter (low field weak inversion gate drive coefficient) to less than 200. This model provides the following MOSFET models:

- Wire (resistor) model, compatible with the SPICE BSIM interconnect model for polysilicon and metal layers. Simulates resistors and capacitors with interconnects.
- Capacitor model. Simulates only capacitors with interconnects.
- Diffusion model, compatible with SPICE BSIM diffusion models.

To set the model parameters, either:

- Enter model parameters as numbers (as in SPICE), or
- Assign the model parameters.

If you convert from SPICE to the Synopsys models, use the `S` keyletter for SPICE BSIM, or `M` for the Synopsys model. (see [BSIM Model – Compatibility Notes: IDS and VGS Curves for PMOS and NMOS](#)).

This section covers the following topics:

- [BSIM Model Features](#)
- [BSIM Model Parameters](#)
- [BSIM Model — Sensitivity Factors](#)
- [BSIM Models — .MODEL VERSION Changes](#)
- [BSIM Model Equations](#)
- [BSIM Model — Charge-Based Capacitance Model](#)
- [BSIM Model — Preventing Negative Output Conductance](#)
- [BSIM Model Calculations](#)
- [BSIM Model — Compatibility Notes](#)

BSIM Model Features

- Vertical field dependence of the carrier mobility
- Carrier velocity saturation
- Drain-induced barrier lowering

- Depletion charge sharing by source and drain
- Non-uniform doping profile for ion-implanted devices
- Channel length modulation
- Subthreshold conduction
- Geometric dependence of electrical parameters

BSIM Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this model also uses the following parameters:

Note:

When you read parameter names, be careful about the difference in appearance between the uppercase letter O, the lowercase letter o, and the number zero (0).

For reference purposes only, simulation obtains the following default values from a medium size n-channel MOSFET device.

To specify the model parameters, use NMOS conventions, even for PMOS (for example, `ETA0=0.02`, not `ETA0=-0.02`).

This section covers the following topics:

- [BSIM Model — Transistor Parameters](#)
- [BSIM Model — Diffusion Layer Parameters](#)
- [BSIM Model — Temperature Parameters](#)

BSIM Model — Transistor Parameters

Name (Alias)	Units	Default	Description
LEVEL		1	MOSFET model level selector. 13 is the BSIM model
CGBOM, (CGBO)	F/m	2.0e-10	Gate-to-bulk parasitic capacitance (F/m of length)
CGDOM, (CGDO)	F/m	1.5e-9	Gate-to-drain parasitic capacitance (F/m of width)
CGSOM, (CGSO)	F/m	1.5e-9	Gate-to-source parasitic capacitance (F/m of width)

Name (Alias)	Units	Default	Description
DL0	μm	0.0	Difference between drawn poly and electrical
DUM1		0.0	Dummy (not used)
DUM2		0.0	Dummy (not used)
DW0	μm	0.0	Difference between drawn diffusion and electrical
ETA0		0.0	Linear vds threshold coefficient
K1	$V^{1/2}$	0.5	Root-vsb threshold coefficient
K2		0.0	Linear vsb threshold coefficient
LETA	mm	0.0	Length sensitivity
LK1	$V^{1/2} \cdot \mu\text{m}$	0.0	Length sensitivity
LK2	μm	0.0	Length sensitivity
LMS (LMUS)	$\mu\text{m} \cdot \text{cm}^2 / (\text{V} \cdot \text{s})$	0.0	Length sensitivity
LMUZ	$\mu\text{m} \cdot \text{cm}^2 / (\text{V} \cdot \text{s})$	0.0	Length sensitivity
LN0		0.0	Length sensitivity
LNB		0.0	Length sensitivity
LND		0.0	Length sensitivity
LPHI	$\text{V} \cdot \mu\text{m}$	0.0	Length sensitivity
LU0	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LU1	$\mu\text{m}^2/\text{V}$	0.0	Length sensitivity
LVFB	$\text{V} \cdot \mu\text{m}$	0.0	Length sensitivity
LX2E	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LX2M (LX2MZ)	$\mu\text{m} \cdot \text{cm}^2 / (\text{V}^2 \cdot \text{s})$	0.0	Length sensitivity
LX2MS	$\mu\text{m} \cdot \text{cm}^2 / (\text{V}^2 \cdot \text{s})$	0.0	Length sensitivity
LX2U0	$\mu\text{m}/\text{V}^2$	0.0	Length sensitivity
LX2U1	$\mu\text{m}^2/\text{V}^2$	0.0	Length sensitivity
LX3E	$\mu\text{m}/\text{V}$	0.0	Length sensitivity

Name (Alias)	Units	Default	Description
LX3MS	$\mu\text{m}\cdot\text{cm}^2/(\text{V}^2\cdot\text{s})$	0.0	Length sensitivity
LX3U1	$\mu\text{m}^2/\text{V}^2$	0.0	Length sensitivity
MUS	$\text{cm}^2/(\text{V}\cdot\text{s})$	600	High drain field mobility
MUZ	$\text{cm}^2/(\text{V}\cdot\text{s})$	600	Low drain field first order mobility
N0		0.5	Low field weak inversion gate drive coefficient (a value of 200 for N0 disables the weak inversion calculation)
NB0		0.0	Vsb reduction to the low field weak inversion gate drive coefficient
ND0		0.0	Vds reduction to the low field weak inversion gate drive coefficient
PHI0	V	0.7	Two times the Fermi potential
TOXM, (TOX)	μm , (m)	0.02	Gate oxide thickness (simulation interprets TOXM or TOX >1 as Angstroms)
TREF	°C	25.0	Reference temperature of model (local override of TNOM)
U00	1/V	0.0	Gate field mobility reduction factor
U1	$\mu\text{m}/\text{V}$	0.0	Drain field mobility reduction factor
VDDM	V	50	Critical voltage for the high-drain field mobility reduction
VFB0 (VFB)	V	-0.3	Flatband voltage
WETA	μm	0.0	Width sensitivity
WK1	$\text{V}^{1/2}\cdot\mu\text{m}$	0.0	Width sensitivity
WK2	μm	0.0	Width sensitivity
WMS (WMUS)	$\mu\text{m}\cdot\text{cm}^2/(\text{V}\cdot\text{s})$	0.0	Width sensitivity
WMUZ	$\mu\text{m}\cdot\text{cm}^2/(\text{V}\cdot\text{s})$	0.0	Width sensitivity
WN0		0.0	Width sensitivity
WNB		0.0	Width sensitivity

Name (Alias)	Units	Default	Description
WND		0.0	Width sensitivity
WPHI	$V \cdot \mu m$	0.0	Width sensitivity
WU0	$\mu m/V$	0.0	Width sensitivity
WU1	$\mu m^2/V$	0.0	Width sensitivity
WVFB	$V \cdot \mu m$	0.0	Width sensitivity
WX2E	$\mu m/V$	0.0	Width sensitivity
WX2M (WX2MZ)	$\mu m \cdot cm^2/(V^2 \cdot s)$	0.0	Width sensitivity
WX2MS	$\mu m \cdot cm^2/(V^2 \cdot s)$	0.0	Width sensitivity
WX2U0	$\mu m/V^2$	0.0	Width sensitivity
WX2U1	$\mu m^2/V^2$	0.0	Width sensitivity
WX3E	$\mu m/V$	0.0	Width sensitivity
WX3MS	$\mu m \cdot cm^2/(V^2 \cdot s)$	0.0	Width sensitivity
WX3U1	$\mu m^2/V^2$	0.0	Width sensitivity
X2E	$1/V$	0.0	Vsb correction to the linear vds threshold coefficient
X2M (X2MZ)	$cm^2/(V^2 \cdot s)$	0.0	Vsb correction to the low field first-order mobility
X2MS	$cm^2/(V^2 \cdot s)$	0.0	Vbs reduction to the high-drain field mobility
X2U0	$1/V^2$	0.0	Vsb reduction to the GATE field mobility reduction factor
X2U1	$\mu m/V^2$	0.0	Vsb reduction to the DRAIN field mobility reduction factor
X3E	$1/V$	0.0	Vds correction to the linear vds threshold coefficient
X3MS	$cm^2/(V^2 \cdot s)$	5.0	Vds reduction to the high-drain field mobility
X3U1	$\mu m/V^2$	0.0	Vds reduction to the drain field mobility reduction factor

Name (Alias)	Units	Default	Description
XPART		1.0	Selects a gate capacitance charge-sharing coefficient

BSIM Model — Diffusion Layer Parameters

Name (Alias)	Units	Default	Description
CJW, (CJSW)	F/m	0.0	Zero-bias bulk junction sidewall capacitance
CJM, (CJ)	F/m ²	4.5e-5	Zero-bias bulk junction bottom capacitance
DS	m	0.0	Average size variation due to the side etching or the mask compensation (not used)
IJS, (JS)	A/m ²	0	Bulk junction saturation current
JSW	A/m	0.0	Sidewall bulk junction saturation current
MJ0, (MJ)		0.5	Bulk junction bottom grading coefficient
MJW, (MJSW)		0.33	Bulk junction sidewall grading coefficient
PJ, (PB)	V	0.8	Bulk junction bottom potential
PJW, (PHP)	V	0.8	Bulk junction sidewall potential
RSHM, (RSH)	ohm/sq	0.0	Sheet resistance/square
WDF	m	0.0	Default width of the layer (not used)

BSIM Model — Temperature Parameters

Name (Alias)	Units	Default	Description
BEX		-1.5	Temperature exponent for the MUZ and MUS mobility parameters.
FEX		0.0	Temperature exponent for the U1 mobility reduction factor.
TCV	V/°K	0.0	Flat-band voltage temperature coefficient.
TREF	°C	25	Temperature at which simulation extracts parameters. This parameter defaults to the TNOM option, which defaults to 25 °C.

BSIM Model — Sensitivity Factors

To denote the L (channel length) and W (channel width) sensitivity factors of a basic electrical parameter in a transistor, add L and W characters at the start of the name. For example, V_{FB0} sensitivity factors are LV_{FB} and WV_{FB} . If $A0$ is a basic parameter, then LA and WA are the corresponding L and W sensitivity factors of this parameter. Do not use the `SCALM` option to scale LA and WA .

This model uses the following equation to obtain this parameter value:

$$A = A0 + LA \cdot \left(\frac{1}{L_{eff}} - \frac{1}{LREF_{eff}} \right) + WA \cdot \left(\frac{1}{W_{eff}} - \frac{1}{WREF_{eff}} \right)$$

Specify LA and WA in units of microns times the units of $A0$.

The left side of the equation represents the effective model parameter value after you adjust the device size. All effective model parameters are in lowercase and start with the z character, followed by the parameter name.

Example:

$$V_{FB0} = -0.350v, LV_{FB} = -0.1v\mu$$

$$WV_{FB} = 0.08v \cdot \mu, L_{eff} = 1 \cdot 10^{-6}m = 1\mu$$

$$W_{eff} = 2 \cdot 10^{-6}m = 2\mu, LREF_{eff} = 2 \cdot 10^{-6}m = 2\mu$$

$$WREF_{eff} = 1 \cdot 10^{-5}m = 10\mu$$

$$z_{vfb} = V_{FB0} + LV_{FB} \cdot \left(\frac{1}{L_{eff}} - \frac{1}{LREF_{eff}} \right) + WV_{FB} \cdot \left(\frac{1}{W_{eff}} - \frac{1}{WREF_{eff}} \right)$$

$$z_{vfb} = -0.35v + -0.1v \cdot \mu \cdot \left(\frac{1}{1\mu} - \frac{1}{2\mu} \right) + 0.08v \cdot \mu \cdot \left(\frac{1}{2\mu} - \frac{1}{10\mu} \right)$$

$$z_{vfb} = -0.35v - 0.05v + 0.032v, z_{vfb} = -0.368v$$

BSIM Models — .MODEL VERSION Changes

You can use the `VERSION` parameter in the `.MODEL` statement to move LEVEL 13: BSIM and LEVEL 39: BSIM2 models between versions. Using the `VERSION` parameter in a LEVEL 13 `.MODEL` statement results in the following changes to the BSIM model.

Model Version	Effect of VERSION on BSIM model
9007B	Introduced the LEVEL 13 BSIM model: no changes
9007D	Removed the K2 limit

Model Version	Effect of VERSION on BSIM model
92A	Changed the TOX parameter default from 1000 Å to 200 Å
92B	Added the K2LIM parameter, which specifies the K2 limit
93A	Introduced the gds constraints
93A.02	Introduced the VERSION parameter
95.1	Fixed the nonprinting TREF and incorrect GMBS problems
96.1	Changed the flatband voltage temperature adjustment

BSIM Model Equations

This section covers the following topics:

- [BSIM Model — Effective Channel Length and Width](#)
- [BSIM Model — IDS Equations](#)
- [BSIM Model — Cutoff Region, \$v_{gs} \leq v_{th}\$](#)
- [BSIM Model — On Region, \$v_{gs} > v_{th}\$](#)
- [BSIM Model — Threshold Voltage](#)
- [BSIM Model — Saturation Voltage \(\$v_{dsat}\$ \)](#)
- [BSIM Model — \$i_{ds}\$ Subthreshold Current](#)
- [BSIM Model — Resistors and Capacitors Generated with Interconnects](#)
- [BSIM Model — Resistances](#)
- [BSIM Model — Capacitances](#)
- [BSIM Model — Temperature Effect](#)

BSIM Model — Effective Channel Length and Width

The effective channel length and width for Level 13 depends on the specified model parameters.

If you specify $DL0$, then:

$$L_{eff} = L_{scaled} \cdot L_{MLT-DL0} \cdot 1e-6$$

$$L_{REF_{eff}} = L_{REF_{scaled}} \cdot L_{MLT-DL0} \cdot 1e-6$$

Otherwise, if you specify X_L or L_D :

$$L_{eff} = L_{scaled} \cdot L_{MLT} + X_L_{scaled} - 2 \cdot L_D_{scaled}$$

$$L_{REF_{eff}} = L_{REF_{scaled}} \cdot L_{MLT} + X_L_{scaled} - 2 \cdot L_D_{scaled}$$

If you specify DW_0 , then:

$$W_{eff} = W_{scaled} \cdot W_{MLT} - DW_0 \cdot 1e-6$$

$$W_{REF_{eff}} = W_{REF_{scaled}} \cdot W_{MLT} - DW_0 \cdot 1e-6$$

Otherwise, if you specify X_W or W_D , then:

$$W_{eff} = W_{scaled} \cdot W_{MLT} + X_W_{scaled} - 2 \cdot W_D_{scaled}$$

$$W_{REF_{eff}} = W_{REF_{scaled}} \cdot W_{MLT} + X_W_{scaled} - 2 \cdot W_D_{scaled}$$

BSIM Model — IDS Equations

Process-oriented model parameters model the device characteristics. Simulation maps these parameters into model parameters at a specific bias voltage. The ids equations are as follows:

BSIM Model — Cutoff Region, $v_{gs} \leq v_{th}$

$ids = 0$ (See [BSIM Model — ids Subthreshold Current](#))

BSIM Model — On Region, $v_{gs} > v_{th}$

For the $v_{ds} < v_{dsat}$ triode region:

$$ids = \frac{\beta}{1 + x_{u1} \cdot v_{ds}} \cdot \left[(v_{gs} - v_{th}) \cdot v_{ds} - \frac{body}{2} \cdot v_{ds}^2 \right]$$

For the $v_{ds} > v_{dsat}$ saturation region:

$$ids = \frac{\beta}{2 \cdot body \cdot arg} \cdot (v_{gs} - v_{th})^2$$

The following equations calculate values used in the preceding equation:

$$\beta = u_{eff} \cdot COX \cdot \frac{W_{eff}}{L_{eff}}$$

$$u_{eff} = \frac{u_0}{1 + x_{u0} \cdot (v_{gs} - v_{th})}$$

$$x_{u0} = z_{u0} - z_x \cdot 2 \cdot u_0 \cdot v_{sb}$$

Simulation uses quadratic interpolation, through three data points to calculate the u_0 carrier mobility.

$$u_0|_{v_{ds}=0} = MUZ - z \cdot x2mz \cdot v_{sb}$$

$$u_0|_{v_{ds}=V_{DDM}} = zmus - z \cdot x2ms \cdot v_{sb}$$

Simulation also calculates the sensitivity of u_0 to v_{ds} at $v_{ds}=V_{DDM}$, which is $zx3ms$.

The following equation calculates the body factor:

$$body = 1 + \frac{g \cdot zk1}{2 \cdot (zphi + v_{sb})^{\frac{1}{2}}}$$

The following equation calculates the g value used in the preceding equation:

$$g = 1 - \frac{1}{1.744 + 0.8364 \cdot (zphi + v_{sb})}$$

The following equation calculates the arg term in the saturation region:

$$arg = \frac{1}{2} \cdot \left[1 + vc + (1 + 2 \cdot vc)^{\frac{1}{2}} \right]$$

The following equations calculate values used in the preceding equation:

$$vc = \frac{xu1 \cdot (v_{gs} - v_{th})}{body}$$

$$xu1 = zu1 - z \cdot x2u1 \cdot v_{sb} + z \cdot x3u1 \cdot (v_{ds} - V_{DDM}), \text{ UPDATE}=2$$

$$xu1 = \frac{zu1 - z \cdot x2u1 \cdot v_{sb} + z \cdot x3u1 \cdot (v_{ds} - V_{DDM})}{L_{eff}}, \text{ UPDATE}=0, 1$$

BSIM Model — Threshold Voltage

You can express the threshold voltage as:

$$v_{th} = zvf_b + zphi + gamma \cdot (zphi + v_{sb})^{\frac{1}{2}} - xeta \cdot v_{ds}$$

The following equations calculate values used in the preceding equation:

$$gamma = zk1 - zk2 \cdot (zphi + v_{sb})^{\frac{1}{2}}$$

$$xeta = zeta - z \cdot x2e \cdot v_{sb} + z \cdot x3e \cdot (v_{ds} - V_{DDM}), \text{ UPDATE}=0, 2$$

$$xeta = zeta + z \cdot x2e \cdot (zphi + v_{sb}) + z \cdot x3e \cdot (v_{ds} - V_{DDM}), \text{ UPDATE}=1$$

BSIM Model — Saturation Voltage (vdsat)

The following equation calculates the saturation voltage in the BSIM Level 13 model:

$$vdsat = \frac{vgs - vth}{body \cdot arg_2^1}$$

BSIM Model — ids Subthreshold Current

Simulation calculates the isub subthreshold current if zn0 is less than 200:

$$isub = \frac{Ilim \cdot Iexp}{Ilim + Iexp}$$

The following equations calculate values used in the preceding equation:

$$Iexp = \beta_o \cdot vt^2 \cdot e^{1.8} \cdot e^{\frac{vgs - vth}{xnvt}} \cdot \left(1 - e^{\frac{vds}{vt}}\right)$$

$$Ilim = 4.5 \cdot \beta_o \cdot vt^2, \beta_o = uo \cdot COX \cdot \frac{Weff}{Leff}$$

$$xn = zn0 - znbpvsvb + znd \cdot vds$$

Simulation also adds the isub current to the ids current in the strong inversion.

BSIM Model — Resistors and Capacitors Generated with Interconnects

Refer to the wire model table (resistor element) for the model parameters that you used. For an example, see [BSIM Model – Compatibility Notes: IDS and VGS Curves for PMOS and NMOS](#).

BSIM Model — Resistances

$$r = RSH \cdot \frac{Leff}{Weff}$$

BSIM Model — Capacitances

$$c = COX \cdot Leff \cdot Weff + 2 \cdot CAPSW \cdot (Leff + Weff)$$

BSIM Model — Temperature Effect

$$MUZ(t) = MUZ \cdot \left(\frac{t}{tnom}\right)^{BEX}, \text{UPDATE}=0, 1$$

$$zmus(t) = zmus \cdot \left(\frac{t}{tnom}\right)^{BEX}, \text{UPDATE}=0, 1$$

$$u_o(t) = u_o\left(\frac{t}{t_{nom}}\right)^{BEX}, \text{UPDATE}=2$$

$$x_{u1}(t) = x_{u1} \cdot \left(\frac{t}{t_{nom}}\right)^{FEX}$$

$$z_{vfb}(t) = z_{vfb} - \Delta t \cdot TCV$$

The following equation calculates the Δt value used in the preceding equations:

$$\Delta t = t - t_{nom}$$

BSIM Model — Charge-Based Capacitance Model

The Level 13 capacitance model conserves charge and has nonreciprocal attributes. Using charge as the state variable guarantees charge conservation. To obtain the total stored charge in each of the gate, bulk, and channel regions, integrate the distributed charge densities/area of the active region.

The `XPART` 40/60 model parameter or 0/100 in the saturation region, partitions the channel charge into drain and source components. This partitioning smoothly changes to 50/50 in the triode region.

- `XPART=0` selects 40/60 drain/source charge-partitioning in the saturation region. That is, 40% of the channel charge in the saturation region is at the drain, and 60% is at the source.
- `XPART=1` selects 0/100 for drain/source charge-partitioning in the saturation region. That is, 100% of the channel charge in the saturation region is at the source; there is no drain charge.
- `XPART=0.5` selects 50/50 partitioning. Half of the channel charge in the saturation region is at the source, and half is at the drain.

Define:

$$v_{tho} = z_{vfb} + z_{phi} + z_{k1} \cdot (z_{phi} + v_{sb})^{\frac{1}{2}}$$

$$cap = COX \cdot L_{eff} \cdot W_{eff}, v_{pof} = \frac{v_{gs} - v_{tho}}{body}$$

$$argx = \frac{body \cdot v_{ds}}{12 \cdot (v_{gs} - v_{tho} - 0.5 \cdot body \cdot v_{ds})}$$

$$\text{If } (v_{gs} - v_{tho} - 0.5 \cdot body \cdot v_{ds}) \leq 1e-8 \text{ then: } argx = \frac{1}{6}$$

$$argy = \frac{(vgs-vtho)^2 - 0.75 \cdot body \cdot (vgs-vtho) \cdot vds + 0.15 \cdot body^2 \cdot vds^2}{6 \cdot (vgs-vtho - 0.5 \cdot body \cdot vds)^3}$$

If $(vgs-vtho - 0.5 \cdot body \cdot vds) \leq 1e-8$ then: $argy = \frac{4}{15}$

This section covers the following topics:

- BSIM Model—Regions Charge Expressions: Accumulation Region, $vgs \leq vtho$, $vgs \leq zvfb - vsb$
- BSIM Model—Regions Charge Expressions: Subthreshold Region, $vgs \leq vtho$, $vgs > zvfb - vsb$
- BSIM Model—Regions Charge Expressions: 50/50 Channel-Charge Partitioning for Drain and Source, $XPART=0.5$; Triode Region, $vgs > vtho$, $vds \leq vpof$
- BSIM Model—Regions Charge Expressions: Saturation Region, $vgs > vtho$, $vds > vpof$
- BSIM Model—Regions Charge Expressions: 40/60 Channel-Charge Partitioning for Drain and Source, $XPART=0$; Triode Region, $vgs > vtho$, $vds \leq vpof$
- BSIM Model—Regions Charge Expressions: Saturation Region, $vgs > vtho$, $vds > vpof$
- BSIM Model—Regions Charge Expressions: 0/100 Channel-Charge Partitioning for Drain and Source, $XPART=1$; Triode Region, $vgs > vtho$, $vds \leq vpof$
- BSIM Model—Regions Charge Expressions: Saturation Region, $vgs > vtho$, $vds > vpof$

BSIM Model—Regions Charge Expressions: Accumulation Region, $vgs \leq vtho$, $vgs \leq zvfb - vsb$

$$Qg = cap \cdot (vgs - zvfb + vsb)$$

$$Qb = -qg, Qs = 0, Qd = 0$$

BSIM Model—Regions Charge Expressions: Subthreshold Region, $vgs \leq vtho$, $vgs > zvfb - vsb$

$$Qg = \frac{cap \cdot zk1}{2} \cdot \left\{ \left[(zk1)^2 + 4(vgs - zvfb + vsb) \right]^{1/2} - zk1 \right\}$$

$$Qb = -qg, Qs = 0$$

**BSIM Model—Regions Charge Expressions: 50/50
Channel-Charge Partitioning for Drain and Source, XPART=.5;
Triode Region, $v_{gs} > v_{tho}$, $v_{ds} \leq v_{pof}$**

$$Q_g = cap \cdot (v_{gs} - z_{vfb} - z_{phi} - 0.5 \cdot v_{ds} + v_{ds} \cdot argx)$$

$$Q_b = cap \cdot [-v_{tho} + z_{vfb} + z_{phi} + (1 - body) \cdot (0.5 - argx) \cdot v_{ds}]$$

$$Q_d = -0.5 \cdot P(q_g + q_b), Q_s = Q_d$$

**BSIM Model—Regions Charge Expressions: Saturation Region,
 $v_{gs} > v_{tho}$, $v_{ds} > v_{pof}$**

$$Q_g = cap \cdot \left(v_{gs} - z_{vfb} - z_{phi} - \frac{v_{gs} - v_{tho}}{3 \cdot body} \right)$$

$$Q_b = cap \cdot \left[z_{vfb} + z_{phi} - v_{tho} + (1 - body) \cdot \frac{(v_{gs} - v_{tho})}{3 \cdot body} \right]$$

$$Q_d = -\frac{cap}{3} \cdot P(v_{gs} - v_{tho}), Q_s = Q_d$$

**BSIM Model—Regions Charge Expressions: 40/60
Channel-Charge Partitioning for Drain and Source, XPART=0;
Triode Region, $v_{gs} > v_{tho}$, $v_{ds} \leq v_{pof}$**

$$Q_g = cap \cdot (v_{gs} - x_{vfb} - z_{phi} - 0.5 \cdot v_{ds} + argx \cdot v_{ds})$$

$$Q_b = cap \cdot [-v_{tho} + z_{vfb} + z_{phi} + (1 - body) \cdot (0.5 - argx) \cdot v_{ds}]$$

$$Q_d = -(cap \cdot [0.5 \cdot (v_{gs} - v_{tho} - body \cdot v_{ds}) + body \cdot argy \cdot v_{ds}])$$

$$Q_s = -(Q_g + Q_b + Q_d)$$

**BSIM Model—Regions Charge Expressions: Saturation Region,
 $v_{gs} > v_{tho}$, $v_{ds} > v_{pof}$**

$$Q_g = cap \cdot \left(v_{gs} - z_{vfb} - z_{phi} - \frac{v_{gs} - v_{tho}}{3 \cdot body} \right)$$

$$Q_b = cap \cdot \left[z_{vfb} + z_{phi} - v_{tho} + (1 - body) \cdot \frac{(v_{gs} - v_{tho})}{3 \cdot body} \right]$$

$$Q_d = -\frac{4 \cdot cap}{15} \cdot (v_{gs} - v_{tho}), Q_s = \frac{3}{2} \cdot Q_d$$

BSIM Model—Regions Charge Expressions: 0/100 Channel-Charge Partitioning for Drain and Source, XPART=1; Triode Region, $v_{gs} > v_{th0}$, $v_{ds} \leq v_{pof}$

$$Q_g = cap \cdot (v_{gs} - z_{vfb} - z_{phi} - 0.5 \cdot v_{ds} + v_{ds} \cdot argx)$$

$$Q_b = cap \cdot [-v_{th0} + z_{vfb} + z_{phi} + (1 - body) \cdot (0.5 - argx) \cdot v_{ds}]$$

$$Q_d = -(cap \cdot [0.5 \cdot (v_{gs} - v_{th0}) - body \cdot v_{ds} \cdot (0.75 - 1.5 \cdot argx)])$$

$$Q_s = -(Q_g + Q_b + Q_d)$$

BSIM Model—Regions Charge Expressions: Saturation Region, $v_{gs} > v_{th0}$, $v_{ds} > v_{pof}$

$$Q_g = cap \cdot \left(v_{gs} - z_{vfb} - z_{phi} - \frac{v_{gs} - v_{th0}}{3 \cdot body} \right)$$

$$Q_b = cap \cdot \left[z_{vfb} + z_{phi} - v_{th0} + (1 - body) \cdot \frac{(v_{gs} - v_{th0})}{3 \cdot body} \right]$$

$$Q_d = 0, Q_s = -Q_g - Q_b$$

BSIM Model — Preventing Negative Output Conductance

The Level 13 model internally protects against conditions that might cause convergence problems, due to negative output conductance. The constraints imposed are:

$$ND \geq 0$$

$$MUS \geq MUZ + X3MS + VDD(M / 2)$$

This model imposes these constraints after adjusting the length and width and setting the VBS dependence. This feature loses some accuracy in the saturation region, particularly at high V_{gs} .

You might need to qualify BSIM1 models again, if the following occur:

- Devices exhibit self-heating during characterization, which causes declining I_{ds} at high V_{ds} . This does not occur if the device characterization measurement sweeps V_{ds} .
- Extraction produces parameters that result in negative conductance.
- This model attempts voltage simulation outside the characterized range of the device.

BSIM Model Calculations

To verify the equations, start some simple simulation and analysis tests, and check the results with a hand calculator. Check the threshold, v_{dsat} , and i_{ds} for a very simple model with many parameters set to zero:

- series resistance, $R_{SH}=0$.
- Turn off diode current, $J_S=J_{SW}=I_S=0$.
- Turn off the Level 13 subthreshold current, $n_0=200$.
- Set the geometry parameters to zero so $L_{eff}=L=1\mu$, $W_{eff}=W=1\mu$.

This test returns the following τ_{OX} value:

$$c_{ox} = \frac{2.00000e-3F}{m^2}$$

The test is at $v_{bs}=-0.35$ so that $\phi-v_{bs}=1.0$. The netlist for this test is located in directory `$installdir/demo/hspice/mos/t1.sp`.

```
$ t1
.option ingold=2 numdgt=6
vd d 0 5
vg g 0 5
vb b 0 -0.35
m1 d g 0 b nch w=10u L=1u
.dc vd 4 5 1
.print ids=LX4(m1) vth=lv9(m1) vdsat=lv10(m1)
.model nch nmos LEVEL=13
+ vfb0=-0.4 lvfb=0 wvfb=0
+ phi0=0.65 lphi=0 wphi=0
+ k1=0.5 lk1=0 wk1=0
+ k2=0 lk2=0 wk2=0
+ eta0=1e-3 leta=0 weta=0
+ muz=600 mus=700 x3ms=10
+ x1=0 ld=0 xw=0 wd=0
+ u00=0 lu0=0 wu0=0
+ u1=0 lu1=0 wu1=0
+ tox=172.657
+ acm=2 rsh=0 js=0 jsw=0 is=0 n0=200
.end
```

Simulation Results

ids	vth	vdsat
1.09907e-02	7.45000e-01	3.69000e+00

Calculations at $v_{gs}=v_{ds}=5$, $v_{bs}=-0.35$

$$\phi_{vbs} = 1$$

$$v_{th} = -0.4 + 0.65 + (0.5 \cdot 1) - (ETA \cdot v_{ds}) = 0.75 - (0.001 \cdot v_{ds}) = 0.745$$

$$g = 1 - \frac{1}{(1.744 + 0.8364 \cdot 1)} = 0.612463$$

$$body = 1 + \frac{g \cdot 0.5}{(2 \cdot 1)} = 1 + 0.25 \cdot g = 1.153116$$

$$vc = 0 \text{ arg} = 1$$

$$v_{dsat} = \frac{(v_{gs} - v_{th})}{body \cdot \sqrt{arg}} = \frac{(5 - 0.745)}{body} = 3.69000$$

Calculations at $v_{ds}=VDDM$ (default $VDDM=5$), mobility= $\mu_{s}=700$

$$ids = cox \cdot \left(\frac{W_{eff}}{L_{eff}} \right) \cdot 700 \cdot \frac{(v_{gs} - v_{th})^2}{(2 \cdot body \cdot arg)}$$

$$ids = \left(\frac{10 \cdot 700 \cdot 4.255^2}{2 \cdot 1.15311 \cdot 1} \right) \cdot cox = 54953.36 \cdot cox$$

$$ids = 1.09907e-2$$

These calculations agree with the above simulation results.

BSIM Model — Compatibility Notes

This section covers the following topics:

- [BSIM Model—Compatibility Notes: Model Parameter Naming](#)
- [BSIM Model—Compatibility Notes: SPICE/Synopsys Model Parameter Differences](#)
- [BSIM Model—Compatibility Notes: Parasitics](#)
- [BSIM Model—Compatibility Notes: Temperature Compensation](#)
- [BSIM Model—Compatibility Notes: BEX Usage](#)
- [BSIM Model—Compatibility Notes: UPDATE Parameter](#)
- [BSIM Model—Compatibility Notes: Explanations](#)
- [BSIM Model — Compatibility Notes: IDS and VGS Curves for PMOS and NMOS](#)

BSIM Model—Compatibility Notes: Model Parameter Naming

The following names are the PrimeSim Continuum simulator -specific: `U00`, `DL0`, `DW0`, `PHI0`, `ETA0`, `NB0`, and `ND0`. A zero was added to the SPICE names to avoid conflicts with other standard parameter names. For example, you cannot use `U0` because it is an alias for `UB`, the mobility parameter in many other levels. You cannot use `DL`, because it is an alias for `XL`, a geometry parameter available in all levels.

You can use `DL0` and `DW0` with this model, but you should use `XL`, `LD`, `XW`, and `WD` instead (noting the difference in units).

Watch the units of `TOX`. It is safest to enter a number greater than one, which simulation always interprets as Angstroms.

To avoid negative gds:

- 1. Set `X3U1`, `LX3U1`, and `WX3U1` to zero.
- 2. Check that
 $zx3ms \geq 0$, where $zx3ms = X3MS$ with `L`, `W` adjustment
- 3. Check that
 $zmuz + VDDM \cdot zx3ms < zmus$

BSIM Model—Compatibility Notes: SPICE/Synopsys Model
Parameter Differences

The following table compares the UCB BSIM1 and the Synopsys Level 13 model parameters. Units in this table are in brackets. This comparison uses the model parameter name only if it differs from the SPICE name. The model specifies parameter units only if they differ from SPICE units. These aliases are in parentheses. Some parameter aliases match the SPICE names.

An asterisk (*) in front of a UCB SPICE name denotes an incompatibility between the parameter name in the Synopsys Level 13 MOSFET device model and the UCB SPICE name (that is, the parameter alias does not match or the units are different).

Even if the parameter name in this model is not the same as in SPICE, the corresponding `L` and `W` sensitivity parameter names might not differ. The following table lists the `L` and `W` sensitivity parameters only for the few cases where the parameters are different.

UC Berkeley SPICE 2, 3	Synopsys Device Model
VFB [V]	VFB0 (VFB)
PHI [V]	PHI0
K1 [V ^{1/2}]	same

UC Berkeley SPICE 2, 3	Synopsys Device Model
K2	same
* ETA	ETA0
MUZ [$\text{cm}^2/\text{V}\cdot\text{s}$]	same
* DL [μm]	DL0
* DW [μm]	DW0
* U0 [$1/\text{V}$]	U00
* U1 [μ/V]	same
X2MZ [$\text{cm}^2/\text{V}^2\cdot\text{s}$]	X2M (X2MZ)
LX2MZ [$\mu\text{m}\cdot\text{cm}^2/\text{V}^2\cdot\text{s}$]	X2M (LX2MZ)
WX2MZ [$\mu\text{m}\cdot\text{cm}^2/\text{V}^2\cdot\text{s}$]	WX2M (WX2MZ)
X2E [$1/\text{V}$]	same
X3E [$1/\text{V}$]	same
X2U0 [$1/\text{V}^2$]	same
X2U1 [$\mu\text{m}/\text{V}^2$]	same
MUS [$\text{cm}^2/\text{V}\cdot\text{s}$]	same
LMUS [$\mu\text{m}\cdot\text{cm}^2/\text{V}\cdot\text{s}$]	LMS (LMUS)
WMUS [$\mu\text{m}\cdot\text{cm}^2/\text{V}\cdot\text{s}$]	WMS (WMUS)
X2MS [$\text{cm}^2/\text{V}^2\cdot\text{s}$]	same
X3MS [$\text{cm}^2/\text{V}^2\cdot\text{s}$]	same
X3U1 [$\mu\text{m}/\text{V}^2$]	same
* TOX [μm]	TOXM[μ] (TOX[m])
* TEMP [$^{\circ}\text{C}$]	TREF
* VDD [V]	VDDM
CGDO [F/m]	CGDOM (CGDO)
CGSO [F/m]	CGSOM (CGSO)
CGBO [F/m]	CGBOM (CGBO)

UC Berkeley SPICE 2, 3	Synopsys Device Model
XPART	same
N0	same
* NB	NB0
* ND	ND0
RSH [ohm/sq]	RSHM (RSH)
JS [A/m ²]	IJS (JS)
PB [V]	PJ (PB)
MJ	MJ0 (MJ)
* PBSW [V]	PJW (PHP)
MJSW	MJW (MJSW)
CJ [F/m ²]	CJM (CJ)
CJSW [F/m]	CCJW (CJSW)
* WDF [m]	—
* DELL [m]	—

In UCB SPICE, you must specify all BSIM model parameters. The Synopsys model provides default values for the parameters.

BSIM Model—Compatibility Notes: Parasitics

ACM >0 invokes parasitic diode models. ACM=0 (default) is SPICE style.

BSIM Model—Compatibility Notes: Temperature Compensation

The default `TNOM` model reference temperature is 25°C, unless you use `.OPTION SPICE` to set the default `TNOM` value to 27°C. This option also sets some other SPICE compatibility parameters. You set `TNOM` in an `.OPTION` line in the netlist, and you can always use the `TREF` model parameter to override it locally (that is, for a model). (The model “reference temperature” means that the model parameters were extracted at and are valid at that temperature.

UCB SPICE does not use `TNOM` (default 27°C) for the BSIM models. Instead, you must specify the `TEMP` model parameter as both the model reference temperature and the analysis temperature. Analysis at `TEMP` applies only to thermally-activated exponentials in the model equations. You cannot adjust model parameter values when you use `TEMP`.

Simulation assumes that you extracted the model parameters at `TEMP`, because `TEMP` is both the reference and the analysis temperature.

In contrast to UCB SPICE's BSIM, the Synopsys Level 13 model does provide for temperature analysis. The default analysis temperature is 25°C (and 27°C in UCB SPICE for all model levels except for BSIM as explained in the previous paragraph). Use a `.TEMP` statement in the netlist to change the analysis temperature.

The Level 13 model provides two temperature coefficients: `TCV` and `BEX`. The following equation adjusts the threshold voltage:

$$vth(t) = vth - TCV \cdot (t - tnom)$$

This model includes two implementations of the `BEX` factor. To select a `BEX` version, use the `UPDATE` parameter, described in the next section. The mobility in BSIM is a combination of five quantities: `MUZ`, `zmus`, `z3ms`, `zx2mz`, and `zx2ms`.

BSIM Model—Compatibility Notes: BEX Usage

$$MUZ(t) = MUZ \cdot \left(\frac{t}{tnom}\right)^{BEX}$$

$$zmus(t) = zmus \cdot \left(\frac{t}{tnom}\right)^{BEX}$$

$$zx3ms(t) = zx3ms \cdot \left(\frac{t}{tnom}\right)^{BEX}$$

$$zx2mz(t) = zx2mz \cdot \left(\frac{t}{tnom}\right)^{BEX}$$

$$zx2ms(t) = zx2ms \cdot \left(\frac{t}{tnom}\right)^{BEX}$$

This is equivalent to multiplying the final mobility by the factor:

$$\left(\frac{t}{tnom}\right)^{BEX}$$

BSIM Model—Compatibility Notes: UPDATE Parameter

The `UPDATE` parameter selects between variations of the BSIM equations. `UPDATE=0` (default) is consistent with UCB SPICE3. `UPDATE=3` also is consistent with UCB SPICE3 and `BEX` usage.

Parameter	Description
<code>UPDATE=0</code>	UCB compatible, previous <code>BEX</code> usage
<code>UPDATE=1</code>	Special X2E equation, previous <code>BEX</code> usage

Parameter	Description
UPDATE=2	Remove $1/L_{eff}$ in U1 equation, present BEX usage
UPDATE=3	UCB compatible, present BEX usage

BSIM Model—Compatibility Notes: Explanations

The normal X2E equation is:

$$\eta = \zeta - (z\chi_2e \cdot v_{sb}) + z\chi_3e \cdot (v_{ds} - V_{DDM})$$

The special X2E equation for UPDATE=1 only, is:

$$\eta = \zeta + z\chi_2e \cdot (z\phi + v_{sb}) + z\chi_3e \cdot (v_{ds} - V_{DDM})$$

The special X2E equation was developed to match a parameter extraction program. If you use a parameter extraction program, check the equations carefully.

The original U1 equation divides by L_{eff} in microns:

$$x_{u1} = \frac{(z_{u1} - (z\chi_2u1 \cdot v_{sb}) + z\chi_3u1 \cdot (v_{ds} - V_{DDM}))}{L_{eff}}$$

This is one of the few places where L_{eff} explicitly enters into the BSIM equations; usually, the L-adjustment model parameters (such as `LU1`) handles the L_{eff} variation.

Physically x_{u1} should decrease as $1/L_{eff}$ at long channels, but when dealing with short-channel devices, you can turn off this variation. Set UPDATE=2 to remove the $1/L_{eff}$ factor in the x_{u1} equation.

UPDATE=2 introduces the present BEX usage as the $1/L_{eff}$ removal ability. UPDATE=3 provides the present BEX using the previous x_{u1} equation.

BSIM Model – Compatibility Notes: IDS and VGS Curves for PMOS and NMOS

The netlists for the `IDS` and `VGS` curves for PMOS and NMOS are located in directory `$installdir/demo/hspice/mos/ml13iv.sp`. This file contains examples of the following model parameter and curve descriptions:

- Two Types of Model Parameter Formats Used
- VGS Curves
- GM Test
- GM B CVN7 5 37 0

- .PROCESS PC Filename=M57R
- N-channel Devices
- First Model Parameter Format
- PMOS Model
- Second Model Parameter Format
- N+ Diffusion Layer
- PMOS Model
- Wire Model for Poly and Metal Layers

LEVEL 27: SOSFET Model

This model is a three-terminal silicon-on-sapphire (SOS) FET transistor model [4]. This SOSFET model is based on a sapphire insulator that isolates the substrate and models the behavior of SOS devices more accurately than standard MOSFET models with physically unreal parameter values. The SOSFET model also includes a charge conservation model (based on the Ward and Dutton model).

Because the defaults of the SOSFET model parameters depend on the channel length, you must specify the `SOSLEV` model parameter to select either the 5 μm or 3 μm processing model.

`SOSLEV=1` selects the 5 μm model; otherwise, this model automatically uses the 3 μm value, including the second-order effects (default=3 μm).

Note:

This model does not include bulk nodes. If you specify bulk nodes, simulation ignores them.

This model does not use the `ACM` model parameter, because it does not include any junction diodes. Also, the only value that the `CAPOP` model parameter accepts is 7. Seven is its own charge conservation model, which you cannot use in other MOSFET models.

Temperature compensation equations for the `VTO` and `UO` SOSFET model parameters are the same as those in the MOSFET model.

Note:

This model includes a special option for bulk nodes for silicon on sapphire. In the model definition, if you specify -1 for the bulk node, this model generates a special node for each element. This bulk node is named in the form, `B#<element_name>`, where the element name is the name of the defined element. Use this name in any statement, such as a `.PRINT` statement to refer to the bulk node in the element.

Syntax:

```
.MODEL mname PMOS <LEVEL=27> <SOSLEV=val> <pname1=val1>
.MODEL mname NMOS <LEVEL=27> <SOSLEV=val> <pname=val1>
```

You can use this `.MODEL` syntax to include a MOSFET Level 27 model in your PrimeSim HSPICE netlist. For a general description of the `.MODEL` statement, see the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

Parameter	Description
mname	Model name.

Parameter	Description
PMOS	Identifies a p-channel MOSFET model.
NMOS	Identifies an n-channel MOSFET model.
LEVEL	Model level selector.
SOSLEV	Selects the processing model. If you set SOSLEV=1, the default=5 μ m. The automatic default=3 μ m.
pname	Parameter model.

This section covers the following topics:

- [SOSFET Model Parameters](#)
- [SOSFET Model — Non-Fully Depleted SOI Model](#)
- [SOSFET Model — Components](#)
- [SOSFET Model — Obtaining Model Parameters](#)
- [SOSFET Model — Fully Depleted SOI Model Considerations](#)
- [SOSFET Model — Example](#)

SOSFET Model Parameters

This section covers the following topics:

- [SOSFET Model — 5- \$\mu\$ m Model Parameters](#)
- [SOSFET Model — 3- \$\mu\$ m Model Parameters](#)

SOSFET Model — 5- μ m Model Parameters

Name (Alias)	Units	Default	Description
CGDO	F/m		Gate-drain overlap capacitance per unit channel width. Default=3.1e-10 (n-type), 2.2e-10 (p-type).
CGSO	F/m		Gate-source overlap capacitance per unit channel width. Default=3.1e-10 (n-type), 2.2e-10 (p-type).
LD	m		Lateral diffusion. The default=0.6 μ (n- type), 0.3 μ (p-type).
RSH	ohm/sq		Drain and source diffusion sheet resistance. The default=25 (n _i -type), 100 (p-type).

Name (Alias)	Units	Default	Description
SOSLEV		1	Model index.
TOX	m	7.0e-8	Oxide thickness.
UO	$\text{cm}^2/(\text{V}\cdot\text{s})$		Surface mobility. Default=350 (n-type), 220 (p-type).
VTO	V		Threshold voltage. Default=1.25 (n-type), -1.25 (p-type).

SOSFET Model — 3- μm Model Parameters

Name (Alias)	Units	Default	Description
A	m/V	0.1 μm	Channel length shortening coefficient (2nd effect)
ALPHA	V/m		Threshold voltage length dependence. Default=0.15 μ (n-type), 0.18 μ (p-type).
CAPOP		7	Capacitance model selector.
CGDO	F/m		Gate-drain overlap capacitance per unit channel width. Default=4.6e-10 (n-type), 3.6e-10 (p-type).
CGSO	F/m		Gate-source overlap capacitance per unit channel width. The default=4.6e-10 (n-type), 3.6e-10 (p-type).
EC	V/m		Critical electric field for velocity saturation (2nd effect). The default=3.0e6 (n-type), 7.5e6 (p-type).
FB			Body effect coefficient (2nd effect). Default=0.15 (n-type), 0 (p-type).
LD	m		Lateral diffusion. Default=0.3 μ (n-type), 0.2 μ (p-type).
LEVEL		27	Model level selector.
RSH	ohm/sq		Drain and source diffusion sheet resistance. Default=25 (n-type), 80 (p-type).
SOSLEV		2	Model index.
THETA	1/V		Mobility degradation coefficient (2nd effect). Default=0.055 (n-type), 0.075 (p-type).
TOX	m	3.4e-8	Oxide thickness.
UO	$\text{cm}^2/(\text{V}\cdot\text{s})$		Surface mobility. Default=370 (n-type), 215 (p-type).

Name (Alias)	Units	Default	Description
VTO	V		Threshold voltage. Default=0.83 (n-type), -0.74 (p-type).

SOSFET Model — Non-Fully Depleted SOI Model

Several MOSFET models are currently available for SOS/SOI applications. The 3-terminal SOS model (`LEVEL=27`) is stable for circuit design usage, but has some limitations. This model does not provide for depleted bulk. Use it only with applications that are not fully depleted and that do not consider kink effects.

The following circuit example is a 4-terminal SOI model for incompletely depleted bulk with the kink effect. Its sub-circuit allows a parasitic capacitance to the substrate. In this example, the bulk is the region under the channel. This model assumes that the substrate is the conductive layer under the insulator.

- For SOI, the insulator is usually silicon dioxide and the substrate is silicon.
- For SOS, the insulator is sapphire and the substrate is the metal that contacts the back of the integrated circuit die.

SOSFET Model — Components

This model consists of the following subcomponents:

- Core IDS model: any level works because the impact ionization and weak inversion models are common to all DC levels. The example uses a `LEVEL=3` DC MOS model.
- Subthreshold model: the `WIC=3` model parameter allows the older models to use the more advanced models found in the BSIM (`LEVEL=13`, `LEVEL=28`) models. The `N0` model parameter should have a typical value around 1.0.
- Impact ionization model: set `ALPHA` and `VCR` parameters to enable the impact ionization model, which is available to all MOS DC equations. Typical values are `ALPHA=0.1` and `VCR=18`.
- Charge conservation gate cap model (`CAPOP=9`, `XQC=.4`) prevents the floating bulk node from obtaining extreme values.
- The automatic periphery diode area calculation method (`ACM`) is set to 3 to automatically calculate the source and drain resistances and diode junction leakage,

and the capacitance. (ACM=3 CJ=0 CJSW=0 CJGATE=4e-10 JS=0 JSW=1e-9 LD=.1u HDIF=1.5u RS=40 RD=40 N=1).

Note:

These models assume that the source/drain diffusions extend to the buried oxide. The area part of the diode has no capacitance to bulk. However, the subcircuit includes linear capacitors to the substrate.

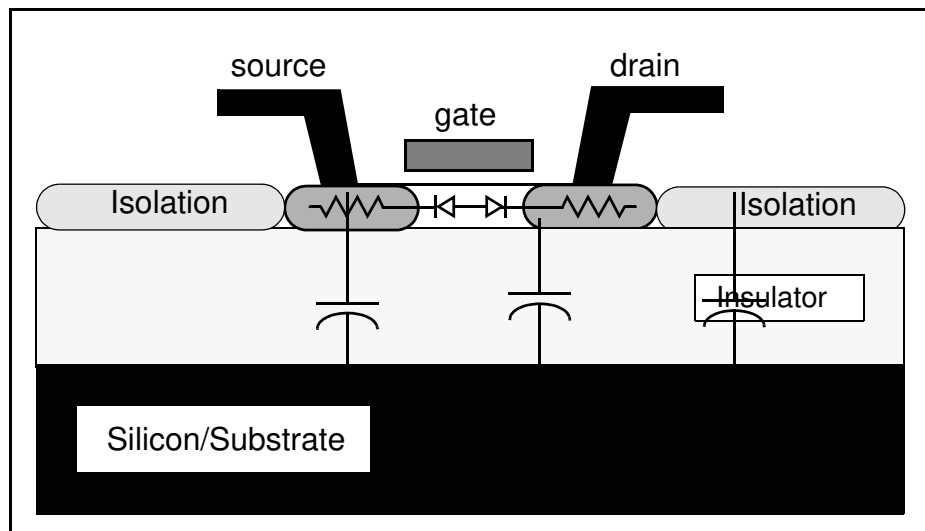
SOSFET Model — Obtaining Model Parameters

Use the optimizing capabilities in the Level 27 MOSFET model to obtain the core IDS model parameters.

Use the optimizer to obtain the core model, subthreshold, and impact ionization parameters. The subthreshold model selected is an improved BSIM type of model that was altered for the older models. The charge conservation model is more charge conserving than the original Ward-Dutton model in SPICE 2G6.

Calculating the automatic diode area and the resistance estimates the junction capacitance, saturation current, and resistance as a function of the transistor width. Use the `VNDS` and `NDS` parameters for a piecewise linear approximation to reverse the junction current characteristics.

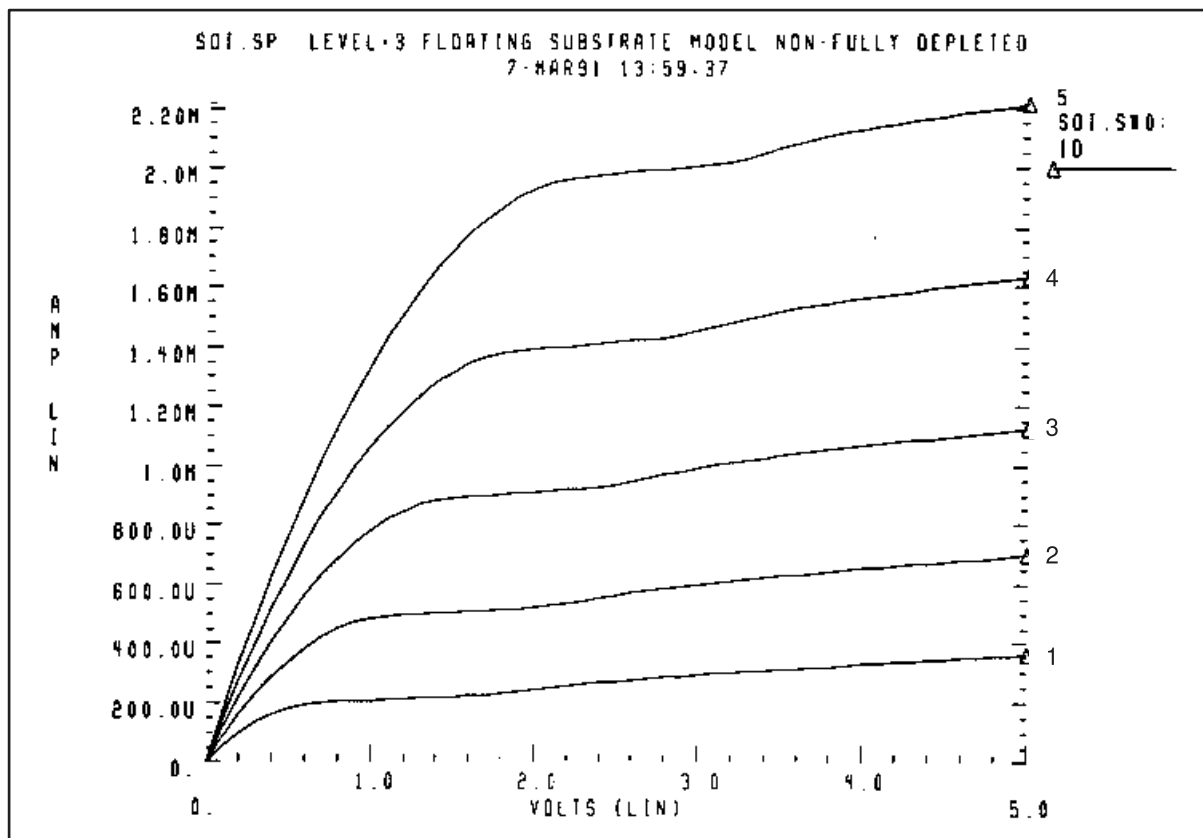
Figure 7 Non Fully Depleted SOI Model



The following example is based on demonstration netlist `ssoi.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
ssoi.sp LEVEL=3 floating bulk model
** non-fully depleted
* test 1st order soi model with floating substrate
.option nomod post
* substrate capacitance 3.45e-11 is for SiO2
.param t_sub_ox=.5u subcap='3.45e-11/t_sub_ox' hdif=1.5u
.global substrate
.dc vd 0 5 0.1 sweep vg 1.5 3.5 0.5
.print id=i(xml.m) vds=v(d) vgs=v(g)
.param vds=0 vgs=0 vbs=0
vd d gnd vds
vg g gnd vgs
vs s gnd 0
vsub substrate gnd vbs
xml d g s nch w=50u L=5u
.macro nch d g s w=10u l=2u
* macro definition for fet+ parasitic cap to substrate
* assumes existence of undepleted bulk
m d g s b nch w=w L=L
cx1 d substrate c='w*2*hdif*subcap'
cx2 s substrate c='w*2*hdif*subcap'
cx3 b substrate c='w*L*subcap'
.eom
.model nch nmos LEVEL=3
+ lmin=.5u lmax=100u wmin=.5u wmax=500u $model selector
+ ld=0.1u wd=.15u xl=0 xw=0
$diffusion+photobias
+ acm=3 hdif=hdif rsh=30 rs=10k rd=10k $resistors
+ ldif=0.1u
$junction cap (ACM=3 (h9007 only) allows diode on gate edge
+ cj=0 cjsw=0 cgate=0.4e-9 mjsw=0.33 php=0.6
+ js=0 jsw=1e-9 n=1 vnds=.5 nds=1 $junction leakage
+ bex=-1.5 tcv=2m $temperature
+ tox=200 capop=9 xqc=.4 meto=0.08u $gate cap
+ alpha=0.1 vcr=18 $impact ionization
+ vto=0.7 phi=1 gamma=1 $threshold
+ eta=10 xj=0.1u $threshold
+ wic=3 n0=0.9 nd=0 $subthreshold
+ uo=400 theta=1m $dc mobility
+ vmax=100k kappa=0 $dc saturation
.end
```

Figure 8 LEVEL 3 Floating Bulk Model



SOSFET Model — Fully Depleted SOI Model Considerations

Fully depleted transistors require additional modeling equations. The first-order effects are:

- Threshold sensitivity to the substrate.
- No kink current.
- Silicon thickness limits the minimum depletion capacitance.

Lack of these effects does not seriously affect an inverter, because the source-to-substrate voltage does not move. Digital circuits with good gate drive are not seriously affected, because a large gate voltage renders a small V_{th} shift to a small change in the I_{DS} current.

The substrate threshold sensitivity can affect circuits such as analog amplifiers that include transistors at back-bias and low gate voltages.

SOSFET Model — Example

This example is based on demonstration netlist `m127iv.sp`, which is available in directory `$installdir/demo/hspice/mos:`

```
*file: m127iv.sp ids and vgs curves for nmos and pmos soscets.
*mosfet level=27 p and n
```

```
.options acct list nopage nomod post
.op
.dc vddn 0 5.0 .1
```

```
* n-channel ids curves (vd=0-&lt;&lt;5, vg=1,2,3,4,5)
.print dc i(vn1) i(vn2) i(vn3) i(vn4) i(vn5)
.probe dc i(vn1) i(vn2) i(vn3) i(vn4) i(vn5)
```

```
* p-channel ids curves (vd=0-&lt;&lt;-5,vg=-1,-2,-3,-4,-5)
.print dc i(vp1) i(vp2) i(vp3) i(vp4) i(vp5)
.probe dc i(vp1) i(vp2) i(vp3) i(vp4) i(vp5)
```

```
* v g s curves
.print dc i(vn6) i(vp6)
.probe dc i(vn6) i(vp6)
```

```
* n-channel LX7=gm (vd=5, vg=0-&lt;&lt;5, vs=0)
* n-channel LX8=gd (vd=0-&lt;&lt;5, vg=5, vs=0)
* n-channel LX9=gb (vd=5, vg=5, vs=0)
.print dc LX7 (m21) LX8 (m5) LX9 (m31)
```

```
* p-channel LX7=gm (vd=0, vg=0-&lt;&lt;-5, vs=-5)
* p-channel LX8=gd (vd=0-&lt;&lt;-5, vg=-5, vs=-5)
* p-channel LX9=gb (vd=0, vg=0, vs=-5)
.print dc LX7 (m22) LX8 (m15) LX9 (m32)
*
```

```
vddn 99 0 5.0
epd 98 0 99 0 -1
```

```
v1 1 0 1
v2 2 0 2
v3 3 0 3
v4 4 0 4
v5 5 0 5
v11 11 0 -1
v12 12 0 -2
v13 13 0 -3
v14 14 0 -4
v15 15 0 -5
*
vn1 99 31 0
vn2 99 32 0
vn3 99 33 0
vn4 99 34 0
```

```

vn5 99 35 0

m1 31 1 0 n1 8u 8u
m2 32 2 0 n1 8u 8u
m3 33 3 0 n1 8u 8u
m4 34 4 0 n1 8u 8u
m5 35 5 0 n1 8u 8u
*
vp1 98 41 0
vp2 98 42 0
vp3 98 43 0
vp4 98 44 0
vp5 98 45 0

m11 41 11 0 p1 8u 8u
m12 42 12 0 p1 8u 8u
m13 43 13 0 p1 8u 8u
m14 44 14 0 p1 8u 8u
m15 45 15 0 p1 8u 8u
*
* g m test
vn6 5 36 0
vp6 0 46 0
m21 36 99 0 n1 8u 8u
m22 46 98 15 p1 8u 8u
*
* g m b test
vn7 5 37 0
vp7 0 47 0
m31 37 5 0 98 n1 8u 8u
m32 47 0 15 99 p1 8u 8u
*
.model n1 nmos level=27 soslev=2
+vto=0.814 tox=0.34e-7 theta=0.55e-1
+fb=0.15 ec=0.3e7 a=0.1e-6
+uo=370 cgso=0.46e-9 cgdo=0.46e-9
+rsh=25 ld=0.3e-6
*
.model p1 pmos level=27 soslev=2
+vto=-0.7212 tox=0.34e-7 theta=0.75e-1
+fb=0.0 ec=0.75e7 a=0.1e-6
+uo=215 cgso=0.36e-9 cgdo=0.36e-9
+rsh=80 ld=0.2e-6
*
.end

```

LEVEL 28: BSIM Derived Model

This model is derived from the base BSIM Model. The following are the significant features of this model:

- Vertical field dependence of the carrier mobility
- Carrier velocity saturation
- Drain-induced barrier lowering
- Depletion charge sharing by the source and drain
- Non-uniform doping profile for ion-implanted devices
- Channel length modulation
- Subthreshold conduction
- Geometric dependence of the electrical parameters

This section covers the following topics:

- [BSIM Derived Model Parameters](#)
- [BSIM Derived Model Equations](#)

BSIM Derived Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this model also uses the following parameters:

This section covers the following topics:

- [BSIM Derived Model — Transistor Process Parameters](#)
- [BSIM Derived Model — Temperature Parameters:](#)
- [BSIM Derived Model — Sensitivity Factors of Model Parameters](#)

BSIM Derived Model — Transistor Process Parameters

Name (Alias)	Units	Default	Description
LEVEL		1	MOSFET model level selector. Set this parameter to 28 for this model.
B1		0.0	Lower vdsat transition point
B2		1	Upper vdsat transition point

Name (Alias)	Units	Default	Description
CGBO	F/m	2.0e-10	Gate-to-bulk parasitic capacitance (F/m of length)
CGDO	F/m	1.5e-9	Gate-to-drain parasitic capacitance (F/m of width)
CGSO	F/m	1.5e-9	Gate-to-source parasitic capacitance (F/m of width)
ETA0		0.0	Linear vds threshold coefficient
ETAMN		0.0	Minimum linear vds threshold coefficient
GAMMN	$V^{1/2}$	0.0	Minimum root-vsbs threshold coefficient
K1	$V^{1/2}$	0.5	Root-vsbs threshold coefficient
K2		0.0	Linear vsbs threshold coefficient
LB1	μm	0.0	Length sensitivity
LB2	μm	0.0	Length sensitivity
LETA	μm	0.0	Length sensitivity
LETAMN	μm	0.0	Length sensitivity
LGAMN	$V^{1/2} \cdot \mu\text{m}$	0.0	Length sensitivity
LK1	$V^{1/2} \cdot \mu\text{m}$	0.0	Length sensitivity
LK2	μm	0.0	Length sensitivity
LMUZ	$\mu\text{m} \cdot \text{cm}^2 / \text{V} \cdot \text{s}$	0.0	Length sensitivity
LN0	μm	0.0	Length sensitivity
LNB	μm	0.0	Length sensitivity
LND	μm	0.0	Length sensitivity
LPHI	$\text{V} \cdot \mu\text{m}$	0.0	Length sensitivity
LU0	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LU1	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LVFB	$\text{V} \cdot \mu\text{m}$	0.0	Length sensitivity
LWFAC	μm	0.0	Length sensitivity

Name (Alias)	Units	Default	Description
LWFACU	μm	0.0	Length sensitivity
LX2E	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LX2M (LX2MZ)	$\mu\text{m} \cdot \text{cm}^2/\text{V}^2 \cdot \text{s}$	0.0	Length sensitivity
LX2U0	$\mu\text{m}/\text{V}^2$	0.0	Length sensitivity
LX2U1	$\mu\text{m}^2/\text{V}^2$	0.0	Length sensitivity
LX33M	$\mu\text{m} \cdot \text{cm}^2/\text{V}^2 \cdot \text{s}$	0.0	Length sensitivity
LX3E	$\mu\text{m}/\text{V}$	0.0	Length sensitivity
LX3MS	$\mu\text{m} \cdot \text{cm}^2/\text{V}^2 \cdot \text{s}$	0.0	Length sensitivity
LX3U1	$\mu\text{m}/\text{V}^2$	0.0	Length sensitivity
MUZ	$\text{cm}^2/\text{V} \cdot \text{s}$	600	Low drain field first order mobility
N0		200	Low field weak inversion gate drive coefficient (value of 200 for N0 disables the weak inversion calculation)
NB0		0.0	Vsb reduction to the low field weak inversion gate drive coefficient
ND0		0.0	Vds reduction to the low field weak inversion gate drive coefficient
PHI0	V	0.7	Two times the Fermi potential
TOXM (TOX)	μm (m)	0.02	Gate oxide thickness (if TOXM or TOX > 1, uses Angstroms)
U00	1/V	0.0	Gate field mobility reduction factor
U1	1/V	0.0	Drain field mobility reduction factor
VDDM	V	5.0	Critical voltage for the high-drain field mobility reduction
VFB0 (VFB)	V	-0.3	Flatband voltage
WB1	μm	0.0	Width sensitivity
WB2	μm	0.0	Width sensitivity

Name (Alias)	Units	Default	Description
WETA	μm	0.0	Width sensitivity
WETAMN	μm	0.0	Width sensitivity
WFAC		4	Weak inversion factor
WFACU		0.0	Second weak inversion factor
WGAMN	$V^{1/2} \cdot \mu\text{m}$	0.0	Width sensitivity
WK1	$V^{1/2} \cdot \mu\text{m}$	0.0	Width sensitivity
WK2	μm	0.0	Width sensitivity
WMUZ	$\mu\text{m} \cdot \text{cm}^2 / V \cdot \text{s}$	0.0	Width sensitivity
WN0	μm	0.0	Width sensitivity
WNB	μm	0.0	Width sensitivity
WND	μm	0.0	Width sensitivity
WPHI	$V \cdot \mu\text{m}$	0.0	Width sensitivity
WU0	$\mu\text{m}/V$	0.0	Width sensitivity
WU1	$\mu\text{m}/V$	0.0	Width sensitivity
WVFB	$V \cdot \mu\text{m}$	0.0	Width sensitivity
WWFAC	μm	0.0	Width sensitivity
WWFACU	μm	0.0	Width sensitivity
WX2E	$\mu\text{m}/V$	0.0	Width sensitivity
WX2M (WX2MZ)	$\mu\text{m} \cdot \text{cm}^2 / V^2 \cdot \text{s}$	0.0	Width sensitivity
WX2U0	$\mu\text{m}/V^2$	0.0	Width sensitivity
WX2U1	$\mu\text{m}^2 / V^2$	0.0	Width sensitivity
WX33M	$\mu\text{m} \cdot \text{cm}^2 / V \cdot \text{s}$	0.0	Width sensitivity
WX3E	$\mu\text{m}/V$	0.0	Width sensitivity
WX3MS	$\mu\text{m} \cdot \text{cm}^2 / V \cdot \text{s}$	0.0	Width sensitivity

Name (Alias)	Units	Default	Description
WX3U1	$\mu\text{m}/\text{V}^2$	0.0	Width sensitivity
X2E	$1/\text{V}$	0.0	Vsb correction to the linear vds threshold coefficient
X2M (X2MZ)	$\text{cm}^2/\text{V}^2\cdot\text{s}$	0.0	Vsb correction to the low field first order mobility
X2U0	$1/\text{V}^2$	0.0	Vsb reduction to the GATE field mobility reduction factor
X2U1	$\mu\text{m}/\text{V}^2$	0.0	Vsb reduction to the DRAIN field mobility reduction factor
X33M	$\text{cm}^2/\text{V}^2\cdot\text{s}$	0.0	Gate field reduction of X3MS
X3E	$1/\text{V}$	0.0	Vds correction to the linear vds threshold coefficient
X3MS	$\text{cm}^2/\text{V}^2\cdot\text{s}$	5.0	Vds correction for the high-drain field mobility
X3U1	$1/\text{V}^2$	0.0	Vds reduction to the drain field mobility reduction factor
XPART		1.0	Selects the coefficient for sharing the gate capacitance charge

Note:

Following are some important points:

- When you read parameter names, be careful about the difference in appearance between the capital letter O and the number zero (0).
- Use NMOS conventions to specify all Level 28 parameters, even for PMOS—for example, `ETA0=0.02`, not `ETA0=-0.02`
- You can use the `WL`-product sensitivity parameter for any parameter with an `L` and `W` sensitivity. Replace the leading “L” of the `L` sensitivity parameter name with a “P”.

BSIM Derived Model — Temperature Parameters:

Name (Alias)	Units	Default	Description
BEX		-1.5	Temperature exponent for the MUZ, X2M, X3MS, and X33M mobility parameters

Name (Alias)	Units	Default	Description
FEX		0.0	Temperature exponent for the U1 mobility reduction factor
TCV	V/°K	0.0	Flat-band voltage temperature coefficient

BSIM Derived Model — Sensitivity Factors of Model Parameters

For transistors, drop the 0 from the end of the parameter name, and add one of the following product sensitivity factors for a basic electrical parameter:

- L (channel length)
- W (channel width)
- WL (width and length)

For example, the $VFB0$ sensitivity factors are $LVFB$, $WVFB$, and $PVFB$. If $A0$ is a basic parameter, LA , WA and PA are the corresponding sensitivity factors for this parameter (you cannot use the $SCALM$ option to scale LA , WA , and PA). Then the model uses the following general formula to obtain the parameter value.

The left side of the equation represents the effective model parameter value after you adjust the device size. All effective model parameters are in lower case and start with the z character, followed by the parameter name.

$$za = A0 + LA \cdot \left[\frac{1}{Leff} - \frac{1}{LREFeff} \right] + WA \cdot \left[\frac{1}{Weff} - \frac{1}{WREFeff} \right]$$

$$PA \cdot \left[\frac{1}{Leff} - \frac{1}{LREFeff} \right] \cdot \left[\frac{1}{Weff} - \frac{1}{WREFeff} \right]$$

Specify LA and WA in units of microns times the units of $A0$. Specify PA in units of square microns times the units of $A0$.

If you set $LREF$ or $WREF=0$, you effectively set the parameter value to infinity. This is the default.

Example:

$$VFB0 = -0.350v$$

$$LVFB = -0.1v\mu$$

$$WVFB = 0.08v \cdot \mu$$

$$Leff = 1 \cdot 10^{-6}m = 1\mu$$

$$W_{eff} = 2 \cdot 10^{-6} m = 2\mu$$

$$L_{REFeff} = 2 \cdot 10^{-6} m = 2\mu$$

$$W_{REFeff} = 1 \cdot 10^{-5} m = 10\mu$$

$$z_{vfb} = V_{FB0} + LV_{FB} \cdot \left(\frac{1}{L_{eff}} - \frac{1}{L_{REFeff}} \right) + WV_{FB} \cdot \left(\frac{1}{W_{eff}} - \frac{1}{W_{REFeff}} \right)$$

$$z_{vfb} = -0.35v + -0.1v \cdot \mu \cdot \left(\frac{1}{1\mu} - \frac{1}{2\mu} \right) + 0.08v \cdot \mu \cdot \left(\frac{1}{2\mu} - \frac{1}{10\mu} \right)$$

$$z_{vfb} = -0.35v - 0.05v + 0.032v$$

$$z_{vfb} = -0.368v$$

BSIM Derived Model Equations

This section covers the following topics:

- [BSIM Derived Model — Effective Channel Length and Width](#)
- [BSIM Derived Model — Threshold Voltage](#)
- [BSIM Derived Model — Effective Mobility](#)
- [BSIM Derived Model — Saturation Voltage \(vdsat\)](#)
- [BSIM Derived Model — Transition Points](#)
- [BSIM Derived Model — Strong Inversion Current](#)
- [BSIM Derived Model — Weak Inversion Current](#)
- [BSIM Derived Model — \$x_{weak} < -z_{wfac} + A_0\$](#)
- [BSIM Derived Model — \$-z_{wfac} + A_0 < x_{weak} < 0\$](#)
- [BSIM Derived Model — \$0 < x_{weak} < A_0\$](#)
- [BSIM Derived Model — \$A_0 < x_{weak}\$](#)

BSIM Derived Model — Effective Channel Length and Width

The effective channel length and width for Level 28 is consistent with the LEVEL 3 model.

L , W , and the M multiplier are from the `.MODEL` statement in the netlist. `SCALE` and `SCALM` are options. If you do not specify any scaling options or multipliers, then:

$$L_{eff} = L + XL - 2 \cdot LD \quad W_{eff} = W + XW - 2 \cdot WD$$

Note:

If you specify LDAC and WDAC in the .MODEL statement, $L_{eff}=L+XL-2 \cdot LDAC$
 $W_{eff}=W+XW-2 \cdot WDAC$

```
Lscaled=L · SCALE
Wscaled=W · SCALE
XLscaled=XL · SCALM
LDscaled=LD · SCALM
XWscaled=XW · SCALM
WDscaled=WD · SCALM
Leff=Lscaled · LMLT+XLscaled-2 · LDscaled
LREFeff=LREFscaled · LMLT+XLREFscaled-2 · LDscaled
Weff=M · (Wscaled · WMLT+XWREFscaled-2 · WDscaled)
WREFeff=M · (WREFscaled · WMLT+XWscaled-2 · WDscaled)
```

BSIM Derived Model — Threshold Voltage

Effective model parameter values for the threshold voltage, after you adjust the device size, are zphi, zvfb, zk1, zk2, zeta, zx2e, zx3e, zgammn, and zetamn. Simulation calculates these values from the PHI0, VFB0, K1, K2, ETA0, X2E, X3E, GAMMN, and ETAMN model parameters, and from their respective length and width sensitivity parameters.

$$xbs=(zphi-vbs)^{1/2}$$

$$xeta=zeta+zx2e \cdot vbs+zx3e \cdot vds$$

$$vth=zvfb+zphi+zk1 \cdot xbs-zk2 \cdot xbs^2-xeta \cdot vds$$

This equation is quadratic in xbs and vds. It is joined to linear equations at $d(vth)/d(xbs)=zgammn$ and at $d(vth)/d(vds)=-zetamn$, which prevents the quadratics from going in the wrong direction.

Both gammn and etamn default to zero, and typically do not affect behavior in the normal operating region.

BSIM Derived Model — Effective Mobility

The effective model parameter values for mobility, after you adjust the device size, are zmuz, zx2m, zx3m, zx33m, zu0, and zx2u0. Simulation calculates these values from the MUZ, X2M, X3m, X33M, U00, and X2U0 model parameters, and from their respective length and width sensitivity parameters.

$$v_{gst} = v_{gs} - v_{th}$$

$$m_{eff} = (zmuz + zx2m \cdot v_{bs})$$

$$cx3ms = \frac{zx3ms}{(muz + zx33m \cdot v_{gst})}$$

$$\left(1 + cx3ms \cdot \left(VDDM + v_{ds} \cdot (VDDM \cdot VDDM + v_{ds} \cdot v_{ds})^{1/2}\right)\right)$$

$$xu0 = zu0 + zx2u0 \cdot v_{bs}$$

$$u_{eff} = \frac{m_{eff}}{(1 + xu0 \cdot v_{gst})}, \beta = u_{eff} \cdot COX \cdot \frac{W_{eff}}{L_{eff}}$$

BSIM Derived Model — Saturation Voltage (vdsat)

The effective model parameter values for the saturation voltage, after you adjust the device size, are zu1, zx2u1, and zx3u1. Simulation calculates these values from the U1, X2U1, and X3U1 model parameters, and from their length and width sensitivity parameters.

$$xbs = (zphi - v_{bs})^{1/2}, g = 1 - \frac{1}{(1.744 + 0.8364 \cdot xbs^2)}$$

$$body = \frac{1 + g \cdot zk1}{(2 \cdot xbs)}, xu1 = zu1 + v_{bs} \cdot zx2u1$$

$$rx = (body^2 + zu1 \cdot 2 \cdot body \cdot v_{gst} + zx3u1 \cdot 4 \cdot v_{gst}^2)^{1/2}$$

$$v_{dsat} = \frac{2 \cdot v_{gst}}{(body + rx)}$$

This vds value generates the partial derivative of:

$$f(v_{ds}, v_{gst}, v_{bs}) = (v_{gst} - body / 2 \cdot v_{ds}) \cdot \frac{v_{ds}}{(1 + (xu1 + zx3u1 \cdot v_{ds}) \cdot v_{ds})}$$

In the preceding equation, vds=zero.

BSIM Derived Model — Transition Points

The effective model parameter values for the transition points, after you adjust the device size, are zb1 and zb2. Simulation calculates these values from the B1 and B2 model parameters, and from their respective length and width sensitivity parameters.

$$v1 = v_{dsat} - zb1 \cdot \frac{v_{dsat}}{1 + v_{dsat}}, v2 = v_{dsat} + zb2 \cdot v_{gst}$$

BSIM Derived Model — Strong Inversion Current

For vds < v1:

$$I_{ds} = \beta \cdot (v_{gst} - body / 2 \cdot v_{ds}) \cdot \frac{v_{ds}}{(1 + (zu1 + zx3u1 \cdot v_{ds}) \cdot v_{ds})}$$

The vds derivative varies approximately linearly between v1 and v2.

For vds > v2, ids is a function of beta and vgst only. If zb1 and zb2 are both positive, their main effect is to increase the saturation current.

BSIM Derived Model — Weak Inversion Current

The effective model parameter values for weak inversion current, after you adjust the device size, are $zn0$, znb , znd , $zwfac$, and $zwfacu$. Simulation calculates these values from the NO , NDO , $NB0$, $WFAC$, and $WFACU$ model parameters, and from their respective length and width sensitivity parameters.

Simulation calculates the weak inversion current when $zn0$ is less than 200, and adds it to the strong inversion current:

$$I_{total} = I_{strong} + I_{weak} \cdot \left(1 - \exp\left(\frac{-v_{ds}}{v_{therm}}\right) \right)$$

In deep subthreshold:

$$\chi_n = zn0 + znb \cdot v_{bs} + znd \cdot v_{ds}$$

$$v_{therm} = \frac{KT}{Q}, \chi_{weak} = \frac{(v_{gs} - v_t)}{(\chi_n \cdot v_{therm})}$$

$$I_{weak} = const \cdot \exp(\chi_{weak})$$

$zwfac$ and $zwfacu$ control the modification of this formula near the threshold. Just above threshold, the device is in saturation:

$$I_{strong} = const \cdot \chi_{weak}^2$$

I_{weak} needs an χ_{weak}^2 term to cancel the kink in gm at the threshold. Then I_{weak} goes to zero for $\chi_{weak} > A0$, which is at a small voltage above the threshold. I_{weak} has four regions.

BSIM Derived Model — $\chi_{weak} < -zwfac + A0$

$$I_{weak} = const \cdot \exp(\chi_{weak})$$

BSIM Derived Model — $-zwfac + A0 < \chi_{weak} < 0$

$$I_{weak} = const \cdot \exp(\chi_{weak} - const \cdot wf)$$

In the preceding equation, wf is the integral with respect to the χ_{weak} value of:

$$dwf = \frac{(\chi_{weak} + zwfac - A0)^2}{[(1 + \chi_{weak} + zwfac - A0)(1 + zwfacu \cdot (\chi_{weak} + zwfac - A0))]}$$

BSIM Derived Model — $0 < \chi_{weak} < A0$

$$I_{weak} = (\text{same formula as in region 2}) - const \cdot \chi_{weak}^2$$

BSIM Derived Model — $A0 < x_{weak}$

$$I_{weak} = 0$$

$A0$, and the constants in the preceding equations, are not model parameters. Continuity conditions, at the boundaries between regions, uniquely determine these constants.

LEVEL 38: Cypress Depletion Model

This model is a further development of the Synopsys [LEVEL 5: AMI-ASPEC Depletion and Enhancement \(Taylor-Huang\) Model](#). This model provides the following features:

- BSIM-style length and width sensitivities
- Degraded body effect at high substrate bias (second γ)
- Empirical fitting parameters for I_{ds} current calculations in the depletion mode of operations
- A comprehensive surface mobility equation
- Drain-induced barrier lowering

At the default parameter settings, the LEVEL 38 model is basically backwards-compatible with LEVEL 5 /ZENH=0.0 with the exception of the surface mobility degradation equation (see the discussion on the next page). Refer to the documentation for LEVEL 5 for the underlying physics that forms the foundation for the Huang-Taylor construct.

In LEVEL 38, the temperature compensation for threshold is ASPEC-style, concurring with the default in LEVEL 5. This section describes the model parameters that are unique to this depletion model. It also describes additional temperature compensation parameters.

LEVEL 38 lets you use all Synopsys device model capacitance options (CAPOP). CAPOP=2 is the default setting for LEVEL 38. If you set CAPOP=6 (AMI capacitance model), LEVEL 38 capacitance calculations become identical to those of LEVEL 5.

The ACM default parameter (ACM=0 in LEVEL 38) invokes SPICE-style parasitics. You can set ACM to 1 (ASPEC), or to 2 (Synopsys device model). All MOSFET models follow this convention.

You can use .OPTION SCALE with the LEVEL 5 MOSFET device model. However, you cannot use the SCALM option, due to the difference in units. You also cannot use the DERIV option.

You must specify the following parameters for MOS LEVEL 38: VTO (VT), TOX, UO (UB), FRC, ECV, and NSUB (DNB).

As with LEVEL 5, this model calculates the I_{ds} current according to three gate voltage regions:

- *Depletion Region, $v_{gs} - v_{fb} < 0$*
The low gate voltage region, which the bulk channel dominates.
- *Enhancement Region, $v_{gs} - v_{fb} > 0, v_{ds} < v_{gs} - v_{fb}$*

The region defined by high gate voltage and low drain voltage. In the enhancement region, both channels are fully turned on.

- *Partial enhancement region, $v_{gs} - v_{fb} > 0$, $v_{ds} > v_{gs} - v_{fb}$*

This region has high gate and drain voltages so the surface region is partially turned on, and the bulk region is fully turned on.

To better model depletion region operations, empirical fitting constants have been added to the original Huang-Taylor mechanism to account for the effects caused by nonuniform channel implants and also to make up for an oversight in the average capacitance construct [5]. The enhancement region uses a significantly more elaborate surface mobility model.

Body effect in LEVEL 38 is calculated in two regions [6].

- *Bulk body effect, $v_{sb} - v_{sbc} > 0$*

With sufficiently high (and negative) substrate bias (exceeding v_{sbc}), the depletion region at the implanted channel-substrate junction reaches the Si-oxide interface. Under such circumstances, the free carriers can accumulate only at the interface (as in an enhancement device) and the bulk doping level determines the body effect.

- *Implant-dominated body effect, $v_{sb} - v_{sbc} < 0$*

Before reaching v_{sbc} , and as long as the implant dose overwhelms the substrate doping level, the deeply-buried transistor (due to the implant) dominates the body effect of the depletion mode device. The γ body effect coefficient is proportional to both the substrate doping and to the first-order implant depth. In Level 38, the `BetaGam` empirical parameter amplifies the body effect due to deep implant.

Model parameters that start with L or W represent geometric sensitivities. In the model equations, three model parameters determine the zX quantity (X is the variable name):

- Large-and-wide channel case value (X).
- Length sensitivity (LX).
- Width sensitivity (WX).

The model calculates these parameters according to $zX = X + LX/L_{eff} + WX/W_{eff}$. For example, the following equation calculates the zero field surface mobility:

$$zUO = UO + \frac{LUO}{l_{eff}} + \frac{WUO}{w_{eff}}$$

Note:

This model uses mostly micrometer units rather than meter units. Units and defaults are often unique in LEVEL 38. The finite difference method calculates the I_{ds} derivatives that define the gm, gds, and gmbs small signal gains. This model does not use the `SCALM` and `DERIV` options.

This section covers the following topics:

- [Cypress Depletion Model Parameters](#)
- [Cypress Depletion Model Equations](#)
- [Cypress Depletion Model — Sample Model File](#)

Cypress Depletion Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this level also uses the following parameters:

Name (Alias)	Units	Default	Description
AFC		1.0	Area factor for MOSFET capacitance
CAPOP		6	Gate capacitance selector
METO	μm	0.0	Metal overlap on gate

Cypress Depletion Model Equations

This section covers the following topics:

- [Cypress Depletion Model — IDS Equation](#)
- [Cypress Depletion Model — Threshold Voltage, vth](#)
- [Cypress Depletion Model — Saturation Voltage, vdsat](#)
- [Cypress Depletion Model — Mobility Reduction, UBeff](#)
- [Cypress Depletion Model — Channel Length Modulation](#)
- [Cypress Depletion Model — Subthreshold Current, ids](#)

Cypress Depletion Model — IDS Equation

Depletion, $v_{gs}-v_{fb} < 0$

$$ids = \beta 1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde + cav \cdot \left[(vgs - vfb) \cdot vde - \frac{vde^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \gamma \cdot \left[(vde + vsb + Phid)^{\frac{3}{2}} - (vsb + Phid)^{\frac{3}{2}} \right] + Icrit \right\}$$

Enhancement, $vgs - vfb > 0$

$$ids = \beta 1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde - \frac{2}{3} \beta cav \gamma \left[(vde + vsb + Phid)^{\frac{3}{2}} - (vsb + Phid)^{\frac{3}{2}} \right] + Icrit \right\} + \beta \cdot \left[(vgs - vfb) \cdot vde - \frac{vde^2}{2} \right]$$

Partial Enhancement, $vgs - vfb < vde$

$$ids = \beta 1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde + cav \cdot \left[(vgs - vfb) \cdot vde - \frac{vde^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \gamma \cdot \left[(vde + vsb + phid)^{\frac{3}{2}} - (vsb + phid)^{\frac{3}{2}} \right] + icrit \right\} + \left(\frac{1}{2} \beta - \frac{1}{2} \beta 1 \beta cav \right) \cdot (vgs - vfb)^2$$

The following equations calculate values used in the preceding equations:

$$\beta 1 = \frac{zKBeta1}{1 + UHSAT \cdot \frac{vde}{Leff}} \cdot UH \cdot \frac{Weff}{Leff}$$

$$\beta = UBeff \cdot cox \cdot \frac{Weff}{Leff}, \quad cav = \frac{cox \cdot cs}{cox + cs}$$

$$cs = \frac{KCS \cdot \epsilon Si}{DP \cdot 1e-4}, \quad Phid = vt \cdot \ln \left(\frac{DNB \cdot nd}{ni^2} \right)$$

$$nd = \frac{NI \cdot 1e4}{DP}, \quad vde = \min(vds, vdsat)$$

The temperature dependence of the mobility terms assume the ordinary exponential form:

$$UH(t) = UH(tnom) \cdot \left(\frac{t}{tnom} \right)^{TUH}$$

$$zUO(t) = zUO(tnom) \cdot \left(\frac{t}{tnom} \right)^{TUH}$$

The following equation calculates the continuity term at the body effect transition point:

$$Icrit = -\frac{2}{3} \cdot cav \cdot \left[(vde + vsbc + Phid)^{\frac{3}{2}} - (vsbc + Phid)^{\frac{3}{2}} \right] \cdot \gamma \cdot \left(\frac{1}{zBetaGam} - 1 \right)$$

This model uses the preceding equation if $vsb > vsbc$. Otherwise,

$$Icrit = 0$$

The following sections describe saturation voltage, threshold voltage, body effect transition voltage, and the γ body effect coefficient.

Cypress Depletion Model — Threshold Voltage, v_{th}

The v_{TO} model parameter, often called the “pinch-off,” is a zero-bias threshold voltage extrapolated from a large device operating in the depletion mode. The following equation calculates the effective pinch-off threshold voltage, including the device size effects and the terminal voltages:

$$v_{th} = v_{fb} - \beta_d \left[v_{ch} - \gamma \left(Phid + v_{sb} \right)^{\frac{1}{2}} + v_{crit} \right]$$

The following equations calculate values used in the preceding equation:

$$v_{fb} = zV_{TO} - zETA \beta_d v_{ds} + \beta_d \cdot \left(v_{ch} - \gamma_0 \left(Phid \right)^{\frac{1}{2}} \right)$$

$$v_{crit} = \left(\gamma - \frac{\gamma}{zBetaGam} \right) \left(Phid + v_{sbc} \right)^{\frac{1}{2}}$$

for $v_{sb} > v_{sbc}$; 0 otherwise.

$$\beta_d = \frac{UH \cdot c_{av}}{zUO \cdot c_{ox}}, \quad v_{ch} = \frac{q \cdot NI}{c_{av}}$$

$$\gamma_0 = \frac{(2 \cdot \epsilon_{si} \cdot q \cdot na1)^{\frac{1}{2}}}{c_{av}}$$

$$na1 = \frac{nd \cdot DNB}{nd + DNB}, \quad nd = \frac{NI}{DP \cdot 1e-4}$$

The following equation computes the effective γ , including small device size effects:

$$\gamma = \frac{\gamma}{zBetaGam} \text{ for } v_{sb} > v_{sbc}, \text{ and } = \gamma; \text{ otherwise,}$$

$$\gamma = \gamma_0 \cdot (1 - scf) \cdot (1 + ncf)$$

The following equations calculate values used in the preceding equations:

If $SCM \leq 0$, then $scf = 0$; otherwise,

$$scf = \frac{XJ}{L_{eff}} \cdot \left\{ \left[1 + \frac{2xd}{XJ} \cdot (SCM \cdot v_{ds} + v_{sb} + Phid)^{\frac{1}{2}} \right]^{\frac{1}{2}} - 1 \right\}$$

If $NWM \leq 0$, then $ncf = 0$; otherwise,

$$ncf = \frac{NWM \cdot xd \cdot (Phid)^{\frac{1}{2}}}{W_{eff}}$$

This equation calculates the x_d value used in the preceding equation:

$$x_d = \left(\frac{2 \cdot \epsilon_{si}}{q \cdot DNB} \right)^{\frac{1}{2}}$$

The following equation calculates the body effect transition point:

$$V_{sbc} = \frac{qDP^2}{2\epsilon_{si}} \left(\frac{NI}{DP \cdot 1e-4} \cdot DNB \right) + zDV_{SBC} + TDV_{SBC} \ln(t - t_{nom}) \cdot Phid$$

If $v_{gs} \leq v_{th}$, this model inverts the surface and includes a residual DC current. If v_{sb} is large enough to make $v_{th} > v_{inh}$, then v_{th} is the inversion threshold voltage.

To determine the residual current, simulation inserts v_{inh} into the i_{ds} , v_{sat} , and mobility equation in place of v_{gs} (except for v_{gs} in the exponential term of the subthreshold current). The following equation computes the inversion threshold voltage (v_{inh}) at a specified v_{sb} :

$$v_{inh} = v_{fb} - \frac{q \cdot NI}{C_{OX}} \cdot v_{sb} + DV_{IN} - zETA \ln v_{ds}$$

Cypress Depletion Model — Saturation Voltage, v_{dsat}

This equation determines the v_{sat} saturation voltage:

$$v_{sat} = v_{gs} - v_{fb} + v_{ch} + \frac{v^2}{2} \cdot \left\{ 1 - \left[1 + \frac{4}{v^2} \cdot (v_{gs} - v_{fb} + v_{ch} + v_{sb} + Phid) \right]^{\frac{1}{2}} \right\}$$

$$v_{dsat} = v_{sat}$$

Simulation modifies v_{sat} to include the carrier velocity saturation:

$$v_{dsat} = v_{sat} + v_c \cdot (v_{sat}^2 + v_c^2)^{\frac{1}{2}}$$

The following equation calculates the v_c value used in the preceding equation:

$$v_c = ECV \cdot Le_{ff}$$

Cypress Depletion Model — Mobility Reduction, U_{Beff}

The U_B surface mobility depends on the terminal voltages:

$$U_{Beff} = \frac{1}{\frac{1}{zU_O} + \frac{(zFRC + zVFRC \cdot v_{de} + zBFRC \cdot v_{sb})(v_{gs} - v_{fb})}{TOX} + \frac{v_{de}}{V_{ST} \cdot Le} + zFSB \cdot v_{sb}^{\frac{1}{2}}}$$

The following equations calculate Le for the preceding equation:

$$Le = Le_{ff}$$

Linear region

$$Le = Leff - \Delta L$$

Saturation region

At elevated temperatures, the following equation calculates the zFRC value used in the preceding equation:

$$zFRC(t) = zFRC(tnom) \cdot \left(\frac{t}{tnom} \right)^{FRCEX}$$

ΔL is the channel length modulation effect, defined in the next section. v_{fb} assumes the role of v_{th} in the LEVEL 5 mobility equation. The degradation parameters are semi-empirical, and are grouped according to their (linearized) mathematical dependencies instead of their physical origin to better provide parameter extraction.[\[7\]](#)

Cypress Depletion Model — Channel Length Modulation

To include the channel length modulation, modify the ids current:

$$ids = \frac{ids}{1 - \frac{\Delta L}{Leff}}$$

The following equation calculates ΔL for the preceding equation:

$$\Delta L = 1e4 \cdot \left[\frac{2.73e3 \cdot XJ}{na1 \cdot \ln\left(\frac{1e20}{na1}\right)} \right]^{\frac{1}{3}} \cdot \left[(vds - vdsat + PHI)^{\frac{1}{3}} - PHI^{\frac{1}{3}} \right]$$

ΔL is in microns, if XJ is in microns and $na1$ is in cm^{-3} .

Cypress Depletion Model — Subthreshold Current, ids

If device leakage currents become important for operation near or below the normal threshold voltage, the model considers the subthreshold characteristics. In the presence of surface states, this equation determines the effective threshold voltage (von):

$$von = \max(vth, vinth) + fast$$

The following equation calculates the $fast$ value used in the preceding equation:

$$fast = vt \cdot \left[1 + \frac{q \cdot FSS}{COX} + \frac{\gamma}{2 \cdot (Phid + vsb)^{\frac{1}{2}}} \right]$$

If $vgs < von$, then:

Partial Enhancement, $0 < vgs - vfb < vde$

$$ids = \beta_1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde + cav \cdot \left[(von-vfb) \cdot vde - \frac{vde^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \bar{\gamma} \cdot \left[(vde + vsb + Phid)^{\frac{3}{2}} - (vsb + Phid)^{\frac{3}{2}} \right] + Icrit \right\} + \frac{1}{2} \cdot \left(\beta \cdot e^{\frac{vgs-von}{fast}} - \beta_1 pcav \right) \cdot (von-vfb)^2$$

Full Enhancement, $vgs-vfb - vde > 0$

$$ids = \beta_1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde - \frac{2}{3} pcav \bar{\gamma} \bar{\gamma} \left[(vde + vsb + Phid)^{\frac{3}{2}} - (vsb + Phid)^{\frac{3}{2}} \right] + Icrit \right\} \beta \cdot \left[(von-vfb) \cdot vde - \frac{vde^2}{2} \right] \cdot e^{\frac{vgs-von}{fast}}$$

Depletion, $vgs-vfb < 0$

$$ids = \beta_1 \cdot \left\{ q \cdot zKI0 \cdot NI \cdot vde + cav \cdot \left[(vgs-vfb) \cdot vde - \frac{vde^2}{2} \right] - \frac{2}{3} \cdot cav \cdot \bar{\gamma} \cdot \left[(vde + vsb + Phid)^{\frac{3}{2}} - (vsb + Phid)^{\frac{3}{2}} \right] + Icrit \right\} \cdot e^{\frac{vgs-von}{fast}}$$

Cypress Depletion Model — Sample Model File

```
$ file Depstor.mod
.MODEL DEPSTOR NMOS LEVEL=38
* PARASITIC ELEMENTS
+ ACM=1
+ LD=0.15u WD=0.2u $ for LEFF AND WEFF
+ CJ=0.3E-16 MJ=0.4 PB=0.8 JS=2.0E-17 $ INTRINSIC DIODE
+ CJSW=0 MJSW=0.3
+ BULK=98 $ DEFAULT NODE FOR SUBSTRATE
* THRESHOLD
+ VTO=-2.5 LVT=-0.25 WVT=0
+ leta=0.02 eta=0.0 weta=0.0
+ TCV=0.003 $ TEMPERATURE COEFFICIENT
* MISC
+ DVIN=0.5 PHI=0.75
+ NFS=2e10 DNB=3.0E16
```

Mobility Model

```
+ UH=1300
+ UO=495 FRC=0.020 FSB=5e-5 VFRC=-1e-4 BFRC=-0
+ LUO=-100 LFRFC=.03 LFSB=-1e-5 LVFRC=-.002 LBFRFC=-1e-3
+ WUO=-30 WFRFC=-0.01 WFSB=5e-5 WVFRC=-0.00 + WBFRFC=-0.4e-3
+ KIO=.9 KBETA1=.5 LKIO=0.16 LKBETA1=-0.15
```



```
+ WKI0=0.0    WKBETA1=-0.0
+ BEX=-1.3    TUH=-1.0    Frcex=1.0
```

Body Effect

```
+ NWM=0.5    SCM=.1
+ DVSBBC=0.1    LDVSBBC=0    WDVSBBC=0
+ TDVSBBC=.002
+ BetaGam=0.9    LBetaGam=-.2    WBetaGam=.1
```

Saturation

```
+ ECV= 2.9    VST=8000    UHSAT=0
* CHANNEL LENGTH MODULATION
+ XJ=0.1
* OXIDE THICKNESS AND CAPACITANCE
+ TOX=165    CGSO=0    CAPOP=2
* CHANNEL IMPLANT
+ NI=1.5e12    KCS=3    DP=0.25
*.END
```

LEVEL 39: BSIM2 Model

BSIM2 (Berkeley Short-Channel IGFET Model 2) is the LEVEL 39 MOSFET model. The Synopsys implementation of this model is based on Berkeley SPICE 3E2.

To provide input to the Level 39 device model, assign the model parameters as for other device models. You can use a tabular model entry without model parameter names in BSIM1, but *not* in BSIM2.

This section covers the following topics:

- [BSIM2 Model Parameters](#)
- [BSIM2 Model Equations](#)
- [BSIM2 Model — Geometry and Bias of Model Parameters](#)
- [BSIM2 Model — Compatibility Notes](#)
- [BSIM2 Model — Preventing Negative Output Conductance](#)
- [BSIM2 Model — Charge-based Gate Capacitance Model \(CAPOP=39\)](#)
- [BSIM2 Model — Synopsys Device Model Enhancements](#)
- [BSIM2 Model — Modeling Example](#)
- [BSIM2 Model — Typical Listing](#)

BSIM2 Model Parameters

This model uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section. In addition, this model also uses the following parameters:

Note:

The Synopsys model does not use `TEMP`, `DELL`, and `DFW` parameters. The width and length sensitivity parameters are associated with the remaining parameters, except for `TOX`, `VDD`, `VGG`, `VBB`, `DL`, and `DW`. Unlike Berkeley SPICE, the Synopsys MOSFET model has `L` and `w` sensitivity for `MU0`. This does not include the *generic* MOS parameters or the `wL`-product sensitivity parameters, which are Synopsys enhancements.

Name (Alias)	Units	Default	Description
AI0	-	0	Impact ionization coefficient
AIB	V-1	0	Sensitivity of the impact ionization coefficient to V_{bs}

Name (Alias)	Units	Default	Description
BI0	V	0	Impact ionization exponent
BIB	-	0	Sensitivity of the impact ionization exponent to V_{bs}
DELL	m	-	Length reduction of the source drain diffusion (not used in the Level 39 MOSFET model)
DL	m	0	Channel length reduction
DW	m	0	Channel width reduction
ETA0	-	0	Drain-induced barrier lowering coefficient
ETAB	V-1	0	Sensitivity of the drain-induced barrier lowering coefficient to V_{bs}
K1	V-1	0.5	Body effect coefficient
K2	-	0	Second-order body effect coefficient (for nonuniform channel doping)
MU0	$\text{cm}^2/\text{V}\cdot\text{s}$	400	Low-field mobility
MU0B	$\text{cm}^2/\text{V}^2\cdot\text{s}$	0	Sensitivity of low-field mobility to V_{bs}
MU20	-	0	Empirical parameter for the output resistance
MU2B	V-1	0	Sensitivity of the empirical parameter to V_{bs}
MU2G	V-1	0	Sensitivity of the empirical parameter to V_{gs}
MU30	$\text{cm}^2/\text{V}^2\cdot\text{s}$	0	Empirical parameter for the output resistance
MU3B	$\text{cm}^2/\text{V}^3\cdot\text{s}$	0	Sensitivity of the empirical parameter to V_{bs}
MU3G	$\text{cm}^2/\text{V}^3\cdot\text{s}$	0	Sensitivity of the empirical parameter to V_{gs}
MU40	$\text{cm}^2/\text{V}^3\cdot\text{s}$	0	Empirical parameter for the output resistance
MU4B	$\text{cm}^2/\text{V}^4\cdot\text{s}$	0	Sensitivity of the empirical parameter to V_{bs}
MU4G	$\text{cm}^2/\text{V}^4\cdot\text{s}$	0	Sensitivity of the empirical parameter to V_{gs}
MUS0	$\text{cm}^2/\text{V}\cdot\text{s}$	600	High-drain field mobility
MUSB	$\text{cm}^2/\text{V}^2\cdot\text{s}$	0	Sensitivity of the high-drain field mobility to V_{bs}
N0	-	0.5	Subthreshold swing coefficient
NB	V1/2	0	Sensitivity of the subthreshold swing to V_{bs}

Name (Alias)	Units	Default	Description
ND	V-1	0	Sensitivity of the subthreshold swing to V_{ds}
PHI	V	0.8	Surface potential
TEMP	C	-	Not used in Level 39 (see BSIM2 Model — Compatibility Notes on page 213)
TOX	m	0.02	Gate oxide thickness (assumes that TOX>1 is in Angstroms)
U10	V-1	0	High-drain field (velocity saturation) mobility reduction factor
U1B	V-2	0	Sensitivity of the mobility reduction factor to V_{bs}
U1D	V-2	0	Sensitivity of the mobility reduction factor to V_{ds}
UA0	V-1	0	First-order vertical-field mobility reduction factor
UAB	V-2	0	Sensitivity of the first-order factor to V_{bs}
UB0	V-2	0	Second-order vertical-field mobility reduction factor
UBB	V-3	0	Sensitivity of the second-order factor to V_{bs}
VBB	V	-5	Body supply voltage (NMOS convention)
VDD	V	5	Drain supply voltage (NMOS convention)
VFB	V	-0.3	Flat band voltage
VGG	V	5	Gate supply voltage (NMOS convention)
VGHIGH	V	0	Upper bound of the weak-strong inversion transition region
VGLOW	V	0	Lower bound of the weak-strong inversion transition region
VOF0	-	0	Threshold offset (normalized to NKT/q) for the subthreshold
VOFB	V-1	0	Sensitivity of the offset to V_{bs}
VOFD	V-1	0	Sensitivity of the offset to V_{ds}
WDF	m	-	Default width (not used in the Level 39 MOSFET model); use <code>.OPTION DEFW=#</code> in the netlist instead

Specify all BSIM2 parameters according to the NMOS convention, even for a PMOS model. Examples: $V_{DD}=5$, not -5 ; $V_{BB}=-5$, not 5 ; and $\text{ETA0}=0.02$, not -0.02 . See [BSIM2 Model — Compatibility Notes](#).

This MOSFET model also includes the $J_{SW}[\text{A/m}]$ source/drain bulk diode sidewall reverse saturation current density.

Other Device Model Parameters that Affect BSIM2

You must specify the following MOSFET model parameters before you can use some Synopsys enhancements, such as

- LDD-compatible parasitics.
- Adjusts the model parameter geometry, relative to a reference device.
- Impact ionization modeling with bulk-source current partitioning.
- Element temperature adjustment of the key model parameters.

This is a partial list. For complete information, see the following:

- [Calculating Effective Length and Width for AC Gate Capacitance](#)
- [MOSFET Diode Model — Drain and Source Resistance Parameters](#)
- [MOSFET Impact Ionization](#)

BSIM2 Model — Compatibility Notes: [.MODEL VERSION Changes](#) describes how the `VERSION` parameter in the `.MODEL` statement changes the BSIM2 model, depending on the model version number.

BSIM2 Model Equations

In the following expressions, model parameters are in all upper case Roman. These expressions assume that you have already adjusted all model parameters for geometry, and that you have already adjusted parameters without a trailing 0 for the bias as appropriate. The exceptions are U_1 and N for which the following equations explicitly calculate the bias dependences.

This section covers the following topics:

- [BSIM2 Model — Threshold voltage, \$V_{th}\$](#)
- [BSIM2 Model — Strong inversion \(\$V_{gs} > V_{th} + V_{GHIGH}\$ \)](#)
- [BSIM2 Model — Weak Inversion \(\$V_{gs} < V_{th} + V_{GLOW}\$; \[\$V_{GLOW} < 0\$ \]\)](#)
- [BSIM2 Model — Strong inversion-to-weak inversion transition region \(\$V_{th} + V_{GLOW} \leq V_{gs} \leq V_{th} + V_{GHIGH}\$ \)](#)

- [BSIM2 Model — Effective Length and Width: If DL is nonzero](#)
- [BSIM2 Model — Effective Length and Width: If DW is nonzero](#)

BSIM2 Model — Threshold voltage, V_{th}

$$V_{th} = V_{bi} + K1\sqrt{PHI - V_{bs}} - K2(PHI - V_{bs}) - ETA \cdot V_{ds}$$

The following equation calculates the v_{bi} value used in the preceding equation:

$$V_{bi} = VFB + PHI$$

BSIM2 Model — Strong inversion ($V_{gs} > V_{th} + V_{GHIGH}$)

Linear region ($V_{ds} < V_{dsat}$) drain-source current I_{DS} :

$$I_{DS} = \frac{\beta' (V_{gs} - V_{th} - \frac{a}{2} V_{ds}) V_{ds}}{1 + UA(V_{gs} - V_{th}) + UB(V_{gs} - V_{th})^2 + U1 \cdot V_{ds}}$$

The following equations calculate values used in the preceding equation:

$$V_{dsat} = \frac{V_{gs} - V_{th}}{a\sqrt{K}},$$

$$K = \frac{1 + V_c + \sqrt{1 + 2V_c}}{2},$$

$$V_c = \frac{U_{1S}(V_{gs} - V_{th})}{a[1 + UA(V_{gs} - V_{th}) + UB(V_{gs} - V_{th})^2]},$$

$$U_{1S} = U10 + U1B \cdot V_{bs},$$

$$U1 = U_{1S} \left[1 - \Theta(V_{dsat} - V_{ds}) \frac{U1D(V_{ds} - V_{dsat})^2}{V_{dsat}^2} \right]$$

In the preceding equations, $\Theta(x)$ is the usual unit step function:

$$\beta' = \beta_0 + \beta_1 \tanh \left(MU2 \frac{V_{ds}}{V_{dsat}} \right) + \beta_3 V_{ds} - \beta_4 V_{ds}^2$$

$$\beta_0 = \frac{W_{eff}}{L_{eff}} MU \cdot C_{ox},$$

$$\beta_1 = \beta_s - (\beta_0 + \beta_3 VDD - \beta_4 VDD^2),$$

$$\beta_i = \frac{W_{eff}}{L_{eff}} MUi \cdot C_{ox}, i = S, 3, 4, a = 1 + \frac{gK1}{2\sqrt{PHI-V_{bs}}},$$

$$g = 1 - \frac{1}{1.744 + 0.8364(PHI-V_{bs})}$$

Saturation ($V_{ds} > V_{dsat}$) drain-source current, I_{DS} :

$$I_{DS} = \frac{\beta' (V_{gs} - V_{th})^2}{2aK [1 + UA(V_{gs} - V_{th}) + UB(V_{gs} - V_{th})^2]} \cdot (1 + f)$$

In the preceding equation, the f impact ionization term is:

$$f = AI \cdot e^{\frac{-BI}{V_{ds} - V_{dsat}}}$$

BSIM2 Model — Weak Inversion ($V_{gs} < V_{th} + VGLOW$; $[VGLOW < 0]$)

Subthreshold drain-source current, I_{ds} :

$$I_{DS} = \beta' \cdot V_{tm}^2 \cdot \exp\left(\frac{V_{gs} - V_{th}}{N \cdot V_{tm}} + VOFF\right) \cdot \left[1 - \exp\left(-\frac{V_{ds}}{V_{tm}}\right)\right] \cdot (1 + f)$$

The following equations calculate the V_{tm} and N values used in the preceding equation:

$$V_{tm} = \frac{kT}{q} \text{ and } N = N0 + \frac{NB}{\sqrt{PHI - V_{bs}}} + ND \cdot V_{ds}$$

BSIM2 Model — Strong inversion-to-weak inversion transition region ($V_{th} + VGLOW \leq V_{gs} \leq V_{th} + VGHIGH$)

$$V_{geff}(V_{gst}) = \sum_{j=0}^3 C_j V_{gst}^j$$

The preceding equation replaces $V_{gst} = V_{gs} - V_{th}$ in the linear or saturation drain currents, based on V_{dsat} (V_{geff}).

At the lower boundary ($V_{gs} - V_{th} = VGLOW$), the saturation equation is valid for all V_{ds} (that is, $V_{dsat}(V_{geff}(VGLOW)) \geq 0$) to allow a match to the above subthreshold equation.

To internally determine the C_j coefficients of the V_{geff} cubic spline, the I_{DS} and dI_{ds}/dV_{gs} conditions must both be continuous at the $V_{gs} = V_{th} + VGLOW$ and $V_{gs} = V_{th} + VGHIGH$ boundaries.

BSIM2 Model — Effective Length and Width: If DL is nonzero

$$L_{eff} = L_{scaled} \cdot LMLT - DL$$

$$LREF_{eff} = LREF_{scaled} \cdot LMLT - DL$$

Otherwise:

$$L_{eff} = L_{scaled} \cdot LMLT + XL_{scaled}^{-2} \cdot LD_{scaled}$$

$$LREF_{eff} = LREF_{scaled} \cdot LMLT + XLREF_{scaled}^{-2} \cdot LD_{scaled}$$

BSIM2 Model — Effective Length and Width: If DW is nonzero

$$W_{eff} = (W_{scaled} \cdot WMLT - DW) \cdot M$$

$$WREF_{eff} = (WREF_{scaled} \cdot WMLT - DW) \cdot M$$

Otherwise:

$$W_{eff} = (W_{scaled} \cdot WMLT + XW^{-2} \cdot WD_{scaled}) \cdot M$$

$$WREF_{eff} = (WREF_{scaled} \cdot WMLT + XWREF_{scaled}^{-2} \cdot WD_{scaled}) \cdot M$$

BSIM2 Model — Geometry and Bias of Model Parameters

Most of the BSIM2 parameters include width and length sensitivity parameters. You can also specify Synopsys-proprietary WL -product sensitivity parameters. If P is a parameter, then its associated width, length, and WL -product sensitivity parameters are WP , LP , and PP .

The value of the P' parameter, adjusted for width, length, and WL -product, is:

$$P' = P + WP \cdot \left(\frac{1}{W_{eff}} - \frac{1}{WREF_{eff}} \right) + LP \cdot \left(\frac{1}{L_{eff}} - \frac{1}{LREF_{eff}} \right) + PP \cdot \left(\frac{1}{W_{eff}} - \frac{1}{WREF_{eff}} \right) \cdot \left(\frac{1}{L_{eff}} - \frac{1}{LREF_{eff}} \right)$$

Berkeley SPICE does not use the $WREF$ and $LREF$ terms. They are effectively infinite, which is the default in the Level 39 MOSFET model.

The following BSIM2 parameters do not have associated geometry-sensitivity parameters:

TOX

TEMP (not used)

VDD

VGG

VBB

DL

DW

BSIM2 parameters ending in 0 are valid at zero bias, and they have associated bias sensitivities, listed in the BSIM2 parameter table.

If P_B , P_D , and P_G are the geometry-adjusted v_{bs} -, v_{ds} -, and v_{gs} - sensitivity parameters, and if they are associated with the P_0 geometry-adjusted zero-bias parameter, then the following equation calculates the P bias-dependent parameter:

$$P = P_0 + P_B \cdot V_{bs} + P_D \cdot V_{ds} + P_G \cdot V_{gs}$$

The exceptions are the U_1 velocity saturation factor and the N subthreshold swing coefficient. [BSIM2 Model — Synopsys Device Model: PrimeSim Continuum Optimizer](#) shows expressions for their bias dependences.

BSIM2 Model — Compatibility Notes

This section covers the following topics:

- [BSIM2 Model — Compatibility Notes: SPICE3 Flag](#)
- [BSIM2 Model — Compatibility Notes: Temperature](#)
- [BSIM2 Model — Compatibility Notes: Parasitics](#)
- [BSIM2 Model — Compatibility Notes: Selecting Gate Capacitance](#)
- [BSIM2 Model — Compatibility Notes: Unused Parameters](#)
- [BSIM2 Model — Compatibility Notes: .MODEL VERSION Changes](#)

BSIM2 Model — Compatibility Notes: SPICE3 Flag

If you specify the `SPICE3=0` (default) model parameter, certain Synopsys corrections to the BSIM2 equations are effective. If you set the `SPICE3` value to 1, the equations are as faithful as possible to the BSIM2 equations for SPICE3E2. Even in this mode, certain numerical problems have been addressed and should not normally be noticeable.

BSIM2 Model — Compatibility Notes: Temperature

The default model reference temperature (T_{NOM}) is 25°C in the Level 39 MOSFET model, unless you set `.OPTION SPICE`, which sets the T_{NOM} default to 27° C. This option also sets some other SPICE compatibility parameters. In the Level 39 model, you set T_{NOM} in an `.OPTION` line in the netlist; to override this locally (that is, for a model), use the `TREF` model parameter. (“Reference temperature” means that the model parameters were extracted at, and are therefore valid at, that temperature.)

UCB SPICE 3 does not use `TNOM` (default 27° C) for the BSIM models. Instead, you must specify the `TEMP` model parameter as both the model reference temperature and the analysis temperature. Analysis at `TEMP` applies only to thermally-activated exponentials in the model equations. You cannot adjust the model parameter values if you use `TEMP`. Simulation assumes that you extracted the model parameters at `TEMP`, because `TEMP` is both the reference and analysis temperature.

For model levels *other than* 4 (BSIM1) and 5 (BSIM2) in UCB SPICE3, simulation adjusts the key model parameters for the difference between `TEMP` (default 27°C) and `TNOM`. To specify `TEMP` in the netlist, use `.TEMP #` as in the Level 39 MOSFET model.

In contrast to UCB SPICE's BSIM models, the Synopsys Level 39 MOSFET model does provide for temperature analysis. The default analysis temperature is 25°C in the Level 39 model. Set `.TEMP #` in your netlist to change the analysis temperature (you cannot use `TEMP` as a model parameter). The Level 39 MOSFET model adjusts the temperature of the key model parameters as explained in [BSIM Model — Temperature Effect](#).

BSIM2 Model — Compatibility Notes: Parasitics

`ACM > 0` invokes the MOS source-drain parasitics in the Level 39 MOSFET device model. `ACM=0` (default) is SPICE style. See [BSIM2 Model — Synopsys Device Model Enhancements](#).

BSIM2 Model — Compatibility Notes: Selecting Gate Capacitance

`CAPOP=39` selects the BSIM2 charge-conserving capacitance model as shipped with Berkeley SPICE 3E2. This is the default selection if you set `SPICE3=1`.

- `XPART` (charge-sharing flag) is currently not a BSIM2 model parameter, despite its specification in the sample BSIM2 input decks shipped with Berkeley SPICE 3E. It appears that its use in SPICE 3E was as a printback debug aid.
- Saturation charge sharing appears to be fixed at 60/40 (S/D) in the BSIM2 capacitance model. For the charge equations, see [BSIM2 Model — Charge-based Gate Capacitance Model \(CAPOP=39\)](#). See also [BSIM2 Model — Synopsys Device Model: PrimeSim Continuum Optimizer](#).

You can choose other `CAPOP` values. `CAPOP=13` (recommended) selects the BSIM1-based charge-conserving capacitance model for the MOSFET LEVEL 13 (BSIM1) or LEVEL 28 (modified BSIM1) device models. This option is the default selection if `SPICE3=0`. If you use this capacitance model, you can use the `XPART` or `XQC` model parameters to adjust charge sharing. See [LEVEL 13: BSIM Model](#) for more information.

BSIM2 Model — Compatibility Notes: Unused Parameters

The Level 39 MOSFET model does not use the `DELL` (S/D diode length reduction) and `WDF` (default device width) SPICE model parameters. SPICE 3E does not use the `DELL`

function. You can specify a default width in the Level 39 MOSFET model, on the `.OPTION` line as `DEFW` (which defaults to 100 μ).

BSIM2 Model — Compatibility Notes: .MODEL VERSION Changes

The Level 39 MOSFET model provides a `VERSION` parameter to the `.MODEL` statement, which lets you move LEVEL 13 BSIM and LEVEL 39 BSIM2 models between device model versions. Use the `VERSION` parameter in a LEVEL 13 `.MODEL` statement. The following table lists the changes in the BSIM model:

Model Version	Effect of VERSION on BSIM2 Model
92A	LEVEL 39 BSIM2 model introduced: no changes
92B	No changes
93A	Introduced <code>gds</code> constraints, fixed a defect in the <code>WMU3B</code> parameter, and introduced a defect in the <code>MU4</code> parameter
93A.02	Introduced the <code>VERSION</code> parameter, and fixed an <code>MU4</code> parameter defect
95.1	Fixed defects that caused <code>PMUSB</code> , <code>LDAC</code> , and <code>WDAC</code> parameter problems, fixed the <code>GMBS</code> defect if you used <code>gds</code> constraints
96.1	Limited $ETA + ETAB \cdot vb5 \geq 0$

BSIM2 Model — Preventing Negative Output Conductance

The Level 39 MOSFET model internally protects against conditions in the LEVEL 13 model that cause convergence problems due to negative output conductance. This model imposes the following constraints:

$$MU2 \geq 0 \quad ND \geq 0 \quad AI \geq 0$$

Simulation imposes these constraints after adjusting the length and width and setting the `VBS` dependence. This feature loses some accuracy in the saturation region, particularly at high V_{gs} .

Consequently, you might need to requalify the BSIM2 models in the following situations:

- Devices exhibit self-heating during characterization, which causes declining I_{ds} at high V_{ds} . This does not occur if the device characterization measurement sweeps V_{ds} .
- The extraction technique produces parameters that result in negative conductance.
- This model simulates the voltage outside the device's characterized range.

BSIM2 Model — Charge-based Gate Capacitance Model (CAPOP=39)

The BSIM2 gate capacitance model conserves charge and has non-reciprocal attributes. Using charges as state variables guarantees charge conservation. Charge partitioning is fixed at 60/40 (S/D) in saturation and is 50/50 in the linear region. $Q_s = -(Q_g + Q_d + Q_b)$ in all regions.

This section covers the following topics:

- [BSIM2 Model — CAPOP=39: Accumulation region \(\$V_{gs} < V_{bs} + V_{FB}\$ \)](#)
- [BSIM2 Model — CAPOP=39: Subthreshold region \(\$V_{bs} + V_{FB} < V_{gs} < V_{th} + V_{GLOW}\$ \)](#)
- [BSIM2 Model — CAPOP=39: Saturation region \(\$V_{ds} > V_{dsat}\$ \)](#)

BSIM2 Model — CAPOP=39: Accumulation region ($V_{gs} < V_{bs} + V_{FB}$)

$$Q_g = C_{ox} W_{eff} \cdot L_{eff} (V_{gs} - V_{bs} - V_{FB})$$

$$Q_b = -Q_g$$

$$Q_d = 0$$

BSIM2 Model — CAPOP=39: Subthreshold region ($V_{bs} + V_{FB} < V_{gs} < V_{th} + V_{GLOW}$)

$$Q_g = C_{ox} W_{eff} \cdot L_{eff} (V_{gs} - V_{bs} - V_{FB})$$

$$\left[1 - \frac{V_{gs} - V_{bs} - V_{FB}}{V_{gs} - V_{bs} - V_{FB} - V_{gst}} + \frac{1}{3} \left\{ \frac{V_{gs} - V_{bs} - V_{FB}}{V_{gs} - V_{bs} - V_{FB} - V_{gst}} \right\}^2 \right]$$

$$Q_b = -Q_g$$

$$Q_d = 0$$

BSIM2 Model — CAPOP=39: Saturation region ($V_{ds} > V_{dsat}$)

$$Q_g = \frac{2}{3} C_{ox} W_{eff} \cdot L_{eff} \cdot V_{gst} + Q_{bulk}$$

The following equations calculate values used in the preceding equation:

$$Q_{bulk} = \frac{1}{3} C_{ox} W_{eff} \cdot L_{eff} [V_{th} - V_{bs} - V_{FB}]$$

$$Q_b = -Q_{bulk}$$

$$Q_d = -\frac{4}{10} \cdot \frac{2}{3} C_{ox} W_{eff} \cdot L_{eff} \cdot V_{gst} = \left(-\frac{4}{15}\right) C_{ox} W_{eff} \cdot L_{eff} \cdot V_{gst}$$

Linear region ($V_{ds} < V_{dsat}$):

$$Q_g = \frac{2}{3} C_{ox} W_{eff} \cdot L_{eff} \cdot V_{gst} \cdot \left[\frac{3 \left(1 - \frac{V_{ds}}{V_{dsat}}\right) + \left(\frac{V_{ds}}{V_{dsat}}\right)^2}{2 - \frac{V_{ds}}{V_{dsat}}} \right] + Q_{bulk}$$

$$Q_b = -Q_{bulk}, Q_d = -\frac{1}{3} C_{ox} W_{eff} \cdot L_{eff} \cdot V_{gst}$$

$$\left[\frac{3 \left(1 - \frac{V_{ds}}{V_{dsat}}\right) + \left(\frac{V_{ds}}{V_{dsat}}\right)^2}{2 - \frac{V_{ds}}{V_{dsat}}} + \frac{\frac{V_{ds}}{V_{dsat}} \left(1 - \frac{V_{ds}}{V_{dsat}}\right) + 0.2 \left(\frac{V_{ds}}{V_{dsat}}\right)^2}{\left(2 - \frac{V_{ds}}{V_{dsat}}\right)^2} \right] + Q_{bulk}$$

BSIM2 Model — Synopsys Device Model Enhancements

In the following expressions, model parameters are in all upper case Roman. Simulation assumes that you have already adjusted all model parameters without a trailing 0 for both geometry and bias as appropriate.

This section covers the following topics:

- [BSIM2 Model — Synopsys Device Model: Alternate Gate Capacitance Model](#)
- [BSIM2 Model — Synopsys Device Model: Modeling Guidelines, Removing Mathematical Anomalies](#)

BSIM2 Model — Synopsys Device Model: Temperature Effects

LEVEL=39 enforces TLEV=1. You cannot currently use any other TLEV value. The following equation adjusts the threshold voltage for Level 39 TLEV=1:

$$V_{th}(T) = V_{bi}(T) + K1 \cdot \sqrt{\phi(T) - V_{bs}} - K2 \cdot (\phi(T) - V_{bs}) - ETA \cdot V_{ds}$$

The following equations calculate values used in the preceding equation:

$$V_{bi}(T) = V_{to}(T) - K1 \cdot \sqrt{\phi(T)} + K2 \cdot \phi(T)$$

$$V_{to}(T) = V_{to} - TCV \cdot (T - T_{nom})$$

In the preceding equations, the nominal-temperature, zero-bias threshold voltage is:

$$\begin{aligned} V_{to} &= V_{bi} + K1 \cdot \sqrt{PHI} - K2 \cdot PHI \\ &= VFB + PHI + K1 \cdot \sqrt{PHI} - K2 \cdot PHI, \end{aligned}$$

Simulation calculates $\phi(T)$ according to the specified `TLEV` value.

The following equation adjusts the mobility:

$$\mu(T) = \mu(T_{nom}) \cdot \left(\frac{T}{T_{nom}} \right)^{BEX} \text{ where } \mu = \frac{\beta'}{C_{ox}(W_{eff}/L_{eff})}.$$

UIS adjusts the velocity saturation:

$$UIS(T) = UIS \cdot \left(\frac{T}{T_{nom}} \right)^{FEX}$$

This model also includes all of the usual Synopsys model adjustments to capacitances, parasitics, diodes, and resistors.

BSIM2 Model — Synopsys Device Model: Alternate Gate Capacitance Model

Select `CAPOP=13` for the charge-conserving capacitance model, widely used with `LEVEL=13` (BSIM1) and `LEVEL=28` (improved BSIM1). See [LEVEL 13: BSIM Model](#) for more details.

BSIM2 Model — Synopsys Device Model: Impact Ionization

To select impact ionization modeling (instead of BSIM2), keep the `AI0=0` value, and specify the `ALPHA` [`ALPHA` · ($V_{ds} - V_{dsat}$) replaces `AI` in equation for f in the BSIM2 equations section], `VCR` (replaces `BI`), and `IIRAT` (multiplies f) model parameters.

Synopsys impact ionization modeling differs from BSIM2 modeling in two ways:

- A bias term ($V_{ds} - V_{dsat}$) multiplies the exponential and `ALPHA` values.
- You can use the `IIRAT` model parameter to partition the impact ionization component of the drain current, between the source and the bulk. `IIRAT` multiplies f in the saturation I_{ds} equation. Thus, the `IIRAT` fraction of the impact ionization current goes to the source, and the $1 - \text{IIRAT}$ fraction goes to the bulk, adding to IDB . `IIRAT` defaults to zero (that is, 100% of impact ionization current goes to the bulk).

BSIM2's impact ionization assumes that all of the impact ionization current is part of I_{ds} . In other words, it flows to the source. This assumption can lead to inaccuracies, for example, in cascode circuits.

BSIM2 Model — Synopsys Device Model: Parasitic Diode for Proper LDD Modeling

The Level 39 MOSFET model includes alternative MOS parasitic diodes to replace SPICE-style MOS parasitic diodes. You can use these alternatives to geometrically scale the parasitics with MOS device dimension, properly modeling the LDD parasitic

resistances, shared sources and drains, and select different diode sidewall capacitances along the gate edge and field edge.

To select the MOS parasitic diode, use the `ACM` model parameter.

- `ACM=0` (default) chooses SPICE style. The alternatives likely to be of most interest to the BSIM2 user are `ACM=2` and `3`.
- `ACM=2` calculates the diode area based on `W`, `XW`, and `HDIF` (contact to gate spacing). You can override the calculation from the element line. You can specify `LDIF` (spacer dimension); `RS` and `RD` (source and drain sheet resistance under the spacer) for LDD devices, and `RSH` (sheet resistance of the heavily-doped diffusion). Thus, simulation properly calculates the total parasitic resistance of the LDD devices.
- `ACM=3` uses all features of `ACM=2`. Its calculations of diode parasitics take into account the sharing of source/drains, and different junction sidewall capacitances along the gate and field edges. Use the `GEO` parameter to specify source/drain sharing from the element line. See [Using an ACM=3 MOS Diode](#) for details.

BSIM2 Model — Synopsys Device Model: Skewing of Model Parameters

As in any other Synopsys model, you can set up the BSIM2 model file for skewing to reflect the process variation. You can perform Worst-Case or Monte-Carlo analysis, based on fab statistics. For more information, see Variation Analysis or Monte Carlo Analysis—Flow and Outputs in the *PrimeSim HSPICE User Guide: Simulation and Analysis*.

BSIM2 Model — Synopsys Device Model: PrimeSim Continuum Optimizer

You can tie the BSIM2 model, like any other the PrimeSim Continuum solution model, into the optimizer in a Synopsys circuit simulator to fit to actual device data. An example fit appears at the end of this chapter.

BSIM2 Model — Synopsys Device Model: Modeling Guidelines, Removing Mathematical Anomalies

Because of the somewhat arbitrary geometric and bias adjustments made in the original BSIM2 parameters, they can take on non-physical values or values that are not mathematically allowed in Berkeley SPICE 3. This can lead to illegal function arguments, program crashes, and unexpected model behavior (for example, negative conductance). You must satisfy the following guidelines and corrections at all geometries of interest, and at biases up to double the supply voltages (that is, to $V_{ds}=2 \cdot V_{DD}$, $V_{gs}=2 \cdot V_{GG}$, and $V_{bs}=2 \cdot V_{BB}$).

To avoid a drain current discontinuity at $V_{ds}=V_{dsat}$, be sure that $BI \neq 0$ if $AI0 \neq 0$.

To prevent negative g_{ds} , be sure that $\text{ETA} > 0$, $\text{MU3} > 0$, and $\text{MU4} < \text{MU3} / (4 * \text{VDD})$. This should ensure a positive g_{ds} value at biases up to double the supply voltages. To simplify matters, set all MU4 parameters to zero. You can obtain reasonably good fits to submicron devices without using MU4 [10].

In the Level 39 MOSFET model, U1S cannot become negative. A negative U1S is physically meaningless, and causes negative arguments in a square root function in one of the BSIM2 equations. The U1D value should be less than unity (between 0 and 1).

For reasonable V_{th} behavior, make sure that:

$$K1 - 2K2 \cdot \sqrt{\text{PHI} - V_{bs}} \geq 0$$

For the equations to make sense, the following must hold: $N > 0$, $\text{VGLow} \leq 0$, and $\text{VGHIGH} \geq 0$.

The BSIM2 gate capacitance model in SPICE 3E tends to display negative C_{gs} in the subthreshold. This is due to $C_{gg} \rightarrow 0$ as $V_{gs} \rightarrow V_{th}$ by construction of the gate charge equation so that $C_{gs} = C_{gg} - C_{gd} - C_{gb} \rightarrow -C_{gd} - C_{gb} \approx -C_{gb}$. Therefore, use $\text{CAPOP}=13$ (default) until UC Berkeley releases an improved BSIM2 gate capacitance model.

BSIM2 Model — Modeling Example

The following is the result of fitting data from a submicron channel-length NMOS device to BSIM2. To fit this data, this example uses the Synopsys ATEM characterization software and the Synopsys simulation optimizer.

Figure 9 I_{DS} vs. V_{ds} for $V_{gs}=1, 2, 3, 4, 5V$; BSIM2 Model vs. Data

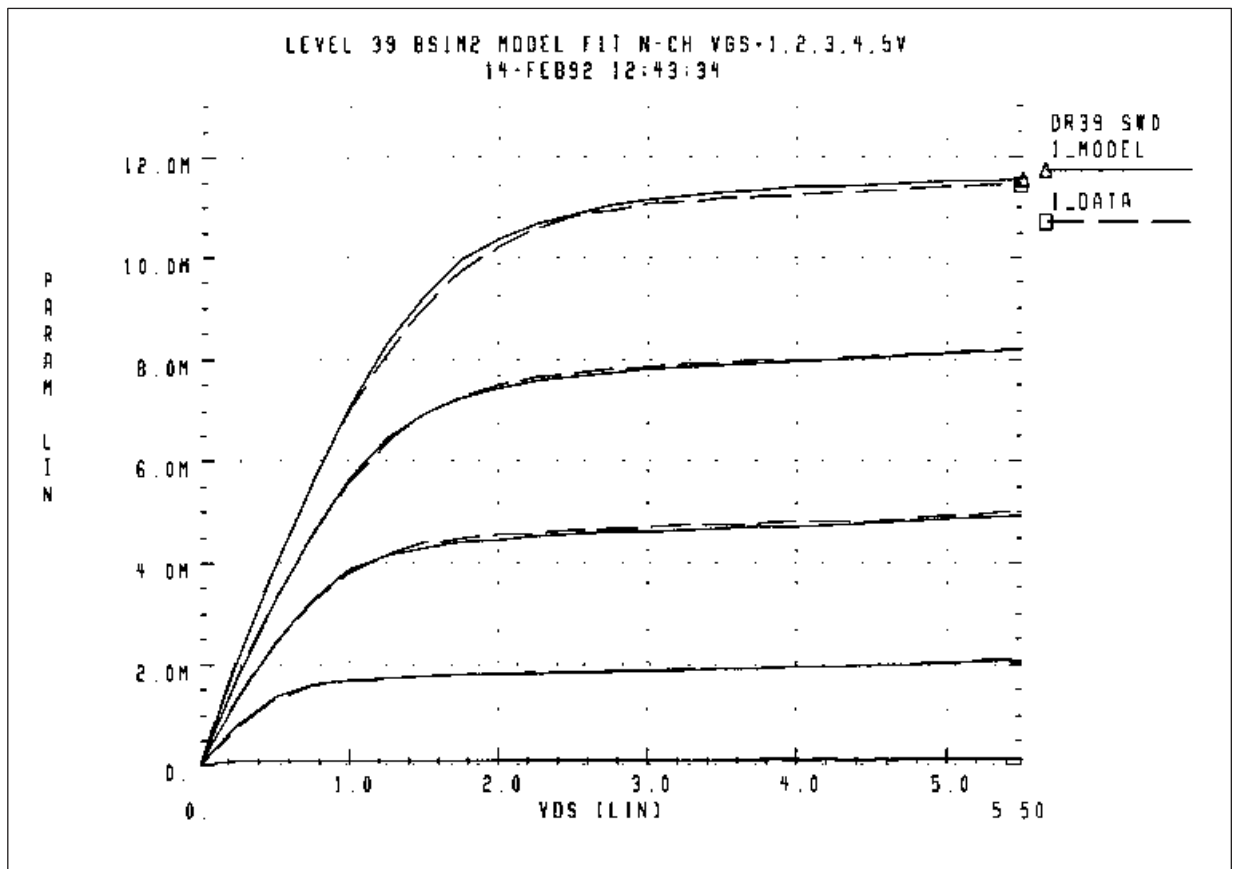


Figure 10 g_{ds} vs. V_{ds} for $V_{gs}=2, 3, 4, 5V$; BSIM2 Model vs. Data, LOG scale

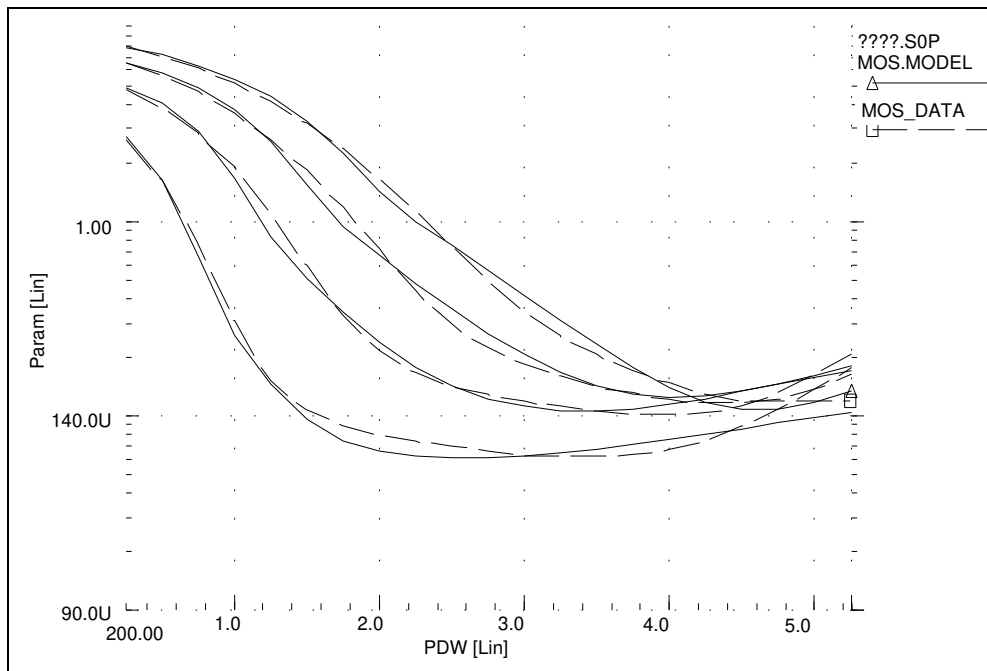


Figure 11 I_{DS} vs. V_{gs} for $V_{ds}=0.1V$, $V_{bs}=0, -1, -2, -3, -4V$, Showing Subthreshold Region; Model vs. Data

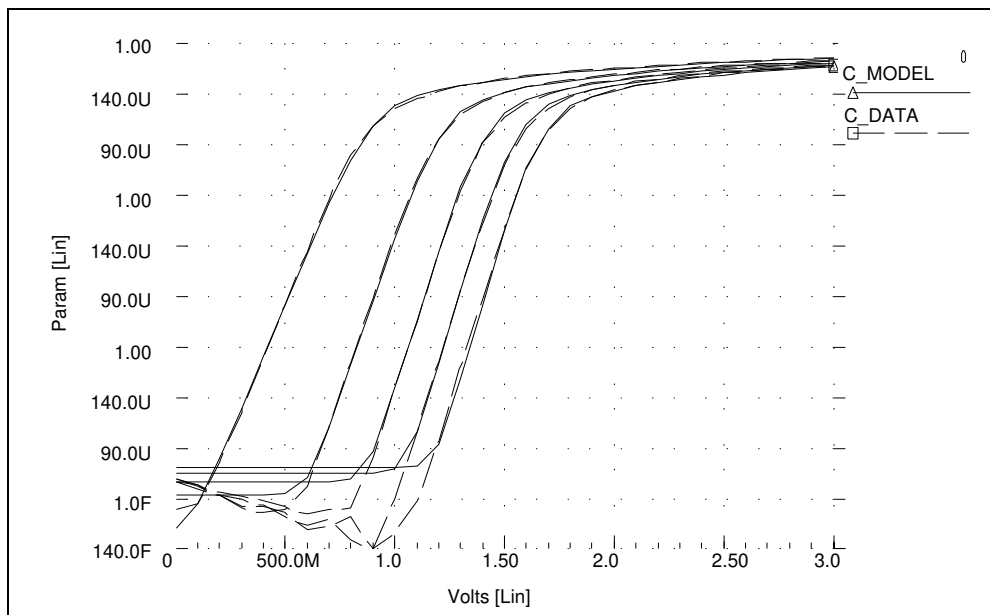
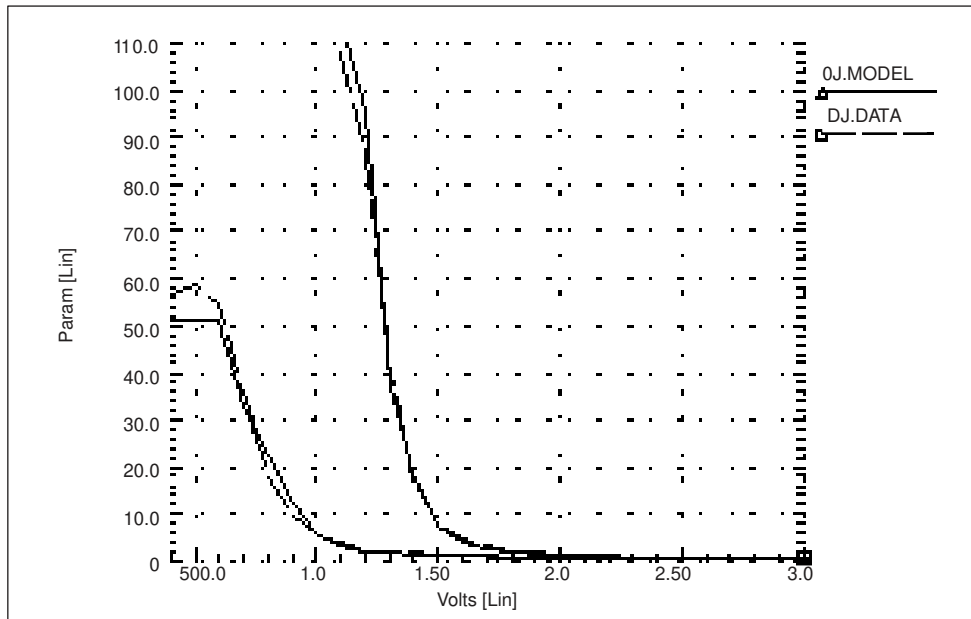


Figure 12 g_m/I_{DS} vs. V_{GS} for $V_{DS}=0.1V$, $V_{BS}=0, -2V$; BSIM2 Model vs. Data



BSIM2 Model — Typical Listing

In this example, geometry sensitivities are set to zero because a fit at only one geometry has been performed. This example includes extra the PrimeSim Continuum solution parameters for LDD, temperature, and geometry.

```
.MODEL NCH NMOS LEVEL=39
+ TOX=2.000000E-02 TEMP=2.500000E+01
+ VDD=5.000000E+00 VGG=5.000000E+00
+ VBB =-5.000000E+00 DL =0.000000E+00
+ DW=0.000000E+00 VGHIGH=1.270000E-01
+ LVGHIGH=0.000000E+00 WVGHIGH=0.000000E+00
+ VGLOW =-7.820000E-02 LVGLOW=0.000000E+00
+ WVGLOW=0.000000E+00 VFB =-5.760000E-01
+ LVFB=0.000000E+00 WVFB=0.000000E+00
+ PHI=6.500000E-01 LPHI=0.000000E+00
+ WPHI=0.000000E+00 K1=9.900000E-01
+ LK1=0.000000E+00 WK1=0.000000E+00
+ K2=1.290000E-01 LK2=0.000000E+00
+ WK2=0.000000E+00 ETA0=4.840000E-03
+ LETA0=0.000000E+00 WETA0=0.000000E+00
+ ETAB =-5.560000E-03 LETAB=0.000000E+00
+ WETAB=0.000000E+00 MU0=3.000000E+02
+ MU0B=0.000000E+00 LMU0B=0.000000E+00
+ WMU0B=0.000000E+00 MUS0=7.050000E+02
+ LMUS0=0.000000E+00 WMUS0=0.000000E+00
```

```
+ MUSB=0.000000E+00 LMUSB=0.000000E+00
+ WMUSB=0.000000E+00 MU20=1.170000E+00
+ LMU20=0.000000E+00 WMU20=0.000000E+00
+ MU2B=0.000000E+00 LMU2B=0.000000E+00
+ WMU2B=0.000000E+00 MU2G=0.000000E+00
+ LMU2G=0.000000E+00 WMU2G=0.000000E+00
+ MU30=3.000000E+01 LMU30=0.000000E+00
+ WMU30=0.000000E+00 MU3B=0.000000E+00
+ LMU3B=0.000000E+00 WMU3B=0.000000E+00
+ MU3G=-2.970000E+00 LMU3G=0.000000E+00
+ WMU3G=0.000000E+00 MU40=0.000000E+00
+ LMU40=0.000000E+00 WMU40=0.000000E+00
+ MU4B=0.000000E+00 LMU4B=0.000000E+00
+ WMU4B=0.000000E+00 MU4G=0.000000E+00
+ LMU4G=0.000000E+00 WMU4G=0.000000E+00
+ UA0=0.000000E+00 LUA0=0.000000E+00
+ WUA0=0.000000E+00 UAB=0.000000E+00
+ LUAB=0.000000E+00 WUAB=0.000000E+00
+ UB0=7.450000E-03 LUB0=0.000000E+00
+ WUB0=0.000000E+00 UBB=0.000000E+00
+ LUBB=0.000000E+00 WUBB=0.000000E+00
+ U10=0.000000E+00 LU10=7.900000E-01
+ WU10=0.000000E+00 U1B=0.000000E+00
+ LU1B=0.000000E+00 WU1B=0.000000E+00
+ U1D=0.000000E+00 LU1D=0.000000E+00
+ WU1D=0.000000E+00 N0=8.370000E-01
+ LN0=0.000000E+00 WN0=0.000000E+00
+ NB=6.660000E-01 LNB=0.000000E+00
+ WNB=0.000000E+00 ND=0.000000E+00
+ LND=0.000000E+00 WND=0.000000E+00
+ VOF0=4.770000E-01 LVOF0=0.000000E+00
+ WVOF0=0.000000E+00 VOFB=-3.400000E-02
+ LVOFB=0.000000E+00 WVOFB=0.000000E+00
+ VOFD=-6.900000E-02 LVOFD=0.000000E+00
+ WVOFD=0.000000E+00 AI0=1.840000E+00
+ LAI0=0.000000E+00 WAI0=0.000000E+00
+ AIB=0.000000E+00 LAIB=0.000000E+00
+ WAIB=0.000000E+00 BIO=2.000000E+01
+ LBIO=0.000000E+00 WBIO=0.000000E+00
+ BIB=0.000000E+00 LBIB=0.000000E+00
+ WBIB=0.000000E+00 DELL=0.000000E+00
+ WDF=0.000000E+00
```

Common SPICE Parameters

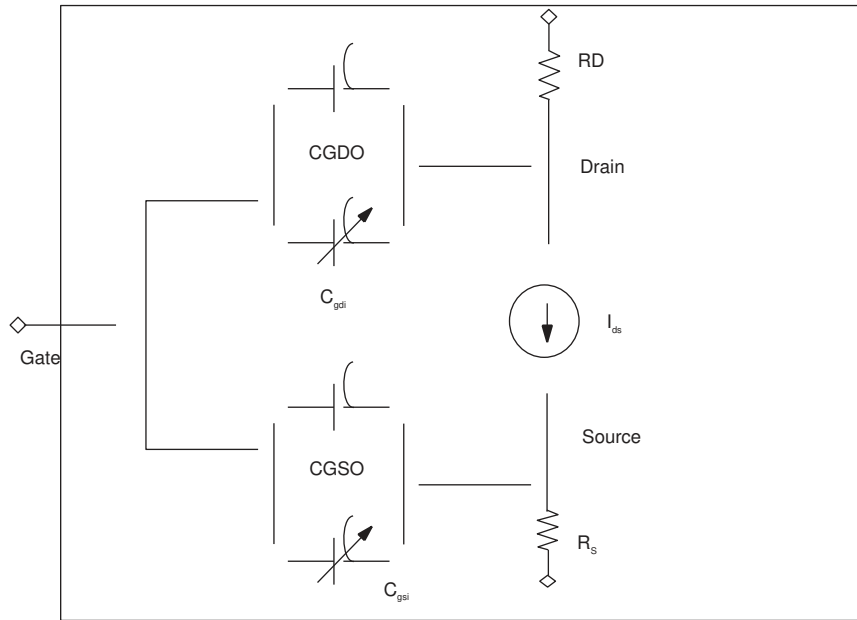
```
+ CGDO=1.000000E-09 CGSO=1.000000E-09
+ CGBO=2.500000E-11
+ RSH=3.640000E+01 JS=1.380000E-06
+ PB=8.000000E-01 PBSW=8.000000E-01
+ CJ=4.310000E-04 CJSW=3.960000E-10
+ MJ=4.560000E-01 MJSW=3.020000E-01
```

Synopsys Parameters

```
+ ACM=3  LMLT=8.500000E-01
+ WMLT=8.500000E-01
+ XL =-5.000000E-08  LD=5.000000E-08
+ XW=3.000000E-07  WD=5.000000E-07
+ CJGATE=2.000000E-10  HDIF=2.000000E-06
+ LDIF=2.000000E-07
+ RS=2.000000E+03  TRS=2.420000E-03
+ RD=2.000000E+03  TRD=2.420000E-03
+ TCV=1.420000E-03  BEX =-1.720000E+00
+ FEX =-2.820000E+00  LMU0=0.000000E+00
+ WMU0=0.000000E+00  JSW =2.400000E-12
```

LEVEL 40: HP a-Si TFT Model

The Synopsys Level 40 MOSFET device model represents a Hewlett-Packard amorphous silicon thin-film transistor model.



This model uses only the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#) section.

This section covers the following topics:

- [Using the HP a-Si TFT Model](#)
- [HP a-Si TFT Model Equations](#)
- [HP a-Si TFT Model — Device Model and Element Statement Example](#)

Using the HP a-Si TFT Model

To use the HP a-Si TFT model:

1. Set `LEVEL=40` to identify the model as the HP a-Si TFT model.
2. Default value for `L` is 10 μ m, and the default value for `W` is 40 μ m.
3. Use the “M” designation for MOSFET rather than the “A” designation for a-Si TFT in the netlist.

4. Use the “NMOS” or “PMOS” designation for device type rather than the “NAT” or “PAT” designation.

Note:

Because of the unavailability of p-channel TFTs, PMOS model testing has been limited.

5. LEVEL 40 is a three-terminal model. It lacks bulk nodes so simulation does not append any parasitic drain-bulk or source-bulk diodes to this model. You can specify a fourth node, but it does not affect the simulation results (except for GMIN terms).
6. Parasitic resistances and overlap capacitances are constant. They are not scaled with width, length, and temperature.
7. Capacitance expressions in this model do not conserve charge.
8. The `TREF` parameter is an exponent in an expression for mobility temperature dependence.
9. Other models use the `BEX` parameter for similar mobility temperature dependence expressions. The HP a-Si TFT `TREF` model parameter is not the same as the `TREF` reference temperature in other models. The reference temperature for the HP a-Si TFT model is 312 K (or 38.85 °C); you cannot modify it. Experimental results from TFT manufacturers indicate that amorphous silicon materials are most stable at this temperature.
10. The default room temperature is 25° C in Synopsys circuit simulators, but is 27° C in most other simulators. When comparing to other simulators, set the nominal simulation temperature to 27° C by using either `.TEMP 27` or `.OPTION TNOM=27` in the netlist. Although the reference temperature of the HP a-Si TFT model is fixed at 312° K (or 38.85° C), the behavior of the model adjusts to other simulation temperatures that you specify, or that are defaults in Synopsys circuit simulators. In the Level 40 MOSFET model, temperature dependency is enabled.
11. The default `CAPOP` value is 40, which is the HP a-Si TFT non-charge-conserving capacitance model. `CAPOP` values of 0, 1, 2, 3, 4, 5, 9, 12, or 13 have not been thoroughly tested.
12. The `DERIV` default is zero, which selects the analytical method. Set `DERIV` to 1 to select the finite difference method.

This section covers the following topics:

- [HP a-Si TFT Model — Effect of SCALE and SCALM](#)
- [HP a-Si TFT Model — Noise Model](#)
- [HP a-Si TFT Model — DELVTO Element](#)

HP a-Si TFT Model — Effect of SCALE and SCALM

`.OPTION SCALE` has the same effect for LEVEL 40 as for other Synopsys device models, such as MOSFET Level 3 or Level 28. If you specify the L and W values in microns rather than meters (for example, L=1 rather than L=1m or 1e-6), set `.OPTION SCALE=1e-6`.

The `SCALM` option is disabled in the LEVEL 40 model. For standard MOSFET models (such as LEVEL 3), `SCALM` affects the scale of model parameters such as `XL`, `XW`, `LD`, `WD`, `CJ`, and `CJSW`.

Because the LEVEL 40 model ignores the `SCALM` option, you can mix LEVEL 40 models in a simulation with other models that use `SCALM`.

In general, netlists for Synopsys simulators should be as standard as possible. Also, you should convert L and W to meters scale instead of microns scale so that you can use the netlist without `.OPTION SCALE=1E-6`. If you follow these recommendations, a system-level simulation can use I/O sub-circuits from different vendors.

HP a-Si TFT Model — Noise Model

The LEVEL 40 model uses the standard `NLEV=0` noise model inherited from other Synopsys MOSFET models.

HP a-Si TFT Model — DELVTO Element

You can use `DELVTO` and `DTEMP` on the element line with LEVEL 40.

HP a-Si TFT Model Equations

The following equations show model parameters in all capital letters; working variables are in lower case. Model parameters and the `vgs` and `vds` bias voltages are inputs. `Ids`, `gm`, and `gds` are the DC outputs. The `Cgs` gate-to-source capacitance and the `Cgd` gate-to-drain capacitance are the AC outputs. The electron charge is `q`, the Boltzmann's constant is `k`, and the permittivity of a vacuum is `ε0`.

This model applies the `SCALE` value before evaluating the equations, and scales by `M` after evaluation.

gm_{tft} and gds_{tft} variables are intermediate, not final, quantities.

For a complete description of TFT technology and the device physics underlying these equations, see the Hewlett-Packard HP IC-CAP manual.

Initially, $Cgdi = 0$, $Cgsi = 0$, $phi = PHI$, $vto = VTO$, and $uo = UO$.

If $uo = 0$, then $uo = 1$.

The following equation computes the Cfm dielectric capacitance per unit area:

$$\text{If } T1 \neq 0 \text{ and } T2 \neq 0, \text{ then } Cfm = \frac{(\epsilon 0 \cdot E1 \cdot E2)}{((T2 \cdot E1) + (T1 \cdot E2))}$$

$$\text{If } T1 = 0 \text{ and } T2 \neq 0, \text{ then } Cfm = \frac{(\epsilon 0 \cdot E2)}{T2}$$

$$\text{If } T2 = 0 \text{ and } T1 \neq 0, \text{ then } Cfm = \frac{(\epsilon 0 \cdot E1)}{T1}$$

$$kp = uo \cdot Cfm \cdot 10^{-4}$$

TEMP is the Synopsys device simulation temperature, specified in oC, but converted to o K internally to evaluate these equations.

$$vt = \frac{(k \cdot TEMP)}{q}$$

$$eg = (2 \cdot 10^4 \cdot (TEMP - 312)) + 1.4$$

$$vto = vto + (DELVTOModel \cdot type) + (DELVTOelement \cdot type)$$

$$vbi = vto, \text{ ratio} = \frac{TEMP}{312}$$

$$\text{If } VTIME \leq 1, \text{ then } uo = uo \cdot (\text{ratio}^{TREF}) \text{ and } kp = kp \cdot (\text{ratio}^{TREF})$$

Note:

T_{REF} is an exponent in adjusting the temperature. It is not the reference temperature of this device model.

$$vfb = vto - (0.5 \cdot PHI) + (0.5 \cdot (1.4 - eg))$$

$$vbi = vfb + (0.5 + PHI \cdot \text{ratio})$$

$$vto = vbi \text{ (printback definition)}$$

$$phi = phi \cdot \text{ratio} \text{ (printback definition)}$$

$$vfb = vbi - phi \text{ (printback definition)}$$

$$vdsat = 0$$

$$beta = kp \cdot W \cdot L$$

$$vth = vbi + (ETA \cdot vds)$$

If $NU \neq 0$ $K2 \neq 0$ $PSI \neq 0$ and $VTIME > 1$, then:

$$vth = vth + f(vgs, vds, NU, K2, PSI, CHI, VTIME, TEMP)$$

$$v_{on} = v_{th}$$

If $NFS \neq 0$, then:

$$x_n = 1 + \left(\frac{(q \cdot NFS \cdot 10^4 \cdot W \cdot L)}{C_{fm}} \right)$$

$$v_{on} = f(v_{th}, (v_t \cdot x_n))$$

This section covers the following topics:

- [HP a-Si TFT Model — Cutoff Region \(\$NFS=0\$, \$v_{gs} \leq v_{on}\$ \)](#)
- [HP a-Si TFT Model — Noncutoff Region \(\$NFS \neq 0\$ \)](#)
- [HP a-Si TFT Model — Cgd, Cgs](#)

HP a-Si TFT Model — Cutoff Region ($NFS=0$, $v_{gs} \leq v_{on}$)

If $NFS = 0$ and $v_{gs} \leq v_{on}$, then:

$$C_{gdi} = 0$$

$$C_{gsi} = 0$$

$$I_{ds} = GO \cdot f(v_{gs}, (DEFF \cdot v_{ds}))$$

$$g_m = GO. g_{ds} = GO \cdot DEFF$$

HP a-Si TFT Model — Noncutoff Region ($NFS \neq 0$)

- If $v_{gs} > v_{on}$, then $v_{gsx} = v_{gs}$.
- If $v_{gs} \leq v_{on}$, then $v_{gsx} = v_{on}$.

Mobility modulation by v_{gs} :

$$u_{eff} = f(u_o, \eta, v_{gs}, THETA)$$

If $V_{MAX} > 0$, then:

$$v_{dsc} = \frac{L \cdot V_{MAX}}{u_{eff}}$$

$$v_{dsat} = (v_{gsx} - v_{th}) + v_{dsc} \sqrt{((v_{gsx} - v_{th})^2 + v_{dsc}^2)}$$

$$C_{fmlw} = \frac{(C_{fm} \cdot CSC)}{(C_{fm} + CSC)} \cdot L \cdot W$$

$Cfmlw$ is the series combination of the dielectric and space charge capacitance for the MIS structure.

If $vds < vdsat$, then:

$$vdsx = vds$$

$$epsfm = Cfmlw \cdot \frac{(T2 + T1)}{\epsilon_0}$$

$epsfm$ is the effective equivalent dielectric constant of the insulator layers.

$$fval = 0.8 + \left(\frac{epsfm - 0.8}{1 + (2 \cdot \pi \cdot FREQ \cdot TAU)^2} \right)$$

$$Cgdi = f(Cfmlw \cdot f(efm, 0.8) \cdot (\exp(fval, FEFF, vgs - vth - vds)))$$

$$Cgsi = f(Cfmlw \cdot f(efm, 0.8) \cdot (\exp(fval, FEFF, (vgs - vth), vds)))$$

Otherwise, $vds \geq vdsat$:

$$vdsx = vdsat$$

$$Cgdi = Cfmlw$$

$$Cgsi = \frac{Cfmlw}{2}$$

If $vdsx \neq 0$, then

$$cdnorm = vdsx \cdot \left(vgsx - vth - \frac{vdsx}{2} \right)$$

Normalized drain current:

$$gm_{tft} = vdsx$$

$$gds_{tft} = vgsx - vth - vdsx$$

$$cd1 = beta \cdot cdnorm$$

Drain current without velocity saturation effect:

$$beta = beta \cdot fgate, idrain = beta \cdot cdnorm$$

$$gm_{tft} = (beta \cdot gm_{tft}) + (dfgdvg \cdot cd1)$$

Velocity saturation factor—if $V_{MAX} \neq 0$, then:

$$f_{drain} = \frac{1}{\left(1 + \left(\frac{v_{dsx}}{v_{dsc}}\right)\right)}$$

$$df_{ddv_g} = -df_{gdv_g} \cdot \frac{((f_{drain})^2) \cdot v_{dsx}}{(v_{dsc} \cdot f_{gate})}$$

$$df_{ddv_d} = \frac{-(f_{drain})^2}{v_{dsc}}$$

Strong inversion current:

$$gm_{tft} = (f_{drain} \cdot gm_{tft}) + (df_{ddv_g} \cdot idrain)$$

$$gds_{tft} = (f_{drain} \cdot gds_{tft}) + (df_{ddv_d} \cdot idrain)$$

$$idrain = f_{drain} \cdot idrain, \beta = \beta \cdot f_{drain}$$

$$Ids = idrain \cdot f(GO, v_{gs}, DEFF, v_{ds})$$

$$gm = f(gm_{tft}, GO)$$

$$gds = f(gds_{tft}, GO, DEFF)$$

Weak inversion current—if $v_{gs} < v_{on}$, then:

$$idrain = idrain \cdot \exp\left(\frac{(v_{gs} - v_{on})}{(v_t \cdot \chi_n)}\right)$$

$$Ids = idrain + f(GO, v_{gs}, DEFF, v_{ds})$$

$$gm_{tft} = \frac{idrain}{(v_t \cdot \chi_n)}, gm = f(gm_{tft}, GO)$$

$$gds_{tft} = gds_{tft} \cdot \exp\left(\frac{(v_{gs} - v_{on})}{(v_t \cdot \chi_n)}\right)$$

$$gds = gds_{tft} + f(GO, DEFF)$$

$v_{dsx} = 0$:

$$Ids = f(GO \cdot v_{gs}, DEFF, v_{ds})$$

$$gm = GO$$

$$gds_{tft} = \beta \cdot (v_{gsx} - v_{th})$$

If $NFS \neq 0$ and $vgs < von$, then:

$$gds_{tft} = gds_{tft} \cdot \exp\left(\frac{(vgs - von)}{(vt \cdot xn)}\right), gds = f(gds_{tft}, GO, DEFF)$$

HP a-Si TFT Model — Cgd, Cgs

$$Cgd = Cgdi + CGDO, Cgs = Cgsi + CGSO$$

HP a-Si TFT Model — Device Model and Element Statement Example

```
.MODEL nch nmos LEVEL=40 UO=0.4229 VTO=1.645 PHI=1.25 NSS=0
+ NFS=2.248E+21 VMAX=1231
+ THETA=-0.01771 ETA=0.0002703 T1=2.6E-07 T2=0 E1=3.9 E2=0
+ GO=9.206E-15 NU=0 K2=2 CHI=0.5
+ PSI=1E-20 VTIME=0.01 TREF=1.5 CGSO=5.203E-14 CGDO=4.43E-14
+ CSC=0.0001447 RD=5097
+ RS=5097 FREQ=1E+06 DEFF=2.15 TAU=1.64E-07 FEFF=0.5
MCKT 1 2 3 nch L=1e-05 W=4e-05
```

LEVEL 47: BSIM3 Version 2 MOSFET Model

The BSIM3 version 2.0 MOS model from UC Berkeley is available as the Synopsys Level 47 model.

The following sections discuss these topics:

- [Using the BSIM3 Version 2 MOSFET Model](#)
- [BSIM3 Version 2 MOSFET Model Parameters](#)
- [BSIM3 Version 2 MOSFET Model Equations](#)
- [BSIM3 Version 2 MOSFET Model — Usage Notes](#)
- [BSIM3 Version 2 MOSFET Model — PMOS Model Example](#)

Using the BSIM3 Version 2 MOSFET Model

The Level 47 model uses the same model parameters for the source/drain diode current, capacitance, and resistance as used in the other supported MOS levels. The `ACM` model parameter controls the choice of source/drain equations.

The Level 47 model also uses the same noise equations as the other MOSFET model levels. The `NLEV` parameter controls the choice of noise equations.

This model, like all Synopsys simulation device models, can include parameters. You can use these parameters to model the process skew, either by worst-case corners or by Monte Carlo. For information about Worst-Case and Monte Carlo analysis, see Worst Case Analysis and Variation Analysis in the *PrimeSim HSPICE User Guide: Simulation and Analysis*.

BSIM3 Version 2 MOSFET Model Parameters

Name	Unit	Default	Description
A0		1	Bulk charge effect. Default is 4.4 for PMOS.
A1	1/V	0	First nonsaturation factor (0 for NMOS, 0.23 for PMOS)
A2		1.0	Second nonsaturation factor (1.0 for NMOS, 0.08 for PMOS)
AT	m/sec	3.3e4	Temperature coefficient of VSAT
BULKMOD		1	Bulk charge model selector

Name	Unit	Default	Description
CDSC	F/m ²	2.4e-4	Drain/source and channel coupling capacitance
CDSCB	F/Vm ²	0	Body coefficient for CDSC
CIT	F/m ²	0.0	Interface state capacitance
DL	m	0.0	Channel length reduction on one side (multiplied by SCALM)
DSUB		DROUT	DIBL coefficient in the subthreshold region
DVT0		2.2	Short-channel effect coefficient 0
DVT1		0.53	Short-channel effect coefficient 1
DVT2	1/V	-0.032	Short-channel effect coefficient 2
DW	m	0.0	Channel width reduction on one side (multiplied by SCALM)
EM	V/m	4.1e7	Electrical field in the channel above which the hot carrier effect dominates
ETA		0.3	Coefficient of the drain voltage reduction
ETA0		0.08	DIBL (Drain Induced Barrier Lowering) coefficient for the subthreshold region
ETAB	1/V	-0.07	Subthreshold region DIBL coefficient
GAMMA1	V ^{1/2}	See BSIM3 Version 2 MOSFET Model Equations.	Body effect coefficient, near interface
GAMMA2	V ^{1/2}	See BSIM3 Version 2 MOSFET Model Equations.	Body effect coefficient in the bulk
K1	V ^{1/2}	0.53	First-order body effect coefficient
K2		-0.0186	Second-order body effect coefficient
K3		80.0	Narrow width effect coefficient
K3B	1/V	0	Body width coefficient of the narrow width effect
KETA	1/V	-0.047	Body bias coefficient of the bulk charge effect

Name	Unit	Default	Description
KT1	V	-0.11	Temperature coefficient for the threshold voltage
KT1L	Vm	0	Channel length sensitivity of the temperature coefficient for the threshold voltage
KT2		0.022	Body bias coefficient of the threshold temperature effect
LDD	m	0.0	Total length of the LDD region
LITL	m		Characteristic length. Default is: $LITL = \left(\frac{\epsilon_{si} T_{ox} X_j}{\epsilon_{ox}} \right)^{1/2}$
NFACTOR		1.0	Subthreshold region swing
NLX	m	1.74e-7	Lateral nonuniform doping along the channel
NPEAK	cm ⁻³ See (8)	1.7e17	Peak doping concentration near the interface
NSUB	cm ⁻³	6.0e16	Substrate doping concentration
PHI	V	See BSIM3 Version 2 MOSFET Model Equations .	Surface potential under strong inversion
PSCBE1	V/m	4.24e8	Exponent 1 for the substrate current induced body effect
PSCBE2	m/V	1.0e-5	Coefficient 2 for the substrate current induced body effect
PVAG		0	Gate dependence of the output resistance
RDS0	ohm	0.0	Source drain contact resistance
RDSW	ohm · μm	0.0	Source drain resistance per unit width
SATMOD		2	Saturation model selector
SUBTHMOD		2	Subthreshold model selector
TNOM (TREF)	°C	25	Temperature at which simulation extracts parameters. This parameter defaults to the TNOM option, which defaults to 25°C. See 4 and 5 in BSIM3 Version 2 MOSFET Model — Usage Notes .

Name	Unit	Default	Description
TOX	m	150e-10	Gate oxide thickness
U0	m^2/Vsec See (8)	0.067	Low field mobility at $T=T_{\text{REF}}$ • 0.067 for n-channel • 0.025 for p-channel
UA	m/V	2.25e-9	First-order mobility degradation coefficient
UA1	m/V	4.31e-9	Temperature coefficient of UA
UB	m^2/V^2	5.87e-19	Second-order mobility degradation coefficient
UB1	m^2/V^2	-7.61e-18	Temperature coefficient of UB
UC	1/V	0.0465	Body bias sensitivity coefficient of mobility
UC0*	$(\text{V/m})^2$		Temperature coefficient
UC1	1/V	-0.056	Temperature coefficient of UC
UTE		-1.5	Mobility temperature exponent
VBM	V	-5.0	Maximum substrate bias
VBX	V	See BSIM3 Version 2 MOSFET Model Equations.	V_{bs} at which the depletion width equals XT
VFB	V		Flat-band voltage
VGHIGH	V	0.12	Upper bound of the weak-strong inversion transition region
VGLOW	V	-0.12	Lower bound of the weak-strong inversion transition region
VOFF	V	-0.11	Offset voltage in the subthreshold region
VSAT	cm/sec	8e6	Saturation velocity of the carrier at $T=T_{\text{REF}}$
VTH0	V	0.7	Threshold voltage of the long channel at $V_{\text{bs}}=0$ and small V_{ds} • 0.7 for n-channel. • - 0.7 for p-channel.
W0	m	2.5e-6	Narrow width effect coefficient
XJ	m	0.15e-6	Junction depth
XPART		1	Charge partitioning flag

Name	Unit	Default	Description
XT	m	1.55e-7	Doping depth
* UC0 has no effect on the model			

BSIM3 Version 2 MOSFET Model Equations

The following model equations are based on the BSIM3 source code:

- [BSIM3 Version 2 Model — \$L_{eff}\$ and \$W_{eff}\$ Equations](#)
- [BSIM3 Version 2 Model — Threshold Voltage Parameters](#)
- [BSIM3 Version 2 Model — Mobility Parameters](#)
- [BSIM3 Version 2 Model — Drain Saturation Voltage Parameters](#)
- [BSIM3 Version 2 Model — Linear Region](#)
- [BSIM3 Version 2 Model — Saturation Region Parameters](#)
- [BSIM3 Version 2 Model — Drain Current Parameters](#)
- [BSIM3 Version 2 Model — Subthreshold Region Parameters](#)
- [BSIM3 Version 2 Model — Transition Region Parameters \(subthMod=2\)](#)
- [BSIM3 Version 2 Model — Temperature Compensation Parameters](#)

BSIM3 Version 2 Model — L_{eff} and W_{eff} Equations

The standard equations for L_{eff} and W_{eff} in the Synopsys model are:

$$L_{eff} = L + XL - (2 \cdot LD)$$

$$W_{eff} = W + XW - (2 \cdot WD)$$

BSIM3 uses the following UCB SPICE3 equations:

$$L_{eff} = L - (2 \cdot DL)$$

$$W_{eff} = W - (2 \cdot DW)$$

The units for these parameters are meters with defaults of zero.

Simulation uses the standard Synopsys model equation for both cases, and accepts $DL(DW)$ as the value for $LD(WD)$. If you specify both $LD(WD)$ and $DL(DW)$ in a `.MODEL` statement, simulation uses the $LD(WD)$ value.

If you specify $LDAC$ and $WDAC$ in the `.MODEL` statement, then:

$$L_{eff} = L + XL - 2 \cdot LDAC, W_{eff} = W + XW - 2 \cdot WDAC$$

This model uses the $LD(DL)$ and $WD(DW)$ values with the RS and RD parameters for $ACM > 0$.

In the following two models return the same simulation results:

```
* HSPICE style:
.MODEL n1 nmos Level=47 XL=0.1e6 LD=0.15e-6
+ SatMod=2 SubthMod=2 BulkMod=1
+ CGSO=0.3e-9 CGDO=0.3e-9 CGBO=0
* SPICE3 style:
.MODEL n2 nmos Level=47 LD=0.1e-6
+ SatMod=2 SubthMod=2 BulkMod=1
+ CGSO=0.3e-9 CGDO=0.3e-9 CGBO=0
```

BSIM3 Version 2 Model — Threshold Voltage Parameters

$$V_{th0}, K1, K2, \phi_s, N_{lx}, K3, W_0, T_{ox}, V_{bi}, D_{vt0}, D_{vt1},$$

$$D_{vt2}, N_{peak}, N_{sub}, Y_1, Y_2, V_{bx}, V_{bm}, V_{bi}, X_t, TREF$$

$$V_{th} = V_{th0} + K1 \left(\sqrt{\phi_s - V_{bs}} - \sqrt{\phi_s} \right) - K2 V_{bs} + K1 \left(\sqrt{1 + \frac{N_{lx}}{L_{eff}}} \sqrt{\frac{\phi_s}{\phi_s - V_{bs}}} - 1 \right) \sqrt{\phi_s}$$

$$+ (K3 + K3B \cdot V_{bs}) \cdot \left(\frac{T_{ox}}{W_{eff} + W_0} \right) \phi_s - \Delta V_{th}$$

$$T_{ratio} = \frac{(TEMP + DTEMP + 273.15)}{(TREF + 273.15)}$$

$$\Delta V_{th} = \theta_{th}(L_{eff}) \cdot (V_{bi} - \phi_s)$$

$$\theta_{th}(L_{eff}) = D_{vt0} \cdot \left[\exp \left(\frac{-D_{vt1} \cdot L_{eff}}{2l_t} \right) + 2 \exp \left(\frac{-D_{vt1} \cdot L_{eff}}{l_t} \right) \right]$$

$$l_t = \sqrt{3 \cdot T_{ox} \cdot X_{dep} \cdot (1 + D_{vt2} \cdot V_{bs})}$$

$$X_{dep} = \sqrt{\frac{2 \cdot \epsilon_{si} \cdot (\phi_s - V_{bs})}{q \cdot N_{peak}}}$$

If you do not specify Φ_{hi} as a model parameter, then:

$$\phi_s = 2 \cdot V_{tm} \cdot \ln \left(\frac{N_{peak}}{n_i} \right) \quad (N_{peak} \text{ and } n_i \text{ in cm}^{-3})$$

$$V_{tm} = K \cdot T / q$$

$$n_i = 1.45e10 \cdot \left(\frac{T}{300.15} \right)^{1.5} \cdot \exp(21.5565981 - E_g / (2 \cdot V_{tm}))$$

$$E_g = 1.16 - (7.02e-4) \cdot T^2 / (T + 1108.0)$$

If you do not specify K_1 and K_2 as model parameters, simulation calculates them as follows:

$$K_1 = Y_2 - 2 \cdot K_2 \cdot \sqrt{\phi_s - V_{bm}}$$

$$K_2 = (Y_1 - Y_2) \cdot \frac{\sqrt{\phi_s - V_{bx}} - \sqrt{\phi_s}}{2 \cdot \sqrt{\phi_s} \cdot (\sqrt{\phi_s - V_{bm}} - \sqrt{\phi_s}) + V_{bm}}$$

$$Y_1 = \frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot N_{peak}}}{C_{ox}}, Y_2 = \frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot N_{sub}}}{C_{ox}}$$

$$V_{bx} = \phi_s - \left(\frac{q \cdot N_{peak} \cdot X_t^2}{2 \cdot \epsilon_{si}} \right)$$

If you do not specify V_{bi} as a model parameter, then:

$$V_{bi} = \frac{k \cdot T}{q} \cdot \ln \left(\frac{1.0e22 \cdot N_{peak}}{n_i^2} \right)$$

BSIM3 Version 2 Model — Mobility Parameters

$$\mu_0, U_a, U_b, U_c$$

$$\mu_{eff} = \frac{\mu_0}{1 + U_a \cdot \left(\frac{V_{gs} + V_{th}}{T_{ox}} \right) + U_b \cdot \left(\frac{V_{gs} + V_{th}}{T_{ox}} \right)^2 + U_c \cdot V_{bs}}$$

BSIM3 Version 2 Model — Drain Saturation Voltage Parameters

$$A_0, v_{sat}, X_j, A_1, A_2, R_{ds0}, R_{dsw}$$

R_{ds} and $Pfactor$:

$$R_{ds} = R_{ds0} + R_{dsw} / (1e6 \cdot W_{eff})$$

$$Pfactor = A_1 \cdot V_{gst} + A_2$$

If $Pfactor > 1$, simulation sets it to $Pfactor = 1$.

$$V_{gst} = V_{gs} - V_{th}$$

If $R_{ds}=0$ and $Pfactor=1$, then:

$$V_{dsat} = \frac{E_{sat} \cdot L_{eff} \cdot V_{gst}}{A_{bulk} \cdot E_{sat} \cdot L_{eff} + V_{gst}}$$

For BULKMOD=1:

$$A_{bulk} = \left(1 + \frac{K1 \cdot A_0 \cdot L_{eff}}{(L_{eff} + T1) \cdot T1s \cdot 2} \right) / (1 + KETA \cdot V_{bs})$$

For BULKMOD=2:

$$A_{bulk} = \left(\frac{K1 \cdot A_0 \cdot L_{eff}}{(L_{eff} + T1) \cdot \sqrt{\phi_s} \cdot 2} \right) / (1 + KETA \cdot V_{bs})$$

$$T1 = 2 \cdot \sqrt{X_j \cdot X_{dep}}$$

For $V_{bs} \leq 0$:

$$T1s = \sqrt{\phi_s \cdot V_{bs}}$$

For $V_{bs} \geq 0$:

$$T1s = \frac{\phi_s \cdot \sqrt{\phi_s}}{\phi_s + \frac{V_{bs}}{2}}, E_{sat} = 2 \cdot \frac{v_{sat}}{\mu_{eff}}$$

In general, V_{dsat} solves $Tmpa \cdot V_{dsat} \cdot V_{dsat} - Tmpb \cdot V_{dsat} + Tmpc = 0$:

$$V_{dsat} = \left(Tmpb - \sqrt{Tmpb^2 - 4 \cdot Tmpa \cdot Tmpc} \right) / (2 \cdot Tmpa)$$

$$Tmpa = A_{bulk} \cdot (A_{bulk} \cdot W_{eff} \cdot v_{sat} \cdot C_{ox} \cdot R_{ds} - 1 + 1 / Pfactor)$$

$$Tmpb = V_{gst} \cdot (2 / Pfactor - 1) + (A_{bulk} \cdot E_{sat} \cdot L_{eff}) + (3 \cdot A_{bulk} \cdot V_{gst} \cdot W_{eff} \cdot v_{sat} \cdot C_{ox} \cdot R_{ds})$$

$$Tmpc = (V_{gst} \cdot E_{sat} \cdot L_{eff}) + (V_{gst}^2 \cdot 2 \cdot W_{eff} \cdot v_{sat} \cdot C_{ox} \cdot R_{ds})$$

BSIM3 Version 2 Model — Linear Region

$$I_{dslin0} = \mu_{eff} \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot \frac{1}{1 + V_{ds} / (E_{sat} \cdot L)} \cdot \left(V_{gs} - V_{th} - A_{bulk} \cdot \frac{V_{ds}}{2} \right) \cdot V_{ds}$$

$$I_{ds} = \frac{I_{dslin0}}{1 + \frac{R_{ds} \cdot I_{dslin0}}{V_{ds}}}$$

BSIM3 Version 2 Model — Saturation Region Parameters

$litl, eta, L_{dd}, E_m, D_{rout}, P_{clm}, P_{dibl1}, P_{dibl2}, P_{scbe1}, P_{scbe2}$

V_{asat} and F_{vag} :

$$V_{asat} = \frac{E_{sat} \cdot L_{eff} + V_{dsat} + 2R_{ds} \cdot v_{sat} \cdot C_{ox} \cdot W_{eff} \cdot \left(V_{gst} - \frac{A_{bulk} \cdot V_{dsat}}{2} \right)}{2 / Pfactor - 1 + R_{ds} \cdot v_{sat} \cdot C_{ox} \cdot W_{eff} \cdot A_{bulk}}$$

$$F_{vag} = 1 + \frac{P_{vag} \cdot V_{gst}}{E_{sat} \cdot L_{eff}}$$

Early Voltage, satMod=1

$$V_A = V_{asat} + F_{vag} \cdot \left(\frac{1 + eta \cdot \frac{L_{dd}}{litl}}{P_{clm} \cdot A_{bulk}} \right) \cdot \left(\frac{(A_{bulk} \cdot E_{sat} \cdot L_{eff} + V_{gst} - \lambda(V_{ds} - V_{dsat}))(V_{ds} - V_{dsat})}{E_{sat} \cdot litl} \right)$$

$$\lambda = \frac{A_{bulk} \cdot E_{sat} \cdot L_{eff} + (V_{gst})}{2 \cdot litl \cdot E_m}$$

Early Voltage, satMod=2

$$V_A = V_{asat} + F_{vag} \cdot U_{vds} \cdot \left(\frac{1}{V_{aclm}} + \frac{1}{V_{adibl}} \right)^{-1}$$

$$U_{vds} = 1 + eta \cdot \frac{L_{dd}}{litl}$$

$$V_{aclm} = \frac{1}{P_{clm}} \cdot \frac{A_{bulk} \cdot E_{sat} \cdot L_{eff} + V_{gst}}{A_{bulk} \cdot E_{sat} \cdot litl} \cdot (V_{ds} - V_{dsat})$$

$$V_{adibl} = \frac{1}{\theta_{rout}} \cdot \left[(V_{gs} - V_{th}) - \left(\frac{1}{A_{bulk} \cdot V_{dsat}} + \frac{1}{V_{gst}} \right)^{-1} \right]$$

$$\theta_{rout} = P_{dibl1} \cdot \left[\exp \left(\frac{-D_{rout} \cdot L_{eff}}{2 \cdot l_t} \right) + 2 \exp \left(\frac{-D_{rout} \cdot L_{eff}}{l_t} \right) \right] + P_{dibl2}$$

$$V_{ahce} = \left[\frac{P_{scbe2}}{L_{eff}} \cdot \exp \left(\frac{-P_{scbe1} \cdot litl}{V_{ds} - V_{dsat}} \right) \right]^{-1}$$

BSIM3 Version 2 Model — Drain Current Parameters

$$I_{dsat} = W_{eff} \cdot v_{sat} \cdot C_{ox} \cdot (V_{gs} - V_{th} - A_{bulk} \cdot V_{dsat}) \cdot Pfactor$$

$$Pfactor = A_1 \cdot V_{gst} + A_2$$

$$I_{ds} = I_{dsat} \cdot \left(1 + \frac{V_{ds} - V_{dsat}}{V_A} \right) \cdot \left(1 + \frac{V_{ds} - V_{dsat}}{V_{ahce}} \right)$$

BSIM3 Version 2 Model — Subthreshold Region Parameters

$$Nfactor, C_{dsc}, C_{dscb}, V_{off}, C_{it}, D_{sub}, eta_0, eta_b$$

n and $DIBL$:

$$n = 1 + \frac{Nfactor \cdot 1.034e-10}{X_{dep} C_{ox}} + \frac{(C_{dsc} + C_{dscb} V_{bs}) \left[\exp\left(\frac{-L_{eff}}{2l_t}\right) + 2 \exp\left(\frac{-L_{eff}}{l_t}\right) \right] + C_{it}}{C_{ox}}$$

$$DIBL = (eta_0 + eta_b \cdot V_{bs}) \cdot \left[\exp\left(\frac{-D_{sub} \cdot L_{eff}}{2 \cdot l_{t0}}\right) + 2 \exp\left(\frac{-D_{sub} \cdot L_{eff}}{l_{t0}}\right) \right]$$

$$l_{t0} = \sqrt{3 \cdot T_{ox} \cdot X_{dep0}}, X_{dep0} = \sqrt{\frac{2 \cdot \epsilon_{si} \cdot \phi_s}{q \cdot N_{peak}}}$$

subthMod=0

$$I_{ds} = g_m = g_{ds} = g_{mb} = 0$$

subthMod=1

$$I_{ds} = \frac{I_{limit} \cdot I_{exp}}{I_{limit} + I_{exp}} \cdot \left[1 - \exp\left(\frac{-V_{ds}}{V_{tm}}\right) \right], I_{limit} = \frac{9}{2} \cdot u0 \cdot \sqrt{\frac{q \epsilon_{si} \cdot N_{peak}}{2 \cdot \phi_s}} \cdot \frac{W_{eff}}{L_{eff}} \cdot V_{tm}^2$$

$$I_{exp} = u0 \cdot \sqrt{\frac{q \cdot \epsilon_{si} \cdot N_{peak}}{2 \cdot \phi_s}} \cdot \frac{W_{eff}}{L_{eff}} \cdot V_{tm}^2 \cdot \exp\left(\frac{V_{gs} - V_{th} - V_{off} + DIBL \cdot V_{ds}}{n \cdot V_{tm}}\right)$$

subthMod=2

$$I_{ds} = u0 \cdot \sqrt{\frac{q \epsilon_{si} N_{peak}}{2 \phi_s}} \cdot \frac{W_{eff}}{L_{eff}} \cdot V_{tm}^2 \cdot \left[1 - \exp\left(\frac{-V_{ds}}{V_{tm}}\right) \right] \cdot \exp\left(\frac{V_{gs} - V_{th} - V_{off} + DIBL \cdot V_{ds}}{n V_{tm}}\right)$$

BSIM3 Version 2 Model — Transition Region Parameters (subthMod=2)

$$V_{gshigh}, V_{gslow}$$

$$I_{ds} = (1-t)^2 \cdot I_{dslow} + 2 \cdot (1-t) \cdot t \cdot I_p + t^2 \cdot I_{dshigh}$$

$$t = \left(\frac{V_p - V_{gslow}}{V_{gslow} - 2V_p + V_{gshigh}} \right) \cdot \left(\sqrt{1 + \frac{(V_{gslow} - 2V_p + V_{gshigh})(V_{gs} - V_{th} - V_{gslow})}{(V_p - V_{gslow})^2}} - 1 \right)$$

$$V_p = \frac{(g_{mhigh} \cdot V_{gshigh} - g_{mlow} \cdot V_{gslow}) - (I_{dshigh} - I_{dslow})}{g_{mhigh} - g_{mlow}}$$

$$I_p = I_{dslow} + g_{mlow} \cdot (V_p - V_{gslow})$$

BSIM3 Version 2 Model — Temperature Compensation Parameters

$A_t, U_{a1}, U_{b1}, U_{c1}, KT1, KT2, UTE$

$$V_{th}(temp) = V_{th}(tref) + (KT1 + KT2 \cdot V_{bs}) \cdot (T_{ratio} - 1)$$

$$u0(temp) = u0(tref) \cdot (T_{ratio})^{UTE}$$

$$V_{sat}(temp) = V_{sat}(tref) - A_t \cdot (T_{ratio} - 1)$$

$$U_a(temp) = U_a(tref) + U_{a1} \cdot (T_{ratio} - 1)$$

$$U_b(temp) = U_b(tref) + U_{b1} \cdot (T_{ratio} - 1)$$

$$U_c(temp) = U_c(tref) + U_{c1} \cdot (T_{ratio} - 1)$$

BSIM3 Version 2 MOSFET Model — Usage Notes

The following are usage notes regarding Level 47 MOSFETs:

1. Set `LEVEL=47` to identify the model as a BSIM3 model.
2. This model is based on BSIM3 version 2.0 from UC Berkeley. Code was received from UC Berkeley in July 1994 in the form of SPICE3e2. Changes announced in a letter from UCB September 13, 1994, have been included. DC sweeps have been checked against SPICE3e2.
3. The default setting for `CAPOP` is `CAPOP=13`, which is the BSIM1 charge-conserving capacitance model. This model does not use the BSIM3 capacitance model.
4. The Level 47 model supports the `TNOM` model parameter name as an alias for `TREF`. The conventional terminology is `TREF`, which is supported as a model parameter in all

Synopsys MOS levels. Level 47 supports the `TNOM` alternative name for compatibility with SPICE3.

5. The default room temperature is 25°C in Synopsys simulators, but is 27°C in SPICE3. If you specify the BSIM3 model parameters at 27°C, add `TREF=27` to the model so that simulation correctly interprets the model parameters. To set the nominal simulation temperature to 27, add `.OPTION TNOM=27` to the netlist when you test the Synopsys model versus SPICE3.
6. The default of `DERIV` is zero, the analytical method. You can set `DERIV` to 1 for the finite difference method. Analytic derivatives in the SPICE3e2 code are not exact in some regions. Setting `DERIV=1` returns more accurate derivatives (`GM`, `GDS`, and `GMBS`), but consumes more CPU time.
7. You can select one of three ways to calculate V_{th} .
 - Using `K1` and `K2` values that you specify.
 - Using `GAMMA1`, `GAMMA2`, `VBM`, and `VBX` values that you enter in the `.MODEL` statement.
 - Using `NPEAK`, `NSUB`, `XT`, and `VBM` values that you specify.
8. You can enter the `NPEAK` and `U0` model parameters in meters or centimeters. Simulation converts `NPEAK` to cm^{-3} as follows: if `NPEAK` is greater than $1\text{e}20$, simulation multiplies it by $1\text{e}-6$, and converts `U0` to m^2/Vsec as follows: if `U0` is greater than 1, simulation multiplies it by $1\text{e}-4$. You must enter the `NSUB` parameter in cm^{-3} units.
9. The specified value of the threshold decreases with increasing temperature for NMOS and PMOS.
10. The default value of `KT1` is -0.11. The negative sign ensures that the absolute value of the threshold decreases with increasing temperature for NMOS and PMOS.
11. You cannot set the `LITL` model parameter below a minimum value of $1.0\text{e}-9$ m to avoid a possible divide by zero error.
12. After you adjust the temperature, `VSAT` cannot go below a minimum value of $1.0\text{e}4$ m/sec to assure that it is positive after temperature compensation.
13. Seven model parameters accommodate the temperature dependencies of six temperature-dependent model variables. These parameters are `KT1` and `KT2` for V_{th} , `UTE` for `U0`, `AT` for `VSAT`, `UA1` for `UA`, `UB1` for `UB`, and `UC1` for `UC`.
14. Set up the temperature conversion between this model and SPICE3 as follows:

```
SPICE3:  .OPTION TEMP=125
         .MODEL NCH NMOS Level=8
         + TNOM=27 ...
HSPICE:  .TEMP 125
         .MODEL NCH NMOS Level=47
         + TREF=27 ...
```

15. The `SCALM` option does not affect parameters that are unique to this model, but it does affect the common MOS parameters, such as `XL`, `LD`, `XW`, `WD`, `CJ`, `CJSW`, `JS`, and `JSW`.
16. Level 47 uses the common Synopsys MOS parasitic models, which `ACM` specifies.
17. Level 47 uses the common Synopsys MOS noise models, which `NLEV` specifies.
18. You can use `DELVTO` and `DTEMP` on the element line with MOSFET Level 47.
19. The `PSCBE1` and `PSCBE2` model parameters determine the impact ionization current, which contributes to the drain-source current. Impact ionization does not contribute to the bulk current.

BSIM3 Version 2 MOSFET Model — PMOS Model Example

In the following example of a PMOS model for the Level 47 MOSFET, `VTH0` is negative.

```
.model pch PMOS Level=47
+ Tnom=27.0
+ Npeak=1.5E+23 Tox=7.0E-09 Xj=1.0E-07
+ dl=0.2E-06 dw=-0.1E-06
+ SatMod=2 SubthMod=2 BulkMod=1
+ Vth0=-.8 Phi=.7 K1=.5 K2=0.03 K3=0
+ Dvt0=48 Dvt1=.6 Dvt2=-5e-4
+ Nlx=0 W0=0
+ Vsat=9E6 Ua=1E-09 Ub=0 Uc=-3E-02
+ Rds0=180 Rdsw=0 U0=7E-03
+ A0=.87
+ Voff=-.07 NFactor=1.5 Cit=-3E-05
+ Cdsc=6E-02 Vglow=-.12 Vghigh=.12
+ Pclm=77 Pdibl1=0 Pdibl2=2E-011
+ Drout=0 Pscbe1=0 Pscbe2=1E-28
+ Eta=0 Lit1=4.5E-08
+ Em=0 Ldd=0
+ kt1=-.3 kt2=-.03
+ At=33000
+ Ua1=4E-09 Ub1=7E-18 Uc1=0
```

LEVEL 49 and 53: BSIM3 Version 3 MOSFET Model

The Synopsys Level 49 and Level 53 models are based on the BSIM3 version 3 MOSFET model from UC Berkeley.

- Level 49 of the PrimeSim Continuum solution is the -enhanced version of BSIM3 version 3. Level 49 maintains compliance with the UC Berkeley release of BSIM3 version 3 with the following three exceptions:
 - Default parameter values — To eliminate differences in default parameter values, Level 49 explicitly assigns the `CAPMOD` and `XPART` parameters, and sets `ACM=10`. In addition, the default in Level=53 for `XPART` is “0” whereas in Level=49 it is “1”.
 - Parameter range limits — Provides parameter range limits that are identical to that of the Berkeley release. Differences occur only in the severity of the warning for five parameters. Level 49 issues a warning that the model exceeded the parameter range, but continues with the simulation.
 - However, the Berkeley release issues a fatal error, and aborts the simulation. These five parameters are `NGATE`, `DVT1W`, `DVT1`, `DSUB`, and `DROUT`. (See [BSIM3 Version 3 MOSFET Model — Parameter Range Limits](#) for more details.)
 - Improvements in numerical stability—Provides improvements in numerical stability. In most practical situations, these improvements do not affect compliance with the Berkeley release, but improve the convergence and the simulation time.
- Level 53 maintains full compliance with the Berkeley release, including numerically-identical model equations, identical parameter default values, and identical parameter range limits.
- Level 49 and 53 both support the following instance parameters, along with the `DELVTO` instance parameter for local mismatch and NBTI (negative bias temperature instability) modeling:
 - `MULU0`, low-field mobility (μ_0) multiplier. Default=1.0.
 - `MULUA`, first-order mobility degradation coefficient (μ_A) multiplier.
 - `MULUB`, second-order mobility degradation coefficient (μ_B) multiplier.

Both Levels 49 and 53 support a superset of model parameters that include the PrimeSim Continuum solution -specific parameters. For Level 53, in all cases, the PrimeSim Continuum solution -specific parameters default to OFF. The single exception in Level 49 is that `ACM` defaults to 0. To achieve Level 49 compliance with Berkeley BSIM3 version 3, set `ACM=10`.

If you set any of the following parameter values for MOSFET Level 49 and 53, simulation reports a warning:

- $L_{\text{eff}} \leq 5\text{e-}8$
- $W_{\text{eff}} \leq 1\text{e-}7$
- $L_{\text{eff}}\text{CV} \leq 5\text{e-}8$
- $W_{\text{eff}}\text{CV} \leq 1\text{e-}7$

Simulation aborts if you set L_{eff} or $W_{\text{eff}} \leq 0.0$.

For Level=49 and 53, LINT is used in the ACM model to calculate RD_{eff} and RS_{eff} for BSIM3 version 3 (LD gets the LINT value).

The following sections discuss these topics:

- [BSIM3 Version 3 MOSFET Model — Selecting Model Versions](#)
- [BSIM3 Version 3.x MOSFET Model Features](#)
- [Using BSIM3 Version 3 MOSFET Model](#)
- [BSIM3 Version 3 MOSFET Model — Usage Notes](#)
- [BSIM3 Version 3 MOSFET Model Parameters](#)
- [BSIM3 Version 3 MOSFET Model — Parameter Range Limits](#)
- [BSIM3 Version 3 MOSFET Model Equations](#)
- [BSIM3 Version 3 MOSFET Model — Additional Information](#)
- [BSIM3 Version 3 MOSFET Model Examples](#)

BSIM3 Version 3 MOSFET Model — Selecting Model Versions

The recommended BSIM3 version 3 model specification is `LEVEL=49, VERSION=3.3.0`. This version provides the most stable and up-to-date representation of the BSIM3 version 3 model.

However, do not change the `VERSION` specification in existing model cards without consulting the foundry or model extraction group that created the model cards.

The following table lists the available BSIM3 version 3 models and their release dates from UC Berkeley and the equivalent the PrimeSim Continuum solution versions.

Berkeley Release	Model Version	PrimeSim Continuum Version
Version 3.3.0 (July, 2005)	3.3.0	06.03
Version 3.2.2 (April 20, 1999)	3.22 3.23 3.24	99.2 01.4 02.2
Version 3.2.1 (April 20, 1999)	3.21	99.2
Version 3.2 (June 16, 1998)	3.2	98.2
Version 3.1 with June 1998 bug fixes	3.1	98.2
Version 3.1 (December 1996)	3.1	98.0
Version 3.0 with June 1998 bug fixes	3.0	98.2
Version 3.0 (October 1995)	3.0	98.0

As of the 99.2 release, there are multiple BSIM3 version 3 releases from Berkeley and several Level 49 releases. For additional release information from the UCB group, see the BSIM3 home page: <http://www-device.EECS.Berkeley.EDU/~bsim3/>

To minimize confusion and maintain backward compatibility, you can select the `VERSION` and `HSPVER` model parameters. `VERSION` selects the Berkeley release version; `HSPVER` selects the Synopsys release version. For example, `HSPVER=97.2` and `VERSION=3.1` reproduce results from the PrimeSim Continuum solution 97.2 using the BSIM3 Version 3.1 model. For detailed discussion of how `HSPVER` relates to the UCB `VERSION`, see [Using BSIM3 Version 3 MOSFET Model](#).

`HSPVER` defaults to the current release that you are using. The `VERSION` model parameter selects among the various Berkeley releases of BSIM3 version 3 as follows:

- Version 3.0 Berkeley release (October 30, 1995) defaults for the PrimeSim Continuum solution 96.1, 96.2, 96.3.

Simulation invokes this version if you specify `VERSION=3.0` and `HSPVER=98.0`. To invoke the Synopsys model version that most accurately represents the Berkeley release of October 1995, specify the `VERSION=3.0` and `HSPVER=98.0` parameters.

- Version 3.1 Berkeley (December 9, 1997) defaults for the PrimeSim Continuum solution 97.1, 97.2, 97.4.

Simulation invokes this version if you specify `VERSION=3.1` or `3.11` and `HSPVER=98.0`. To invoke the Synopsys model version that most accurately represents the Berkeley release of December 1996, specify the `VERSION=3.1` or `3.11` and `HSPVER=98.0` parameters.

- Berkeley Version 3.0, 3.1 bug fixes. Berkeley corrected several Version 3.0 and 3.1 bugs in the June, 1998 release. These fixes are incorporated into the PrimeSim Continuum solution 98.2, which simulation uses if you specify `VERSION=3.0` or `VERSION=3.1` and `HSPVER=98.2`. As a result of bug fixes, you might notice some differences between Version 3.0/3.1 in PrimeSim Continuum 98.2 and previous Version 3.0/3.1 releases. Notably, differences occur if you specify `PD` and `PS` perimeter factors that are less than W_{eff} (`PD,PS < W_{eff}` no longer clamp to W_{eff} in Version 3.1) and if `DLC` and `LINT` are not identical ($L_{eff}CV$ calculation bug in Versions 3.0 and 3.1).

For a complete list of bug fixes, see the BSIM3 web site: <http://www-device.eecs.berkeley.edu/~bsim3>

Note:

Version 3.11 was introduced in the PrimeSim Continuum solution. This version represents Berkeley Version 3.1 (Dec., 1996) with the PrimeSim Continuum solution bug fixes. This model maintains backward compatibility. Starting with 98.2, Version 3.1 and 3.11 are identical, and represent Version 3.1 with Berkeley June, 1998 bug fixes.

- Version 3.2 Berkeley release (June 16, 1998). Simulation invokes this version if you specify `VERSION=3.2` and `HSPVER=98.2`.
- Version 3.2.1 Berkeley release (April 20, 1999). Simulation invokes this version if you specify `VERSION=3.21` and `HSPVER=99.2`.
- Version 3.2.2 Berkeley release (April 20, 1999). Simulation invokes this version if you specify `VERSION=3.22` and `HSPVER=99.2`.
- For the latest the PrimeSim Continuum solution improvements, use `VERSION=3.3.0` and `HSPVER=06.03`.

Note:

Versions 3.2.1 and 3.2.2 are identical, except BSIM3 version 3.2.1 uses a bias-dependent V_{fb} and BSIM3 version 3.2.2 uses a bias-independent V_{fb} for the `CAPMOD=1` and `2` capacitance models.

BSIM3 Version 3.x MOSFET Model Features

This section covers the following topics:

- [BSIM3 Version 3.3 Model Features](#)
- [BSIM3 Version 3.2 Model Features](#)

BSIM3 Version 3.3 Model Features

In July 2005, Berkeley released BSIM3 Version 3.3, which includes the following new features:

- A new small-signal AC charge-deficit ACNQS model that enables the NQS effect in AC simulation. This can be turned on by setting `acnqsMod=1` and off by setting `acnqsMod=0`.
- A new channel thermal noise model “SPICE2new” that the thermal noise coefficient varies smoothly between 4 to 8/3 as the device moves from a linear to a saturation region. The new `noiMod` model flags are:
 - `noiMod=5`: Flicker noise model — SPICE2; Thermal noise model — SPICE2new
 - `noiMod=6`: Flicker noise model — BSIM3 version 3; Thermal noise model — SPICE2new
- Addition of a `LINTNOI` parameter that introduces an offset to the length reduction parameter (`Lint`) to improve the accuracy of the flicker noise model.
- Changed the `Rds0` source drain current contact resistance check model.
- Fixed bugs in BSIM3 version 3.24.

For more information about the Berkeley releases, see the BSIM3 web site: <http://www-device.eecs.berkeley.edu/~bsim3>

BSIM3 Version 3.2 Model Features

In June 1998, Berkeley released BSIM3 Version 3.2, which includes the following new features:

- A new intrinsic capacitance model, `CAPMOD=3`, includes finite charge layer thickness effects; `CAPMOD` now defaults to 3 (new parameters: `CAPMOD=3`, `ACDE`, and `MOIN`).
- Improved modeling of C-V characteristics at the weak-to-strong inversion transition (new parameters: `NOFF` and `VOFFCV`).
- V_{th} dependence on T_{ox} (new parameter: `TOXM`).
- Flatband voltage parameter more accurately models different gate materials (new parameter: `VFB`).
- Improved substrate current scalability with channel length (new parameter: `APLHA1`).
- Restructured nonquasi-static (NQS) model includes pole-zero analysis and bug fixes. `NQSMOD` is a BSIM3 element parameter. This is supported as an element parameter too.
- Junction diode model temperature dependence (new parameters: `TCJ`, `TCJSW`, `TCJSWG`, `TPB`, `TPBSW`, and `TPBSWG`).

- Adjustable current limiting in the junction diode current model (new parameter: `IJTH`).
- Use C-V inversion charge equations (`CAPMOD=0,1,2,3`) to calculate the thermal noise if `NOIMOD=2` or `4`.
- Eliminated the small negative capacitance values (`Cgs` and `Cgd`) in the accumulation-depletion regions.
- Separate set of length/width dependence parameters for the CV model (`LLC`, `LWC`, `LWLC`, `WLC`, `WWC`, and `WWLC` parameters).
- Additional parameter checking.
- Bug fixes.

Note:

If you use the defaults for all new Version 3.2 parameters, Version 3.2 and Version 3.1 (with June, 1998 bug fixes) return identical DC results. However, transient and AC results generally differ. This discrepancy arises only from differences in the flatband voltage calculations used in the intrinsic charge/capacitance models. These differences occur in all `CAPMOD` models 1-3.

- Level 53 resets `HSPVER < 98.0` to `98.0`.
- `HSPVER < 98.2` resets to `98.2` if `VERSION >= 3.2` for Levels 49/53.
- Version 3.0, 3.1, and 3.1.1 in the PrimeSim Continuum solution do not support `NQSMOD` and `CAPMOD=3`. Only Version 3.2 supports them.
- Version 3.24 added `Rds` noise to the thermal noise model. Simulation smooths out the unified flicker noise, from the linear region to the saturation region. You might need to re-extract the parameters for the unified flicker noise model.

For more information about the Berkeley releases, see the BSIM3 website: <http://www-device.eecs.berkeley.edu/~bsim3>

Using BSIM3 Version 3 MOSFET Model

Note the following points when you use BSIM3 version 3 with a Synopsys circuit simulator:

- Use either the Level 49 or Level 53 model. Level 53 fully complies with the Berkeley BSIM3 version 3 release. In most cases Level 49 returns the same results as Level 53, runs as fast or faster, shows better convergence, and allows a wider range of parameter specifications.
- Explicitly set all Berkeley-specific BSIM3 model parameters in the model card. This minimizes problems resulting from version changes and compatibility with other simulators. You do not explicitly set all lwp binning parameters.

- To match results with simulations from previous the PrimeSim Continuum solution versions, use the `HSPVER=YY.N` model parameter, such as `HSPVER=97.4`. Do not use the full year specification (such as 1997.4). The patch version number format is `HSPVER=YY.NN` (for example, `HSPVER=98.21` is release 98.2.1).
- Levels 49 and 53 support the `TNOM` model parameter name as an alias for `TREF`. The conventional terminology in the PrimeSim Continuum solution is `TREF`, which all Synopsys model MOS levels support as a model parameter. Both Levels 49 and 53 support the `TNOM` alternative name for compatibility with SPICE3.

The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in SPICE3. If you specify the BSIM3 model parameters at 27°C, add `TNOM=27` to the model so that simulation correctly interprets the model parameters. To set the nominal simulation temperature to 27, add `.OPTION TNOM=27` to the netlist when you test the Synopsys model versus SPICE3.

You can use `DELVTO` and `DTEMP` on the element line with Levels 49 and 53. The following equation converts the temperature setup between the Synopsys model and SPICE3:

```
SPICE3:      .OPTION TEMP=125
             .MODEL NCH NMOS Level=8
             + TNOM=27 ...
Synopsys Model:      .TEMP 125
             .MODEL NCH NMOS Level=49
             + TNOM=27 ...
```

- To automatically calculate the drain, source area, and perimeter factors for Berkeley junction diode models, use `ACM=12` with `CALCACM=1`. Normally, `ACM=10-13` defaults the area and perimeter factors to 0. To override this value for `ACM=12`, specify `CALCACM=1`. Define the PrimeSim Continuum solution -specific parameter (`HDIF`) in the model card. If you do not want parasitic `Rs` and `Rd` with the BSIM3 version 3 internal `Rsd`, either do not specify the `RSH`, `RSC`, `RDC`, `RS`, and `RD` in the PrimeSim Continuum solution parameters (default is 0) or set them to 0.
- Simulation and analysis either warns or aborts with a fatal error if certain model parameter values are out of a normal range. To view all warnings, you might need to increase the `.OPTION WARNLIMIT` value (default=1). To turn on full parameter range checking, set the `PARAMCHK=1` model parameter (default is 0). If you use `PARAMCHK=0`, simulation checks a smaller set of parameters. (See Note below; also see [BSIM3 Version 3 MOSFET Model — Parameter Range Limits](#) for more details about parameter limits.) Use the `APWARN=1` model parameter (default=0) to turn off `PS`, `PD` < W_{eff} warnings.
- Use `NQSMOD` only with Version 3.2. This is supported as an element parameter too.

Note:

The default setting of `PARAMCHK` can influence MOSFET behavior in the PrimeSim Continuum solution in not checking and/or enforcing the range limits for some MOSFET model parameters. `PARAMCHK` selects between a default and enhanced level of MOSFET parameter checks. If set to 0, only default checks are performed.

If set to 1, an enhanced set of parameter checks are performed and certain parameters will be reset to nominal values if they are found to be out of range.

Because the default setting of `PARAMCHK` can vary between model levels (e.g., 49/53=0 and 54=1), it is important to implicitly set `PARAMCHK` in your model file or insure it has been set by the model vendor. A failure to do so can cause unexpected variations in simulation results and a failure to correlate with other simulators that may have different default settings.

BSIM3 Version 3 MOSFET Model — Usage Notes

- [BSIM3 Version 3 Model — Usage Notes: Enhanced Diode Model DC Equations](#)
- [BSIM3 Version 3 Model — Usage Notes: Nonquasi-Static \(NQS\) Model](#)
- [BSIM3 Version 3 Model — Usage Notes: PrimeSim Continuum Junction Diode Model and Area Calculation Method](#)
- [BSIM3 Version 3 Model — Usage Notes: Reverse Junction Breakdown Model](#)
- [BSIM3 Version 3 Model — Usage Notes: TSMC Diode Model](#)
- [BSIM3 Version 3 Model — Usage Notes: STI / LOD Parameters](#)
- [BSIM3 Version 3 Model — Usage Notes: Parameter Differences](#)
- [BSIM3 Version 3 Model — Usage Notes: Noise Model](#)
- [BSIM3 Version 3 Model — Usage Notes: Performance Improvements](#)
- [BSIM3 Version 3 Model — Usage Notes: Reduced Parameter Set \(BSIM3-lite\)](#)
- [BSIM3 Version 3 Model — Usage Notes: Parameter Binning](#)
- [BSIM3 Version 3 Model — Usage Notes: WPE Model](#)
- [BSIM3 Version 3 Model — Usage Notes: Ig Model](#)
- [BSIM3 Version 3 Model — Usage Notes: Charge Models](#)

- [BSIM3 Version 3 Model — Usage Notes: Printback Options](#)
- [BSIM3 Version 3 Model — Usage Notes: Mobility Multiplier](#)

BSIM3 Version 3 Model — Usage Notes: Enhanced Diode Model DC Equations

BSIM3 diode IV equations, considering carrier recombination and trap-assisted tunneling current to improve the I_{off}-related value. This section describes the use model for this enhancement.

The model parameter `bsim4diode` invokes the feature. In addition, 30 related model parameters are also added for this enhancement: `jtss`, `jtsd`, `jtssws`, `jtsswd`, `jtsswgs`, `jtsswgd`, `njts`, `njtssw`, `njtsswg`, `xtss`, `xtsd`, `xtssws`, `xtsswd`, `xtsswgs`, `xtsswgd`, `tnjts`, `tnjtssw`, `tnjtsswg`, `vtss`, `vtssd`, `vtssws`, `vtsswd`, `vtsswgs`, `vtsswgd`, `njtssd`, `njtsswd`, `njtsswgd`, `tnjtssd`, `tnjtsswd`, and `tnjtsswgd`. This diode model is only available under `AMC=12` for version `>=3.2`.

Syntax:

```
.model nch nmos level=49...
+ bsim4diode=1 acm=12 version=3.2....
```

These equations come from BSIM4.

$$\begin{aligned}
 I_{bs_total} &= I_{bs} \cdot W_{effcj} \cdot JTSSWGS(T) \cdot \left(e^{\frac{-V_{bs}}{NJTSSWG(T)V_{tm0}}} \frac{VTSSWGS}{VTSSWGS - V_{bs}} - 1 \right) \\
 &\quad - P_{s_eff} \cdot JTSSWS(T) \cdot \left(e^{\frac{-V_{bs}}{NJTSSW(T)V_{tm0}}} \frac{VTSSWS}{VTSSWGS - V_{bs}} - 1 \right) \\
 &\quad - A_{s_eff} \cdot JTSS(T) \cdot \left(e^{\frac{-V_{bs}}{NJTS(T)V_{tm0}}} \frac{VTSS}{VTSS - V_{bs}} - 1 \right) \\
 I_{bd_total} &= I_{bd} \cdot W_{effcj} \cdot JTSSWGD(T) \cdot \left(e^{\frac{-V_{bd}}{NJTSSWG(T)V_{tm0}}} \frac{VTSSWGD}{VTSSWGS - V_{bd}} - 1 \right) \\
 &\quad - P_{d_eff} \cdot JTSSWD(T) \cdot \left(e^{\frac{-V_{bd}}{NJTSSW(T)V_{tm0}}} \frac{VTSSWD}{VTSSWD - V_{bd}} - 1 \right) \\
 &\quad - A_{s_eff} \cdot JTSD(T) \cdot \left(e^{\frac{-V_{bd}}{NJTS(T)V_{tm0}}} \frac{VTSS}{VTSD - V_{bd}} - 1 \right)
 \end{aligned}$$

where:

$$JTSSWGS(T) = JTSSWGS \cdot e^{\frac{JTSSWGS \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$JTSSWS(T) = JTSSWS \cdot e^{\frac{JTSSWS \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$JTSS(T) = JTSS \cdot e^{\frac{JTSSWGS \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$JTSSWGD(T) = JTSSWGD \cdot e^{\frac{JTSSWGD \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$JTSSWD(T) = JTSSWD \cdot e^{\frac{JTSSWD \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$JTSSD = JTSD \cdot e^{\frac{JTSD \cdot \frac{Eg0}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}{V_{tm} \cdot \frac{Temp}{Tnom} - 1}}$$

$$NJTSSWG(T) = NJTSSWG \cdot \left[1 + TN_{JTSSWG} \cdot \left(\frac{Temp}{Tnom} - 1 \right) \right]$$

$$NJTSSW(T) = NJTSSW \cdot \left[1 + TN_{JTSSW} \cdot \left(\frac{Temp}{Tnom} - 1 \right) \right]$$

$$NJTS(T) = NJTS \cdot \left[1 + TN_{JTS} \cdot \left(\frac{Temp}{Tnom} - 1 \right) \right]$$

Note:

These notes apply for this upgrade:

- The BSIM3 model does not have BSIM4-like equations for Aseff/ADeff/ Pseff/Pdeff equations. In the above equations, they are replaced by ->BSIM3sourceArea, ->BSIM3drainArea, ->BSIM3sourcePerimeter, and ->BSIM3drainPerimeter.
- Model parameter JTWEFF and related equations are not implemented for BSIM3.
- Eg0-related model parameters including MTRLMOD and BG0SUB / BGASUB / BGBSUB are not implemented for BSIM3.

BSIM3 Version 3 Model — Usage Notes: Nonquasi-Static (NQS) Model

You can also select the Berkeley NonQuasi-Static (NQS) model for Levels 49 and 53. This model provides a first-order correction to the quasi-static charge models. See M.Chan, K.- Y. Hui, C. Hu, and P.-K. Ko, IEEE Trans. Electron Devices, vol. ED-45, pp.834-841, 1998. The Level 49/53 MOSFET model supports only the model parameter implementation.

To invoke the NQS model, specify the `NQSMOD=1` parameter in the model card. You can use `NQSMOD` with any of the `CAPMOD` Levels (0-3) but only with Version 3.2. Version 3.0 and 3.1 do not support NQS.

Note:

The PrimeSim Continuum solution supports both NDQSMOD model and element parameters.

BSIM3 Version 3 Model — Usage Notes: PrimeSim Continuum Junction Diode Model and Area Calculation Method

You can use two junction diode models with both Levels 49 and 53: the PrimeSim Continuum solution junction model and the Berkeley junction model.

- For the PrimeSim Continuum solution junction model, specify the `ACM=0,1,2, or 3` model parameter value.
- For the Berkeley junction model, specify `ACM=10,11,12, or 13`.

The default `ACM` value is 0 for Levels 49 or 10 for Level 53. For the junction current, junction capacitance, and parasitic resistance equations corresponding to `ACM=0,1,2,3` see [MOS Diode Equations](#).

Set `ACM=10,11,12, or 13` to enable the Berkeley junction diodes and to add parasitic resistors to the MOSFET. The parasitic resistor equations for `ACM=10-13` correspond to the `ACM=0-3` parasitic resistor equations. `ACM=10-13` all use the Berkeley junction capacitance model equations:

```
(Bulk-source capacitance)
if (Ps > Weff)
  Cbs=AS * Cjbs + (PS - Weff) * Cjbssw + Weff *
  Cjbsswg
else
  Cbs=AS * Cjbs + PS * Cjbsswg
```

The `AS` and `PS` area and perimeter factors default to 0, if you do not specify them on the element line.

```
if (Vbs < 0)
  Cjbs=Cj * (1 - (Vbs/Pb)) -Mj
  Cjbssw=Cjsw * (1 - (Vbs/Pbsw)) -Mjsw
  Cjbsswg=Cjswg * (1 - (Vbs/Pbswg)) -Mjswg
else
  Cjbs=Cj * (1 + Mj * (Vbs/Pb))
  Cjbssw=Cjsw * (1 + Mjsw * (Vbs/Pbsw))
  Cjbsswg=Cjswg * (1 + Mjswg * (Vbs/Pbswg))
```

Bulk-drain equations are analogous. `ACM=10,11,12,13` do not use the PrimeSim Continuum solution equations for `AS`, `PS`, `AD`, and `PD`. In accordance with the BSIM3 version 3 model, the default values for these area and perimeter factors are zero. To invoke the PrimeSim

Continuum solution calculations for AS, PS, AD, and PD, specify the `CALCACM=1` model parameter.

Note:

Simulation invokes `CALCACM` only if `ACM=12`. The calculations used in `ACM=10, 11, and 13` are not consistent with the Berkeley diode calculations.

`CALCACM=1` and `ACM=12` invoke the following area and perimeter calculations:

If you do not specify `AD` on the element line:

```
AD=2 * HDIFeff * Weff
else:
AD=AD * WMLT^2
```

If you do not specify `AS` on the element line:

```
AS=2 * HDIFeff * Weff
else:
AS=AS * WMLT^2
```

If you do not specify `PS` on the element line:

```
PS=4 * HDIFeff + 2 * Weff
else:
PS=PS * WMLT
```

If you do not specify `PD` on the element line:

```
PD=4 * HDIFeff + 2 * Weff
else:
PD=PD * WMLT
```

Note:

W_{eff} is not the same W_{eff} used in the BSIM3 version 3, and Levels 49 and 53 I-V, C-V model equations.

The preceding equations use the following simple forms:

```
Weff=W * WMLT + XW
HDIFeff=HDIF * WMLT
```

Parameter	Description
HDIF	Heavy diffusion length specified in the model card
W	Width specified on the element line
WMLT	Shrink factor specified in the model card
XW	Etch/mask effect factor specified in the model card

Note:

These equations ignore the `SCALM`, `SCALE`, and `M` factor effects. See [Using an ACM=2 MOS Diode](#) (`ACM=2`) for further details.

BSIM3 Version 3 Model — Usage Notes: Reverse Junction Breakdown Model

The PrimeSim Continuum solution of MOSFET Level 49-53 (BSIM Level 3.2 or later) supports the Reverse Junction Breakdown model (introduced with BSIM4.6.4). This functionality uses similar diode model equations as for BSIM4.6.4 `DIOMOD=2`. Refer to Equation (11.5) of the BSIM4.6.4 manual for details. Note that the equation for forward biased region is affected accordingly.

Parameter	Default	Description
BREAKMOD	0 (off)	Flag to turn on/off reverse junction breakdown model, set to 1 to turn on
BVD	bvs	Drain diode breakdown voltage
BVS	10 (V)	Source diode breakdown voltage
IJTHDREV	ijthsrev	Reverse drain diode forward limiting current
IJTHSREV	0.1 (A)	Reverse source diode forward limiting current
XJBVD	xjbvs	Fitting parameter for drain diode breakdown current
XJBVS	1.0 (no unit)	Fitting parameter for source diode breakdown current

If the above parameters are less than or equal to zero, the parameters are reset to 0.0 and warnings are issued as follows:

- If (`bvs` ≤ 0.0), “`bvs` reset to 0.0”.
- If (`bvd` ≤ 0.0), “`bvd` reset to 0.0”.
- If (`ijthsrev` ≤ 0.0), “`ijthsrev` reset to 0.0”.
- If (`ijthdrev` ≤ 0.0), “`ijthdrev` reset to 0.0”.
- If (`xjbvs` ≤ 0.0), “`xjbvs` reset to 0.0”.
- If (`xjbvd` ≤ 0.0), “`xjbvd` reset to 0.0”.

BSIM3 Version 3 Model — Usage Notes: TSMC Diode Model

The PrimeSim Continuum solution of MOSFET Level 49 (`ACM=12`, BSIM3 version 3.2 or later) supports a TSMC diode model. You can use this TSMC diode model to simulate the

breakdown effect, the resistance-induced non-ideality factor, and geometry-dependent reverse current of a diode.

Order this model directly from Taiwan Semiconductor Manufacturing Company (TSMC), not from Synopsys. See the TSMC web site: <http://www.tsmc.com>

BSIM3 Version 3 Model — Usage Notes: STI / LOD Parameters

The PrimeSim Continuum solution of BSIM3 version 3 (Level=49 and 53) supports UC Berkeley's STI/LOD stress effect model. To turn on this stress effect model in BSIM3 version 3, specify `STIMOD=1` in your model cards.

Parameter	Unit	Default	Bin	Description
STIMOD (Also instance parameter)		0.0		STI model selector, which gives priority to the instance parameter. • 0: No STI effect. • 1: IDB's STI model • 2: GSHUNT's STI model
KU0	M	0.0	No	Mobility degradation/enhancement coefficient for stress effect
KVSAT	M	0.0	No	Saturation velocity degradation/enhancement parameter for stress effect. $1.0 \leq kvsat \leq 1.0$
KVTH0	V*m	0.0	No	Threshold shift parameter for stress effect
LKU0		0.0	No	Length dependence of KU0
LLODKU0		0.0	No	Length parameter for U0 stress effect, >0
LLODVTH		0.0	No	Length parameter for Vth stress effect, >0
LODETA0	M	1.0	No	ETA0 shift modification factor for stress effect, >0
LODK2	m	1.0	No	K2 shift modification factor for stress effect, >0
PKVTH0		0.0	No	Cross-term dependence of KVTH0
SA (instance parameter)		0.0		Distance between S/D diffusion edge to poly gate edge from one side. If not given, or, if (≤ 0), the stress effect is turned off.
SAREF	M	1e-06	No	Reference distance for SA, >0.0
SB (instance parameter)		0.0		Distance between S/D diffusion edge to poly gate edge from the other side. If not given, or, if (≤ 0), the stress effect is turned off
SBREF	M	1e-06	No	Reference distance for SB, >0.0

Parameter	Unit	Default	Bin	Description
STETA0		0.0	No	ETA0 shift factor related to VTH0 change
STK2		0.0	No	K2 shift factor related to VTh0 change
TKU0		0.0	No	Temperature coefficient of KU0
WKU0		0.0	No	Width dependence of KU0
WKVTH0		0.0	No	Width dependence of KVTH0
WLOD	M	0.0	No	Width parameter for stress effect
WLODKU0		0.0	No	Width parameter for U0 stress effect, >0
WLODVTH		0.0	No	Width parameter for Vth stress effect, >0

BSIM3 Version 3 Model — Usage Notes: Parameter Differences

Some parameter names differ between the Synopsys model and the Berkeley junction models. The Synopsys models ($ACM=0-3$) do not recognize the following BSIM3 version 3 parameters:

- **CJSWG** (ignored, use **CJGATE** instead)
- **MJSWG** (ignored; the PrimeSim Continuum solution has no equivalent parameter, and simulation sets the gate sidewall grading coefficient=**MJSW**)
- **PBSW** (ignored, use **PHP** instead)
- **PBSWG** (ignored; the PrimeSim Continuum solution has no equivalent parameter, and simulation sets the gate sidewall contact potential=**PHP**)

The Berkeley model ($ACM=10,11,12,13$) does not recognize the following parameters:

- **CJGATE** (ignored, use **CJSWG** instead)
- **PHP** (ignored, use **PBSW** instead)

BSIM3 Version 3 Model — Usage Notes: Noise Model

The PrimeSim Continuum solution **NLEV** parameter overrides the BSIM3 version 3 **NOIMOD** parameter. Specifying **NLEV** invokes the PrimeSim Continuum solution noise model.

See [MOSFET Noise Models](#) for more information. If you do not specify **NLEV**, simulation invokes the Berkeley noise equations.

BSIM3 Version 3 Model — Usage Notes: Performance Improvements

To improve the performance of Levels 49 and 53 reduce the complexity of model equations, replacing some calculations with spline functions and compiler optimization. For Level 49, the result is a reduction in simulation time of up to 40% compared to releases before 97.4 while maintaining accuracy to 5 digits or better. To use the spline functions, set the `SFVTFLAG=1` model parameter in the model card. `SFVTFLAG=0`, the default value, disables the spline functions. For Level 53, all BSIM3 version 3 non-compliant features default to off.

BSIM3 Version 3 Model — Usage Notes: Reduced Parameter Set (BSIM3-lite)

Setting the `LITE=1` model parameter in Level 49 to invoke the BSIM3 version 3-lite reduced parameter set model. Use it with model binning. Without binning to account for geometry effects, the full BSIM3 version 3 model specifies several model parameters. However, it is often difficult to extract a “global” BSIM3 version 3 model that is accurate over the entire geometry range.

To improve accuracy over a range of geometries, you can bin the model parameters. That is, this model divides the entire length-width geometry range into rectangular regions or bins. Simulation extracts a different set of parameters for each bin. The built-in bilinear parameter interpolation scheme maintains continuity (over length-width) at the boundaries between bins. Because many BSIM3 model parameters account for MOSFET geometry effects, these geometry-effect parameters are redundant. You can eliminate them when you use binning.

The BSIM3-lite model parameter set was created in response to the question: What BSIM3 parameters should be excluded when using a binned model? To invoke the BSIM3-lite model, specify the `LITE=1` model parameter in the model card.

Simulation checks the model card to determine if it conforms to the BSIM3-lite parameter set. BSIM3-lite takes advantage of the smaller number of calculations, and reduces simulation times by up to 10% compared to the full parameter set BSIM3 model. Only Level 49 supports `LITE=1`.

The following table lists model parameters (total 49) that the BSIM3-lite model excludes. Either exclude all parameters in this list from the model card or explicitly set them to the default value specified in the list. In some cases, as noted, the BSIM3-lite default value differs from the standard BSIM3 version 3 default value. You should also exclude `WR`, `ALPHA0`, and `CIT`, but the BSIM3-lite model card does not require this exclusion.

Parameter	Comments
MOBMOD	Recommended default or set=1

Parameter	Comments
NQSMOD	Recommended default or set=0
ALPHA0	Recommended default or set=0 for Version 3.2
B0	default=0
B1	default=0
CIT	Recommended default or set=0
DROUT	default=0, std default=0.56
DSUB	default=0
DVT0	default=0, std default=2.2
DVT0W	default=0
DVT1	default=0, std default=0.53
DVT1W	default=0, std default=5.3e6
DVT2	default=0, std default=-0.032
DVT2W	default=0, std default=-0.032
DWB	default=0
DWG	default=0
GAMMA1	do not define
GAMMA2	do not define
K3	default=0, std default=80
K3B	default=0
KT1L	default=0
LL	default=0
LLC	default=0
LLN	default=1
LW	default=0
LWC	default=0
LWL	default=0

Parameter	Comments
LWLC	default=0
LWN	default=1
NGATE	Recommended default or set=0
NLX	default=0, std default=1.74e-7
NSUB	do not define
PDIBLC1	default=0, std default=0.39
PRWB	default=0
PRWG	default=0
TOXM	default=tox
VBM	do not define
VBX	do not define
W0	no effect
WL	default=0
WLC	default=0
WLN	default=1
WR	Recommended default or set=1
WW	default=0
WWC	default=0
WWL	default=0
WWLC	default=0
WWN	default=1
XT	do not define

BSIM3 Version 3 Model — Usage Notes: Parameter Binning

To support parameter binning, Berkeley BSIM3 version 3 specifies `LWP` parameters. To bilinearly interpolate a subset of model parameters over $1/L_{\text{eff}}$ and $1/W_{\text{eff}}$, you specify four terms:

- `Xo` parameter
- `X` length term
- `Xw` width term
- `Xp` product term

The simulation then interpolates the parameter value at specified `L,W`.

$$X = X_o + X_l / L_{\text{eff}} + X_w / W_{\text{eff}} + X_p / L_{\text{eff}} / W_{\text{eff}}$$

See [BSIM3 Version 3 MOSFET Model — Parameter Range Limits](#) to determine whether you can bin a parameter. Simulation adds the `LMIN`, `LMAX`, `WMIN`, `WMAX`, `LREF`, and `WREF` parameters to allow multiple cell binning. `LMIN`, `LMAX`, `WMIN`, `WMAX` define the cell boundary. `LREF` and `WREF` are offset values that provide a convenient interpolation scheme. The simulation uses the `LREF` and `WREF` offsets if you define both values and you specify the `BINFLAG > 0.9` model parameter.

The simulation then interpolates the parameter value at a specified `L,W`:

$$X = X_o + X_l * (1 / L_{\text{eff}} - 1 / L_{\text{REF}}) + X_w * (1 / W_{\text{eff}} - 1 / W_{\text{REF}}) + X_p / (1 / L_{\text{eff}} - 1 / L_{\text{REF}}) / (1 / W_{\text{eff}} - 1 / W_{\text{REF}})$$

To select micron units for the `lwp` geometry parameters, set the `BINUNIT=1` model parameter. For other choices of `BINUNIT`, the lengths are in units of meters. Simulation handles the `XL`, `XLREF`, `XW`, and `XWREF` parameters in a manner consistent with other Synopsys MOSFET models, and they produce shifts in parameter values without disrupting the continuity across the bin boundaries.

BSIM3 Version 3 Model — Usage Notes: WPE Model

The PrimeSim Continuum solution BSIM3 version 3 (Level=49, BSIM3 Version 3.2.2 or later) supports UC Berkeley's BSIM4.5 WPE (well-proximity effects) model. To turn on this WPE model in BSIM3 version 3, specify `WPEMOD=1` in your model cards.

WPE Model Parameters:

Name	Default	Min	Max	Bin	Description
WPEMOD	0	0	1	No	Flag for WPE model (WPEMOD=1 to activate this model)
K2WE	0.0			Yes	K2 shift factor for well-proximity effect

Name	Default	Min	Max	Bin	Description
KU0WE	0.0			Yes	Mobility degradation factor for well-proximity effect
KVTH0WE	0.0			Yes	Threshold shift factor for well-proximity effect
SCREF	1.0e-6	0.0		No	Reference distance to calculate SCA, SCB and SCC
WEB	0.0			No	Coefficient for SCB
WEC	0.0			No	Coefficient for SCC

WPE Instance Parameters:

Name	Default	Min	Max	Description
SC	0.0	0.0		Distance to a single well edge
SCA	0.0	0.0		Integral of the first distribution function for scattered well dopant
SCB	0.0	0.0		Integral of the second distribution function for scattered well dopant
SCC	0.0	0.0		Integral of the third distribution function for scattered well dopant

BSIM3 Version 3 Model — Usage Notes: Ig Model

The PrimeSim Continuum solution of BSIM3 version 3 (Level=49, BSIM3 Version 3.2.2 or later) supports UC Berkeley's BSIM4.5 Ig (gate direct tunneling currents) model. To turn on this Ig model in BSIM3 version 3, specify IGCMOD=1 and/(or) IGBMOD=1 in your model cards.

Ig Model Parameters:

Name	Default	Min	Max	Bin	Description
AIGBACC	1.36e-2			Yes	Parameter for Igb
AIGBINV	1.11e-2			Yes	Parameter for Igb
AIGC	NMOS:1.36e-2/PMOS:9.80e-3			Yes	Parameter for Igc

Name	Default	Min	Max	Bin	Description
AIGSD	NMOS:1.36e-2/PMOS:9.80e-3			Yes	Parameter for Igs,d
BIGBACC	1.71e-3			Yes	Parameter for Igb
BIGBINV	9.49e-4			Yes	Parameter for Igb
BIGC	NMOS:1.71e-3/PMOS:7.59e-4			Yes	Parameter for Igc
BIGSD	NMOS:1.71e-3/PMOS:7.59e-4			Yes	Parameter for Igs,d
CIGBACC	0.075			Yes	Parameter for Igb
CIGBINV	0.006			Yes	Parameter for Igb
CIGC	NMOS:0.075/P MOS:0.03			Yes	Parameter for Igc
CIGSD	NMOS:0.075/P MOS:0.03			Yes	Parameter for Igs,d
DLCIG	LINT			No	Source/Drain overlap length for Igs and Igd
EIGBINV	1.1			Yes	Parameter for the Si bandgap for Igbinv
IGBMOD	0	0	1	No	Gate-to-body Ig model selector
IGCMOD	0	0	2	No	Gate-to-channel Ig model selector
NIGBACC	1.0	0.0		Yes	Parameter for Igbacc slope
NIGBINV	3.0	0.0		Yes	Parameter for Igbinv slope
NIGC	1.0	0.0		Yes	Parameter for Igc slope
NSD	1.0e20			Yes	S/D doping concentration
NTOX	1.0			Yes	Exponent for Tox ratio
PIGCD	1.0	0.0		Yes	Parameter for Igc partition
POXEDGE	1.0	0.0		Yes	Factor for the gate edge Tox
TEMPMOD	0	0	2	No	Temperature model selector

Name	Default	Min	Max	Bin	Description
TOXE	TOX	0.0		No	Electrical gate oxide thickness in meters
TOXREF	30.0e-10	0.0		No	Target tox value
TVFBSD OFF	0.0			Yes	Temperature parameter for vfbsdoff
VFBSDOFF	0.0			Yes	S/D flatband voltage offset

BSIM3 Version 3 Model — Usage Notes: Charge Models

In BSIM3 version 3, the BSIM1 capacitance model is $CAPMOD=0$. Simulation replaces this with a modified BSIM1 capacitance model, based on the $CAPOP=13$ model in Level 49. Level 53 uses the Berkeley BSIM1 capacitance model for $CAPMOD=0$. The following table lists $CAPMOD$ defaults for the Berkeley BSIM3 version 3 model, and for Levels 49 and 53.

Version	BSIM3 version 3	Level 49	Level 53
3.0	1	1	1
3.1	2	0	2
3.2	3	3	3
3.3	3	3	3

VFFLAG: The $CAPMOD=0$ capacitance model normally calculates the threshold voltage as $V_{th}=v_{fb} + \phi + k_1 * \sqrt{\phi - v_{bs}}$, where v_{fb} is the $VFBCV$ model parameter. This eliminates any dependence on the V_{TH0} parameter. To allow capacitance dependence on V_{TH0} , set the $VFFLAG=1$ model parameter. The $CAPMOD=0$ capacitance model calculates the threshold voltage as $V_{th}=v_{th0} + k_1 * \sqrt{\phi - v_{bs}} - k_1 * \sqrt{\phi}$. The $VFFLAG$ default value is 0.

BSIM3 Version 3 Model — Usage Notes: Printback Options

You can printback all model parameters with units. The printback also indicates whether Berkeley or Synopsys model junction diodes and noise models are invoked, and which parameters are not used (for example, simulation does not use $CJGATE$ if $ACM=0-3$).

BSIM3 Version 3 Model — Usage Notes: Mobility Multiplier

You can define mobility multiplier parameters in the BSIM3 version 3 instance line.

Name	Default	Description
MULU0	1.0	Low-field mobility (U0) multiplier
MULUA	1.0	First-order mobility degradation coefficient (UA) multiplier
MULUB	1.0	Second-order mobility degradation coefficient (UB) multiplier

When the PrimeSim Continuum solution prints back a MOSFET element summary (`.OPTION LIST`), it identifies the BSIM3 version 3 MOSFET, and prints back these three additional instance parameters.

BSIM3 Version 3 MOSFET Model Parameters

These tables are a superset of the BSIM3 version 3 model parameter set, and include the PrimeSim Continuum solution parameters. These PrimeSim Continuum solution parameters are noted in the description column, and always default (for Level 53) to maintain compliance with the BSIM3 version 3 standard. These parameters also apply to Level 49 with the following exceptions:

- [BSIM3 Version 3 Model — Model Flags](#)
- [BSIM3 Version 3 Model — Basic Parameters](#)
- [BSIM3 Version 3 Model — AC and Capacitance Parameters](#),
- [BSIM3 Version 3 Model — Length and Width Parameters](#)
- [BSIM3 Version 3 Model — Temperature Parameters](#)
- [BSIM3 Version 3 Model — Bin Description Parameters](#)
- [BSIM3 Version 3 Model — Process Parameters](#)
- [BSIM3 Version 3 Model — Noise Parameters](#)
- [BSIM3 Version 3 Model — Junction Parameters](#)
- [BSIM3 Version 3 Model — NonQuasi-Static \(NQS\) Parameters](#)
- [BSIM3 Version 3.2 Model — Parameters](#)

BSIM3 Version 3 Model — Model Flags

Name	Unit	Default	Bin	Description
VERSION	-	3.3	No	Selects from BSIM3 Versions 3.0, 3.1, 3.2, and 3.3. Issues a warning if you do not explicitly set it.

Name	Unit	Default	Bin	Description
HSPVER	-	98.2	No	Selects from the PrimeSim Continuum solution versions: 98.2, 97.4, 97.2, 96.4, 96.3, 96.1
PARAMCHK	-	0	No	PARAMCHK=1 checks the model parameters for range compliance
APWARN	-	0	No	When >0 turns off the warning message for PS,PD < Weff (PrimeSim Continuum specific)
BINFLAG	-	0	No	Uses wref, lref if you set this flag >0.9 (PrimeSim Continuum)
MOBMOD	-	1	No	Selects a mobility model
CAPMOD	-	3	No	Selects from the 0,1,2,3 charge models Level 49 CAPMOD defaults to 0.
CAPOP	-	-	No	Obsolete for Levels 49 and 53. The PrimeSim Continuum solution ignores it (PrimeSim Continuum specific) in all versions.
NOIMOD	-	1	No	Berkeley noise model flag
NLEV	-	-(off)	No	The noise model flag (non-zero overrides NOIMOD) (PrimeSim Continuum specific). See MOSFET Noise Models for more information.
NQSMOD	-	0 (off)	No	NQS Model flag
SFVTFLAG	-	0 (off)	No	Spline function for Vth (PrimeSim Continuum specific)
VFBFLAG	-	0 (off)	No	VFB selector for CAPMOD=0 (PrimeSim Continuum specific)
ACNQS MOD	-	0	No	AC small-signal NQS model selector

BSIM3 Version 3 Model — Basic Parameters

Name	Unit	Default	Bin	Description
A0	-	1.0	Yes	Bulk charge effect coefficient, channel length
A1	1/V	0	Yes	First nonsaturation factor
A2	-	1.0	Yes	Second nonsaturation factor
AGS	1/V	0.0	Yes	Gate bias coefficient of Abulk

Name	Unit	Default	Bin	Description
ALPHA0	m/V	0	Yes	First parameter of the impact ionization current
B0	m	0.0	Yes	Bulk charge effect coefficient, channel width
B1	m	0.0	Yes	Bulk charge effect width offset
BETA0	V	30	Yes	Second parameter of the impact ionization current
CDSC	F/m ²	2.4e-4	Yes	Drain/source and channel coupling capacitance
CDSCB	F/Vm ²	0	Yes	Body coefficient for CDSC
CDSCD	F/Vm ²	0	Yes	Drain bias sensitivity of CDSC
CIT	F/m ²	0.0	Yes	Interface state capacitance
DELTA	V	0.01	Yes	Effective Vds parameter
DROUT	-	0.56	Yes	Length dependence coefficient of the DIBL correction parameter in R _{out}
DSUB	-	DROUT	Yes	DIBL coefficient exponent in the subthreshold region
DVT0	-	2.2	Yes	Short channel effect coefficient 0 for V _{th}
DVT0W	1/m	0	Yes	Narrow width coefficient 0 for V _{th} , small L
DVT1	-	0.53	Yes	Short channel effect coefficient 1 for V _{th}
DVT1W	1/m	5.3e6	Yes	Narrow width coefficient 1 for V _{th} , small L
DVT2	1/V	-0.032	Yes	Short channel effect coefficient 2 for V _{th}
DVT2W	1/V	-0.032	Yes	Narrow width coefficient 2 for V _{th} , small L
ETA0	-	0.08	Yes	DIBL (drain induced barrier lowering) coefficient for the subthreshold region
ETAB	1/V	-0.07	Yes	DIBL coefficient for the subthreshold region
K1	V ^{1/2}	0.50	Yes	First-order body effect coefficient
K2	-	-0.0186	Yes	Second-order body effect coefficient
K3	-	80.0	Yes	Narrow width effect coefficient
K3B	1/V	0	Yes	Body width coefficient, narrow width effect
KETA	1/V	-0.047	Yes	Body-bias coefficient of the bulk charge effect

Name	Unit	Default	Bin	Description
NCH	cm ⁻³ See (6)	1.7e17	Yes	Peak doping concentration near the interface
NFACTOR	-	1.0	Yes	Subthreshold region swing
NGATE	cm ⁻³	0	Yes	Poly gate doping concentration
NLX	m	1.74e-7	Yes	Lateral nonuniform doping along the channel
NSUB	cm ⁻³	6.0e16	Yes	Substrate doping concentration
PCLM	-	1.3	Yes	Coefficient of the channel length modulation values ≤ 0 result in an error message and program exit.
PDIBLC1	-	0.39	Yes	Coefficient 1 for the DIBL (drain-induced barrier lowering) effect
PDIBLC2	-	0.0086	Yes	Coefficient 2 for the DIBL effect
PDIBLCB	1/V	0	Yes	Body effect coefficient of the DIBL effect coefficients
PRWB	1/V ^{1/2}	0	Yes	Body effect coefficient of RDSW
PRWG	1/V	0	Yes	Gate bias effect coefficient of RDSW
PSCBE1	V/m	4.24e8	Yes	Exponent 1 for the substrate current induced body effect
PSCBE2	m/V	1.0e-5	Yes	Coefficient 2 for the substrate current induced body effect
PVAG	-	0	Yes	Gate dependence of Early voltage
RDSW	ohm · μm	0.0	Yes	Parasitic source drain resistance per unit width
RSH	0.0	ohm/squ are	No	Source/drain sheet resistance in ohm per square
TOX	m	150e-10	No	Gate oxide thickness
U0	cm ² /V/ sec	670 nmos 250 pmos	Yes	Low field mobility at T=TREF=TNOM
UA	m/V	2.25e-9	Yes	First-order mobility degradation coefficient
UB	m ² /V ²	5.87e-19	Yes	Second-order mobility degradation coefficient

Name	Unit	Default	Bin	Description
UC	1/V	-4.65e-11 or -0.0465	Yes	Body bias sensitivity coefficient of mobility-4.65e-11 for MOBMOD=1,2 or, -0.0465 for MOBMOD=3
VBM	V	-3.0	Yes	Maximum substrate bias for calculating V_{th}
VGSLIM	V	0	No	Asymptotic Vgs value, The Min value is 5V. 0—value indicates an asymptote of infinity. (PrimeSim Continuum and Level 49 specific)
VOFF	V	-0.08	Yes	Offset voltage in the subthreshold region
VSAT	m/sec	8e4	Yes	Saturation velocity of the carrier at $T=T_{REF}=T_{NOM}$
VTH0 (VTHO)	V	0.7 NMOS -0.7 PMOS	Yes	Threshold voltage of the long channel device at $V_{bs}=0$ and small V_{ds}
W0	m	2.5e-6	Yes	Narrow width effect coefficient
WR	-	1.0	Yes	Width offset from Weff for the Rds calculation
XJ	m	0.15e-6	Yes	Junction depth

BSIM3 Version 3 Model — AC and Capacitance Parameters,

Name	Unit	Default	Bin	Description
XPART	-	0	No	Charge partitioning rate flag (default deviates from BSIM3 version 3=0) Level 49 XPART defaults to 1
CGBO	F/m	0	No	Gate-bulk overlap capacitance per unit channel length
CGDO	F/m	p2 See (2)	No	Non-LDD region source-gate overlap capacitance per unit channel length
CGS1	F/m	0.0	Yes	Lightly-doped source-gate overlap region capacitance
CGSO	F/m	p1 See (1)	No	Non-LDD region source-gate overlap capacitance per unit channel length
CF	F/m	See (3)	Yes	Fringing field capacitance
CGD1	F/m	0.0	Yes	Lightly-doped drain-gate overlap region capacitance
CKAPPA	F/m	0.6	Yes	Coefficient for the lightly-doped region overlap capacitance fringing field capacitance

Name	Unit	Default	Bin	Description
CLC	m	0.1e-6	Yes	Constant term for short channel model
CLE	-	0.6	Yes	Exponential term, short channel model
VFBCV	V	-1.0	Yes	Flat band voltage, used only in CAPMOD=0 C-V calculations

BSIM3 Version 3 Model — Length and Width Parameters

Name	Unit	Default	Bin	Description
DLC	m	LINT	No	Length offset fitting parameter from CV
DWB	$m/V^{1/2}$	0.0	Yes	Coefficient of the substrate body bias dependence for Weff
DWC	m	WINT	No	Width offset fitting parameter from CV
DWG	m/V	0.0	Yes	Coefficient of the gate dependence for Weff
LINT	m	0.0	No	Length offset fitting the parameter from the I-V without the bias
LL	m^{LLN}	0.0	No	Coefficient of the length dependence for the length offset
LLN	-	1.0	No	Power of the length dependence of the length offset
LW	m^{LWN}	0.0	No	Coefficient of the width dependence for the length offset
LWL	$m^{LWN} * m^{LLN}$	0.0	No	Coefficient of the length and width cross term for the length offset
LWN	-	1.0	No	Power of the width dependence of the length offset
WINT	m	0.0	No	Width offset fitting parameter from I-V without bias
WLN	-	1.0	No	Power of the length dependence of the width offset
WW	m^{WWN}	0.0	No	Coefficient of the width dependence for the width offset
WWL	$m^{WWN} * m^{WLN}$	0.0	No	Coefficient of the length and width cross term for the width offset
WWN	-	1.0	No	Power of the width depends on the width offset.

BSIM3 Version 3 Model — Temperature Parameters

Name	Unit	Default	Bin	Description
AT	m/sec	3.3e4	Yes	Temperature coefficient for the saturation velocity
KT1	V	-0.11	Yes	Temperature coefficient for Vth
KT1L	m-V	0.0	Yes	Temperature coefficient for the channel length dependence of Vth
KT2	-	0.022	Yes	Body bias coefficient of the Vth temperature effect
PRT	ohm-um	0	Yes	Temperature coefficient for RDSW
UA1	m/V	4.31e-9	Yes	Temperature coefficient for UA
UB1	(m/V) ²	-7.61e-18	Yes	Temperature coefficient for UB
UC1	m/V ²	-5.69e-11	Yes	Temperature coefficient for UC
UTE	-	-1.5	Yes	Mobility temperature exponent
XTI	-	3.0	No	Junction current temperature exponent

BSIM3 Version 3 Model — Bin Description Parameters

Name	Unit	Default	Bin	Description
BINUNIT				Assumes that W _{EFF} , L _{EFF} , W _{REF} , I _{ref} units are in microns if BINUNIT=1, or, in meters, otherwise
LMAX	m	1.0	No	Maximum channel length
LMIN	m	0.0	No	Minimum channel length
WMAX	m	1.0	No	Maximum channel width
WMIN	m	0.0	No	Minimum channel width

BSIM3 Version 3 Model — Process Parameters

Name	Unit	Default	Bin	Description
DTOXCV (capmod=3 only)				Difference between the electrical and physical gate oxide thicknesses, due to the effects of the gate poly-depletion and finite channel charge layer thickness.

Name	Unit	Default	Bin	Description
GAMMA1	$V^{1/2}$	See (8)	Yes	Body effect coefficient near the surface
GAMMA2	$V^{1/2}$	See (9)	Yes	Body effect coefficient in the bulk
VBX	V	See (10)	Yes	VBX at which the depletion region width equals XT
XT	m	1.55e-7	Yes	Doping depth

BSIM3 Version 3 Model — Noise Parameters

Name	Unit	Default	Bin	Description
AF	-	1.0	No	Flicker noise exponent
EF	-	1.0	No	Flicker noise frequency exponent
EM	V/m	4.1e7	No	Flicker noise parameter
KF	-	0.0	No	Flicker noise coefficient
LINTNOI	m	0.0	No	Length reduction parameter offset
NIOA	-	1.0e20 nmos 9.9e18 pmos	No	Body effect coefficient near the surface
NOIB	-	5.0e4 nmos 2.4e3 pmos	No	Body effect coefficient in the bulk
NOIC	-	-1.4e-12 nmos 1.4e-12 pmos	No	VBX at which the depletion region width equals XT

Note:

See also [MOSFET Noise Models](#) for the PrimeSim Continuum solution noise model usage (the `NLEV` parameter for the PrimeSim Continuum solution overrides the Berkeley `NOIMOD` parameter).

BSIM3 Version 3 Model — Junction Parameters

Name	Unit	Default	Bin	Description
ACM	-	10	No	Area calculation method selector (PrimeSim Continuum specific) • ACM=0-3 uses the PrimeSim Continuum solution junction models • ACM=10-13 uses the Berkeley junction models Level 49 ACM defaults to 0
CJ	F/m ²	5.79e-4	No	Zero-bias bulk junction capacitance (Default deviates from BSIM3 version 3=5.0e ⁻⁴)
CJGATE	F/m	CJSW	No	Zero-bias gate-edge sidewall bulk junction capacitance (PrimeSim Continuum-specific) (use only if ACM=3)
CJSW	F/m	0.0	No	Zero-bias sidewall bulk junction capacitance (Default deviates from BSIM3 version 3=5.0e ⁻¹⁰)
CJSWG	F/m	CJSW	No	Zero-bias gate-edge sidewall bulk junction capacitance (use only with the Berkeley junction model, ACM=10-13)
JS	A/m ²	0.0	No	Bulk junction saturation current. (Default deviates from BSIM3 version 3=1.0e ⁻⁴)
JSW	A/m	0.0	No	Sidewall bulk junction saturation current
MJ	-	0.5	No	Bulk junction grading coefficient
MJSW	-	0.33	No	Sidewall bulk junction grading coefficient
MJSWG	-	MJSW	No	Gate-edge sidewall bulk junction grading coefficient (use only with the Berkeley junction model: ACM=10-13) the PrimeSim Continuum solution has no equivalent parameter. Always set the gate-edge grading coefficient to MJSW for the PrimeSim Continuum solution junction model.
N	-	1	No	Emission coefficient (PrimeSim Continuum-specific), (use only with the PrimeSim Continuum solution junction model, ACM=0-3)
NJ	-	1	No	Emission coefficient (use only with the Berkeley junction model: ACM=10-13)
PB, PHIB	V	1.0	No	Bulk junction contact potential
PBSW	V	1.0	No	Sidewall bulk junction contact potential

Name	Unit	Default	Bin	Description
PBSWG	V	PBSW	No	Gate-edge sidewall bulk junction contact potential (use only with the Berkeley junction model, ACM=10-13). The PrimeSim Continuum solution has no equivalent parameter. Gate-edge contact potential is always set to PHP for the PrimeSim Continuum solution junction model.
PHP	V	1.0	No	Sidewall bulk junction contact potential (PrimeSim Continuum) (use only with the PrimeSim Continuum solution junction model: ACM=0-3)

Note:

See [MOSFET Noise Models](#) for the PrimeSim Continuum solution junction diode model usage.

BSIM3 Version 3 Model — NonQuasi-Static (NQS) Parameters

Name	Unit	Default	Bin	Description
ELM	-	5.0	Yes	Elmore constant

BSIM3 Version 3.2 Model — Parameters

Name	Unit	Default	Bin	Description
ACDE	m/V	1.0	Yes	Exponential coefficient for the charge thickness in the accumulation and depletion regions
ALPHA1	V-1	0.0	Yes	Substrate current parameter
JTH	A	0.1	No	Diode limiting current
LLC	mlIn	LL	No	Coefficient of the length dependence for the C-V channel length offset
LWC	mlwn	LW	No	Coefficient of the width dependence for the C-V channel length offset
LWLC	mlIn+lwN	LWL	No	Coefficient of the length and width for the C-V channel length offset
MOIN	m/V	15.0	Yes	Coefficient, gate-bias dependent surface potential
NOFF	-	1.0	Yes	I-V parameter, weak to strong inversion transition
TCJ	V/K	0.0	No	Temperature coefficient of CJ

Name	Unit	Default	Bin	Description
TCJSW	V/K	0.0	No	Temperature coefficient of CJSW
TCJSWG	V/K	0.0	No	Temperature coefficient of CJSWG
TOXM	m	TOX	No	Reference gate oxide thickness
TPB	V/K	0.0	No	Temperature coefficient of PB
TPBSW	V/K	0.0	No	Temperature coefficient of PBSW
TPBSWG	V/K	0.0	No	Temperature coefficient of PBSWG
VFB	V	See (11)	Yes	DC flatband voltage
VOFFCV	-	0.0	Yes	C-V parameter, weak to strong inversion transition
WLC	mwln	WL	No	Coefficient of the length dependence for the C-V channel width offset
WWC	mwwn	WW	No	Coefficient of the width dependence for the C-V channel width offset
WWLC	mwln+ wwn	WWL	No	Coefficient of the length and width cross terms for the C-V channel width offset

BSIM3 Version 3 MOSFET Model — Parameter Range Limits

Simulation reports either a warning or a fatal error if BSIM3 version 3 parameters fall outside predefined ranges. These range limitations prevent (or at least warn of) potential numerical problems. Level 53 follows exactly the BSIM3 version 3 range limit reporting scheme. Level 49 deviates from the BSIM3 version 3 scheme as noted in the comments column in the following tables.

To control the maximum number of simulation warning messages printing to the output file, use:

```
.OPTION WARNLIMIT=#
```

In the preceding `.OPTION` statement, `#` is the maximum number of warning messages that simulation reports. The default `WARNLIMIT` value is 1. In some cases, simulation checks parameters only if you set the `PARAMCHK=1` model parameter.

This section covers the following topics:

- [BSIM3 Version 3 Model — Model Parameter Range Limit](#)
- [BSIM3 Version 3 Model — Element Parameter Range Limit](#)

BSIM3 Version 3 Model — Model Parameter Range Limit

Name	Limits	Comments
A1	-	See A2 conditions on the next line
A2	< 0.01 Warn and reset a2=0.01 if paramchk=1 > 1 Warn and reset a2=1,a1=0 if paramchk=1	
ACDE	< 0.4, >1.6 Warn	
B1	=-Weff Fatal B1 + Weff < 10^{-7} Warn if paramchk=1	
CDSC	< 0 Warn if paramchk=1	
CDSCD	< 0 Warn if paramchk=1	
CGB0	< 0 Warn and reset to 0 if paramchk=1	
CGD0	< 0 Warn and reset to 0 if paramchk=1	
CGS0	< 0 Warn and reset to 0 if paramchk=1	
DELTA	< 0 Fatal	
DROUT	< 0 Fatal if paramchk=1	Level 49 reports a warning
DSUB	< 0 Fatal	< 0 Level 49 reports a warning
DVT0	< 0 Warn if paramchk=1	
DVT1	< 0 Fatal	< 0 Level 49 reports a warning
DVT1W	< 0 Fatal	< 0 Level 49 reports a warning
ETA0	<=0 Warn if paramchk=1	
IJTH	< 0 Fatal	
MOIN	< 5.0, >25 Warn	
NCH	<=0 Fatal <= 10^{15} Warn if parmchk=1 >= 10^{21} Warn if parmchk=1	if > 10^{20} simulation multiplies NCH by 10^{-6} before the other limit checks.
NFACTOR	< 0 Warn if paramchk=1	

Name	Limits	Comments
NGATE	< 0 Fatal $> 10^{25}$ Fatal $\leq 10^{18}$ Fatal if parmchk=1	if $> 10^{23}$ simulation multiplies NGATE by 10^{-6} before the other limit checks. Level 49 returns: < 0 Fatal $> 10^{25}$ Warn $\leq 10^{18}$ Warn if parmchk==1
NLX	$< -Leff$ Fatal < 0 Warn if parmchk=1	
NOFF	$< 0.1, > 4.0$ Warn	
NSUB	≤ 0 Fatal $\leq 10^{14}$ Warn if parmchk=1 $\geq 10^{21}$ Warn if parmchk=1	Ignores NSUB if k1,k2 are defined
PCLM	≤ 0 Fatal	
PDIBLC1	< 0 Warn if parmchk=1	
PDIBLC2	< 0 Warn if parmchk=1	
PS	$< Weff$ Warn	
PSCBE2	≤ 0 Warn if parmchk=1	
RDSW	< 0 Warn if parmchk=1 and reset rds=0	
TOX	Min value=5e-10 ≤ 0 Fatal $< 10^{-9}$ Warn if parmchk=1	BSIM3tox $< 5e-10$ is automatically reset to min value 5e-10
TOXM	≤ 0 Fatal $< 10^{-9}$ Warn if parmchk=1	
U0	≤ 0 Fatal	
VBM		Ignores if you defined K1 and K2
VSAT	≤ 0 Fatal $< 10^3$ Warn if parmchk==1	
W0	$= -Weff$ Fatal $w0 + Weff < 10^{-7}$ Warn if parmchk==1	
XJ	≤ 0 Fatal	

BSIM3 Version 3 Model — Element Parameter Range Limit

Name	Limits	Comments
Leff	$< 5.0 \times 10^{-8}$ Fatal	
LeffCV	$< 5.0 \times 10^{-8}$ Fatal	

Name	Limits	Comments
PD	< Weff, Warn	
PS	< Weff, Warn	
Weff	< 1.0 x 10 ⁻⁷ Fatal	
WeffCV	< 1.0 x 10 ⁻⁷ Fatal	

For a list of output template parameters in the MOSFET models, and which parameters this model supports, see [MOSFET Output Templates](#).

BSIM3 Version 3 MOSFET Model Equations

The effective channel length and width in all model equations are:

$$L_{eff} = L_{drawn} - 2dL$$

$$W_{eff} = W_{drawn} - 2dW$$

$$W'_{eff} = W'_{drawn} - 2dW'$$

$$W_{drawn} = W * WMULT + XW \quad L_{drawn} = L * LMULT + XL$$

- The unprimed W_{eff} is bias-dependent.
- The primed quantity is bias-independent.

$$dW = dW' + dW_g V_{gsteff} + dW_b (\sqrt{\phi_s - V_{bseff}} - \sqrt{\phi_s})$$

$$dW' = W_{int} + \frac{W_L}{L^{WLN}} + \frac{W_W}{W^{WWN}} + \frac{W_{WL}}{L^{WLN} W^{WWN}}$$

$$dL = L_{int} + \frac{L_L}{L^{LLN}} + \frac{L_W}{W^{LWN}} + \frac{L_{WL}}{L^{LLN} W^{LWN}}$$

C-V calculations replace dW' with:

$$dW' = DWC + \frac{W_{LC}}{L^{WLN}} + \frac{W_{WC}}{W^{WWN}} + \frac{W_{WLC}}{L^{WLN} W^{WWN}}$$

C-V also replaces dL' with:

$$dL' = DLC + \frac{L_{LC}}{L^{LLN}} + \frac{L_{WC}}{W^{LWN}} + \frac{L_{WLC}}{L^{LLN} W^{LWN}}$$

Note:

For details of BSIM3 Version 3 equations, see the web site: <http://www-device.eecs.berkeley.edu/~bsim3/get.html>

BSIM3 Version 3 MOSFET Model — Additional Information

1. If you do not specify C_{gs0} , simulation calculates it as follows:

- If you specify a d/c value that is greater than 0.0, then,

$$cgs0 = p1 = \max(0, dlc * cox - cgs1)$$

$$\text{Otherwise, } cgs0 = 0.6 * xj * cox$$

2. If you do not specify C_{gd0} , simulation calculates it as follows:

- if you specify a d/c value that is greater than 0.0, then,

$$cgdo = p2 = \max(0, dlc * cox - cgdl)$$

$$\text{Otherwise } cgdo = 0.6 * xj * cox$$

3. If you do not specify C_f , simulation calculates it using:

$$C_f = \frac{2\epsilon_{ox}}{H} \log \left(1 + \frac{4 \times 10^{-7}}{T_{ox}} \right)$$

4. If you do not specify V_{th0} in the .MODEL statement, simulation calculates it with $V_{fb} = -1$, using:

$$V_{th0} = V_{fb} + \phi_s + K_1 \sqrt{\phi_s}$$

5. If you do not specify K_1 and K_2 , simulation calculates it using:

$$K_1 = GAMMA_2 + 2K_2 \sqrt{\phi_s - V_{bs}}$$

$$K_2 = \frac{(GAMMA_2 - GAMMA_1) (\sqrt{\phi_s - V_{bs}} - \sqrt{\phi_s})}{2\sqrt{\phi_s} (\sqrt{\phi_s - V_{bm}} - \sqrt{\phi_s}) + V_{bm}}$$

6. If you do not specify n_{ch} , but you specify $GAMMA_1$, then simulation calculates n_{ch} from:

$$n_{ch} = \frac{GAMMA_1^2 C_{ox}^2}{2q\epsilon_{si}}$$

7. If you do not specify n_{ch} or $GAMMA_1$, then n_{ch} defaults to $1.7e17$ per cubic meter and simulation calculates $GAMMA_1$ from n_{ch} .

8. If you do not specify ϕ_{HI} , simulation calculates it using:

$$\phi_s = 2 \frac{k_B T}{q} \log \left(\frac{n_{ch}}{n_i} \right)$$

$$n_i = 1.45 \times 10^{10} \left(\frac{T}{300.15} \right)^{1.5} \exp \left(21.5565981 - \frac{q E_g(T)}{2 k_B T} \right)$$

$$E_g(T) = 1.16 - \frac{7.02 \times 10^{-4} T^2}{T + 1108}$$

9. If you do not specify GAMMA_1 , simulation calculates it using:

$$\text{GAMMA}_1 = \frac{\sqrt{2 q \epsilon_{si} n_{ch}}}{C_{ox}}$$

10. If you do not specify GAMMA_2 , simulation calculates it using:

$$\text{GAMMA}_2 = \frac{\sqrt{2 q \epsilon_{si} n_{sub}}}{C_{ox}}$$

11. If you do not specify V_{bx} , simulation calculates it using:

$$V_{bx} = \phi_s - \frac{q n_{ch} X_t^2}{2 \epsilon_{si}}$$

12. The BSIM3 model can calculate v_{th} in any of three ways:

- Using K_1 and K_2 values that you specify
- Using GAMMA_1 , GAMMA_2 , VBM , and VBX values that you enter in the `.MODEL` statement
- Using NPEAK , NSUB , XT , and VBM values that you specify

You can enter the U_0 model parameter in meters or centimeters. Simulation converts U_0 to m^2/Vsec as follows: if U_0 is greater than 1, it is multiplied by $1\text{e-}4$. You must enter the NSUB parameter in cm^{-3} units.

Specify a negative value of VTH0 for the p-channel in the `.MODEL` statement.

The PSCBE1 and PSCBE2 model parameters determine the impact ionization current, which contributes to the bulk current.

BSIM3 Version 3 MOSFET Model Examples

The following is an example of a NMOS model for the Level 49 MOSFET. VTH0 is positive.

```
.model nch nmos Level=49
+ Tnom=27.0
```


Chapter 2: MOSFET Models

LEVEL 49 and 53: BSIM3 Version 3 MOSFET Model

```

+ nch=1.024685E+17 tox=1.00000E-08 xj=1.00000E-07
+ lint=3.75860E-08 wint=-2.02101528644562E-07
+ vth0=.6094574 k1=.5341038 k2=1.703463E-03 k3=-17.24589
+ dvt0=.1767506 dvt1=.5109418 dvt2=-0.05
+ nlx=9.979638E-08 w0=1e-6
+ k3b=4.139039
+ vsat=97662.05 ua=-1.748481E-09 ub=3.178541E-18 uc=1.3623e-10
+ rdsw=298.873 u0=307.2991 prwb=-2.24e-4
+ a0=.4976366
+ keta=-2.195445E-02 a1=.0332883 a2=.9
+ voff=-9.623903E-02 nFactor=.8408191 cit=3.994609E-04
+ cdsc=1.130797E-04
+ cdsch=2.4e-5
+ eta0=.0145072 etab=-3.870303E-03
+ dsub=.4116711
+ pclm=1.813153 pdiblc1=2.003703E-02 pdiblc2=.00129051
+ pdiblc3=-1.034e-3
+ drout=.4380235 psche1=5.752058E+08 psche2=7.510319E-05
+ pvag=.6370527 prt=68.7 ngate=1.e20 alpha0=1.e-7 beta0=28.4
+ prwg=-0.001 ags=1.2
+ dvt0w=0.58 dvt1w=5.3e6 dvt2w=-0.0032
+ kt1=-.3 kt2=-.03
+ at=33000
+ ute=-1.5
+ ua1=4.31E-09 ub1=7.61E-18 uc1=-2.378e-10
+ kt1l=1e-8
+ wr=1 b0=1e-7 b1=1e-7 dwg=5e-8 dwb=2e-8 delta=0.015
+ cgdl=1e-10 cgsl=1e-10 cgbo=1e-10 xpart=0.0
+ cgdo=0.4e-9 cgso=0.4e-9
+ clc=0.1e-6
+ cle=0.6
+ ckappa=0.6

```

The following is an example of a PMOS model for the Level 49 MOSFET. V_{TH0} is negative.

```

.model pch PMOS Level=49
+ Tnom=27.0
+ nch=5.73068E+16 tox=1.00000E-08 xj=1.00000E-07
+ lint=8.195860E-08 wint=-1.821562E-07
+ vth0=-.86094574 k1=.341038 k2=2.703463E-02 k3=12.24589
+ dvt0=.767506 dvt1=.65109418 dvt2=-0.145
+ nlx=1.979638E-07 w0=1.1e-6
+ k3b=-2.4139039
+ vsat=60362.05 ua=1.348481E-09 ub=3.178541E-19 uc=1.1623e-10
+ rdsw=498.873 u0=137.2991 prwb=-1.2e-5
+ a0=.3276366
+ keta=-1.8195445E-02 a1=.0232883 a2=.9
+ voff=-6.623903E-02 nFactor=1.0408191 cit=4.994609E-04
+ cdsc=1.030797E-3
+ cdsch=2.84e-4
+ eta0=.0245072 etab=-1.570303E-03
+ dsub=.24116711
+ pclm=2.6813153 pdiblc1=4.003703E-02 pdiblc2=.00329051

```

Chapter 2: MOSFET Models

LEVEL 49 and 53: BSIM3 Version 3 MOSFET Model

```
+ pdib1cb=-2.e-4
+ drout=.1380235 psche1=0 psche2=1.e-28
+ pvag=-.16370527
+ prwg=-0.001 ags=1.2
+ dvt0w=0.58 dvt1w=5.3e6 dvt2w=-0.0032
+ kt1=-.3 kt2=-.03 prt=76.4
+ at=33000
+ ute=-1.5
+ ua1=4.31E-09 ub1=7.61E-18 uc1=-2.378e-10
+ kt1l=0
+ wr=1 b0=1e-7 b1=1e-7 dwg=5e-8 dwb=2e-8 delta=0.015
+ cgd1=1e-10 cgs1=1e-10 cgbo=1e-10 xpart=0.0
+ cgdo=0.4e-9 cgso=0.4e-9
+ clc=0.1e-6
+ cle=0.6
+ ckappa=0.6
```

LEVEL 50: Philips MOS9 Model

This model defaults to version 903. If you set the model parameter `VERSION`, you can control it to either version 902 or 903. The 903 version includes parameters `NFMOD`, `NFAR`, `NFBR`, `NFCR`, `SL3VT0`. If you set `version=902`, it will select the 902 model. The Philips MOS Model 9 is available as Level 50 in the Synopsys models (based on the “Unclassified Report NL-UR 003/94” by R.M.D.A. Velghe, D.B.M. Klaassen, and F.M. Klaassen).

MOSFET Level 50 incorporates all features of Philips MOS 9, except for the gate noise current. You can select either of two MOSFET Level 50 diode models:

- **ACM Parasitic Diode Model** by using the `JS`, `JSW`, `N`, `CJ`, `CJSW`, `CJGATE`, `MJ`, `MJSW`, `PB`, `PHP`, `ACM`, and `HDIF` parameters. This version does not use the older `IS` parameter. To use this model, select the `JUNCAP=0` (default) parameter.
- **Philips JUNCAP Parasitic Diode Model**. To use this model, select the `JUNCAP=1` model parameter.

This section covers the following topics:

- [Philips MOS9 Model Parameters](#)
- [Philips MOS9 Model — JUNCAP Parameters](#)
- [Using the Philips MOS9 Model](#)
- [Philips MOS9 Model — Example](#)

Philips MOS9 Model Parameters

Name	Unit	Default (N)	Default (P)	Description
A1R	-	6.0	10.0	Weak-avalanche current factor
A2R	V	38.0	59.0	Exponent of weak-avalanche current
A3R	-	650.0e-3	520.0e-3	Factor of minimum drain bias, above which avalanche sets in
ALPR	-	3.0e-3	44.0e-3	Channel length modulation factor
BETSQ	AV^{-2}	83.0e-6	26.1e-6	Gain factor of infinite square transistor
COL	F/m	320.0e-12	320.0e-12	Gate overlap capacitance per unit width
ETAALP	-	150.0e-3	170.0e-3	Exponent of length dependence of ALPR

Name	Unit	Default (N)	Default (P)	Description
ETABET	-	1.6	1.6	Exponent of temperature dependence of gain factor
ETADSR	-	600.0e-3	600.0e-3	Exponent of drain dependence of GAM1R
ETAGAMR	-	2.0	1.0	Exponent of back-bias dependence of zero gate-drive, drain-induced threshold shift
ETAMR	-	2.0	1.0	Exponent of back-bias dependence of the subthreshold slope
ETAZET	-	170.0e-3	30.0e-3	Exponent of length dependence of ZET1R
FBET1	-	0	0	Relative mobility decrease due to the first profile
FBET2	-	0	0	Relative mobility decrease due to the second profile
FTHE1	-	0	0	Coefficient describing the width dependence of THE1 for $W < W_{DOG}$
GAM1R	-	145.0e-3	77.0e-3	Drain-induced threshold shift coefficient for high gate drive
GAM0OR	-	18.0e-3	7.0e-3	Drain-induced threshold shift coefficient, at zero gate drive, and zero back-bias
GTHE1	-	0	0	Parameter that selects either the old (=0) or the new (=1) scaling rule of θ_1
KOR	$V^{-1/2}$	650.0e-3	470.0e-3	Low-back-bias body factor
KR	$V^{-1/2}$	110.0e-3	470.0e-3	High-back-bias body factor
LAP	m	100.0e-9	25.0e-9	Lateral diffusion per side
LER	m	1.1e-6	1.25e-6	Reference L_{eff}
LP1	M	1E-6	1E-6	Characteristic length of the first profile
LP2	M	1E-8	1E-8	Characteristic length of the second profile
LVAR	m	-220.0e-9	-460.0e-9	Variation in gate length
MOR	-	500.0e-3	375.0e-3	Subthreshold slope factor
NFAR	$V^{-1}m^{-2}$	7.15e+22	1.53xe+22	1st flicker noise coefficient added in release 98.4

Name	Unit	Default (N)	Default (P)	Description
NFBR	$V^{-1}m^{-2}$	2.16e+06	4.06e+06	2nd flicker noise coefficient added in release 98.4
NFCR	V^{-1}	0.0	2.92e-10	3rd flicker noise coefficient added in release 98.4
NFMOD		0		Flicker noise selector. 0 selects the old flicker noise model added in release 98.4
NFR	V^2	70.0e-12	21.4e-12	Flicker noise coefficient
NTR	J	24.4e-21	21.1e-21	Thermal noise coefficient
PHIBR	V	650.0e-3	650.0e-3	Strong inversion surface potential
SL2GA MOO	-	0	0	Second coefficient of the γ_{00} length dependence
SL2K	$V^{1/2}m^2$	0	0	Second coefficient of the length dependence of K
SL2KO	$V^{1/2}m^2$	0	0	Second coefficient of the length dependence of K_0
SL2VTO	Vm^2	0.0	0.0	Second length dependence of VTO
SL3VTO	V	0	0	Third coefficient of the length dependence of VTO
SLA1	m	1.3e-6	-15.0e-6	Length dependence of A1R
SLA2	Vm	1.0e-6	-8.0e-6	Length dependence of A2R
SLA3	m	-550.0e-9	-450.0e-9	Length dependence of A3R
SLALP	-	-5.65e-3	9.0e-3	Coefficient of length dependence of ALPR
SLGAM1	-	160.0e-9	105.0e-9	Length dependence of GAM1R
SLGAMOO	m^2	20.0e-15	11.0e-15	Length dependence of GAMOOR
SLK	$V^{-1/2}m$	-280.0e-9	-200.0e-9	Length dependence of K
SLKO	$V^{-1/2}m$	-130.0e-9	-200.0e-9	Length dependence of k_0
SLMO	$m^{1/2}$	280.0e-6	47.0e-6	Length dependence coefficient of MOR
SLTHE1R	$V^{-1}m$	140.0e-9	70.0e-9	Length dependence coefficient of THE1R
SLTHE2R	$V^{-1/2}m$	-33.0e-9	-75.0e-9	Length dependence coefficient of THE2R

Name	Unit	Default (N)	Default (P)	Description
SLTHE3R	V^{-1m}	185.0e-9	27.0e-9	Length dependence coefficient of THE3R
SLVSBT	Vm	-4.4e-6	0.0	Length dependence of VSBTR
SLVSBX	Vm	0.0	0.0	Length dependence of VSBX
SLVTO	Vm	-135.0e-9	35.0e-9	Length dependence of VTO
SLZET1	-	-390.0e-3	-2.8	Length dependence coefficient of ZET1R
STA1	K^{-1}	0.0	0.0	Temperature coefficient of A1R
STLTHE1	$V^{-1m/K}$	0.0	0.0	Temperature dependence of the length dependence for THE1R
STLTHE2	$V^{-1/2m/K}$	0.0	0.0	Temperature dependence of the length dependence for THE2R
STLTHE3	$V^{-1m/K}$	-620.0e-12	0.0	Temperature dependence of the length dependence for THE3R
STMO	K^{-1}	0.0	0.0	Temperature dependence coefficient MOR
STTHE1R	$V^{-1/K}$	0.0	0.0	Temperature dependence coefficient THE1R
STTHE2R	$V^{-1m/K}$	0.0	0.0	Temperature dependence coefficient THE2R
STTHE3R	$V^{-1/K}$	-660.0e-6	0.0	Temperature dependence coefficient of THE3R
STVTO	V/K	-1.2e-3	-1.7e-3	Temperature dependence of VTO
SWA1	m	3.0e-6	30.0e-6	Width dependence of A1R
SWA2	Vm	2.0e-6	15.0e-6	Width dependence of A2R
SWA3	m	0.0	-140.0e-9	Width dependence of A3R
SWALP	m	1.67e-9	180.0e-12	Coefficient of width dependence of ALPR
SWGAM1	-	-10.0e-9	-11.0e-9	Width dependence of GAM1R
SWK	$V^{-1/2m}$	275.0e-9	115.0e-9	Width dependence of K
SWKO	$V^{-1/2m}$	2.0e-9	115.0e-9	Width dependence of k_0
SWTHE1	V^{-1m}	-58.0e-9	-80.0e-9	Width dependence coefficient of THE1R
SWTHE2	$V^{-1/2m}$	30.0e-9	20.0e-9	Width dependence coefficient of THE2R

Name	Unit	Default (N)	Default (P)	Description
SWTHE3	V^{-1m}	20.0e-9	11.0e-9	Width dependence coefficient of THE3R
SWVSBX	Vm	-675.0e-9	0.0	Width dependence of VSBX
SWVTO	Vm	130.0e-9	50.0e-9	Width dependence of VTO
THE1R	V^{-1}	190.0e-3	190.0e-3	Gate-induced mobility reduction coefficient
THE2R	$V^{-1/2}$	12.0e-3	165.0e-3	Back-bias induced mobility reduction coefficient
THE3R	V^{-1}	145.0e-3	27.0e-3	Lateral field induced mobility reduction coefficient
TOX	m	25.0e-9	25.0e-9	Oxide thickness
TR	°C	21.0	21.0	Reference temperature for model
VPR	V	340.0e-3	235.0e-3	Characteristic voltage for channel length modulation
VSBTR	V	2.1	100.0	Limiting voltage for back-bias dependence
VSBXR	V	660.0e-3	0.0	Transition voltage for dual-k-factor model
VTOR	V	730.0e-3	1.1	Threshold voltage at zero bias
WDOG	m	0	0	Characteristic drawn gate width, below which dogboning appears
WER	m	20.0e-6	20.0e-6	Reference Weff
WOT	m	0.0	0.0	Channel-stop diffusion per side
WVAR	m	-25.0e-9	-130.0e-9	Variation in active width
ZET1R	-	420.0e-3	1.3	Weak-inversion correction factor

Philips MOS9 Model — JUNCAP Parameters

The following are JUNCAP model parameters specific to this model:

Name	Unit	Default	Description
JUNCAP	-	0	JUNCAP flag: 0-off, 1-on.
CJBR	$F \cdot m^{-2}$	1.00e-12	Bottom junction capacitance at $V=V_R$.

Name	Unit	Default	Description
CJGR	$F \cdot m^{-1}$	1.00e-12	Gate edge junction capacitance at $V=V_R$.
CJSR	$F \cdot m^{-1}$	1.00e-12	Sidewall junction capacitance at $V=V_R$.
DTA	$^{\circ}C$	0.0	Temperature offset of JUNCAP element relative to T_A .
JSDBR	$A \cdot m^{-2}$	1.00e-3	Bottom saturation-current density due to diffusion from back contact.
JSDGR	$A \cdot m^{-1}$	1.00e-3	Gate edge saturation-current density due to diffusion from back contact.
JSDSR	$A \cdot m^{-1}$	1.00e-3	Sidewall saturation-current density due to diffusion from back contact.
JSGBR	$A \cdot m^{-2}$	1.00e-3	Bottom saturation-current density due to electron-hole generation at $V=V_R$.
JSGGR	$A \cdot m^{-1}$	1.00e-3	Gate edge saturation-current density due to electron-hole generation at $V=V_R$.
JSGSR	$A \cdot m^{-1}$	1.00e-3	Sidewall saturation-current density due to electron-hole generation at $V=V_R$.
NB	-	1.00	Emission coefficient of the bottom forward current.
NG	-	1.00	Emission coefficient of the gate edge forward current.
NS	-	1.00	Emission coefficient of the sidewall forward current.
PB	-	0.40	Bottom-junction grading coefficient.
PG	-	0.40	Gate edge-junction grading coefficient.
PS	-	0.40	Sidewall-junction grading coefficient.
TH3MOD	-	1	Switch that activates THE3-clipping: • If TH3MOD ==1 (default), effective THE3 can be slightly negative, and clipping does not occur. • If TH3MOD ==0, this model clips the effective THE3 to more than zero.
VDBR	v	1.00	Diffusion voltage of the bottom junction at $T=T_R$.
VDGR	v	1.00	Diffusion voltage of the gate edge junction at $T=T_R$.
VDSR	v	1.00	Diffusion voltage of the sidewall junction at $T=T_R$.
VR	V	0.0	Voltage at which simulation determines the parameters.

Using the Philips MOS9 Model

To use the Philips MOS9 model:

1. Set LEVEL=50 to identify the model as the Philips MOS Model 9.
2. The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in most other simulators. When comparing to other simulators, set the simulation temperature to 27 use `.TEMP 27` or `.OPTION TNOM=27`.
3. The model parameter set must include the TR model reference temperature, which corresponds to `TREF` in other model levels. The default for TR is 21.0°C to match the Philips simulator.
4. This model has its own charge-based capacitance model. Level 50 ignores the `CAPOP` parameter, which selects different capacitance models.
5. This model uses analytical derivatives for the conductances. This model ignores the `DERIV` parameter, which selects the finite difference method.
6. `DTEMP` increases the temperature of individual elements relative to the circuit temperature. Set `DTEMP` on the element line.
7. Defaults are nonzero so use the `.MODEL` statement to set every model parameter listed in the Level 50 Model Parameters table.
8. Use the JUNCAP model parameter to select one of two available parasitic junction diode models, ACM or JUNCAP. `JUNCAP=1` selects the Philips JUNCAP model, `JUNCAP=0` (default) selects the ACM model.
9. Philips added a switch named `TH3MOD` to MOS Model 9. You can use this switch to re-activate effective `THE3` clipping, which was removed in an earlier version of this model.
 - If `TH3MOD==1` (default), effective `THE3` can be slightly negative, and clipping does not occur.
 - If `TH3MOD==0`, this model clips the effective `THE3` to more than zero.

Philips MOS9 Model — Example

```
.model nch nmos Level=50
+ ler =1e-6 wer=10e-6
+ lvar =0.0 lap=0.05e-6
+ wvar =0.0 wot=0.0
+ tr = 27.00 vtor=0.8
+ stvto =0 slvto=0
+ sl2vto =0 swvto=0
+ kor =0.7 slko=0
+ swko =0 kr=0.3
```

Chapter 2: MOSFET Models

LEVEL 50: Philips MOS9 Model

```

+ slk =0 swk=0
+ phibr =0.65 vsbxb=0.5
+ slvsbx =0 swvsbx=0
+ betsq =120e-6 etabet=1.5
+ the1r =0.3 stthe1r=0
+ slthe1r =0 stlthe1=0
+ swthe1 =0 the2r=0.06
+ stthe2r =0 slthe2r=0
+ stlthe2 =0 swthe2=0
+ the3r =0.1 stthe3r=0
+ slthe3r =0 stlthe3=0
+ swthe3 =0 gam1r=0.02
+ slgam1 =0 swgam1=0
+ etadsr =0.60 alpr=0.01
+ etaalp =0 slalp=0
+ swalp =0 vpr=0.4
+ gamoor =0.006 slgamoo=0
+ etagamr =2.0 mor=0.5
+ stmo =0 slmo=0
+ etamr =2.0 zet1r=1.0
+ etazet =0.5 slzet1=0
+ vsbtr =2.5 slvsbt=0
+ alr =10 stal=0
+ sla1 =0 swa1=0
+ a2r =30 sla2=0
+ swa2 =0 a3r=0.8
+ sla3 =0 swa3=0
+ tox =15.00e-9 col=0.3e-9
+ ntr =2.0e-20 nfr=5.0e-11
+ acm =2 hdif=1u
+ js =1e-3 cj=1e-3
+ mj =0.5 pb=0.8
+ cjsw =1e-9 cjgate=1e-9
+ mjsw =0.3 php=0.8

```

LEVEL 54: BSIM4 MOSFET Model

The UC Berkeley BSIM4 model explicitly addresses many issues in modeling sub-0.13 micron CMOS technology and RF high-speed CMOS circuit simulation. The Level 54 model is based on the UC Berkeley BSIM4 MOS model. BSIM4.5 is fully supported in this release. For details, see the BSIM web site: <http://www-device.eecs.berkeley.edu/~bsim3/bsim4.html>

The following sections discuss these topics:

- [BSIM4 MOSFET Model — General Syntax](#)
- [BSIM4 MOSFET Model Parameters](#)
- [BSIM4 MOSFET Model — Usage Notes](#)
- [BSIM4 MOSFET Model — Version Updates](#)
- [BSIM4 MOSFET Model — Output Templates](#)

BSIM4 MOSFET Model — General Syntax

The general syntax for including a BSIM4 model element in a netlist is:

```
Mxxx nd ng ns [nb] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val]
+ [RGATEMOD=val] [RBODYMOD=val] [TRNQSMOD=val]
+ [ACNQSMOD=val] [GEOMOD=val] [RGEOMOD=val]
+ [NRS=val] [NRD=val] [RBPB=val] [RBPD=val]
+ [RBPS=val] [RBDB=val] [RBSB=val] [NF=val]
+ [MIN=val] [RDC=val] [RSC=val] [DELVTO=val]
+ [MULU0=val] [DELK1=val] [DELNFCT=val]
+ [DELTOX=val] [OFF] [IC=Vds, Vgs, Vbs]
+ [WNFLAG=val] [MULID0=val]
```

BSIM4 MOSFET Model Parameters

Parameter	Description
<i>MNAME</i>	MOSFET model name reference.
Mxxx	
ACNQSMOD	AC small-signal NQS model selector.
GEOMOD	Geometry-dependent parasitics model selector—specifies how the end S/D diffusions connect.

Parameter	Description
RBODYMOD	Substrate resistance network model selector.
RGATEMOD	Gate resistance model selector.
RGEOMOD	Source/drain diffusion resistance and contact model selector—specifies the end S/D contact type: point wide or merged) and how to compute the S/D parasitics resistance.
TRNQSMOD	Transient NQS model selector.
WNFLAG	Turn on to select bin model based on width per NF for multi-finger devices.
AD	Drain diffusion area.
AS	Source diffusion area.
DELK1	Shift in body bias coefficient (K1).
DELNFCT	Shift in subthreshold swing factor (NFACTOR).
DELTOX	Shift in gate electrical and physical oxide thickness (TOXE and TOXP). That is, the difference between the electrical and physical gate oxide/insulator thicknesses.
DELVTO (DELVT0)	Shift in the VTH0 zero-bias threshold voltage.
IC	Initial guess in the order
L	BSIM4 MOSFET channel length in meters.
MIN	Whether to minimize the number of drain or source diffusions for even-number fingered device.
MULID0	Scaling factor of drain current, the default is 1.0.
MULU0	Low-field mobility (U0) multiplier.
NB	Bulk terminal node name.
ND	Drain terminal node name.
NF	Number of device fingers.
NG	Gate terminal node name.
NRD	Number of drain diffusion squares.
NRS	Number of source diffusion squares.
NS	Source terminal node name.

Parameter	Description
OFF	Sets the initial condition to OFF in DC analysis.
PD	Perimeter of the drain junction: • If PERMOD=0, it excludes the gate edge. • Otherwise, it includes the gate edge.
PS	Perimeter of the source junction: • If PERMOD=0, it excludes the gate edge. • Otherwise, it includes the gate edge.
RBDB	Resistance connected between dbNode and bNode.
RBPB	Resistance connected between bNodePrime and bNode.
RBPD	Resistance connected between bNodePrime and dbNode.
RBPS	Resistance connected between bNodePrime and sbNode.
RBSB	Resistance connected between sbNode and bNode.
RDC	Drain contact resistance for per-finger device.
RSC	Source contact resistance for per-finger device.
W	BSIM4 MOSFET channel width in meters.

BSIM4 MOSFET Model — Usage Notes

- [BSIM4 Model — Usage Notes: Improvements Over BSIM3 Version 3](#)
- [BSIM4 Model — Usage Notes: Parameter Range Limit](#)
- [BSIM4 Model — Usage Notes: TSMC Diode Model](#)
- [BSIM4 Model — Usage Notes: JUNCAP2 Model Support](#)
- [BSIM4 Model — Usage Notes: STI / LOD Parameters](#)
- [BSIM4 Model — Usage Notes: LMLT and WMLT Parameters](#)
- [BSIM4 Model — Usage Notes: PrimeSim Continuum Junction Diode Model and ACM](#)

BSIM4 Model — Usage Notes: Improvements Over BSIM3 Version 3

BSIM4 includes the following major improvements and additions over BSIM3 version 3:

- An accurate new model of the intrinsic input resistance (R_{ii}) for both RF, high-frequency analog, and high-speed digital applications
- A flexible substrate resistance network for RF modeling
- A new accurate channel thermal noise model, and a noise partition model for the induced gate noise
- A non-quasi-static (NQS) model consistent with the R_{ii} -based RF model, and a consistent AC model that accounts for the NQS effect in both transconductances and capacitances
- An accurate gate direct tunneling model
- A comprehensive and versatile geometry-dependent parasitics model for source/drain connections and multi-finger devices
- An improved model for steep vertical retrograde doping profiles
- A better model for pocket-implanted devices in V_{th} , the bulk charge effect model, and R_{out}
- An asymmetrical and bias-dependent source/drain resistance, either internal or external to the intrinsic MOSFET
- Accepts either the electrical or physical gate oxide thickness as the model input in a physically accurate manner
- A quantum mechanical charge-layer-thickness model for both IV and CV
- A more accurate mobility model for predictive modeling
- A gate-induced drain leakage (G_{IDL}) current model, not available in earlier BSIM models
- An improved unified flicker ($1/f$) noise model, which is smooth over all bias regions, and which considers the bulk charge effect
- Different diode IV and CV characteristics for the source and drain junctions
- A junction diode breakdown with or without current limiting
- A dielectric constant of the gate dielectric as a model parameter
- `.OPTION LIST` now prints the total capacitances, instead of just the intrinsic capacitances for the BSIM4 (Level 54) MOSFET model

BSIM4.2.1 has the following improvements over BSIM4.2.0:

- A new `GISL` (Gate Induced Source Leakage) current component corresponds to the same current at the drain side (`GIDL`).
- The warning limits for effective channel length, channel width, and gate oxide thickness have been reduced to avoid unnecessary warnings if you use BSIM4 aggressively, beyond the desired model card application ranges.
- The `DELTOX` parameter in the MOS active element (M) models the relative variation on the trans conductance (oxide thickness) of the MOS in Monte Carlo analysis.
- `.OPTION LIST` can now print an element summary for the MOSFET Level=54 model

BSIM4 Model — Usage Notes: Parameter Range Limit

Simulation reports either a warning or a fatal error if BSIM3 version 3 parameters fall outside predefined ranges. These range limitations prevent (or at least warn of) potential numerical problems.

To control the maximum number of simulation warning messages printing to the output file, use:

```
.OPTION WARNLIMIT=#
```

In the preceding `.OPTION` statement, `#` is the maximum number of warning messages that simulation reports. The default `WARNLIMIT` value is 1. In some cases, simulation checks parameters only if you set the `PARAMCHK=1` model parameter.

Parameter	Limits	Comment
ACDE	< 0.1, > 1.6 warn	if (version < 4.299999) acde<0.4 warn
B1	b1=-weff fatal	
CKAPPAD	< 0.02 warn and set 0.02	if (version >= 4.299999 &&stiMod != 2) version < 4.299999)
CKAPPAS	< 0.02 warn and set 0.02	if (version >= 4.299999 &&stiMod != 2) version < 4.299999)
CLC	< 0 fatal	
DELTA	< 0 fatal	
DROUT	< 0 fatal	
DSUB	< 0 fatal	
DVT1	< 0 fatal	

Parameter	Limits	Comment
DVT1W	< 0 fatal	
GBMIN	< 1.0e-20 warn	
LINTNOI	> leff/2, fatal	if (version>=4.39999)
LPE0	< -leff, fatal	
LPEB	< -leff, fatal	
MOIN	< 5.0, > 25 warn	
NDEP	<= 0 fatal	
NF	< 1 fatal	
NGATE	< 0,> 1.0e25 fatal	
NGCON	< 1 fatal	If ngcon !=1,2 warn and set 1
NOFF	=0 warn and set 1; < 0.1 warn	if (version < 4.2999 stiMod=2) noff > 4 warn
NSUB	<= 0 fatal	
PCLM	<= 0 fatal	
PHIN	< -0.4 fatal	if (version>=4.4 phin <=0 fatal
PSCBE2	<= 0 fatal	
TOXREF	<= 0 fatal	
VOFFCV	< -0.5 warn	if (version<4.2999 stiMod=2) voffcv > 0.5 warn
W0	w0=-weff fatal	
XJ	<= 0 fatal	
If PARAMCHK=1 the following parameter limit range is added		
A2	< 0.01 warn, set 0.01; >1 .0 warn, set 1 and a1 set 0	• If the total calculated A2 (including binnings) is smaller than 0.01, then it is set to 0.01. • If the total A2 is larger than 1, it is set to 1.
B1	b1+weff< 1e-7 warn	if (version>=4.39999) b1+weff<1e-9 warn
CDSC	< 0 warn	

Parameter	Limits	Comment
CDSCD	< 0 warn	
CGBO	< 0 warn	if (version >4.4999)
CGDO	< 0 warn	
CGSO	< 0 warn	
DVT0	< 0 warn	
ETA0	< 0 warn	
KVSAT	< -1.0, > 1.0 fatal	
LAMBDA	> 1e-9 warn	if (version>=4.3)
LC	< 0	if (version>= 4.3)
LEFF	<= 1e-9 warn	
LEFFCV	<= 1e-9 warn	
LODETA0	<= 0 warn	if (version>= 4.4)
LODETA0	<= 0	
LODK2	<= 0 warn	if (version>= 4.4)
LODK2	<= 0	
MJD	>= 0.99 warn	
MJS	>= 0.99 warn	
MJSWD	>= 0.99 warn	
MJSWGD	>= 0.99 warn	
MJSWGS	>= 0.99 warn	
MJSWS	>= 0.99 warn	
NDEP	<= 1e12, >= 1e21 warn	
NF	>1 fatal	if (stiMod == 2)
NFACTOR	< 0 warn	
NGATE	0 <ngate <= 1e18 warn	
NIGBACC	<= 0 fatal	

Parameter	Limits	Comment
NIGBINV	≤ 0 fatal	
NIGC	≤ 0 fatal	
NJD	< 0 warn	
NJS	< 0 warn	
NJTS	< 0 warn	
NJTSD	< 0 warn	
NJTSSW	< 0 warn	
NJTSSWD	< 0 warn	
NJTSSWG	< 0 warn	
NJTSSWGD	< 0 warn	
NSUB	$\leq 1e14, \geq 1e21$ warn	
NTNOI	< 0 warn	
PDIBL1	< 0 warn	
PDIBL2	< 0 warn	
PDITS	< 0 fatal	
PDITSL	< 0 fatal	
PIGCD	≤ 0 fatal	
POXEDGE	≤ 0 fatal	
PROUT	< 0 fatal	
PRWG	< 0 warn, set 0	
RDS0	< 0 warn, set 0	
RDSW	< 0 warn, set 0	
RDSWMIN	< 0 warn, set 0	
RNOIA	< 0 warn	
RNOIB	< 0 warn	
RSHG	≤ 0 warn	if (version ≥ 4.299999 && stiMod $\neq 2$)

Parameter	Limits	Comment
SA	< 0 fatal	
SA0	<= 0 fatal	
SB	< 0 fatal	
SB0	<= 0 fatal	
SC	< 0 warn	if (wpemod=1)
SCA	< 0 warn	if (wpemod=1)
SCB	< 0 warn	if (wpemod=1)
SCC	< 0 warn	if (wpemod=1)
SCREF	< 0 warn	if (wpemod=1)
SD	< 0 fatal	If nf>1
SL	< 1e-7 warn; < 0 fatal	
SW	< 1e-7 warn; < 0 fatal	
TNOB	< 0 warn	
TNOIA	< 0 warn	
TOXE	Toxe + Delttox < 1e-10 warn	
TOXM	< 1e-10 warn	
TOXP	Toxp + Delttox < 1e-10 warn	
VSATTEMP	<1e3 warn	
VTL	< 6e4 warn	if (version>=4.3)
VTSD	< 0 warn	
VTSS	< 0 warn	
VTSSWD	< 0 warn	
VTSSWGD	< 0 warn	
VTSSWGS	< 0 warn	
VTSSWS	< 0 warn	

Parameter	Limits	Comment
W0	w0+weff< 1e-7 warn	if (version>=4.39999) w0+weff < 1e-9 warn
WEFF	<= 1e-9 warn	
WEFFCV	<= 1e-9 warn	
WLOD	< 0 fatal	
XN	< 3	if (version>= 4.3)
XRCRG1	<=0 warn	if (version >= 4.299999 && stiMod != 2)

BSIM4 Model — Usage Notes: TSMC Diode Model

The PrimeSim Continuum solution of MOSFET Level 54 (BSIM4) supports a TSMC junction diode model. You can use this TSMC junction diode model to simulate the temperature dependence, source/body, and drain/body currents of a junction diode.

Note:

For a complete description of this effect model, visit the official UCB BSIM web site:

<http://www-device.eecs.berkeley.edu/~bsim3/>

You can order either of these models directly from Taiwan Semiconductor Manufacturing Company (TSMC)—not from Synopsys. See the TSMC web site: <http://www.tsmc.com>.

BSIM4 Model — Usage Notes: JUNCAP2 Model Support

The PrimeSim Continuum solution BSIM4 support for the juncap2 junction model is based on Philips JUNCAP2 model in BSIM4 version 4.2 and later. You use the flag `JUNCAP=0` to turn on the built-in BSIM4 default junction model. You can use `JUNCAP=1` to switch to the Juncap1 model and `JUNCAP=2` to access the Juncap2 model. The Juncap2 model has 5 versions: 200.0, 200.1, 200.2, 200.3, and 200.33, which can be toggled using the `JCAP2VERSION` flag. A new express option of the JUNCAP2 model, invoked by setting `SWJUNEXP=1`, is also supported in BSIM4 model.

Parameter	Unit	Default	Description
JUNCAP	-	0	Flag to turn on juncap diode model, 1 for juncap1, 2 for juncap2
JCAP2VERSION	-	200.1	Juncap2 model version (200.0, 200.1)

Parameter	Unit	Default	Description
CBBTBOT	AV^{-3}	1.00E-012	Band-to-band tunneling pre-factor of bottom component
CBBTGAT	$AV^{-3}m$	1.00E-018	Band-to-band tunneling pre-factor of gate-edge component
CBBTSTI	$AV^{-3}m$	1.00E-018	Band-to-band tunneling pre-factor of STI-edge component
CJORBOT	F/m^2	1.00E-003	Zero-bias capacitance per unit-of-area of bottom component
CJORGAT	F/m^2	1.00E-009	Zero-bias capacitance per unit-of-length of gate-edge component
CJORSTI	F/m^2	1.00E-009	Zero-bias capacitance per unit-of-length of STI-edge component
CSRHBOT	A/m^2	1.00E+002	Shockley-Read-Hall pre-factor of bottom component
CSRHGAT	A/m^2	1.00E-004	Shockley-Read-Hall pre-factor of gate-edge component
CSRHSTI	A/m^2	1.00E-004	Shockley-Read-Hall pre-factor of STI-edge component
CTATBOT	A/m^3	1.00E+002	Trap-assisted tunneling pre-factor of bottom component
CTATGAT	A/m^2	1.00E-004	Trap-assisted tunneling pre-factor of gate-edge component
CTATSTI	A/m^2	1.00E-004	Trap-assisted tunneling pre-factor of STI-edge component
FBBTBOT	V/m	1.00E+009	Normalization field at the reference temperature for band-to-band tunneling of bottom component
FBBTGAT	V/m	1.00E+009	Normalization field at the reference temperature for band-to-band tunneling of gate-edge component
FBBTSTI	V/m	1.00E+009	Normalization field at the reference temperature for band-to-band tunneling of STI-edge component
IDSATRBOT	A/m^2	1.00E-012	Saturation current density at the reference temperature of bottom component
IDSATRGAT	A_m	1.00E-018	SATURATION current density at the reference temperature of gate-edge component

Parameter	Unit	Default	Description
IDSATRSTI	A/m	1.00E-018	Saturation current density at the reference temperature of STI-edge component
IMAX	A	1000	Maximum current up to which forward current behaves exponentially
MEFFTATBOT	-	0.25	Effective mass (in units of m_0) for trap-assisted tunneling of bottom component
MEFFTATGAT	-	0.25	Effective mass (in units of m_0) for trap-assisted tunneling of gate-edge component
MEFFTATSTI	-	0.25	Effective mass (in units of m_0) for trap-assisted tunneling of STI-edge component
PBOT	-	0.5	Grading coefficient of bottom component
PBRBOT	V	4	Breakdown onset tuning parameter of bottom component
PBRGA	V	4	Breakdown onset tuning parameter of gate-edge component
PBRSTI	V	4	Breakdown onset tuning parameter of STI-edge component
PGAT	-	0.5	Grading coefficient of gate-edge component
PHIGBOT	V	1.16	Zero-temperature bandgap voltage of bottom component
PHIGGAT	V	1.16	Zero-temperature bandgap voltage of gate-edge component
PHIGSTI	V	1.16	Zero-temperature bandgap voltage of STI-edge component
PSTI	-	0.5	Grading coefficient of STI-edge component
STFBBTBOT	1/K	-1.00E-003	Temperature scaling parameter for band-to-band tunneling of bottom component
STFBBTGAT	1/K	-1.00E-003	Temperature scaling parameter for band-to-band tunneling of gate-edge component
STFBBTSTI	1/K	-1.00E-003	Temperature scaling parameter for band-to-band tunneling of STI-edge component
VBIRBOT	V	1	Built-in voltage at the reference temperature of bottom component

Parameter	Unit	Default	Description
VBIRGAT	V	1	Built-in voltage at the reference temperature of gate-edge component
VBIRSTI	V	1	Built-in voltage at the reference temperature of TI-edge component
VBRBOT	V	10	Breakdown voltage of bottom component
VBRGAT	V	10	Breakdown voltage of gate-edge component
VBRSTI	V	10	Breakdown voltage of STI-edge component
XJUNGAT	m	1.00E-007	Junction depth of gate-edge component
XJUNSTI	m	1.00E-007	Junction depth of STI-edge component
SWJUNEXP	-	0	Flag for JUNCAP-express; 0=full model,1=express model
VJUNREF	-	2.5	Typical maximum junction voltage; usually about 2*VSUP
FJUNQ	-	0.03	Fraction below which junction capacitance components are considered negligible

BSIM4 Model — Usage Notes: STI / LOD Parameters

The PrimeSim Continuum solution BSIM4 supports the full STI (Shallow Trench Isolation) or LOD (Length of Oxide Definition) induced mechanical stress-effect model (for version 4.3 or later), which was first released in the UCB BSIM4.3.0 model version. The PrimeSim Continuum solution BSIM4 turns on the simulation of this effect when the following conditions (consistent with those of the UCB BSIM4 model) are satisfied:

```
if (VERSION >=4.3)
{
if ((SA > 0 and SB > 0 and NF==1) or ( SA > 0 and SB > 0 and ( NF >1 and
SD > 0 )))
{ UCB's STI/LOD model is turned on}
}
```

If `VERSION>=4.3`, the STI model is not dependent on `STIMOD=0` or `1`. In this case, the STI model is applied to the model similar to the UCB STI model. When parameter values that satisfy `SA>0`, `SB>0`, `NF>1`, and `SD=0` are given in model cards, no evaluation of such effect is performed.

Parameter	Unit	Default	Bin?	Description
STIMOD (Also instance parameter)		0.0 (V < 4.3) 1.0 (V >=4.3)		STI model selector, which gives priority to the instance parameter. • 0: No STI effect. • 1: UCB's STI model • 2: TSMC's STI model
KU0	M	0.0	No	Mobility degradation/enhancement coefficient for stress effect
KVSAT	M	0.0	No	Saturation velocity degradation/enhancement parameter for stress effect. $1.0 \leq kvsat \leq 1.0$
KVTH0	V*m	0.0	No	Threshold shift parameter for stress effect
LKU0		0.0	No	Length dependence of KU0
LLODKU0		0.0	No	Length parameter for U0 stress effect, >0
LLODVTH		0.0	No	Length parameter for Vth stress effect, >0
LODETA0	M	1.0	No	ETA0 shift modification factor for stress effect, >0
LODK2	m	1.0	No	K2 shift modification factor for stress effect, >0
PKVTH0		0.0	No	Cross-term dependence of KVTH0
SA (instance parameter)		0.0		Distance between S/D diffusion edge to poly gate edge from one side. If not given, or, if (≤ 0), the stress effect is turned off.
SAREF	M	1e-06	No	Reference distance for SA, >0.0
SB (instance parameter)		0.0		Distance between S/D diffusion edge to poly gate edge from the other side. If not given or, if (≤ 0), the stress effect is turned off
SBREF	M	1e-06	No	Reference distance for SB, >0.0
SD (instance parameter)		0.0		Distance between neighboring fingers. For NF >1, If not given or (≤ 0), stress effect is turned off

Parameter	Unit	Default	Bin?	Description
STETA0		0.0	No	ETA0 shift factor related to VTH0 change
STK2		0.0	No	K2 shift factor related to VTh0 change
TKU0		0.0	No	Temperature coefficient of KU0
WKU0		0.0	No	Width dependence of KU0
WKVTH0		0.0	No	Width dependence of KVTH0
WLOD	M	0.0	No	Width parameter for stress effect
WLODKU0		0.0	No	Width parameter for U0 stress effect, >0
WLODVTH		0.0	No	Width parameter for Vth stress effect, >0

BSIM4 Model — Usage Notes: LMLT and WMLT Parameters

You can use `LMLT` and `WMLT` to shrink the length and width in memory design. The `LMLT` and `WMLT` parameters are unitless, and are used to scale (usually scale down) MOSFET drawn length and width (specified in BSIM4 MOSFET instance lines), respectively. This makes memory design and netlist creation quite convenient because most (if not all) memory circuits use the smallest feature sizes as the process capability improves, even within the same generation of CMOS technology.

The shrunken device length and width will then be further offset (by `XL` and `XW`, respectively) to the actual device size in lithography and etching process steps, and finally to the electrical size as a result of subsequent ion implementation and annealing steps.

Name (Alias)	Default	Description
LMLT	1.0	Channel length shrinking factor
WMLT	1.0	Device width shrinking factor

Both `LMLT` and `WMLT` must be greater than 0; if not, simulation resets them to 1.0 (default) and issues a warning message.

To use these two parameters, add them in the model cards, without any other modifications. For example:

```
.model nmos nmos
+ level=54 lmlt=0.85 wmlt=0.9
```

- The drawn channel length (L) is multiplied by L_{MLT} .
- The drawn channel width (W) is multiplied by W_{MLT} ,

BSIM4 evaluates the effective length and width, L_{eff} and W_{eff} , as:

$$L_{eff} = L_{new} - 2.0 * dL$$

$$W_{eff} = W_{new} - 2.0 * dW$$

L_{new} and W_{new} are evaluated as:

$$L_{new} = L + XL$$

$$W_{new} = W + XW$$

dL and dW are evaluated with L_{new} and W_{new} :

$$T0 = \text{pow}(L_{new}, LLN)$$

$$T1 = \text{pow}(W_{new}, LWN)$$

$$dL = LINT + LL/T0 + LW / T1 + LWL / (T0*T1)$$

$$T2 = \text{pow}(L_{new}, WLN)$$

$$T3 = \text{pow}(W_{new}, WWN)$$

$$dW = WINT + WL / T2 + WW / T3 + WWL / (T2*T3)$$

Similarly, the preceding equations determine the L_{dlc} , L_{dlcig} , L_{effCV} , W_{dwc} , W_{dwcig} , W_{effCV} , W_{effCJ} , $grgeltd$, and W_{new} model variables and quantities.

When multiplied by the L_{MLT} and W_{MLT} parameters, L_{eff} and W_{eff} become:

$$L_{eff} = L_{new}' - 2.0 * dL$$

$$W_{eff} = W_{new}' - 2.0 * dL$$

L_{new}' and W_{new}' are evaluated as:

$$L_{new}' = L * L_{MLT} + XL$$

$$W_{new}' = W * W_{MLT} + XW$$

Similarly, dL , dW , L_{dlc} , L_{dlcig} , L_{effCV} , W_{dwc} , W_{dwcig} , W_{effCV} , W_{effCJ} , and $grgeltd$ are all evaluated from L_{new}' and W_{new}' .

BSIM4 Model — Usage Notes: PrimeSim Continuum Junction Diode Model and ACM

BSIM4 now supports Area Calculation Method (ACM) similar to BSIM3 version 3 for the following models and corresponding ACM values:

- For the PrimeSim Continuum solution junction model, specify $ACM=0,1,2$, or 3 .
- For the Berkeley BSIM4 junction model, specify $ACM=10,11,12$, or 13 .

For the junction current, junction capacitance, and parasitic resistance equations corresponding to $ACM=0,1,2,3$ see [MOSFET Diode Models](#).

JUNCAP Model Parameters:

Parameter	Description
nf	Number of device fingers
min	Whether to minimize the number of drain or source diffusions for even-number fingered device
rbdb	Resistance connected between the internal drain-side body node and the external body node
rbsb	Resistance connected between the internal source-side body node and the external body node
rbpb	Resistance connected between the internal reference body node and the external body node
rbps	Resistance connected between the internal reference body node and the internal drain-side body node
rbpd	Resistance connected between the internal reference body node and the internal source-side body node
trnqsmode	Transient NQS model selector
acnqsmode	AC small-signal NQS model selector
rbodmode	Substrate resistance network model selector
rgatmode	Gate resistance model selector
geomode	Geometry-dependent parasitics model selector
rgeomode	Source/Drain diffusion resistance and contact model selector

MOSFET Level 54 uses the generic MOSFET model parameters described in [Common MOSFET Model Parameters](#). It also uses the parameters described in this section, which apply only to MOSFET Level 54.

The simulation calculates R_d and R_s as follows:

$$R_d(TEMP) = R_d(TNOM) * (1 + TRD * (TEMP - TNOM))$$

$$R_s(TEMP) = R_s(TNOM) * (1 + TRS * (TEMP - TNOM))$$

JUNCAP Instance Parameters:

Parameter	Unit	Default	Description
RDC	ohm	0.0	Drain contact resistance for the per-finger device

Parameter	Unit	Default	Description
RSC	ohm	0.0	Source contact resistance for the per-finger device
DELVTO (DELVT0)	V	0.0	Shift in the zero-bias threshold voltage (VTH0)
MULU0		1.0	Low-field mobility (U0) multiplier
DELK1	$V^{1/2}$		Shift in the body bias coefficient (K1)
DELNFCT		0.0	Shift in subthreshold swing factor (NFACTOR)

Model Selectors / Controllers:

Parameter	Default	Bin	Description
VERSION	4.5	NA	Model version number
BINUNIT	1	NA	Binning unit selector
PARAMCHK	1	NA	Switch for the parameter check
MOBMOD	1	NA	Mobility model selector
RDSMOD	0	NA	Bias-dependent source/drain resistance model selector
IGCMOD	0	NA	Gate-to-channel tunneling current model selector
IGBMOD	0	NA	Gate-to-substrate tunneling current model selector
CAPMOD	2	NA	Capacitance model selector
RGATEMOD	0 (no gate resistance)		Gate resistance model selector
RBODYMOD	0 (network off)	NA	Substrate resistance network model selector
TRNQSMOD	0	NA	Transient NQS model selector
ACNQSMOD	0	NA	AC small-signal NQS model selector
FNOIMOD	1	NA	Flicker noise model selector
TNOIMOD	0	NA	Thermal noise model selector
DIOMOD	1	NA	Source/drain junction diode IV model selector

Parameter	Default	Bin	Description
TEMPMOD	0	NA	Temperature mode selector
PERMOD	1	NA	PS/PD includes/excludes the gate-edge perimeter
GEOMOD	0 (isolated)	NA	Geometry-dependent parasitics model selector
RGEOMOD	0 (no S/D diffusion resistance)	NA	Source/drain diffusion resistance and contact model selector
WPEMOD	0	NA	Flag for WPE model (WPEMOD=1 to activate this model)

Process Parameters:

Parameter	Default	Bin	Description
EPSROX	3.9 (SiO ₂)	No	Gate dielectric constant relative to vacuum
TOXE	3.0e-9m	No	Electrical gate equivalent oxide thickness
TOXP	TOXE	No	Physical gate equivalent oxide thickness
TOXM	TOXE	No	Tox at which simulation extracts parameters
DTOX	0.0m	No	Defined as (TOXE-TOXP)
XJ	1.5e-7m	Yes	S/D junction depth
GAMMA1 (Y ₁ in equation)	calculated (V ^{1/2})	Yes	Body-effect coefficient near the surface
GAMMA2 (Y ₂ in equation)	calculated (V ^{1/2})	Yes	Body-effect coefficient in the bulk
NDEP	1.7e17cm ⁻³	Yes	Channel doping concentration at the depletion edge for the zero body bias
NSUB	6.0e16cm ⁻³	Yes	Substrate doping concentration
NGATE	0.0cm ⁻³	Yes	Poly Si gate doping concentration
NSD	1.0e20cm ⁻³	Yes	Source/drain doping concentration. If (NSD > 1.0e-23) NSD*=1.0e-6
VBX	calculated (v)	No	V _{bs} at which the depletion region width equals X _T
XT	1.55e-7m	Yes	Doping depth

Parameter	Default	Bin	Description
RSH	0.0ohm/square	No	Source/drain sheet resistance
RSHG	0.1ohm/square	No	Gate electrode sheet resistance

Basic Model Parameters:

Parameter	Default	Bin	Description
A0	1.0	Yes	Coefficient of the channel-length dependence of the bulk charge effect
A1	$0.0V^{-1}$	Yes	First non-saturation effect parameter
A2	1.0	Yes	Second non-saturation factor
AGS	$0.0V^{-1}$	Yes	Coefficient of the V_{gs} dependence of the bulk charge effect
B0	0.0m	Yes	Bulk charge effect coefficient for the channel width
B1	0.0m	Yes	Bulk charge effect width offset
CDSC	$2.4e-4F/m^2$	Yes	Coupling capacitance between the source/drain and the channel
CDSCB	$0.0F/(Vm^2)$	Yes	Body-bias sensitivity of CDSC
CDSCD	$0.0(F/Vm^2)$	Yes	Drain-bias sensitivity of DCSC
CIT	$0.0F/m^2$	Yes	Interface trap capacitance
DELTA δ (in equation)	0.01V	Yes	Parameter for DC V_{dseff}
DROUT	0.56	Yes	Channel-length dependence of the DIBL effect on R_{out}
DSUB	DROUT	Yes	DIBL coefficient exponent in the subthreshold region
DVT0	2.2	Yes	First coefficient of the short-channel effect on V_{th}
DVT0W	0.0	Yes	First coefficient of the narrow width effect on V_{th} for a small channel length
DVT1	0.53	Yes	Second coefficient of the short-channel effect on V_{th}

Parameter	Default	Bin	Description
DVT1W	$5.3\text{e}6\text{m}^{-1}$	Yes	Second coefficient of the narrow width effect on V_{th} for a small channel length
DVT2	-0.032V^{-1}	Yes	Body-bias coefficient of the short-channel effect on V_{th}
DVT2W	-0.032V^{-1}	Yes	Body-bias coefficient of narrow width effect for small channel length
DVTP0	0.0m	Yes	First coefficient of the drain-induced V_{th} shift due to long-channel pocket devices
DVTP1	0.0V^{-1}	Yes	First coefficient of the drain-induced V_{th} shift due to long-channel pocket devices
DWB	$0.0\text{m}/\text{V}^{1/2}$	Yes	Coefficient of the body bias dependence of the W_{eff} bias dependence
DWG	$0.0\text{m}/\text{V}$	Yes	Coefficient of gate bias dependence of W_{eff}
ETA0	0.08	Yes	DIBL coefficient in the subthreshold region
ETAB	-0.07V^{-1}	Yes	Body-bias coefficient for the DIBL effect for the subthreshold
EU	1.67 (NMOS); 1.0 (PMOS)	No	Exponent for the mobility degradation of $MOBMOD=2$
FPROUT	$0.0\text{V}/\text{m}^{0.5}$	Yes	Effect of the pocket implant on R_{out} degradation
K1	$0.5\text{V}^{1/2}$	Yes	First-order body bias coefficient
K2	0.0	Yes	Second-order body bias coefficient
K3	80.0	Yes	Narrow width coefficient
K3B	0.0V^{-1}	Yes	Body effect coefficient of K3
KETA	-0.047V^{-1}	Yes	Body-bias coefficient of the bulk charge effect
LINT	0.0m	No	Channel-length offset parameter
LP	$1\text{e}-8(\text{m})$	Yes	Mobility channel length degradation of $MOBMOD=2$
LPE0	$1.74\text{e}-7\text{m}$	Yes	Lateral non-uniform doping parameter
LPEB	0.0m	Yes	Lateral non-uniform doping effect on K1

Parameter	Default	Bin	Description
MINV	0.0	Yes	V_{gsteff} fitting parameter for the moderate inversion condition
NFACTOR	1.0	Yes	Subthreshold swing factor
PCLM	1.3	Yes	Channel-length modulation parameter
PDIBLC1	0.39	Yes	Parameter for the DIBL effect on R_{out}
PDIBLC2	0.0086	Yes	Parameter for the DIBL effect on R_{out}
PDIBLCB	$0.0V^{-1}$	Yes	Body bias coefficient of the DIBL effect on R_{out}
PDITS	$0.0V^{-1}$	Yes	Impact of the drain-induced V_{th} shift on R_{out}
PDITSD	$0.0V^{-1}$	Yes	V_{ds} dependence of the drain-induced V_{th} shift for R_{out}
PDITSL	$0.0m^{-1}$	No	Channel-length dependence of the drain-induced V_{th} shift for R_{out}
PHIN	0.0V	Yes	Non-uniform vertical doping effect on the surface potential
PSCBE1	4.24e8V/m	Yes	First substrate current induced body-effect parameter
PSCBE2	1.0e-5m/V	Yes	Second substrate current induced body-effect parameter
PVAG	0.0	Yes	Gate-bias dependence of Early voltage
U0	$0.067m^2/(Vs)$ (NMOS); $0.025m^2/(Vs)$ (PMOS)	Yes	Low-field mobility
UA	1.0e-9m/V for MOBMOD=0 and 1; 1.0e-15m/V for MOBMOD=2	Yes	Coefficient of the first-order mobility degradation due to the vertical field
UB	$1.0e-19m^2/V^2$	Yes	Coefficient of the second-order mobility degradation due to the vertical field
UC	-0.0465 V^{-1} for MOBMOD=1; -0.0465e-9 m/V^2 for MOBMOD=0 and 2	Yes	Coefficient of the mobility degradation due to the body-bias effect
UD	$1e14(1/m^2)$	Yes	Mobility coulomb scattering coefficient

Parameter	Default	Bin	Description
UP	$0(1/m^2)$	Yes	Mobility channel length coefficient
VBM	-3.0V	Yes	Maximum applied body bias in the VTH0 calculation
VFB	-1.0V	Yes	Flat-band voltage (PHIN)
VOFF	-0.08V	Yes	Offset voltage in subthreshold region for large W and L
VOFFL	0.0mV	No	Channel-length dependence of VOFF
VSAT	8.0e4m/s	Yes	Saturation velocity
VTH0 or VTHO	0.7V (NMOS) -0.7V (PMOS)	Yes	Long-channel threshold voltage at $V_{bs}=0$
W0	2.5e-6m	Yes	Narrow width parameter
WINT	0.0m	No	Channel-width offset parameter

Asymmetric and Bias-Dependent Rds Model Parameters:

Parameter	Default	Bin	Description
PRWB	0.0V-0.5	Yes	Body-bias dependence of the LDD resistance
PRWG	1.0V-1	Yes	Gate-bias dependence of the LDD resistance
RDSW	200.0 ohm(μ m)WR	Yes	Zero bias LLD resistance per unit width for RDSMOD=0
RDSWMIN	0.0 ohm(μ m)WR	No	LDD resistance per unit width at high V_{gs} and zero V_{bs} for RDSMOD=0
RDW	100.0 ohm(μ m)WR	Yes	Zero bias lightly-doped drain resistance $R_d(v)$ per unit width for RDSMOD=1
RSW	100.0 ohm(μ m)WR	Yes	Zero bias lightly-doped source resistance $R_s(V)$ per unit width for RDSMOD=1
RSWMIN	0.0 ohm(μ m)WR	No	Lightly-doped source resistance per unit width at high V_{gs} and zero V_{bs} for RDSMOD=1
WR	1.0	Yes	Channel-width dependence of the LDD resistance
NRS	1.0	No	Number of source diffusion squares
NRD	1.0	No	Number of drain diffusion squares

Impact Ionization Current Model Parameters:

Parameter	Default	Bin	Description
ALPHA0	0.0Am/V	Yes	First parameter of the impact ionization current
ALPHA1	0.0A/V	Yes	Isub parameter for length scaling
BETA0	30.0V	Yes	Second parameter for the impact ionization current

Gate-Induced Drain Leakage Model Parameters:

Parameter	Default	Bin	Description
AGIDL	0.0ohm	Yes	Pre-exponential coefficient for GIDL
AGISL	0.0	Yes	Pre-exponential coefficient for GISL
BGIDL	2.3e9V/m	Yes	Exponential coefficient for GIDL
BGISL	2.3e9V/m	Yes	Exponential coefficient for GISL
CGIDL	0.5V3	Yes	Parameter for the body-bias effect on GIDL
EGIDL	0.8V	Yes	Fitting parameter for band bending for GIDL
EGISL	0.8V	Yes	Fitting parameter for Band bending

Gate Dielectric Tunneling Current Model Parameters:

Parameter	Default	Bin	Description
AIGS	1.36e-2(NMOS) and 9.8e-3(PMOS)	Yes	Parameter for I_{gs}
AIGBACC	$0.43 (F_s^2 / \epsilon_g)^{0.5} m^{-1}$	Yes	Parameter for I_{gb} in the accumulation
AIGBINV	$0.35 (F_s^2 / \epsilon_g)^{0.5} m^{-1}$	Yes	Parameter for I_{gb} in the inversion
AIGC	0.054 (NMOS) and 0.31 (PMOS) $(F_s^2 / \epsilon_g)^{0.5} m^{-1}$	Yes	Parameter for I_{gcs} and I_{gcd}
AIGD	1.36e-2(NMOS) and 9.8e-3(PMOS)	Yes	Parameter for I_{gd}
AIGSD	0.43 (NMOS) and 0.31 (PMOS) $(F_s^2 / \epsilon_g)^{0.5} m^{-1}$	Yes	Parameter for I_{gs} and I_{gd}

Parameter	Default	Bin	Description
BIGBACC	$0.054 (F_s^2 / \epsilon_g)^{0.5} \text{ m}^{-1} \text{ V}^{-1}$	Yes	Parameter for I_{gb} in the accumulation
BIGBINV	$0.03 (F_s^2 / \epsilon_g)^{0.5} \text{ m}^{-1} \text{ V}^{-1}$	Yes	Parameter for I_{gb} in the inversion
BIGC	0.054 (NMOS) and 0.024 (PMOS) $(F_s^2 / \epsilon_g)^{0.5} \text{ m}^{-1} \text{ V}^{-1}$	Yes	Parameter for I_{gcs} and I_{gcd}
BIGD	1.71e-3(NMOS) and 7.59e-4(PMOS)	Yes	Parameter for I_{gd}
BIGS	1.71e-3(NMOS) and 7.59e-4(PMOS)	Yes	Parameter for I_{gs}
BIGSD	0.054 (NMOS) 0.024 (PMOS) $(F_s^2 / \epsilon_g)^{0.5}$ $\text{m}^{-1} \text{ V}^{-1}$	Yes	Parameter for I_{gs} and I_{gd}
CIGBACC	0.075 V^{-1}	Yes	Parameter for I_{gb} in the accumulation
CIGBINV	0.0006 V^{-1}	Yes	Parameter for I_{gb} in the inversion
CIGC	0.075 (NMOS) and 0.03(PMOS) V^{-1}	Yes	Parameter for I_{gcs} and I_{gcd}
CIGD	0.075(NMOS) and 0.03(PMOS)	Yes	Parameter for I_{gd}
CIGS	0.075(NMOS) and 0.03(PMOS)	Yes	Parameter for I_{gs}
CIGSD	0.075 (NMOS) and 0.03 (PMOS) V^{-1}	Yes	Parameter for I_{gs} and I_{gd}
DLCIG	LINT	No	Source/drain overlap length for I_{gs} and I_{gd}
EIGBINV	1.1V	Yes	Parameter for I_{gb} in the inversion
NIGBACC	1.0	Yes	Parameter for I_{gb} in the accumulation
NIGBINV	3.0	Yes	Parameter for I_{gb} in the inversion
NIGC	1.0	Yes	Parameter for I_{gcs} , I_{gcd} , I_{gs} and I_{gd}
NTOX	1.0	Yes	Exponent for the gate oxide ratio
PIGCD	1.0	Yes	V_{ds} dependence of I_{gcs} and I_{gcd}
POXEDGE	1.0	Yes	Factor for the gate oxide thickness in the source/drain overlap regions

Parameter	Default	Bin	Description
TOXREF	3.0e-9m	No	Nominal gate oxide thickness for the gate dielectric tunneling current model only
VFBSDOFF	0.0V	Yes	Flatband voltage offset parameter

Charge/Capacitance Model Parameters:

Parameter	Default	Bin	Description
ACDE	1.0m/V	Yes	Exponential coefficient for the charge thickness in CAPMOD=2 for the accumulation and depletion regions
CF	calculated (F/m)	Yes	Fringing field capacitance
CGBO	0.0 (F/m)	No	Gate-bulk overlap capacitance per unit channel length
CGDL	0.0F/m	Yes	Overlap capacitance between gate and lightly-doped source region
CGDO	calculated (F/m)	No	Non LDD region drain-gate overlap capacitance per unit channel width
CGSL	0.0F/m	Yes	Overlap capacitance between the gate and the lightly-doped source region
CGSO	calculated (F/m)	No	Non LDD region source-gate overlap capacitance per unit channel width
CKAPPAD	CKAPPAS	Yes	Coefficient of bias-dependent overlap capacitance for the drain side
CKAPPAS	0.6V	Yes	Coefficient of the bias-dependent overlap capacitance for the source side
CLC	1.0e-7m	Yes	Constant term for the short channel model
CLE	0.6	Yes	Exponential term for the short channel model
DLC	LINT (m)	No	Channel-length offset parameter for the CV model
DWC	WINT (m)	No	Channel-width offset parameter for the CV model
MOIN	15.0	Yes	Coefficient for the gate-bias dependent surface potential
NOFF	1.0	Yes	CV parameter in $V_{gsteff,CV}$ for weak to strong inversion
VFBCV	-1.0V	Yes	Flat-band voltage parameter (for CAPMOD=0 only)

Parameter	Default	Bin	Description
VOFFCV	0.0V	Yes	CV parameter in $V_{gsteff,CV}$ for weak to strong inversion
XPART	0.0	No	Charge partition parameter

High-Speed / RF Model Parameters:

Parameter	Default	Bin	Description
GBMIN	1.0e-12mho	No	Conductance in parallel with each of the five substrate resistances to avoid potential numerical instability due to an unreasonably large substrate resistance
RBDB	50.0 ohm	No	Resistance between dbNode and dbNode
RBDBX0	100 ohms	No	Scaling pre-factor for RBDBX
RBDBY0	100 ohms	No	Scaling pre-factor for RBDBY
RBPB	50.0 ohm	No	Resistance between bNodePrime and bNode
RBPBX0	100 ohms	No	Scaling pre-factor for RBPBX
RBPBXL	0.0	No	Length Scaling parameter for RBPBX
RBPBXNF	0.0	No	Number of fingers Scaling parameter for RBPBX
RBPBXW	0.0	No	Width Scaling parameter for RBPBX
RBPBY0	100 ohms	No	Scaling pre-factor for RBPBY
RBPBYL	0.0	No	Length Scaling parameter for RBPBY
RBPBYNF	0.0	No	Number of fingers Scaling parameter for RBPBY
RBPBYW	0.0	No	Width Scaling parameter for RBPBY
RBPDP	50.0 ohm	No	Resistance between bNodePrime and dbNode
RBPDP0	50 ohms	No	Scaling pre-factor for RBPDP
RBPDDL	0.0	No	Length Scaling parameter for RBPDP
RBPDPNF	0.0	No	Number of fingers Scaling parameter for RBPDP
RBPDPW	0.0	No	Width Scaling parameter for RBPDP
RBPS	50.0 ohm	No	Resistance between bNodePrime and sbNode
RBPS0	50 ohms	No	Scaling pre-factor for RBPS

Parameter	Default	Bin	Description
RBPSL	0.0	No	Length Scaling parameter for RBPS
RBPSNF	0.0	No	Number of fingers Scaling parameter for RBPS
RBPSW	0.0	No	Width Scaling parameter for RBPS
RBSB	50.0 ohm	No	Resistance between sbNode and bNode
RBSBX0	100 ohms	No	Scaling pre-factor for RBSBX
RBSBY0	100 ohms	No	Scaling pre-factor for RBSBY
RBSDBXL	0.0	0.0	Length Scaling parameter for RBSBX and RBDBX
RBSDBXNF	0.0	No	Number of fingers Scaling parameter for RBSBX and RBDBX
RBSDBXW	0.0	No	Width Scaling parameter for RBSBX and RBDBX
RBSDBYL	0.0	No	Length Scaling parameter for RBSBY and RBDBY
RBSDBYNF	0.0	No	Number of fingers Scaling parameter for RBSBY and RBDBY
RBSDBYW	0.0	No	Width Scaling parameter for RBSBY and RBDBY
XRCRG1	12.0	Yes	Parameter for the distributed channel-resistance effect for both intrinsic-input resistance and charge-deficit NQS models
XRCRG2	1.0	Yes	Parameter to account for the excess channel diffusion resistance for both intrinsic input resistance and charge-deficit NQS models

Flicker and Thermal Noise Model Parameters:

Parameter	Default	Bin	Description
AF	1.0	No	Flicker noise exponent
EF	1.0	No	Flicker noise frequency exponent
EM	4.1e7V/m	No	Saturation field
KF	$0.0 \text{ A}^{2-EF} \text{ s}^{1-EF} \text{ F}$	No	Flicker noise coefficient
NOIA	$6.25e41 \text{ (eV)}^{-1} \text{ s}^{1-EF} \text{ m}^{-3}$ for NMOS: $6.188e40 \text{ (eV)}^{-1} \text{ s}^{1-EF} \text{ m}^{-3}$ for PMOS	No	Flicker noise parameter A

Parameter	Default	Bin	Description
NOIB	$3.125e26 \text{ (eV)}^{-1} \text{ s}^{1-E_F} \text{ m}^{-1}$ for NMOS; $1.5e25 \text{ (eV)}^{-1} \text{ s}^{1-E_F} \text{ m}^{-1}$ for PMOS	No	Flicker noise parameter B
NOIC	$8.75 \text{ (eV)}^{-1} \text{ s}^{1-E_F} \text{ m}$	No	Flicker noise parameter C
NTNOI	1.0	No	Noise factor for short-channel devices for TNOIMOD=0 only
TNOIA	1.5	No	Coefficient of the channel-length dependence of the total channel thermal noise
TNOIB	3.5	No	Channel-length dependence parameter for partitioning the channel thermal noise

Layout-Dependent Parasitics Model Parameters:

Parameter	Default	Bin	Description
DMCG	0.0m	No	Distance from the S/D contact center to the gate edge
DMCGT	0.0m	No	DMCG of the test structures
DMCI	DMCG	No	Distance from the S/D contact center to the isolation edge in the channel-length direction
DMDG	0.0m	No	Same as DMCG, but for merged devices only
DWJ	DWC (in CVmodel)	No	Offset of the S/D junction width
MIN	0	No	Minimize the number of drain or source diffusions for even-number fingered device
NF	1	No	Number of device figures
NGCON	1	No	Number of gate contacts
XGL	0.0m	No	Offset of the gate length due to variations in patterning
XGW	0.0m	No	Distance from the gate contact to the channel edge

Asymmetric Source/Drain Junction Diode Model Parameters:

Parameter	Default	Bin	Description
BVD	BVD=BVS	No	Breakdown voltage

Parameter	Default	Bin	Description
BVS	BVS=10.0V	No	Breakdown voltage
CJD	CJD=CJS	No	Bottom junction capacitance per unit area at zero bias
CJS	CJS=5.0e-4 F/m ²	No	Bottom junction capacitance per unit area at zero bias
CJSWD	CJSWD=CJSWS	No	Isolation-edge sidewall junction capacitance per unit length
CJSWGD	CJSWGD=CJSWS	No	Gate-edge sidewall junction capacitance per unit length
CJSWGS	CJSWGS=CJSWS	No	Gate-edge sidewall junction capacitance per unit length
CJSWS	CJSWS=5.0e-10 F/m	No	Isolation-edge sidewall junction capacitance per unit length
IJTHDFWD	IJTHDFWD=IJTHSFWD	No	Limiting current in the forward bias region
IJTHDREV	IJTHDREV=IJTHSREV	No	Limiting current in the reverse bias region
IJTHSFWD	IJTHSFWD=0.1A	No	Limiting current in the forward bias region
IJTHSREV	IJTHSREV=0.1A	No	Limiting current in the reverse bias region
JSD	JSD=JSS	No	Bottom junction reverse saturation current density
JSS	JSS=1.0e-4A/m ²	No	Bottom junction reverse saturation current density
JSWD	JSWD=JSWS	No	Isolation-edge sidewall reverse saturation current density
JSWGD	JSWGD=JSWGS	No	Gate-edge sidewall reverse saturation current density
JSWGS	JSWGS=0.0A/m	No	Gate-edge sidewall reverse saturation current density
JSWS	JSWS=0.0A/m	No	Isolation-edge sidewall reverse saturation current density
MJD	MJD=MJS	No	Bottom junction capacitance grading coefficient
MJS	MJS=0.5	No	Bottom junction capacitance grading coefficient

Parameter	Default	Bin	Description
MJSWD	MJSWD=MJSWS	No	Isolation-edge sidewall junction capacitance grading coefficient
MJSWGD	MJSWGD=MJSWS	No	Gate-edge sidewall junction capacitance grading coefficient
MJSWGS	MJSWGS=MJSWS	No	Gate-edge sidewall junction capacitance grading coefficient
MJSWS	MJSWS=0.33	No	Isolation-edge sidewall junction capacitance grading coefficient
PBD	PBD=PBS	No	Bottom junction built-in potential
PBS	PBS=1.0V	No	Bottom junction built-in potential
PBSWD	PBSWD=PBSWS	No	Isolation-edge sidewall junction built-in potential
PBSWGD	PBSWGD=PBSWS	No	Gate-edge sidewall junction built-in potential
PBSWGS	PBSWGS=PBSWS	No	Gate-edge sidewall junction built-in potential
PBSWS	PBSWS=1.0V	No	Isolation-edge sidewall junction built-in potential
XJBVD	XJBVD=XJBVS	No	Fitting parameter for the diode breakdown
XJBVS	XJBVS=1.0	No	Fitting parameter for the diode breakdown

Temperature Dependence Parameters:

Parameter	Default	Bin	Description
AT	3.3e4m/s	Yes	Temperature coefficient for the saturation velocity
KT1	-0.11V	Yes	Temperature coefficient for the threshold voltage
KT1L	0.0Vm	Yes	Channel length dependence of the temperature coefficient for the threshold voltage
KT2	0.022	Yes	Body-bias coefficient of the V_{th} temperature effect
NJS, NJD	NJS=1.0; NJD=NJS	No	Emission coefficients of junction for the source and drain junctions
PRT	0.0ohm-m	Yes	Temperature coefficient for R_{dsw}
TCJ	0.0K ⁻¹	No	Temperature coefficient of C_J
TCJSW	0.0K ⁻¹	No	Temperature coefficient of C_{JSW}

Parameter	Default	Bin	Description
TCJSWG	$0.0K^{-1}$	No	Temperature coefficient of CJSWG
TNOM	27°X	No	Temperature at which simulation extracts parameters
TPB	0.0V/K	No	Temperature coefficient of PB
TPBSW	0.0V/K	No	Temperature coefficient of PBSW
TPBSWG	0.0V/K	No	Temperature coefficient of PBSWG
TRD	$0.0\ 1K^{-1}$	No	Temperature coefficient for the drain diffusion and the Rd contact resistances.
TRS	$0.0\ 1/K$	No	Temperature coefficient for the source diffusion and the Rs contact resistances.
TVFBSD OFF	$0.0K^{-1}$	Yes	Temperature coefficient of VFBSDOFF
TVOFF	$0.0K^{-1}$	Yes	Temperature coefficient of VOFF
UA1	1.0e-9m/V	Yes	Temperature coefficient for UA
UB1	-1.0e-18 (m/V ²)	Yes	Temperature coefficient for UB
UC1	0.056/V ⁻¹ for MOBMOD=1; 0.056e-9m/V ² for MOBMOD=0 and 2	Yes	Temperature coefficient for UC
UD1	$0.0(1/m)^2$	Yes	Temperature coefficient for UD
UTE	-1.5	Yes	Mobility temperature exponent
XTIS, XTID	XTIS=3.0; XTID=XTIS	No	Junction current temperature exponents for the source and drain junctions

Well Proximity Effect Parameters:

Parameter	Default	Bin	Description
K2WE	0.0	Yes	K2 shift factor for well proximity effect
KU0WE	0.0	Yes	Mobility degradation factor for well proximity effect
KVTH0WE	0.0	Yes	Threshold shift factor for well proximity effect
SC	0.0[m]	No	Distance to a single well edge

Parameter	Default	Bin	Description
SCA	0.0	No	Integral of the first distribution function for scattered well dopant
SCB	0.0	No	Integral of the second distribution function for scattered well dopant
SCC	0.0	No	Integral of the third distribution function for scattered well dopant
SCREF	1e-6[m]	No	Reference distance to calculate SCA, SCB, and SCC
WEB	0.0	No	Coefficient for SCB
WEC	0.0	No	Coefficient for SCC

dW and dL Parameters:

Parameter	Default	Bin	Description
LL	0.0mLLN	No	Coefficient of the length dependence for the length offset
LLC	LL	No	Coefficient of the length dependence for the CV channel length offset
LLN	1.0	No	Power of the length dependence for the length offset
LW	0.0mLWN	No	Coefficient of the width dependence for the length offset
LWC	LW	No	Coefficient of the width dependence for the CV channel length offset
LWL	0.0 mLWN+LLN	No	Coefficient of the length and width cross term dependence for the length offset
LWLC	LWL	No	Coefficient of the length and width cross-term dependence for the CV channel length offset
LWN	1.0	No	Power of the width dependence, length offset
WL	0.0 $\mu\Omega/\text{N}$	No	Coefficient of the length dependence of the width offset
WLC	WL	No	Coefficient of the length dependence for the CV channel width offset
WLN	1.0	No	Power of the length dependence of the width offset

Parameter	Default	Bin	Description
WW	0.0mWWN	No	Coefficient of the width dependence of the width offset
WWC	WW	No	Coefficient of the width dependence for the CV channel width offset
WWL	0.0 mWWN+WLN	No	Coefficient of the length and width cross term dependence for the width offset
WWLC	WWL	No	Coefficient of the length and width cross-term dependence for the CV channel width offset
WWN	1.0	No	Power of the width dependence of the width offset

Range Parameters for Model Application:

Parameter	Default	Binnable	Description
LMIN	0.0μ	No	Minimum channel length
LMAX	1.0m	No	Maximum channel length
WMIN	0.0m	No	Minimum channel width
WMAX	1.0m	No	Maximum channel width

BSIM4 MOSFET Model — Version Updates

- [BSIM4 Model Version 4.8.2 Updates](#)
- [BSIM4 Model Version 4.8.1 Updates](#)
- [BSIM4 Model Version 4.8 Updates](#)
- [BSIM4 Model Version 4.7 Updates](#)
- [BSIM4 Model Version 4.6.6 Updates](#)
- [BSIM4 Model Version 4.6.5 Updates](#)
- [BSIM4 Model Version 4.6.3 Updates](#)
- [BSIM4 Model Version 4.6.2 Updates](#)
- [BSIM4 Model Version 4.6.1 Updates](#)

- [BSIM4 Model Version 4.6 Updates](#)
- [BSIM4 Model Version 4.5 Updates](#)

BSIM4 Model Version 4.8.2 Updates

Compared with version 4.8.1, the following changes were made.

- Added new parameter with name BSIM4gidlclamp to provide tuning flexibility of clamping value to limit the arbitrary large current of IGIDL/ IGISL.
- Added new parameter with name BSIM4idovvdsc to provide more tuning flexibility of BSIM4idovvds for noise sensitive application.
- CF is calculated based on the updated value of TOXE.

BSIM4 Model Version 4.8.1 Updates

Compared with version 4.8, the following changes were made.

- Added warning for TNOIMOD=1.
- Updated the default value of UA, UC, and UC1 for MOBMOD =6 and MOBMOD =5.
- Removed the limitation on TNOIA, TNOIB, TNOIC, and RNOIA greater than zero and limited the entire function.
- When DLCIG<0 or DLCIGD<0, clamped the value to zero and issue a warning message.
- Some minor bug fixes.

BSIM4 Model Version 4.8 Updates

Compared with version 4.7, the following changes were made:

- BSIM4mobMod=4, 5, and 6 are introduced as modified forms of mobMod=0, 1, and 2 respectively.
- Fixed derivative issue in igcMod=2.
- Updated the default values of GISL-related parameters.
- Added body resistances checks.
- Added binning equations for TVOFFCV, AGISD, BGISD, and CGISD.
- Some minor bug fixes.

BSIM4 Model Version 4.7 Updates

Compared with version 4.6.6, the following changes were made:

- Removed $\text{tnoiMod}=2$ in BSIM4.6.6, changing $\text{tnoiMod}=3$ in BSIM4.6.6 to $\text{tnoiMod}=2$ in BSIM4.7
- Implemented GF's DIBL/Rout model in $\text{mtrlMod}=0$ as well (it was implemented in $\text{mtrlMod}=1$ only in BSIM4.6.6)
- Added compatibility mode between $\text{mtrlMod}=0$ and 1 (in this mode, $\text{mtrlMod}=0$ with $k=3.9$ is the same in IV and CV as $\text{mtrlMod}=1$)

BSIM4 Model Version 4.6.6 Updates

Compared with BSIM4.6.5, several new features were added.

- Enhancement of GIDL/GISL Model
 - $\text{GIDLMod}=1$ is introduced to decouple V_d from V_g through new parameters RGIDL , KGIDL and FGIDL (same for GISL)
 - Binnable parameters RGIDL , KGIDL , FGIDL are introduced with new Mod selector, $\text{gidlMod}=1$
- Enhancement of DIBL/Rout Model
 - Existing DIBL/Rout model in BSIMSOI is enhanced with an additional term, DVTP5 , to better capture the V_{ds} effect in long channel device.
 - The model is implemented with Binnable parameters DVTP2 , DVTP3 , DVTP4 , and DVTP5 .
- Temperature Dependence of Sub-threshold Leakage Current
 - Improved formulations to capture temperature dependence of Leakage Current
 - Enhanced temperature dependence model has been implemented with new Binnable parameters TNFACTOR , TETA0 , TVOFFCV .
- Enhanced Thermal Noise Model for BSIM4
 - $\text{tnoiMod}=2$ is introduced
 1. Drain and gate noise follows $\text{tnoiMod}=1$
 2. A new expression is used for the correlation coefficient.

- Introduced $\text{tnoiMod}=3$, with New expressions for drain noise (S_{id}), induced gate noise (S_{ig}) and correlation coefficient (c)
- Frequency-dependent formula implemented for $\text{tnoiMod}=2$ and $\text{tnoiMod}=3$ ensures that the gate noise never exceeds the drain noise.
- Additional big fixes.

BSIM4 Model Version 4.6.5 Updates

In the prior version, source and drain diode current showed unphysical and high values for $A_{\text{seff}}=P_{\text{seff}}=A_{\text{deff}}=P_{\text{deff}}=0$, although it correctly predicts positive values of A_{seff} , P_{seff} , A_{deff} , and P_{deff} .

In version 4.6.5:

- If A_{seff} , P_{seff} , A_{deff} , P_{deff} are negative, their values are set to '0'.
- For A_{seff} and $P_{\text{seff}}=0$, the source side diode is turned off by setting $\text{SourceSaturationCurrent}=0.0$
- For A_{deff} and $P_{\text{deff}}=0$, the drain side diode is turned off by setting $\text{DrainSaturationCurrent}=0.0$.

BSIM4 Model Version 4.6.3 Updates

The only change in BSIM4.6.3 compared with BSIM4.6.2 was the refinement of the calculation below.

```
pParam->BSIM4litl = sqrt(3.0 * 3.9 / epsrox * pParam->BSIM4xj * toxe)
```

BSIM4 Model Version 4.6.2 Updates

The v. 4.6.2 release added the following features and bug fixes:

- Mobility model ($\text{mobMod}=3$) added to enhance the modeling of Coulombic scattering in the high-k/metal gate transistors. Trap assisted tunneling (TAT) has been improved to include the width dependence. A new model parameter JTWEFF is introduced and set to zero to maintain the backward compatibility.
- Bug fixes include:
 - Output Conductance Model: VASCBE: A division by zero bug in the output conductance calculation is fixed.
 - Thermal Noise Model ($\text{tnoiMod}=0$): The scaling factor NF is added to the equivalent resistance R_{ds}/NF .
 - Negative Thermal Noise ($\text{tnoiMod}=1$) The noise spectral density will not be negative now.

- Source/Drain Bulk Junction Capacitance: The source/drain bulk junction capacitance is not discontinuous when V_{bs}/V_{bd} is zero. The S/D junction sidewall capacitance along the isolation edge is set to zero if they are negative.
- Derivative Issue in Capacitance Model (capMod=0): An error in the derivative calculation of dV_{gs_eff}/dV_g has been fixed.
- Toxp Calculation (mtrlMod=1): The physical oxide thickness should be layout independent. Source/Drain Resistance: A division by zero bug in the R_{end} calculation is fixed.
- Typo of SC: SCA is a typo of SC in b4.
- Drain/Body Breakdown Voltage: The reset value of drain/body breakdown voltage does not cause non-convergence now. Some other similar bugs have also been fixed.

BSIM4 Model Version 4.6.1 Updates

Compared with BSIM4.6.0, several new features are added in this version.

- New material model is introduced for the predictive modeling of Non-SiO₂ insulator, Non-Poly Silicon gate and Non-silicon channel. The following new parameters are added:
 - MTRLMOD: New material model selector
 - PHIG, EPSRGATE: Non-poly silicon gate parameters
 - EOT, VDDEOT: Non-SiO₂ gate dielectric
 - EASUB, EPSRSUB, NI0SUB, BG0SUB, TBGASUB, TBGBSUB, ADOS, BDOS: Non-silicon channel parameters
- Mobility model (MOBMOD = 0 and MOBMOD = 1) has been improved through predictive modeling of vertical electric field. The improved mobility model is selected through MTRLMOD = 1 for backward compatibility.
- GIDL/GISL models are improved through an improved definition of flatband voltages at S/D ends. The improved GISL/GIDL model is selected through MTRLMOD = 1 for backward compatibility.
- The Poly-depletion model is modified to account for new gate and gate-insulator materials.
- C-V model has been improved by adding a new $V_{gsteffCV}$ definition through CVCHARGEMOD = 1. Six new parameters have been added: CVCHARGEMOD, MINVCV, LMINVCV, WMINCV, PMINVCV and VOFFCVL

- The following bugs have been fixed:
 - An error in the derivative calculation of $dV_{dseffCV_dVb}$ has been fixed for CAPMOD = 1 and CAPMOD = 2 in b4ld.c
 - The warning limits for NOFF and VOFFCV have been removed from b4check.c.

BSIM4 Model Version 4.6 Updates

The current PrimeSim Continuum solution release supports BSIM version 4.6.0, including the following features:

- The IGISL and IGIDL modules have independent model parameters (as opposed to v. 4.5, in which the GIDL and GISL leakage current modules shared the parameter set).
- The parameters for the source and drain side junction diode current due to the trap-assisted tunneling current in space-change region are fully separated.
- The parameters for the gate tunneling current in the S/D overlap diffusion regions (I_{gs}/I_{gd}) are separated.

In addition the three bug fixes are implemented in this version.

- The implementation of coulomb scattering in the mobility model has been changed to avoid the non-monotonicity in drain current as a function of gate voltage. Also, the default value of parameter has been changed to 0.
- The accuracy of RgateMod = 2 has been improved by accounting correctly the contribution from Rgate to overall noise. The change has been made in b4noi.c. The default value for the model parameter VFB was missing. The default value of the parameter VFB has been now set to -1.0V in b4set.c

BSIM4 Model Version 4.5 Updates

In July 2005, Berkeley released BSIM4 Version 4.5 includes the following features:

- The gate resistance parameters X_{GW} and $NGCON$ are now available as both model and instance parameters.
- Four modes are added or enhanced:
 - $R_{BODYMOD}=2$: substrate resistance model that is scalable with channel length, channel width and number fingers
 - $IGCMOD=2$: implements full BSIM4 V_{th} model into IGC enables accurate predictions of IGC V_{bs} dependence

- `TEMPMOD=2`: enhances `TEMPMOD` so `Vth(DITS)` and gate tunneling models are functions of nominal temperature and adds the temperature dependence of zero-bias flat-band voltage
- `WPEMOD=1`: adds a new well-proximity effect model developed by CMC.
- Adds a new mobility model that accounts for Coulomb scattering effect as well as the channel length dependence of mobility due to heavy halo-doping.
- Adds additional temperature dependence of model parameters `VOFF` and `VFBSDOFF`.
- A fatal error message is issued when a model parameter `VTSS`, `VTSD`, `VTSSWS`, or `VTSSWGD` is negative. A warning message was issued previously.
- A warning message is issued if model parameter `CGBO` is negative. In which case, `CGBO` is set to 0.
- Fixed bugs in BSIM 4.4.

Note:

BSIM4 Version 4.5 officially supports instance parameter `DELVT0`, which is used to represent threshold variations.

See also:

`.OPTION VER_CONTROL`

`.OPTION MODPARCHK`

`.OPTION PHD` for the PHD flow that can generally help large case in the PrimeSim Continuum solution DC OP convergence for BSIM4 test cases.

Other Noise Sources Modeled in BSIM4 Model Version 4.5:

BSIM4 models the thermal noise due to the substrate, electrode gate, and source/drain resistances. Shot noise due to various gate tunneling components is modeled as well.

The PrimeSim Continuum solution can print out `rg`, `rbps`, `rbpd`, `rbpb`, `rsb`, `rdb`, `igs`, `igd`, `igb` noise information. You can add the following noise parameters to the model card.

Output Parameters	turns on...
+ <code>rgatemod=1</code>	the <code>rg</code> noise output
+ <code>rbodymod=1</code>	the <code>rbps</code> , <code>rbpb</code> noise output
+ <code>igbmod=1</code>	the <code>igb</code> noise output
+ <code>igcmmod=1</code>	the <code>igs</code> and <code>igd</code> noise output

Output Parameters	turns on...
+ rbsbx0=1 rbsby0=1 rdbdx0=1 rdbdy0=1 rbodymod=1	the rbsb and rbdb noise output

BSIM4 MOSFET Model — Output Templates

For a list of output template parameters in the MOSFET models, and which parameters this model supports, see [MOSFET Output Templates](#).

LEVEL 55: EPFL-EKV MOSFET Model

The EPFL-EKV MOSFET model is a scalable and compact simulation model built on fundamental physical properties of the MOS structure. This model is dedicated to the design and simulation of low-voltage, low-current analog, and mixed analog-digital circuits using submicron CMOS technologies.

- The intrinsic part of the MOSFET includes the equations and parameters used to simulate the EPFL-EKV MOSFET model.
- The extrinsic part of the MOSFET model includes series resistances of the source and drain diffusions, junction currents, and capacitances.

This section covers the following topics:

- [EPFL-EKV MOSFET Model — Single Equation Model](#)
- [EPFL-EKV MOSFET Model — Effects Modeled](#)
- [EPFL-EKV MOSFET Model — Coherence of Static and Dynamic Models](#)
- [EPFL-EKV MOSFET Model — Bulk Reference and Symmetry](#)
- [EPFL-EKV MOSFET Model — EKV Intrinsic Model Parameters](#)
- [EPFL-EKV MOSFET Model — Static Intrinsic Model Equations](#)
- [EPFL-EKV MOSFET Model — Quasi-static Model Equations](#)
- [EPFL-EKV MOSFET Model — Intrinsic Noise Model Equations](#)
- [EPFL-EKV MOSFET Model — Operating Point Information](#)
- [EPFL-EKV MOSFET Model — Estimation and Limits of Static Intrinsic Model Parameters](#)
- [EPFL-EKV MOSFET Model Updates](#)

EPFL-EKV MOSFET Model — Single Equation Model

The EPFL-EKV MOSFET model is a *single expression*, which preserves continuity of first-order and higher-order derivatives with respect to any terminal voltage in the entire range of validity of the model. This section describes the analytical expressions of first-order derivatives as transconductances and transcapacitances, you can use them in simulation.

EPFL-EKV MOSFET Model — Effects Modeled

The EPFL-EKV MOSFET model version 2.6 models the following physical effects:

- Basic geometrical and process-related aspects, such as oxide thickness, junction depth, and effective channel length and width.
- Effects of the doping profile.
- Substrate effect.
- Modeling of weak, moderate, and strong inversion behavior.
- Modeling of mobility effects due to vertical and lateral fields, and velocity saturation.
- Short-channel effects, such as channel-length modulation (CLM), source and drain charge-sharing (including for narrow channel widths), and the reverse short channel effect (RSCE).
- Modeling of the substrate current due to impact ionization.
- Quasi-static charge-based dynamic model.
- Thermal and flicker noise modeling.
- Short-distance geometry-dependent and bias-dependent device matching.

EPFL-EKV MOSFET Model — Coherence of Static and Dynamic Models

Simulation derives all aspects of the static, the quasi-static, and the non-quasi-static (NQS) dynamic and noise models, from the normalized transconductance-to-current ratio. These expressions use symmetric normalized forward and reverse currents.

- For quasi-static dynamic operations, you can use a charge-based model for the node charges and trans-capacitances or a simpler capacitances model.
- The dynamic model, including the time constant for the NQS model, is described in symmetrical terms of the forward and reverse normalized currents.

The charge formulation also expresses the effective mobility dependence of a local field.

EPFL-EKV MOSFET Model — Bulk Reference and Symmetry

Voltages are all referred to the local substrate:

$V_G = V_{GB}$ Intrinsic gate-to-bulk voltage

$V_S = V_{SB}$ Intrinsic source-to-bulk voltage

$V_D = V_{DB}$ Intrinsic drain-to-bulk voltage

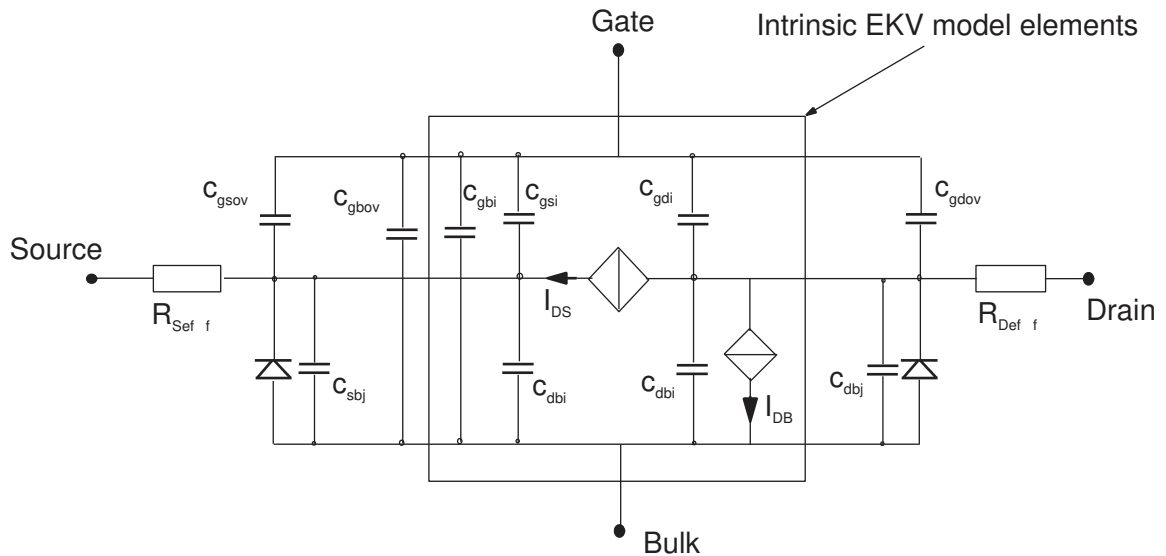
V_S and V_D are the intrinsic source and drain voltages so the voltage drop over the extrinsic resistive elements must already be accounted for externally.

V_D is the electrical drain voltage, where $V_D > V_S$. Bulk reference handles the model symmetrically with respect to source and drain. This symmetry is inherent in common MOS technologies (excluding non-symmetric source-drain layouts).

Note:

Intrinsic model equations are presented for an N-channel MOSFET. P-channel MOSFETs are dealt with as pseudo-N-channel. That is, simulation reverses the polarity of the voltages (V_G , V_S , V_D , V_{FB} , V_{TO} , and τ_{CV}) before computing the P-channel current, which has a negative sign. No other distinctions are made between N-channel and P-channel, except the η factor for calculating the effective mobility.

Figure 13 Level 55 Equivalent Circuit



The following table represents the intrinsic and extrinsic elements of the MOS transistor. For quasi-static dynamic operation, this figure shows only the intrinsic capacitances from the simpler capacitances model. However, you can also use a charge-based transcapacitances model in simulation.

Name	Unit	Default	Description
L	m	-	Channel length
W	m	-	Channel width
M or NP	-	1.0	Parallel multiple device number
N or NS	-	1.0	Series multiple device number

EPFL-EKV MOSFET Model — EKV Intrinsic Model Parameters

Name	Unit	Default	Range	Description
COX ¹	F/m ²	0.7e-3	-	Gate oxide capacitance per unit area
DL	m	0	-	Channel length correction
DW ²	m	0	-	Channel width correction
XJ	m	0.1e-6	1.0e-9	Junction depth

Name	Unit	Default ³	Range	Description
E0 (EO)	V/m	1.0e12	$\geq 1E5$	Mobility reduction coefficient
GAMMA	$\sqrt{V}$	1.0	≥ 0	Body effect parameter
KP	A/V ⁴	50.0e-6	-	Transconductance parameter
PHI	V	0.7	≥ 0.1	Bulk Fermi potential (*2)
UCRIT	V/m	2.0e6	$\geq 1E5$	Longitudinal critical field
VTO ⁴	V	0.5	-	Long-channel threshold voltage

Name	Unit ⁵	Default	Range	Description
NSUB ⁶	cm ⁻³	-	≥ 0	Channel doping

1. This model can calculate the default value of COX as a function of TOX.
2. DL and DW parameters are usually negative; see the effective length and width calculation.
3. This model can calculate the default values of VTO, GAMMA, PHI, and KP as a function of TOX, NSUB, UO, and VFB for statistical circuit simulation.
4. As VG, VTO also references the bulk.
5. In this example, cm is the basic unit for NSUB and UO. TOX and VMAX are in m.
6. Optional parameter for the dependence of GAMMA on COX, and for calculating PHI.

Name	Unit ⁵	Default	Range	Description
UO ⁷	cm ⁶ /(Vs)	-	≥ 0	Low-field mobility
THETA ⁸	1/V	0	≥ 0	Mobility reduction coefficient
TOX ⁹	m	-	≥ 0	Oxide thickness
VFB ¹⁰	V	-	-	Flat-band voltage
VMAX ¹¹	m/s	-	≥ 0	Saturation velocity

The preceding parameters accommodate the scaling behavior of the process and basic intrinsic model parameters, and statistical circuit simulation. Simulation uses the `TOX`, `NSUB`, `VFB`, `UO`, and `VMAX` parameters only if you did not specify `COX`, `GAMMA`, `PHI`, `VTO`, `KP`, or `UCRIT`. You can also use a simpler mobility reduction model, due to the vertical field. Simulation uses the `THETA` mobility reduction coefficient only if you did not specify `E0`.

Name	Unit	Default	Range	Description
LAMBDA	-	0.5	≥ 0	Depletion length coefficient (channel length modulation)
LETA	-	0.1	-	Short-channel effect coefficient
WETA	-	0.25	-	Narrow-channel effect coefficient

Name	Unit	Default	Range	Description
LK	m	0.29e-6	$\geq 1.0\text{e-}8$	Reverse short channel effect characteristic length
Q0 (QO)	$a \cdot s / m^2$	0	-	Reverse short channel effect peak charge density

Name	Unit	Default	Range	Description
IBA	1/m	0	-	First impact ionization coefficient
IBB	V/m	3.0e8	$\geq 1.0\text{e}8$	Second impact ionization coefficient

5. In this example, cm is the basic unit for `NSUB` and `UO`. `TOX` and `VMAX` are in m.

7. Optional parameter for the dependence of `KP` on `COX`.

8. Optional parameter for mobility reduction due to the vertical field.

9. Optional parameter for calculating `COX`.

10. Optional parameter for calculating `VTO` as a function of `COX`, `GAMMA`, or `PHI`.

11. Optional parameter for calculating `UCRIT`.

Name	Unit	Default	Range	Description
IBN	-	1.0	≥ 0.1	Saturation voltage factor for impact ionization

Name	Unit	Default	Description
BEX	-	-1.5	Mobility temperature exponent
IBBT	1/K	9.0e-4	Temperature coefficient for IBB
TCV	V/K	1.0e-3	Threshold voltage temperature coefficient
UCEX	-	0.8	Longitudinal critical field temperature exponent

Name	Unit	Default	Description
AGAMMA	$\sqrt{V} m$	0	Area related body effect mismatch parameter
AKP	m	0	Area related gain mismatch parameter
AVTO	Vm	0 ¹²	Area related threshold voltage mismatch parameter

Name	Unit	Default	Description
AF	-	1	Flicker noise exponent
KF	- ¹³	0	Flicker noise coefficient

Name	Unit	Default	Description
NQS ¹⁴	-	0	Non-Quasi-Static (NQS) operation switch
XQC ¹⁵	-	0.4	Charge/capacitance model selector

12. Only DEV values apply to the statistical matching parameters (AVTO, AGAMMA, AKP) for Monte-Carlo type simulations. Default is 1E-6 for all three parameters in some implementations to allow sensitivity analysis of the matching parameters. Do not specify LOT for AVTO, AGAMMA, or AKP.
13. The unit for KF might depend on the flicker noise model that you select, if these options are available.
14. NQS=1 switches Non-Quasi-Static operation on, default is off (the NQS model option might not be available in all implementations).
15. Selects either the charges/transcapacitances (default) or the capacitances-only model. XQC=0.4: charges/transcapacitances model; XQC=1: capacitances only model. (the XQC model option might not be available in all implementations).

Name	Unit	Default	Description
SATLIM ¹⁶	-	exp(4)	Ratio defining the i_f/i_r saturation limit.

EPFL-EKV MOSFET Model — Static Intrinsic Model Equations

This section covers the following topics:

- [EPFL-EKV MOSFET Model — Basic Definitions](#)
- [EPFL-EKV MOSFET Model — Parameter Preprocessing: Handling of Model Parameters for P-Channel MOSFETs](#)
- [EPFL-EKV MOSFET Model — Parameter Preprocessing: Initializing Intrinsic Parameters](#)
- [EPFL-EKV MOSFET Model — Parameter Preprocessing: Default Values and Parameter Ranges](#)
- [EPFL-EKV MOSFET Model — Parameter Preprocessing: Intrinsic Parameters Temperature Dependence](#)
- [EPFL-EKV MOSFET Model — Bulk Referenced Intrinsic Voltages](#)
- [EPFL-EKV MOSFET Model — Effective Channel Length and Width](#)
- [EPFL-EKV MOSFET Model — Short Distance Matching](#)
- [EPFL-EKV MOSFET Model — Reverse Short-channel Effect \(RSCE\)](#)
- [EPFL-EKV MOSFET Model — Effective Gate Voltage Including RSCE](#)
- [EPFL-EKV MOSFET Model — Effective substrate factor including charge-sharing for short and narrow channels](#)
- [EPFL-EKV MOSFET Model — Pinch-off Voltage Including Short-Channel and Narrow-Channel Effects](#)
- [EPFL-EKV MOSFET Model — Slope Factor](#)
- [EPFL-EKV MOSFET Model — Large Signal Interpolation Function](#)
- [EPFL-EKV MOSFET Model — Forward Normalized Current](#)
- [EPFL-EKV MOSFET Model — Velocity Saturation Voltage](#)
- [EPFL-EKV MOSFET Model — Drain-to-source Saturation Voltage for Reverse Normalized Current](#)

16. Only used for operating point information. (the SATLIM option might not be available in all implementations).

- [EPFL-EKV MOSFET Model — Channel-length Modulation](#)
- [EPFL-EKV MOSFET Model — Equivalent Channel Length Including Channel-length Modulation and Velocity Saturation](#)
- [EPFL-EKV MOSFET Model — Reverse Normalized Current](#)
- [EPFL-EKV MOSFET Model — Transconductance Factor and Mobility Reduction Due to Vertical Field](#)
- [EPFL-EKV MOSFET Model — Simple Mobility Reduction Model](#)
- [EPFL-EKV MOSFET Model — Specific Current](#)
- [EPFL-EKV MOSFET Model — Drain-to-source Current](#)
- [EPFL-EKV MOSFET Model — Transconductances](#)
- [EPFL-EKV MOSFET Model — Impact Ionization Current](#)

EPFL-EKV MOSFET Model — Basic Definitions

$\epsilon_{si} = SCALE \cdot 1.045 \times 10^{-12} [F / m]$ Permittivity of silicon

$\epsilon_{ox} = SCALE \cdot 34.5 \times 10^{-12} [F / m]$ Permittivity of silicon dioxide

$q = 1.602 \times 10^{-19} [C]$ Magnitude of electron charge

$k = 1.3807 \times 10^{-23} [JK^{-1}]$ Boltzmann constant

$T_{ref} = 300.15 [K]$ Reference temperature

$T_{nom} [K]$ Nominal temperature of model parameters

$T [K]$ Model simulation temperature

$V_t(T) = \frac{k \cdot T}{q}$ Thermal voltage

$E_g(T) = \left(1.16 - 0.000702 \cdot \frac{T^2}{T + 1108} \right) [eV]$ Energy gap

$n_i(T) = 1.45 \times 10^{16} \cdot \left(\frac{T}{T_{ref}} \right) \cdot \exp \left(\frac{E_g(T_{ref})}{2 \cdot V_t(T_{ref})} - \frac{E_g(T)}{2 \cdot V_t(T)} \right) [m^{-3}]$ Intrinsic carrier concentration

EPFL-EKV MOSFET Model — Parameter Preprocessing: Handling of Model Parameters for P-Channel MOSFETs

For P-channel devices, simulation reverses the sign of v_{FB} , v_{TO} , and TCV before processing. Therefore, v_{TO} and TCV are usually positive and v_{FB} is usually negative for N-channel, and vice versa for P-channel MOSFETs.

EPFL-EKV MOSFET Model — Parameter Preprocessing: Initializing Intrinsic Parameters

The basic intrinsic model parameters are related to the fundamental process parameters as in early SPICE models:

$COX \rightarrow TOX$

$GAMMA$ and $PHI \rightarrow NSUB$

$VTO \rightarrow VFB$

$KP \rightarrow UO$

$UCRIT \rightarrow VMAX$

For statistical circuit simulation, you should introduce parameter variations on the level of the latter parameters. You can also use these dependencies to analyze device scaling and to obtain parameter sets from other MOSFET models. Therefore, you can use the following relations:

If you do not specify COX , simulation initializes it as:

$$COX = \begin{cases} \epsilon_{ox} / TOX & \text{for: } TOX > 0 \\ \text{default} & \text{otherwise} \end{cases}$$

If you do not specify $GAMMA$, simulation initializes it as:

$$GAMMA = \begin{cases} \frac{\sqrt{2q\epsilon_{si}(NSUB \cdot 10^6)}}{COX} & \text{for: } NSUB > 0 \\ \text{default} & \text{otherwise} \end{cases}$$

If you do not specify PHI , simulation initializes it as:

$$PHI = \begin{cases} 2V_t(T_{nom}) \cdot \ln \left(\frac{NSUB \cdot 10^6}{n_i(T_{nom})} \right) & \text{for: } NSUB > 0 \\ \text{default} & \text{otherwise} \end{cases}$$

If you do not specify VTO , simulation initializes it as:

$$VTO = \begin{cases} VFB + PHI + GAMMA \cdot \sqrt{PHI} & \text{if you specify VFB} \\ \text{default} & \text{otherwise} \end{cases}$$

If you do not specify KP , simulation initializes it as:

$$KP = \begin{cases} (UO \cdot 10^{-4}) \cdot COX & \text{for: } UO > 0 \\ \text{default} & \text{otherwise} \end{cases}$$

If you do not specify $UCRIT$, simulation initializes it as:

$$UCRIT = \begin{cases} VMAX / (UO \cdot 10^4) & \text{for: } VMAX > 0, UO > 0 \\ default & \text{otherwise} \end{cases}$$

If you do not specify $E0$, simulation uses a simplified mobility model with the $THETA$ parameter:

$$E0 = \begin{cases} 0 & \text{if you specify } THETA \\ default & \text{otherwise} \end{cases}$$

Note:

The $E0$ value is zero, indicating to use the simplified mobility model is used in conjunction with $THETA$, instead of the standard mobility model.

Optional parameters might not be available in all implementations.

EPFL-EKV MOSFET Model — Parameter Preprocessing: Default Values and Parameter Ranges

If you do not define a specific model parameters, simulation either initializes it according to the above relations, or sets it to its default value. Certain parameters restrict their numerical range to avoid numerical problems, such as divisions by zero. If you specify a parameter value outside the specified range (see the range column in the parameter tables), then simulation sets its value to the closest acceptable value.

EPFL-EKV MOSFET Model — Parameter Preprocessing: Intrinsic Parameters Temperature Dependence

$$VTO(T) = VTO - TCV \cdot (T - T_{nom})$$

$$KP(T) = KP \cdot \left(\frac{T}{T_{nom}} \right)^{BEX}$$

$$UCRIT(T) = UCRIT \cdot \left(\frac{T}{T_{nom}} \right)^{UCEX}$$

$$PHI(T) = PHI \cdot \frac{T}{T_{nom}} - 3 \cdot V_t \cdot \ln \left(\frac{T}{T_{nom}} \right) - E_g(T_{nom}) \cdot \frac{T}{T_{nom}} + E_g(T)$$

$$IBB(T) = IBB \cdot [1.0 + IBBT \cdot (T - T_{nom})]$$

EPFL-EKV MOSFET Model — Bulk Referenced Intrinsic Voltages

Simulation refers all voltages to the local substrate (see [EPFL-EKV MOSFET Model — Bulk Reference and Symmetry](#)):

$$V_G = V_{GB} = V_{GS} - V_{BS} \text{ Intrinsic gate-to-bulk voltage}$$

$$V_S = V_{SB} = -V_{BS} \text{ Intrinsic source-to-bulk voltage}$$

$$V_D = V_{DB} = V_{DS} - V_{BS} \text{ Intrinsic drain-to-bulk voltage}$$

For P-channel devices, simulation inverts all signs before processing.

EPFL-EKV MOSFET Model — Effective Channel Length and Width

$$W_{eff} = W + DW$$

$$L_{eff} = L + DL$$

Note:

Unlike the convention in other MOSFET models, DL and DW usually permit a negative value due to the above definition.

EPFL-EKV MOSFET Model — Short Distance Matching

The inverse of the square root of the transistor area usually suitably describes a random mismatch between two transistors with an identical layout and close to each other. The following relationships have been adopted:

$$VTO_a = VTO + \frac{AVTO}{\sqrt{NP \cdot W_{eff} \cdot NS \cdot L_{eff}}}$$

$$KP_a = KP \cdot \left(1 + \frac{AKP}{\sqrt{NP \cdot W_{eff} \cdot NS \cdot L_{eff}}} \right)$$

$$GAMMA_a = GAMMA + \frac{AGAMMA}{\sqrt{NP \cdot W_{eff} \cdot NS \cdot L_{eff}}}$$

These model equations apply only in Monte-Carlo and sensitivity simulations. Because negative values for both KP_a and $GAMMA_a$ are not physically meaningful, simulation clips them at zero.

EPFL-EKV MOSFET Model — Reverse Short-channel Effect (RSCE)

$$C_\varepsilon = 4 \cdot (22 \times 10^{-3})^2 C_A = 0.028$$

$$\xi = C_A \cdot \left(10 \cdot \frac{L_{eff}}{LK} - 1 \right)$$

$$\Delta V_{RSCE} = \frac{2 \cdot Q_0}{COX} \cdot \frac{1}{\left[1 + \frac{1}{2} \cdot \left(\xi + \sqrt{\xi^2 + C_\epsilon} \right) \right]^2}$$

EPFL-EKV MOSFET Model — Effective Gate Voltage Including RSCE

$$V'_G = V_G - VTO_a - \Delta V_{RSCE} + PHI + GAMMA_a \sqrt{PHI}$$

EPFL-EKV MOSFET Model — Effective substrate factor including charge-sharing for short and narrow channels

Pinch-off voltage for narrow-channel effect:

$$V_{P0} = \begin{cases} V'_G - PHI - GAMMA_a \left(\sqrt{V'_G + \left(\frac{GAMMA_a}{2} \right)^2} - \frac{GAMMA_a}{2} \right) & \text{for: } V'_G > 0 \\ -PHI & \text{for: } V'_G \leq 0 \end{cases}$$

Effective substrate factor accounting for charge-sharing:

$$V'_{SD} = \frac{1}{2} \cdot \left[V_{SD} + PHI + \sqrt{(V_{SD} + PHI)^2 + (4V_t)^2} \right]$$

Note:

This equation prevents a negative value in the square roots argument in the subsequent code.

$$\gamma^\circ = GAMMA_a - \frac{\epsilon_{si}}{COX} \cdot \left[\frac{LETA}{L_{eff}} \cdot \left(\sqrt{V'_S} + \sqrt{V'_D} \right) - \frac{3 \cdot WETA}{W_{eff}} \cdot \sqrt{V_{P0} + PHI} \right]$$

$$\gamma' = \frac{1}{2} \cdot \left(\gamma^\circ + \sqrt{\gamma^{\circ 2} + 0.1 \cdot V_t} \right)$$

Note:

This equation prevents the effective substrate factor from becoming negative.

EPFL-EKV MOSFET Model — Pinch-off Voltage Including Short-Channel and Narrow-Channel Effects

$$V_P = \begin{cases} V'_G - PHI - \gamma' \cdot \left(\sqrt{V'_G + \left(\frac{\gamma'}{2} \right)^2} - \frac{\gamma'}{2} \right) & \text{for: } V'_G > 0 \\ -PHI & \text{for: } V'_G \leq 0 \end{cases}$$

Note:

The pinch-off voltage accounts for channel doping effects, such as the threshold voltage and the substrate effect.

For long-channel devices, V_p is a function of gate voltage; for short-channel devices, it also becomes a function of the source and drain voltage due to the charge-sharing effect.

EPFL-EKV MOSFET Model — Slope Factor

$$n = 1 + \frac{GAMMA_a}{2 \cdot \sqrt{V_p + PHI + 4V_t}}$$

Note:

The slope factor (or body effect factor) is primarily a function of the gate voltage, and links to the weak inversion slope.

EPFL-EKV MOSFET Model — Large Signal Interpolation Function

$F(v)$ is the large-signal interpolation function, relating normalized currents to normalized voltages. A simple and accurate expression for the transconductance-to-current ratio consistently formulates:

- The static large-signal interpolation function.
- The dynamic model for the intrinsic charges (and capacitances).
- The intrinsic time constant and the thermal noise model for the whole range of current, from weak to strong inversion.

$$\frac{g_{ms} \cdot V_t}{I_{DS}} = \frac{1}{\sqrt{0.25 + i} + 0.5}$$

Large-signal interpolation function:

$$y = \sqrt{0.25 + i} - 0.5$$

$$v = 2y + \ln(y)$$

You cannot analytically invert this equation. However, you can use a Newton-Raphson iterative scheme to invert this equation. Currently, this model uses a simplified algorithm that avoids iteration, leading to a continuous expression for the large signal interpolation function.

The (inverted) large signal interpolation function has the following asymptotes in strong and weak inversion:

$$F(v) = \begin{cases} (v/2)^2 & \text{for: } v \gg 0 \\ \exp(v) & \text{for: } v \ll 0 \end{cases}$$

EPFL-EKV MOSFET Model — Forward Normalized Current

$$i_f = F\left[\frac{V_P - V_S}{V_t}\right]$$

EPFL-EKV MOSFET Model — Velocity Saturation Voltage

$$V_C = UCRIT \cdot NS \cdot L_{eff}$$

Note:

This equation accounts for the NS multiple series device number:

$$V_{DSS} = V_C \cdot \left[\sqrt{\frac{1}{4} + \frac{V_t}{V_C} \cdot \sqrt{i_f}} - \frac{1}{2} \right]$$

Note:

The V_{DSS} variable is half the value of the actual saturation voltage.

EPFL-EKV MOSFET Model — Drain-to-source Saturation Voltage for Reverse Normalized Current

$$V_{DSS'} = V_C \cdot \left[\sqrt{\frac{1}{4} + \frac{V_t}{V_C} \cdot \left(\sqrt{i_f} - \frac{3}{4} \cdot \ln(i_f) \right)} - \frac{1}{2} \right] + V_t \cdot \left[\ln\left(\frac{V_C}{2V_t}\right) - 0.6 \right]$$

EPFL-EKV MOSFET Model — Channel-length Modulation

$$\Delta V = 4 \cdot V_t \cdot \sqrt{LAMBDA \cdot \left(\sqrt{i_f} - \frac{V_{DSS}}{V_t} \right) + \frac{1}{64}}$$

$$V_{ds} = \frac{V_D - V_S}{2}$$

$$V_{ip} = \sqrt{V_{DSS}^2 + \Delta V^2} - \sqrt{(V_{ds} - V_{DSS})^2 + \Delta V^2}$$

$$L_C = \sqrt{\frac{\epsilon_{si}}{COX}} \cdot XJ$$

$$\Delta L = LAMBDA \cdot L_C \cdot \ln\left(1 + \frac{V_{ds} - V_{ip}}{L_C \cdot UCRIT}\right)$$

EPFL-EKV MOSFET Model — Equivalent Channel Length Including Channel-length Modulation and Velocity Saturation

$$L' = NS \cdot L_{eff} \cdot \Delta L + \frac{V_{ds} + V_{ip}}{UCRIT}$$

$$L_{min} = NS \cdot L_{eff} / 10$$

Note:

These equations also account for the NS multiple series device number.

$$L_{eq} = \frac{1}{2} \cdot \left(L' + \sqrt{L'^2 + L_{min}^2} \right)$$

Note:

This equation prevents the equivalent channel length from becoming zero or negative.

EPFL-EKV MOSFET Model — Reverse Normalized Current

Reverse normalized current:

$$i_r' = F \left[\frac{V_P - V_{ds} - V_S - \sqrt{V_{DSS}'^2 + \Delta V^2} + \sqrt{(V_{ds} - V_{DSS}')^2 + \Delta V^2}}{V_t} \right]$$

Reverse normalized current for the mobility model, intrinsic charges/capacitances, the thermal noise model, and the NQS time-constant:

$$i_r = F \left[\frac{V_P - V_D}{V_t} \right]$$

EPFL-EKV MOSFET Model — Transconductance Factor and Mobility Reduction Due to Vertical Field

$$\beta_0 = KP_a \cdot \frac{NP \cdot W_{eff}}{L_{eq}}$$

Note:

The NP (or M) device parameter returns accurate results for simulating parallel devices. Using NS (or N) for series devices is only approximate.

L_{eq} accounts for multiple NS series device numbers.

$$\eta = \begin{cases} 1 / 2 & \text{for NMOS} \\ 1 / 3 & \text{for PMOS} \end{cases}$$

$$q_{B0} = GAMMA_A \cdot \sqrt{PHI}$$

$$\beta_0' = \beta_0 \cdot \left(1 + \frac{COX}{E0 \cdot \epsilon_{si}} \cdot q_{B0} \right)$$

$$\beta = \frac{\beta_0'}{1 + \frac{COX}{E0 \epsilon_{si}} \cdot V_t \cdot |q_B + \eta \cdot q_I|}$$

For the definition of the q_B normalized depletion and the q_I inversion charges refer to the [EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges - Normalized Intrinsic Node Charges](#). Use β_0' to ensure that

$$\beta \approx \beta_0$$

when

$$q_I \ll q_B$$

The formulation of β arises from the integration of the local effective field as a function of depletion and inversion charge densities along the channel. You do not need to specify the substrate bias dependency, because the model includes the depletion charge.

Note:

The resulting mobility expression also depends on V_{DS} .

EPFL-EKV MOSFET Model — Simple Mobility Reduction Model

For compatibility with the former EKV model versions (before v2.6), you can choose the simpler mobility reduction model, which uses the `THETA` parameter.

If you do *not* specify the `E0` model parameter (see [EPFL-EKV MOSFET Model — Static Intrinsic Model Equations](#)), simulation uses the simpler mobility model:

$$V_P' = \frac{1}{2} \cdot \left(V_P + \sqrt{V_P^2 + 2V_t^2} \right)$$

$$\beta = \frac{\beta_0}{1 + THETA \cdot V_P'}$$

EPFL-EKV MOSFET Model — Specific Current

$$I_S = 2 \cdot n \cdot \beta \cdot V_t^2$$

EPFL-EKV MOSFET Model — Drain-to-source Current

$$I_{DS} = I_S \cdot (i_f - i_r)$$

For P-channel devices, I_{DS} has a negative sign.

This drain current expression is a single equation, valid in all operating regions:

- Weak, moderate, and strong inversion
- Nonsaturation
- Saturation

This current is therefore not only continuous among all of these regions, but also continuously derivable.

EPFL-EKV MOSFET Model — Transconductances

Simulation derives the transconductances from the drain current:

$$g_{mg} \equiv \frac{\partial}{\partial V_G} I_{DS} g_{ms} \equiv -\frac{\partial}{\partial V_S} I_{DS} g_{md} \equiv \frac{\partial}{\partial V_{DS}} I_{DS}$$

In the following relationships, the source for the derivatives is a reference:

$$g_m \equiv \frac{\partial}{\partial V_{GS}} I_{DS} = g_{mg} g_{mbs} \equiv \frac{\partial}{\partial V_{BS}} I_{DS} = g_{ms} - g_{mg} - g_{md} g_{ds} \equiv \frac{\partial}{\partial V_{DS}} I_{DS} = g_{md}$$

This model includes the analytic derivatives.

EPFL-EKV MOSFET Model — Impact Ionization Current

$$V_{ib} = V_D - V_S - IBN \cdot 2 \cdot V_{DSS}$$

$$I_{DB} = \begin{cases} I_{DS} \cdot \frac{IBA}{IBB} \cdot V_{ib} \cdot \exp\left(\frac{-IBBL_C}{V_{ib}}\right) & \text{for: } V_{ib} > 0 \\ 0 & \text{for: } V_{ib} \leq 0 \end{cases}$$

Note:

The factor 2 in the v_{ib} expression accounts for the fact that the numerical value of V_{DSS} is half of the actual saturation voltage. The substrate current is a component of the total extrinsic drain current, flowing from the drain to the bulk. This model therefore expresses the total drain current as

$$I_D = I_{DS} + I_{DB}$$

Note:

The substrate current therefore also affects the total extrinsic conductances, especially the drain conductance.

EPFL-EKV MOSFET Model — Quasi-static Model Equations

MOSFET Level 55 includes both a charge-based model for transcapacitances, allowing charge-conservation during transient analysis, you can select a simpler capacitances-based model instead.

Note:

The charges model is symmetrical in terms of the forward and reverse normalized currents—that is, the model is symmetrical for both the drain and source sides.

The pinch-off voltage in the dynamic model provides the short-channel effects (such as charge-sharing and reverse short-channel effects).

This section covers the following topics:

- [EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges](#)
- [EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges - Normalized Intrinsic Node Charges](#)
- [EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges - Total Charges](#)
- [EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances -Transcapacitances](#)
- [EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances -Normalized Intrinsic Capacitances](#)
- [EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances - Total Intrinsic Capacitances](#)

EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges

$$n_q = 1 + \frac{GAMMA A_a}{2 \cdot \sqrt{V_P + PHI + 10^{-6}}}$$

EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges - Normalized Intrinsic Node Charges

$$x_f = \sqrt{\frac{1}{4} + i_f}$$

$$x_r = \sqrt{\frac{1}{4} + i_r}$$

$$q_D = -n_q \cdot \left(\frac{4}{15} \cdot \frac{3x_r^3 + 6x_r^2x_f + 4x_rx_f^2 + 2x_f^3}{(x_f + x_r)^2} - \frac{1}{2} \right)$$

$$q_S = -n_q \cdot \left(\frac{4}{15} \cdot \frac{3x_f^3 + 6x_f^2x_r + 4x_fx_r^2 + 2x_r^3}{(x_f + x_r)^2} - \frac{1}{2} \right)$$

$$q_I = q_S + q_D = -n_q \cdot \left(\frac{4}{3} \cdot \frac{x_f^2 + x_fx_r + x_r^2}{x_f + x_r} - 1 \right)$$

$$q_B = \begin{cases} \left(-GAMMA_a \cdot \sqrt{V_P + PHI + 10^{-6}} \right) \cdot \frac{1}{V_t} - \left(\frac{n_q}{n_q} \right) \cdot q_I & \text{for: } V'_G > 0 \\ -V'_G \cdot \frac{1}{V_t} & \text{for: } V'_G \leq 0 \end{cases}$$

$$q_G = -q_I - q_{OX} - q_B$$

q_{ox} is a fixed oxide charge, which simulation assumes is zero. The preceding equations express the charge conservation among the four nodes of the transistor.

EPFL-EKV Model — Quasi-static Model: Dynamic Model for the Intrinsic Node Charges - Total Charges

$$C_{ox} = COX \cdot NP \cdot W_{eff} \cdot NS \cdot L_{eff}$$

$$Q_{(I,B,D,S,G)} = C_{ox} \cdot V_t \cdot q_{(I,B,D,S,G)}$$

EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances -Transcapacitances

Simulation derives the intrinsic capacitances from the node charges for the terminal voltages:

$$C_{xy} = \pm \frac{\partial}{\partial V_y} (Q_x) \quad x, y = G, D, S, B$$

The preceding equation uses the positive sign if $x=y$ or the negative sign otherwise. This equation produces simple, continuous analytical expressions for all transcapacitances in terms of the x_f pinch-off voltage, the x_r slope factor, and derivatives thereof, from weak to strong inversion, and from non-saturation to saturation.

EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances - Normalized Intrinsic Capacitances

Set $x_{QC}=1$ to select a simplified capacitive dynamic model that uses the five intrinsic capacitances corresponding to the [Figure 13](#). This model ignores the slight bias dependence of the n slope factor. The result is the following simple set of functions:

$$C_{gs} = \frac{2}{3} \cdot \left(1 - \frac{x_r^2 + x_r + \frac{1}{2}x_f}{(x_f + x_r)^2} \right)$$

$$C_{gd} = \frac{2}{3} \cdot \left(1 - \frac{x_f^2 + x_f + \frac{1}{2}x_r}{(x_f + x_r)^2} \right)$$

$$C_{gb} = \left(\frac{n_q - 1}{n_q} \right) \cdot (1 - C_{gs} - C_{gd})$$

$$C_{sb} = (n_q - 1) \cdot C_{gs}$$

$$C_{db} = (n_q - 1) \cdot C_{gd}$$

EPFL-EKV Model — Quasi-static Model: Intrinsic Capacitances - Total Intrinsic Capacitances

$$C_{(gs,gd,gb,sb,db)} = C_{ox} \cdot C_{(gs,gd,gb,sb,db)}$$

EPFL-EKV MOSFET Model — Intrinsic Noise Model Equations

The $INDS$ current source models the noise between the intrinsic source and the drain. This noise includes a thermal noise component and a flicker noise component, and has the following Power Spectral Density (PSD):

$$S_{INDS} = S_{thermal} + S_{flicker}$$

This section covers the following topics:

- [EPFL-EKV Model — Intrinsic Noise Model: Thermal Noise](#)
- [EPFL-EKV Model — Intrinsic Noise Model: Flicker Noise](#)

EPFL-EKV Model — Intrinsic Noise Model: Thermal Noise

The following equation calculates the PSD thermal noise component:

$$S_{thermal} = 4kT \cdot \frac{\mu_{eff}}{(NS \cdot L_{eff})^2} \cdot |Q_I| = 4kT \cdot \beta \cdot |q_I|$$

The preceding thermal noise expression is valid in all regions of operation, including for small V_{DS} .

EPFL-EKV Model — Intrinsic Noise Model: Flicker Noise

The following equation calculates the PSD flicker noise component:

$$S_{flicker} = \frac{KF \cdot g_{mg}^2}{NP \cdot W_{eff} \cdot NS \cdot L_{eff} \cdot COX \cdot f^{AF}}$$

In some implementations, you can select different expressions.

EPFL-EKV MOSFET Model — Operating Point Information

At operating points, the following information displays to help in circuit design.

This section covers the following topics:

- [EPFL-EKV Model — Operating Point: Numerical values of model internal variables](#)
- [EPFL-EKV Model — Operating Point: Transconductance efficiency factor](#)
- [EPFL-EKV Model — Operating Point: Early voltage](#)
- [EPFL-EKV Model — Operating Point: Overdrive voltage](#)
- [EPFL-EKV Model — Operating Point: SPICE-like threshold voltage](#)
- [EPFL-EKV Model — Operating Point: Saturation voltage](#)
- [EPFL-EKV Model — Operating Point: Saturation / non-saturation flag](#)

EPFL-EKV Model — Operating Point: Numerical values of model internal variables

The following are the intrinsic charges and capacitances:

$V_G, V_S, V_D, I_{DS}, I_{DB}, g_{mg}, g_{ms}, g_{mbs}, g_{md}, V_P, n, \beta, I_S, if, ir', t, t_0$.

EPFL-EKV Model — Operating Point: Transconductance efficiency factor

$$tef = g_{ms} \cdot V_t / I_{DS}$$

EPFL-EKV Model — Operating Point: Early voltage

$$VM = I_{DS} / g_{md}$$

EPFL-EKV Model — Operating Point: Overdrive voltage

$$n \cdot (V_P - V_S) \approx V_G - V_{TO_a} - n \cdot V_S$$

For P-channel devices, $n \cdot (v_p - v_s)$ has a negative sign.

EPFL-EKV Model — Operating Point: SPICE-like threshold voltage

$$V_{TH} = V_{TO_a} + \Delta V_{RSC} + \gamma \cdot \sqrt{V'_S - GAMMA_a} \cdot \sqrt{PHI}$$

This expression is the SPICE-like threshold voltage (the source). It also accounts for charge-sharing and reverse short-channel effects on the threshold voltage.

For P-channel devices, V_{TH} has a negative sign.

EPFL-EKV Model — Operating Point: Saturation voltage

$$V_{DSAT} = 2V_{DSS} + 4V_t$$

For P-channel devices, V_{DSAT} has a negative sign.

EPFL-EKV Model — Operating Point: Saturation / non-saturation flag

$$'SAT' \text{ or } '1' \text{ for } \frac{i_f}{i_r} > SATLIM$$

$$'LIN' \text{ or } '0' \text{ for } \frac{i_f}{i_r} \leq SATLIM$$

Note:

Some simulators implement the operating point differently (some information might not be available).

EPFL-EKV MOSFET Model — Estimation and Limits of Static Intrinsic Model Parameters

If you do not have access to a parameter extraction facility, simulation can roughly estimate the EKV intrinsic model parameters from the SPICE Level 2/3 parameters as indicated in the following table. Pay attention to the units of the parameters. This estimation method generally returns reasonable results. Nevertheless, be aware that the underlying modeling in SPICE Level 2/3 and in the EKV model is not the same, even if the names and the function of several parameters are similar. Therefore, it is preferred if parameter extraction is made directly from measurements.

Lower and upper limits indicated in the table should give an idea on the order of magnitude of the parameters but do not necessarily correspond to physically meaningful

limits, nor to the range specified in the parameter tables. These limits may be helpful for obtaining physically meaningful parameter sets when using nonlinear optimization techniques to extract EKV model parameters.

Name	Unit	Default	Example	Lower	Upper	Estimation ¹
COX	F/m ¹⁷	0.7E-3	3.45E-3	-	-	ϵ_{ox}/TOX
DL	m	0	$-0.15 \cdot L_{min}$	$-0.5 \cdot L_{min}$	$0.5 \cdot L_{min}$	$\chi l - 2 \cdot ld$
DW	m	0	$-0.1 \cdot W_{min}$	$-0.5 \cdot W_{min}$	$0.5 \cdot W_{min}$	$\chi w - 2 \cdot wd$
E0	V/m	1.0E12	200E6	$0.1 / (0.4 \cdot tox)$	-	$0.2 / (theta \cdot tox)$
GAMMA	$\sqrt{v}$	1.0	0.7	0	2	$\sqrt{2q\epsilon_{si} \cdot n_{sub}} / cox$
IBA	1/m	0.0	2.0E8	0.0	5.0E8	$alpha \cdot vcr / l_c$
IBB	V/m	3.0E8	2.0E8	1.8E8	4.0E8	vcr / l_c
IBN	-	1.0	0.6	0.4	1.0	-
KP	A/V ²	50E-6	150E-6	10E-6	-	$uo \cdot cox$
LAMBDA	-	0.5	0.8	0	3	-
LETA	-	0.1	0.3	0	2	-
LK	m	0.29E-6	0.4E-6	0.05E-6	2E-6	-
PHI ¹⁷	V	0.7	0.5	0.3	2	$2v_t \cdot \ln(n_{sub} / n_i)$
Q0	As/m ¹⁷	0.0	230E-6	0	-	-
UCRIT	V/m	2.0E6	2.3E6	1.0E6	25E6	V_{MAX}/UO
VTO	V	0.5	0.7	0	2	VTO
WETA	-	0.25	0.2	0	2	-
XJ	m	0.1E-6	0.15E-6	0.01E-6	1E-6	XJ

$$\epsilon_{ox}=0.0345E-9 \text{ F/m} \quad q=1.609E-19 \text{ C} \quad k=1.381E-23 \text{ J/K} \quad L_c = \sqrt{\epsilon_{si} \cdot XJ / COX}$$

$$\epsilon_{si}=0.104E-9 \text{ F/m} \quad n_i=1.45E16 \text{ m}^{-3} \quad V_t=kT/q=0.0259 \text{ V (at room temperature)}$$

17. The minimum value of PHI also determines the minimum value of the pinch-off voltage. Due to the intrinsic temperature dependence of PHI, higher temperatures use a lower value, which limits the range of simulation for small currents.

Note:

Parameters in this table use m (meter) as the length unit. L_{min} is the minimum drawn length of transistors. W_{min} is the minimum drawn width of transistors. Example values are shown for enhancement N-channel devices.

EPFL-EKV MOSFET Model Updates

Synopsys has made several improvements to the original EKV v2.6 MOSFET model. Wherever possible, these enhancements maintain backward compatibility with previous versions.

This section covers the following topics:

- [EPFL-EKV Model Updates](#)
- [EPFL-EKV Model Corrections](#)

EPFL-EKV Model Updates

Update	Description	Consequence
Revision I, Sep 1997	The narrow channel effect on the substrate factor was revised to improve the transcapacitances behavior. The narrow channel effect is no longer a function of the v_s source voltage, but of the v_p pinch-off voltage.	The $WETA$ and DW narrow channel effect parameters require different numerical values to achieve the same effect.
Revision II, Jul 1998	Intrinsic Time Constant: Simulation calculates the τ_0 intrinsic time constant as a function of the effective β factor (including vertical field dependent mobility and short-channel effects), instead of the maximum mobility using the KP parameter. Thermal noise: Simulation calculates the $S_{thermal}$ thermal noise power spectral density as a function of the effective β factor (including vertical field dependent mobility and short-channel effects), instead of the maximum mobility using the KP parameter.	The NQS time constant has an additional gate voltage dependence, resulting in more conservative (lower) estimation of the NQS time constant at high V_G, and additional dependence on short-channel effects. $S_{thermal}$ has an additional gate voltage and short-channel effect dependence.

Update	Description	Consequence
	Optional process parameters to calculate electrical intrinsic parameters: This option calculates the electrical parameters as a function of the optional parameters: COX→TOX GAMMA and PHI→NSUB VTO→VFB KP→UO UCRIT→VMAX cm is the length unit for the NSUB and UO parameters.	These parameters accommodate scaling behavior and allow meaningful statistical circuit simulation, due to de-correlation of physical effects. If you use these optional parameters, this version is compatible with former revisions, except for the default calculation of the parameters.
	Optional simplified mobility model: The simple mobility model used in former model versions by using the THETA parameter, was reinstated as an option.	This mobility model simplifies adaptation from earlier model versions to the current version.
	Parameter synonyms: You can use E0 and Q0 as synonyms for the EO and QO parameters.	This option accommodates some simulators that use only alphabetic characters.
	Operating point information: This enhancement models the analytical expression for the SPICE-like VTH threshold voltage in the operating point information to include the charge-sharing and reverse short-channel effects. This option modifies the analytical expression for the VDSAT saturation voltage in the operating point information so it has a non-zero value in weak inversion.	This enhancement improves design information.

EPFL-EKV Model Corrections

Correction	Description
EPFL R11, March, 1999	Equation 45, Equation 53, Equation 54, and Equation 58 were corrected for multiple series device behavior by using the NS parameter.

Correction	Description
EPFL R12, July 30, 1999	EPFL released the following corrections: 1.99/07/30 mb (r12) corrected dGAMMAprime_dVG (narrow channel). An error in the analytical model derivatives of the GAMMAprime variable affected the transconductances and transcapacitances. 2.99/07/30 mb (r12) prevents PHI from being smaller than 0.2, both at init and after updating the temperature. For some CMOS technologies, PHI parameter values are as low as 400mV, required to account for particular process details. If you increase the temperature from room temperature, PHI decreases due to its built-in temperature dependence. As a result, PHI attains very low values or even becomes negative when it reaches 100degC. For the model to function at these temperatures, PHI has a lower limit of 200mV. The usual range for this parameter is well above this value (600mV to 1V). 3.99/06/28 mb (r12) fixed COX/KP initialization (rg). 4.99/05/04 mb (r12) completed parameter initialization for XQC, DL, and DW, and removed IBC and ibc (cd).

LEVEL 57: BSIM3 SOI Model

The UC Berkeley SOI model (BSIM3SOI) supports Fully Depleted (FD), Partially Depleted (PD), and Dynamically Depleted (DD) SOI devices of which BSIM3PD2.0.1 for PD SOI devices is Synopsys MOSFET Level 57. For a description of this model, see the *BSIM3PD2.0 MOSFET MODEL User's Manual*.

Level 57 uses the UCB Version 2.2.3 model, which includes a separate set of the geometry-dependence parameters (LLC , LWC , $LWLC$, WLC , WWC , and $WWLC$) to calculate $L_{eff}CV$ and $W_{eff}CV$.

Level 57 also includes a new Full-Depletion (FD) module ($soiMod=1$). This module provides a better fit to FD SOI devices. As $soiMod=0$ (default), the model is identical to previous BSIMSOI PD models. This module also includes gate to channel/drain/source tunneling currents and overlap components.

See also: `.OPTION VER_CONTROL`

The following enhancements to the BSIMSOI PD version were made starting in the BSIMSOI 3.0 version:

- If the self-heating model is on, simulation calculates the channel surface potential.
- NDIF includes parameter Compared with BSIM4.6.1, several new features are added in this version. An error in the derivative calculation of dV_{gs_eff}/dV_g has been fixed. The reset value of drain/body breakdown voltage does not cause non-convergence now. Some other similar bugs have also been fixed.
- The Δ_{tox} parameter in the MOS active element (M) models the relative variation on the trans conductance (oxide thickness) of the MOS in Monte Carlo analysis.

The following sections discuss these topics:

- [BSIM3 SOI Model — General Syntax](#)
- [BSIM3 SOI Model Parameters](#)
- [BSIM3 SOI Model — Range Limit Parameters](#)
- [BSIM3 SOI Model — Version Updates](#)
- [BSIM3 SOI Model — PD Version Updates](#)
- [BSIM3 SOI Model — Output Templates](#)

BSIM3 SOI Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] [nb] [nT] mname [L=val] [W=val]
+ [M=val] [NRD=val] [AD=val] [AS=val] [PD=val] [PS=val]
+ [NRS=val] [NRB=val] [RTH0=val] [NBC=val]
+ [NSEG=val] [PDBCP=val] [PSBCP=val] [AGBCP=val]
+ [AEBCP=val] [VBSUSR=val] [DELTOX=val] [TNODEOUT | TNODEOUT=val]
+ [Off] [FRBODY] [BJToff=val] [IC=Vds, Vgs, Vbs, Ves]
+ [SOIMOD=val] [SOIQ0=val] [MULID0=val]
```

Parameter	Description
AD	Drain diffusion area. Overrides .OPTION DEFAD statement. Default=DEFAD.
AEBCP	Parasitic body-to-substrate overlap area for the body contact.
AGBCP	Parasitic gate-to-body overlap area for the body contact.
AS	Source diffusion area. Overrides .OPTION DEFAS statement. Default=DEFAS.
BJTOFF	Turning off BJT if equal to 1.
CTH0	Thermal capacitance per unit width: - If you do not specify CTH0, simulation extracts it from the model card. - If you specify CTH0, it overrides CTH0 in the model card.
DELTOX	Shift in gate oxide thickness (TOX). That is, the difference between the electrical and physical gate oxide/insulator thicknesses.
FRBODY	Coefficient of the distributed body resistance effects. Default=1.0
IC	Initial guess in the order (drain, front gate, internal body, back gate, external voltage). (ignores Vps for 4-terminal devices) Use these only if you specify UIC in the .TRAN statement. The .IC statement overrides it.
L	SOI MOSFET channel length in meters. This parameter overrides DEFL in an OPTIONS statement. Default=DEFL with a maximum of 0.1m.
M	Multiplier to simulate multiple SOI MOSFETs in parallel. The M setting affects all channel widths, diode leakages, capacitances, and resistances. Default=1.
mname	MOSFET model name reference.
Mxxx	SOI MOSFET element name. Must begin with M, followed by up to 1023 alphanumeric characters.
nb	Internal body node name or number.
NBC	Number of body contact isolation edge.
nd	Drain terminal node name or number.

Parameter	Description
<i>ne</i>	Back gate (or substrate) node name or number.
<i>ng</i>	Front gate node name or number.
<i>np</i>	External body contact node name or number.
NRB	Number of squares for the body series resistance.
NRD	NRD (Number of squares of drain diffusion for resistance calculations) overrides .OPTION DEFNRD. For nonCMI models such as BSIM3 etc... Default=DEFNRD, if you set ACM=0 or 1 model parameter. Default=0.0, if you set ACM=2 or 3. For CMI models such as BSIM4 and so on. Default=1.0
NRS	NRS (Number of squares of source diffusion for resistance calculations) overrides .OPTION DEFNRS. For nonCMI models such as BSIM3 etc... Default=DEFNRS, if you set ACM=0 or 1 model parameter. Default=0.0, if you set ACM=2 or 3. For CMI models such as BSIM4 and so on. Default=1.0
ns	Source terminal node name or number.
NSEG	Number of segments for partitioning the channel width.
nT	Temperature node name or number.
OFF	Sets the initial condition of the element to OFF in DC analysis.
PD	Drain junction perimeter, including channel edge. Overrides .OPTION DEFDPD.
PDBCP	Parasitic perimeter length for the body contact at the drain side.
PS	Source junction perimeter including channel edge. Overrides .OPTION DEFPS.
PSBCP	Parasitic perimeter length for the body contact at the source side.
RTH0	Thermal resistance per unit width: • If you do not specify RTH0, simulation extracts it from the model card. • If you specify RTH0, it overrides RTH0 in the model card.
SOIMOD	SOI model selector • If SOIMOD is not specified, it's extracted from the model card. • If SOIMOD is specified, it overrides SOIMOD in the model card.
SOIQ0	Floating body charge initialization. This parameter is set for the BQI algorithm of a floating body node.
TNODEOUT	Temperature node flag indicating the use of the T node.
VBSUSR	Optional initial Drain/Body Breakdown Voltage of Vbs that you specify for transient analysis.
W	MOSFET channel width in meters. This parameter overrides DEFW in an OPTIONS statement. Default=DEFW.
MULID0	Scaling factor of drain current, the default is 1.0.

- If you do not set `TNODEOUT` or `TNODEOUT = 0`, you can specify four nodes for a device to float the body. Specifying five nodes implies that the fifth node is the external body contact node with a body resistance between the internal and external terminals. This configuration applies to a distributed body resistance simulation.
- If you set `TNODEOUT` or `TNODEOUT = 0`, simulation interprets the last node as the temperature node. You can specify five nodes to float the device. Specifying six nodes implies body contact. Seven nodes is a body-contacted case with an accessible internal body node. You can use the temperature node to simulate thermal coupling.

BSIM3 SOI Model Parameters

This section covers the following topics:

- [BSIM3 SOI Model — Control Parameters](#)
- [BSIM3 SOI Model — Process Parameters](#)
- [BSIM3 SOI Model — DC Parameters](#)
- [BSIM3 SOI Model — AC and Capacitance Parameters](#)
- [BSIM3 SOI Model — Temperature Parameters](#)
- [BSIM3 SOI Model — Avalanche Impact Ionization Parameters](#)
- [BSIM3 SOI Model — Notes](#)

BSIM3 SOI Model — Control Parameters

Parameter	Unit	Default	Description
CAPMOD	-	2	Flag for the short channel capacitance model
MOBMOD	-	1	Mobility model selector
NOIMOD	-	1	Flag for the noise model
SHMOD	-	0	Flag for self-heating:0 - no self-heating 1 - self-heating

BSIM3 SOI Model — Process Parameters

Parameter	Unit	Default	Description
DXOXCV (capmod=3 only)			Difference between the electrical and physical gate oxide thicknesses, due to the effects of the gate poly-depletion and the finite channel charge layer thickness.

Parameter	Unit	Default	Description
NCH	1/cm3	1.7e17	Channel doping concentration
NGATE	1/cm3	0	Poly gate doping concentration
NSUB	1/cm3	6.0e16	Substrate doping concentration
TBOX	m	3.0e-7	Buried oxide thickness
TOX	m	1.0e-8	Gate oxide thickness
TSI	m	1.0e-7	Silicon film thickness
XJ	m	-	S/D junction depth

BSIM3 SOI Model — DC Parameters

Parameter	Unit	Default	Description
A0	-	1.0	Bulk charge effect coefficient for the channel length
A1	1/V	0.0	First non-saturation effect parameter
A2	-	1.0	Second non-saturation effect parameter
AELY	V/m	0	Channel length dependency of the early voltage for the bipolar current
AGIDL	1/W	0.0	GIDL constant
AGS	1/V	0.0	Gate bias coefficient of A_{bulk}
AHLI	-	0	High-level injection parameter for the bipolar current
ALPHA0	m/V	0.0	First parameter of the impact ionization current
B0	m	0.0	Bulk charge effect coefficient for the channel width
B1	m	0.0	Bulk charge effect width offset
BETA0	1/V	0.0	First V_{ds} dependence parameter of the impact ionization current
BETA1	-	0.0	Second V_{ds} dependence parameter of the impact ionization current

Parameter	Unit	Default	Description
BETA2	V	0.1	Third V_{ds} dependence parameter of the impact ionization current
BGIDL	V/m	0.0	GIDL exponential coefficient
CDSC	F/m ²	2.4e-4	Drain/source to the channel coupling capacitance
CDSCB	F/m ²	0	Body-bias sensitivity of cdsc
CDSCD	F/m ²	0	Drain-bias sensitivity of cdsc
CIT	F/m ²	0.0	Interface trap capacitance
DELTA	-	0.01	Effective V_{ds} parameter
DROUT	-	0.56	L dependence coefficient of the DIBL correction parameter in Rout
DSUB	-	0.56	DIBL coefficient exponent
DVT0	-	2.2	First coefficient of the short-channel effect on Vth
DVT0W	-	0	First coefficient of the narrow width effect on Vth for a small channel length
DVT1	-	0.53	Second coefficient of the short-channel effect on Vth
DVT1W	-	5.3e6	Second coefficient of the narrow width effect on Vth for a small channel length
DVT2	1/V	-0.032	Body-bias coefficient of the short-channel effect on Vth
DVT2W	1/V	-0.032	Body-bias coefficient of the narrow width effect on Vth for a small channel length
DWB	m/V ^{1/2}	0.0	Coefficient of the substrate body bias dependence of Weff
DWBC	m	0.0	Width offset for the body contact isolation edge
DWG	m/V	0.0	Coefficient of the gate dependence of Weff
ESATI	V/m	1.e7	Saturation channel electric field for the impact ionization current
ETA0	-	0.08	DIBL coefficient in the subthreshold region
ETAB	1/V	-0.07	Body-bias coefficient for the DIBL effect in the subthreshold region

Parameter	Unit	Default	Description
FBJTII	-	0.0	Fraction of the bipolar current affecting the impact ionization
ISBJT	A/m2	1.0e-6	BJT injection saturation current
ISDIF	A/m2	0	Body to source/drain injection saturation current
ISREC	A/m2	1.0e-5	Recombination in the depletion saturation current
ISTUN	A/m2	0.0	Reverse tunneling saturation current
K1	V ^{1/2}	0.6	First-order body effect coefficient
K1W1	m	0	First-order effect width dependent parameter
K1W2	m	0	Second-order effect width dependent parameter
K2	-	0	Second-order body effect coefficient
K3	-	0	Narrow coefficient
K3B	1/V	0	Body effect coefficient of k3
KB1	-	1	Backgate body charge coefficient
KETA	1/V	-0.6	Body-bias coefficient of the bulk charge effect
KETAS	V	0.0	Surface potential adjustment for the bulk charge effect
LBJT0	m	0.2e-6	Reference channel length for the bipolar current
LII	-	0	Channel length dependence parameter for the impact ionization current
LINT	m	0.0	Length offset fitting parameter from I-V without bias
LN	m	2.0e-6	Electron/hole diffusion length
NBJT	-	1	Power coefficient of the channel length dependency for the bipolar current
NDIODE	-	1.0	Diode non-ideality factor
NFACTOR	-	1	Subthreshold swing factor
NGIDL	V	1.2	GIDL V _{ds} enhancement coefficient
NLX	m	1.74e-7	Lateral non-uniform doping parameter

Parameter	Unit	Default	Description
NRECF0	-	2.0	Recombination non-ideality factor at the forward bias
NRECR0	-	10	Recombination non-ideality factor at the reversed bias
NTUN	-	10.0	Reverse tunneling non-ideality factor
PCLM	-	1.3	Channel length modulation parameter
PDIBLC1	-	0.39	Correction parameter for the DIBL effect of the first output resistance
PDIBLC2	-	0.0086	Correction parameter for the DIBL effect of the second output resistance
PRWB	1/V1	0	Body effect coefficient of Rdsw
PRWG	1/V1/2	0	Gate-bias effect coefficient of Rdsw
PVAG	-	0.0	Gate dependence of the Early voltage
RBODY	ohm/m ⁻²	0.0	Intrinsic body contact sheet resistance
RBSH	ohm/m ⁻²	0.0	Extrinsic body contact sheet resistance
RDSW	$\Omega^* \mu m^{wr}$	100	Parasitic resistance per unit width
RSH	ohm/square	0.0	Source/drain sheet resistance in ohm per square
SII0	1/V	0.5	First V _{gs} dependence parameter for the impact ionization current
SII1	1/V	0.1	Second V _{gs} dependence parameter for the impact ionization current
SII2	1/V	0	Third V _{gs} dependence parameter for the impact ionization current
SIID	1/V	0	V _{ds} dependence parameter of the drain saturation voltage for the impact ionization current
TII	-	0	Temperature dependence parameter for the impact ionization current
U0	cm ² /(V-s ec)	NMOS-670 PMOS-250	Mobility at Temp=Tnom
UA	m/V	2.25e-9	First-order mobility degradation coefficient

Parameter	Unit	Default	Description
UB	$(\text{m/V})^2$	5.87e-19	Second-order mobility degradation coefficient
UC	1/V	-0.0465	Body-effect of the mobility degradation coefficient
VABJT	V	10	Early voltage for the bipolar current
VDSATII0	V	0.9	Nominal drain saturation voltage at threshold for the impact ionization current
VECB	v	0.026v	Electron tunneling from the conduction band
VEVB	v	0.075v	Electron tunneling from the valence band
voff	v	-0.08	Offset voltage in the subthreshold region for large W and L
Vrec0	V	0.0	Voltage dependent parameter for the recombination current
vsat	m/sec	8e4	Saturation velocity at Temp=Tnom
vth0	v	NMOS 0.7 PMOS -0.7	Threshold voltage @ Vbs=0 for a long, wide device
Vtun0	V	0.0	Voltage dependent parameter for the tunneling current
w0	m	0	Narrow width parameter
wint	m	0.0	Width offset fitting parameter from I-V without bias
wr	-	1	Width offset from Weff for the Rds calculation

BSIM3 SOI Model — AC and Capacitance Parameters

Parameter	Unit	Default	Description
ACDE	m/V	1.0	Exponential coefficient for the charge thickness in the CapMod=3 for the accumulation and depletion regions
ASD	V	0.3	Smoothing parameter for the source/drain bottom diffusion
CF	F/m	cal.	Fringing field capacitance of the gate-to-source/drain
CGDL	F/m	0.0	Overlap capacitance for the lightly-doped drain-gate region

Parameter	Unit	Default	Description
CGDO	F/m	0	Non LDD region drain-gate overlap capacitance per channel length
CGEO	F/m	0	Gate substrate overlap capacitance per unit channel length
CGSL	F/m	0.0	Overlap capacitance for the lightly-doped source-gate region
CGSO	F/m	calculated	Non LDD region source-gate overlap capacitance per channel length
CJSWG	F/m ²	1.e-10	Source/drain (gate side) sidewall junction capacitance per unit width (normalized to 100nm Tsi)
CKAPPA	F/m	0.6	Coefficient for the fringing field capacitance for the overlap capacitance in the lightly-doped region
CLC	m	0.1e-7	Constant term for the short-channel model
CLE	-	0.0	Exponential term for the short-channel model
CSDESW	F/m	0.0	Fringing capacitance per unit length for the source/drain sidewall
CSDMIN	V	cal.	Minimum capacitance for the source/drain bottom diffusion
DELVT	V	0.0	Threshold voltage adjustment for C-V
DLBG	m	0	Length offset fitting parameter for the backgate charge
DLC	m	lint	Length offset fitting parameter for the gate charge
DLCB	m	lint	Length offset fitting parameter for the body charge
DWC	m	wint	Width offset fitting parameter from C-V
FBODY	-	1.0	Scaling factor for the body charge
LDIF0	-	1	Channel length dependency coefficient of the diffusion cap.
MJSWG	V	0.5	Grading coefficient of the source/drain (gate side) sidewall junction capacitance
MOIN	V ^{1/2}	15.0	Coefficient for the gate-bias dependent surface potential
NDIF	-	-1	Power coefficient of the channel length dependency for the diffusion capacitance

Parameter	Unit	Default	Description
PBSWG	V	0.7	Built-in potential of the source/drain (gate side) sidewall junction capacitance
TT	second	1ps	Diffusion capacitance transit time coefficient
VSDFB	V	cal.	Flatband voltage for the source/drain bottom diffusion capacitance
VSDTH	V	cal.	Threshold voltage for the source/drain bottom diffusion capacitance
XPART	-	0	Charge partitioning rate flag

BSIM3 SOI Model — Temperature Parameters

Parameter	Unit	Default	Description
AT	m/sec	3.3e4	Temperature coefficient for U_a
CTH0	(W*S)/m°C	0	Normalized thermal capacity
KT1	V	-0.11	Temperature coefficient for the threshold voltage
KT2	-	0.022	Body-bias coefficient of the threshold voltage temperature effect
KTIL	V*m	0	Channel length dependence of the temperature coefficient for the threshold voltage
NTRECF	-	0	Temperature coefficient for N_{ref}
NTRECR	-	0	Temperature coefficient for N_{recr}
PRT	Ω_{um}	0	Temperature coefficient for R_{dsw}
RTH0	m°C/W	0	Normalized thermal resistance
TCJSWG	1/K	0	Temperature coefficient of C_{jswg}
TNOM	°C	25	Temperature at which simulation expects parameters
TPBSWG	V/K	0	Temperature coefficient of P_{bswg}
UA1	m/V	4.31e-9	Temperature coefficient for U_a
UB1	(m/V) ²	-7.61e-18	Temperature coefficient for U_b
UC1	1/V	-0.056	Temperature coefficient for U_c

Parameter	Unit	Default	Description
UTE	-	-1.5	Mobility temperature exponent
XBJT	-	1	Power dependence of j_{bit} on the temperature
XDIF	-	XBJT	Power dependence of j_{dif} on the temperature
XREC	-	1	Power dependence of j_{rec} on the temperature
XTUN	-	0	Power dependence of j_{tun} on the temperature

BSIM3 SOI Model — Avalanche Impact Ionization Parameters

The following parameters improve the BSIM3SOI model to address parasitic BJT-induced avalanche impact ionization current.

Parameter	Unit	Default	Description	Notes
ABJTII	-	0.0	Exponent factor for avalanche current	-
CBJTII	m/V	0.0	Length scaling parameter for II BJT part	-
EBJTII	1/V	0.0	Impact ionization parameter for BJT part	-
IIMOD	-	0 (original II model)	Impact ionization model selector	=1 selects the new II model
MBJTII	-	0.4	Internal B-C grading coefficient	-
TVBCI	-	0.0	Temperature coefficient for VBCI	-
VBCI	V	0.7	Internal B-C built-in potential	-

BSIM3 SOI Model — Notes

- BSIMPD2.01 supports capmod=2 and 3 only. It does not support capmod=0 and 1.
- Modern SOI technology commonly uses source/drain extension or LDD. The source/drain junction depth (X_j) can be different from the silicon film thickness (T_{si}). By default, if you do not specify X_j , simulation sets it to T_{si} . X_j cannot be greater than T_{si} .
- BSIMPD refers the substrate to the silicon below the buried oxide (not to the well region in BSIM3) to calculate the backgate flatband voltage (V_{fb}) and the parameters related to the source/drain diffusion bottom capacitance (V_{sdth} , V_{sdfb} , C_{sdmin}).
 - Positive n_{sub} means the same type of doping as the body.
 - Negative n_{sub} means the opposite type of doping.

- **New `WOFLK` Parameter:**

The following equation models the SPICE2 flicker noise current density, used in both UCB SOI code and the Synopsys Level=57 MOSFET model for `noiMod=1` and 4:

$$S_{id}, f[I^2/Hz] = K_F \cdot I_{ds}^{AF} / (C_{ox} \cdot L_{eff}^2 \cdot f^{EF}) \quad -- \quad (1)$$

However, if AF is not equal to unity, it does not scale properly with W_{eff} , because I_{ds} is approximately proportional to W_{eff} . Also, without the PrimeSim Continuum solution multiplicity factor (M factor) in equation (1), this model cannot simulate multiple transistors in parallel.

To solve these problems, the PrimeSim Continuum solution added a `WOFLK` (width normalizing) parameter, and corrects equation (1) as:

$$M \cdot K_F \cdot [(W_{eff}/WOFLK)^{(1-AF)}] \cdot I_{ds}^{AF} / (C_{ox} \cdot L_{eff}^2 \cdot f^{EF}) \quad -- \quad (2)$$

The default value of `WOFLK` is -1.0 to switch off the new width-scaling model. The unit is in meters.

The next equation handles the flicker noise model (`noimod=1` & 4), depending on whether you specify `WOFLK`.

If `WOFLK` ≤ 0.0 (the default case), then the flicker noise model of `noiMod=1` and 4 uses this equation for backward compatibility.

$$\begin{aligned} & M \cdot K_F \cdot I_{ds}^{AF} / (C_{ox} \cdot L_{eff}^2 \cdot f^{EF}) \quad -- \quad (3) \\ \text{ELSE} \\ & M \cdot K_F \cdot [(W_{eff}/WOFLK)^{(1-AF)}] \cdot I_{ds}^{AF} / (C_{ox} \cdot L_{eff}^2 \cdot f^{EF}) \end{aligned}$$

BSIM3 SOI Model — Range Limit Parameters

Simulation reports either a warning or a fatal error if BSIM3 version 3 parameters fall outside predefined ranges. These range limitations prevent (or at least warn of) potential numerical problems.

To control the maximum number of simulation warning messages printing to the output file, use:

```
.OPTION WARNLIMIT=#
```

In the preceding `.OPTION` statement, `#` is the maximum number of warning messages that simulation reports. The default `WARNLIMIT` is 10. The source/drain bulk junction capacitance is not be discontinuous when V_{bs}/V_{bd} is zero. The S/D junction sidewall capacitance along the isolation edge is set to zero if they are negative. value is 1. In some cases, simulation checks parameters only if you set the `PARMAMCHK=1` model parameter.

Parameter	Limits	Comment
ACDE	< 0.1, > 1.6 warn	if (2.2299 < version < 3.1099) acde < 0.4 warn
B1	b1+weff =0 fatal	
CLC	< 0 fatal	
DELTA	< 0 fatal	
DROUT	< 0 fatal	
DSUB	< 0 fatal	
DVT1	< 0 fatal	
DVT1W	< 0 fatal	
MOIN	< 5, > 25 warn	
MOINFD	< 5.0 warn	if (version > 2.9999)
NGATE	< 0, > 1e25 fatal	
NLX	< -leff fatal	
NOFF	=0 warn, set 1.0	If (version>3.1999) < 0.1, > 4.0 warn
NPEAK	<= 0 fatal	
PCLM	<= 0 fatal	
TBOX	<= 0 fatal	
TOX	tox+deltox < 0 fatal	
TOXM	toxm+deltox < 0 fatal	
U0	<= 0 fatal	
VSAT	<= 0 fatal	
W0	w0+weff =0 fatal	
if paramchk=1 following parameter limit range is added		
A2	< 0.01, > 1 warn	
ACDE	< 0.4, > 1.6 warn	if (version > 3.1099) acde < 0.1 warn
AELY	< 0 warn	
AGIDL	< 0 warn	

Parameter	Limits	Comment
AHLI	< 0 warn	
ALPHAGB1	< 0 warn	
ALPHAGB2	< 0 warn	
ASD	< 0 warn	
B1	$b1 + w_{eff} < 1e-7$ warn	
BETA0	< 0 warn	
BETA1	< 0 warn	
BETA2	< warn	
BETAGB1	< 0 warn	
BGIDL	< 0 warn	
CAPMOD	< 2 warn	
CDSC	< 0 warn	
CDSCD	< 0 warn	
CGDOI	< 0 warn	
CGEO	< 0 warn	
CGSOI	< 0 warn	
CSDESW	< 0 warn	
CSDMIN	< 0 warn	
CTH0	< 0 warn	
DELTA VOX	≤ 0 warn	
DLBG	< 0 warn	
DVT0	< 0 warn	
DWBC	< 0 warn	
EBG	< 0 warn	
ESATII	< 0 warn	Should be within (0,1)
ETA0	< 0 warn	

Parameter	Limits	Comment
FBJTII	< 0 warn	
ISBJT	< 0 warn	
ISDIF	< 0 warn	
ISREC	< 0 warn	
ISTUN	< 0 warn	
K1W1	< 0 warn	
K1W2	< 0 warn	
KETAS	< 0 warn	
LEFF	<= 5e-8 warn	
LEFFCV	<= 5e-8 warn	
LII	< 0 warn	
MOIN	< 5, > 25 warn	
NBJT	< 0 warn	
NDIF	< 0 warn	if (version <= 2.9999)
NDIODE	< 0 warn	
NECB	< 0 warn	
NEVB	< 0 warn	
NFACTOR	< 0 warn	
NGATE	0 < ngate <= 1e18 warn	
NGIDL	< 0 warn	
NIGC	<= 0 fatal	if (version > 2.9999)
NLX	< 0 warn	
NPEAK	<= 1e15, >= 1e21 warn	
NSUB	>= 1e21 warn	
NTOX	< 0 warn	
NTRECF	< 0 warn	

Parameter	Limits	Comment
NTRED	< 0 warn	
NTUN	< 0 warn	
PDIBL1	< 0 warn	
PDIBL2	< 0 warn	
PIGCD	<= 0 fatal	if (version > 2.9999)
POXEDGE	<= 0 fatal	if (version > 2.9999)
RBODY	< 0 warn	
RBODY	< 0 warn	
RBSH	< 0 warn	
RBSH	< 0 warn	
RDS0	> 0, < 0.001 warn	
RDSW	< 0 warn	
RHALO	< 0 warn	
RTH0	< 0 warn	
SII1	< 0 warn	
SII2	< 0 warn	
SIID	< 0 warn	
TCJSWG	< 0 warn	if (version <= 2.9999)
TII	< 0 warn	
TOX	tox+deltox < 0 warn	
TOXQM	<= 0 warn	
TOXTREF	< 0 warn	
TPBSWG	< 0 warn	if (version <= 2.9999)
TT	< 0 warn	
VECB	< 0 warn	
VEVB	< 0 warn	

Parameter	Limits	Comment
VGB1	< 0 warn	
VGB2	< 0 warn	
VOXH	< 0 warn	
VREC0	< 0 warn	
VSATTEMP	< 1e3 warn	
VTUN0	< 0 warn	
W0	$w0 + weff < 1e-7$ warn	
WEFF	$\leq 1e-7$ warn	
WEFFCV	$\leq 1e-7$ warn	
WTH0	< 0 warn	
XJ	> tsi warn	

BSIM3 SOI Model — Version Updates

This section covers the following topics:

- [BSIM3 SOI Version 3.2 Updates](#)
- [BSIM3 SOI Version 3.1 Updates](#)

BSIM3 SOI Version 3.2 Updates

The model selector, `SoiMod`, is an instance parameter and a model parameter. `SoiMod` determines the operation of BSIMSOI. If `SoiMod=0` (default), the model equation is identical to the BSIMPD equation. If `SoiMod=1` (unified model for PD&FD) or `SoiMod=2` (ideal FD), the following equations (FD module) are added on top of BSIMPD.

$$V_{bs0} = \frac{C_{Si}}{C_{Si} + C_{BOX}} \left(\phi_{hi} - \frac{qN_{ch}(1 + N_{LX}/L_{eff})}{2\epsilon_{Si}} \cdot T_{Si}^2 + V_{nonideal} + \Delta V_{DIBL} \right) + \eta_e \frac{C_{BOX}}{C_{Si} + C_{BOX}} \cdot (V_{es} - V_{FBb})$$

$$\text{where } C_{Si} = \frac{\epsilon_{Si}}{T_{Si}}, C_{BOX} = \frac{\epsilon_{OX}}{T_{BOX}}, C_{OX} = \frac{\epsilon_{OX}}{T_{OX}}$$

$$\Delta V_{DIBL} = D_{vbd0} \left(\exp \left(-D_{vbd1} \frac{L_{eff}}{2l} \right) + 2 \exp \left(-D_{vbd1} \frac{L_{eff}}{l} \right) \right) \cdot (V_{bi} - 2\Phi_B)$$

$$\eta_e = K_{1b} - K_{2b} \cdot \left(\exp \left(-D_{k2b} \frac{L_{eff}}{2l} \right) + 2 \exp \left(-D_{k2b} \frac{L_{eff}}{l} \right) \right)$$

$$\phi_{hi} = \phi_{hiON} - \frac{C_{OX}}{C_{OX} + (C_{Si}^{-1} + C_{BOX}^{-1})^{-1}} \cdot N_{OFF,FD} V_t \cdot \ln \left(1 + \exp \left(\frac{V_{th,FD} - V_{gs,eff} - V_{OFF,FD}}{N_{OFF,FD} V_t} \right) \right)$$

$$\phi_{hiON} = 2\Phi_B + V_t \ln \left(1 + \frac{V_{gs,eff,FD} (V_{gs,eff,FD} + 2K1\sqrt{2\Phi_B})}{MoinFD \cdot K1 \cdot V_t^2} \right),$$

$$V_{gs,eff,FD} = N_{OFF,FD} V_t \cdot \ln \left(1 + \exp \left(\frac{V_{gs,eff} - V_{th,FD} - V_{OFF,FD}}{N_{OFF,FD} V_t} \right) \right)$$

Here N_{ch} is the channel doping concentration. N_{LX} is the lateral non-uniform doping coefficient to account for the lateral non-uniform doping effect. V_{FBb} is the backgate flatband voltage. $V_{th,FD}$ is the threshold voltage at $V_{bs}=V_{bs0}(\phi=2\Phi_B)$. V_t is thermal voltage. $K1$ is the body effect coefficient.

If $SoiMod=1$, the lower bound of V_{bs} (SPICE solution) is set to V_{bs0} . If $SoiMod=2$, V_{bs} is pinned at V_{bs0} . Notice that there is no body node and body leakage/charge calculation in $SoiMod=2$.

The zero field body potential that determines the transistor threshold voltage, V_{bsmos} , is then calculated by

$$\begin{aligned} V_{bsmos} &= V_{bs} - \frac{C_{Si}}{2qN_{ch}T_{Si}} (V_{bs0}(T_{OX} \rightarrow \infty) - V_{bs}) \text{ if } V_{bs} \leq V_{bs0}(T_{OX} \rightarrow \infty) \\ &= V_{bs} \text{ else} \end{aligned}$$

The subsequent clamping of V_{bsmos} uses the same equation that was utilized in BSIMPD. You can download the BSIMPD manual at:

<http://www-device.eecs.Berkeley.edu/~bsimsoi>

If SoiMod=3 is specified, BSIMSOI selects the operation mode for the user based on the estimated value of V_{bs0} at $\phi_i=2\Phi_B$ (bias independent), V_{bs0t} :

- If $V_{bs0t} > V_{bs0fd}$, BSIMSOI is in the ideal FD mode (SoiMod=2).
- If $V_{bs0t} < V_{bs0pd}$, BSIMSOI is in the BSIMPD mode (SoiMod=0).

Otherwise, BSIMSOI is operated under SoiMod=1.

Notice that both V_{bs0fd} and V_{bs0pd} are model parameters.

A new model parameter, T_{OXM} , is introduced to represent the T_{OX} dependence for the model parameters K1 and K2 (compatible to BSIM3 version 3.2). A new model parameter, N_{OFF} , is introduced in $V_{gsteff,cv}$ to adjust the CV curve around the threshold (compatible to BSIM3 version 3.2).

This section covers the following topics:

- [BSIM3 SOI Version 3.2 Updates — Noise Model](#)
- [BSIM3 SOI Version 3.2 Updates — Simple and Unified Flicker Noise Models](#)
- [BSIM3 SOI Version 3.2 Updates — Thermal Noise Models](#)
- [BSIM3 SOI Version 3.2 Updates — Model Parameters](#)

BSIM3 SOI Version 3.2 Updates — Noise Model

The BSIMSOI3.2 version implements a flicker noise and thermal noise model compatible with BSIM4. In addition, the new noise model includes gate tunneling-induced shot noise and thermal noise due to gate electrode resistance. (1) Flicker noise models

Note:

For output noise parameters, see [Chapter 6, MOSFET Noise Models](#).

BSIM3 SOI Version 3.2 Updates — Simple and Unified Flicker Noise Models

BSIMSOI3.2 provides two flicker noise models. When the model selector $fnoiMod$ is set to 0, a simple flicker noise model which is convenient for hand calculation is invoked. A unified physical flicker noise model, which is the default model, is used if $fnoiMod=1$. These two modes come from BSIMSOI3.1, but the unified model has many improvements. For instance, it is now smooth over all bias regions and considers the bulk charge effect.

- $fnoiMod = 0$ (simple model)

$$S_{id}(f) = \frac{K_f I_{ds}^{af}}{C_{OX} L_{eff}^2 f^{ef}}$$

The noise density is:

- $fnoiMod = 1$ (unified model)

The physical mechanism for the flicker noise is trapping/de-trapping related charge fluctuation in oxide traps, which results in fluctuations of both mobile carrier numbers and mobility in the channel. The unified flicker noise model captures this physical process.

The noise density in inversion region is given by:

$$S_{id,inv}(f) = \frac{k_B T q^2 \mu_{eff} I_{ds}}{C_{oxe} L_{eff}^2 A_{bulk} f^{ef} \cdot 10^{10}} \left(NOIA \log \left(\frac{N_0 + N^*}{N_l + N^*} \right) + NOIB (N_0 - N_l) + \frac{NOIC}{2} (N_0^2 - N_l^2) \right) \\ + \frac{k_B T I_{ds}^2 \Delta L_{clm}}{W_{eff} L_{eff}^2 f^{ef} \cdot 10^{10}} \frac{NOIA + NOIB \cdot N_l + NOIC \cdot N_l^2}{(N_l + N^*)^2}$$

where μ_{eff} is the effective mobility at the given bias condition, and L_{eff} and W_{eff} are the effective length and width, respectively. The parameter N_0 is the charge density at the source side given by:

$$N_0 = \frac{C_{ox} V_{gsteff}}{q}$$

The parameter N_l is the charge density at the source side given by:

$$N_l = \frac{C_{ox} V_{gsteff}}{q} \left(1 - \frac{A_{bulk} V_{dseff}}{V_{gsteff} + 2V_t} \right)$$

N^* is given by:

$$N^* = K_B T \cdot \frac{(C_{ox} + C_d + CIT)}{q_2}$$

where CIT is a model parameter from DC IV and C_d is the depletion capacitance. ΔL_{clm} is the channel length reduction due to channel length modulation and given by:

$$\Delta L_{clm} = Litl \cdot \log \frac{\frac{V_{ds} V_{dseff}}{Litl} + EM}{E_{sat}} \\ E_{sat} = \frac{2V_{SAT}}{\mu_{eff}}$$

In the subthreshold region, the noise density is written as:

$$S_{id,subvt}(f) = \frac{NOIA \cdot k_B T \cdot I_{ds}^2}{W_{eff} L_{eff} f^{EF} N^{*2} \cdot 10^{10}}$$

The total flicker noise density is:

$$S_{id}(f) = \frac{S_{id,inv}(f) \times S_{id,insubvt}(f)}{S_{id,inv}(f) + S_{id,insubvt}(f)}$$

BSIM3 SOI Version 3.2 Updates — Thermal Noise Models

There are two channel thermal noise models in BSIMSOI3.2 version. One is the charge based model (default) similar to that used in BSIMSOI3.1. The other is the holistic model. These two models can be selected through the model selector `tnoiMod`.

- `tnoiMod` = 0 (charge based)

The noise current is given by:

$$\bar{i}_d^2 = \frac{4k_B T \Delta f}{L_{eff}^2} \cdot NTNOI, \\ R_{ds} + \frac{1}{\mu_{eff} |Q_{inv}|}$$

where R_{ds} is the source/drain resistance, and the parameter `NTNOI` is introduced from more accurate fitting of short-channel devices. Q_{inv} is the inversion channel charge computed from the capacitance models.

- `tnoiMod` = 1 (holistic)

In this thermal noise model, all the short-channel effects and velocity saturation effects incorporated in the IV model are automatically included, hence the name “holistic thermal noise model.” In addition, the amplification of the channel thermal noise through G_m and G_{mbs} as well as the induced- gate noise with partial correlation to the channel thermal noise are all captured in the new “noise partition” model. The noise voltage source partitioned to the source side is given by:

$$\bar{v}_d^2 = 4k_B T \cdot \theta_{moi}^2 \frac{V_{dseff} \Delta f}{I_{ds}}$$

and the noise current source put in the channel region with gate and body amplification is given by:

$$\bar{v}_d^2 = 4k_B T \frac{V_{dseff} \Delta f}{I_{ds}} [G_{ds} + \beta_{moi} \cdot (G_m + G_{mbs})]^2 \bar{v}_d^2 \cdot (G_m + G_{ds} + G_{mbs})^2$$

where:

$$\theta_{moi} = RNOIB \left[1 + TNOIB \cdot L_{eff}^2 \left(\frac{V_{gsteff}}{E_{sat} L_{eff}} \right) \right] \\ \beta_{moi} = RNOIA^2 \left[1 + TNOIA \cdot L_{eff} \left(\frac{V_{gsteff}}{E_{sat} L_{eff}} \right) \right]$$

BSIM3 SOI Version 3.2 Updates — Model Parameters

Symbols Used in Equation	Symbol used in SPICE	Description	Unit	Default
SoiMod	SoiMod	SOI model selector (instance) • SoiMod=0 BSIMPD • SoiMod=1 Unified model for PD & FD • SoiMod=2 ideal FD • SoiMod=3 Auto selection by SIMSOI	-	0
T_{OXM}	toxm	Gate oxide thickness used in extraction	μ	T_{ox}
N_{OFF}	noff	CV parameter for $V_{gesteff,cv}$	-	1.0
V_{bs0fd}	vbs0fd	Upper bound of built-in potential lowering for ideal FD operation	V	0.0
V_{bs0fd}	vbs0fd	Lowering bound of built- in potential lowering for ideal FD operation	V	0.5
fnoiMod	fnoiMod	Flicker noise model selector	-	1
tnoiMod	tnoiMod	Thermal noise model selector	-	0
NTNOI	ntnoi	Noise factor for short-channel devices for TNOIMOD=0 only	-	1.0
TNOIA	tnoia	Coefficient of channel-length dependence of total channel thermal noise	-	1.5
TNOIB	tnoib	Channel-length dependence parameter for channel thermal noise partitioning	-	3.5
RNOIA	rnoia	Thermal noise parameter		0.577
RNOIB	rnoib	Thermal noise parameter		0.37

BSIM3 SOI Version 3.1 Updates

In addition to BSIMSOI3.0, the MOSFET Level 57 model also supports the UCB BSIMSOI3.1 model version, which includes the following new features that are not available in BSIMSOI3.0.

This section covers the following topics:

- [BSIM3 SOI Version 3.1 Updates — Ideal Full-Depletion \(FD\) Modeling](#)
- [BSIM3 SOI Version 3.1 Updates — Gate Resistance Modeling](#)

- [BSIM3 SOI Version 3.1 Updates — Gate Resistance Equivalent Circuit](#)
- [BSIM3 SOI Version 3.1 Updates — Enhanced Binning Capability](#)

BSIM3 SOI Version 3.1 Updates — Ideal Full-Depletion (FD) Modeling

BSIMSOI3.0 supports the modeling of these two families of SOI MOSFETs with a `SOIMOD` switching model flag.

- `SOIMOD=0` for partially depleted devices (PD).
- `SOIMOD=1` for devices that tend to operate in a mixed mode of PD and FD.

V3.1 also provides an ideal full-depletion (FD) module (`SOIMOD=2`), not available in V3.0 to model FD SOI devices that literally exhibit no floating-body behavior. As in BSIMSOI3.0, the default `SOIMOD` value is 0 for BSIMSOI3.1.

The following physical modeling components, related to the internal SOI body node, are critical for accurately modeling PD SOI devices, but are not needed for the ideal FD module. Thus, for the ideal FD module, the PrimeSim Continuum solution ignores these components, which makes SOI MOSFET modeling much easier than for PD devices, or non-ideal FD devices.

- Source/Drain to body diode currents
- Source-Body-Drain parasitic BJT currents
- Impact ionization currents
- Gate-body direct currents
- Body-related capacitances.

BSIM3 SOI Version 3.1 Updates — Gate Resistance Modeling

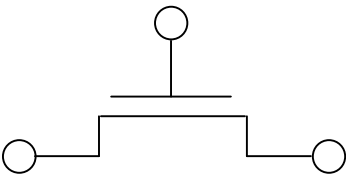
BSIMSOI3.1 uses the same gate resistance models as in the BSIM4 model with four options for various gate-resistance modeling topologies.

Name	Unit	Default	Bin	Description
NGCON	-	-	N	Number of gate contacts
RGATEMOD	-	0	N	Gate resistance model selector • RGATEMOD=0: No gate resistance • RGATEMOD=1: Constant gate resistance • RGATEMOD=2: Rii model with variable resistance • RGATEMOD=3: Rii model with two nodes
RSHG	Ohm/Sq	0.1	N	Gate electrode sheet resistance

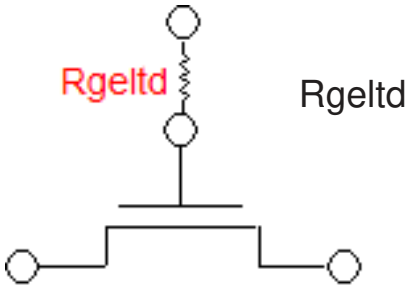
Name	Unit	Default	Bin	Description
XGL	m	0	N	Offset of the gate length due to variations in patterning
XGW	m	0	N	Distance from the gate contact to the channel edge in the W direction
XRCRG1	-	12	Y	Parameter for distributed channel-resistance effect for intrinsic input resistance
XRCRG2	-	1	Y	Parameter to account for the excess channel diffusion resistance for intrinsic input resistance

BSIM3 SOI Version 3.1 Updates — Gate Resistance Equivalent Circuit

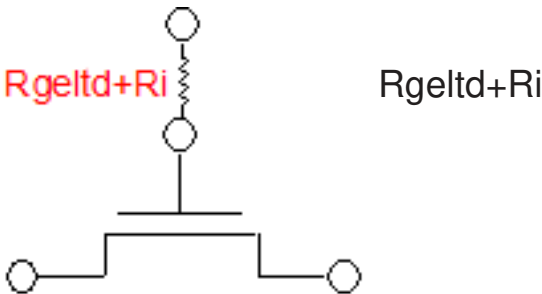
RGATEMOD=0: No gate resistance (default)



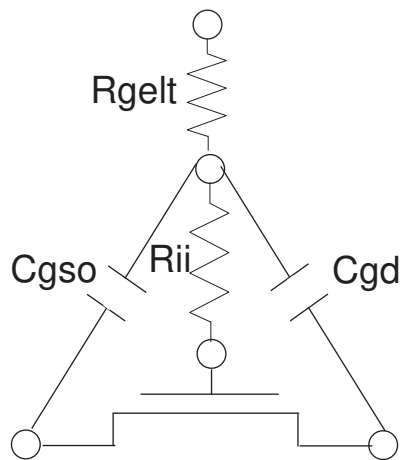
RGATEMOD=1: Constant gate resistance



RGATEMOD=2: variable resistance with Rii model



RGATEMOD=3: Rii model with two nodes



Rgelt: (Poly) gate electrode resistance, bias independent.

Rii: Intrinsic input gate resistance, reflected to the gate from the intrinsic channel region. It is bias-dependent and a first-order non-quasi static model for RF and rapid transient MOSFET operations.

BSIM3 SOI Version 3.1 Updates — Enhanced Binning Capability

Model parameters for the following components are now binnable for better accuracy and scalability:

- Junction depth
- Gate-tunneling current
- Temperature dependence of threshold voltage, mobility, saturation velocity, parasitic resistance, and diode currents.

Bug Fixes:

The BSIMSOI3.1 from UC Berkeley fixes the following bugs in the BSIMSOI3.0 model:

- The model now takes NSEG into account when calculating the gate-channel tunneling current.
- The G_{MIN} connecting gate and drain is now multiplied by $1e-6$ to reduce false leakage current.
- A swapping error in the source/drain overlap capacitance stamping.
- The bulk charge effect coefficient (A_{bulkCV}) is now corrected for Q_{inv} and its derivatives in CAPMOD=2.

BSIM3 SOI Model — PD Version Updates

- BSIM PD version 2.2 enhances the model flexibility and accuracy from PD version 2.0, and the following are its major features.
 - Gate-body tunneling (substrate current) enhances the model accuracy.
 - Body contact resistance improves the modeling accuracy.
 - Binning enhances the model flexibility.
- BSIM PD version 2.21 updates the PD version 2.2 for bug fixes and S/D swapping for the gate current components.
- BSIM PD version 2.22 updates the 2.21 version for bug fixes and enhancements. The major features include:
 - FRBODY instance parameter
 - Improved temperature dependence of the gate direct tunneling model
 - Two new model parameters, `VEVB` and `VECB`
 - UC Berkeley code no longer supports the `NECB` and `NEVB` model parameters. Version 2.22 accepts these parameters for backwards compatibility, but they have no effect.
- `.OPTION LIST` prints an element summary for the MOSFET Level=57 model.

Element	Description
<code>bjtoff</code>	BJT on/off flag (Turn off BJT if equal to 1)
<code>rth0</code>	Thermal Resistance per unit width
<code>cth0</code>	Thermal Capacitance per unit width
<code>nrb</code>	Number of squares for the body series resistance
<code>frbody</code>	Coefficient of the distributed body resistance effects
<code>nbc</code>	Number of body contact isolation edge
<code>nseg</code>	Number of segments for channel width partitioning
<code>pdbcp</code>	Parasitic perimeter length for the body contact at the drain side
<code>psbcp</code>	Parasitic perimeter length for the body contact at the source side
<code>agbcp</code>	Parasitic gate-to-body overlap area for the body contact

Element	Description
aebcp	Parasitic body-to-substrate overlap area for the body contact
vbsusr	Optional initial value of Vbs, which you specify for transient analysis

- BSIM PD version 2.23 includes several bug fixes and enhancements from version 2.2.
 - Adds geometric dependency in CV delta L and delta W .
 - Fixes a gate-body-tunneling residue problem in the low-bias region.
 - Provides an additional parameter (dtox $_{cv}$) in CAPMOD=3 for flexibility
 - Other bug fixes

Using BSIM3 SOI PD Model

To use BSIM3-SOI PD versions 2.0, 2.2, 2.21, or 2.22 in simulation, apply the VERSION model parameter. For example:

- Invokes PD2.0 if VERSION=2.0
- Invokes PD2.2 and PD2.21 if VERSION=2.2
- Invokes PD2.22 if VERSION=2.22
- Invokes PD2.23 if VERSION=2.23.

For gate-body tunneling, set the IGMOD model parameter to 1.

Example:

```
mckt drain gate source bulk nch L=10e-6 W=10e-6
.model nch nmos Level=57 igmod=1 version=2.2
+ tnom=27 tox=4.5e-09 tsi=.0000001 tbox=8e-08
+ mobmod=0 capmod=2 shmod=0 paramchk=0
+ wint=0 lint=-2e-08 vth0=.42 k1=.49
+ k2=.1 k3=0 k3b=2.2 nlx=2e-7
+ dvt0=10 dvt1=.55 dvt2=-1.4 dvt0w=0
+ dvt1w=0 dvt2w=0 nch=4.7e+17 nsub=-1e+15
+ ngate=1e+20 agidl=1E-15 bgidl=1E9 ngidl=1.1
+ ndiode=1.13 ntun=14.0 nrecf0=2.5 nreocr=4
+ vrec0=1.2 ntrecf=.1 ntrecr=.2 isbjt=1E-4
+ isdif=1E-5 istun=2E-5 isrec=4E-2 xbjt=.9
+ xdif=.9 xrec=.9 xtun=0.01 ahli=1e-9
+ lbjt0=0.2e-6 ln=2e-6 nbjt=.8 ndif=-1
+ aely=1e8 vabjt=0 u0=352 ua=1.3e-11
+ ub=1.7e-18 uc=-4e-10 w0=1.16e-06 ags=.25
+ A1=0 A2=1 b0=.01 b1=10
+ rdsw=0 prwg=0 prwb=-.2 wr=1
```

```
+ rbody=1E0    rbsh=0.0    a0=1.4    keta=0.1
+ ketas=0.2    vsat=135000    dwg=0    dwb=0
+ alpha0=1e-8    beta0=0    beta1=0.05    beta2=0.07
+ vdsatii0=.8    esatii=1e7    voff=-.14    nfactor=.7
+ cdsc=.00002    cdsch=0    cdschd=0    cit=0
+ pclm=2.9    pvag=12    pdiblc1=.18    pdiblc2=.004
+ pdiblc3=-.234    drout=.2    delta=.01    eta0=.05
+ etab=0    dsub=.2    rth0=.005    clc=.0000001
+ cle=.6    cf=1e-20    ckappa=.6    cgdl=1e-20
+ cgsl=1e-20    kt1=-.3    kt1l=0    kt2=.022
+ ute=-1.5    ual=4.31e-09    ub1=-7.61e-18    uc1=-5.6e-11
+ prt=760    at=22400    cgso=1e-10    cgdo=1e-10
+ cjswg=1e-12    tt=3e-10    asd=0.3    csdesw=1e-12
+ tcjswg=1e-4    mjswg=.5    pbswg=1
```

BSIM3 SOI Model — Output Templates

For a list of output template parameters in the MOSFET models, and which parameters this model supports, see [MOSFET Output Templates](#).

LEVEL 58: UFSOI MOSFET Model

University of Florida SOI includes non-fully depleted (NFD) and fully depleted (FD) SOI models (a dynamic mode must not operate between NFD and FD) that separately describe two main types of SOI devices. The UFSOI version 4.5F model is Level 58 in the Synopsys MOSFET models.

Some processes use an external contact to the body of the device. The Synopsys MOSFET model supports only a 4-terminal device, which includes drain, front gate, source, and back gate (or substrate). The additional body contact is currently not supported so it floats.

The effects of parasitic diodes in SOI are different from those in the bulk MOSFET. The SOI model does not include the MOSFET *junction* model (ACM), developed for bulk MOSFETs.

The general syntax for MOSFET Level 58 in a netlist is:

```
Mxxx nd ngf ns [ngb] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val] [NRD=val]
+ [NRS=val] [NRB=val] [RTH=val] [CTH=val] [off]
+ [IC=Vds,Vgfs,VGbs]
```

In the preceding syntax, square brackets denote optional parameters. The arguments are identical to those for the BSIM3-SOI model, but the thermal resistance and capacitance have different names.

Table 1 Thermal Resistance and Capacitance Names

Name	Description
CTH	Thermal capacitance , unit in $W \cdot s \cdot K^{-1}$, default is 0.0.
RTH	Thermal resistance, unit in $K \cdot W^{-1}$, default is 0.0.

Note:

Following are some important notes:

- The default value for channel length (L) and width (W) is 1.0e-6.
- The present version supports only four nodes (only floating-body devices). AB is typically zero; specify it accordingly.
- If you activate the self-heating option (on the model line), RTH and CTH define the thermal impedance of the device. Typical values are 5e3 (for RTH) and 1e-12 (for CTH), but these can vary widely from one device to another.

- For $M > 1$, you must specify W , AD , AS , NRD , NRS , NRB , PDJ , PSJ , RTH , and CTH per gate finger.
- The initial condition (IC) is in the following order: V_{ds} drain voltage, V_{gfs} front gate voltage, and V_{bgs} back gate voltage.

This sections covers the following topics:

- [UFSOI Model — FDSOI Parameters](#)
- [UFSOI Model — NFDSOI Parameters](#)
- [UFSOI MOSFET Model Template](#)

UFSOI Model — FDSOI Parameters

The following tables describe Level 58 model parameters for the fully depleted (FD) SOI, including parameter names, descriptions, units, defaults, and typical notes.

This section covers the following topics:

- [FDSOI Flag Parameters](#)
- [FDSOI Structural Parameters](#)
- [FDSOI Electrical Parameters](#)
- [FDSOI Optional Parameters](#)

FDSOI Flag Parameters

Parameter	Unit	Default	Typical Value	Description
Level	-	-	-	Level 57 for UFSOI
NFDMOD	-	0	0	Model selector (0: FD)
BJT	-	1	1	Parasitic bipolar flag (0: off; 1: on)
SELFT	-	0	0	Self-heating flag: • 0: no self-heating • 1: approximate model • 2: full self-heating
TPG	-	1	-	Type of gate polysilicon: • +1: opposite to body • -1: same as body
TPS	-	-1	-	Type of substrate: • +1: opposite to body • -1: same as body

FDSOI Structural Parameters

Parameter	Unit	Default	Typical Value	Description
DL	m	0.0	(0.05-0.15)x10 ⁻⁶	Channel-length reduction
DW	m	0.0	(0.1-0.5)x10 ⁻⁶	Channel-width reduction
LLDD	m	0.0	(0.05-0.2)x10 ⁻⁶	LDD/LDS region length (0 for no LDD)
NBODY	cm ⁻³	5.0e16	1017-1018	Film (body) doping density
NDS	cm ⁻³	5.0e19	1019-1020	Source/drain doping density
NGATE	cm ⁻³	0.0	1019-1020	Poly-gate doping density (0 for no poly-gate depletion)
NLDD	cm ⁻³	5.0e19	1x1019	LDD/LDS doping density (>1e19: LDD/LDS treated as D/S extensions)
NSUB	cm ⁻³	1.0e15	1015-1017	Substrate doping density
TB	m	0.1e-6	(30-100) x10 ⁻⁹	Film (body) thickness
TOXB	m	0.5e-6	(80-400)x10 ⁻⁹	Back-gate oxide thickness
TOXF	m	1.0e-8	(3-8)x10 ⁻⁹	Front-gate oxide thickness

FDSOI Electrical Parameters

Parameter	Unit	Default	Typical Value	Description
ALPHA	cm ⁻¹	0.0	2.45x106	Impact-ionization coefficient (0 for no impact ionization)
BETA	V·cm ⁻¹	0.0	1.92x106	Impact-ionization exponential factor (0 for no impact ionization)
BGIDL	V·cm ⁻¹	0.0	(4-8)x109	Exponential factor for gate-induced drain leakage (0 for no GIDL)
CGFBO	F·m ⁻¹	0.0	0.0	Gate-body overlap capacitance
CGFDO	F·m ⁻¹	0.0	1x10 ⁻¹⁰	Gate-drain overlap capacitance

Parameter	Unit	Default	Typical Value	Description
CGFSO	F·m-1	0.0	1x10-10	Gate-source overlap capacitance
FNA	-	1.0	0.5-2	Flicker noise exponent
FNK	F·A	0.0	0-10-2	Flicker noise coefficient
GAMMA	-	0.3	0.3-1.0	BOX fringing field weighting factor
JRO	A·m ⁻² -1	1.0e-10	10-11-10-9	Body-source/drain junction recombination current coefficient
KAPPA	-	0.5	0.5-1.0	BOX fringing field weighting factor
LDIFF	m	1.0e-7	(0.1-0.5)x10-6	Effective diffusion length in source/drain
M	-	2.0	1.0-2.0	Body-source/drain junction recombination ideality factor
NQFB	cm ⁻²	0.0	~ 1011	Back oxide fixed charge (normalized)
NQFF	cm ⁻²	0.0	~ 1010	Front oxide fixed charge (normalized)
NQFSW	cm ⁻²	0.0	~ ±1012	Effective sidewall fixed charge (0 for no narrow-width effect)
NSB	cm ⁻² ·eV-1	0.0	~ 1011	Back surface state density
NSF	cm ⁻² ·eV-1	0.0	~1010	Front surface state density
QM	-	0.0 -0.5		Energy quantization parameter (0 for no quantization)
RD	ohm·m	0.0	200-1000	Specific drain parasitic resistance
RHOB	ohm/sq.	0.0	30x103	Body sheet resistance
RS	ohm·m	0.0	200-1000	Specific source parasitic resistance

Parameter	Unit	Default	Typical Value	Description
SEFF	cm·s-1	1.0e5	(0.5-5)x10 ⁵	Effective recombination velocity in source/drain
THETA	cm·V-1	7.0e2	(0.1-3)x10 ⁻⁶	Mobility degradation coefficient
UO	cm ² ·V ⁻¹ ·s-1	7.0e2	200-700 (nMOS) 70-400 (pMOS)	Low-field mobility
VSAT	cm·s-1	1.0e-6	(0.5-1)x10	Carrier saturated drift velocity

FDSOI Optional Parameters

Parameter	Unit	Default	Typical Value	Description
BFACT	-	0.3	0.1-0.5	V _{DS} averaging factor for mobility degradation
FVBJT	-	0.0	0-1	BJT current directional partitioning factor (0 for lateral 1D flow)
RHOSD	ohm/sq.	0.0	50	Source/drain sheet resistance
TAUO	s	calc.	10-7-10-5	Carrier lifetime in lightly doped regions
VFBB	V	calc.	-	Back-gate flatband voltage
VFBF	V	calc.	-1 (nMOS) 1 (pMOS)	Front-gate flatband voltage
WKB	V	calc.	-	Back-gate work function difference
WKF	V	calc.	~ VFBF	Front-gate work function difference

UFSOI Model — NFDSOI Parameters

The following tables describe the Level 58 model parameters for non fully depleted (NFD) SOI, including parameter names, descriptions, units, defaults, and typical notes.

This section covers the following topics:

- [NFDSOI Flag Parameters](#)
- [NFDSOI Structural Parameters](#)

- [NFDSOI Electrical Parameters](#)
- [NFDSOI Optional Parameters](#)

NFDSOI Flag Parameters

Parameter	Unit	Default	Typical Value	Description
Level	-	-	-	Level 57 for UFSOI
BJT	-	1	1	Parasitic bipolar flag: • 0: off • 1: on
NFDMOD	-	0	-	Model selector (1: NFD)
SELFT	-	0	0	Self-heating flag: • 0: no self-heating • 1: approximate model • 2: full self-heating
TPG	-	1	1	Type of gate polysilicon • +1: opposite to body • -1: same as body
TPS	-	-1	-1	Type of substrate • +1: opposite to body • -1: same as body)

NFDSOI Structural Parameters

Parameter	Unit	Default	Typical Value	Description
DL	m	0.0	(0.05-0.15)x10 ⁻⁶	Channel-length reduction
DW	m	0.0	(0.1-0.5)x10 ⁻⁶	Channel-width reduction
LLDD	m	0.0	(0.05-0.2)x10 ⁻⁶	LDD/LDS region length (0 for no LDD)
LRSCE	m	0.0	~0.1x10 ⁻⁶	Characteristic length for reverse short-channel effect (0 for no RSCE)
NBH	cm ⁻³	5.0e17	1019-1020	Halo doping density
NBL	cm ⁻³	5.0e16	1017-1018	Low body doping density
NDS	cm ⁻³	5.0e19	1019-1020	Source/drain doping density

Parameter	Unit	Default	Typical Value	Description
NGATE	cm-3	0.0	1019-1020	Poly-gate doping density (0 for no poly-gate depletion)
NHALO	cm-3	-	~1018	Halo doping density
NLDD	cm-3	5.0e19	1x1019	LDD/LDS doping density (>1e19: LDD/LDS treated as D/S extensions)
NSUB	cm-3	1.0e15	1015-1017	Substrate doping density
TB	m	0.1e-6	(30-100)x10-9	Film (body) thickness
TF	m	0.2e-6	(3-8)x10-9	Silicon film thickness
THALO	m	0.0	-	Halo thickness (0 for no halo)
TOXB	m	0.5e-6	(80-400)x10-9	Back-gate oxide thickness
TOXF	m	1.0e-8	(3-8)x10-9	Front-gate oxide thickness

NFDSOI Electrical Parameters

Parameter	Unit	Default	Typical Value	Description
ALPHA	cm-1	0.0	2.45x106	Impact-ionization coefficient (0 for no impact ionization)
BETA	V·cm-1	0.0	1.92x106	Impact-ionization exponential factor (0 for no impact ionization)
BGIDL	V·cm-1	0.0	(4-8)x109	Exponential factor for gate-induced drain leakage (0 for no GIDL)
CGFBO	F·m-1	0.0	0.0	Gate-body overlap capacitance
CGFDO	F·m-1	0.0	1x10-10	Gate-drain overlap capacitance
CGFSO	F·m-1	0.0	1x10-10	Gate-source overlap capacitance
FNA	-	1.0	0.5-2	Flicker noise exponent
FNK	F·A	0.0	0-10-2	Flicker noise coefficient

Parameter	Unit	Default	Typical Value	Description
JRO	A·m ⁻¹	1.0e-10	10 ⁻¹¹ -10 ⁻⁹	Body-source/drain junction recombination current coefficient
LDIFF	m	1.0e-7	(0.1-0.5)x10 ⁻⁶	Effective diffusion length in source/drain
M	-	2.0	1.0-2.0	Body-source/drain junction recombination ideality factor
NQFB	cm ⁻²	0.0	~ 10 ¹¹	Back oxide fixed charge (normalized)
NQFF	cm ⁻²	0.0	~ 10 ¹⁰	Front oxide fixed charge (normalized)
NQFSW	cm ⁻²	0.0	~ ±10 ¹²	Effective sidewall fixed charge (0 for no narrow-width effect)
NTR	cm ⁻³	0.0	10 ¹⁴ -10 ¹⁵	Effective trap density for trap-assisted junction tunneling (0 for no tunneling)
QM	-	0.0 -0.5		Energy quantization parameter (0 for no quantization)
RD	ohm·m	0.0	200-1000	Specific drain parasitic resistance
RHOB	ohm/sq.	0.0	30x10 ³	Body sheet resistance
RS	ohm·m	0.0	200-1000	Specific source parasitic resistance
SEFF	cm·s ⁻¹	1.0e5	(0.5-5)x10 ⁵	Effective recombination velocity in source/drain
THETA	cm·V ⁻¹	7.0e2	(0.1-3)x10 ⁻⁶	Mobility degradation coefficient
UO	cm ² ·V ⁻¹ ·s ⁻¹	7.0e2	200-700 (nMOS) 70-400 (pMOS)	Low-field mobility
VSAT	cm·s ⁻¹	1.0e-6	(0.5-1)x10	Carrier saturated drift velocity

NFDSOI Optional Parameters

Parameter	Unit	Default	Typical Value	Description
VFBF	V	calc.	-1 (nMOS) 1 (pMOS)	Front-gate flatband voltage
VFBB	V	calc.	-	Back-gate flatband voltage
WKF	V	calc.	~ VFBF	Front-gate work function difference
WKB	V	calc.	-	Back-gate work function difference
TAUO	s	calc.	10-7-10-5	Carrier lifetime in lightly doped regions
BFACT	-	0.3	0.1-0.5	V_{DS} -averaging factor for mobility degradation
FVBJT	-	0.0	0-1	BJT current directional partitioning factor (0 for lateral 1D flow)
RHOSD	ohm/sq.	0.0	50	Source/drain sheet resistance

Note:

Following are some important notes:

- The model line must include LEVEL=58 and NFDMOD=0 for FD, or NFDMOD=1 for NFD devices.
- Specifying VFBF turns off the narrow-width effect defined by NQFSW (which can be positive or negative) and the reverse short-channel effect defined by LRSCE (and NBH or NHALO if specified); the latter effect is also turned off if you specify WKF.
- For floating-body devices, CGFBO is small; you should set it to 0.
- JRO and SEFF influence the gain of the BJT, but LDIFF affects only bipolar charge storage in the source/drain. If you specify THALO, then NBH and NHALO also influence the BJT gain.
- Loosely correlate the TAUO value with JRO in accord with basic pn-junction recombination/generation properties. Its default value is based on JRO, which is appropriate for short L; for long L, body generation predominates over that in the junctions so specify TAUO.

- The non-local impact-ionization model is physical so do not arbitrarily vary its parameters.
- The `LDD` option intensifies the model so set `LLDD` to 0 for large-scale circuit simulation, and add the unbiased `LDD` resistance to `RD`; this simplification stops if you specify `NLDD > 1e19`.

UFSOI MOSFET Model Template

For a list of output template parameters in the MOSFET models, and which parameters this model supports, see [MOSFET Output Templates](#).

LEVEL 59: BSIM3 SOI FD Model

The UC Berkeley SOI (BSIM3-SOI) Fully Depleted (FD) model is Level 59 in the Synopsys MOSFET models. For a description of this model, see the *BSIM3SOI FD2.1 MOSFET MODEL User Manual*, at:

<http://www-device.eecs.berkeley.edu/~bsim3soi>

The following sections discuss these topics:

- [BSIM3 SOI FD Model — General Syntax](#)
- [BSIM3 SOI FD Model Parameters](#)
- [BSIM3 SOI FD Model — Template Output](#)

BSIM3 SOI FD Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] mname [L=val]
+ [W=val] [M=val] [AD=val] [AS=val] [PD=val] [PS=val]
+ [NRD=val] [NRS=val] [NRB=val] [RTH0=val] [CTH0=val]
+ [off] [BJToff=val] [IC=Vds, Vgs, Vbs, Ves, Vps]
```

Parameter	Description
Mxxx	SOI MOSFET element name. Must begin with M, followed by up to 1023 alphanumeric characters.
nd	Drain terminal node name or number.
ng	Front gate node name or number.
ns	Source terminal node name or number.
ne	Back gate (or substrate) node name or number.
np	Optional external body contact node name or number.
mname	MOSFET model name reference.
L	SOI MOSFET channel length in meters. This parameter overrides DEFL in an OPTIONS statement. Default=DEFL with a maximum of 0.1m.
W	MOSFET channel width in meters. This parameter overrides DEFW in an OPTIONS statement. Default=DEFW.
M	Multiplier to simulate multiple SOI MOSFETs in parallel. The M setting affects all channel widths, diode leakages, capacitances, and resistances. Default=1.

Parameter	Description
AD	Drain diffusion area. Overrides DEFAD in the OPTIONS statement. Default=DEFAD.
AS	Source diffusion area. Overrides DEFAS in the OPTIONS statement. Default=DEFAS.
PD	Perimeter of the drain junction, including the channel edge. Overrides DEFPD in the OPTIONS statement.
PS	Perimeter of the source junction, including the channel edge. Overrides DEFPS in the OPTIONS statement.
NRD	Number of squares of drain diffusion for the drain series resistance. Overrides DEFNRD in the OPTIONS statement.
NRS	Number of squares of source diffusion for the source series resistance. Overrides DEFNRS in the OPTIONS statement.
NRB	Number of squares for the body series resistance.
RTH0	Thermal resistance per unit width: • If you do not specify RTH0, simulation extracts it from the model card. • If you specify RTH0, it overrides RTH0 in the model card.
CTH0	Thermal capacitance per unit width • If you do not specify CTH0, simulation extracts it from the model card. • If you specify CTH0, it overrides CTH0 in the model card.
OFF	Sets the initial condition to OFF for this element in DC analysis.
BJTOFF	Turns off BJT if equal to 1.
IC	Initial guess in the order (drain, front gate, internal body, back gate, external voltage). (ignores Vps in a 4-terminal device) Simulation uses these settings if you specify UIC in the .TRAN statement. The .IC statement overrides them.

BSIM3 SOI FD Model Parameters

This section covers the following topics:

- [BSIM3 SOI FD Model — Control Parameters](#)
- [BSIM3 SOI FD Model — Process Parameters](#)
- [BSIM3 SOI FD Model — DC Parameters](#)
- [BSIM3 SOI FD Model — AC and Capacitance Parameters](#)
- [BSIM3 SOI FD Model — Temperature Parameters](#)

BSIM3 SOI FD Model — Control Parameters

Parameter	Unit	Default	Description
CAPMOD	-	2	Flag for the short channel capacitance model
Level	-	-	Level 59 for BSIM3SOI
MOBMOD	-	1	Mobility model selector
NOIMOD	-	1	Flag for the Noise model
SHMOD	-	0	Flag for self-heating: • 0=no self-heating • 1=self-heating

BSIM3 SOI FD Model — Process Parameters

Parameter	Unit	Default	Description
NCH	$1/\text{cm}^3$	$1.7\text{e}17$	Channel doping concentration
NGATE	$1/\text{cm}^3$	0	Poly gate doping concentration
NSUB	$1/\text{cm}^3$	$6.0\text{e}16$	Substrate doping concentration
TBOX	m	$3.0\text{e}-7$	Buried oxide thickness
TOX	m	$1.0\text{e}-8$	Gate oxide thickness
TSI	m	$1.0\text{e}-17$	Silicon film thickness

BSIM3 SOI FD Model — DC Parameters

Parameter	Unit	Default	Description
A0	-	1.0	Bulk charge effect coefficient for the channel length
A1	$1/V$	0.0	First non-saturation effect parameter
A2	-	1.0	Second non-saturation effect parameter
ABP	-	1.0	Coefficient of A_{beff} dependency on V_{gst}
ADICE0	-	1	DICE bulk charge factor
AGIDL	$1/W$	0.0	GIDL constant

Parameter	Unit	Default	Description
AGS	1/V	0.0	Gate bias coefficient of A_{bulk}
All	1/V	0.0	First V_{dsat} parameter for the L_{eff} dependence
ALPHA0	m/V	0.0	First parameter of the impact ionization current
ALPHA1	1/V	1.0	Second parameter of the impact ionization current
B0	m	0.0	Bulk charge effect coefficient for the channel width
B1	m	0.0	Width offset for the bulk charge effect
BGIDL	V/m	0.0	GIDL exponential coefficient
BII	m/V	0.0	Second V_{dsat} parameter for the L_{eff} dependence
CDSC	F/m ²	2.4e-4	Drain/source to the channel coupling capacitance
CDSCB	F/m ²	0	Body-bias sensitivity of cdsc
CDSCD	F/m ²	0	Drain-bias sensitivity of cdsc
CII	-	0.0	First V_{dsat} parameter for the V_{ds} dependence
CIT	F/m ²	0.0	Interface trap capacitance
DELP	V	0.02	Constant for limiting V_{bseff} to the surface potential
DELTA	-	0.01	Effective V_{ds} parameter
DII	V	-1.0	Second V_{dsat} parameter for the V_{ds} dependence
DROUT	-	0.56	L dependence coefficient of the DIBL correction parameter in R_{out}
DSUB	-	0.56	DIBL coefficient exponent
DVBD0	V	0	First coefficient of the V_{ds} 0 dependency on L_{eff}
DVBD1	V	0	Second coefficient of the V_{ds} 0 dependency on L_{eff}
DVT0	-	2.2	First coefficient of the short-channel effect on V_{th}
DVT0W	-	0	First coefficient of the narrow-width effect on V_{th} for a small channel length
DVT1	-	0.53	Second coefficient of the short-channel effect on V_{th}
DVT1W	-	5.3e6	Second coefficient of the narrow-width effect on V_{th} for a small channel length

Parameter	Unit	Default	Description
DVT2	1/V	-0.032	Body-bias coefficient of the short-channel effect on V_{th}
DVT2W	1/V	-0.032	Body-bias coefficient of the narrow width effect on V_{th} for small channel length
DWB	$m/V^{1/2}$	0.0	Coefficient of the substrate body bias dependence for W_{eff}
DWG	m/V	0.0	Coefficient of the gate dependence for W_{eff}
EDL	m	2e-6	Electron diffusion length
ETA0	-	0.08	DIBL coefficient in the subthreshold region
ETAB	1/V	-0.07	Body-bias coefficient for the DIBL effect in the subthreshold region
ISBJT	A/m^2	1.0e-6	BJT injection saturation current
ISDIF	A/m^2	0	Body to source/drain injection saturation current
ISREC	A/m^2	1.0e-5	Recombination in the depletion saturation current
ISTUN	A/m^2	0.0	Reverse tunneling saturation current
K1	$V^{1/2}$	0.6	Coefficient for the first-order body effect
K2	-	0	Coefficient for the second-order body effect
K3	-	0	Narrow coefficient
K3B	1/V	0	Body-effect coefficient of k3
KB1	-	1	Coefficient of the V_{bs} 0 dependency on V_{gbs}
KB3	-	1	Coefficient of the V_{bs} 0 dependency on V_{gs} at subthreshold region
KBJT1	m/V	0	Parasitic bipolar Early effect coefficient
KETA	m	-0.6	Body-bias coefficient of the bulk charge effect
LINT	m	0.0	Length-offset fitting parameter from I-V without bias
MXC	-	-0.9	Fitting parameter for calculating A_{beff}
NDIODE	-	1.0	Diode non-ideality factor
NFACTOR	-	1	Subthreshold swing factor

Parameter	Unit	Default	Description
NGIDL	V	1.2	GIDL V_{ds} enhancement coefficient
NLX	m	1.74e-7	Lateral non-uniform doping parameter
NTUN	-	10.0	Reverse tunneling non-ideality factor
PCLM	-	1.3	Channel length modulation parameter
PDIBL1	-	0.39	First correction parameter for the DIBL effect of the output resistance
PDIBL2	-	0.0086	Second correction parameter for the DIBL effect of the output resistance
PRWB	$1/V^1$	0	Body effect coefficient of R_{dsw}
PRWG	$1/V^{1/2}$	0	Gate bias effect coefficient of R_{dsw}
PVAG		0.0	Gate dependence of the Early voltage
RBODY	ohm/m^2	0.0	Intrinsic body contact sheet resistance
RBSH	ohm/m^2	0.0	Extrinsic body contact sheet resistance
RDSW	$\Omega * \mu\text{m}^{wr}$	100	Parasitic resistance per unit width
RSH	ohm/square	0.0	Source/drain sheet resistance in ohm per square
U0	$\text{cm}^2/(\text{V}\cdot\text{s})$	NMOS-670 PMOS-250	Mobility at Temp=Tnom
UA	m/V	2.25e-9	First-order coefficient for mobility degradation
UB	$(\text{m}/\text{V})^2$	5.87e-19	Second-order coefficient for mobility degradation
UC	1/V	-0.0465	Body-effect coefficient for mobility degradation
VBSA	V	0	Transition body voltage offset
VOFF	v	-0.08	Offset voltage in the subthreshold region for large W and L values
VSAT	m/sec	8e4	Saturation velocity at Temp=Tnom
VTH0	v	NMOS 0.7 PMOS -0.7	Threshold voltage @ $V_{bs}=0$ for a long, wide device
W0	m	0	Narrow width parameter
WINT	m	0.0	Width offset fitting parameter from I-V without bias

Parameter	Unit	Default	Description
WR	-	1	Width offset from Weff for calculating R_{ds}

BSIM3 SOI FD Model — AC and Capacitance Parameters

Parameter	Unit	Default	Description
ASD	V	0.3	Smoothing parameter for the source/drain bottom diffusion
CF	F/m	cal.	Gate to source/drain fringing field capacitance
CGDL	F/m	0.0	Lightly-doped drain-gate region overlap capacitance
CGDO	F/m	calculated	Non-LDD region drain-gate overlap capacitance per channel length
CGEO	F/m	0.0	Gate-substrate overlap capacitance per channel length
CGSL	F/m	0.0	Lightly-doped source-gate region overlap capacitance
CGSO	F/m	calculated	Non-LDD region source-gate overlap capacitance per channel length
CJSWG	F/m ²	1.e-10	Source/drain (gate side) sidewall junction capacitance per unit width (normalized to 100nm Tsi)
CKAPPA	F/m	0.6	Coefficient for lightly-doped region overlap capacitance fringing field capacitance
CLC	m	0.1e-7	Constant term for the short channel model
CLE	-	0.0	Exponential term for the short channel model
CSDSW	F/m	0.0	Source/drain sidewall fringing capacitance per unit length
CSDMIN	V	cal.	Source/drain bottom diffusion minimum capacitance
DLC	m	lint	Length offset fitting parameter for the gate charge
DWC	m	wint	Width offset fitting parameter from C-V
MJSWG	V	0.5	Source/drain (gate side) sidewall junction capacitance grading coefficient

Parameter	Unit	Default	Description
PBSWG	V	0.7	Built-in potential for the source/drain (gate side) sidewall junction capacitance
TT	second	1ps	Diffusion capacitance transit time coefficient
VSDFB	V	cal.	Flatband voltage for the source/drain bottom diffusion capacitance
VSDTH	V	cal.	Threshold voltage for the source/drain bottom diffusion capacitance

BSIM3 SOI FD Model — Temperature Parameters

Parameter	Unit	Default	Description
XPART	-	0	Charge partitioning rate flag
AT	m/sec	3.3e4	Temperature coefficient for U_a
CTH0	(W*S)/m°C	0	Normalized thermal capacity
KT1	V	-0.11	Temperature coefficient for the threshold voltage
KT2	-	0.022	Body-bias coefficient for the temperature effect of the threshold voltage
KTIL	V*m	0	Channel length dependence of the temperature coefficient for the threshold voltage
PRT	$\Omega - \mu m$	0	Temperature coefficient for R_{dsw}
RTH0	m°C/W	0	Normalized thermal resistance
TNOM	°C	25	Temperature at which simulation expects parameters
UA1	m/V	4.31e-9	Temperature coefficient for U_a
UB1	(m/V) ²	-7.61e-18	Temperature coefficient for U_b
UC1	1/V	-0.056	Temperature coefficient for U_c
UTE	-	-1.5	Mobility temperature exponent
XBJT	-	1	Power dependence of j_{bjt} on the temperature
XDIF	-	XBJT	Power dependence of j_{dif} on the temperature
XREC	-	1	Power dependence of j_{rec} on the temperature

Parameter	Unit	Default	Description
XTUN	-	0	Power dependence of j_{tun} on the temperature

Note:

BSIM3 SOI FD refers the substrate to the silicon below the buried oxide, not to the well region in BSIM3. It calculates the backgate flatband voltage (V_{fbb}) and the parameters related to the bottom capacitance of the source/drain diffusion (V_{sdth} , V_{sdfb} , C_{sdmin}).

- Positive n_{sub} means the same type of doping as the body.
- Negative n_{sub} means opposite type of doping.

BSIM3 SOI FD Model — Template Output

For a list of output template parameters in the MOSFET models, and which parameters this model supports, see [MOSFET Output Templates](#).

LEVEL 60: BSIM3 SOI DD Model

The UC Berkeley SOI model (BSIM3 SOI) supports Fully Depleted (FD), Partially Depleted (PD), and Dynamically Depleted (DD) SOI devices. BSIM3 DD V2.2 for DD SOI devices is Level 60 in the Synopsys MOSFET models. For a description of this model, see the *BSIM3DD2.1 MOSFET MODEL User's Manual*, at <http://www-device.eecs.berkeley.edu/~bsim3>

BSIM3 DD V2.1 includes many advanced concepts for dynamic and continuous transition between PD and FD operation. These concepts are collectively named Dynamic Depletion.

The following sections discuss these topics:

- [BSIM3 SOI DD Model — Features](#)
- [BSIM3 SOI DD Model — General Syntax](#)
- [BSIM3 SOI DD Model Parameters](#)

BSIM3 SOI DD Model — Features

- Simulation applies dynamic depletion to both I-V and C-V. Tbox and Tsi continuously scale the charge and drain current.
- Supports external body bias and backgate bias; a total of 6 nodes.
- Real floating body simulation in both I-V and C-V. Diode and C-V formulation properly bind the body potential.
- Improved self-heating.
- Improved impact ionization current model.
- Various diode leakage components and parasitic bipolar current.
- Depletion charge model (EBCI) for better accuracy in predicting capacitive coupling. The BSIM3 version 3 based model is also improved.
- Dynamic depletion can suit different requirements for SOI technologies.
- Single I-V expression as in BSIM3 version 3.1 to assure continuities of I_{ds} , G_{ds} , G_m and their derivatives for all bias conditions.

BSIM3 SOI DD Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val] [NRD=val] [NRS=val]
+ [NRB=val] [RTHO=val] [CTHO=val] [off] [BJToff=val]
+ [IC=Vds, Vgs, Vbs, Ves, Vps] [SOIQ0=val]
```

Parameter	Description
Mxxx	SOI MOSFET element name. Must begin with M, followed by up to 1023 alphanumeric characters.
nd	Drain terminal node name or number.
ng	Front gate node name or number.
ns	Source terminal node name or number.
ne	Back gate (or Substrate) node name or number.
np	External body contact node name or number.
mname	MOSFET model name reference.
L	SOI MOSFET channel length in meters. This parameter overrides DEFL in an .OPTION statement. Default=DEFL with a maximum of 0.1m.
W	SOI MOSFET channel width in meters. This parameter overrides DEFW in an .OPTION statement. Default=DEFW with a maximum of 0.1m.
M	Multiplier to simulate multiple SOI MOSFETs in parallel. The M setting affects all channel widths, diode leakages, capacitances, and resistances. Default=1.
AD	Drain diffusion area. Overrides DEFAD in the .OPTION statement: Default=DEFAD
AS	Source diffusion area. Overrides DEFAS in the .OPTION statement: Default=DEFAS
PD	Perimeter of the drain junction, including the channel edge. Overrides DEFDP in the .OPTION statement.
PS	Perimeter of the source junction, including the channel edge. Overrides DEFPS in the .OPTION statement.
NRD	Number of squares of the drain diffusion for the drain series resistance. Overrides DEFNRD in the .OPTION statement.
NRS	Number of squares of the source diffusion for the source series resistance. Overrides DEFNRS in the .OPTION statement.
NRB	Number of squares for the body series resistance.
RDC	Additional drain resistance due to the contact resistance in units of ohms. This value overrides the RDC setting in the model specification. Default=0.0.

Parameter	Description
RSC	Additional source resistance due to the contact resistance in units of ohms. This value overrides the RDC setting in the model specification. Default=0.0.
RTHO	Thermal resistance per unit width: - If you do not specify RTHO, simulation extracts it from the model card. - If you specify RTHO, it overrides RTHO in the model card.
CTHO	Thermal capacitance per unit width: - If you do not specify CTHO, simulation extracts it from the model card. - If you specify CTHO, it overrides CTHO in the model card.
OFF	Sets the initial condition to OFF for this element in DC analysis.
BJTOFF	Turns off BJT if equal to 1.
IC	Initial guess in the order (drain, front gate, internal body, back gate, external voltage). Simulation ignores Vps in a 4-terminal device. Use these settings if you specify UIC in the .TRAN statement. The .IC statement overrides it.
SOIQ0	Floating body charge initialization. This parameter is set for the BQI algorithm of a floating body node.

BSIM3 SOI DD Model Parameters

This section covers the following topics:

- [BSIM3 SOI DD Model — Control Parameters](#)
- [BSIM3 SOI DD Model — Process Parameters](#)
- [BSIM3 SOI DD Model — DC Parameters](#)
- [BSIM3 SOI DD Model — AC and Capacitance Parameters](#)
- [BSIM3 SOI DD Model — Temperature Parameters](#)
- [BSIM3 SOI DD Model — Notes](#)

BSIM3 SOI DD Model — Control Parameters

SPICE Symbol	Description	Unit	Default	Notes
CAPMOD	Flag for the short channel capacitance model	-	2	nl-1
MOBMOD	Mobility model selector	-	1	-
NOIMOD	Flag for the noise model	-	1	-

SPICE Symbol	Description	Unit	Default	Notes
SHMOD	Flag for self-heating • 0=no self-heating • 1=self-heating	-	0	

For more information on the Notes, see [BSIM3 SOI DD Model — Notes](#) section.

BSIM3 SOI DD Model — Process Parameters

SPICE Symbol	Description	Unit	Default	Notes
NCH	Channel doping concentration	1/cm ³	1.7x10 ¹⁷	-
NGATE	Poly gate doping concentration	1/cm ³	0	-
NSUB	Substrate doping concentration	1/cm ³	6x10 ¹⁶	nl-2
TBOX	Buried oxide thickness	m	3x10 ⁻⁷	-
TOX	Gate oxide thickness	m	1x10 ⁻⁸	-
TSI	Silicon film thickness	m	10 ⁻⁷	-

For more information on the Notes, see [BSIM3 SOI DD Model — Notes](#) section.

BSIM3 SOI DD Model — DC Parameters

SPICE Symbol	Description	Unit	Default	Notes
vth0	Threshold voltage @ V _{bs} =0 for the long and wide device	V	0.7 for N -0.7 for P	nl-3
A0	Bulk charge effect coefficient for the channel length	-	1.0	-
A1	First non-saturation effect parameter	1/V	0.0	-
A2	Second non-saturation effect parameter	0	1.0	-
ABP	Coefficient of the Abeff dependency on V _{gst}	-	1.0	-
ADICE0	DICE bulk charge factor	-	1	-
AGIDL	GIDL constant	Ω-1	0.0	-
AGS	Gate bias coefficient of A _{bulk}	1/V	0.0	-

SPICE Symbol	Description	Unit	Default	Notes
All	First Vdsatii parameter for the Leff dependence	1/V	0.0	-
ALPHA0	First parameter of the impact ionization current	m/V	0.0	-
ALPHA1	Second parameter of the impact ionization current	1/V	1.0	-
B0	Bulk charge effect coefficient for the channel width	m	0.0	-
B1	Bulk charge effect width offset	m	0.0	-
BETA0	Third parameter of the impact ionization current	V	30	-
BGIDL	GIDL exponential coefficient	V/m	0.0	-
BII	Second Vdsatii parameter for the Leff dependence	m/V	0.0	-
CDSC	Drain/Source to the channel coupling capacitance	F/m ²	2.4e-4	-
CDSCB	Body-bias sensitivity of Cdsc	F/m ²	0	-
CDSCD	Drain-bias sensitivity of Cdsc	F/m ²	0	-
CII	First Vdsatii parameter for the Vds dependence	-	0.0	-
CIT	Interface trap capacitance	F/m ²	0.0	-
DELP	Constant for limiting Vbseff to fs	V	0.02	-
DELTA	Effective Vds parameter	-	0.01	-
DII	Second Vdsatii parameter for the Vds dependence	V	-1.0	-
DROUT	L dependence coefficient of the DIBL correction parameter in Rout	-	0.56	-
DSUB	DIBL coefficient exponent	-	0.56	-
DVBD0	First coefficient of Vbs0, Leff dependency	V	0	-
DVBD1	Second coefficient of Vbs0, Leff dependency	V	0	-

SPICE Symbol	Description	Unit	Default	Notes
DVT0	First coefficient of the short-channel effect on V_{th}	-	2.2	-
DVT0W	First coefficient of the narrow-width effect on V_{th} for a small channel length	-	0	-
DVT1	Second coefficient of the short-channel V_{th} effect	-	0.53	-
DVT1W	Second coefficient of the narrow-width effect on V_{th} for a small channel length	-	5.3e6	-
DVT2	Body-bias coefficient of the short-channel V_{th} effect	1/V	-0.032	-
DVT2W	Body-bias coefficient of the narrow-width effect on V_{th} for a small channel length	1/V	-0.032	-
DWB	Coefficient, substrate body bias dependence, W_{eff}	$m/V^{1/2}$	0.0	
DWG	Coefficient of the gate dependence of W_{eff}	m/V	0.0	
EDL	Electron diffusion length	m	2e-6	-
ETA0	DIBL coefficient in the subthreshold region	-	0.08	-
ETAB	Body-bias coefficient for subthreshold DIBL effect	1/V	-0.07	-
ISBJT	BJT injection saturation current	A/m ²	1e-6	-
ISDIF	Body to source/drain injection saturation current	A/m ²	0.0	-
ISREC	Recombination in the depletion saturation current	A/m ²	1e-5	-
ISTUN	Reverse tunneling saturation current	A/m ²	0.0	-
K1	First order body effect coefficient	$V^{1/2}$	0.5	-
K2	Second order body-effect coefficient	-	0	-
K3	Narrow width coefficient	-	0	-
K3B	Body-effect coefficient of k3	1/V	0	-
KB1	Coefficient of V_{bs0} dependency on V_{es}	-	1	-
KBJT1	Parasitic bipolar early effect coefficient	m/V	0	-

SPICE Symbol	Description	Unit	Default	Notes
KETA	Body-bias coefficient of the bulk charge effect	m	-0.6	-
LINT	Length offset fitting parameter of I-V without bias	m	0.0	-
MXC	Fitting parameter for calculating Abeff	-	-0.9	-
NDIODE	Diode non-ideality factor	-	1.0	-
NFACTOR	Subthreshold swing factor	-	1	-
NGIDL	GIDL Vds enhancement coefficient	V	1.2	-
NLX	Lateral non-uniform doping parameter	m	1.74e-7	-
NTUN	Reverse tunneling non-ideality factor	-	10.0	-
PCLM	Channel length modulation parameter	-	1.3	-
PDIBL1	Correction parameter for the DIBL effect of the first output resistance	-	.39	-
PDIBL2	Correction parameter for the DIBL effect of the second output resistance	-	0.086	-
PRWB	Body-effect coefficient of Rds	1/V	0	-
PRWG	Gate bias effect coefficient of Rds	$1/V^{1/2}$	0	-
PVAG	Gate dependence of the Early voltage	-	0.0	-
RBODY	Intrinsic body contact sheet resistance	ohm/m ²	0.0	-
RBSH	Extrinsic body contact sheet resistance	ohm/m ²	0.0	-
RDSW	Parasitic resistance per unit width	$\Omega\text{-mm}^{W_r}$	100	-
RSH	Source drain sheet resistance in ohm per square	Ω/square	0.0	-
U0	Mobility at Temp=Tnom • NMOSFET • PMOSFET	cm ² /V-sec	670250	-
UA	First-order mobility degradation coefficient	m/V	2.25e-9	-

SPICE Symbol	Description	Unit	Default	Notes
UB	Second-order mobility degradation coefficient	$(\text{m/V})^2$	5.9e-19	-
UC	Body-effect of the mobility degradation coefficient	1/V	-.0465	-
VBSA	Transition body voltage offset	V	0	-
VOFF	Offset voltage in the subthreshold region for large W and L values	V	-0.08	-
VSAT	Saturation velocity at Temp=Tnom	m/sec	8e4	-
W0	Narrow width parameter	m	0	-
WINT	Width offset fitting parameter of I-V without bias	m	0.0	-
WR	Width offset from Weff for calculating Rds	-	1	-

For more information on the Notes, see [BSIM3 SOI DD Model — Notes](#) section.

BSIM3 SOI DD Model — AC and Capacitance Parameters

SPICE Symbol	Description	Unit	Default	Notes
ASD	Smoothing parameter for the source/drain bottom diffusion	-	0.3	-
CGD1	Overlap capacitance for the lightly-doped drain-gate region	F/m	0.0	-
CGDO	Non-LDD region drain-gate overlap capacitance per channel length	F/m	calculated nC-2	-
CGEO	Gate substrate overlap capacitance per unit channel length	F/m	0.0	-
CGS1	Overlap capacitance for the lightly-doped source-gate region	F/m	0.0	-
CGSO	Non-LDD region source-gate overlap capacitance per channel length	F/m	calculated nC-1	-
CJSWG	Source/Drain (gate side) sidewall junction Capacitance per unit width (normalized to 100nm T_{si})	F/m^2	1e-10	-

SPICE Symbol	Description	Unit	Default	Notes
CKAPPA	Coefficient of the fringing field capacitance for the overlap capacitance in the lightly-doped region	F/m	0.6	-
CSDESW	Source/drain sidewall fringing capacitance per unit length	F/m	0.0	-
CSDMIN	Minimum capacitance of the source/drain bottom diffusion	V	calculated	nC-5
MJSWG	Grading coefficient for the Source/Drain (gate side) sidewall junction capacitance	V	0.5	-
PBSWG	Built-in potential for the Source/Drain (gate side) sidewall junction capacitance	V	.7	-
TT	Coefficient for the diffusion capacitance transit time	second	1ps	-
VSDFB	Flatband voltage for the source/drain bottom diffusion capacitance	V	calculated	nC-3
VSDTH	Threshold voltage for the source/drain bottom diffusion capacitance	V	calculated	nC-4
XPART	Charge partitioning rate flag	-	0	
CF	Fringing field capacitance for the gate-to-source/drain	F/m	calculated	nC-6
CLC	Constant term for the short-channel mode	m	0.1×10^{-7}	-
CLE	Exponential term for the short-channel mode	none	0.0	-
DLC	Length offset fitting parameter from C-V	m	lint	-
DWC	Width offset fitting parameter from C-V	m	wint	-

For more information on the Notes, see [BSIM3 SOI DD Model — Notes](#) section.

BSIM3 SOI DD Model — Temperature Parameters

SPICE Symbol	Description	Unit	Default	Notes
AT	Temperature coefficient for the saturation velocity	m/sec	3.3e4	-
CTH0	Normalized thermal capacity	(W*S)/m°C	0	-

SPICE Symbol	Description	Unit	Default	Notes
KT1	Temperature coefficient for the threshold voltage	V	-0.11	-
KT11	Channel length dependence of the temperature coefficient for the threshold voltage	V*m	0.0	
KT2	Body-bias coefficient of the V_{th} temperature effect	none	0.022	-
PRT	Temperature coefficient for R_{dsw}	$\Omega\text{-}\mu\text{m}$	0	-
RTH0	Normalized thermal resistance	mC/W	0	-
TNOM	Temperature at which simulation expects parameters	$^{\circ}\text{C}$	27	-
UA1	Temperature coefficient for U_a	m/V	4.31e-9	-
UB1	Temperature coefficient for U_b	$(\text{m/V})^2$	-7.61e-18	-
UC1	Temperature coefficient for U_c	1/V	-0.056	nT-1
UTE	Mobility temperature exponent	none	-1.5	-
XBJT	Power dependence of j_{bjt} on the temperature	none	2	-
XDIF	Power dependence of j_{dif} on the temperature	none	2	-
XREC	Power dependence of j_{rec} on the temperature	none	20	-
XTUN	Power dependence of j_{tun} on the temperature	none	0	-

For more information on the notes, see [BSIM3 SOI DD Model — Notes](#) section.

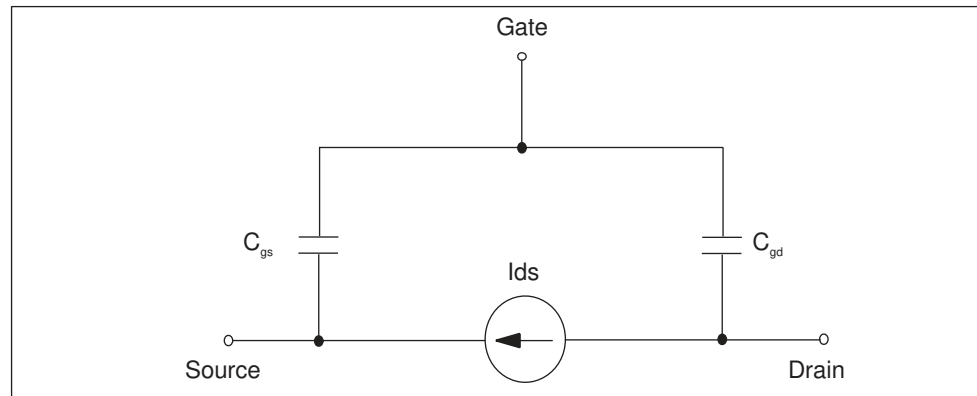
BSIM3 SOI DD Model — Notes

Note	Explanation
nl-1	<i>Capmod</i> 0 and 1 do not calculate the dynamic depletion. Therefore, ddMod does not work with <i>capmod</i> .

Note	Explanation
nI-2	BSIMSOI refers to a substrate of the silicon below the buried oxide, not the well region in BSIM3. It calculates the backgate flatband voltage (V_{fbb}) and the parameters related to the source/drain diffusion bottom capacitance (V_{sdth} , V_{sdfb} , C_{sdmin}). Positive n_{sub} is the same type of doping as the body. Negative n_{sub} is the opposite type of doping.
nC-1	If you do not specify $cgso$, simulation calculates it: if you specify d/c greater than 0, then $cgso=pl=(dlc*cox)-cgs1$ if the previously-calculated $cgso<0$, then $cgso=0$ else $cgso=0.6*Tsi*cox$
nC-2	Calculates $Cgdo$ similar to $Csdo$
nC-3	If $nsub$ is positive, then: $V_{sdfb} = -\frac{kT}{q} \log \left(\frac{10^{20} \cdot n_{sub}}{n_i \cdot n_i} \right) - 0.3$ else $V_{sdfb} = -\frac{kT}{q} \log \left(\frac{10^{20}}{n_{sub}} \right) + 0.3$
nC-4	If $nsub$ is positive, then: $\phi_{sd} = 2\frac{kT}{q} \log \left(\frac{n_{sub}}{n_i} \right), Y_{sd} = \frac{5.753 \times 10^{-12} \sqrt{n_{sub}}}{C_{box}}$ $V_{sdth} = V_{sdfb} + \phi_{sd} + Y_{sd} \sqrt{\phi_{sd}}$ else: $\phi_{sd} = 2\frac{kT}{q} \log \left(-\frac{n_{sub}}{n_i} \right), Y_{sd} = \frac{5.753 \times 10^{-12} \sqrt{n_{sub}}}{C_{box}}$ $V_{sdth} = V_{sdfb} - \phi_{sd} - Y_{sd} \sqrt{\phi_{sd}}$
nC-5	$X_{sddep} = \sqrt{\frac{2\epsilon_{si}\phi_{sd}}{q n_{sub} \cdot 10^6 }}, C_{sddep} = \frac{\epsilon_{si}}{X_{sddep}}, C_{sdmin} = \frac{C_{sddep}C_{box}}{C_{sddep} + C_{box}}$
nC-6	If you do not specify cf , then simulation calculates it: $CF = \frac{2\epsilon_{ox}}{\pi} \ln \left(1 + \frac{4 \times 10^{-7}}{T_{ox}} \right)$
nT-1	For $mobmod=1$ and 2 , the unit is m/V^2 . Default is $-5.6E-11$. For $mobmod=3$, the unit is $1/V$ and the default is -0.056 .

LEVEL 61: RPI a-Si TFT Model

This is a Synopsys MOSFET model which is based on the AIM-SPICE MOS15 amorphous silicon (a-Si) thin-film transistor (TFT) model, developed by Rensselaer Polytechnic Institute.



This section covers the following topics:

- [RPI a-Si TFT Model Features](#)
- [Using RPI a-Si TFT Model with Synopsys Simulators](#)
- [RPI a-Si TFT Model Parameters](#)
- [RPI a-Si TFT Model Equations](#)

RPI a-Si TFT Model Features

Features of the AIM-SPICE MOS15 a-Si TFT model include:

- Modified charge control model; induced charge trapped in localized states
- Above threshold includes:
 - Field effect mobility becoming a function of gate bias
 - Band mobility dominated by lattice scattering
- Below threshold includes:
 - Fermi level located in deep localized states
 - Relate position of Fermi level, including the deep DOS, back to the gate bias

- Empirical expression for current at large negative gate biases for hole-induced leakage current
- Applies interpolation techniques to equations to unify the model

Using RPI a-Si TFT Model with Synopsys Simulators

To simulate using the AIM-SPICE MOS15 a-Si TFT model:

1. Set Level=61 to identify the model as the AIM-SPICE MOS15 a-Si TFT model.
2. The default value for L is 100 μm , and the default value for W is 100 μm .
3. Level 61 is a 3-terminal model. This model does not include a bulk node; therefore simulation does not append parasitic drain-bulk or source-build diodes are appended to the model. You can specify a fourth node, but it does not affect simulation results.
4. The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in most other simulators. When comparing to other simulators, use `.TEMP 27` or `.OPTION TNOM=27` to set the simulation temperature to 27 in the netlist.

The following is an example of how Level 61 modifies the Synopsys MOSFET model and element statement.

```
mckt drain gate source nch L=10e-6 W=10e-6
.MODEL nch nmos Level=61
+ alphasat=0.6 cgdo=0.0 cgso=0.0 def0=0.6
+ delta=5.0 el=0.35 emu=0.06 eps=11
+ epsi=7.4 gamma=0.4 gmin=1e23 iol=3e-14
+ kasat=0.006 kvt=-0.036 lambda=0.0008 m=2.5
+ muband=0.001 rd=0.0 rs=0.0 sima0=1e-14
+ tnom=27 tox=1.0e-7 v0=0.12 vaa=7.5e3
+ vds1=7 vfb=-3 vgs1=7 vmin=0.3 vto=0.0
```

RPI a-Si TFT Model Parameters

Name	Unit	Default	Description
ALPHASAT	-	0.6	Saturation modulation parameter
CGDO	F/m	0.0	Gate-drain overlap capacitance per meter channel width
CGSO	F/m	0.0	Gate-source overlap capacitance per meter channel width
DEF0	eV	0.6	Dark Fermi level position
DELTA	-	5	Transition width parameter

Name	Unit	Default	Description
EL	eV	0.35	Activation energy of the hole leakage current
EMU	eV	0.06	Field effect mobility activation energy
EPS	-	11	Relative dielectric constant of the substrate
EPSI	-	7.4	Relative dielectric constant of the gate insulator
GAMMA	-	0.4	Power law mobility parameter
GMIN	$\text{m}^{-3}\text{eV}^{-1}$	1E23	Minimum density of deep states
IOL	A	3E-14	Zero-bias leakage current parameter
KASAT	$1/^{\circ}\text{X}$	0.006	Temperature coefficient of ALPHASAT
KVT	$\text{V}/^{\circ}\text{X}$	-0.036	Threshold voltage temperature coefficient
LAMBDA	$1/\text{V}$	0.0008	Output conductance parameter
M	-	2.5	Knee shape parameter
MUBAND	m^2/Vs	0.001	Conduction band mobility
RD	m	0.0	Drain resistance
RS	m	0.0	Source resistance
SIGMA0	A	1E-14	Minimum leakage current parameter
TNOM	oC	25	Parameter measurement temperature
TOX	m	1E-7	Thin-oxide thickness
V0	V	0.12	Characteristic voltage for deep states
VAA	V	7.5E3	Characteristic voltage for field effect mobility
VDSL	V	7	Hole leakage current drain voltage parameter
VFB	V	-3	Flat band voltage
VGSL	V	7	Hole leakage current gate voltage parameter
VMIN	V	0.3	Convergence parameter
VTO	V	0.0	Zero-bias threshold voltage

RPI a-Si TFT Model Equations

This section covers the following topics:

- [RPI a-Si TFT Model — Drain Current](#)
- [RPI a-Si TFT Model — Temperature Dependence](#)
- [RPI a-Si TFT Model — Capacitance](#)

RPI a-Si TFT Model — Drain Current

$$I_{ds} = I_{leakage} + I_{ab}, \quad I_{ab} = g_{ch} V_{dse} (1 + LAMBDA \cdot V_{ds})$$

$$V_{dse} = \frac{V_{ds}}{\left[1 + (V_{ds} / V_{sate})^M\right]^{1/M}}$$

$$V_{sate} = \alpha_{sat} V_{gte}$$

$$g_{ch} = \frac{g_{chi}}{1 + g_{chi} (RS + RD)}, \quad g_{chi} = q n_s W \cdot MUBAND / L$$

$$n_s = \frac{n_{sa} n_{sb}}{n_{sa} + n_{sb}}$$

$$n_{sa} = \frac{EPSI \cdot V_{gte}}{(q \cdot TOX)} \cdot \left(\frac{V_{gte}}{V_{aat}} \right)^{GAMMA}$$

$$n_{sb} = n_{so} \left(\frac{t_m}{TOX} \frac{V_{gfbe}}{V_0} \frac{EPSI}{EPS} \right)^{\frac{2V_0}{V_e}}$$

$$n_{so} = N_c t_m \frac{V_e}{V_0} \exp \left(-\frac{DEF0}{V_{th}} \right), \quad N_c = 3.0 \cdot 10^{25} m^{-3}$$

$$V_e = \frac{2 \cdot V_0 \cdot V_{tho}}{2 \cdot V_0 - V_{th}}, \quad t_m = \sqrt{\frac{EPS}{2q \cdot GMIN}}$$

$$V_{gte} = \frac{VMIN}{2} \left[1 + \frac{V_{gt}}{VMIN} + \sqrt{DELTA^2 + \left(\frac{V_{gt}}{VMIN} - 1 \right)^2} \right]$$

$$V_{gt} = V_{gs} - V_T$$

$$V_{gfb} = \frac{VMIN}{2} \left[1 + \frac{V_{gfb}}{VMIN} + \sqrt{DELTA^2 + \left(\frac{V_{gfb}}{VMIN} - 1 \right)^2} \right]$$

$$V_{gfb} = V_{gs} - V_{FB}, I_{leakage} = I_{hl} + I_{min}$$

$$I_{hl} = IOL \left[\exp \left(\frac{V_{ds}}{V_{DSL}} \right) - 1 \right] \exp \left(-\frac{V_{gs}}{V_{GSL}} \right) \exp \left[\frac{EL}{q} \left(\frac{1}{V_{tho}} - \frac{1}{V_{th}} \right) \right]$$

$$I_{min} = SIGMA0 \cdot V_{ds}$$

RPI a-Si TFT Model — Temperature Dependence

$$V_{tho} = k_B \cdot TNOM / q, V_{th} = k_B \cdot (TEMP) / q$$

$$V_{aat} = VAA \exp \left[\frac{EMU}{q \cdot GAMMA} \left(\frac{1}{V_{th}} - \frac{1}{V_{tho}} \right) \right]$$

$$V_T = VTO + KVT (TEMP - TNOM)$$

$$\alpha_{sat} = ALPHASAT + KASAT (TEMP - TNOM)$$

RPI a-Si TFT Model — Capacitance

$$C_{gs} = C_f + \frac{2}{3} C_{gc} \left[1 - \left(\frac{V_{sate} - V_{dse}}{2V_{sate} - V_{dse}} \right)^2 \right]$$

$$C_{gd} = C_f + \frac{2}{3} C_{gc} \left[1 - \left(\frac{V_{sate}}{2V_{sate} - V_{dse}} \right)^2 \right]$$

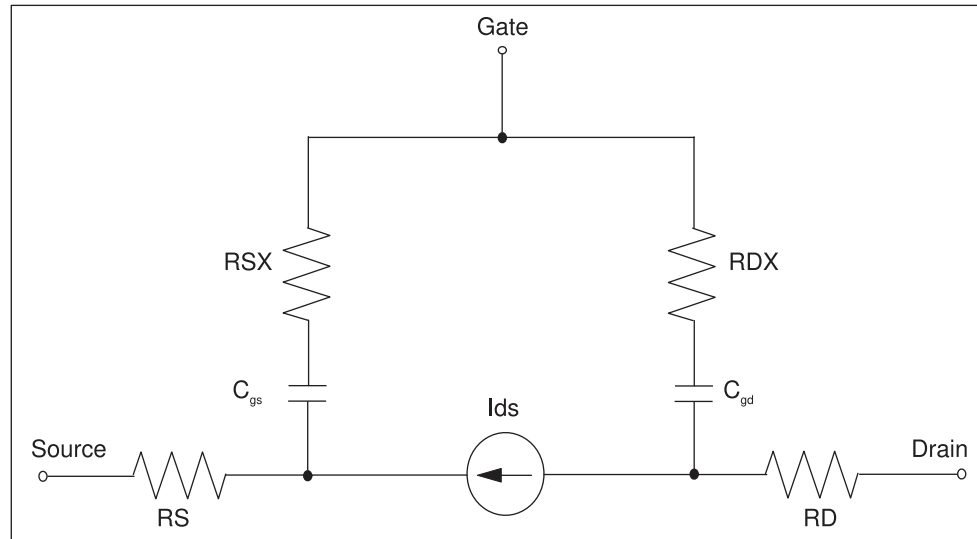
$$C_f = 0.5 \cdot EPS \cdot W, \left(C_{gs} = q \frac{dn_{sc}}{dV_{gs}} \right)$$

$$n_{sc} = \frac{n_{sac} n_{sbc}}{n_{sac} + n_{sbc}}, \left(n_{sa} = \frac{EPSI \cdot V_{gte}}{(q \cdot TOX)} \cdot \left(\frac{V_{gte}}{V_{aat}} \right)^{GAMMA} \right)$$

$$n_{sbc} = n_{sb}$$

LEVEL 62: RPI Poli-Si TFT Model

This is an AIM-SPICE MOS16 poly-silicon (Poli-Si) thin-film transistor (TFT) model, developed by Rensselaer Polytechnic Institute.



The following topics are discussed in the next sections.

- [RPI Poli-Si TFT Model Features](#)
- [Using RPI Poli-Si TFT Model with Synopsys Simulators](#)
- [RPI Poli-Si TFT Model Parameters](#)
- [RPI Poli-Si TFT Model Equations](#)
- [RPI Poli-Si TFT Model Version 2 Equations](#)

RPI Poli-Si TFT Model Features

Features of the AIM-SPICE MOS16 Poli-Si TFT model include:

- A design based on the crystalline MOSFET model
- Field effect mobility that becomes a function of the gate bias
- Effective mobility that accounts for trap states:
 - For low V_{gs} , it is the power law
 - For high V_{gs} , it is the constant

- Reverse bias drain current function of the electric field near the drain and the temperature
- A design independent of the channel length
- A unified DC model that includes all four regimes for channel lengths down to 4 μm :
 - Leakage (thermionic emission)
 - Subthreshold (diffusion-like model)
 - Above threshold (c-Si-like with mFet)
 - Kink (impact ionization with feedback)
- An AC model accurately reproduces the C_{gc} frequency dispersion
- Automatic scaling of model parameters that accurately model a wide range of device geometries

Using RPI Poli-Si TFT Model with Synopsys Simulators

To simulate using the AIM-SPICE MOS16 Poli-Si TFT model:

1. Set `LEVEL=62` to identify the model as the AIM-SPICE MOS16 Poli-Si TFT model.
2. The default value for `L` is 100 μm , and the default value for `W` is 100 μm .
3. Level 62 is a 3-terminal model. This model does not include a bulk node; therefore, simulation does not append a parasitic drain-bulk or source-bulk diode to the model. You can specify a fourth node, but it does not affect the simulation results.
4. The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in most other simulators. When comparing to other simulators, use `.TEMP 27` or `.OPTION TNOM=27` to set the simulation temperature to 27°C in the netlist.

The following is an example of how Level 62 modifies a MOSFET device model and element statement:

```
mckt drain gate source nch L=10e-6 W=10e-6
.MODEL nch nmos Level=62
+ asat=1 at=3e-8 blk=0.001 bt=0.0 cgdo=0.0
+ cgso=0.0 dasat=0.0 dd=1.4e-7 delta=4.0
+ dg=2.0e-7 dmul=0.0 dvt=0.0 dvto=0.0 eb=0.68
+ eta=7 etac0=7 etac00=0 i0=6.0 i00=150
+ lasat=0.1 kink=19e-6 mc=3.0 mk=1.3 mmu=3.0
+ mu0=100 mu1=0.0022 mus=1.0 rd=0.0 rdx=0.0
+ rs=0.0 rsx=0.0 tnom=27 tox=1.0e7 vfb=-0.1
+ vkink=9.1 von=0.0 vto=0.0
```

RPI Poli-Si TFT Model Parameters

This section covers the following topics:

- [RPI Poli-Si TFT Model — General Parameters](#)
- [RPI Poli-Si TFT Model — Version 2 Parameters](#)
- [RPI Poli-Si TFT Model — Self-heating Parameters](#)
- [RPI Poli-Si TFT Model — ACM Parameters for Drain and Source Resistance Calculus Specific to PrimeSim Continuum](#)

RPI Poli-Si TFT Model — General Parameters

Name	Unit	Default	Description
ASAT	-	1	Proportionality constant of Vsat
AT	m/V	3E-8	DIBL parameter 1
BLK	-	0.001	Leakage barrier lowering constant
BT	m · V	1.9E-6	DIBL parameter 2
CAPMOD	-	0	Model capacitance selector (zero recommended)
CGDO	F/m	0	Gate-drain overlap capacitance per meter channel width
CGSO	F/m	0	Gate-source overlap capacitance per meter channel width
DASAT	1/°C	0	Temperature coefficient of ASAT
DD	m	1400 Å	Vds field constant
DELTA	-	4.0	Transition width parameter
DG	m	2000 Å	Vgs field constant
DMU1	$\chi\mu^2/\zeta\sigma\times C$	0	Temperature coefficient of MU1
DVT	V	0	The difference between VON and the threshold voltage
DVTO	V/°C	0	Temperature coefficient of VTO
EB	EV	0.68	Barrier height of the diode
EPS	-	11.7	Constant of the substrate (default user-specifiable)

Name	Unit	Default	Description
EPSI	-	3.9	Constant of the gate insulator (deflt. user-specifiable)
ETA	-	7	Subthreshold ideality factor
ETAC0	-	ETA	Capacitance subthreshold ideality factor at zero-drain bias
ETAC00	1/V	0	Capacitance subthreshold coefficient of the drain bias
I0	A/m	6.0	Leakage scaling constant
I00	A/m	150	Reverse diode saturation current
KSS	-	0	Small signal parameter (zero is recommended)
LASAT	M	0	Coefficient for length dependence of ASAT
LKINK	M	19E-6	Kink effect constant
MC	-	3.0	Capacitance knee shape parameter
MK	-	1.3	Kink effect exponent
MMU	-	3.0	Low field mobility exponent
MU0	cm ² /Vs	100	High field mobility
MU1	cm ² /Vs	0.0022	Low field mobility parameter
MUS	cm ² /Vs	1.0	Subthreshold mobility
RD	μ	0	Drain resistance
RDX	Ω	0	Resistance in series with C _{gd}
RS	μ	0	Source resistance
RSX	Ω	0	Resistance in series with C _{gs}
TNOM	°C	25	Parameter measurement temperature
TOX	m	1E-7	Thin-oxide thickness
V0	V	0.12	Characteristic voltage for deep states
VFB	V	-0.1	Flat band voltage
VKINK	V	9.1	Kink effect voltage

Name	Unit	Default	Description
VON	V	0	On-voltage
VSI	V	2.0	vgs dependence parameter
VST	V	2.0	vgs dependence parameter
VTO	V	0	Zero-bias threshold voltage
ZEROC	-	0	Flag for capacitance calculations in capmod=1 - capmod=1: set the 0 capacitance value - capmod=0: calculation capacitance
XL	m	0	Length bias accounts for the masking and etching effects
XW	m	0	Width bias accounts for the masking and etching effects
LMLT	-	1	Length shrink factor
WMLT	-	1	Width shrink factor
COMPATIBLE=0 1	-	0	- COMPATIBLE=0: The PrimeSim Continuum solution implementation of AIM-SPICE MOS 16 RPI TFT model. - COMPATIBLE=1: Enables compatibility to a commonly-used TFT model by the industry.

RPI Poli-Si TFT Model — Version 2 Parameters

Name	Unit	Default	Description
VERSION	-	1	1=Version 1 2=Version 2
DIBLMOD	-	1	DIBL model selector
INTDSNOD	-	1 if VERSION=1 0 if VERSION=2	Extrinsic series resistance mode selector: 0 - uses internal approximation for drain and source resistances 1 – uses drain and source resistances as extrinsic elements
ISUBMOD	-	0	Channel length modulation selector: 0 – uses LAMBDA 1 – uses LS and VP
LAMBDA	1/V		Channel length modulation parameter

Name	Unit	Default	Description
LME	m	0	ME length dependence parameter
LMU0	m	0	MU0 length dependence parameter
LMU1	m	0	MU1 length dependence parameter
LS	-	3.50E-008	Channel length modulation parameter
ME (MS)	-	2.5	Long channel saturation transition parameter
META	-	1	ETA floating-body parameter
MSS	-	1.5	Vdse transition parameter
THETA	M/V	0	Mobility degradation parameter
VMAX	m/s	4.00E+004	Saturation velocity
VP	V	0.2	Channel length modulation parameter
VSIGMA	V	VSI, if specified 0.2, otherwise	VGS dependence parameter
VSIGMA	V	VSI, if specified, 0.2 otherwise	VGS dependence parameter
VSIGMAT	V	VST, if specified 1.7, otherwise	VGS dependence parameter
VSIGMAT	V	VST, if specified, 1.7 otherwise	VGS dependence parameter

RPI Poli-Si TFT Model — Self-heating Parameters

Name	Unit	Default	Description
CTH0	(W*S)/m°C	1.00E-005	Thermal capacitance per unit width
RTH0	m°C/W	0	Thermal resistance per unit width
SHMOD	-	0	Self-heating flag
WTH0	m	0	Width offset for thermal resistance and capacitance scaling

RPI Poli-Si TFT Model — ACM Parameters for Drain and Source Resistance Calculus Specific to PrimeSim Continuum

Name	Unit	Default	Description
ACM	-	-	Area calculation method
HDIF	m	0	Length of heavily doped diffusion
LD	m	0	Lateral diffusion into channel from source and drain diffusion
LDIF	m	0	Length of heavily doped region adjacent to gate
RDC	ohm	0	Additional drain resistance due to contact
RSC	ohm	0	Additional source resistance due to contact
RSH	ohm/sq	0	Drain and source diffusion sheet resistance
WD	m	0	Lateral diffusion into channel from bulk along width

Note:

Source and drain resistances are calculated similarly to the other PrimeSim Continuum solution MOSFET models, which are based on the value of `ACM`.

RPI Poli-Si TFT Model Equations

This section covers the following topics:

- [RPI Poli-Si TFT Model — Drain Current: Total Current](#)
- [RPI Poli-Si TFT Model — Drain Current: Subthreshold Current](#)
- [RPI Poli-Si TFT Model — Drain Current: Subthreshold Leakage Current](#)
- [RPI Poli-Si TFT Model — Drain Current: Impact Ionization Effect](#)
- [RPI Poli-Si TFT Model — Threshold Voltage](#)
- [RPI a-Si TFT Model — Temperature Dependence](#)
- [RPI Poli-Si TFT Model — Capacitance: CAPMOD=0](#)
- [RPI Poli-Si TFT Model — Capacitance: CAPMOD=1](#)
- [RPI Poli-Si TFT Model — Geometry Effect](#)
- [RPI Poli-Si TFT Model — Self Heating](#)

RPI Poli-Si TFT Model — Drain Current: Total Current

$$I_{ds} = \frac{I_a \cdot I_{sub}}{I_a + I_{sub}} \cdot (1 + I_{kink}) + I_{leak}$$

RPI Poli-Si TFT Model — Drain Current: Subthreshold Current

The following is the expression for the subthreshold current:

$$I_{sub} = MUS \cdot C_{ox} \frac{W}{L} V_{sth}^2 \exp\left(\frac{V_{GT}}{V_{sth}}\right) \left[1 - \exp\left(-\frac{V_{DS}}{V_{sth}}\right)\right]$$

$$I_{sub} = MUS \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot V_{sth}^2 \cdot \exp\left(\frac{V_{GT}}{V_{sth}}\right) \cdot \left(1 - \exp\left(-\frac{V_{DS}}{V_{sth}}\right)\right)$$

$$C_{OX} = \frac{\epsilon_i}{T_{OX}}, V_{sth} = ETA \cdot V_{th}, V_{th} = \frac{k_B \cdot TEMP}{q}$$

$$V_{GT} = V_{GS} - V_{Teff}$$

$$V_{Teff} = V_{TX} - \frac{AT \cdot V_{DS}^2 + BT}{L_{eff} \cdot \left(1 + \exp\left(\frac{V_{GS} - V_{ST} - V_{TX}}{V_{SI}}\right)\right)}$$

In the preceding equations, ϵ_i is the dielectric constant of the oxide, and k_B is the Boltzmanns constant.

Above the $V_{GT} > 0$ threshold, the following equation calculates the conduction current:

$$I_a = \frac{\mu_{FET} \cdot C_{OX} \cdot W_{eff}}{L_{eff}} \left(V_{GTE} \cdot V_{DS} - \frac{V_{DS}^2}{2 \cdot \alpha_{sat}} \right) \text{ for } V_{DS} \leq \alpha_{sat} V_{GTE}$$

$$\frac{\mu_{FET} \cdot C_{OX} \cdot W_{eff} \cdot V_{GTE}^2 \cdot \alpha_{sat}}{2 \cdot L_{eff}} \quad \text{for } V_{DS} > \alpha_{sat} V_{GTE}$$

$$\frac{1}{\mu_{FET}} = \frac{1}{MUO} + \frac{1}{\mu1 \cdot \left(\frac{2V_{GTE}}{V_{sth}}\right)^{MMU}}$$

$$V_{GTE} = V_{sth} \cdot \left[1 + \frac{V_{GT}}{2 \cdot V_{sth}} + \sqrt{DELTA^2 + \left(\frac{V_{GT}}{2 \cdot V_{sth}} - 1\right)^2} \right]$$

RPI Poli-Si TFT Model — Drain Current: Subthreshold Leakage Current

Subthreshold leakage current is the result of the thermionic field emission of carriers through the grain boundary trap states as described in the following equations.

$$I_{leak} = IO \cdot W_{eff} \cdot \left[\exp \left(\frac{q \cdot BLK \cdot V_{DS}}{k \cdot T} \right) - 1 \right] \cdot (X_{TFE} + X_{TE}) + I_{diode}$$

$$X_{TFE} = \frac{X_{TFE, lo} \cdot X_{TFE, hi}}{X_{TFE, lo} + X_{TFE, hi}}$$

The following equations calculate values for the preceding equations:

$$X_{TE} = \exp(-W_C)$$

$$W_C = \frac{E_C - E_t}{k \cdot T} = \frac{0.55 eV}{k \cdot T}$$

$$X_{TFE, lo} = \frac{4\sqrt{\pi}}{3} \cdot f \cdot \exp\left(\frac{4}{27} \cdot f^2 - W_c\right) \quad \text{for } f \leq f_{lo}$$

$$X_{TFE, lo}(f_{lo}) \cdot \exp\left[\left(\frac{1}{f_{lo}} + \frac{8}{27} \cdot f_{lo}\right) \cdot (f - f_{lo})\right] \quad \text{for } f > f_{lo}$$

$$X_{TFE, hi} = \frac{2W_c}{3} \cdot \exp\left(1 - \frac{2W_c}{3}\right) \quad \text{for } f > f_{hi}$$

$$\left(1 - \frac{3\sqrt{W_c}}{2 \cdot f}\right)^{-1} \exp\left[\frac{-W_c^{3/2}}{f}\right] \quad \text{for } f \geq f_{hi}$$

$$f_{hi} = 3 \cdot \left(\frac{W_c^{3/2}}{2W_c - 3}\right)$$

$$f = \frac{F_{min}}{2} \left[1 + \frac{\frac{F_f}{F_{fo}}}{F_{min}} + \sqrt{DELTA^2 + \left(\frac{\frac{F_f}{F_{fo}}}{F_{min}} - 1\right)^2} \right]$$

$$F_{min} = 1e^{-4}$$

$$X_{TFE, lo}(f_{lo}) = \frac{2\sqrt{4\pi}}{3} \cdot f_{lo} \cdot \exp\left(\frac{4}{27} f_{lo}^2 - W_c\right)$$

$$F_f = \left(\frac{V_{DS}}{DD} - \frac{V_{GS} - V_{FB}}{DG} \right)$$

$$F_{fo} = (k \cdot T)^{3/2} \cdot \left(\frac{4}{3} \cdot \frac{2\pi\sqrt{2m^*}}{q \cdot h} \right)$$

$$m^* = 0.27 \cdot m_0$$

$$f_{lo} = \frac{3}{2} \cdot (\sqrt{W_c + 1} - 1)$$

$$I_{diode} = IOO \cdot W_{eff} \cdot \exp\left(-\frac{EB}{k \cdot T}\right) \left[1 - \exp\left(-\frac{q \cdot V_{DS}}{k \cdot T}\right) \right]$$

RPI Poli-Si TFT Model — Drain Current: Impact Ionization Effect

$V_{GT} > 0$ very large drain biases include the kink effect. Level 62 models this effect as impact ionization in a narrow region near the drain, and adds the I_{kink} impact ionization current to the drain current. The expression is:

$$I_{kink} = A_{kinkt} \cdot (V_{DS} - V_{DSE}) \cdot \exp\left(-\frac{VKINK}{V_{DS} - V_{DSE}}\right)$$

$$A_{kinkt} = \frac{1}{VKINK} \left(\frac{LKINK}{L_{eff}} \right)^{MK}$$

$$V_{DSE} = \frac{V_{DS}}{\left(1 + \left(\frac{V_{DS}}{V_{DSAT}} \right)^3 \right)^{1/3}} - V_{th}$$

$$V_{DSAT} = \alpha_{sat} \cdot V_{GTE}$$

RPI Poli-Si TFT Model — Threshold Voltage

If you do not specify V_{TO} , then $V_T = V_{ON} - DVT$. Otherwise, $V_T = V_{TO}$.

RPI Poli-Si TFT Model — Temperature Dependence

$$V_{TX} = V_T - DV_{TO} \cdot (TEMP - TNOM)$$

$$\mu_1 = MU1 + DMU1 \cdot (TEMP - TNOM)$$

$$\alpha_{sat} = ASAT - \frac{LASAT}{L_{eff}} - DASAT \cdot (TEMP - TNOM)$$

RPI Poli-Si TFT Model — Capacitance: CAPMOD=0

$$C_{gs} = C_f + \frac{2}{3} \cdot C_{gcs} \cdot \left[1 - \left(\frac{V_{DSAT} - V_{DSEX}}{2V_{DSAT} - V_{DSEX}} \right)^2 \right]$$

$$C_{gd} = C_f + \frac{2}{3} \cdot C_{gcd} \cdot \left[1 - \left(\frac{V_{DSAT}}{2V_{DSAT} - V_{DSEX}} \right)^2 \right]$$

$$C_f = 0.5 \cdot EPS \cdot W_{eff}$$

$$C_{gcd} = \frac{C_{OX}}{1 + \eta_{cd} \cdot \exp\left(-\frac{V_{GTX}}{\eta_{cd} V_{th}}\right)}$$

$$C_{gcs} = \frac{C_{OX}}{1 + ETAC0 \cdot \exp\left(-\frac{V_{GTX}}{ETAC0 V_{th}}\right)}$$

$$C_{OX} = \frac{W_{eff} \cdot L_{eff} \cdot \epsilon_i}{TOX}, \eta_{cd} = ETAC0 + ETAC00 \cdot V_{DSEX}$$

$$V_{DSEX} = \frac{V_{DS}}{\left[1 + \left(\frac{V_{DS}}{V_{DSAT}} \right)^{MC} \right]^{\frac{1}{MC}}}$$

$$V_{GTX} = V_{GS} - V_{TX}$$

RPI Poli-Si TFT Model — Capacitance: CAPMOD=1

If ZERO equals 1:

$$V_{gst} = V_{GS} - V_{TX}$$

$$\Phi = 0.6$$

$$V_{gst} < \frac{-\Phi}{2} C_{gs} = C_{gd} = 0$$

$$\frac{-\Phi}{2} \leq V_{gst} < 0 C_{gs} = \frac{4 \cdot V_{gst} \cdot C_{OX}}{3 \cdot \Phi} + \frac{2 \cdot C_{OX}}{3}, C_{gd} = 0$$

$$V_{gst} \geq 0 \left\{ \begin{array}{l} V_{DS} \geq V_{DSAT} \quad C_{gs} = \frac{2C_{OX}}{3}, \quad C_{gd} = 0 \\ V_{DS} < V_{DSAT} \quad C_{gd} = \frac{2}{3} \cdot C_{OX} \cdot \left[1 - \left(\frac{V_{DSAT}}{2(V_{DSAT} - V_{DSEX})} \right)^2 \right] \\ C_{gs} = \frac{2}{3} \cdot C_{OX} \cdot \left[1 - \left(\frac{V_{DSAT} - V_{DSE}}{2(V_{DSAT} - V_{DSEX})} \right)^2 \right] \end{array} \right.$$

$$C_{OX} = \frac{\epsilon_i \cdot W_{eff} \cdot L_{eff}}{T_{OX}}, \quad V_{DSEX} = \frac{V_{DS}}{\left[1 + \left(\frac{V_{DS}}{V_{DSAT}} \right)^{MC} \right]^{\frac{1}{MC}}}$$

RPI Poli-Si TFT Model — Geometry Effect

$$W_{eff} = W + XW$$

$$L_{eff} = L + XL$$

RPI Poli-Si TFT Model — Self Heating

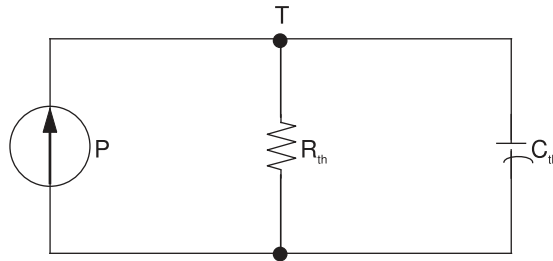
Self heating is turned on if self-heating parameters $SHMOD=1$ and $RTH0 > 0$.

$SHMOD$ and $RTH0$ are also instance parameters. They override the corresponding model parameters.

The effective thermal resistance and capacitance equations and equivalent circuit are:

$$R_{th} = \frac{RTH0}{W_{eff} + WTH0_{eff}}$$

$$C_{th} = CTH0 \cdot (W_{eff} + WTH0_{eff})$$



RPI Poli-Si TFT Model Version 2 Equations

The RPI TFT poly-Si Version 2 model is available in the PrimeSim Continuum solution. Invoke the model by setting the VERSION parameter to 2.

This section covers the following topics:

- [RPI Poli-Si TFT Model — Version 2 Equations: Threshold Voltage and VGS - VT](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Mobility](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Channel Conductance](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Saturation Voltage and Effective Vds](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Drain Current](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Total Drain Current, including Kink Effect and Leakage](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Additional Geometry Scaling](#)
- [RPI Poli-Si TFT Model — Version 2 Equations: Temperature Dependence](#)

RPI Poli-Si TFT Model — Version 2 Equations: Threshold Voltage and VGS - VT

$$V_{Teff} = V_{TX} - \Delta V_{T,DIBL}$$

$$\text{If DIBLMOD} = 0, \text{ then } \Delta V_{T,DIBL} = \frac{AT \cdot V_{DS}^2 + BT}{L_{eff}}$$

$$\text{else, if DIBLMOD} = 1, \text{ then } \Delta V_{T,DIBL} = \frac{AT \cdot V_{DS}^2 + BT}{L_{eff} \cdot \left[1 + \exp \left(\frac{V_{GS} - V_{SIGMAT} - V_{TX}}{VSIGMA} \right) \right]}$$

otherwise

$$\Delta V_{T,DIBL} = \frac{AT \cdot V_{DS}^2}{L_{eff} \cdot \left[1 + \exp \left(\frac{V_{GS} - V_{SIGMAT} - V_{TX}}{VSIGMA} \right) \right]} + \frac{BT}{L_{eff}}$$

$$V_{GT} = V_{GS} - V_{Teff}$$

$$V_{GTE} = ETA \cdot V_{th} \cdot \left[1 + \frac{V_{GT}}{2 \cdot ETA \cdot V_{th}} + \sqrt{DELTA^2 + \left(\frac{V_{GT}}{2 \cdot ETA \cdot V_{th}} - 1 \right)^2} \right]$$

$$V_{th} = \frac{k_B \cdot TEMP}{q}$$

RPI Poli-Si TFT Model — Version 2 Equations: Mobility

$$\Delta V_{T,DIBL} = \frac{AT \cdot V_{DS}^2}{L_{eff} \cdot \left[1 + \exp \left(\frac{V_{GS} - V_{SIGMAT} - V_{TX}}{V_{SIGMA}} \right) \right]} + \frac{BT}{L_{eff}}$$

$$\mu_{eff} = MUS + \frac{\mu_{FET}}{1 + \frac{THETA}{T_{OX}} \cdot V_{GTE}}$$

$$\frac{1}{\mu_{FET}} = \frac{1}{\mu_0} + \frac{1}{\mu_1 \cdot \left(\frac{2V_{GTE}}{\eta_f V_{th}} \right)^{MMU}}$$

$$\eta_f = \frac{ETA}{1 + META \cdot \frac{ETA_1}{ETA} \cdot r_{i1}}$$

$$r_{i1} = \frac{i_1}{1 + i_1}$$

$$i_1 = \left(\frac{LKINK}{L_{eff}} \right)^{MKINK} \cdot \frac{W_{eff}}{L_{eff}} \cdot \frac{V_{DS} - V_{DSE}}{VKINK} \cdot \exp \left(-\frac{VKINK}{V_{DS} - V_{DSE}} \right)$$

$$V_{DSE} = \frac{V_{DS}}{\left[1 + \left(\frac{V_{DS}}{V_{GTE}} \right)^{MSS} \right]^{\frac{1}{MSS}}} - V_{th}$$

RPI Poli-Si TFT Model — Version 2 Equations: Channel Conductance

$$g_{ch} = \frac{g_{chi}}{1 + g_{chi} \cdot (R_D + R_S)}, g_{chi} = q \cdot n_s \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}}$$

$$n_s = \frac{C_{OX} \cdot ETA \cdot V_{TH}}{q} \cdot \log \left[1 + \frac{1}{2} \cdot \exp \left(\frac{V_{GT}}{\eta_f \cdot V_{th}} \right) \right], C_{OX} = \frac{\epsilon_i}{T_{OX}}$$

RPI Poli-Si TFT Model — Version 2 Equations: Saturation Voltage and Effective V_{ds}

$$V_L = \frac{VMAX_L}{\mu_{eff}}$$

$$I_{sat} = \frac{g_{chi} \cdot V_{GTE}}{1 + \frac{V_{GTE}}{V_L} + g_{chi} \cdot R_S + \sqrt{1 + 2 \cdot g_{chi} \cdot R_S + \left(1 + \frac{V_{GTE}}{V_L} \right)^2}}$$

$$V_{Dsat} = \frac{I_{sat}}{g_{ch}}$$

$$I_{choo} = g_{ch} \cdot V_{DS} \cdot \frac{(1 + \lambda \cdot V_{DS})}{\left[1 + \left(\frac{V_{DS}}{V_{Dsat}}\right)^{ME_L}\right]^{\frac{1}{ME_L}}}$$

If ISUBMOD=1, then $\lambda = 0$

otherwise $\lambda = LAMBDA$

$$q_s = q \cdot \left(n_s \cdot I_{choo} \cdot \frac{C_{OX} \cdot R_d}{q} \right)$$

$$V_{satnew} = \frac{2 \cdot VMAX_L \cdot q_s}{q_s \cdot \mu_{eff} + 2 \cdot VMAX_L \cdot \frac{C_{OX}}{\alpha_{SAT}}}$$

$$V_{DS0} = V_{DS} - I_{choo} \cdot (R_D + R_S)$$

$$V_{DSEnew} = \frac{V_{DS0}}{\left[1 + \left(\frac{V_{DS0}}{V_{satnew}}\right)^{MSS}\right]^{\frac{1}{MSS}}} - V_{th}$$

RPI Poli-Si TFT Model — Version 2 Equations: Drain Current

If ISUBMOD = 0, then $I_{dp} = I_{choo}$

otherwise:

$$I_{dp} = \frac{I_{choo}}{1 - \frac{\Delta_L}{L_{eff}}}, \Delta_L = \frac{LS \cdot \log\left(1 + \frac{V_{DS0} V_{DSEnew}}{VP}\right)}{1 + \frac{V_{DSEnew}}{VP} + C_{OX} \cdot \mu_{eff} \cdot R_s \cdot \frac{W_{eff}}{L_{eff}}}$$

RPI Poli-Si TFT Model — Version 2 Equations: Total Drain Current, including Kink Effect and Leakage

$$m = \left(\frac{LKINK}{L_{eff}}\right)^{MKINK} \cdot \frac{V_{DS} - V_{DSEnew}}{VKINK} \cdot \exp\left(-\frac{VKINK}{V_{DS} - V_{DSEnew}}\right)$$

$$I_{ds} = I_{dp} \cdot (1 + m) + I_{leak}$$

The subthreshold leakage current equation is the same as for VERSION=1.

RPI Poli-Si TFT Model — Version 2 Equations: Additional Geometry Scaling

$$\mu_0 = MU0 \cdot \left[1 + \left(\frac{LMU0}{L_{eff}} \right)^2 \right]$$

$$\mu_{1L} = MU1 \cdot \left[1 - \left(\frac{LMU1}{L_{eff}} \right)^2 \right]$$

$$ME_L = ME \cdot \left[1 - \left(\frac{LME}{L_{eff}} \right)^2 \right]$$

RPI Poli-Si TFT Model — Version 2 Equations: Temperature Dependence

$$V_{TX} = VTO - DVTO \cdot (TEMP - TNOM)$$

$$\mu_1 = \mu_{1L} + DMU1 \cdot (TEMP - TNOM)$$

$$\alpha_{sat} = ASAT - \frac{LASAT}{L_{eff}} - DASAT \cdot (TEMP - TNOM)$$

LEVEL 63: Philips MOS11 Model

The Philips MOS Model 11, Level 1101 and 1102, are available as Level 63 in the Synopsys MOSFET models.

Philips MOS Model 11, Level 1102, is an updated version of Level 1101. It uses slightly different equations than Level 1101. The surface potential is calculated iteratively using a second-order Newton-Raphson procedure, resulting in a more accurate description of the surface potential. It includes two types of geometrical scaling rules: physical rules and binning rules. You use the `BINFLAG` parameter to select these rules.

The parasitic diode model includes the Philips JUNCAP Parasitic Diode Model.

The implementation history of the Philips MOS Model 11 is as follows:

Level	Implementation Description
1102.1	Default of NT (noise parameter) changed from 1.656e-20 to 1.624e-20 Bug in noise equation for induced gate noise fixed. Maximum temperature limiting for self-heating implemented.
1102	Iterative solution of surface potential. More accurate, physics-based implementation of velocity saturation effects. More accurate, physics-based equations for thermal noise, induced gate noise, and their correlations. Self-heating implemented. Temperature dependence parameter of the thermal resistance, <code>ATH</code> , added. Internal variable <code>VDBt</code> was simplified, which results in a more physical description for $V_{DS} < 0$. Definition of starting condition of surface potential Ψ_s and Y_{ov} rewritten to increase numerical efficiency and to avoid possible floating-point exceptions.
1101.6	Default value of NT (noise parameter) changed from 1.656e-20 to 1.624e-20. Handling of low limit of internal variable, <code>G_{mob}</code> , simplified. Maximum temperature limiting for self-heating implemented.
1101.5	Added temperature dependence parameter of the thermal resistance (<code>ATH</code>).
1101.4	Improved thermal noise density.
1101.2	Self-heating implementation.

This section covers the following topics:

- [Using the Philips MOS11 Model](#)
- [Philips MOS11 Model Parameters](#)
- [Philips MOS11 Model Examples](#)

Using the Philips MOS11 Model

To use the Philips MOS11 model:

1. Set `LEVEL=63` to identify the model as Philips MOS Model 11.
2. Set the MOS11 version:
 - Set `VERSION=1102.1` (default) to identify the model as Level 1102.1
 - Set `VERSION=1102.0` to identify the model as Level 1102.0
 - Set `VERSION=1101.6` to identify the model as Level 1101.6
 - Set `VERSION=1101.5` to identify the model as Level 1101.5
 - Set `VERSION=1101.4` to identify the model as Level 1101.4
 - Set `VERSION=1101.2` to identify the model as Level 1101.2
 - Set `VERSION=1101.1` to identify the model as Level 1101.1
 - Set `VERSION=1101.0` to identify the model as Level 1101.0
 - Set `VERSION=1100.3` to identify the model as Level 1100.3
 - Set `VERSION=1100.2` to identify the model as Level 1100.2
 - Set `VERSION=1100.1` to identify the model as Level 1100.1
 - Set `VERSION=1100.0` to identify the model as Level 1100.0
3. Set the flag for binned model:
 - Set `BINFLAG=0` (default) to select the physical geometry scaling rules.
 - Set `BINFLAG=1` to select the binning geometry scaling rules.
4. The existing version is internally switched for backward compatibility as follows:
 - `VERSION=11010` switched to `VERSION=1101.1` and `BINFLAG=0`
 - `VERSION=11011` switched to `VERSION=1101.1` and `BINFLAG=1`

5. Set the flag for self-heating:
 - Set `SHMOD=0` (default) to select no self-heating.
 - Set `SHMOD=1` to select self-heating.
6. The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in most other simulators. When comparing to other simulators, use `.TEMP 27` or `.OPTION TNOM=27` to set the simulation temperature to 27 in the netlist.
7. The set of model parameters should always include the `TR` model reference temperature which corresponds to `TREF` in other levels in the Synopsys MOSFET model levels. The default for `TR` is 21.0 to match the Philips simulator.
8. This model has its own charge-based capacitance model. This model ignores the `CAPOF` parameter, which selects different capacitance models.
9. This model uses analytical derivatives for the conductances. This model ignores the `DERIV` parameter, which selects the finite difference method.
10. You can use `DTEMP` with this model to increase the temperature of individual elements, relative to the circuit temperature. Set `DTEMP` on the element line.
11. Because the defaults are non-zero, you should set every Level 63 model parameter in the Model Parameters table in the `.MODEL` statement.
12. The general syntax for the MOSFET element is the same as the other standard MOSFET models, other than `PS` and `PD`. In Level 63, `PS` and `PD` are the length of the sidewall of the source/drain, which is not under the gate.
13. Philips MOS11 has its own `LMIN` parameter, which has a different definition from that of the PrimeSim Continuum solution. To avoid the conflict with `LMIN` in simulation, Synopsys changed the `LMIN` parameter in the Level 63 MOSFET model to `LLMIN`.

Philips MOS11 Model Parameters

This section covers the following topics:

- [Philips MOS11 Model — Control, Geometry, and Self-Heating Parameters](#)
- [Philips MOS11 Model — Physical Geometry Scaling](#)
- [Philips MOS11 Model — Binning Geometry Scaling](#)
- [Philips MOS11 Model — JUNCAP Parameters](#)

Philips MOS11 Model — Control, Geometry, and Self-Heating Parameters

Name (Alias)	Description	Units	NMOS	PMOS
LEVEL	Level of this model	-	63	63
VERSION	Version of this model		1102.1	1102.1
BINFLAG	Flag for binned model • 0: Physical model • 1: Binned model	-	0	0
ATH	Temperature coefficient of thermal resistance	-	0	0
CTH	Thermal capacitance	J/K-	3e-9	3e-9
DTA	Temperature offset of the device	°C	0	0
LAP	Effective channel length reduction per side, due to the lateral diffusion of source/drain dopant ions	m	4e-8	4e-8
LER	Effective channel length, reference transistor	m	1e-6	1e-6
LVAR	Difference between the actual and programmed poly silicon gate length	m	0	0
RTH	Thermal resistance	K/W	300	300
SHMOD	Flag for self-heating • 0: No self-heating • 1: Self-heating	-	0	0
TR	Temperature at which simulation determines the reference transistor parameters	°C	21	21
WER	Effective channel width, reference transistor	m	1e-5	1e-5
WOT	Effective reduction of the channel width per side due to the lateral diffusion of the channel stop dopant ions	m	0	0
WVAR	Difference between the actual and programmed field oxide opening	m	0	0

Philips MOS11 Model — Physical Geometry Scaling

Name (Alias)	Description	Units	NMOS	PMOS
A1R	Weak-avalanche current factor for reference transistor at reference temperature	-	6	6
A2R	Exponent of the weak-avalanche current for the reference transistor	V	38	38
A3R	Drain-source voltage factor, above which weak-avalanche occurs for reference transistor	-	1	1
AGIDLR	Gain factor for gate-induced drain leakage current for a channel width of 1 μm	AV^{-3}	0	0
ALPEXP	Exponent of the length dependence of α	-	1	1
ALPR	Factor of the channel length modulation for the reference transistor	-	1e-2	1e-2
BACC	Probability factor for intrinsic gate tunneling current in accumulation	V	48	48
BETSQ	Gain factor for an infinite square transistor at the reference temperature	AV^{-2}	3.709e-4	1.15e-4
BGIDL	Probability factor for gate-induced drain leakage current at the reference temperature	V	41.0	41.0
BINV	Probability factor for intrinsic gate tunneling current in inversion	V	48	87.5
CGIDL	Factor for the lateral field dependence of the gate-induced drain leakage current		0	0
COL	Gate overlap capacitance per unit channel length	F	3.2e-16	3.2e-16
ETABETR (ETABET)	Exponent of the temperature dependence of the gain factor	-	1.3	0.5
ETAMOBR	Effective field parameter, depletion/ inversion charge dependence for reference transistor	-	1.4	3
ETAPH	Exponent of the temperature dependence of θ_{sr} for the reference temperature	-	1.35	3.75
ETAR	Exponent of temperature dependence of θ_{R}	-	0.95	0.4
ETASAT	Exponent of temperature dependence of θ_{sat}	-	1.04	0.86

Name (Alias)	Description	Units	NMOS	PMOS
ETASR	Exponent of temperature dependence of θ_{sr}	-	0.65	0.5
FBET1	Relative mobility decrease due to first lateral profile	-	0	0
FBET2	Relative mobility decrease due to second lateral profile	-	0	0
GATENOISE	In/exclusion flag of induced gate thermal noise	-	0	0
IGACCR	Gain factor for intrinsic gate tunneling current in accumulation for reference transistor	AV^{-2}	0	0
IGINVR	Gain factor for intrinsic gate tunneling current in inversion for reference transistor	AV^{-2}	0	0
IGOVR	Gain factor for Source/Drain overlap tunneling current for reference transistor	AV^{-2}	0	0
KOR	Body-effect factor for the reference transistor	$V^{1/2}$	0.5	0.5
KOV	Body-effect factor for the Source/Drain overlap extensions	$V^{1/2}$	2.5	2.5
KPINV	Inverse of the body-effect factor, poly-silicon gate	$V^{-1/2}$	0	0
LLMIN	Minimum effective channel length in technology, calculates smoothing factor m	M	1.5e-7	1.5e-7
LP1	Characteristic length of first lateral profile	m	8e-7	8e-7
LP2	Characteristic length of second lateral profile	M	8e-7	8e-7
MOEXP	Exponent of the length dependence of m_0	-	1.34	1.34
MOO	Parameter for short-channel subthreshold slop		0	0
MOR	Parameter for short-channel subthreshold slope for the reference transistor	-	0	0
NFAR	First coefficient of the flicker noise for the reference transistor	$V^{-1}m^{-4}$	1.573e23	3.825e24
NFBR	Second coefficient of the flicker noise for the reference transistor	$V^{-1}m^{-2}$	4.752e9	1.015e9
NFCR	Third coefficient of the flicker noise for the reference transistor	V^{-1}	0	7.3e-8

Name (Alias)	Description	Units	NMOS	PMOS
NT	Thermal noise coefficient at the actual temperature	J	1.624e-20	1.624e-20
NU (NUR)	Exponent of field dependence, mobility model minus 1 (such as $v-1$) at reference temperature	-	2	2
NUEXP	Exponent of the temperature dependence of parameter v	-	5.25	3.23
PHIBR	Surface potential at the onset of strong inversion at the reference temperature	V	0.95	0.95
SDIBLEXP	Exponent of length dependence of σ_{DIBL}	-	1.35	1.35
SDIBLO	Drain-induced barrier-lowering parameter for the reference transistor	$V^{-1/2}$	1e-4	1e-4
SL2KO	Second coefficient of the length dependence of k_0	-	0	0
SL2PHIB	Second coefficient of length dependence of ϕ_B	-	0	0
SLA1	Coefficient of the length dependence of a^1	-	0	0
SLA2	Coefficient of the length dependence of a^2	-	0	0
SLA3	Coefficient of the length dependence of a^3	-	0	0
SLALP	Coefficient of the length dependence of α	-	1	1
SLETABET	Length dependence coefficient of $\eta\beta_R$	-	0	0
SLKO	Coefficient of the length dependence of k_0	-	0	0
SLPHIB	Coefficient of the length dependence of ϕ_B	-	0	0
SLSSF	Length dependence coefficient of σ_{sf}	-	1.0	1.0
SLTHESAT	Length dependence coefficient of θ_{sat}	-	1	1
SSFR	Static feedback parameter, reference transistor	$V^{-1/2}$	6.25e-3	6.25e-3
STA1	Temperature dependence coefficient of a^1	K^{-1}	0	0
STBGIDL	Coefficient of the temperature dependence of $VK^{-1} B_{GIDL}$	K^{-1}	-3.638e-4	-3.638e-4
STETAMOB	Temperature dependence coefficient of η_{mob}	K^{-1}	0	0

Name (Alias)	Description	Units	NMOS	PMOS
STPHIB	Temperature dependence coefficient of ϕ_B	VK^{-1}	-8.5e-4	-8.5e-4
STVFB	Temperature dependence coefficient of V_{FB}	V/K	5e-4	5e-4
SWA1	Coefficient of the width dependence of a^1	-	0	0
SWA2	Coefficient of the width dependence of a^2	-	0	0
SWA3	Coefficient of the width dependence of a^3	-	0	0
SWALP	Coefficient of the width dependence of α	-	0	0
SWETAMOB	Width dependence coefficient of η_{mob}	-	0	0
SWKO	Coefficient of the width dependence of k_0	-	0	0
SWPHIB	Coefficient of the width dependence of ϕ_B	-	0	0
SWSSF	Coefficient of the width dependence of σ_{sf}	-	0	0
SWTHEPH	Coefficient of the width dependence of θ_{sr}	-	0	0
SWTHER	Coefficient of the width dependence of θ_R	-	0	0
SWTHESAT	Width dependence coefficient of θ_{sat}	-	0	0
SWTHESR	Coefficient of the width dependence of θ_{sr}	-	0	0
SWTHETH	Coefficient of the width dependence of θ_{TH}	-	0	0
THEPHR	Coefficient of the mobility reduction due to phonon scattering for the reference transistor at the reference temperature	V^{-1}	1.29e-2	1e-3
THER1	Numerator of gate voltage dependent part of series resistance for reference transistor	V	0	0
THER2	Denominator of gate voltage dependent part of series resistance for the reference transistor	V	1	1
THERR	Series resistance coefficient for reference transistor at reference temperature	V^{-1}	0.155	0.08
THESATEXP	Exponent of length dependence of θ_{sat}	-	1	1
THESATR	Velocity saturation parameter due to optical/acoustic phonon scattering for the reference transistor at the reference temperature	V^{-1}	0.5	0.2

Name (Alias)	Description	Units	NMOS	PMOS
THESRR	Mobility reduction coefficient, due to surface roughness scattering for reference transistor at reference temperature	V^{-1}	0.4	0.73
THETHEXP	Exponent of the length dependence of θ_{TH}	-	1	1
THETHR	Coefficient of self-heating for the reference transistor at the reference temperature	V^{-3}	1e-3	0.5e-3
TOX	Thickness of the gate oxide layer	M	3.2e-9	3.2e-9
VFBOV	Flat-band voltage for the Source/Drain overlap extensions	V	0	0
VFBR (VFB)	Flat-band voltage for the reference transistor at the reference temperature	V	-1.05	-1.05
VP	Characteristic voltage of the channel length modulation	V	5e-2	5e-2

Philips MOS11 Model — Binning Geometry Scaling

Name (Alias)	Description	Units	NMOS	PMOS
GATENOISE	Inclusion/exclusion flag of the induced gate thermal noise	-	0	0
KOV	Body-effect factor for the source/drain overlap extensions	$V^{1/2}$	2.5	2.5
KPINV	Inverse of body-effect factor, poly-silicon gate	$V^{-1/2}$	0	0
NT	Coefficient for the thermal noise at the reference temperature	J	1.656e-20	1.656e-20
NU	Exponent of the field dependence of the mobility model at the reference temperature	-	2	2
PLA1	Coefficient for the length-dependent part of a_1	-	0	0
PLA2	Coefficient for the length-dependent part of a_2	V	0	0
PLA3	Coefficient for the length-dependent part of a_3	-	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PLAGIDL	Coefficient for the length dependence of A_{GIDL}	AV^{-3}	0	0
PLALP	Coefficient for the length-dependent part of α	-	0	0
PLBACC	Coefficient for the length-dependent part of B_{ACC}	V	0	0
PLBET	Coefficient for the length dependent of β	AV^{-2}	0	0
PLBGIDL	Coefficient for the length dependence of B_{GIDL}	V	0	0
PLBINV	Coefficient for the length-dependent part of I_{GINV}	V	0	0
PLCGDO	Coefficient for the length-dependent part of C_{GDO}	F	0	0
PLCGIDL	Coefficient for the length dependence of C_{GIDL}		0	0
PLCGSO	Coefficient for the length-dependent part of C_{GSO}	F	0	0
PLCOX	Coefficient for the length-dependent part of C_{OX}	F	0	0
PLETAMOB	Coefficient, length-dependent η_{mob} part	-	0	0
PLIGACC	Coefficient for the length-dependent part of I_{GACC}	AV^{-2}	0	0
PLIGINV	Coefficient for the length-dependent part of I_{GINV}	AV^{-2}	0	0
PLIGOV	Coefficient for the length-dependent part of I_{GOV}	AV^{-2}	0	0
PLKO	Coefficient for the length dependent of k_0	$V^{1/2}$	0	0
PLMEXP	Coefficient for the length-dependent part of $1/m$	-	0	0
PLMO	Coefficient for the length-dependent part of m_0	-	0	0
PLNFA	Coefficient for the length-dependent part of N_{FA}	$V^{-1}m^{-4}$	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PLNFB	Coefficient for the length-dependent part of N_{FB}	$V^{-1}m^{-2}$	0	0
PLNFC	Coefficient for the length-dependent part of N_{FC}	V^{-1}	0	0
PLPHIB	Coefficient for the length dependent of Φ_B	V	0	0
PLSDIBL	Coefficient for the length-dependent part of σ_{dibl}	$V^{-1/2}$	0	0
PLSSF	Coefficient for the length-dependent part of σ_{sf}	$V^{-1/2}$	0	0
PLTA1	Coefficient for the length-dependent part of $S_T:a$	K^{-1}	0	0
PLTBGIDL	Coefficient for the length dependence of $S_T:B_{GIDL}$	VK^{-1}	0	0
PLTETABET	Coefficient for the length-dependent part of $\eta\beta$	-	0	0
PLTETAMOB	Coefficient for the length-dependent part of η_{ph}	K^{-1}	0	0
PLTETAPH	Coefficient for the length-dependent part of η_{ph}	-	0	0
PLTETAR	Coefficient for the length-dependent part of η_R	-	0	0
PLTETASAT	Coefficient for the length-dependent part of η_{sat}	-	0	0
PLTETASR	Coefficient for the length-dependent part of η_{sr}	-	0	0
PLTHEPH	Coefficient for the length dependent of θ_{ph}	V^{-1}	0	0
PLTHER	Coefficient for the length-dependent part of θ_R	V^{-1}	0	0
PLTHESAT	Coefficient for length-dependent part of θ_{sat}	V^{-1}	0	0
PLTHESR	Coefficient for the length dependent part of θ_{sr}	V^{-1}	0	0
PLTHETH	Coefficient for the length-dependent part of θ_{TH}	V^{-3}	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PLTNUEXP	Coefficient for the length-dependent part of V_{exp}	-	0	0
PLTPHIB	Coefficient for the length-dependent part of $ST; \phi_B$	VK^{-1}	0	0
PLTVFB	Coefficient for the length-dependent part of $ST; V_{FB}$	VK^{-1}	0	0
PLWA1	Coefficient for the length times width a^1 dependent part	-	0	0
PLWA2	Coefficient of the length times width a^2 dependent part	V	0	0
PLWA3	Coefficient for the length times width a^3 dependent part	-	0	0
PLWAGIDL	Coefficient for the width over length dependence of A_{GIDL}	AV^{-3}	0	0
PLWALP	Coefficient for the length times width dependent part of α	-	0	0
PLWBACC	Coefficient for the length times width dependent part of B_{ACC}	V	0	0
PLWBET	Coefficient, width over length dependent of β	AV^{-2}	0	0
PLWBGIDL	Coefficient for the length times width dependence of B_{GIDL}	V	0	0
PLWBINV	Coefficient for the length times width dependent part of I_{GINV}	V	0	0
PLWCGDO	Coefficient for the width over length dependent part of C_{GDO}	F	0	0
PLWCGIDL	Coefficient for the length times width dependence of C_{GIDL}		0	0
PLWCGSO	Coefficient for the width over length dependent part of C_{GSO}	F	0	0
PLWCOX	Coefficient for the length times width dependent part of C_{OX}	F	0	0
PLWETAMOB	Coefficient, length times width dependent η_{mob} part	-	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PLWHEPH	Coefficient, length times width, θ_{ph} dependent	V^{-1}	0	0
PLWIGACC	Coefficient for the length times width dependent part of I_{GACC}	AV^{-2}	0	0
PLWIGINV	Coefficient for the length times width dependent part of I_{GINV}	AV^{-2}	0	0
PLWIGOV	Coefficient for the width over length dependent part of I_{GOV}	AV^{-2}	0	0
PLWKO	Coefficient, length times width k_0 dependent	$V^{1/2}$	0	0
PLWMEXP	Coefficient of the length times width 1/m dependent part	-	0	0
PLWMO	Coefficient for the length times width m_0 dependent part	-	0	0
PLWNFA	Coefficient for the length times width dependent part of N_{FA}	$V^{-1}m^{-4}$	0	0
PLWNFB	Coefficient for the length times width dependent part of N_{FB}	$V^{-1}m^{-2}$	0	0
PLWNFC	Coefficient for the length times width dependent part of N_{FC}	V^{-1}	0	0
PLWPHIB	Coefficient, length times width Φ_B dependent	V	0	0
PLWSDIBL	Coefficient, length times width dependent of σ_{dibl}	$V^{-1/2}$	0	0
PLWSSF	Coefficient for the length times width σ_{sf} dependent part	$V^{-1/2}$	0	0
PLWTA1	Coefficient for the length times width dependent part of $S_T:a_1$	K^{-1}	0	0
PLWTBGIDL	Coefficient for the length times width dependence of $S_T:B_{GIDL}$	VK^{-1}	0	0
PLWTETA BET	Coefficient for the length times width η_{sr} dependent part	-	0	0
PLWTETA MOB	Coefficient for the length times width dependent part of η_{ph}	K^{-1}	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PLWTETAPH	Coefficient for the length times width η_{ph} dependent part	-	0	0
PLWTETAR	Coefficient for the length times width dependent part of η_R	-	0	0
PLWTETASAT	Coefficient for the length times width dependent part of η_{sat}	-	0	0
PLWTETASR	Coefficient for the length times width dependent part of η_{sr}	-	0	0
PLWOTHER	Coefficient, length times width θ_R dependent	V^{-1}	0	0
PLWTHESAT	Coefficient, length times width θ_{sat} dependent	V^{-1}	0	0
PLWTHESR	Coefficient, length times width, θ_{sr} dependent	V^{-1}	0	0
PLWTHETH	Coefficient for the length times width θ_{TH} dependent part	V^{-3}	0	0
PLWTNUEXP	Coefficient fore the length times width dependent part of v_{exp}	-	0	0
PLWTPHIB	Coefficient for the length times width dependent part of $ST; \phi_B$	VK^{-1}	0	0
PLWTVFB	Coefficient for the length times width dependent part of $ST; V_{FB}$	VK^{-1}	0	0
POA1	Coefficient of the geometry-independent a^1 part	-	6.022	6.858
POA2	Coefficient for the geometry-independent a^2 part	V	3.802e1	5.732e1
POA3	Coefficient of the geometry-independent a^3 part	-	6.407e-1	4.254e-1
POAGIDL	Coefficient for the geometry-independent part of A_{GIDL}	AV^{-3}	0	0
POALP	Coefficient for the geometry-independent part of α	-	2.5e-2	2.5e-2
POBACC	Coefficient for the geometry-independent part of B_{ACC}	V	48	48

Name (Alias)	Description	Units	NMOS	PMOS
POBET	Coefficient, geometry independent β part	AV^{-2}	1.922e-3	3.814e-4
POBGIDL	Coefficient for the geometry independent part of B_{GIDL}	V	41.0	41.0
POBINV	Coefficient for the geometry-independent part of I_{GINV}	V	48	87.5
POCGDO	Coefficient for the geometry-independent C_{GDO} part	F	6.392e-15	6.358e-15
POCGIDL	Coefficient for the geometry independent part of C_{GIDL}		0	0
POCGSO	Coefficient for the geometry-independent part of C_{GSO}	F	6.392e-15	6.358e-15
POCOX	Coefficient for the geometry-independent C_{OX} part	F	2.98e-14	2.717e-14
POETAMOB	Coefficient, geometry independent η_{mob} part	-	1.40	3
POIGACC	Coefficient for the geometry-independent part of I_{GACC}	AV^{-2}	0	0
POIGINV	Coefficient for the geometry-independent part of I_{GINV}	AV^{-2}	0	0
POIGOV	Coefficient for the geometry-independent part of I_{GOV}	AV^{-2}	0	0
POKO	Coefficient, geometry independent k_0 part	$V^{1/2}$	0.5	0.5
POMEXP	Coefficient for the geometry-independent $1/m$ part	-	0.2	0.2
POMO	Coefficient for the geometry-independent part of m_0	-	0	0
PONFA	Coefficient for the geometry-independent N_{FA} part	$V^{-1}m^{-4}$	8.323e22	1.90e22
PONFB	Coefficient for the geometry-independent N_{FB} part	$V^{-1}m^{-2}$	2.514e7	5.043e6
PONFC	Coefficient for the geometry-independent N_{FC} part	V^{-1}	0	3.627e-10
POPHIB	Coefficient, geometry independent Φ_B part	V	0.950	0.950

Name (Alias)	Description	Units	NMOS	PMOS
POSDIBL	Coefficient for the geometry-independent part of σ_{dibl}	$V^{-1/2}$	8.53e-4	3.551e-5
POSSF	Coefficient for the geometry-independent part of σ_{sf}	$V^{-1/2}$	1.2e-2	1.0e-2
POTA1	Coefficient for the geometry-independent $S_T:a$ part	K^{-1}	0	0
POTBGIDL	Coefficient for the geometry independent part of $S_T:B_{GIDL}$	VK^{-1}	-3.638e-4	-3.638e-4
POTETABET	Coefficient for the geometry-independent part of $\eta\beta$	-	1.30	0.5
POTETAMOB	Coefficient for the geometry-independent part of $ST:\eta_{ph}$	K^{-1}	0	0
POTETAPH	Coefficient for the geometry-independent η_{ph} part	-	1.35	3.75
POTETAR	Coefficient for the geometry-independent η_R - part	-	0.95	0.4
POTETASAT	Coefficient for the geometry-independent part of η_{sat}	-	1.04	0.86
POTETASR	Coefficient for the geometry-independent η_{sr} part	-	0.65	0.5
POTHEPH	Coefficient, geometry independent θ_{ph} part	V^{-1}	1.290e-2	1.0e-3
POTHER	Coefficient for the geometry-independent part of θ_R	V^{-1}	8.12e-2	7.9e-2
POTHSAT	Coefficient for the geometry-independent part of θ_{sat}	V^{-1}	2.513e-1	1.728e-1
POTHSR	Coefficient, geometry independent θ_{sr} part	V^{-1}	3.562e-1	7.30e-1
POTHETH	Coefficient for the geometry-independent part of θ_{TH}	V^{-3}	1.0e-5	0
POTNUEXP	Coefficient for the geometry-independent v_{exp} part	-	5.25	3.23
POTPHIB	Coefficient for the geometry-independent part of $ST;\phi_B$	VK^{-1}	-8.5e-4	-8.5e-4
POTVFB	Coefficient for the geometry-independent part of $ST;V_{FB}$	VK^{-1}	5.0e-4	5.0e-4

Name (Alias)	Description	Units	NMOS	PMOS
PWA1	Coefficient for the width-dependent part of a_1	-	0	0
PWA2	Coefficient for the width-dependent part of a_2	V	0	0
PWA3	Coefficient for the width-dependent part of a_3	-	0	0
PWAGIDL	Coefficient for the width dependence of A_{GIDL}	AV^{-3}	0	0
PWALP	Coefficient for the width-dependent part of α	-	0	0
PWBACC	Coefficient for the width-dependent part of B_{ACC}	V	0	0
PWBET	Coefficient for the width dependent of β	AV^{-2}	0	0
PWBGIDL	Coefficient for the width dependence of B_{GIDL}	V	0	0
PWBINV	Coefficient for the width-dependent part of I_{GINV}	V	0	0
PWCGDO	Coefficient for the width dependent part of C_{GDO}	F	0	0
PWCGIDL	Coefficient for the width dependence of C_{GIDL}		0	0
PWCGSO	Coefficient for the width-dependent part of C_{GSO}	F	0	0
PWCOX	Coefficient for the width-dependent part of C_{OX}	F	0	0
PWETAMOB	Coefficient, width-dependent η_{mob} part	-	0	0
PWIGACC	Coefficient for the width-dependent part of I_{GACC}	AV^{-2}	0	0
PWIGINV	Coefficient for the width-dependent part of I_{GINV}	AV^{-2}	0	0
PWIGOV	Coefficient for the width-dependent part of I_{GOV}	AV^{-2}	0	0
PWKO	Coefficient for the width dependent of k_0	$V^{1/2}$	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PWMEXP	Coefficient for the width dependent part of $1/m$	-	0	0
PWMO	Coefficient for the width-dependent part of m_0	-	0	0
PWNFA	Coefficient for the width-dependent part of N_{FA}	$V^{-1}m^{-4}$	0	0
PWNFB	Coefficient for the width-dependent part of N_{FB}	$V^{-1}m^{-2}$	0	0
PWNFC	Coefficient for the width-dependent part of N_{FC}	V^{-1}	0	0
PWPHIB	Coefficient for the width dependent of Φ_B	V	0	0
PWSDIBL	Coefficient for the width-dependent part of σ_{dibl}	$V^{-1/2}$	0	0
PWSSF	Coefficient for the width-dependent part of σ_{sf}	$V^{-1/2}$	0	0
PWTA1	Coefficient for the width-dependent part of $S_T:a$	K^{-1}	0	0
PWTBGIDL	Coefficient for the width dependence of $S_T:B_{GIDL}$	VK^{-1}	0	0
PWTETABET	Coefficient for the width-dependent part of $\eta\beta$	-	0	0
PWTETAMOB	Coefficient for the width-dependent part of η_{ph}	K^{-1}	0	0
PWTETAPH	Coefficient for the width-dependent part of η_{ph}	-	0	0
PWTETAR	Coefficient for the width-dependent part of η_R	-	0	0
PWTETASAT	Coefficient for the width-dependent part of η_{sat}	-	0	0
PWTETASR	Coefficient for the width-dependent part of η_{sr}	-	0	0
PWTHEPH	Coefficient for the width dependent of θ_{ph}	V^{-1}	0	0
PWOTHER	Coefficient for the width dependent part of θ_R	V^{-1}	0	0

Name (Alias)	Description	Units	NMOS	PMOS
PWTHESAT	Coefficient for the width-dependent part of θ_{sat}	V^{-1}	0	0
PWTHESR	Coefficient for the width dependent part of θ_{sr}	V^{-1}	0	0
PWTHETH	Coefficient for the width-dependent part of θ_{TH}	V^{-3}	0	0
PWTNUEXP	Coefficient for the width-dependent part of V_{exp}	-	0	0
PWTPHIB	Coefficient for the width-dependent part of $ST; \phi_B$	VK^{-1}	0	0
PWTVFB	Coefficient for the width-dependent part of $ST; V_{FB}$	VK^{-1}	0	0
THER1	Numerator of the gate voltage dependent part of the series resistance for all transistors in bin	V	0	0
THER2	Denominator of the gate voltage dependent part of the series resistance for all transistors in the bin	V	1.0	1.0
TOX	Thickness of the gate oxide layer	m	3.2e-9	3.2e-9
VFB	Flat-band voltage for the reference transistor at the reference temperature	V	-1.05	-1.05
VFBOV	Flatband voltage for the source/drain overlap extensions	V	0	0
VP	Characteristic voltage of the channel length modulation	V	5e-2	5e-2

Philips MOS11 Model — JUNCAP Parameters

Name	Description	Units	Default
CJBR	Bottom junction capacitance at $V=V_R$	Fm^{-2}	1e-12
CJGR	Gate edge junction capacitance at $V=V_R$	Fm^{-1}	1e-12
CJSR	Sidewall junction capacitance at $V=V_R$	Fm^{-1}	1e-12
DTA	Temperature offset of the JUNCAP element with respect to T_A	$^{\circ}C$	0

Name	Description	Units	Default
JSDBR	Bottom saturation-current density due to diffusion from back contact	Am^{-2}	1e-03
JSDGR	Gate edge saturation-current density due to back-contact diffusion	Am^{-1}	1e-03
JSDSR	Sidewall saturation-current density due to back-contact diffusion	Am^{-1}	1e-03
JSGBR	Bottom saturation-current density due to generating an electron hole at $V=V_R$	Am^{-2}	1e-03
JSGGR	Gate edge saturation-current density due to generating an electron hole at $V=V_R$	Am^{-1}	1e-03
JSGSR	Sidewall saturation-current density due to generating an electron hole at $V=V_R$	Am^{-1}	1e-03
NB	Emission coefficient of the bottom forward current	-	1
NG	Emission coefficient of the gate edge forward current	-	1
NS	Emission coefficient of the sidewall forward current	-	1
PB	Bottom junction grading coefficient	-	0.4
PG	Gate edge junction grading coefficient	-	0.4
PS	Sidewall junction grading coefficient	-	0.4
VDBR	Diffusion voltage of the bottom junction at $T=T_R$	V	1
VDGR	Diffusion voltage of the gate-edge junction at $T=T_R$	V	1
VDSR	Diffusion voltage of the sidewall junction at $T=T_R$	V	1
VR	Voltage at which simulation determines parameters	V	0

Note:

All symbols refer to “Unclassified Report NL-UR 2001/813”.

Philips MOS11 Model Examples

Following are some examples of the Philips MOS11 Model implementation:

Example 4

```
.model nch nmos level=63
+ VERSION=1100
```

Chapter 2: MOSFET Models

LEVEL 63: Philips MOS11 Model

```

+ LER=1E-06 WER=1E-05 LAP=-1.864E-08
+ TR=21 VFBR=-1.038 SLPHIB=-1.024E-08
+ SL2PHIB=1.428E-14 KOR=5.763E-01 SLKO=2.649E-08
+ SL2KO=-1.737E-14 KPINV=2.2E-01 PHIBR=0.85
+ BETSQ=1.201E-04 ETABET=1.3 FBET1=-3.741000E-01
+ LP1=2.806E-06 LP2=1E-10 THESATEXP=2
+ THESRR=7.109E-01 THEPHR=1E-03 TOX=3.2E-09
+ ETAPH=1.75E+00 ETAMOB=2.825 NUR=1
+ NUEXP=3.228 THERR=1.267E-01 ETAR=0.4
+ THER2=1 THESATR=6.931E-02 SLTHESAT=1
+ ETASAT=8.753E-01 SSFR=2.304E-03 VP=5E-02
+ SLSSF=1.002E-06 ALPR=1.062E-02 SLALP=9.957E-01
+ ALPEXP=1.039 THETHR=2.413E-03 THETHEXP=1
+ SDIBLO=1.06E-06 SDIBLEXP=6.756 LLMIN=2E-07
+ MOR=1.05E-03 MOEXP=3.146
+ A1R=9.938E+04 STA1=9.3E-02 SLA1=-2.805E-03
+ A2R=4.047E+01 SLA2=1E-15
+ A3R=7.54E-01 SLA3=-8.705E-08
+ COL=3.2E-10
+ NTR=1.6237E-20 NFAR=1 NFBR=0
+ NFCR=0 GATENOISE=0
+ CJSR=1.347E-3 CJSR=0.183E-9 CJGR=0.374E-9
+ JSDBR=0.027E-6 JSDBR=0.040E-12 JSDBR=0.100E-12
+ VR=0.000
+ JSGBR=1.900E-6 JSGBR=78.000E-12 JSGBR=54.000E-12
+ VB=20.000
+ VDBR=0.828 VDSR=0.593 VDGR=0.500
+ PB=0.394 PS=0.171 PG=0.193
+ NB=1.000 NS=1.000 NG=1.000

```

Example 5

```

.model nch nmos level=63
+ VERSION=11010
+ LVAR=0.000000E+00
+ LAP=-1.864000E-08 WVAR=0.000000E+00 WOT=0.000000E+00
+ TR=2.100000E+01 VFB=-1.038000E+00 STVFB=0.000000E+00
+ SLPHIB=-1.024000E-08 SL2PHIB=1.428000E-14 SWPHIB=0.000000E+00
+ KOR=5.763000E-01 SLKO=2.649000E-08 SL2KO=-1.737000E-14
+ SWKO=0.000000E+00 KPINV=2.200000E-01 PHIBR=8.500000E-01
+ BETSQ=1.201000E-04 ETABETR=1.300000E+00 FBET1=-3.741000E-01
+ LP1=2.806000E-06 FBET2=0.000000E+00 LP2=1.000000E-10
+ THESRR=7.109000E-01 SWTHESR=0.000000E+00 THEPHR=1.000000E-03
+ ETAPH=1.750000E+00 SWTHEPH=0.000000E+00 ETAMOB=2.825000E+00
+ STETAMOB=0.000000E+00 SWETAMOB=0.000000E+00 NU=1.000000E+00
+ NUEXP=3.228000E+00 THERR=1.267000E-01 ETAR=4.000000E-01
+ SWTHER=0.000000E+00 THER1=0.000000E+00 THER2=1.000000E+00
+ THESATR=6.931000E-02 SLTHESAT=1.000000E+00 THESATEXP=2.000000E+00
+ ETASAT=8.753000E-01 SWTHESAT=0.000000E+00 SSFR=2.304000E-03
+ SLSSF=1.002000E-06 SWSSF=0.000000E+00 ALPR=1.062000E-02
+ SLALP=9.957000E-01 ALPEXP=1.039000E+00 SWALP=0.000000E+00
+ VP=5.000000E-02 THETHR=2.413000E-03 THETHEXP=1.000000E+00
+ SWTHETH=0.000000E+00 SDIBLO=1.060000E-06 SDIBLEXP=6.756000E+00

```

```
+ MOR=1.050000E-03 MOEXP=3.146000E+00 LLMIN=2.000000E-07
+ A1R=9.938000E+04 STA1=9.300000E-02 SLA1=-2.805000E-03
+ SWA1=0.000000E+00 A2R=4.047000E+01 SLA2=1.000000E-15
+ SWA2=0.000000E+00 A3R=7.540000E-01 SLA3=-8.705000E-08
+ SWA3=0.000000E+00 TOX=3.200000E-09 COL=3.200000E-10
+ NT=1.623700E-20 NFAR=1.000000E-00 NFBR=0.000000E+00
+ NFCR=0.000000E+00 GATENOISE =0.000000E-00 DTA=0.000000E-00
```

Example 6

```
.model nch nmos level=63
+ LEVEL =63
+ VERSION=11011
+ LVAR=0 LAP=4.0E-08 WVAR=0.0 WOT=0.0 TR=21 VFB=-0.105E+01
+ POKO=0.5 PLKO=0.0 PWKO=0.0 PLWKO=0.0 KPINV=0.0
+ POPHIB=0.95 PLPHIB=0.0 PWPHIB=0.0 POBET=1.922E-03
+ POTHSR=3.562E-01 POTHEPH=1.29E-02 POETAMOB=1.4 POTHER=8.120E-02
+ THER1=0.0 THER2=0.1E+01 POTHSAT=0.2513 POTHETH =1.0E-5
+ POSDIBL=8.530E-4 POSSF=1.2E-2 POALP=2.5E-2 VP=5.0E-2
+ POMEXP=0.2 POA1=6.022 POA2=38.02 POA3=0.6407
+ POBINV=48 POBACC=48 KOV=2.5 TOX=3.2E-09
+ POCOX=2.980E-14 POCGDO=6.392E-15 POCGSO=6.392E-15
+ NT=1.656E-20 PONFA=8.323E+22 PONFB=2.514E+7
+ POTVFB=5.0E-4 POTPHIB=-8.5E-4
+ POTETABET=1.30 POTETASR=0.65 POTETAPH=1.350 NU=2.0
+ POTNUEXP=5.25 POTETAR=0.95 POTETASAT=1.040
```

LEVEL 64: STARC HiSIM MOSFET Model

HiSIM (Hiroshima-University STARC IGFET Model) is a publicly-available MOSFET model for circuit simulation. It uses drift-diffusion approximation, and a channel-surface-potential description. You can model all MOSFET characteristics closely, based on their physical origins by using fewer model parameters (about 90 model parameters); each parameter set is sufficient for all gate lengths. These model parameters are directly related to the MOSFET physics that a simulator can easily extract according to its physical meanings.

The STARC HiSIM model is Level 64 in the Synopsys MOSFET models. To use this model, specify:

```
M1 drain gate source bulk NCH w=4u l=1u
.MODEL NCH NMOS LEVEL=64
```

Since the STARC HiSIM1.1.0 release, the Synopsys version of the HiSIM model has included a `VERSION` number parameter to facilitate backward compatibility. Starting in the 2003.03 release, Synopsys uses the STARC version control mechanism so you must enter an integer for the `VERSION` model parameter. For example, to specify HiSIM version 1.0.0, set the `VERSION` model parameter to 100. If you do not set the `VERSION` parameter, simulation issues a warning and automatically sets this parameter to 100.

You can set the `VERSION` value to:

- 100, 101, 102 (HiSIM1.0.* series)
- 110, 111, 112 (HiSIM1.1.* series)
- 120 (HiSIM1.2.*)

To turn off the model effects, use the following settings:

- Short-Channel Effect `SC1=SC2=SC3=0`
- Reverse-Short-Channel Effect `LP=0`
- Quantum-Mechanical Effect `QME1=QME2=QME3=0`
- Poly-Depletion Effect `PGD1=PGD2=PGD3=0`
- Channel-Length Modulation `CLM1=CLM2=CLM3=0`
- Narrow-Channel Effect `WFC=MUEPH2=0`

This section covers the following topics:

- [STARC HiSIM Model Parameters](#)

STARC HiSIM Model Parameters

This section covers the following topics:

- [STARC HiSIM Model — Model Selectors](#)
- [STARC HiSIM Model — Technological Parameters](#)
- [STARC HiSIM Model — Temperature Dependence Parameters](#)
- [STARC HiSIM Model — Quantum Effect Parameters](#)
- [STARC HiSIM Model — Poly Depletion Parameters](#)
- [STARC HiSIM Model — Short Channel Parameters](#)
- [STARC HiSIM Model — Narrow Channel Parameters](#)
- [STARC HiSIM Model — Mobility Parameters](#)
- [STARC HiSIM Model — Channel Length Modulation Parameters](#)
- [STARC HiSIM Model — Substrate Current Parameters](#)
- [STARC HiSIM Model — Current Parameters](#)
- [STARC HiSIM Model — GIDL Current Parameters](#)
- [STARC HiSIM Model — 1/f Noise Parameters](#)
- [STARC HiSIM Model — Conserving Symmetry at \$V_{ds}=0\$ for Short-Channel MOSFETS](#)
- [STARC HiSIM Model — MOS Diode](#)
- [STARC HiSIM Model — Subthreshold Swing](#)
- [STARC HiSIM Model — Parameter Version Defaults](#)

STARC HiSIM Model — Model Selectors

Parameter	Default	Description
LEVEL	64	Model selector
VERSION	100	Model version number
CORSRD	0(no)	Flag. Indicates whether to include the R_s and R_d contact resistors, and whether to solve equations iteratively. CORSRD=1(yes)

Parameter	Default	Description
COGIDL	0	Selects the gate induced drain leakage (GIDL) current model. • COGIDL=0 (yes) • COGIDL=1 (no) VERSION < 111 does not support this model.
COIIGS	0	Selects the gate tunneling current model. • COIIGS=0 (yes), • COIIGS=1 (no) VERSION < 111 does not support this model.
COISTI	0	Selects the shallow-trench-isolation (STI) leakage current. COISTI=0 (no) COISTI=1 (yes), only if VERSION ≥ 110.
COISUB	0	Substrate current model selector. • for VERSION < 110, COISUB=0 (yes), • otherwise, COISUB=1 (no)
CONOIS	0	1/f noise model selector. • CONOIS=0 (no) • CONOIS=1 (yes)
COOVLP	0	Overlap capacitance model selector. • COOVLP=-1, constant value • COOVLP=0, approximating the field linear reduction • COOVLP=1, considering the lateral impurity profile.
NOISE	5	Channel thermal and flicker noises combination selector. • NOISE=1 Channel thermal noise=SPICE2 model Flicker noise=SPICE2 model • NOISE=2 Channel thermal noise=HiSIM1 model for the BSIM3 model Flicker noise=HiSIM1 model • NOISE=3 Channel thermal noise=SPICE2 model Flicker noise=HiSIM1 model • NOISE=4 Channel thermal noise=HiSIM1 model for the BSIM3 model Flicker noise=SPICE2 model • NOISE=5 Channel thermal noise=NONE Flicker noise=HiSIM1 model

STARC HiSIM Model — Technological Parameters

Parameter	Default	Description
LP	0.0m	Pocket penetration length
NSUBC	5.94e+17cm-3	Substrate-impurity concentration
NSUBP	5.94e+17cm-3	Maxim pocket concentration

Parameter	Default	Description
RD	0.0ohm*m	Drain-contact resistance
RS	0.0ohm*m	Source-contact resistance
TOX	3.6e-9m	Oxide thickness
TPOLY	0.0m	Height of the gate poly-Si
VFBC	-0.722729V	Flat-band voltage
XJ	0.0m	Junction depth (if VERSION < 110)
XLD	0.0m	Gate-overlap length
XPOLYD	0.0m	Difference between the gate-poly and the design lengths
XQY	0.0m	Distance from the drain junction to the maximum electric field point (if VERSION ≥ 110)
XWD	0.0m	Gate-overlap width

STARC HiSIM Model — Temperature Dependence Parameters

Parameter	Default	Description
BGTMP1	9.03e-5eVK ⁻¹	Bandgap narrowing
BGTMP2	3.05e-7eVK ⁻²	Bandgap narrowing

STARC HiSIM Model — Quantum Effect Parameters

Parameter	Default	Description
QME1	0.0mV	Coefficient for the quantum mechanical effect
QME2	0.0V	Coefficient for the quantum mechanical effect
QME3	0.0m	Coefficient for the quantum mechanical effect

STARC HiSIM Model — Poly Depletion Parameters

Parameter	Default	Description
PGD1	0.0V	Strength of the poly depletion

Parameter	Default	Description
PGD2	0.0V	Threshold voltage of the poly depletion
PGD3	0.0	V_{ds} dependence of the poly depletion

STARC HiSIM Model — Short Channel Parameters

Parameter	Default	Description
PARL1	1.0	Strength of the lateral-electric-field gradient
PARL2	2.2e-8m	Depletion width of the channel/contact junction
SC1	13.5V ⁻¹	Short-channel coefficient 1
SC2	1.8V-2	Short-channel coefficient 2
SC3	0.0V ⁻² m	Short-channel coefficient 3
SCP1	0.0V ⁻¹	Short-channel coefficient 1 for the pocket
SCP2	0.0V-2	Short-channel coefficient 2 for the pocket
SCP3	0.0V ⁻² m	Short-channel coefficient 3 for the pocket

STARC HiSIM Model — Narrow Channel Parameters

Parameter	Default	Description
WFC	0.0m*F/cm ²	Voltage reduction
MUEPH2	0.0	Mobility reduction
NSTI	0.0cm-3	Substrate-impurity concentration at the shallow-trench-isolation (STI) edge, if VERSION ≥ 110
W0	0.0log(cm)	Minimum gate width
WSTI	0.0m	Width of the high-field region at the shallow-trench-isolation (STI), if VERSION ≥ 110
WVTHSC	0.0	Short-channel effect at the shallow-trench-isolation (STI) edge, if VERSION ≥ 110

STARC HiSIM Model — Mobility Parameters

Parameter	Default	Description
BB	2.0(NMOS) 1.0(PMOS)	High-field-mobility degradation
MUECB0	300.0cm ² /Vs	Coulomb scattering
MUECB1	30.0cm ² /Vs	Coulomb scattering
MUEPH0	0.295	Phonon scattering
MUEPH1	1.0e7	Phonon scattering
MUESR0	1.0	Surface-roughness scattering
MUESR1	7.0e8	Surface-roughness scattering
MUETMP	0.0	Temperature dependence of phonon scattering
NDEP	1.0	Coefficient of the effective-electric field
NINV	0.5	Coefficient of the effective-electric field
NINVD	0.0V ⁻¹	Modification of NINV
RPOCK1	0.0V ² *m ^{1/2} /A	Resistance coefficient caused by the potential barrier
RPOCK2	0.0V	Resistance coefficient caused by the potential barrier
RPOCP1	0.0	Resistance coefficient caused by the potential barrier, if VERSION ≥ 110
RPOCP2	0.0	Resistance coefficient caused by the potential barrier, if VERSION ≥ 110
VDS0	0.05V	Drain voltage for extracting low-field mobility
VMAX	1.0e7cm/s	Maximum saturation velocity
VOVER	0.0	Velocity overshoot effect
VOVERP	0.0	L _{gate} dependence of the velocity overshoot

STARC HiSIM Model — Channel Length Modulation Parameters

Parameter	Default	Description
CLM1	0.3	Hardness coefficient of the channel/contact junction

Parameter	Default	Description
CLM2	0.0	Coefficient for the Q_B contribution
CLM3	0.0	Coefficient for the Q_I contribution

STARC HiSIM Model — Substrate Current Parameters

Parameter	Default	Description
SUB1	$0.0V^{-1}$	Substrate current coefficient 1
SUB2	-70.0	Substrate current coefficient 2
SUB3	1.0	Substrate current coefficient 3

STARC HiSIM Model — Current Parameters

Parameter	Default	Description
GLEAK1	$0.0A \cdot V^{-3/2}/C$	Gate current coefficient 1
GLEAK2	0.0	Gate current coefficient 2
GLEAK3	0.0	Gate current coefficient 3

STARC HiSIM Model — GIDL Current Parameters

Parameter	Default	Description
GIDL1	$0.0A \cdot m \cdot V^{-3/2}/C$	GIDL current coefficient 1
GIDL2	$0.0V^{-1/2}/cm$	GIDL current coefficient 2
GIDL3	0.0	GIDL current coefficient 3

STARC HiSIM Model — 1/f Noise Parameters

Parameter	Default	Description
AF	1.0	SPICE2 flicker noise exponent
CIT	$0.0F/cm^2$	Capacitance caused by the interface trapped carriers

Parameter	Default	Description
EF	0.0	SPICE2 flicker noise frequency exponent
KF	0.0	SPICE2 flicker noise coefficient
NFALP	2.0e-15	Contribution of the mobility fluctuation
NFTRP	1.0e11	Ratio of trap density to the attenuation coefficient

STARC HiSIM Model — Conserving Symmetry at $V_{ds}=0$ for Short-Channel MOSFETS

Parameter	Default	Description
PZADD0	1.0e-3V	Symmetry conservation coefficient
VZADD0	1.0e-2V	Symmetry conservation coefficient

STARC HiSIM Model — MOS Diode

Parameter	Default	Description
CJ	$8.397247e-04Fm^{-2}$	Bottom junction capacitance per unit area at zero bias
CJSW	$5.0e-10Fm^{-1}$	Source/drain sidewall junction capacitance per unit area at zero bias
CJSWG	$5.0e-10Fm^{-1}$	Source/drain gate sidewall junction capacitance per unit area at zero bias
JSO	$1.0e-4Am^{-2}$	Saturation current density
JSOSW	$0.0Am^{-1}$	Sidewall saturation current density
MJ	0.5	Bottom junction capacitance grading coefficient
MJSW	0.33	Source/drain sidewall junction capacitance grading coefficient
MJSWG	0.33	Source/drain gate sidewall junction capacitance grading coefficient
NJ	1.0	Emission coefficient
NJSW	1.0	Sidewall emission coefficient

Parameter	Default	Description
PB	1.0V	Bottom junction build-in potential
PBSW	1.0V	Source/drain sidewall junction build-in potential
PBSWG	1.0V	Source/drain gate sidewall junction build-in potential
VDIFFJ	0.5V	Diode threshold voltage between source/drain and substrate
XTI	3.0	Junction current temperature exponent coefficient

STARC HiSIM Model — Subthreshold Swing

Parameter	Default	Description
PTHROU	0.0	Correction for steep subthreshold swing

Note:

Model parameter defaults in the above tables are valid only for versions 100 and 110. For other versions, please refer to the following table:

STARC HiSIM Model — Parameter Version Defaults

Parameter	Version=100, 110	Others
BGTMP1	9.03e-5	90.25e-6
BGTMP2	3.05e-7	100.0e-9
CJ	8.397247e-04	5.0e-04
CLM1	0.3	0.7
CLM2	0.0	2.0
CLM3	0.0	1.0
GIDL1	0.0	5.0e-3 for HiSIM101, 5.0e-6 for others
GIDL2	0.0	1.0e+6
GIDL3	0.0	0.3
GLEAK1	0.0	0.01e+6 for HiSIM101, 10.0e+3 for others
GLEAK2	0.0	20.0e+6

Parameter	Version=100, 110	Others
GLEAK3	0.0	0.3
LP	0.0	15.0e-9
MUEPH0	0.295	0.300
MUEPH1	1.00e+7	25.0e+3
MUESR0	1.0	2.0
MUESR1	7.00e+8	2.0e+15
MUETMP	0.0	1.5
NFALP	2.00e-15	1.0e-16
NFTRP	100.0e+9	10.0e+9
NINVD	0.0	1.0e-9
NSUBC	5.94e+17	1.0e+17
NSUBP	5.94e+17	1.0e+17
PARL2	2.20e-8	1.0e+17
PGD1	0.0	0.01
PGD2	0.0	1.0
PGD3	0.0	0.8
PZADD0	1.0e-3	5.0e-3
QME1	0.0	40.0e-12
QME2	0.0	300.0e-12
RD	0.0	80.0e-6
RPOCK1	0.0	0.01
RPOCK2	0.0	0.1
RPOCP1	0.0	1.0
RS	0.0	80.0e-6
SC1	13.5	0.0
SC2	1.8	0.0

Parameter	Version=100, 110	Others
SUB1	0.0	10.0
SUB2	-70.0	20.0
SUB3	1.0	0.8
TOX	3.60e-9	5.0e-9
VFBC	-0.722729	-1.0
VMAX	1.00e+7	7.00e+6
VOVER	0.0	0.01
VOVERP	0.0	0.1

LEVEL 65: SSIMSOI Model

Level 65 is a surface-potential, charge-based, and partially depleted SOI MOSFET model developed by Motorola semiconductor.

The following sections discuss these topics:

- [Using SSIMSOI Model with PrimeSim Continuum](#)
- [SSIMSOI Model — General Syntax](#)
- [SSIMSOI Model Parameters](#)

Using SSIMSOI Model with PrimeSim Continuum

To simulate using the SSIMSOI model:

1. Set `LEVEL=65` to identify the model as the SSIMSOI model.
2. Set the correct simulator room temperature.

The default room temperature is 25°C in Synopsys circuit simulators, but is 27°C in most other simulators. When comparing to other simulators, use `TEMP 27` or `.OPTION TNOM=27` to set the simulation temperature to 27°C in the netlist.

3. Set `DTEMP` on the element line.

You can use `DTEMP` with this model to increase the temperature of individual elements, relative to the circuit temperature. If you do not specify `DTEMP`, simulation extracts `TRISE` from the model card. If you do specify `DTEMP`, it overrides `TRISE` in the model card.

SSIMSOI Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] mname [L=val] [W=val]
+ [M=val] [AD=val] [AS=val] [PD=val] [PS=val]
+ [BODYTYPE=val] [IGATE=val] [AB=val] [PB=val] [LXB=val]
+ [WXB=val] [LPE=val] [DTEMP=val]
```

For more information on parameters, see the [SSIMSOI Model Parameters](#) section.

SSIMSOI Model Parameters

This section covers the following topics:

- [SSIMSOI Model — General Parameters](#)
- [SSIMSOI Model — Intrinsic Parameters \(Geometry Modifiers and Threshold Voltage\)](#)
- [SSIMSOI Model — Intrinsic Parameters \(Mobility and Saturation, Output Conductance\)](#)
- [SSIMSOI Model — Parasitic Parameters](#)

SSIMSOI Model — General Parameters

Parameter	Description
Mxxx	SSIMSOI element name. Must begin with M, followed by up to 1023 alphanumeric characters.
nd	Drain terminal node name or number.
ng	Front gate node name or number.
ns	Source terminal node name or number.
ne	Back gate (or Substrate) node name or number.
np	External body contact node name or number.
mname	SSIMSOI model name reference.
L	SSIMSOI channel length in meters. Default is 5.0 um.
W	SSIMSOI channel width in meters. Default is 5.0 um.
M	Multiplier to simulate multiple SSIMSOIs in parallel. Default=1.
AD	Drain diffusion area. Default=0.
AS	Source diffusion area. Default=0.
PD	Drain diffusion perimeter. Default=0.
PS	Source diffusion perimeter. Default=0.
BODYTYPE	Flag to choose floating(0) or Tgate(2). Default=0.
IGATE	Flag to turn on/off(0/1) gate current calculations. Default=1.
AB	Body diffusion area. Default=0.
PB	Body diffusion perimeter (Body Contacted). Default=0.

Parameter	Description
LXB	Extrinsic Gate Length (Body Contacted). Default=0.
WXB	Extrinsic Gate Width (Body Contacted). Default=0.
LPE	Flag to turn on/off lpe-related parasitics.
DTEMP	Increases the temperature.

SSIMSOI Model — Intrinsic Parameters (Geometry Modifiers and Threshold Voltage)

Name	Parameter	Units	Default
ABIAS	Width modifier	micron	0
CS1LE	Exponent for I-dependence of n1		-1
CS1LL	I-dependence parameter of n1		0
CS1WE			-1
CS1WL			0
CS2LE			-1
CS2LL			0
CS2WE			-1
CS2WL			0
DBIAS	Diffusion resistor processing bias	micron	0
DIBLE			-2
DIBLL			0
DLIVCV	Length modified for capacitance model	micron	0
DPHII	Norm. error in phi at extro. Vth0		0
GP1	Bulk charge coefficient		1.744
GP2	I-dependence parameter of n2		0.8364
LG2CT	Distance from contact to poly edge	micron	1.0
LLDD	Ldd spacer width	micron	0

Name	Parameter	Units	Default
N1	Surface region doping density	1/cm ³	5.0e16
N2	Bulk region doping density	1/cm ³	2.0e16
NFS	Fast surface state density	1/V-cm ²	0
NG	Poly gate doping density	1/cm ³	
NGF	Gate oxide fixed charge density	1/cm ²	0
ODIF	Outdiffusion of s/d under gate	micron	0
ODIFS	Outdiffusion of s under gate (asymmetric)	micron	oidf
PBIAS	Length modifier (use with odif)	micron	0
SHRINK	Exponent for l-dependence of n2		0
SHRINK2	W-dependence parameter of n1		0
TBOX	Back oxide thickness.	Angstrom	250
TCV	Temperature coefficient of threshold voltage.	1/K	0
TCVFB	Temperature coefficient of flatband voltage	1/K	
TOX	Gate oxide thickness.	Angstrom	250
TSI	Silicon film thickness.	cm	0.2e-4
VFB	Reference (large) MOSFET flatband voltage	v	calc
VFBLE	Exponent for length dependence of vth0		-1
VFBLL	Length dependence parameter of vth0		0
VFBWE	Exponent for width dependence of vth0		-1
VFBWL	Width dependence parameter of vth0		0
VTH0	Linear region vth, reference (large) MOSFET, Vbs=0	v	0.8 (nmos) -0.8 (pmos)
WBRK	Depth of surface region	micron	0.2

SSIMSOI Model — Intrinsic Parameters (Mobility and Saturation, Output Conductance)

Name	Parameter	Units	Default
UBREF	Mobility parameter	$\text{cm}^2/\text{V}\cdot\text{s}$	700 (nmos) 300 (pmos)
UBRED	Mobility field reduction factor	$(\text{cm}/\text{V})^{\text{egvexp}}$	100
EAVFAC	Effective field coefficient		0.5
EAVFWL	Width dependence of eavfac		0.0
EAVFWE	Exponent of width dependence of eavfac		-2.0
EAVEXP	Exponent of mobility field function		1.0
UBVDS	Drain dependence of eff. field		0.5
VSAT	Channel carrier saturation velocity	cm/sec	$1.0\text{e-}7$
ESAT0	Vsat divisor, velocity field model		2
ESAT1	Divisor, carrier velocity at sat.		1
LC00	Mult. For channel length modulation		0.2
LC01	Length dependence of lc00	$1/\text{micron}$	0
LC1	Bias dependence of channel length modulation	$1/\text{V}$	0
WLMOD	Mult. for channel width modulation		0
DV2	Par. for lin/sat transition region		0.05
DV3	Length dependence of dv2		0
EXB	Temperature exponent of ubref		1.5

SSIMSOI Model — Parasitic Parameters

Name	Parameter	Units	Default
AF	Flicker noise exponent		1.0
AIMP0	Impact ionization parameter		0
AIMPL	Length dependence of aimp0	micron	0

Name	Parameter	Units	Default
AIMPT	Temperature dependence of aimp0	1/K	0
AIMPW	Width dependence of aimp0	micron	0
BIMP0	Exponent for impact ionization	1/V	28.0
BIMP2	Width dependence of bimp0	micron	0
BIMPL	Length dependence of bimp0	micron	0
CCC	Contact-to-contact capacitance	F/micron	0
CCCR	Ccc error in RCE netlists	F/micron	0
CCP	Contact-to-poly capacitance	F/micron	0
CCPR	Ccp error in RCE netlist	F/micron	0
CCX	Contact-to-soisub capacitance	F/micron	0
CEXP	Flicker noise cox exponent		1.0
CFR	Gate to S/D fringing capacitance	F/micron	0
CFRB	Gate to body overlap capacitance	F/micron	0
CFRBOX	Back-gate to S/D fringing capacitance	F/micron	0
CFRS	Gate to s-fringing capacitance (asymmetrical)	F/micron	cfr
CJCH	S/D zero-bias junction channel-side capacitance	F/m	0.0
CPX	Poly-to-soisub capacitance	F/micron	0
FEXP	Flicker noise frequency exponent		1.0
FOC	Bias dependence of overlap capacitance		1.0
FSAT1	Bias dependence of impact ionization	1/V	0
GIDLA	GIDL pre-exponential parameter	A/m	(off)
GIDLB	GIDL exponential parameter	m/V	3.0e9
GIDLC	GIDL bulk-dependence parameter	V ³	8.0
GIDLE	GIDL bandgap	V	calculated
GIDLT	GIDL temperature-dependence parameter		calculated

Name	Parameter	Units	Default
GISLA	GISL pre-exponential parameter	A/m	gidla
GISLB	GISL exponential parameter	m/V	gidlb
GISLC	GISL bulk-dependence parameter	V ³	gidlc
GISLE	GISL bandgap	V	gidle
GISLT	GISL temperature-dependence parameter		gidlt
GTUNDEL TOX	Intrinsic region delta tox (electrical vs physical)	angstrom	(off)
GTUNDTOX OVL	S/D overlap region delta tox (electrical vs physical)	angstrom	(off)
GTUNECBA	ECB fitting parameter		0.6
GTUNECBB	ECB barrier height	V	3.1
GTUNEC BBO	ECB barrier height	V	3.1
GTUNECBM	ECB effective mass		0.4
GTUNEVBA	EVB fitting parameter		0.4
GTUNEVBB	EVB barrier height	V	4.2
GTUNEV BBO	EVB barrier height	V	3.1
GTUNEV BEG	EVB energy bandgap	V	1.12
GTUNEVBM	EVB effective mass		0.32
GTUNHVBA	HVB fitting parameter		1.0
GTUNHVBB	HVB barrier height	V	4.5
GTUNHV BBO	HVB barrier height	V	4.5
GTUNHVBM	HWB effective mass		0.3
GTUNSTOX OVL	S overlap region delta tox (asymmetric)	angstrom	gundtoxovl
GTUNWDEP	Accumulation 2-D fringing parameter	micron	0

Name	Parameter	Units	Default
JGO	S/D diode SCR generation coefficient	A/cm^2	0
JGOLE	Exponent for I-dependence for jgo		0
JGOLL	I-dependence parameter for jgo		0
JGOS	S diode SCR generation coefficient (asymmetrical)	A/cm^2	jgo
JGOSLE	Exponent for I-dependence for jgo		jgole
JGOSLL	I-dependence parameter for jgo		jgoll
JGOST	Temperature adjustment coefficient for jgo		jgot
JGOSWE	Exponent for w-dependence for jgo		jgowe
JGOSWL	W-dependence parameter for jgo		jgowl
JGOT	Temperature adjustment coefficient for jgo		0
JGOWE	Exponent for w-dependence for jgo		0
JGOWL	Exponent for I-dependence for jgo		0
JRO	S/D diode SCR recombination coef	A/cm^2	1e6
JROLE	Exponent for I-dependence for jro		0
JROLL	I-dependence parameter for jro		0
JROS	S diode SCT recombination coefficient (asymmetrical)	A/cm^2	jro
JROSLE	Exponent for I-dependence for jro		jrole
JROSLL	I-dependence parameter for jro		jroll
JROST	Temperature adjustment coefficient for jro		jrot
JROSWE	Exponent for w-dependence for jro		jrowe
JROSWL	W-dependence parameter for jro		jrowl
JROT	Temperature adjustment coefficient for jro		calculated
JROWE	Exponent for w-dependence for jro		0
JROWL	W-dependence parameter for jro		0
KF	Flicker noise coefficient		0.0

Name	Parameter	Units	Default
M	S/D diode recombination slope factor		2.0
MG	S/D diode generation slope factor		2.0
MGS	S diode generation slope factor		mg
MICH	S/D junction channel-side grading coefficient		0.5
MS	S diode recombination slope factor (asymmetrical)		m
NBIT	Effective doping parameter for I-bit		1.0
NDIF	Effective doping parameter for Q-diffusion		1.0
NLEV	Flicker noise equation level		0
ODIFACT	S/D active diffusion	micron	0
ODIFBC	Body contact diffusion	micron	0
ONKINK	Voltage adjustment for onset of the kink	V	0
PBCH	S/D junction channel-side built-in pot.	V	0.8
RSHBODY	Sheet res. of intrinsic body	ohm/sq	3000
RSHBODY EXT	Sheet res. of extrinsic body	ohm/sq	3000
RSHMIN	Sheet res. of s/d-gate overlap	ohm/sq	0
RSHMINS	Sheet res. of s-gate overlap (asymmetrical)	ohm/sq	rshmin
RSHPLS	Sheet res. of heavily doped S/D	ohm/sq	0
SEFF	S/D diode QNR recombination velocity	cm/s	1e5
SEFFL	I-dependence parameter for seff		0
SEFFLE	Exponent for I-dependence for seff		0
SEFFS	S diode QNR recombination velocity (asymmetrical)	cm/s	seff
SEFFSLE	Exponent for dependence for seffs		seffle
SEFFSLL	I-dependence parameter for seffs		seffll
SEFFST	Temperature adjustment coefficient for seffs		sefft

Name	Parameter	Units	Default
SEFFSWE	Exponent for w-dependence for seffs		seffwe
SEFFSWL	W-dependence parameter for seffs		seffwl
SEFFT	Temperature adjustment coefficient for seff		0
SEFFWE	Exponent for w-dependence for seff		0
SEFFWL	W-dependence parameter for seff		0
TCBODY	Temperature coefficient for rshbody and rshbodyext	1/K	0
TCMIN	Temperature coefficient for rshmin	1/K	0
TCMINS	Temperature coefficient for rshims (asymmetrical)	1/K	tcmin
TCPLS	Temperature coefficient for rshpls	1/K	0
TCPBCH	S/D temperature coefficient for pbch	V/K	calculated
VOC	Bias dependence of overlap capacitance	V	0
WFR	Additional contact width	micron	0

LEVEL 66: PrimeSim Continuum HVMOS Model

The PrimeSim Continuum solution level 66 model is Synopsys proprietary model for high-voltage CMOS integrated circuits design and simulation. It is accurate in modeling the high-voltage device physics and robust in SPICE simulation. It is designed for various high-voltage CMOS processes, technology nodes, and device structures, including both widely-employed LDMOS (Laterally Diffused MOS) and EDMOS (Extended Drain MOS) transistors that may have any field plate oxide layer which helps the drain engineering. In addition, applications can be whole power management ICs such as switch power supply controllers, flash memory, hot-swap chips, and chips used in the automotive and medical industries.

This solution of HVMOS model has been widely adopted for production by many semiconductor companies and major foundries.

The model has been developed based on BSIM4, with high accuracy in geometrical, bias, and temperature scaling. It explicitly and accurately considers the following physical effects: independent bias-dependent drain/source resistances, quasi-saturation, self-heating, Gm fall-off in the saturation region, symmetric and asymmetrical source and drain structures, and many other high-voltage operation-related unique device behaviors.

This section covers the following topics:

- [PrimeSim Continuum Solution HVMOS Model — General Syntax](#)
- [PrimeSim Continuum Solution HVMOS Model Parameters](#)

PrimeSim Continuum Solution HVMOS Model — General Syntax

Syntax:

```
Mxxx nd ng ns [nb] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val]
+ [RGATEMOD=val] [RBODYMOD=val]
+ [ACNQSMOD=val] [GEOMOD=val] [RGEOMOD=val]
+ [TRNQSMOD=val] [SA=val] [SB=val] [SD=val] [SHMOD=val]
+ [NRS=val] [NRD=val] [RBPB=val] [RBPD=val]
+ [RBPS=val] [RBSB=val]
+ [MIN=val] [RDC=val] [RSC=val] [DELVTO=val]
+ [MULU0=val] [DELK1=val]
+ [DELTOX=val] [OFF] [IC=Vds, Vgs, Vbs]
+ [WNFLAG=val]
```

PrimeSim Continuum Solution HVMOS Model Parameters

The following list only presents the PrimeSim Continuum solution proprietary model parameters, while the rest are the same as [LEVEL 54: BSIM4 MOSFET Model](#).

Parameter	Description
ACNQSMOD	AC small-signal NQS model selector.
AD	Drain diffusion area.
AS	Source diffusion area.
DELK1	Shift in body bias coefficient (K1).
DELNFCT	Shift in subthreshold swing factor (NFACTOR).
DELTOX	Shift in gate electrical and physical oxide thickness (TOXE and TOXP). That is, the difference between the electrical and physical gate oxide/insulator thicknesses.
DELVTO	Shift in the VTH0 zero-bias threshold voltage.
GEOMOD	Geometry-dependent parasitics model selector—specifies how the end S/D diffusions connect.
IC	Initial guess in the order
L	MOSFET channel length in meters.
MIN	Whether to minimize the number of drain or source diffusions for even-number fingered device.
MNAME	MOSFET model name reference.
MULU0	Low-field mobility (U0) multiplier.
NB	Bulk terminal node name.
ND	Drain terminal node name.
NF	Number of device fingers.
NG	Gate terminal node name.
NRD	Number of drain diffusion squares.
NRS	Number of source diffusion squares.
NS	Source terminal node name.
OFF	Sets the initial condition to OFF in DC analysis.
PD	Perimeter of the drain junction: • If PERMOD=0, it excludes the gate edge. • Otherwise, it includes the gate edge.

Parameter	Description
PS	Perimeter of the source junction: • If PERMOD=0, it excludes the gate edge. • Otherwise, it includes the gate edge.
RBDB	Resistance connected between dbNode and bNode.
RBODYMOD	Substrate resistance network model selector.
RBPB	Resistance connected between bNodePrime and bNode.
RBPD	Resistance connected between bNodePrime and dbNode.
RBPS	Resistance connected between bNodePrime and sbNode.
RBSB	Resistance connected between sbNode and bNode.
RDC	Drain contact resistance for per-finger device.
RGATEMOD	Gate resistance model selector.
RGEOMOD	Source/drain diffusion resistance and contact model selector—specifies the end S/D contact type: point wide or merged) and how to compute the S/D parasitics resistance.
RSC	Source contact resistance for per-finger device.
TRNQSMOD	Transient NQS model selector.
SHMOD	Self-heating model selector.
W	MOSFET channel width in meters.
WNFLAG	Turn on to select bin model based on width per NF for multi-finger devices.

LEVEL 68: STARC HiSIM2 MOSFET Model

HiSIM (Hiroshima University/STARC IGFET Model) is the first complete surface-potential-based MOSFET model for circuit simulation. The most important advantage of the surface-potential-based modeling is the unified description of device characteristics for all bias conditions. The physical reliability of the drift-diffusion approximation has been proved by 2-D device simulations with channel lengths below 0.1 μ m.

Note:

HiSIM2 source code, and all copyrights, trade secrets or other intellectual property rights and the source code in its entirety, is owned by Hiroshima University and STARC.

To obtain analytical solutions for describing device performances, the charge sheet approximation of the inversion layer with zero thickness has been introduced. Together with the gradual-channel approximation all device characteristics are then described analytically by the channel-surface potentials at the source side and at the drain side. These surface potentials are functions of applied voltages on the four MOSFET terminals; the gate voltage V_g , the drain voltage V_d , the bulk voltage V_b and the reference potential of the source V_s . This is the long-channel basis of the HiSIM model, and extensions of the model approximations are done for advanced technologies. All newly appearing phenomena such as short-channel and reverse-short-channel effects are included in the surface potential calculations causing modifications resulting from the features of these advanced technologies.

The section covers the following topics:

- [STARC HiSIM2 Model Parameters](#)
- [STARC HiSIM2 Model Instance Parameters](#)
- [STARC HiSIM2 Model: Parameters Added for Updates from HiSIM2 3.0.0 to HiSIM2 3.1.0](#)
- [STARC HiSIM2 Model Updates](#)

STARC HiSIM2 Model — General Syntax (All Versions)

The following lists and describes the HiSIM Level 68 general parameters.

```
Mxxx nd ng ns [nb] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val]
+ [NRS=val] [NRD=val] [XGW=val] [XGL=val]
+ [RDC=val] [RSC=val]
+ [NF=val] [NGCON=val] [RBPB=val] [RBPD=val]
+ [RBPS=val] [RBDB=val] [RBSB=val] [SA=val]
+ [SB=val] [SD=val] [TEMP=val] [DTEMP=val]
```

```
+ [NSUBCDFM=val]

.MODEL MNAME N(P)MOS LEVEL=68 VERSION=240.0 or 2.40
+ [CORSRD=val] [COOVLDP=val] [COISUB=val]
+ [COIIGS=val] [COGIDL=val] [COISTI=val]
+ [COADOV=val] [CONQS=val] [CORG=val] [CORBNET=val]
+ [COFLICK=val] [COTHRML=val] [COIGN=val] [COIPRV=val]
+ [COPPRV=val] [CODFM=val]
+ ...
```

STARC HiSIM2 Model Parameters

This section lists the STARC HiSIM2 MOSFET Model parameters:

- [STARC HiSIM2 Model — General Parameters](#)
- [STARC HiSIM2 Model — Basic Device Parameters](#)
- [STARC HiSIM2 Model — Velocity Parameters](#)
- [STARC HiSIM2 Model — Quantum-Mechanical Effect Parameters](#)
- [STARC HiSIM2 Model — Poly-Silicon Gate Depletion Effect Parameters](#)
- [STARC HiSIM2 Model — Short Channel Effect Parameters](#)
- [STARC HiSIM2 Model — Mobility Model Parameters](#)
- [STARC HiSIM2 Model — Channel-Length Modulation Parameters](#)
- [STARC HiSIM2 Model — Narrow Channel Effect Parameters](#)
- [STARC HiSIM2 Model — Small Size Effect Parameters](#)
- [STARC HiSIM2 Model — Substrate Current Parameters](#)
- [STARC HiSIM2 Model — Impact- Ionization Induced Bulk Potential Change Parameters](#)
- [STARC HiSIM2 Model — Gate Leakage Current Parameters](#)
- [STARC HiSIM2 Model — GIDL Current Parameters](#)
- [STARC HiSIM2 Model — Conservation of Symmetry at Vds=0 for Short-Channel MOSFETs](#)
- [STARC HiSIM2 Model — Smoothing Coefficient Between Linear and Saturation Region](#)
- [STARC HiSIM2 Model — Source / Bulk and Drain / Bulk Diode Parameters](#)
- [STARC HiSIM2 Model — 1/f Noise Parameters](#)

- [STARC HiSIM2 Model — DFM Support Parameters](#)
- [STARC HiSIM2 Model — Non-Quasi-Static \(NQS\) Parameters](#)
- [STARC HiSIM2 Model — Capacitance Parameters](#)
- [STARC HiSIM2 Model — Parasitic Resistance Parameters](#)
- [STARC HiSIM2 Model — Binning Model Parameters](#)
- [STARC HiSIM2 Model — Depletion Mode Parameters](#)

STARC HiSIM2 Model — General Parameters

Description	Flag
Contact resistances R_s and R_d	CORSRD=0: No (default) CORSRD=1 & RS/RD $\neq$ 0: Yes, as internal resistances of HiSIM CORSRD=2 & RD $\neq$ 0: Yes, simple analytical formulation CORSRD=-1 & RS/RD $\neq$ 0: Yes, as external resistances of HiSIM
Overlap capacitance model	COOVLP=0: Constant overlap capacitance, either given LOVER plus CGSO/CGDO or LOVER only COOVLP=1: Bias dependent overlap capacitance (default: yes). Selecting a model either by defining NOVER value or not
Overlap capacitance for $COOVLP=1$ and $NOVER = 0$	COQOVSM=0: Analytically solved without inversion condition COQOVSM=1: Iteratively solved (default) COQOVSM=2: Analytically solved with inversion condition
Substrate current I_{sub}	COISUB=0: No (default) COISUB=1: Yes
Gate current I_{gate}	COIIGS=0: No (default) COIIGS=1: Yes
GIDL current I_{GIDL}	COGIDL=0: No (default) COGIDL=1: Yes
STI leakage current $I_{ds,STI}$	COISTI=0: No (default) COISTI=1: Yes

Description	Flag
Lateral field induced and overlap charges/capacitances are added to intrinsic ones	COADOV=0: No COADOV=1: Yes (default)
Non-quasi-static mode	CONQS=0: No (default) CONQS=1: Yes
Gate-contact resistance is included (This flag can also be given as an instance parameter)	CORG=0: No (default) CORG=1: Yes
Substrate resistance network is invoked (This flag can also be given as an instance parameter)	CORBNET=0: No (default) CORBNET=1: Yes
1/f noise is calculated	COFLICK=0: No (default) COFLICK=1: Yes
Thermal noise is calculated	COTHRML=0: No (default) COTHRML=1: Yes
Induced gate and cross correlation noise are calculated	COIGN=0 COTHRML=0: No (default) COIGN=1 & COTHRML=1: Yes
Previous I_{ds} is used as an initial value for calculating source/drain resistance effect (R_s and/or R_d^{-1})	COIPRV=0: No COIPRV=1: Yes (default)
Previous ϕ_s is used as an initial value for the iteration	COPPRV=0: No COPPRV=1: Yes (default)
Parameter variations for the DFM support are considered	CODFM=0: No (default) CODFM=1: Yes
Accurate calculation of the capacitance reciprocity is performed	CORECIP=0: No CORECIP=1: Yes (default)
Lateral-Field-Induced Charge is considered.	COQY=0: No (default) COQY=1: Yes

STARC HiSIM2 Model — Basic Device Parameters

Parameter	Description
TOX	Physical oxide thickness
XL	Difference between real and drawn gate length
XW	Difference between real and drawn gate width

Parameter	Description
XLD	Gate-overlap length
XWD	Gate-overlap width
TPOLY	Height of the gate poly-Si for fringing capacitance
LL	Coefficient of gate length modification
LLD	Coefficient of gate length modification
LLN	Coefficient of gate length modification
WL	Coefficient of gate width modification
WLD	Coefficient of gate width modification
WLN	Coefficient of gate width modification
NSUBC	Substrate-impurity concentration
NSUBP	Maximum pocket concentration
LP	Pocket penetration length
*NPEXT	Maximum concentration of pocket tail
*LPEXT	Extension length of pocket tail
VFBC	Flat-band voltage
VFBCCL	Channel length dependence of flat-band voltage
VFBCCLP	Channel length dependence of flat-band voltage
VBI	Built-in potential
KAPPA	Dielectric constant for gate dielectric
EG0	Bandgap
BGTMP1	Temperature dependence of bandgap
BGTMP2	Temperature dependence of bandgap
TNOM	Temperature selected as a nominal temperature value
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Velocity Parameters

Parameter	Description
VMAX	Saturation velocity
VOVER	Velocity overshoot effect
VOVERP	L_{eff} dependence of velocity overshoot
*VTMP	Temperature dependence of the saturation velocity
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Quantum-Mechanical Effect Parameters

Parameter	Description
QME1	V_{gs} dependence of quantum mechanical effect
QME2	V_{gs} dependence of quantum mechanical effect
QME3	Minimum T_{ox} modification

STARC HiSIM2 Model — Poly-Silicon Gate Depletion Effect Parameters

Parameter	Description
PGD1	Strength of poly-depletion effect
PGD2	Threshold voltage of poly-depletion effect
*PGD4	L_{gate} dependence of poly-depletion effect
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Short Channel Effect Parameters

Parameter	Description
PARL2	Depletion width of channel/contact junction
SC1	Magnitude of short-channel effect
SC2	V_{ds} dependence of short-channel effect

Parameter	Description
*SC3	V_{bs} dependence of short-channel effect
*SC3VBS	For diminishing of impurity gradient
**SC4	Coupling of V_{ds} and V_{bs} dependence
SCP1	Magnitude of short-channel effect due to pocket
SCP2	V_{ds} dependence of short-channel due to pocket
*SCP3	V_{bs} dependence of short-channel effect due to pocket
*SCP21	Short-channel-effect modification for small V_{ds}
*SCP22	Short-channel-effect modification for small V_{ds}
*BS1	Body-coefficient modification by impurity profile
*BS2	Body-coefficient modification by impurity profile
*PTL	Strength of punch-through effect
*PTLP	Channel-length dependence of punch-through effect
*PTP	Strength of punch-through effect
*PT2	V_{ds} dependence of punch-through effect
*PT4	V_{bs} dependence of punch-through effect
*PT4P	V_{bs} dependence of punch-through effect
*GDL	Strength of high-field effect
*GDLP	Channel-length dependence of high-field effect
*GDLD	Channel-length dependence of high-field effect
*NSUBPL	Starting channel length of NSUBP reduction
*NSUBPFAC	Minimum factor of reduction
*NSUBPDLT	Delta for smoothness of the NSUBPFAC model
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Mobility Model Parameters

Parameter	Description
MUECB0	Coulomb scattering
MUECB0LP	Length dependence of Coulomb scattering
MUECB1	Coulomb scattering
MUECB1LP	Length dependence of Coulomb scattering
MUEPH0	Phonon scattering
MUEPH1	Phonon scattering
MUETMP	Temperature dependence of phonon scattering
*MUEPHL	Length dependence of phonon mobility reduction
*MUEPLP	Length dependence of phonon mobility reduction
*MUEPLD	Length dependence of phonon mobility reduction
**MUEPHL2	Length dependence of phonon mobility reduction
**MUEPLP2	Length dependence of phonon mobility reduction
MUESR0	Surface-roughness scattering
MUESR1	Surface-roughness scattering
*MUESRL	Length dependence of surface roughness mobility reduction
*MUESLP	Length dependence of surface roughness mobility reduction
*MUEPHS	Mobility modification due to small size
*MUEPSP	Mobility modification due to small size
NDEP	Depletion charge contribution on effective-electric field
*NDEPL	Modification of Q_B contribution for short-channel case
*NDEPLP	Modification of Q_B contribution for short-channel case
NINV	Inversion charge contribution on effective-electric field
*NINVD	Reduced resistance effect for small V_{ds}
BB	High-field-mobility degradation

* indicates minor parameters, and ** indicates minor parameters for special cases.

STARC HiSIM2 Model — Channel-Length Modulation Parameters

Parameter	Description
CLM1	Hardness coefficient of channel/contact junction
CLM2	Coefficient for Q_B contribution
CLM3	Coefficient for Q_I contribution
*CLM4	No more used
*CLM5	Effect of pocket implantation
*CLM6	Effect of pocket imp8l1antation
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Narrow Channel Effect Parameters

Parameter	Description
WFC	Threshold voltage change due to capacitance change
*WVTH0	Threshold voltage shift
**NSUBCW	Modification of substrate concentration for narrow width
**NSUBCWP	Modification of substrate concentration for narrow width
**NSUBCW2	Modification of substrate concentration for narrow width
**NSUBCWP2	Modification of substrate concentration for narrow width
**NSUBCMAX	Upper limit of substrate concentration for narrow width
*NSUBPW	Modification of pocket concentration for narrow width
*NSUBPWP	Modification of pocket concentration for narrow width
*NPEXTW	Modification of expansion concentration for narrow width
*NPEXTWP	Modification of expansion concentration for narrow width
*MUEPHW	Width dependence of phonon mobility reduction
*MUEPWP	Width dependence of phonon mobility reduction
*MUEPWD	Width dependence of phonon mobility reduction
**MUEPHW2	Phonon related mobility reduction

Parameter	Description
**MUEPWP2	Phonon related mobility reduction
*MUESRW	Change of surface roughness related mobility
*MUESWP	Change of surface roughness related mobility
*NDEPW	Modification of universal mobility for narrow width
*NDEPWP	Modification of universal mobility for narrow width
*VTHSTI	Threshold voltage shift due to STI
*VDSTI	V_{ds} dependence of STI subthreshold
*SCSTI1	The same effect as SC1 but at STI edge
*SCSTI2	The same effect as SC2 but at STI edge
NSTI	Substrate-impurity concentration at the STI edge
**NSTIL	Length dependence of STI impurity concentration
**NSTILP	Length dependence of STI impurity concentration
WSTI	Width of the high-field region at STI edge
*WSTIL	Channel-length dependence of WSTI
*WSTILP	Channel-length dependence of WSTI
*WSTIW	Channel-width dependence of WSTI
*WSTIWP	Channel-width dependence of WSTI
WL1	Threshold voltage shift of STI leakage due to small size effect
WL1P	Threshold voltage shift of STI leakage due to small size effect
NSUCPSTI1	Channel concentration change due to diffusion-region length between gate and STI
NSUBCSTI2	Channel concentration change due to diffusion-region length between gate and STI
NSUBCSTI3	Channel concentration change due to diffusion-region length between gate and STI
NSUBPSTI1	Pocket concentration change due to diffusion-region length between gate and STI

Parameter	Description
NSUBPSTI2	Pocket concentration change due to diffusion-region length between gate and STI
NSUBPSTI3	Pocket concentration change due to diffusion-region length between gate and STI
MUESTI1	Mobility change due to diffusion-region length between gate and STI
MUESTI2	Mobility change due to diffusion-region length between gate and STI
MUESTI3	Mobility change due to diffusion-region length between gate and STI
SAREF	Reference length of diffusion between gate and STI
SBREF	Reference length of diffusion between gate and STI

* indicates minor parameters, and ** indicates minor parameters for special cases.

STARC HiSIM2 Model — Small Size Effect Parameters

Parameter	Description
WL2	Threshold voltage shift due to small size effect
WL2P	Threshold voltage shift due to small size effect
*MUEPHS	Mobility modification due to small size
*MUEPSP	Mobility modification due to small size
*VOVERS	Modification of maximum velocity due to small size
*VOVERSP	Modification of maximum velocity due to small size

* indicates minor parameters, and ** indicates minor parameters for special cases.

STARC HiSIM2 Model — Substrate Current Parameters

Parameter	Description
SUB1	Substrate current coefficient of magnitude
SUB1L	L_{gate} dependence SUB1
SUB1LP	L_{gate} dependence SUB1

Parameter	Description
SUB2	Substrate current coefficient of exponential term
SUB2L	L_{gate} dependence of SUB2
SVDS	Substrate current dependence on V_{ds}
SLG	Substrate current dependence on L_{gate}
SLGL	Substrate current dependence on L_{gate}
SLGLP	Substrate current dependence on L_{gate}
SVBS	Substrate current dependence on V_{bs}
SVBSL	L_{gate} dependence of SVBS
SVBSLP	L_{gate} dependence of SVBS
SVGS	Substrate current dependence on V_{gs}
SVGSL	L_{gate} dependence of SVGS
SVGSLP	L_{gate} dependence of SVGS
SVGSW	W_{gate} dependence of SVGS
SVGSWP	W_{gate} dependence of SVGS
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Impact- Ionization Induced Bulk Potential Change Parameters

Parameter	Description
IBPC1	Impact-ionization induced bulk potential change
IBPC2	Impact-ionization induced bulk potential change

STARC HiSIM2 Model — Gate Leakage Current Parameters

Parameter	Description
GLEAK1	Gate to channel current coefficient
GLEAK2	Gate to channel current coefficient

Parameter	Description
GLEAK3	Gate to channel current coefficient
GLEAK4	Gate to channel current coefficient
*GLEAK5	Gate to channel current coefficient (short channel correction)
*GLEAK6	Gate to channel current coefficient (V_{ds} dependence correction)
*GLEAK7	Gate to channel current coefficient (gate length and width dependence correction)
EGIG	Temperature dependence of gate leakage
IGTEMP2	Temperature dependence of gate leakage
IGTEMP3	Temperature dependence of gate leakage
GLKSD1	Gate to source/drain current coefficient
GLKSD2	Gate to source/drain current coefficient
GLKSD3	Gate to source/drain current coefficient
GLKB1	Gate to bulk current coefficient
GLKB2	Gate to bulk current coefficient
GLKB3	Flat-bands shift for gate to bulk current
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — GIDL Current Parameters

Parameter	Description
GIDL1	Magnitude of GIDL
GIDL2	Field dependence of GIDL
GIDL3	V_{ds} dependence of GIDL
*GIDL4	Threshold of V_{ds} dependence
*GIDL5	Correction of high-field contribution
*GIDL6	V_{bs} dependence of GIDL
*GIDL7	Correction of high-field contribution

Parameter	Description
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Conservation of Symmetry at $V_{ds}=0$ for Short-Channel MOSFETs

Parameter	Description
VZADD0	Symmetry conservation coefficient
PZADD0	Symmetry conservation coefficient

STARC HiSIM2 Model — Smoothing Coefficient Between Linear and Saturation Region

Parameter	Description
*DDLTMAX	Smoothing coefficient for V_{ds}
*DDLTSPL	L_{gate} dependence of smoothing coefficient
*DDLTICT	L_{gate} dependence of smoothing coefficient
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Source / Bulk and Drain / Bulk Diode Parameters

Parameter	Description
JS0	Saturation current density
JS0SW	Sidewall saturation current density
JS0SWG	Gate sidewall saturation current density
IJS0SWG	Length dependence of JS0SWG
WJS0SWG	Width dependence of JS0SWG
PJS0SWG	Cross-term dependence of JS0SWG
NJ	Emission coefficient

Parameter	Description
NJSW	Sidewall emission coefficient
NJSWG	Gate sidewall emission coefficient
Subtmp	Temperature dependence of I_{sub}
XTI	Temperature coefficient for forward current densities
XTI2	Temperature coefficient for reverse current densities
DIVX	Reverse current coefficient
CTEMP	Temperature coefficient of reverse currents
CISB	Reverse biased saturation current
CISBK	Reverse biased saturation current (at low temperature)
CVB	Bias dependence coefficient of CISB
CVBK	Bias dependence coefficient of CISB (at low temperature)
CJ	Bottom junction capacitance per unit area at zero bias
CJSW	Source/drain sidewall junction cap. grading coefficient per unit length at zero bias
CJSWG	Source/drain sidewall junction capacitance per unit length at zero bias
MJ	Bottom junction capacitance grading coefficient
MJSW	Source/drain sidewall junction capacitance grading coefficient
MJSWG	Source/drain gate sidewall junction capacitance grading coefficient
PB	Bottom junction build-in potential
PBSW	Source/drain sidewall junction build-in potential
PBSWG	Source/drain gate sidewall junction build-in potential
VDIFFJ	Diode threshold voltage between source/drain and substrate
TCJBD	Temperature dependence of drain-side diode capacitance
TCJBDSW	Temperature dependence of drain-side diode capacitance
TCJBDSWG	Temperature dependence of drain-side diode capacitance
TCJBS	Temperature dependence of source-side diode capacitance

Parameter	Description
TCJBSSW	Temperature dependence of source-side diode capacitance
TCJBSSWG	Temperature dependence of source-side diode capacitance

STARC HiSIM2 Model — $1/f$ Noise Parameters

Parameter	Description
NFALP	Contribution of the mobility fluctuation
NFTRP	Ratio of trap density to attenuation coefficient
*CIT	Capacitance caused by the interface trapped carriers
FALPH	Power of f describing deviation of $1/f$

* indicates minor parameters, and ** indicates minor parameters for special cases.

STARC HiSIM2 Model — DFM Support Parameters

Parameter	Description
MPHDFM	Mobility dependence on NSUBC due to μ_{phonon}

STARC HiSIM2 Model — Non-Quasi-Static (NQS) Parameters

Parameter	Description
DLY1	Coefficient for delay due to diffusion of carriers
DLY2	Coefficient for delay due to conduction of carriers
DLY3	Coefficient for RC delay of bulk carriers

STARC HiSIM2 Model — Capacitance Parameters

Parameter	Description
XQY	Distance from drain junction to maximum electric field point
*XQY1	V_{bs} dependence of Q_y

Parameter	Description
*XQY2	L_{gate} dependence of Q_y
QYRAT	Partitioning ratio of Q_y between source and drain
LOVER	Overlap length
NOVER	Impurity concentration in overlap region
VFBOVER	Flat-band voltage in overlap region
OVSLP	Coefficient for overlap capacitance
OVMAG	Coefficient for overlap capacitance
CGSO	Gate-to-source overlap capacitance
CGDO	Gate-to-drain overlap capacitance
CGBO	Gate-to-bulk overlap capacitance
* indicates minor parameters, and ** indicates minor parameters for special cases.	

STARC HiSIM2 Model — Parasitic Resistance Parameters

Parameter	Description
RS	Source-contact resistance in LDD region
RD	Drain-contact resistance in LDD region
RSH	Source/drain sheet resistance
RSHG	Gate sheet resistance
GBMIN	Substrate resistance network
RBPB	Substrate resistance network
RBPD	Substrate resistance network
RBPS	Substrate resistance network
RBDB	Substrate resistance network
RBSB	Substrate resistance network

STARC HiSIM2 Model — Binning Model Parameters

Parameter	Description
LBINN	Power of L_{drawn} dependence
WBINN	Power of W_{drawn} dependence
LMAX	Maximum length of L_{drawn} valid
LMIN	Minimum length of L_{drawn} valid
WMAX	Maximum length of W_{drawn} valid
WMIN	Minimum length of W_{drawn} valid

The complete binning criteria is as follows:

$$\begin{aligned} L_{\text{MIN}} + X_{\text{LREF}} &\leq L + X_L < L_{\text{MAX}} + X_{\text{LREF}} \\ W_{\text{MIN}} + X_{\text{WREF}} &\leq W + X_W < W_{\text{MAX}} + X_{\text{WREF}} \end{aligned}$$

If you do not specify X_{LREF} , the simulation sets it to X_L .

If you do not specify X_{WREF} , the simulation sets it to X_W .

STARC HiSIM2 Model — Depletion Mode Parameters

Table 2

Parameter	Description
CODEP	Model flag to select the depletion model
DEPBB	High-field mobility degradation in the resistor region
DEPDDLTL	Smoothing coefficient for $V_{\text{ds,res}}$
DEPETA	V_{ds} dependence of the threshold voltage shift
DEPLEAK	Leakage current coefficient
DEPLEAKL	L_{gate} dependence of leakage current coefficient
DEPLEAKLP	L_{gate} dependence of leakage current coefficient
DEPMUE0	Coulomb scattering in the resistor region
DEPMUE0L	L_{gate} dependence of the Coulomb scattering in the resistor region

Table 2 (Continued)

Parameter	Description
DEPMUE0LP	Lgate dependence of the Coulomb scattering in the resistor region
DEPMUE0TMP	Temperature dependence of DEPMUE0
DEPMUE1	Coulomb scattering in the resistor region
DEPMUE1L	Lgate dependence of the Phonon scattering in the resistor region
DEPMUE1LP	Lgate dependence of the Phonon scattering in the resistor region
DEPMUE2	Coulomb scattering of the resistor part
DEPMUE2TMP	Temperature dependence of DEPMUE2
DEPMUEA1	Modification of Ires
DEPMUEBACK0	Coulomb scattering in the back region
DEPMUEBACK0L	Lgate dependence of the Coulomb scattering in the back region
DEPMUEBACK0LP	Lgate dependence of the Coulomb scattering in the back region
DEPMUEBACK1	Coulomb scattering in the back region
DEPMUEBACK1L	Lgate dependence of the Phonon scattering in the back region
DEPMUEBACK1LP	Lgate dependence of the Phonon scattering in the back region
DEPMUEPH0	Phonon scattering in the resistor region
DEPMUEPH1	Phonon scattering in the resistor region
DEPMUETMP	Temperature dependence of the phonon scattering in the resistor region
DEPSUBSL	Factor of the sub-threshold sloop
DEPVDSEF1	Effective drain potential coefficient-1 in the resistor region
DEPVDSEF1L	Lgate dependence of the effective drain potential coefficient-1

Table 2 (Continued)

Parameter	Description
DEPVDSEF1LP	Lgate dependence of the effective drain potential coefficient-1
DEPVDSEF2	Effective drain potential coefficient-2 in the resistor region
DEPVDSEF2L	Lgate dependence of the effective drain potential coefficient-2
DEPVDSEF2LP	Lgate dependence of the effective drain potential coefficient-2
DEPVFBC	Flat-band voltage of the resistor part
DEPVGPSL	Smoothing coefficient for gm
DEPVMAX	Saturation velocity in the resistor region
DEPVMAXL	Lgate dependence of the saturation velocity in the resistor region
DEPVMAXLP	Lgate dependence of the saturation velocity in the resistor region
DEPVSATR	Vbs dependence of Vds,sat of the resistor part
DEPVTMP	Temperature dependence of DEPVMAX
NDEPM	impurity concentration of the surface N layer
NDEPML	Lgate dependence of impurity concentration of the surface N layer
NDEPMLP	Lgate dependence of impurity concentration of the surface N layer
TNDEP	Thickness of the surface N layer
TNDEPV	Vds dependence of the surface N layer thickness
TNDEPVB	Vbs dependence of TNDEPV

STARC HiSIM2 Model Instance Parameters

Parameter	Description
L	Gate length (L_{gate})

Parameter	Description
W	Gate width (W_{gate})
Diode	
AD	Junction area of the drain contact
AS	Junction area of the source contact
PD	Junction periphery of the drain contact
PS	Junction periphery of the source contact
Source/Drain Resistance	
NRD	Number of drain squares
NRS	Number of source squares
RDC	Additional drain resistance due to contact
RSC	Additional source resistance due to contact
Gate Resistance	
NF	Number of gate fingers
NGCON	Number of gate contacts
XGL	Offset of the gate length
XGW	Distance from the gate contact to the channel edge
Substrate Network	
RBDB	Substrate resistance network
RBPB	Substrate resistance network
RBPD	Substrate resistance network
RBPS	Substrate resistance network
RBSB	Substrate resistance network
Length of Diffusion	
SA	Length of diffusion between gate and STI
SB	Length of diffusion between gate and STI
SD	Length of diffusion between gate and gate

Parameter	Description
Temperature	
TEMP	Device temperature (T)
DTEMP	Device temperature change
Design for Manufacture	
NSUBCDFM	Substrate impurity concentration

STARC HiSIM2 Model: Parameters Added for Updates from HiSIM2 3.0.0 to HiSIM2 3.1.0

Parameter	Description	Unit	Min	Max	Default
Depletion mode model					
COVDSRES	Model flag for selection of $V_{ds,sat}$ treatment. COVDSRES=-1 is compatible to 3.0.0	-	-	-	3
TNDEPMIN	Minimum thickness of the surface layer	m	$TNDEP.1 \times 10^{-9}$	TNDEP	$TNDEP.1 \times 10^{-6}$
DEPNINVDC	Modification of V_{dse} dependence on E_{eff} (for μ_{res})	1/V	-	-	0.0
DEPNINVDH	Modification of V_{dse} dependence on E_{eff} (for μ_{res})	1/V	-	-	0.0
DEPNINVDW	Coefficient of modification of V_{dse} dependence on E_{eff} (for μ_{res})	-	-	-	0.0
DEPNINVDWP	Coefficient of modification of V_{dse} dependence on E_{eff} (for μ_{res})	-	-	-	1.0
DEPDVFBC	Flat-band voltage shift of the resistor part	-	-	-	0.1
DEPSUBSL0	Factor of the sub-threshold slope	-	-	-	2.0
DEPVSATA	$V_{ds,sat}$ shift	V	-	-	0
Vertical-doping-inhomogeneity model					
NSUBD	Deep-substrate-impurity concentration	$1/\text{cm}^3$	0	-	NSUBC

Parameter	Description	Unit	Min	Max	Default
VSFTD	Shift along ΔV_{th} for ΔV_{th} vs $\sqrt{2\phi_B - V_{bs}}$ line for NSUBD	V	-	-	0
SC2D	SC2 for NSUBD	1/V	-	-	0
WDEPV	Change-over point in ΔV_{th} vs $\sqrt{2\phi_B - V_{bs}}$ line for NSUBDW0	$V^{1/2}$	-	-	10
NSUBDNW	Deep-narrow-substrate-impurity concentration	$1/\text{cm}^3$	-	-	0
NSUBDW	Coefficient for W-scaling	-	-	-	0
NSUBDWP	Exponent in W-scaling equation	-	-	-	1
NSUBDW0	Primary coefficient for W-scaling	$1/\text{cm}^3$	-	-	NSUBD
Deep punch-through model					
COPT	Model selection flag. COPT=0: without punchthrough effects (default) and COPT = 1: with punchthrough effects	-	0	1	0
COPSPT	Flag for solution method. COPSPT = 0: analytical solution for deep punch-through (default) and COPSPT = 1: iterative solution for deep punch-through	-	0	1	0
XJPT	Effective junction depth for punch-through current calculation	m	-	-	3×10^{-8}
NJUNC	Source/drain diffusion dopant concentration	$1/\text{cm}^3$	-	-	10^{20}
MUPT	Effective mobility for punch-through current	$\text{m}^2/\text{V/s}$	-	-	0
VFBPT	Flatband voltage shifter for punch-through current	V	-	-	0
PSLIMPT	Potential limiter for punch-through current calculation	V	-	-	0
Oxide thickness in the overlap region					
TOXOV	Oxide thickness of the overlap region	m	0	-	TOX

Parameter	Description	Unit	Min	Max	Default
OVINVDLT	Analytical overlap inversion when COQOVSM = 2	-	0	-	75
Lateral diffusion of the source/drain under the gate for better capacitance fitting					
XLDC	Different length dependence of capacitance from currents	m	0	50n	0
XWDC	Different width dependence of capacitance from currents	m	-10n	100n	XWD
Improved diode model selection					
CODIO	Flag. CODIO=0: default. CODIO= 1: Improved diode model	-	0	1	0

STARC HiSIM2 Model Updates

This section provides the HiSIM2 model version updates:

- [HiSIM 3.2.0 Updates](#)
- [HiSIM 3.1.1 Updates](#)
- [HiSIM 3.1.0 Updates](#)
- [HiSIM 3.0.0 Updates](#)
- [HiSIM 2.9.1 Updates](#)
- [HiSIM 2.9.0 Updates](#)
- [HiSIM 2.7.0 Updates](#)
- [HiSIM 2.6.1 Updates](#)
- [HiSIM 2.6.0 Updates](#)
- [HiSIM 2.5.1 Updates](#)
- [HiSIM 2.5.0 Updates](#)
- [HiSIM 2.4.3 Updates](#)
- [HiSIM 2.4.2 Updates](#)
- [HiSIM 2.4.1 Updates](#)

- [HiSIM 240SC2 Updates](#)
- [HiSIM 2.3.1 Updates](#)

HiSIM 3.2.0 Updates

The following is a summary of update information for HiSIM2 3.2.0 from HiSIM2 3.1.1

- Clamp of forward back bias is implemented for VNUD (vertically non-uniform doping effects) model. This bug-fix assures protections against a negative input to sqrt()
- You can now set maximum (farthest from zero) to reverse back bias
 - New model parameter: VBSRFAR (reverse back bias farthest from zero)
- VNUD (vertically non-uniform doping effects) model implementation is extended to cover the cases where NSUBD > NSUBC (retrograde)
- Temperature dependence in $2\Phi_B$ is active for submodels such as V_{th} shift associated with short-channel effects and reverse-short-channel effects, and GID(S)L. In those submodels, $2\Phi_B$ now reflects temperature other than 300 K
- The model flag COPB20COMPAT switches $2\Phi_B$
 - **COPB20COMPAT=0** (recommended): uses the temperature-dependent $2\Phi_B$
 - **COPB20COMPAT=1** (backward compatible): uses $2\Phi_B(\# = 300\text{ K})$

HiSIM 3.1.1 Updates

Qbd_swg and Qbs_swg bug fixes, when CODIO=0, for backward compatibility.

HiSIM 3.1.0 Updates

The following is a summary of bug fixes and enhancements from version 3.0.0:

- Improvement on the depletion mode model.
- Improvement on vertical-doping-inhomogeneity model.
- Added deep punch-through model which gives better capability of subthreshold slope.
- Oxide thickness in the overlap region.
- Lateral diffusion of the S/D under the gate for better capacitance fitting.
- Selector to include improved diode model.
- Bug fixes on the induced gate noise.

HiSIM 3.0.0 Updates

Implemented the depletion mode model (CODEP=3) with improvements on the capacitance characteristics.

HiSIM 2.9.1 Updates

The following is a summary of bug fixes and enhancements from version 2.9.0 to 2.9.1:

- Modification of the RD/RS calculation method in reverse mode.
- Fixes to the negative Cdd/Css issue in CODEP=2.

HiSIM 2.9.0 Updates

The following is a summary of bug fixes and enhancements from version 2.8.0 to 2.9.0 per the HiSIM-STARC release notes:

- Improvement of the depletion mode model with CODEP=2, where CODEP=1 refers the version 2.8.0 model.
- Addition of a model parameter for the junction leakage current at the gate-edge sidewall.
- Improvement of the temperature dependence of I_{sub} .
- Improvement of the ranges in the parameter declaration.
- Bug fixes.

HiSIM 2.8.0 Updates

The following is a summary of bug fixes and enhancements from version 2.7.0 to 2.8.0 per the HiSIM-STARC release notes:

- New depletion mode MOSFET model, if the model flag CODEP is set to 1.

Caution:

STI leakage, channel length modulation, thermal noise, and induced gate noise are not yet supported in this depletion mode model.

For detailed information on a more typical depletion mode MOSFET structure with additional device parameters, refer to the user manual.

- Changed model descriptions.
- Bug fixes.

HiSIM 2.7.0 Updates

The following is a summary of bug fixes and enhancements from version 2.6.1 to 2.7.0 per the HiSIM-STARC release notes.

- Unit Conversion from CGS to MKS
 - In this release, the source code for model evaluation was converted from CKS units to MKS units.
- Model description changes (Refer to *HiSim-STARC User's Manual for detailed information*).
 - Suppression of error messages from the range check when new model flag `COERRREP` is set to 0.
 - Improvement of `GIDL` current model with new model parameters `GIDL6` and `GIDL7`.
 - Improvement of `STI` effect model with new model parameters `NSUBCSTI1`, `NSUBCSTI2`, and `NSUBCSTI3`.
 - Introduction of gate length dependence for `VFBC` with new model parameters `VFBCL` and `VFBCLP`.
 - Improvement of smoothness of the `NSUBPFAC` model with new model parameters `NSUBPDLT`.
 - Invalidation of the `NSUBPFAC` model when `NSUBPFAC=1`.
 - Range check of model parameters `NSUBPFAC` and `NSUBPDLT`.
- Bug fixes:
 - Calculation of `MUEPH1` when `CODFM=1`.
 - Reset of `SC4` value when `CORECIP=1`.
 - Calculation of some derivatives.
 - Floating point exception in the smoothing functions.
 - High-field effect model with model parameter `GDLD`.
 - Code improvements of `Fn_POW` and `Fn_SZ` functions for higher exception robustness.

For details, refer to the official Hiroshima University HiSIM2 release website: <http://home.hiroshima-u.ac.jp/usdl/HiSIM2/C-Code/protect.cgi>

HiSIM 2.6.1 Updates

The following is a summary of bug fixes and enhancements from version 2.6.0 to 2.6.1 per the HiSIM-STARC release notes.

The following bugs were fixed in this release:

- Missing initialization of `NSUBC`, `NSUBP` and `MUEPH1` in the `hsm2temp.c` for repeating simulation.
- Unit converting in the `hsm2temp.c`.
- Condition for generating of the internal drain node and gate node.
- Derivative calculations in the **Qover** model code (C code only).
- Floating point exception in the `QME` model code.
- Floating point exception in the smoothing functions.
- Maximum gate width for the range check when model parameter `NF` $\neq 1$.

For details, refer to the official Hiroshima University HiSIM2 release website: <http://home.hiroshima-u.ac.jp/usdl/HiSIM2/C-Code/protect.cgi>

HiSIM 2.6.0 Updates

The following is a summary of bug fixes and enhancements from version 2.5.1 to 2.6.0 per the HiSIM-STARC release notes.

- Model description changes (Refer to *HiSim-STARC User's Manual for detailed information*).
 - Introduction of the Well Proximity Effect (WPE) model with new model parameters `WEB`, `WEC`, `NSUBCWPE`, `NSUBPWPE`, and `NPEXTWPE`.
 - New instance parameters: `SCA`, `SCB`, and `SCC`.
 - Improvement of the overlap capacitance model.
 - Activation of Induced Gate Noise model for the C code.
- Other changes:
 - New guideline for the `VZADD0` setting. See *User's Manual*.
 - Default value change for instance parameters `NRD` and `NRS`.

For details, refer to the official Hiroshima University HiSIM2 release website: <http://home.hiroshima-u.ac.jp/usdl/HiSIM2/C-Code/protect.cgi>

HiSIM 2.5.1 Updates

The following is a summary of bug fixes and enhancements from version 2.5.0 to 2.5.1 per the HiSIM-STARC release notes. See the *HiSIM2.5.1 User's Manual* for full details.

- Update information of HiSIM_2.5.1-Beta-5 from HiSIM_2.5.1-Beta4
 - Bug fix in the modeling of the short-channel effect induced by the pocket implantation
 - Bug fix in the modeling of the gate leakage currents
 - Change of default values for model flags and parameters as follows:

Parameter	Default
VZADD0	20E-3
PZADD0	20E-3

- $$M_{Coulomb0} = MUECB0 \cdot \left(\frac{L_{gate} \cdot 10^6}{1\mu m} \right)^{MUECB0LP}$$

Introduction of a channel length dependence for the screening effect of the Coulomb scattering with new model parameters MUECB0LP and MUECB1LP.

$$M_{Coulomb1} = MUECB1 \cdot \left(\frac{L_{gate} \cdot 10^6}{1\mu m} \right)^{MUECB1LP}$$

- Change of Min / Max values for model parameter as follows:

Parameter	Min	Max
VOVER	0	50
VMAX	1.00E+05	2.00E+07
VTMP	-5	1
MUESR1	1.00E+13	1.00E+16

Parameter	Min	Max
SC3VBS	-3	0

Note:

Previously (up to beta 4) SC3VBS is not treated as a model parameter. The default value of SC3VBS is now set to zero, which was -2.5 previously.

- Introduction of range check for NDEPW and NDEPWP.
- Removal of warning for SC3.
- Removal of initial jump when COPPRV flag set to 1.
- Improvement of the overlap capacitance model.
- Removal of the warning message for channel length modulation in hsm2eval.c.
- Removal of unnecessary comments in source code.

For details, refer to the official Hiroshima University HiSIM2 release website: <http://home.hiroshima-u.ac.jp/usdl/HiSIM2/C-Code/protect.cgi>

HiSIM 2.5.0 Updates

The following is a summary of bug fixes and enhancements from version 2.4.3 to 2.5.0 per the HiSim-STARC release notes.

- A new highly accurate intrinsic capacitance model is added, which is controlled by new model parameter CORECIP (default=0). In addition, if CORECIP=1 is set and an overlap capacitance is added with COADOV=1, it is recommended to give a non-zero NOVER value. For Igb calculation, model parameter values must preserve that VFBC +GLKB3 <= 0.
- Changed some smoothing parameter values for better fitting.
- Fixed bugs related to poly-depletion model.
- Removed negative-capacitance clamping.
- Fixed several bugs related to QM effects.
- Added reverse-mode treatment for IGISL and IGIDL.
- Fixed bugs related to derivative calculations.

HiSIM 2.4.3 Updates

The following summarizes the bug fixes and changes from HiSIM 2.4.2 to 2.4.3 per the HiSim-STARC release note.

- A bug in hsm2temp was fixed.
- Bugs of the Vbsz dependence were fixed.
- Improved capacitance characteristics.

HiSIM 2.4.2 Updates

The following is a summary of bug fixes and changes over version 2.4.1 per the HiSim-STARC release note.

- The calculation of Taub for NQS model was corrected.
- The avoidance of zero division.
- The deletion of the unnecessary sentence.
- Bug fix of asymmetrical Qover model.
- The initial setting of GLPART1 was missed.
- The default value of model parameter 'VERSION' was corrected.
- The calculation of the induced gate noise with M factor $\neq 1$ was corrected.
- The calculation of Igate model was corrected.
- Changes to the default model parameters as follows:
 - GLKSD2 1e-2
 - GLKSD3 -1e-2
 - FN1 0.0
 - FN2 0.0
 - FVBS 0.0

HiSIM 2.4.1 Updates

The following is a summary of updates and changes from HiSIM 240-SC2 to HiSIM 2.4.1:

- Induced Gate Noise model: bug fix to avoid division by zero and use of uninitialized values.
- Impact-ionization-induced Bulk Potential Change model: bug fix of derivatives when **CORSRD = 1**.

- Bug fix of numerical calculation for surface potentials.
- The W and/or L dependence of the gate resistance model was discontinuous.
- Bug fix to avoid division by zero if **DLY1 = 0** and **DLY2 = 0** in **NQS model**.
- Improved capacitance reciprocity in **AC** analysis.
- Fixed the inconsistency result of **AC** analysis and **TR** analysis in **NQS** model.
- Bug fix of sign of the parasitic capacitances for the small-signal **NQS** model.
- Bug fix of handling the multiplication factor **M**. Bug fix of handling the number of gate fingers **NF**.
- The thermal noise and flicker noise were incorrect in some cases.
- The temperature specification with an instance parameter **TEMP** was invalid.
- The **M * NF** was multiplied twice for output values like **gigds** and **gigss**.
- Bug fix of extrapolation when Vbse less than 10.5 V.
- Changed the minimum and maximum values of the model parameters **CVB**, **CVBK** and **DDLMAX**.
- The parameter **SVBSLP**, **SVGSW** and **SVGSWP** were not declared in hsm2mask.
- Bug fix to avoid the negative **Cgb**.
- Improvements and bug fixes of the overlap charge model.
- Bug fix to avoid division by zero in calculation of **Vdx2**.
- The temperature dependence of the noise value was incorrect.
- Deleted the unnecessary '#ifdef' statements in hsm2acld.c.
- Initialize the pyABb_i for **AC-NQS** model.
- Fixed the gate partitioning model for **Igate**.
- If **GLPART1 != 0.5**, the option calculating explicitly is selected.

For the full release note for the HiSIM 2.4.1, go to: <http://www.starc.jp/other/contactus-e.html>

HiSIM 240SC2 Updates

The following changes and fixed issues are implemented in the SC2 release:

- Multiplication factor #M is introduced.
- Bugs in the derivatives of Ps0 in the accumulation zone.
- Bugs in the derivatives of T10 in Igate model.
- Bug in Vdsat output for users (caused by Qover model).
- ISBS2-related bugs in the junction-diode-current model.
- Default values for NSTI and EGIG were wrong. NSTI = 0.0 changed to NSTI = 5.0e17, EGIG = 1.1 changed to EGIG = 0.0
- Coding bug in the DFM model.

Changed Parameters:

- Removed Parameter: SCSTI3
- Replaced Parameters: CLM4 (replaced for DDLTMAX, DDLTICT and DDLTSLP), #LOD (replaced for #SA, #SB, #SD, SAREF and SBREF)
- Renamed Parameter: IGTEMP1 changed to EGIG

Parameter	Unit	Min	Default	Max	Description
Unit/default value modified parameters					
GIDL4	V	//	0.0	//	Threshold of Vds dependence.
LOVER	//	//	30e-9	//	Overlap length.
NSTI	cm-3	//	5.0e17	//	Substrate-impurity concentration at STI edge.
PGD1	//	//	0.0	//	Strength of poly-Si gate depletion.
RD	$\Omega \cdot m$	//	//	//	Drain-contact resistance in LDD region
RS	$\Omega \cdot m$	//	//	//	Source-contact resistance in LDD region
VBI	//	//	1.1	1.2	Built-in potential.
Added model flag					
CODFM	None	0	0	1	Flag for DFM support.

Parameter	Unit	Min	Default	Max	Description
Added parameters					
#M	None		1.0		Multiplication factor
#NSUBCDFM	None	1.0e16	None	1.0e19	NSUBC as an instance parameter for DFM support
#SA	m		0		Distance from STI edge to Gate edge
#SB	m		0		Distance from STI edge to Gate edge
#SD	m		0		Distance from Gate edge to Gate edge
DDLTICT	None	-3.0	10.0	20.0	Lgate dependence of smoothing coeff.
DDLTMAX	None	0.0	10.0	20.0	Smoothing coefficient for Vds
DDLTSLP	1 / (μm)	0.0	0.0	20.0	Lgate dependence of smoothing coeff.
GLKB3	V		0.0		Flat-band shift for gate to bulk current
MPHDFM	None	-3.0	-0.3	3.0	NSUBC dependence of μ_{phonon} for DFM support
NDEPL	None		0.0		Modification of Qb contribution for short-channel case
NDEPLP	None		1.0		Modification of Qb contribution for short-channel case
NOVER	cm-3		0.0		Impurity concentration in overlap region
SAREF	m		1e-6		Reference distance from STI edge to Gate edge
SBREF	m		1e-6		Reference distance from STI edge to Gate edge
VDSTI	None		0.0		Vds dependence of STI subthreshold
VFBOVER	V		-0.5		Flat-band voltage in overlap region

Parameter	Unit	Min	Default	Max	Description
WSTIW	μm^{WSTIWP}	0.0			Channel-width dependence of WSTI
WSTIWP	None	1.0			Channel-width dependence of WSTI
XL	m	0.0			Gate length offset due to mask/etch effect
XQY1	$F \cdot \mu m^{QXY2-1}$	0.0			Vbs dependence of Qy
XQY2	None	2.0			Lgate dependence of Qy
XW	m	0.0			Gate width offset due to mask/etch effect

The // mark indicates values not changed from HiSIM2.3.1-SC2. The # mark indicates instance parameters.

6 + 74 × 3 = 228 binning parameters are added in addition to the above parameters.

HiSIM 2.3.1 Updates

The STARC HiSIM2 is Level 68 in the Synopsys MOSFET models. HiSIM2.3.1 is an improved version that resolved many issues of previous releases.

HiSIM version 2.3.1 provides:

- Improved Linear Region Modeling
- Improved Small Size Model
- Improved Isub Model
- Inclusion of Induced Gate Noise
- Support for DFM (Interface to Instance Parameters)

LEVEL 69: PSP Model

The PSP model is a compact MOSFET model intended for digital, analog, and RF designs. It has been jointly developed by the Pennsylvania State University and Philips Research. The roots of this model lie in both SP (Penn State) and MOS Model 11 (Philips).

It is a surface-potential based MOS model containing all relevant physical effects (mobility reduction, velocity saturation, DIBL, gate current, lateral doping gradient effects, STI stress, and so forth), to model present-day and upcoming deep-submicron bulk CMOS technologies. The JUNCAP2 source/drain junction model is an integrated part of PSP.

For a full description of the PSP models, see <http://pspmodel.asu.edu/>.

The following sections discuss these topics:

- [PSP Model Features](#)
- [PSP Model Updates](#)
- [Using PSP Model in the PrimeSim Continuum Solution](#)
- [PSP 103.1 Model Parameters](#)
- [PSP 103.1 Instance Parameters](#)
- [PSP Model — Output Templates](#)

PSP Model Features

The PSP general features of this model are:

- Physical surface-potential-based formulation in both intrinsic and extrinsic model modules
- Physical and accurate description of the accumulation region
- Inclusion of all relevant small-geometry effects
- Modeling of the halo implant effects, including the output conductance degradation in long devices
- Coulomb scattering and non-universality in the mobility model
- Non-singular velocity-field relation enabling the modeling of RF distortions, including inter-modulation effects
- Complete Gummel symmetry
- Mid-point bias linearization enabling accurate modeling of the ration-based circuits (for example, R2R circuits)

- Quantum-mechanical corrections
 - Correction for the polysilicon depletion effects
 - Gate-induced drain leakage (GIDL) and gate-induced source leakage current (GISL) model
 - Surface-potential-based noise model including channel thermal noise, flicker noise, and channel-induced gate noise
 - Advanced junction model, including trap-assisted tunneling, band-to-band tunneling, and avalanche breakdown
 - Spline-collocation-based non-quasi-static (NQS) model, including all terminal currents
 - Stress model (based on BSIM4 version)
-

PSP Model Updates

This section provides the PSP100 model version updates:

- [PSP 104.0 Model Updates](#)
- [PSP 103.8.2 Model Updates](#)
- [PSP 103.8.1 Model Updates](#)
- [PSP 103.8 Model Updates](#)
- [PSP 103.7 Model Updates](#)
- [PSP 103.6 Model Updates](#)
- [PSP 103.5 Model Updates](#)
- [PSP 103.4 Model Updates](#)
- [PSP 103.3 Model Updates](#)
- [PSP 103.2 Model Updates](#)
- [PSP 103.1 Model Updates](#)
- [PSP103.0 Model Updates](#)
- [PSP102.3 Model Updates](#)
- [PSP102.2 Model Updates](#)
- [PSP102.1 Model Updates](#)
- [PSP102.0 Model Updates](#)

- [PSP101.0 Model Updates](#)
- [PSP100.1 Model Updates](#)

PSP 104.0 Model Updates

The following updates were made in PSP v104.0:

- Addition of a new DIBL model based on a quasi-Fermi level correction, including the screening effect in inversion regime.
- Addition of new parameters to improve g_m description in saturation regime: `THESATT` (local model), `THESATTO` (global model), and `POTHESTATT` (binning model).
- Bug fix on the $C_{gb}-C_{bg}$ reciprocity in strong inversion regime, when the bias-dependence of interface states model is activated with high value of CTG/CTB parameters.
- New calculation of the drain saturation voltage to improve the drain saturation current of long channel transistors.
- Improvement of S/D symmetry for low value of AX parameter, by introduction of a new mathematical function of linear-saturation transition.
- Bug fix on the source and drain access resistances should be independent to the number of fingers NF.
- Removal of the effective doping bias-dependence effect in the surface potential equation and its associated parameters `VNSUB`, `VNSUBO`, `POVNSUB`, `NSLP`, `NSLPO`, `PONSLP`, `DNSUB`, `DNSUBO`, and `PODNSUB`.
- New binning equations with "hybrid" approach to mix physical scaling rules with binning rules.

PSP 103.8.2 Model Updates

The following updates were made in PSP v103.8.2:

- Addition of a switch parameter SWFIX for backward compatibility.
- Surface potential equation code enhancement to improve convergence.
- Non-reciprocity in C_{gb}/C_{bg} at nonzero CTG and CTB parameters.
- Bug fix on inner fringe capacitance's reciprocity in accumulation regime when `FCINRACC=0`.
- Modifications of if conditions for the calculations of the `c_igid` and `fknee` variables.

PSP 103.8.1 Model Updates

The following updates were made in PSP 103.8.1:

- Additional parameters for temperature control: TRISE, DTEMP and TREF
- Modification of scaling rules of inner fringe charges (physical vs electrical width)
- Modification of $\text{sqrtumax} = \text{sqrt}(\text{abs}(\text{umax}))$ calculation: the absolute function is removed because the variable umax is always positive

PSP 103.8 Model Updates

The following updates were made in PSP v103.8:

- New inner fringe charges model with its associated parameters. CINR, CINRD, DVFBINR, FCINRDEP, FCINRACC, AXINR (local model) CINRW, CINRDW, DVFBINRO, FCINRDEPO, FCINRACCO, AXINRO (global model) POCINR, PLCINR, PWCINR, PLWCINR, CINRD, PLCINRD, PWCINRD, PLWCINRD, POFCINRDEP, POFCINRACC, PODVFBINR, POAXINR (binning model).
- Model of overlaps inversion charge with its associated parameters. FCGOVACC, FCGOVACCD, CGOVACCG (local model) FCGOVACCO, FCGOVACCD, CGOVACCGO (global model) POFCGOVACC, POFCGOVACCD, POCGOVACCG (binning model).
- Modification of gmin implementation.
- Fix on the sign of the flicker noise sources including edge transistor.

PSP 103.7 Model Updates

The following updates were made in PSP v103.7:

- Gate leakage currents: additional parameters for overlap gate leakage currents CG2OV, CG3OV (local model), CG2OVO, CG3OVO (global model), and POCG2OV, POCG3OV (binning model).
- Charge partitioning: new switch parameter SWQPART to modify the charge partitioning between the drain and the source.
- Charge model in saturation: by setting SWQSAT=1, additional parameters for charge model to improve CV description in saturation CFAC, THESATAC, AXAC, ALPAC (local model), CFACL, CFACLEXP, CFACW, THESATACO, THESATACL, THESATACLEXP, THESATACW, THESATACLW, AXACO, AXACL, ALPACL, ALPACLEXP, ALPACW (global model), and POCFAC, PLCFAC, PWCFAC, PLWCFAC, POTHESATAC, PLTHESATAC, PWTHESATAC, PLWTHESATAC, POAXAC, PLAXAC, PWAXAC, LWAXAC, POALPAC, PLALPAC, PWALPAC, PLWALPAC (binning model).
- Juncap2: addition of two multiplier factors, for current IFACTOR and charge CFACTOR.

PSP 103.6 Model Updates

The following updates were made in PSP v103.6:

- Induced gate noise: clipped value of *migid* using the correlation factor *c_igid*.
- Thermal noise of edge transistor: bug fix to avoid possible division by zero during the calculation of *redge*.
- Improvement of *gm/Id* in weak inversion: new model of interface states.
- Addition of new parameter *NSUBEDGELEXP*: exponent for channel length dependence of edge transistor substrate doping.
- Minimum values of calculated local parameters *NOV* and *NOVD* in global mode.

PSP 103.5 Model Updates

The following updates were made in PSP v103.5:

- Addition of new mobility parameters for coulomb scattering effect: THECS, STTHECS (local model), THECSO, STTHECSO (global model), and POTHECS, POSTTHECS (binning model).
- Addition of new parameters for quadratic temperature dependence of flatband voltage: ST2VFB (local model), ST2VFBO (global model), and POST2VFB (binning model).

PSP 103.4 Model Updates

The following updates were made in PSP v103.4:

- Addition of edge MOSFET for modeling of subthreshold hump effect. It can be turned on by setting *SWEDGE*=1.
- New switch to enable/disable the induced gate noise *SWIGN* (default value is 1 for backward compatibility).
- New parameters for modeling of short channel effects on subthreshold slope:
 - Subthreshold slope degradation for short channel transistors: PSCE (local model), PSCEL, PSCELEXP, PSCEW (global model), and POPSCE, PLPSCE, PWPSCE, PLWPSCE (binning model).
 - Subthreshold slope dependence with drain voltage: PSCED (local model), PSCEDO (global model), and POPSCED (binning model).
 - Subthreshold slope dependence with bulk voltage: PSCEB (local model), PSCEBO (global model), and POPSCEB (binning model).

- New parameter of JUNCAP model: Coefficient for reverse breakdown current limitation FREV.
- Minor bugfixes.

PSP 103.3 Model Updates

The following updates were made in PSP v103.3:

- Added excess noise which is important for sub-100-nm channel lengths. This addition is backward compatible with PSP 103.2. New parameters include: FNTexc (local model), FNTexcL (global model), POFNTexc, PLFNTexc, PWFNTexc, and PLWFNTexc (binning model).
- Fixed some minor bugs.

PSP 103.2 Model Updates

The following updates were made in PSP v103.3:

- Added support for self-heating in the PrimeSim Continuum solution. The self-heating version of the model has a fifth terminal (dt) to represent the temperature increase. New parameters include: RTH, CTH, and STRTH (local model), RTHO, CTHO, and STRTHO (global model and binning model).
- Fixed some minor bugs in the calculation of the OP-output.
- Improved the noise-mode by changing the behavior when crossing $V_{ds}=0$ at high frequency.

PSP 103.1 Model Updates

The following updates were made in PSP v103.1:

- Added external sheet resistance R_{SHD} for drain diffusion (used when $SWJUNASYM = 1$).
- Bug-fix and minor implementation change in NUD-model.
- Minor bug-fix in conditional for SP-calculation of overlap areas.

PSP103.0 Model Updates

The PSP103 model is NOT completely backward-compatible with PSP102.3.

The main changes are:

- Global, local and binning models are unified. When `SWGEO = 1` (default) the global model is used. When `SWGEO = 0` the local model is selected. The binning model is invoked if `SWGEO` is set to 2.
- Added non-uniform doping (NUD) model. The model can be invoked on by setting `SWNUD = 1` or 2. When `SWNUD = 1`, a separate surface potential calculation is carried out and the NUD model does not affect the CV results. This avoids non-reciprocal capacitances. When `SWNUD = 2`, the extra surface potential calculation is skipped and this may result in non-reciprocal capacitances.
- Added related model parameters `GFACNUDO`, `GFACNUDL`, `GFACNUDLEXP`, `GFACNUDW`, `GFACNUDLW`, `VSBNUDO` and `DVSBNUDO` to global, `GFACNUD`, `VSBNUD` and `DVSBNUD` to local and `POGFACNUD`, `PLGFACNUD`, `PWGFACNUD`, `PLWGFACNUD`, `POVSBNUD` and `PODVSBNUD` to binning models.
- Added Vth-adjustment model for CV. It can be turned on by setting `SWDELVTAC = 1`. Note that this requires extra computation of surface potentials. Added related model parameters `FACNEFFACO`, `FACNEFFACL`, `FACNEFFACW`, `FACNEFFACLW`, `DELVTACO`, `DELVTACL`, `DELVTACLEXP`, `DELVTACW` and `DELVTACLW` to global, `FACNEFFAC` and `DELVTAC` to local and `POFACNEFFAC`, `PLFACNEFFAC`, `PWFACNEFFAC`, `PLWFACNEFFAC`, `PODELVTAC`, `PLDELVTAC`, `PWDELVTAC` and `PLWDELVTAC` to binning model.
- Added external diffusion resistances to source and drain.
- Added instance parameters `NRS` and `NRD`; added model parameters `RSR` to global and binning, `RSE` and `RDE` to local model.
- Modified the geometrical scaling rules of following parameters: `VFB`, `STVFB`, `DPHIB`, `STBET`, and `STTHESAT`.
- Modified the binning rule of `BETN`.
- Removed the effect of `FETA` from CV.
- Some minor bug-fixes and implementation changes.

PSP102.3 Model Updates

PSP 102.3 is backward compatible with PSP 102.2. The main changes are:

- Added asymmetric junction model for the drain-bulk junction. The new junction parameters have a suffix "D" added to their names. When `SWJUNASYM = 1` the original parameters are used for the sourcebulk junction and the new parameters are used for drain-bulk junction. When `SWJUNASYM = 0` the original junction parameters are used for both source-bulk and drain-bulk junctions as in symmetric case, and the new junction parameters are neglected.

- Added asymmetric models for the overlap region of the drain side. These include:
 - Added related model parameters `TOXOVDO`, `LOVD` and `NOVDO` to global, `TOXOVD` and `NOVD` to local and `POTOXOVD`, `PONOVD`, `PLNOVD`, `PWNOVD` and `PLWNOVD` to binning models.
 - Asymmetric GIDL/GISL model. Added related parameters `AGIDLWD`, `BGIDLDO`, `STBGIDLDO` and `CGIDLDO` to global, `AGIDL`, `BGIDL`, `STBGIDL` and `CGIDL` to local and `POAGIDL`, `PLAGIDL`, `PWAGIDL`, `PLWAGIDL`, `POBGIDL`, `POSTBGIDL` and `POCGIDL` to binning models.
 - Asymmetric overlap gate current model. Added related parameters `IGOVDW` to global, `IGOVD` to local and `POIGOVD`, `PLIGOVD`, `PWIGOVD` and `PLWIGOVD` to binning models.
 - Asymmetric overlap capacitance model. Added related parameters `CGOVD` to local, `POCGOVD`, `PLCGOVD`, `PWCGOVD` and `PLWCGOVD` to binning models.
 - Asymmetric outer fringe capacitance model. Added related parameters `CFRDW` to global, `CFRD` to local and `POCFRD`, `PLCFRD`, `PWCFRD` and `PLWCFRD` to binning models.
- When `SWJUNASYM = 1` the original parameters for the models listed above are used for the source side and the newly added parameters are used for the drain side. When `SWJUNASYM = 0` the original parameters are used for both source and drain sides and the new parameters are ignored.
- Added `EF(local)`, `EFO(global)` and `POEF(binning)` as flicker noise frequency exponent parameters.
- Added noise parameters `LINTNOI` and `ALPNOI` to global model to increase the flexibility of the length scaling of the flicker noise.
- Some minor bug-fixes and implementation changes.

PSP102.2 Model Updates

The PSP102.2 model is backward-compatible with PSP102.1.

The main changes are:

- Added well proximity effect (`WPE`) model according to CMC's specification
- Added parameters `EPSROX (local)`, `EPSROXO (global)`, `POEPSROX (binning)` representing relative dielectric constant of gate oxide
- Added instance parameters `DELVTO` and `FACTUO` to local, global, and binning model
- Added `NF` support to global and binning model
- Extended stress model to support `NF`

- Added substrate resistance network and external gate resistance to PSP model besides `NQS` mode
- Added geometry scaling for gate resistance in global and binning models, resulting in additional model parameters `RSHG`, `RINT`, `DLSIL`, and `RVPOLY` and instance parameters `NGCON` and `XGW`
- Added JUNCAP2 Express as an optional alternative to the full JUNCAP2 model. Added related model parameters `SWJUNEXP`, `VJUNREF`, and `FJUNQ`, using model parameter `SWJUNEXP=1` to enable JUNCAP2 Express model
- Added “dummy” parameters `LMIN`, `LMAX`, `WMIN`, `WMAX` to binning model, which can be used as labels for the binning parameter sets
- Some minor bug fixes and minor implementation changes

PSP102.1 Model Updates

The PSP102.1 model is backward-compatible with PSP102.0.

The main changes include:

- Added clipping boundaries for `SWNQS`
- Made several minor changes and improvements in model implementation
- Solved bug in stress model
- Solved bug in JUNCAP2

PSP102.0 Model Updates

The PSP102.0 model is backward-compatible with PSP101.0 in all practical cases.

The most important changes include:

- The value for LG when `SWJUNCAP=2` was corrected
- The clipping/limiting behavior of NP has been made more transparent
- A minor numerical issue was resolved
- The scaling rule for DPHIB is now correctly implemented

PSP101.0 Model Updates

The PSP101.0 model is *not* backward compatible with PSP100.1.

The most important changes include:

- A complete set of binning scaling rules was added as a phenomenological alternative to the physics-based geometrical scaling rules
- BSIM-like instance parameters `AS`, `AD`, `PS` and `PD` were added for the junction model
- To avoid confusion between zeros and “O”s, zeros no longer occur in parameter names. They have all been replaced by “O”s
- Some global parameter names have an additional “O” in their names in order to avoid duplicate names in the global and local model

PSP100.1 Model Updates

The PSP100.1 model is compatible with the PSP100 model. It improves performance by about a factor of 2, and adds a thermal noise coefficient (`FNT`) parameter. This parameter is used to add in thermal noise. Set `FNT=1` (default) to turn on thermal noise. Set `FNT=0` to turn off thermal noise.

Using PSP Model in the PrimeSim Continuum Solution

The PSP model is Level 69 in the Synopsys MOSFET models. To use this model, specify:

For PSP version < 103:

```
M1 drain gate source bulk NCH w=4u l=1u
.MODEL NCH NMOS LEVEL=69
+ GEOMOD [=1|0] BINMOD [=1|0]
+ VERSION=num
```

Where:

- `GEOMOD=1`: Default, global model
- `GEOMOD=0`: Local model
- `BINMOD=0`: Default, non-binning model
- `BINMOD=1`: Binning model
- `VERSION=num`: Specify version number of the PSP model (<103)

For PSP version >= 103:

```
M1 drain gate source bulk [dt] NCH w=4u l=1u
.MODEL NCH NMOS LEVEL=69
+ SWGEO =[0|1|2]
+ VERSION=num
```


Where:

- `SWGEO=0`: Local model
- `SWGEO=1`: Default, global model
- `SWGEO=2`: Binning model
- `VERSION=num`: Specify version number of the PSP model (≥ 103)
- `dt`: Thermal node

See Also

- [.OPTION VER_CONTROL](#)

PSP 103.1 Model Parameters

This section covers the following PSP model parameters:

PSP 103.1 Model — Basic Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ¹⁸
VERSION	-	103.11	-	-	PSP version selection parameter	0, 1, 2
TYPE	-	1	-1	-	Channel type parameter; 1: NMOS, -1: PMOS ¹⁸	0, 1, 2
TR	°C	21	-273	-	Reference temperature	0, 1, 2
DTA	K	0	-	-	Temperature offset with respect to ambient circuit temperature	0, 1, 2

PSP 103.1 Model — Switch Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ¹⁹
SWGEO	-	1	0	2	Flag for geometrical model (0: local, 1:global, 2: binning)	0, 1, 2
SWIGATE	-	0	0	1	Flag for gate current (0: off)	0, 1, 2
SWIMPACT	-	0	0	1	Flag for impact ionization current (0: off)	

18. Indicates for which value of SWGEO the listed parameter is used.

19. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ¹⁹
SWGIDL	-	0	0	1	Flag for GIDL/GISL current (0: off)	0, 1, 2
SWJUNCAP	-	0	0	3	Flag for JUNCAP (0: off)	0, 1, 2
SWJUNASYM	-	0	-	-	Flag for asymmetric junctions (0: off)	0, 1, 2
SWNUD	-	0	0	2	Flag for NUD-effect (0: off)	0, 1, 2
SWDELVTAC	-	0	0	1	Flag for separate charge calculation (0: off)	0, 1, 2
QMC	-	1	0	-	Quantum-mechanical correction factor	0, 1, 2
SWEDGE	-	0	0	1	Flag for drain current of edge transistors (0: off)	0, 1, 2
SWIGN	-	1	0	1	New switch to active/disable the induced gate noise (0: off)	0, 1, 2

PSP 103.1 Model — Labels for Binning Set

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁰
LMIN	m	0	-	-	Dummy parameter to label binning set	2
LMAX	m	1	-	-	Dummy parameter to label binning set	2
WMIN	m	0	-	-	Dummy parameter to label binning set	2
WMAX	m	1	-	-	Dummy parameter to label binning set	2

PSP 103.1 Model — Process Parameters

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
LVARO	m	0	-	-	Geometry independent difference between actual and programmed poly-silicon gate length	1, 2

19. Indicates for which value of SWGEO the listed parameter is used.

20. Indicates for which value of SWGEO the listed parameter is used.

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
LVARL	-	0	-	-	Length dependence of ΔL_{PS}	1, 2
LVARW	-	0	-	-	Width dependence of ΔL_{PS}	1, 2
LAP	m	0	-	-	Effective channel length reduction per side due to lateral diffusion of source /drain dopant ions	1, 2
WVARO	m	0	-	-	Geometry independent difference between actual and programmed field-oxide opening	1, 2
WVARL	-	0	-	-	Length dependence of ΔW_{OD}	1, 2
WVARW	-	0	-	-	Width dependence of ΔW_{OD}	1, 2
WOT	m	0	-	-	Effective reduction of channel width per side due to lateral diffusion of channel-stop dopant ions	1, 2
DLQ	m	0	-	-	Effective channel length offset for CV	1, 2
DWQ	m	0	-	-	Effective channel width offset for CV	1, 2
VFB	V	-1	-	-	Flat-band voltage at TR	0
VFBO	V	-1	-	-	Geometry-independent part	1
VFBL	V	0	-	-	Length dependence	1
VFBW	V	0	-	-	Width dependence	1
VFBLW	V	0	-	-	Area dependence	1
POVFB	V	-1	-	-	Geometry independent part	2
PLVFB	V	0	-	-	Length dependence	2
PWVFB	V	0	-	-	Width dependence	2
PLWVFB	V	0	-	-	Length times width dependence	2
STVFB	V/K	5.10^{-4}	-	-	Temperature dependence of VFB	0

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
STVFBO	V/K	$5 \cdot 10^{-4}$	-	-	Geometry-independent part	1
STVFB	V/K	0	-	-	Length dependence	1
STVFBW	V/K	0	-	-	Width dependence	1
STVFBW	V/K	0	-	-	Area dependence	1
POSTVFB	V/K	$5 \cdot 10^{-4}$	-	-	Geometry independent part	2
PLSTVFB	V/K	0	-	-	Length dependence	2
PWSTVFB	V/K	0	-	-	Width dependence	2
PLWSTVFB	V/K	0	-	-	Length times width dependence	2
ST2VFB	K ⁻¹	0	-	-	Quadratic temperature dependence of VFB	0
ST2VFBO	K ⁻¹	0	-	-	Geometry independent part	1
POST2VFB	K ⁻¹	0	-	-	Geometry independent part	2
TOX	m	$2 \cdot 10^{-9}$	10^{-10}	-	Gate oxide thickness	0
TOXO	m	$2 \cdot 10^{-9}$	10^{-10}	-	Gate oxide thickness	1
POTOX	m	$2 \cdot 10^{-9}$	-	-	Geometry independent part	2
EPSROX	-	3.9	1	-	Relative permittivity of gate dielectric	0
EPSROXO	-	3.9	1	-	Geometry independent part	1
POEPSROX	-	3.9	1	-	Geometry independent part	2
NEFF	m ⁻³	$5 \cdot 10^{23}$	10^{20}	10^{26}	Substrate doping	0
NSUBO	m ⁻³	$3 \cdot 10^{23}$	10^{20}	-	Geometry independent substrate doping	1
NSUBW	-	0	-	-	Width dependence of substrate doping due to segregation	1
WSEG	m	10^{-8}	10^{-10}	-	Characteristic length for segregation of substrate doping	1
NPCK	m ⁻³	10^{24}	0	-	Pocket doping level	1

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
NPCKW	-	0	-	-	Width dependence of NPCK due to segregation	1
WSEGP	m	10 ⁻⁸	10 ⁻¹⁰	-	Characteristic length for segregation of pocket doping	1
LPCK	m	10 ⁻⁸	10 ⁻¹⁰	-	Characteristic length for lateral doping profile	1
LPCKW	-	0	-	-	Width dependence of LPCK due to segregation	1
FOL1	-	0	-	-	First order length dependence of short channel body-effect	1
FOL2	-	0	-	-	Second order length dependence of short channel body-effect	1
PONEFF	m ⁻³	5.10 ²³	-	-	Geometry independent part	2
PLNEFF	m ⁻³	0	-	-	Length dependence	2
PWNEFF	m ⁻³	0	-	-	Width dependence	2
PLWNEFF	m ⁻³	0	-	-	Length times width dependence	2
FACNEFFAC	-	1	0	-	Pre-factor for effective substrate doping in separate charge calculation when SWDELVTAC = 1	0
FACNEFFACO	-	1	0	-	Geometry independent part	1
FACNEFFACL	-	0	-	-	Length dependence	1
FACNEFFACW	-	0	-	-	Width dependence	1
FACNEFFACLW	-	0	-	-	Area dependence	1
POFACNEFFAC	-	1	-	-	Geometry independent part	2
PLFACNEFFAC	-	0	-	-	Length dependence	2
PWFACNEFFAC	-	0	-	-	Width dependence	2
PLWFACNEF FAC	-	0	-	-	Area dependence	2

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
GFACNUD	-	1	0.01	-	Body factor change due to NUD-effect	0
GFACNUDO	-	1	-	-	Geometry independent part	1
GFACNUDL	-	0	-	-	Length dependence	1
GFACNUDLEXP	-	1	-	-	Exponent for length dependence	1
GFACNUDW	-	0	-	-	Width dependence	1
GFACNUDLW	-	0	-	-	Area dependence	1
POGFACNUD	-	1	-	-	Geometry independent part	2
PLGFACNUD	-	0	-	-	Length dependence	2
PWGFACNUD	-	0	-	-	Width dependence	2
PLWGFACNUD	-	0	-	-	Area dependence	2
VSBNUD	V	0	0	-	Lower V_{SB} -value for NUD-effect	0
VSBNUDO	V	0	-	-	Geometry independent part	1
POVSBNUD	V	0	-	-	Geometry independent part	2
DVSBNUD	V	1	0.1	-	V_{SB} -range for NUD-effect	0
DVSBNUDO	V	1	-	-	Geometry independent part	1
PODVSBNUD	V	1	-	-	Geometry independent part	2
VNSUB	V	0	-	-	Effective doping bias-dependence parameter	0
VNSUBO	V	0	-	-	Geometry independent part	1
POVNSUB	V	0	-	-	Geometry independent part	2
NSLP	V	0.05	10^{-3}	-	Effective doping bias-dependence parameter	0
NSLPO	V	0.05	-	-	Geometry independent part	1
PONSLP	V	0.05	-	-	Geometry independent part	2

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
DNSUB	V ⁻¹	0	0	1	Effective doping bias-dependence parameter	0
DNSUBO	V ⁻¹	0	-	-	Geometry independent part	1
PODNSUB	V ⁻¹	0	-	-	Geometry independent part	2
DPHIB	V	0	-	-	Offset of ϕ_B	0
DPHIBO	V	0	-	-	Geometry independent part	1
DPHIBL	V	0	-	-	Length dependence	1
DPHIBLEXP	-	1	-	-	Exponent for length dependence	1
DPHIBW	V	0	-	-	Exponent for length dependence	1
DPHIBLW	V	0	-	-	Area dependence	1
PODPHIB	V	0	-	-	Geometry independent part	2
PLDPHIB	V	0	-	-	Length dependence	2
PWDPHIB	V	0	-	-	Width dependence	2
PLWDPHIB	V	0	-	-	Length times width dependence	2
DELVTAC	V	0	-	-	Offset of ϕ_B in separate charge calculation when SWDELVTAC = 1	0
DELVTACO	V	0	-	-	Geometry independent part	1
DELVTACL	-	1	-	-	Length dependence	1
DELVTACLEXP	V	0	-	-	Exponent for length dependence	1
DELVTACW	V	0	-	-	Width dependence	1
DELVTACLW	V	0	-	-	Area dependence	1
PODELVTAC	V	0	-	-	Geometry independent part	2
PLDELVTAC	V	0	-	-	Length dependence	2
PWDELVTAC	V	0	-	-	Width dependence	2

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
PLWDELVTAC	V	0	-	-	Length times width dependence	2
NP	m ⁻³	10 ²⁶	0	-	Gate poly-silicon doping	0
NPO	m ⁻³	10 ²⁶	-	-	Geometry-independent part	1
NPL	-	0	-	-	Length dependence	1
PONP	m ⁻³	10 ²⁶	-	-	Geometry independent part	2
PLNP	m ⁻³	-	-	-	Length dependence	2
PWNP	m ⁻³	-	-	-	Width dependence	2
PLWNP	m ⁻³	-	-	-	Length times width dependence	2
CT	-	0	0	-	Interface states factor	0
CTO	-	0	-	-	Geometry-independent part	1
CTL	-	0	-	-	Length dependence	1
CTLEXP	-	1	-	-	Exponent for length dependence	1
CTW	-	0	-	-	Width dependence	1
CTLW	-	0	-	-	Area dependence	1
POCT	-	0	-	-	Geometry independent part	2
PLCT	-	0	-	-	Length dependence	2
PWCT	-	0	-	-	Width dependence	2
PLWCT	-	0	-	-	Length times width dependence	2
TOXOV	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Overlap oxide thickness	0
TOXOVO	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Geometry independent part	1
POTOXOV	m	2.10 ⁻⁹	-	-	Geometry independent part	2
TOXOVD	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Overlap oxide thickness for drain side	0
TOXOVDO	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Geometry independent part	1
POTOXOVD	m	2.10 ⁻⁹	-	-	Geometry independent part	2

21. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ²¹
LOV	m	0	0	-	Overlap length for overlap capacitance	1
LOVD	m	0	0	-	Overlap length for gate/drain overlap capacitance	1
NOV	m ⁻³	5.10 ²⁵	10 ²³	10 ²⁷	Effective doping of overlap region	0
NOVO	m ⁻³	5.10 ²⁵	-	-	Geometry independent part	1
PONOV	m ⁻³	5.10 ²⁵	-	-	Geometry independent part	2
PLNOV	m ⁻³	0	-	-	Length dependence	2
PWNOV	m ⁻³	0	-	-	Width dependence	2
PLWNOV	m ⁻³	0	-	-	Length times width dependence	2
NOVD	m ⁻³	5.10 ²⁵	10 ²³	10 ²⁷	Effective doping of overlap region for drain side	0
NOVDO	m ⁻³	5.10 ²⁵	-	-	Geometry independent part	1
PONOVD	m ⁻³	5.10 ²⁵	-	-	Geometry independent part	2
PLNOVD	m ⁻³	0	-	-	Length dependence	2
PWNOVD	m ⁻³	0	-	-	Width dependence	2
PLWNOVD	m ⁻³	0	-	-	Length times width dependence	2

PSP 103.1 Model — DIBL Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²²
CF	-	0	0	-	DIBL parameter	0
CFL	-	0	-	-	Length dependence	1
CFLEXP	-	2	-	-	Exponent for length dependence	1
CFW	-	0	-	-	Width dependence	1
POCF	-	0	-	-	Geometry independent part	2

21. Indicates for which value of SWGEO the listed parameter is used.

22. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²²
PLCF	-	0	-	-	Length dependence	2
PWCF	-	0	-	-	Width dependence	2
PLWCF	-	0	-	-	Length times width dependence	2
CFB	V^{-1}	0	0	1	Back-bias dependence of CF	0
CFBO	V^{-1}	0	-	-	Geometry independent part	1
POCFB	V^{-1}	0	-	-	Geometry independent part	2
CFD	V^{-1}	0	0	-	Drain voltage dependence of CF	0
CFDO	V^{-1}	0	-	-	Geometry independent part	1
POCFD	V^{-1}	0	-	-	Geometry independent part	2

PSP 103.1 Model — Subthreshold Slope Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²³
PSCE	-	0	0	-	Subthreshold slope coefficient for short channel transistor	0
PSCEL	-	0	-	-	Length dependence	1
PSCELEXP	-	2	-	-	Exponent for length dependence	1
PSCEW	-	0	-	-	Width dependence	1
POPSCE	-	0	-	-	Geometry independent part	2
PLPSCE	-	0	-	-	Length dependence	2
PWPSCE	-	0	-	-	Width dependence	2
PLWPSCE	-	0	-	-	Length times width dependence	2
PSCEB	V^{-1}	0	0	1	Back-bias dependence of PSCE	0
PSCEBO	V^{-1}	0	-	-	Geometry independent part	1
POPSCEB	V^{-1}	0	-	-	Geometry independent part	2

22. Indicates for which value of SWGEO the listed parameter is used.

23. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²³
PSCED	V ⁻¹	0	0	-	Drain voltage dependence of PSCE	0
PSCEDO	V ⁻¹	0	-	-	Geometry independent part	1
POPCED	V ⁻¹	0	-	-	Geometry independent part	2

PSP 103.1 Model — Mobility Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁴
BETN	m ² /V/s	7.10 ⁻²	0	-	Product of channel aspect ratio and zero-field mobility at TR	0
UO	m ² /V/s	5.10 ⁻²	-	-	Zero-field mobility at TR	1
FBET1	-	0	-	-	Relative mobility decrease due to first lateral profile	1
FBET1W	-	0	-	-	Width dependence of FBET1	1
LP1	m	10 ⁻⁸	10 ⁻¹⁰	-	Mobility-related characteristic length of first lateral profile	1
LP1W	-	0	-	-	Width dependence of LP1	1
FBET2	-	0	-	-	Relative mobility decrease due to second lateral profile	1
LP2	m	10 ⁻⁸	10 ⁻¹⁰	-	Mobility-related characteristic length of second lateral profile	1
BETW1	-	0	-	-	First higher-order width scaling coefficient of BETN	1
BETW2	-	0	-	-	Second higher-order width scaling coefficient of BETN	1
WBET	m	10 ⁻⁹	10 ⁻¹⁰	-	Characteristic width for width scaling of BETN	1
POBETN	m ² /V/s	7.10 ⁻²	-	-	Geometry independent part	2
PLBETN	m ² /V/s	0	-	-	Length dependence	2
PWBETN	m ² /V/s	0	-	-	Width dependence	2

23. Indicates for which value of SWGEO the listed parameter is used.

24. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁴
PLWBETN	m ² /V/s	0	-	-	Length times width dependence	2
STBET	0	1	-	-	Temperature dependence of BETN	0
STBETO	0	1	-	-	Geometry independent part	1
STBETL	0	0	-	-	Length dependence	1
STBETW	0	0	-	-	Width dependence	1
STBETLW	0	0	-	-	Area dependence	1
POSTBET	0	1	-	-	Geometry independent part	2
PLSTBET	0	0	-	-	Length dependence	2
PWSTBET	0	0	-	-	Width dependence	2
PLWSTBET	0	0	-	-	Length times width dependence	2
MUE	m/V	0.5	0	-	Mobility reduction coefficient at TR	0
MUEO	m/V	0.5	-	-	Geometry independent part	1
MUEW	-	0	-	-	Width dependence	1
POMUE	m/V	0.5	-	-	Geometry independent part	2
PLMUE	m/V	0	-	-	Length dependence	2
PWMUE	m/V	0	-	-	Width dependence	2
PLWMUE	m/V	0	-	-	Length times width dependence	2
STMUE	-	0	-	-	Temperature dependence of MUE	0
STMUEO	-	0	-	-	Geometry independent part	1
POSTMUE	-	0	-	-	Geometry independent part	2
THEMU	-	1.5	0	-	Mobility reduction exponent at TR	0
THEMUO	-	1.5	0	-	Geometry independent part	1
POTHEMU	-	1.5	-	-	Geometry independent part	2

24. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁴
STTHEMU	-	1.5	-	-	Temperature dependence of THEMU	0
STTHEMUO	-	1.5	-	-	Geometry independent part	1
POSTTHEMU	-	1.5	-	-	Geometry independent part	2
CS	-	0	0	-	Coulomb scattering parameter at TR	0
CSO	-	0	-	-	Geometry independent part	1
CSL	-	0	-	-	Length dependence	1
CSLEXP	-	1	-	-	Exponent for length dependence	1
CSW	-	0	-	-	Width dependence	1
CSLW	-	0	-	-	Area dependence	1
POCS	-	0	-	-	Geometry independent part	2
PLCS	-	0	-	-	Length dependence	2
PWCS	-	0	-	-	Width dependence	2
PLWCS	-	0	-	-	Length times width dependence	2
STCS	-	0	-	-	Temperature dependence of CS	0
STCSO	-	0	-	-	Geometry independent part	1
POSTCS	-	0	-	-	Geometry independent part	2
THECS	-	2	0	-	Coulomb scattering exponent at TR	0
THECSO	-	2	0	-	Geometry independent part	1
POTHECS	-	0	0	-	Geometry independent part	2
STTHECS	-	0	-	-	Temperature dependence of THECS	0
STTTHECSO	-	0	-	-	Geometry independent part	1
POSTTHECS	-	0	-	-	Geometry independent part	2
XCOR	V ⁻¹	0	0	-	Non-universality parameter	0

24. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁴
XCORO	V^{-1}	0	-	-	Geometry independent part	1
XCORL	-	0	-	-	Length dependence	1
XCORW	-	0	-	-	Width dependence	1
XCORLW	-	0	-	-	Area dependence	1
POXCOR	V^{-1}	0	-	-	Geometry independent part	2
PLXCOR	V^{-1}	0	-	-	Length dependence	2
PWXCOR	V^{-1}	0	-	-	Width dependence	2
PLWXCOR	V^{-1}	0	-	-	Length times width dependence	2
STXCOR	-	0	-	-	Temperature dependence of XCOR	0
STXCORO	-	0	-	-	Geometry independent part	1
POSTXCOR	-	0	-	-	Geometry independent part	2
FETA	-	1	0	-	Effective field parameter	0
FETAO	-	1	-	-	Geometry independent part	1
POFETA	-	1	-	-	Geometry independent part	2

PSP 103.1 Model — Series Resistance Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁵
RS	Ω	30	0	-	Source/drain series resistance at TR	0
RSW1	Ω	50	-	-	Source/drain series resistance for a channel width W_{EN}	1
RSW2	-	0	-	-	Higher-order width scaling	1
PORS	Ω	30	-	-	Geometry independent part	2
PLRS	Ω	0	-	-	Length dependence	2
PWRS	Ω	0	-	-	Width dependence	2

24. Indicates for which value of SWGEO the listed parameter is used.

25. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE _{O²⁵}
PLWRS	Ω	0	-	-	Length times width dependence	2
STRS	-	1	-	-	Temperature dependence of RS	0
STRSO	-	1	-	-	Geometry dependent part	1
POSTRS	-	1	-	-	Geometry independent part	2
RSB	V^{-1}	0	-0.5	-	Back-bias dependence of RS	0
RSBO	V^{-1}	0	-	-	Geometry independent part	1
PORSB	V^{-1}	0	-	-	Geometry independent part	2
RSG	V^{-1}	0	-0.5	-	Gate-bias dependence of RS	0
RSGO	V^{-1}	0	-	-	Geometry independent part	1
PORSG	V^{-1}	0	-	-	Geometry independent part	2

PSP 103.1 Model — Velocity Saturation Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁶
THESAT	V^{-1}	1	0	-	Velocity saturation parameter at TR	0
THESATO	V^{-1}	0	-	-	Geometry independent part	1
THESATL	V^{-1}	0.05	-	-	Length dependence	1
THESATLEXP	-	1	-	-	Exponent for length dependence	1
THESATW	-	0	-	-	Width dependence	1
THESATLW	-	0	-	-	Area dependence	1
POTHE SAT	V^{-1}	1	-	-	Geometry independent part	2
PLTHE SAT	V^{-1}	0	-	-	Length dependence	2
PWTHE SAT	V^{-1}	0	-	-	Width dependence	2
PLWTHE SAT	V^{-1}	0	-	-	Length times width dependence	2

25. Indicates for which value of SWGEO the listed parameter is used.

26. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁶
STTHESAT	-	1	-	-	Temperature dependence of THESAT	0
STTHESATO	-	1	-	-	Geometry independent temperature dependence	1
STTHESATL	-	0	-	-	Length dependence	1
STTHESATW	-	0	-	-	Width dependence	1
STTHESATLW	-	0	-	-	Area dependence	1
POSTTHESAT	-	1	-	-	Geometry independent part	2
PLSTTHESAT	-	0	-	-	Length dependence	2
PWSTTHESAT	-	0	-	-	Width dependence	2
PLWSTTHESAT	-	0	-	-	Length times width dependence of temperature dependence	2
THESATB	V ⁻¹	0	-0.5	-	Back-bias dependence of velocity saturation	0
THESATBO	V ⁻¹	0	-	-	Geometry independent part	1
POTTHESATB	V ⁻¹	0	-	-	Geometry independent part	2
PLTHESATB	V ⁻¹	0	-	-	Length dependence	2
PWTHESATB	V ⁻¹	0	-	-	Width dependence	2
PLWTHESATB	V ⁻¹	0	-	-	Length times width dependence	2
THESATG	V ⁻¹	0	-0.5	-	Gate-bias dependence of velocity saturation	0
THESATGO	V ⁻¹	0	-	-	Geometry independent part	1
THESATGO	V ⁻¹	0	-	-	Geometry independent part	2
PLTHESATG	V ⁻¹	0	-	-	Length dependence	2
PWTHESATG	V ⁻¹	0	-	-	Width dependence	2
PLWTHESATG	V ⁻¹	0	-	-	Length times width dependence	2

26. Indicates for which value of SWGEO the listed parameter is used.

PSP 103.1 Model — Saturation Voltage Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁷
AX	-	3	2	-	Linear/saturation transition factor	0
AXO	-	18	-	-	Geometry independent	1
AXL	-	0.4	0	-	Length dependence	1
POAX	-	3	-	-	Geometry independent part	2
PLAX	-	0	-	-	Length dependence	2
PWAX	-	0	-	-	Width dependence	2
PLWAX	-	0	-	-	Length times width dependence	2

PSP 103.1 Model — Channel Length Modulation (CLM) Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁸
ALP	-	0.01	0	-	CLM pre-factor	0
ALPL	-	$5 \cdot 10^{-4}$	-	-	Length dependence	1
ALPLEXP	-	1	-	-	Exponent for length dependence	1
ALPW	-	0	-	-	Width dependence	1
POALP	-	0.01	-	-	Geometry independent part	2
PLALP	-	0	-	-	Length dependence	2
PWALP	-	0	-	-	Width dependence	2
PLWALP	-	0	-	-	Length times width dependence	2
ALP1	V	0	0	-	CLM enhancement factor above threshold	0
ALP1L1	V	0	-	-	Length dependence	1
ALP1LEXP	-	0.5	-	-	Exponent for length dependence	1
ALP1L2	-	0	0	-	Second order length dependence	1
ALP1W	-	0	0	-	Width dependence	1

27. Indicates for which value of SWGEO the listed parameter is used.

28. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE ²⁸ O ²⁸
POALP1	V	0	0	-	Geometry independent part	2
PLALP1	V	0	0	-	Length dependence	2
PWALP1	V	0	0	-	Width dependence	2
PLWALP1	V	0	-	-	Length times width dependence	2
ALP2	V ⁻¹	0	0	-	CLM enhancement factor below threshold	0
ALP2L1	V	0	-	-	Length dependence	1
ALP2LEXP	-	0.5	-	-	Exponent for length dependence	1
ALP2L2	-	0	0	-	Second order length dependence	1
ALP2W	-	0	-	-	Width dependence	1
POALP2	V ⁻¹	0	-	-	Geometry independent part	2
PLALP2	V ⁻¹	0	-	-	Length dependence	2
PWALP2	V ⁻¹	0	-	-	Width dependence	2
PLWALP2	V ⁻¹	0	-	-	Length times width dependence	2
VP	V	0.05	10 ⁻¹⁰	-	CLM logarithmic dependence parameter	0
VPO	V	0.05	-	-	Geometry independent part	1
POVP	V	0.05	-	-	Geometry independent part	2

PSP 103.1 Model — Impact Ionization (II) Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁹
A1	-	1	0	-	Impact-ionization pre-factor	0
A1O	-	1		-	Geometry independent part	1
A1L	-	0		-	Length dependence	1
A1W	-	0		-	Width dependence	1

28. Indicates for which value of SWGEO the listed parameter is used.

29. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁹
POA1	-	1		-	Geometry independent part	2
PLA1	-	0		-	Length dependence	2
PWA1	-	0		-	Width dependence	2
PLWA1	-	0		-	Length times width dependence	2
A2	V	10	0	-	Impact-ionization exponent at TR	0
A2O	V	10	-	-	Geometry independent part	1
POA2	V	10	-	-	Geometry independent part	2
STA2	V	0	-	-	Temperature dependence of A2	0
STA2O	V	0	-	-	Geometry independent part	1
POSTA2	V	0	-	-	Geometry independent part	2
A3	-	1	0	-	Saturation-voltage dependence of I_D	0
A3O	-	1	-	-	Geometry independent part	1
A3L	-	0	-	-	Length dependence	1
A3W	-	0	-	-	Width dependence	1
POA3	-	1	-	-	Geometry independent part	2
PLA3	-	0	-	-	Length dependence	2
PWA3	-	0	-	-	Width dependence	2
PLWA3	-	0	-	-	Length times width dependence	2
A4	$V^{-1/2}$	0	0	-	Back-bias dependence of I_D	0
A4O	$V^{-1/2}$	0	-	-	Geometry independent part	1
A4L	-	0	-	-	Length dependence	1
A4W	-	0	-	-	Width dependence	1
POA4	$V^{-1/2}$	0	-	-	Geometry independent part	2
PLA4	$V^{-1/2}$	0	-	-	Length dependence	2
PWA4	$V^{-1/2}$	0	-	-	Width dependence	2

29. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ²⁹
PLWA4	$V^{-1/2}$	0	-	-	Length times width dependence	2

PSP 103.1 Model — Gate Current Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁰
GCO	-	0	-10	10	Gate tunnelling energy adjustment	0
GCOO	-	0	-	-	Geometry independent part	1
POGCO	-	0	-	-	Geometry independent part	2
IGINV	A	0	0	-	Gate channel current pre-factor	0
IGINVLW	A	0	-	-	Gate channel current pre-factor for a channel area of $W_{EN} \cdot L_{EN}$	1
POIGINV	A	0	-	-	Geometry independent part	2
PLIGINV	A	0	-	-	Length dependence	2
PWIGINV	A	0	-	-	Width dependence	2
PLWIGINV	A	0	-	-	Length times width dependence	2
IGOV	A	0	0	-	Gate overlap current pre-factor	0
IGOVW	A	0	-	-	Gate overlap current pre-factor for a channel width of W_{EN}	1
POIGOV	A	0	-	-	Geometry independent part	2
PLIGOV	A	0	-	-	Length dependence	2
PWIGOV	A	0	-	-	Width dependence	2
PLWIGOV	A	0	-	-	Length times width dependence	2
IGOVD	A	0	0	-	Gate overlap current pre-factor for drain side	0
IGOVDW	A	0	-	-	Gate overlap current pre-factor for a channel width of W_{EN} for drain side	1
POIGOVD	A	0	-	-	Geometry independent part	2

29. Indicates for which value of SWGEO the listed parameter is used.

30. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ³⁰
PLIGOVD	A	0	-	-	Length dependence	2
PWIGOVD	A	0	-	-	Width dependence	2
PLWIGOVD	A	0	-	-	Length times width dependence	2
STIG	-	2	-	-	Temperature dependence of gate current	0
STIGO	-	2	-	-	Geometry independent part	1
POSTIG	-	2	-	-	Geometry independent part	2
GC2	-	0.375	0	10	Gate current slope factor	0
GC2O	-	0.375	-	-	Geometry independent part	1
POGC2	-	3.75×10^{-1}	-	-	Geometry independent part	2
GC3	-	0.063	-2	2	Gate current curvature factor	0
GC3O	-	0.063	-	-	Geometry independent part	1
POGC3	-	6.3×10^{-2}	-	-	Geometry independent part	2
CHIB	V	3.1	1	-	Tunnelling barrier height	0
CHIBO	V	3.1	-	-	Geometry independent part	1
POCHIB	V	3.1	-	-	Geometry independent part	2

PSP 103.1 Model — Gate-Induced Drain Leakage (GIDL) Parameters

Name	Unit	Default	Min.	Max.	Description	SWGE O ³¹
AGIDL	A/V^3	0	0	-	GIDL pre-factor	0
AGIDLW	A/V^3	0	-	-	Width dependence	1
POAGIDL	A/V^3	0	-	-	Geometry independent part	2
PLAGIDL	A/V^3	0	-	-	Length dependence	2

30. Indicates for which value of SWGEO the listed parameter is used.

31. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ³¹
PWAGIDL	A/V^3	0	-	-	Width dependence	2
PLWAGIDL	A/V^3	0	-	-	Length times width dependence	2
AGIDLD	A/V^3	0	0	-	GIDL pre-factor for drain side	0
AGIDLDO	A/V^3	0	-	-	Width dependence	1
POAGIDLD	A/V^3	0	-	-	Geometry independent part	2
PLAGIDLD	A/V^3	0	-	-	Length dependence	2
PWAGIDLD	A/V^3	0	-	-	Width dependence	2
PLWAGIDLD	A/V^3	0	-	-	Length times width dependence	2
BGIDL	V	41	0	-	GIDL probability factor at TR	0
BGIDLO	V	41	-	-	Geometry independent part	1
POBGIDL	V	41	-	-	Geometry independent part	2
BGIDLD	V	41	0	-	GIDL probability factor at TR for drain side	0
BGIDLDO	V	41	-	-	Geometry independent part	1
POBGIDLD	V	41	-	-	Geometry independent part	2
STBGIDL	V/K	0	-	-	Temperature dependence of BGIDL	0
STBGIDLO	V/K	0	-	-	Geometry independent part	1
POSTBGIDL	V/K	0	-	-	Geometry independent part	2
STBGIDLD	V/K	0	-	-	Temperature dependence of BGIDL for drain side	0
STBGIDLDO	V/K	0	-	-	Geometry independent part	1
POSTBGIDLD	V/K	0	-	-	Geometry independent part	2
CGIDL	-	0	-	-	Back-bias dependence of GIDL	0
CGIDLO	-	0	-	-	Geometry independent part	1
POCGIDL	-	0	-	-	Geometry independent part	2

31. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ³¹
CGIDLD	-	0	-	-	Back-bias dependence of GIDL for drain side	0
CGIDLDO	-	0	-	-	Geometry independent part	1
POCGIDLD	-	0	-	-	Geometry independent part	2

PSP 103.1 Model — Charge Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGE O ³²
COX	F	10^{-14}	0	-	Oxide capacitance for intrinsic channel	0
POCOX	F	10^{-14}	-	-	Geometry independent part	2
PLCOX	F	0	-	-	Length dependence	2
PWCOX	F	0	-	-	Width dependence	2
PLWCOX	F	0	-	-	Length times width dependence	2
CGOV	F	10^{-15}	0	-	Oxide capacitance for gate–drain/source overlap	0
POCGOV	F	10^{-15}	-	-	Geometry independent part	2
PLCGOV	F	0	-	-	Length dependence	2
PWCGOV	F	0	-	-	Width dependence	2
PLWCGOV	F	0	-	-	Length times width dependence	2
CGOVD	F	10^{-15}	0	-	Oxide capacitance for gate–drain/source overlap for drain side	0
POCGOVD	F	10^{-15}	-	-	Geometry independent part	2
PLCGOVD	F	0	-	-	Length dependence	2
PWCGOVD	F	0	-	-	Width dependence	2
PLWCGOVD	F	0	-	-	Length times width dependence	2

31. Indicates for which value of SWGEO the listed parameter is used.

32. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGE O ³²
CGBOV	F	0	0	-	Oxide capacitance for gate–bulk overlap	0
CGBOVL	F	0	-	-	Oxide capacitance for gate–bulk overlap for a channel length of L_{EN}	1
POCGBOV	F	0	-	-	Geometry independent part	2
PLCGBOV	F	0	-	-	Length dependence	2
PWCGBOV	F	0	-	-	Width dependence	2
PLWCGBOV	F	0	-	-	Length times width dependence	2
CFR	F	0	-	-	Outer fringe capacitance	0
CFRW	F	0	0	-	Outer fringe capacitance for a channel width of W_{EN}	1
POCFR	F	0	-	-	Geometry independent part	2
PLCFR	F	0	-	-	Length dependence	2
PWCFR	F	0	-	-	Width dependence	2
PLWCFR	F	0	-	-	Length times width dependence	2
CFRD	F	0	0	-	Outer fringe capacitance for drain side	0
CFRDW	F	0	-	-	Outer fringe capacitance for a channel width of W_{EN} for drain side	1
POCFRD	F	0	-	-	Geometry independent part	2
PLCFRD	F	0	-	-	Length dependence	2
PWCFRD	F	0	-	-	Width dependence	2
PLWCFRD	F	0	-	-	Length times width dependence	2

32. Indicates for which value of SWGEO the listed parameter is used.

PSP 103.1 Model — Noise Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ³³
FNT	-	1	0	-	Thermal noise coefficient	0
FNTO	-	1	-	-	Geometry independent part	1
POFNT	-	1	-	-	Geometry independent part	2
FNTEXC	-	0	0	-	Excess noise coefficient	0
FNTEXCL	-	0	0	-	Length dependence of excess noise	1
POFNTEXC	-	0	-	-	Geometry independent part	2
PLFNTEXC	-	0	-	-	Length dependence	2
PWFNTEXC	-	0	-	-	Width dependence	2
PLWFNTEXC	-	0	-	-	Length times width dependence	2
NFA	V^{-1}/m^4	$8 \cdot 10^{22}$	0	-	First coefficient of flicker noise	0
NFALW	V^{-1}/m^4	$8 \cdot 10^{22}$	-	-	First coefficient of flicker noise for a channel area of $W_{EN} \cdot L_{EN}$	1
PONFA	V^{-1}/m^4	$8 \cdot 10^{22}$	-	-	Geometry independent part	2
PLNFA	V^{-1}/m^4	0	-	-	Length dependence	2
PWNFA	V^{-1}/m^4	0	-	-	Width dependence	2
PLWNFA	V^{-1}/m^4	0	-	-	Length times width dependence	2
NFB	V^{-1}/m^2	$3 \cdot 10^7$	0	-	Second coefficient of flicker noise	0
NFBLW	V^{-1}/m^2	$3 \cdot 10^7$	-	-	Second coefficient of flicker noise for a channel area of $W_{EN} \cdot L_{EN}$	1
PONFB	V^{-1}/m^2	$3 \cdot 10^7$	-	-	Geometry independent part	2
PLNFB	V^{-1}/m^2	0	-	-	Length dependence	2
PWNFB	V^{-1}/m^2	0	-	-	Width dependence	2
PLWNFB	V^{-1}/m^2	0	-	-	Length times width dependence	2
NFC	V^{-1}	0	0	-	Third coefficient of flicker noise	0

33. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³³
NFCLW	V^{-1}	0	-	-	Third coefficient of flicker noise for a channel area of $W_{EN} \cdot L_{EN}$	1
PONFC	V^{-1}	0	-	-	Geometry independent part	2
PLNFC	V^{-1}	0	-	-	Length dependence	2
PWNFC	V^{-1}	0	-	-	Width dependence	2
PLWNFC	V^{-1}	0	-	-	Length times width dependence	2
EF	-	1	0	-	Flicker noise frequency exponent	0
EFO	-	1	-	-	Geometry independent part	1
POEF	-	1	-	-	Geometry independent part	2
LINTNOI	m	0	-	-	Length offset for flicker noise	1
ALPNOI	-	2	-	-	Exponent for length offset	1

PSP 103.1 Model — Edge Transistor Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
WEDGE	m	10-8	0	-	Electrical width of edge transistor per side	1
WEDGEW	m	0	0	-	Main transistor width dependence of WEDGE	1
VFBEDGE	V	-1	-	-	Flat band voltage of edge transistor at TR	0
VFBEDGE0	V	-1	-	-	Geometry independent part	1
POVFBEDGE	V	-1	-	-	Geometry independent part	2
STVFBEDGE	V/K	$5 \cdot 10^{-4}$	-	-	Temperature dependence of VFBEDGE	0
STVFBEDGE0	V/K	$5 \cdot 10^{-4}$	-	-	Geometry independent part	1
STVFBEDGE1	V/K	0	-	-	Length dependence	1
STVFBEDGEW	V/K	0	-	-	Width dependence	1

33. Indicates for which value of SWGEO the listed parameter is used.

34. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
STVFBEDGE _{LW}	V/K	0	-	-	Area dependence	1
POSTVFBEDGE	V/K	$5 \cdot 10^{-4}$	-	-	Geometry independent part	2
PLSTVFBEDGE	V/K	0	-	-	Length dependence	2
PWSTVFBEDGE	V/K	0	-	-	Width dependence	2
PLWSTVFBEDGE	V/K	0	-	-	Length times width dependence	2
DPHIBEDGE	V	0	-	-	Offset of ψ_B for edge transistor	0
DPHIBEDGE _{EO}	V	0	-	-	Geometry independent part	1
DPHIBEDGE _L	V	0	-	-	Length dependence	1
DPHIBEDGE _L EXP	-	1	-	-	Exponent for length dependence	1
DPHIBEDGE _W	-	0	-	-	Width dependence	1
DPHIBEDGE _{LW}	-	0	-	-	Area dependence	1
PODPHIBEDGE	V	0	-	-	Geometry independent part	2
PLDPHIBEDGE	V	0	-	-	Length dependence	2
PWDPHIBEDGE	V	0	-	-	Width dependence	2
PLWDPHIBEDGE	V	0	-	-	Length times width dependence	2
NEFFEDGE	m ⁻³	$5 \cdot 10^{23}$	10^{20}	10^{26}	Effective substrate doping of edge transistor	0
NSUBEDGE _{EO}	m ⁻³	$5 \cdot 10^{23}$	10^{20}	-	Geometry independent substrate doping	1
NSUBEDGE _L	-	0	-	-	Length dependence	1
NSUBEDGE _L EXP	-	1	-	-	Exponent for length dependence	1
NSUBEDGE _W	-	0	-	-	Width dependence of edge transistor substrate doping	1
NSUBEDGE _{LW}	-	0	-	-	Area dependence	1
PONEFFEDGE	m ⁻³	$5 \cdot 10^{23}$	-	-	Geometry independent part	2

34. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
PLNEFFEDGE	m ⁻³	0	-	-	Length dependence	2
PWNEFFEDGE	m ⁻³	0	-	-	Width dependence	2
PLWNEFFEDGE	m ⁻³	0	-	-	Length times width dependence	2
CTEDGE	-	0	0	-	Interface states factor of edge transistor	0
CTEDGE0	-	0	-	-	Geometry-independent part	1
CTEDGE1	-	0	-	-	Length dependence	1
CTEDGEEXP	-	1	-	-	Exponent for length dependence	1
POCTEDGE	-	0	-	-	Geometry-independent part	2
PLCTEDGE	-	0	-	-	Length dependence	2
PWCTEDGE	-	0	-	-	Width dependence	2
PLWCTEDGE	-	0	-	-	Length times width dependence	2
BETNEDGE	m ² /V/ s	5 . 10 ⁻⁴	0	-	Product of channel aspect ratio and zero-field mobility of edge transistor at TR	0
FBETEDGE	-	0	-	-	Length dependence	1
LPEDGE	m	10 ⁻⁸	10 ⁻¹⁰	-	Exponent for length dependence	1
BETEDGEW	-	0	-	-	Width scaling coefficient	1
POBETNEDGE	m ² /V/ s	5 . 10 ⁻⁴	-	-	Geometry-independent part	2
PLBETNEDGE	m ² /V/ s	0	-	-	Length dependence	2
PWBETNEDGE	m ² /V/ s	0	-	-	Width dependence	2
PLWBETNEDGE	m ² /V/ s	0	-	-	Length times width dependence	2
STBETEDGE	-	1	-	-	Temperature dependence of BETNEDGE	0

34. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
STBETEDGE0	-	1	-	-	Geometry-independent part	1
STBETEDGEL	-	0	-	-	Length dependence	1
STBETEDGEW	-	0	-	-	Width dependence	1
STBETEDGELW	-	0	-	-	Area dependence	1
POSTBETEDGE	-	1	-	-	Geometry-independent part	2
PLSTBETEDGE	-	0	-	-	Length dependence	2
PWSTBETEDGE	-	0	-	-	Width dependence	2
PLWSTBETEDGE	-	0	-	-	Length times width dependence	2
PSCEEDGE	-	0	0	-	Subthreshold slope coefficient for short channel edge transistor	0
PSCEEDGEL	-	0	-	-	Length dependence	1
PSCEEDGEL EXP	-	2	-	-	Exponent for length dependence	1
PSCEEDGEW	-	0	-	-	Width dependence	1
POPSCEEDGE	-	0	-	-	Geometry-independent part	2
PLPSCEEDGE	-	0	-	-	Length dependence	2
PWPSCEEDGE	-	0	-	-	Width dependence	2
PLWPSCEEDGE	-	0	-	-	Length times width dependence	2
PSCEBEDGE	V ⁻¹	0	0	1	Back-bias dependence of PSCEEDGE	0
PSCEBEDGE0	V ⁻¹	0	-	-	Geometry-independent part	1
POPSCEBEDGE	V ⁻¹	0	-	-	Geometry-independent part	2
PSCEDEDGE	V ⁻¹	0	0	-	Drain voltage dependence of PSCEEDGE	0
PSCEDEDGE0	V ⁻¹	0	-	-	Geometry-independent part	1
POPSCEDEDGE	V ⁻¹	0	-	-	Geometry-independent part	2

34. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
CFEDGE	-	0	0	-	DIBL parameter of edge transistor	0
CFEDGE _L	-	0	-	-	Length dependence	1
CFEDGE _{LEXP}	-	2	-	-	Exponent for length dependence	1
CFEDGE _W	-	0	-	-	Width dependence	1
POCFEDGE	-	0	-	-	Geometry-independent part	2
PLCFEDGE	-	0	-	-	Length dependence	2
PWCFEDGE	-	0	-	-	Width dependence	2
PLWCFEDGE	-	0	-	-	Length times width dependence	2
CFBEDGE	V ⁻¹	0	0	1	Back-bias dependence of CFEDGE	0
CFBEDGE _O	V ⁻¹	0	-	-	Geometry-independent part	1
POCFBEDGE	V ⁻¹	0	-	-	Geometry-independent part	2
CFDEEDGE	V ⁻¹	0	0	-	Drain voltage dependence of CFEDGE	0
CFDEEDGE _O	V ⁻¹	0	-	-	Geometry-independent part	1
POCFDEEDGE	V ⁻¹	0	-	-	Geometry-independent part	2
FNTEDGE	-	1	0	-	Thermal noise coefficient of edge transistor	0
FNTEDGE _O	-	1	-	-	Geometry-independent part	1
POFNTEDGE	-	1	-	-	Geometry-independent part	2
NFAEDGE	V ⁻¹ /m ⁴	8 . 10 ²²	0	-	First coefficient of flicker noise of edge transistor	0
NFAEDGE _{LW}	V ⁻¹ /m ⁴	8 . 10 ²²	-	-	Area dependence	1
PONFAEDGE	V ⁻¹ /m ⁴	8 . 10 ²²	-	-	Geometry-independent part	2
PLNFAEDGE	V ⁻¹ /m ⁴	0	-	-	Length dependence	2

34. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁴
PWNFAEDGE	V^{-1}/m^4	0	-	-	Width dependence	2
PLWNFAEDGE	V^{-1}/m^4	0	-	-	Length times width dependence	2
NFBEDGE	V^{-1}/m^2	$3 \cdot 10^7$	0	-	Second coefficient of flicker noise of edge transistor	0
NFBEDGE _{LW}	V^{-1}/m^2	$3 \cdot 10^7$	-	-	Area dependence	1
PONFBEDGE	V^{-1}/m^2	$3 \cdot 10^7$	-	-	Geometry-independent part	2
PLNFBEDGE	V^{-1}/m^2	0	-	-	Length dependence	2
PWNFBEDGE	V^{-1}/m^2	0	-	-	Width dependence	2
PLWNFBEDGE	V^{-1}/m^2	0	-	-	Length times width dependence	2
NFCEDGE	V^{-1}	0	0	-	Third coefficient of flicker noise of edge transistor	0
NFCEDGE _{LW}	V^{-1}	0	-	-	Area dependence	1
PONFCEDGE	V^{-1}	0	-	-	Geometry-independent part	2
PLNFCEDGE	V^{-1}	0	-	-	Length dependence	2
PWNFCEDGE	V^{-1}	0	-	-	Width dependence	2
PLWNFCEDGE	V^{-1}	0	-	-	Length times width dependence	2
EFEDGE	-	1	0	-	Flicker noise frequency exponent of edge transistor	0
EFEDGE _O	-	1	-	-	Geometry-independent part	1
POEFEDGE	-	1	-	-	Geometry-independent part	2

34. Indicates for which value of SWGEO the listed parameter is used.

PSP 103.1 Model — Stress Model Parameters

The PSP stress model parameters are derived from BSIM4.40 with as little modifications as possible. Parameter names have been copied, but are subjected to PSP conventions by replacing every zero by an 'O'.

Note:

The `STK2` and `LODK2` parameters are not available in PSP. Except for these changes, stress parameters determined for BSIM can be directly applied in PSP.

The parameters listed in the following table are part of PSP global parameter set (both geometrical and binning).

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁵
SAREF	m	10 ⁻⁶	10 ⁻⁹	-	Reference distance between OD edge to Poly from one side	1, 2
SBREF	m	10 ⁻⁶	10 ⁻⁹	-	Reference distance between OD edge to Poly from other side	1, 2
WLOD	m	0	-	-	Width parameter	1, 2
KUO	m	0	-	-	Mobility degradation/enhancement coefficient	1, 2
KVSAT	m	0	-1	1	Saturation velocity degradation/enhancement parameter	1, 2
TKUO	-	0	-	-	Temperature coefficient of KUO	1, 2
LKUO	m ^{LLODKUO}	0	-	-	Length dependence of KUO	1, 2
WKUO	m ^{WLODKUO}	0	-	-	Width dependence of KUO	1, 2
PKUO	m ^{LLODKUO+WLODKUO}	0	-	-	Cross-term dependence of KUO	1, 2
LLODKUO	-	0	0	-	Length parameter for mobility stress effect	1, 2
WLODKUO	-	0	0	-	Width parameter for mobility stress effect	1, 2
KVTHO	Vm	0	-	-	Threshold shift parameter	1, 2
LKVTHO	m ^{LLODVTH}	0	-	-	Length dependence of KVTHO	1, 2

35. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁵
WKVTHO	m ^{WLODVTH}	0	-	-	Width dependence of KVTHO	1, 2
PKVTHO	m ^{LLODVTH+WLODVTH}	0	-	-	Cross-term dependence of KVTHO	1, 2
LLODVTH	-	0	0	-	Length parameter for threshold voltage stress effect	1, 2
WLODVTH	-	0	0	-	Width parameter for threshold voltage stress effect	1, 2
STETAO	m	0	-	-	ETAO shift factor related to threshold voltage change	1, 2
LODETAO	-	1	0	-	ETAO shift modification factor	1, 2

PSP 103.1 Model — Well Proximity Effect (WPE) Model Parameters

The PSP WPE model parameters are derived from BSIM4.5.0 with as little modifications as possible. Parameter names have been copied, but are subjected to PSP conventions by replacing every zero by an 'O'.

Note:

The K_{2WE} parameter are not available in PSP. Except for these changes, stress parameters determined for BSIM can be directly applied in PSP.

The parameters listed in the following table have both geometrical and binning rules included. You can use one of the following parameter sets depending on which scaling rule is selected.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁶
SCREF	m	1.10^{-6}	0	-	Distance between OD-edge and well edge of a reference device	1, 2
WEB	-	0	-	-	Coefficient for SCB	1, 2
WEC	-	0	-	-	Coefficient for SCC	1, 2
KVTHOWEO	-	0	-	-	Geometry independent threshold shift parameter	1
KVTHOWEL	-	0	-	-	Length dependence	1

35. Indicates for which value of SWGEO the listed parameter is used.

36. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁶
KVTHOWEW	-	0	-	-	Width dependence	1
KVTHOWELW	-	0	-	-	Area dependence	1
POKVTHOWE	-	0	-	-	Geometry independent part of threshold shift parameter	2
PLKVTHOWE	-	0	-	-	Length dependence	2
PWKVTHOWE	-	0	-	-	Width dependence	2
PLWKVTHOWE	-	0	-	-	Length times width dependence	2
KUOWEO	-	0	-	-	Geometry independent mobility degradation factor	1
KUOWEL	-	0	-	-	Length dependence	1
KUOWEW	-	0	-	-	Width dependence	1
KUOWELW	-	0	-	-	Area dependence	1
POKUOWE	-	0	-	-	Geometry independent part of mobility degradation factor	2
PLKUOWE	-	0	-	-	Length dependence	2
PWKUOWE	-	0	-	-	Width dependence	2
PLWKUOWE	-	0	-	-	Length times width dependence	2

PSP 103.1 Model — Source-bulk and Drain-bulk Junction Model Parameters

The PSP JUNCAP2 model parameters are part of both the global and the local parameter sets. You can use one of the following parameter sets depending on which scaling rule is selected:

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
TRJ	³⁷ C	21	T _{min}	-	Reference temperature	0, 1

36. Indicates for which value of SWGEO the listed parameter is used.

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
SWJUNEXP	-	0	0	1	Flag for JUNCAP2 Express; 0: full JUNCAP2 model, 1: Express model	0, 1
IMAX	A	1000	10^{-12}	-	Maximum current up to which forward current behaves exponentially	0, 1
Capacitance Parameters						
CJORBOT	F/m ²	10^{-3}	10^{-12}	-	Zero-bias capacitance per unit-of-area of bottom component for source-bulk junction	0, 1
CJORSTI	F/m	10^{-9}	10^{-18}	-	Zero-bias capacitance per unit-of-length of STI-edge component for source-bulk junction	0, 1
CJORGAT	F/m	10^{-9}	10^{-18}	-	Zero-bias capacitance per unit-of-length of gate-edge component for source-bulk junction	0, 1
VBIRBOT	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of bottom component for source-bulk junction	0, 1
VBIRSTI	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of STI-edge component for source-bulk junction	0, 1
VBIRGAT	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of gate-edge component for source-bulk junction	0, 1
PBOT	-	0.5	0.05	0.95	Grading coefficient of bottom component for source-bulk junction	0, 1
PSTI	-	0.5	0.05	0.95	Grading coefficient of STI-edge component for source-bulk junction	0, 1
PGAT	-	0.5	0.05	0.95	Grading coefficient of gate-edge component for source-bulk junction	0, 1
CJORBOTD	F/m ²	10^{-3}	10^{-12}	-	Zero-bias capacitance per unit-of-area of bottom component for drain-bulk junction	1
CJORSTID	F/m	10^{-9}	10^{-18}	-	Zero-bias capacitance per unit-of-length of STI-edge component for drain-bulk junction	1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
CJORGATD	F/m	10^{-9}	10^{-18}	-	Zero-bias capacitance per unit-of-length of gate-edge component for drain-bulk junction	1
VBIRBOTD	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of bottom component for drain-bulk junction	1
VBIRSTID	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of STI-edge component for drain-bulk junction	1
VBIRGATD	V	1	$V_{bi,low}$	-	Built-in voltage at the reference temperature of gate-edge component for drain-bulk junction	1
PBOTD	-	0.5	0.05	0.95	Grading coefficient of bottom component for drain-bulk junction	1
PSTID	-	0.5	0.05	0.95	Grading coefficient of STI-edge component for drain-bulk junction	1
PGATD	-	0.5	0.05	0.95	Grading coefficient of gate-edge component for drain-bulk junction	1
Ideal-Current Parameters						
PHIGBOT	V	1.16	-	-	Zero-temperature band-gap voltage of bottom component for source-bulk junction	0, 1
PHIGSTI	V	1.16	-	-	Zero-temperature band-gap voltage of STI-edge component for source-bulk junction	0, 1
PHIGGAT	V	1.16	-	-	Zero-temperature band-gap voltage of gate-edge component for source-bulk junction	0, 1
IDSATRBOT	A/m ²	10^{-12}	0	-	Saturation current density at the reference temperature of bottom component for source-bulk junction	0, 1
IDSATRSTI	A/m	10^{-18}	0	-	Saturation current density at the reference temperature of STI-edge component for source-bulk junction	0, 1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
IDSATRGAT	A/m	10^{-18}	0	-	Saturation current density at the reference temperature of gate-edge component for source-bulk junction	0, 1
PHIGBOTD	V	1.16	-	-	Zero-temperature band-gap voltage of bottom component for drain-bulk junction	1
PHIGSTID	V	1.16	-	-	Zero-temperature band-gap voltage of STI-edge component for drain-bulk junction	1
PHIGGATD	V	1.16	-	-	Zero-temperature band-gap voltage of gate-edge component for drain-bulk junction	1
IDSATRBOTD	A/m ²	10^{-12}	0	-	Saturation current density at the reference temperature of bottom component for drain-bulk junction	1
IDSATRSTID	A/m	10^{-18}	0	-	Saturation current density at the reference temperature of STI-edge component for drain-bulk junction	1
IDSATRGATD	A/m	10^{-18}	0	-	Saturation current density at the reference temperature of gate-edge component for drain-bulk junction	1
Shockley-Read-Hall Parameters						
CSRHBOT	A/m ³	10^2	0	-	Shockley-Read-Hall pre-factor of bottom component for source-bulk junction	0, 1
CSRHSTI	A/m ²	10^{-4}	0	-	Shockley-Read-Hall pre-factor of STI-edge component for source-bulk junction	0, 1
CSRHGAT	A/m ²	10^{-4}	0	-	Shockley-Read-Hall pre-factor of gate-edge component for source-bulk junction	0, 1
XJUNSTI	m	10^{-7}	10^{-9}	-	Junction depth of STI-edge component for source-bulk junction	0, 1
XJUNGAT	m	10^{-7}	10^{-9}	-	Junction depth of gate-edge component for source-bulk junction	0, 1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
CSRHBOTD	A/m ³	10 ²	0	-	Shockley-Read-Hall pre-factor of bottom component for drain-bulk junction	1
CSRHSTID	A/m ²	10 ⁻⁴	0	-	Shockley-Read-Hall pre-factor of STI-edge component for drain-bulk junction	1
CSRHGATD	A/m ²	10 ⁻⁴	0	-	Shockley-Read-Hall pre-factor of gate-edge component for drain-bulk junction	1
XJUNSTID	m	10 ⁻⁷	10 ⁻⁹	-	Junction depth of STI-edge component for drain-bulk junction	1
XJUNGATD	m	10 ⁻⁷	10 ⁻⁹	-	Junction depth of gate-edge component for drain-bulk junction	1
Trap-assisted Tunneling Parameters						
CTATBOT	A/m ³	10 ²	0	-	Trap-assisted tunneling pre-factor of bottom component for source-bulk junction	0, 1
CTATSTI	A/m ²	10 ⁻⁴	0	-	Trap-assisted tunneling pre-factor of STI-edge component for source-bulk junction	0, 1
CTATGAT	A/m ²	10 ⁻⁴	0	-	Trap-assisted tunneling pre-factor of gate-edge component for source-bulk junction	0, 1
MEFFTATBOT	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of bottom component for source-bulk junction	0, 1
MEFFTATSTI	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of STI-edge component for source-bulk junction	0, 1
MEFFTATGAT	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of gate-edge component for source-bulk junction	0, 1
CTATBOTD	A/m ³	10 ²	0	-	Trap-assisted tunneling pre-factor of bottom component for drain-bulk junction	1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
CTATSTID	A/m ²	10 ⁻⁴	0	-	Trap-assisted tunneling pre-factor of STI-edge component for drain-bulk junction	1
CTATGATD	A/m ²	10 ⁻⁴	0	-	Trap-assisted tunneling pre-factor of gate-edge component for drain-bulk junction	1
MEFFTATB OTD	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of bottom component for drain-bulk junction	1
MEFFTATSTID	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of STI-edge component for drain-bulk junction	1
MEFFTATG ATD	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of gate-edge component for drain-bulk junction	1
Band-to-band Tunneling Parameters						
CBBTBOT	AV ⁻³	10 ⁻¹²	0	-	Band-to-band tunneling prefactor of bottom component for source-bulk junction	0, 1
CBBTSTI	AV ⁻³ m	10 ⁻¹⁸	0	-	Band-to-band tunneling prefactor of STI-edge component for source-bulk junction	0, 1
CBBTGAT	AV ⁻³ m	10 ⁻¹⁸	0	-	Band-to-band tunneling prefactor of gate-edge component for source-bulk junction	0, 1
FBBTRBOT	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of bottom component for source-bulk junction	0, 1
FBBTRSTI	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of STI-edge component for source-bulk junction	0, 1
FBBTRGAT	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of gate-edge component for source-bulk junction	0, 1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
STFBBTBOT	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of bottom component for source-bulk junction	0, 1
STFBBTSTI	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of STI-edge component for source-bulk junction	0, 1
STFBBTGAT	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of gate-edge component for source-bulk junction	0, 1
CBBTBOTD	AV ⁻³	10 ⁻¹²	0	-	Band-to-band tunneling prefactor of bottom component for drain-bulk junction	1
CBBTSTID	AV ⁻³ m	10 ⁻¹⁸	0	-	Band-to-band tunneling prefactor of STI-edge component for drain-bulk junction	1
CBBTGATD	AV ⁻³ m	10 ⁻¹⁸	0	-	Band-to-band tunneling prefactor of gate-edge component for drain-bulk junction	1
FBBTRBOTD	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of bottom component for drain-bulk junction	1
FBBTRSTID	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of STI-edge component for drain-bulk junction	1
FBBTRGATD	Vm ⁻¹	10 ⁹	-	-	Normalization field at the reference temperature for band-to-band tunneling of gate-edge component for drain-bulk junction	1
STFBBTBOTD	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of bottom component for drain-bulk junction	1
STFBBTSTID	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of STI-edge component for drain-bulk junction	1

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
STFBBTGATD	K ⁻¹	-10 ⁻³	-	-	Temperature scaling parameter for band-to-band tunneling of gate-edge component for drain-bulk junction	1
Avalanche and Breakdown Parameters						
VBRBOT	V	10	0.1	-	Breakdown voltage of bottom component for source-bulk junction	0, 1
VBRSTI	V	10	0.1	-	Breakdown voltage of STI-edge component for source-bulk junction	0, 1
VBRGAT	V	10	0.1	-	Breakdown voltage of gate-edge component for source-bulk junction	0, 1
PBRBOT	V	4	0.1	-	Breakdown onset tuning parameter of bottom component for source-bulk junction	0, 1
PBRSTI	V	4	0.1	-	Breakdown onset tuning parameter of STI-edge component for source-bulk junction	0, 1
PBRGAT	V	4	0.1	-	Breakdown onset tuning parameter of gate-edge component for source-bulk junction	0, 1
VBRBOTD	V	10	0.1	-	Breakdown voltage of bottom component for drain-bulk junction	1
VBRSTID	V	10	0.1	-	Breakdown voltage of STI-edge component for drain-bulk junction	1
VBRGATD	V	10	0.1	-	Breakdown voltage of gate-edge component for drain-bulk junction	1
PBRBOTD	V	4	0.1	-	Breakdown onset tuning parameter of bottom component for drain-bulk junction	1
PBRSTID	V	4	0.1	-	Breakdown onset tuning parameter of STI-edge component for drain-bulk junction	1
PBRGATD	V	4	0.1	-	Breakdown onset tuning parameter of gate-edge component for drain-bulk junction	1
JUNCAP Express Parameters						

37. Indicates for which value of SWJUNASYM the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWJUNASYM ³⁷
VJUNREF	V	2.5	0.5	-	Typical maximum source-bulk junction voltage; usually about 2 . V_{sup}	0, 1
FJUNQ	V	0.03	0	-	Fraction below which source-bulk junction capacitance components are neglected	0, 1
VJUNREFD	V	2.5	0.5	-	Typical maximum drain-bulk junction voltage; usually about 2 . V_{sup}	1
FJUNQD	V	0.03	0	-	Fraction below which drain-bulk junction capacitance components are neglected	1

PSP 103.1 Model — Parasitic Resistance Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁸
RG	Ω	0	0	-	Gate resistance R_{gate}	0
RGO	Ω	0	-	-	Gate resistance R_{gate}	1, 2
RINT	$\frac{\Omega}{m^2}$	0	0	-	Contact resistance between silicide and poly	1, 2
RVPOLY	$\frac{\Omega}{m^2}$	0	0	-	Vertical poly resistance	1, 2
RSHG	Ω / μ	0	0	-	Gate electrode diffusion sheet resistance	1, 2
DLSIL	m	0	-	-	Silicide extension over the physical gate length	1, 2
RSE	Ω	0	0	-	External source resistance	0
RDE	Ω	0	0	-	External drain resistance	0
RSH	Ω / μ	0	-	-	Sheet resistance of source diffusion	1, 2
RSHD	Ω / μ	0	-	-	Sheet resistance of drain diffusion	1, 2
RBULK	Ω	0	0	-	Bulk resistance R_{bulk}	0

37. Indicates for which value of SWJUNASYM the listed parameter is used.

38. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁸
RBULKO	Ω	0	-	-	Bulk resistance R_{bulk}	1, 2
RWELL	Ω	0	0	-	Well resistance R_{well}	0
RWELLO	Ω	0	-	-	Well resistance R_{well}	1, 2
RJUNS	Ω	0	0	-	Source-side bulk resistance R_{juns}	0
RJUNSO	Ω	0	-	-	Source-side bulk resistance R_{juns}	1, 2
RJUND	Ω	0	0	-	Drain-side bulk resistance R_{jund}	0
RJUNDO	Ω	0	-	-	Drain-side bulk resistance R_{jund}	1, 2

PSP 103.1 Model — Self-heating Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ³⁹
The following parameters are only available in the self-heating version of the model:						
RTH	K/W	0	0	-	Thermal resistance	0
RTHO	K/W	0	-	-	Geometry-independent	1,2
RTHW1	K/W	0	-	-	Width dependence	1,2
RTHW2	-	0	-	-	Offset in width dependence	1,2
RTHLW	-	0	-	-	Length correction to width dependence	1,2
CTH	J/K	0	0	-	Thermal resistance	0
CTHO	J/K	0	-	-	Geometry-independent	1,2
CTHW1	J/K	0	-	-	Width dependence	1,2
CTHW2	-	0	-	-	Offset in width dependence	1,2
CTHLW	-	0	-	-	Length correction to width dependence	1,2
STRTH	-	0	-	-	Temperature sensitivity of RTH	0
STRTHO	-	0	-	-	Geometry-independent part	1,2

38. Indicates for which value of SWGEO the listed parameter is used.

39. Indicates for which value of SWGEO the listed parameter is used.

PSP 103.1 Model — NQS Model Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ⁴⁰
SWNQS	-	0	0	9	Switch for NQS effects / number of collocation points	0, 1, 2
MUNQS	-	1	0	-	Relative mobility for NQS modeling	0
MUNQSO	-	1	-	-	Relative mobility for NQS modeling	1, 2

PSP 103.1 Instance Parameters

Name	Unit	Default	Min.	Max.	Description	SWGEO ⁴¹
L	m	10^{-6}	10^{-9}	-	Drawn channel length	1, 2
W	m	10^{-6}	10^{-9}	-	Drawn channel width (total width)	1, 2
ABSOU RCE	m ²	10^{-12}	0	-	Source junction area	0, 1, 2
LSSOU RCE	m	10^{-6}	0	-	STI-edge part of source junction perimeter	0, 1, 2
LGSOU RCE	m	10^{-6}	0	-	Gate-edge part of source junction perimeter	0, 1, 2
ABDRAIN	m ²	10^{-12}	0	-	Drain junction area	0, 1, 2
LSDRAIN	m	10^{-6}	0	-	STI-edge part of drain junction perimeter	0, 1, 2
LGDRAIN	m	10^{-6}	0	-	Gate-edge part of drain junction perimeter	0, 1, 2
AS	m ²	10^{-12}	0	-	Source junction area (alternative spec.)	0, 1, 2
PS	m	10^{-6}	0	-	Source STI-edge perimeter (alternative spec.)	0, 1, 2
AD	m ²	10^{-12}	0	-	Drain junction area (alternative spec.)	0, 1, 2

40. Indicates for which value of SWGEO the listed parameter is used.

41. Indicates for which value of SWGEO the listed parameter is used.

Name	Unit	Default	Min.	Max.	Description	SWGEO ⁴¹
PD	m	10^{-6}	0	-	Drain STI-edge perimeter (alternative spec.)	0, 1, 2
JW	m	1.10^{-6}	0	-	Junction width	0
DELVTO	V	0	-	-	Threshold voltage shift parameter	0, 1, 2
DELVTOE DGE	V	0	-	-	Threshold voltage shift parameter of edge transistor	0, 1, 2
FACTUO	-	1	0	-	Zero-field mobility pre-factor	0, 1, 2
FACTUOE DGE	-	1	0	-	Zero-field mobility pre-factor of edge transistor	0, 1, 2
SA	m	0	-	-	Distance between OD-edge and poly at source side	1, 2
SB	m	0	-	-	Distance between OD-edge and poly at drain side	1, 2
SD	m	0	-	-	Distance between neighboring fingers	1, 2
SCA	-	0	0	-	Integral of the first distribution function for scattered well dopant	1, 2
SCB	-	0	0	-	Integral of the second distribution function for scattered well dopant	1, 2
SCC	-	0	0	-	Integral of the third distribution function for scattered well dopant	1, 2
SC	m	0	-	-	Distance between OD edge and nearest well edge	1, 2
NRS	-	0	-	-	Number of squares of source diffusion	1, 2
NRD	-	0	-	-	Number of squares of drain diffusion	1, 2
NGCON	-	1	1	2	Number of gate contacts	1, 2
XGW	m	10^{-7}	-	-	Distance from the gate contact to the channel edge	1, 2
NF	-	1	1	-	Number of fingers; internally rounded to the nearest integer	1, 2
MULT	-	1	0	-	Number of devices in parallel	0, 1, 2

41. Indicates for which value of SWGEO the listed parameter is used.

PSP Model — Output Templates

The following output templates (and recommended templates) are available for use with PSP models. See also: [Additional Output Templates for PSP](#).

Name	Alias
Channel current	LX4
Diode current	LX5, LX6
Junction capacitance	LX28, LX29
l _{ii}	LX69
I _{gidl}	LX70
W _{eff} /L _{eff}	LX62, LX63
Capacitance	LX82~LX87 (recommended, since they include intrinsic, overlap, and fringe capacitances), LX18~LX23

LEVEL 70: BSIMSOI Model

BSIMSOI100 Model

BSIM-SOI 100.0.0 (symmetric BSIM-SOI) is an international standard model for SOI (Silicon-On-Insulator) circuit design. This model offers two different mode of operations: *SOIMOD=1* is dedicated to the dynamic depletion devices and has surface potential based core IV and CV formulation, *SOIMOD=0* is developed for only partial depletion devices and is formulated on top of the BSIM-BULK framework.

The following sections discuss these topics:

- [BSIMSOI100 Model — General Syntax](#)
- [BSIMSOI100 Model — Version Updates](#)

BSIMSOI100 Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [nb] [nT] mname [L=val] [W=val]
+ [M=val] [AD=val] [AS=val] [PD=val] [PS=val] [NRD=val]
+ [NRS=val] [NRB=val] [NBC=val] [NSEG=val] [PDBCP=val]
+ [PSBCP=val] [AGBCP=val] [AEBCP=val] [FRBODY=val] [BJTOFF=val]
+ [DTEMP=val] [RGATEMOD=val] [NF=val] [SA=val] [SB=val] [SD=val]
```

```
+ [DELVTO=val] [AGBCPD=val] [MULU0=val] [VFBSDOFF=val] [MINZ=val]
+ [GEOMOD=val] [RGEOMOD=val] [SCA=val] [SCB=val] [SCC=val]
+ [SC=val] [XGW=val] [NGCON=val] [IDS0MULT=val] [EDGEFET=val]
+ [SSIMOD=val] [AGBCP2=val] [TNODEOUT]
.model mname n(p)mos LEVEL=70 VERSION=val ...
For the BSIMSOI100 model, VERSION can be 100.
```

BSIMSOI100 Model — Version Updates

This section covers the following topics:

- [BSIMSOI100 Version 100 Update](#)

BSIMSOI100 Version 100 Update

BSIMSOI100 release has the following changes:

- Real floating body simulation in both I-V and C-V. The body potential is determined by the balance of all the body current components.
- An improved parasitic bipolar current model. This includes enhancements in the various diode leakage components, second order effects (high-level injection and Early effect), diffusion charge equation, and temperature dependence of the diode junction capacitance.
- An improved impact-ionization current model. The contribution from BJT current is also modeled by the parameter FBJTII.
- An improved GIDL/GISL model.
- Asymmetric current/capacitance model Source/Drain diode and asymmetric Source/Drain resistance.
- Gate tunneling currents with a gate current component in the body contact region.
- Linear and non-linear body resistance model for body-contacted SOI devices.
- Instance parameters (PDBCP, PSBCP, AGBCP, AGBCP2, AEBCP, NBC) are provided to model the parasitics of devices with various body-contact and isolation structures.
- Improved history dependence of the body charges with two new parameters (FBODY,DLCB).

BSIMSOI4 Model

The UC Berkeley SOI model (BSIM SOI 4.0) addresses several issues in modeling sub-0.13 micron CMOS/SOI high-speed and RF circuit simulations. Many inputs from the Compact Model Council (CMC) meetings were incorporated into the model. This model is fully backward compatible with its previous 3.X versions. For a description of this model, see the *BSIMSOI4.0 MOSFET Model User's Manual* at http://www-device.eecs.berkeley.edu/bsim/?page=BSIMSOI_LR

The BSIMSOI4 model provides the following major improvements and additions over the BSIMSOIv3.2 model:

- A scalable stress effect model for process induced stress effect; device performance thus becomes a function of the active area geometry and the location of the device in the active area.
- An asymmetric current/capacitance model S/D diode and asymmetric S/D resistance.
- An improved GIDL model with BSIM4 GIDL compatibility.
- Noise model improvements, such as:
 - Improved width/length dependence on flicker noise
 - SPICE2 thermal noise model is introduced as `TNOIMOD=2` with parameter `NTNOI` that adjusts the magnitude of the noise density
 - Body contact resistance induced thermal noise
 - Thermal noise induced by the body resistance network
 - Shot noises induced by `Ibs` and `Ibd` separated
- A two resistance body resistance network introduced for RF simulation.
- Threshold voltage model enhancement, such as:
 - Long-channel DIBL effect model added
 - Channel-length dependence of body effect improved
- Drain-induced threshold shift (DITS) model introduced in output conductance.
- Improved model accuracy in moderate inversion region with BSIM4 compatible `Vgsteff`.
- Multi-finger device with instance parameter `NF`.
- A new instance parameter `AGBCPD` to improve gate current for body contact.

- A new instance parameter `DELVTO` representing threshold voltage variation.
- Instance `FRBODY` is both an instance and a model parameter.

The following sections discuss these topics:

- [BSIMSOI4 Model — General Syntax](#)
- [BSIMSOI4 Model Parameters](#)
- [BSIMSOI4 Model — Parameter Range Limit](#)
- [BSIMSOI4 Model — Version Updates](#)

BSIMSOI4 Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] [nb] [nT] mname [L=val] [W=val]
+ [M=val] [AD=val] [AS=val] [PD=val] [PS=val] [NRD=val]
+ [NRS=val] [NRB=val] [RTH0=val] [CTH0=val] [NBC=val]
+ [NSEG=val] [PDBCP=val] [PSBCP=val] [AGBCP=val] [AEBCP=val]
+ [VBSUSR=val] [DELTOX=val] [TNODEOUT] [Off] [FRBODY=val]
+ [BJTOFF=val] [IC=vds, vgs, vbs, ves, vps] [dtemp=val]
+ [soimod=val] [rgatemod=val] [nf=val] [sa=val] [sb=val]
+ [sd=val] [rdbb=val] [rbsb=val] [delvto=val] [agbcpd=val]
+ [rbodymod=val] [mulu0=val] [SOIQ0=val]
+ [DELSVAT=val] [MULSVAT=val]
```

BSIMSOI4 Model Parameters

This section covers the following topics:

- [BSIMSOI4 Model — General Parameters](#)
- [BSIMSOI4 Model — Control Parameters](#)
- [BSIMSOI4 Model — Process Parameters](#)
- [BSIMSOI4 Model — DC Parameters](#)
- [BSIMSOI4 Model — Gate-to-body Tunneling Parameters](#)
- [BSIMSOI4 Model — AC and Capacitance Parameters](#)
- [BSIMSOI4 Model — Temperature Parameters](#)
- [BSIMSOI4 Model — Built-in Potential Lowering Model Parameters](#)
- [BSIMSOI4 Model — RF Model Parameters](#)

- [BSIMSOI4 Model — Noise Model Parameters](#)
- [BSIMSOI4 Model — Stress Model Parameters](#)

BSIMSOI4 Model — General Parameters

Parameter	Description
<i>Mxxx</i>	SOI MOSFET element name. Must begin with M, followed by up to 1023 alphanumeric characters.
<i>nd</i>	Drain terminal node name or number.
<i>ng</i>	Front gate node name or number.
<i>ns</i>	Source terminal node name or number.
<i>ne</i>	Back gate (or Substrate) node name or number.
<i>np</i>	External body contact node name or number. If 'body node' is to be checked in a <code>.biaschk</code> command, use keyword 'np' as in the BSIM3 SOI model.
<i>nb</i>	Internal body node name or number.
<i>nT</i>	Temperature node name or number.
<i>mname</i>	SOI MOSFET model name reference.
<i>L</i>	SOI MOSFET channel length in meters. The default is 5e-6.
<i>W</i>	SOI MOSFET channel width in meters. The default is 5e-6.
<i>M</i>	Multiplier to simulate multiple SOI MOSFETs in parallel. The M setting affects all channel widths, diode leakages, capacitances, and resistances. The default is 1.
<i>AD</i>	Drain diffusion area. The default is 0.
<i>AS</i>	Source diffusion area. The default is 0.
<i>PD</i>	Perimeter of the drain junction, including the channel edge. The default is 0.
<i>PS</i>	Perimeter of the source junction, including the channel edge. The default is 0.
<i>NRD</i>	Number of squares of drain diffusion for the drain series resistance. The default is 1.
<i>NRS</i>	Number of squares of source diffusion for the source series resistance. The default is 1.
<i>NRB</i>	Number of squares for the body series resistance. The default is 1.

Parameter	Description
FRBODY	Coefficient of the distributed body resistance effects. • If FRBODY is not specified, it's extracted from the model card • If FRBODY is specified, it overrides FRBODY in the model card
RTH0	Thermal resistance per unit width. • If RTH0 is not specified, it's extracted from the model card • If RTH0 is specified, it overrides RTH0 in the model card
CTH0	Thermal capacitance per unit width. • If CTH0 is not specified, it's extracted from the model card • If CTH0 is specified, it overrides RTH0 in the model card
NBC	Number of body contact isolation edge. The default is 0
NSEG	Number of segments for partitioning the channel width. The default is 1
PDBCP	Parasitic perimeter length for the body contact at the drain side. The default is 0
PSBCP	Parasitic perimeter length for the body contact at the source side. The default is 0
AGBCP	Parasitic gate-to-body overlap area for the body contact. The default is 0
AEBCP	Parasitic body to substrate overlap area for the body contact. The default is 0
VBSUSR	Optional initial value of Vbs that you specify for transient analysis.
DELTOX	Shift in gate oxide thickness (TOX), that is, the difference between the electrical and physical gate oxide/insulator thickness. The default is 0
TNODEOUT	Temperature node flag indicating the use of the T node. If you do not set TNODEOUT, you can specify four nodes for a device to float the body. Specifying five nodes implies that the fifth node is the external body contact node with a body resistance between the internal and external terminals. This configuration applies to a distributed body resistance simulation. If you set TNODEOUT, simulation interprets the last node as the temperature node. You can specify five nodes to float the device. Specifying six nodes implies body contact. Seven nodes is a body-contacted case with a accessible internal body node. You can use the temperature node to simulate thermal coupling.
OFF	Sets the initial condition to OFF for this element in DC analysis.
BJTOFF	Turns off BJT if equal to 1.
IC	Initial guess in the order (drain, front gate, internal body, back gate, external voltage). Simulation ignores Vps in a 4-terminal device. Use these settings if you specify UIC in the .TRAN statement. The .IC statement overrides it.

Parameter	Description
DTEMP	Increase in temperature. The default is 0.
SOIMOD	SOI model selector. - If SOIMOD is not specified, it's extracted from the model card - If SOIMOD is specified, it overrides SOIMOD in the model card
RGATEMOD	Gate resistance model selector. - If RGATEMOD is not specified, it's extracted from the model card - If RGATEMOD is specified, it overrides RGATEMOD in the model card
NF	Number of fingers. The default is 1.
SA	Distance between OD edge to poly from one side. The default is 0.
SB	Distance between OD edge to poly from another side. The default is 0.
SD	Distance between neighboring fingers. The default is 0.
RBDB	Resistance between dbNode and bNode. - If RBDB is not specified, it's extracted from the model card - If RBDB is specified, it overrides RBDB in the model card
RBSB	Resistance between sbNode and bNode. - If RBSB is not specified, it's extracted from the model card - If RBSB is specified, it overrides RBSB in the model card
DELVTO	Zero bias threshold voltage variation. The default is 0.
AGBCPD	Parasitic gate to body overlap area for body contact in DC. The default is 0.
RBODYMOD	Body resistance model selector. - If RBODYMOD is not specified, it's extracted from the model card - If RBODYMOD is specified, it overrides RBODYMOD in the model card
MULU0	Low-field mobility (U_0) multiplier($U_{0_eff} = MULU0 * U_0$), the default is 1.0
SOIQ0	Floating body charge initialization. This parameter is set for the BQI algorithm of a floating body node.
DELSVAT	Shift in Saturation velocity ($VSAT$).
MULSVAT	Scaling factor of Saturation velocity ($VSAT$), the default is 1.0.

BSIMSOI4 Model — Control Parameters

Parameter	Description	Unit	Default
VERSION	Mode version number	-	4.40

Parameter	Description	Unit	Default
BSOIUPDATE	Flag for fixing thermal noise NF issue in versions 4.0-4.3 0: Does not fix noise NF issue 1: Fixes noise NF issue	-	0
SOIMOD	SOI model selector (instance) 0: BSIMPD 1: Unified model for PD&FD 2: Ideal FD 3: Auto selection by BSIMSOI	-	0
SHMOD	Flag for self-heating 0 - no self-heating 1 - self-heating	-	0
MOBMOD	Mobility model selector	-	0
CAPMOD	Flag for the short channel capacitance model	-	1
NOIMOD	Flag for Noise model	-	2
IGCMOD	Gate-to-channel tunneling current model selector	-	1
IGBMOD	Gate-to-body tunneling current model selector	-	0
RDSMOD	Bias-dependent source/drain resistance model selector	-	0
RGATEMOD	Flag for gate resistance model	-	0
RBODYMOD	Flag for body resistance model	-	0
FNOIMOD	Flicker noise model selector	-	1
TNOIMOD	Thermal noise model selector	-	0
MTRLMOD	New material model selector	-	0
VGSTCVMOD	VgsteffCV model selector	-	0
IIIMOD	Impact ionization current model selector	-	0
GIDLMOD	New GIDL/GISL model selector	-	0
FDMOD	New δV_{bi} model selector	-	0

BSIMSOI4 Model — Process Parameters

Parameter	Description	Unit	Default
TSI	Silicon film thickness	m	10^{-7}
TBOX	Buried oxide thickness	m	3×10^{-7}
TOX	Gate oxide thickness	m	1×10^{-8}
TOXM	Gate oxide thickness used in extraction	m	TOX
XJ	S/D junction depth	m	TSI
NCH	Channel doping concentration	$1/\text{cm}^3$	1.7×10^{17}
NSUB	Substrate doping concentration	$1/\text{cm}^3$	6×10^{16}
NGATE	Poly gate doping concentration	$1/\text{cm}^3$	0
EOT	Effective SiO ₂ thickness	nm	10
LEFFEOT	Effective length for extraction of EOT	um	1
WEFFEOT	Effective width for extraction of EOT	um	10
TEMPEOT	Temperature for extraction of EOT	K	300.15

BSIMSOI4 Model — DC Parameters

Parameter	Description	Unit	Default
VTH0	Threshold voltage @ $V_{bs}=0$ for long and wide device	-	0.7
K1	First order body effect coefficient	$V^{1/2}$	0.6
K1W1	First body effect width dependent parameter	m	0
K1W2	Second body effect width dependent parameter	m	0
K2	Second order body effect coefficient	-	0
K3	Narrow width coefficient	-	0
K3B	Body effect coefficient of k3	1/V	0
KB1	Backgate body charge coefficient	-	1
W0	Narrow width parameter	m	0
LPE0/NLX	Lateral non-uniform doping parameter	m	1.74×10^{-7}

Parameter	Description	Unit	Default
LPEB	Lateral non-uniform doping effect on K1	m	0.0
DVT0	First coefficient of short-channel effect on Vth	-	2.2
DVT1	Second coefficient of short-channel effect on Vth	-	0.53
DVT2	Body-bias coefficient of short-channel effect on Vth	1/V	-0.032
DVT0W	First coefficient of narrow width effect on Vth for small channel length	-	0
DVT1W	Second coefficient of narrow width effect on Vth for small channel length	-	5.3e6
DVT2W	Body-bias coefficient of narrow width effect on Vth for small channel length	1/V	-0.032
U0	Mobility at Temp = T _{nom} • NMOSFET • PMOSFET	cm ² /(V-s ec)	670 250
UA	First-order mobility degradation coefficient	m/V	2.25e-9
UB	Second-order mobility degradation coefficient	(m/V) ²	5.9e-19
UC	Body-effect of mobility degradation coefficient	1/V	-.0465
UD	Coulomb scattering factor of mobility	-	0
UCS	Mobility exponent factor in mobMod=4	-	0
EU	Mobility exponent factor in mobMod=4	-	0
VSAT	Saturation velocity at Temp=T _{nom}	m/sec	8e4
A0	Bulk charge effect coefficient for channel length	-	1.0
AGS	Gate bias coefficient of A _{bulk}	1/V	0.0
B0	Bulk charge effect coefficient for channel width	m	0.0
B1	Bulk charge effect width offset	m	0.0
KETA	Body-bias coefficient of bulk charge effect	V ⁻¹	0
KETAS	Surface potential adjustment for bulk charge effect	V	0
A1	First non-saturation effect parameter	1/V	0.0
A2	Second non-saturation effect parameter	0	1.0

Parameter	Description	Unit	Default
RDSW	Parasitic resistance per unit width	$\Omega - \mu\text{m}^{W_r}$	100
PRWB	Body effect coefficient of R_{dsw}	$1/V$	0
PRWG	Gate bias effect coefficient of R_{dsw}	$1/V^{1/2}$	0
WR	Width offset from W_{eff} for R_{ds} calculation	-	1
NFACTOR	Subthreshold swing factor	-	1
WINT	Width offset fitting parameter from I-V without bias	m	0.0
LINT	Length offset fitting parameter from I-V without bias	m	0.0
DWG	Coefficient of W_{eff} 's gate dependence	m/V	0.0
DWB	Coefficient of W_{eff} 's substrate body bias dependence	m/V ^{1/2}	0.0
DWBC	Width offset for body contact isolation edge	m	0.0
VOFF	Offset voltage in the subthreshold region for large W and L	V	-0.08
ETA0	DIBL coefficient in subthreshold region	-	0.08
ETAB	Body-bias coefficient for the subthreshold DIBL effect	$1/V$	-0.07
DSUB	DIBL coefficient exponent	-	0.56
CIT	Interface trap capacitance	F/m ²	0.0
CDSC	Drain/Source to channel coupling capacitance	F/m ²	2.4e-4
CDSCB	Body-bias sensitivity of C_{dsc}	F/m ²	0
CDSCD	Drain-bias sensitivity of C_{dsc}	F/m ²	0
PCLM	Channel length modulation parameter	-	1.3
PDIBL1	First output resistance DIBL effect correction parameter	-	0.39
PDIBL2	Second output resistance DIBL effect correction parameter	-	0.086
DROUT	L dependence coefficient of the DIBL correction parameter in R_{out}	-	0.56

Parameter	Description	Unit	Default
PVAG	Gate dependence of Early voltage	-	0.0
DELTA	Effective V_{ds} parameter	-	0.01
ALPHA0	The first parameter of impact ionization current	m/V	0.0
FBJTII	Fraction of bipolar current affecting the impact ionization	-	0.0
BETA0	First V_{ds} dependent parameter of impact ionization current	V^{-1}	0
BETA1	Second V_{ds} dependent parameter of impact ionization current	-	0
BETA2	Third V_{ds} dependent parameter of impact ionization current	V	0.1
VDSATII0	Nominal drain saturation voltage at threshold for impact ionization current	V	0.9
TII	Temperature dependent parameter for impact ionization current	-	0
LII	Channel length dependent parameter at threshold for impact ionization current	-	0
ESATII	Saturation channel electric field for impact ionization current	V/m	1e7
SIIO	First V_{gs} dependent parameter for impact ionization current	V^{-1}	0.5
SII1	Second V_{gs} dependent parameter for impact ionization current	V^{-1}	0.1
SII2	Third V_{gs} dependent parameter for impact ionization current	-	0
SIID	V_{ds} dependent parameter of drain saturation voltage for impact ionization current	V^{-1}	0
ABJTII	Exponent factor for avalanche current	V^{-1}	0
CBJTII	Length scaling parameter for II BJT part	m	0
EBJTII	Impact ionization parameter for BJT part	-	0
MBJTII	Internal B-C grading coefficient	-	0.4
VBCI	Internal B-C built-in potential	V	0

Parameter	Description	Unit	Default
AGIDL	Pre-exponential GIDL constant	Ω^{-1}	0.0
BGIDL	GIDL exponential coefficient	V/m	2.3e9
CGIDL	Parameter for body bias effect on GIDL	V^3	0.5
EGIDL/NGIDL	Fitting parameter for band bending for GIDL	V	1.2
AGISL	Pre-exponential GISL constant	Ω^{-1}	0.0
BGISL	GISL exponential coefficient	V/m	2.3e9
CGISL	Parameter for body bias effect on GISL	V^3	0.5
EGISL	Fitting parameter for band bending for GISL	V	1.2
RGIDL	Vgs-dependent parameter for GIDL when gidlMod = 1	-	1.0
KGIDL	Vds-dependent parameter for GIDL when gidlMod = 1	V	0.0
FGIDL	Vds-dependent parameter for GIDL when gidlMod = 1	-	-
RGISL	Vgs-dependent parameter for GISL when gidlMod = 1	-	1.0
KGISL	Vbs-dependent parameter for GISL when gidlMod = 1	V	0.0
FGISL	Vbs-dependent parameter for GISL when gildMod = 1	V	0.0
NDIODE	Diode non-ideality factor for source	-	1.0
NDIODED	Diode non-ideality factor for drain	-	Default to its source value
NRECF0	Recombination non-ideality factor at forward bias for source	-	2.0
NRECF0D	Recombination non-ideality factor at forward bias for drain	-	Default to its source value
NRECR0	Recombination non-ideality factor at reversed bias for source	-	10

Parameter	Description	Unit	Default
NRECR0D	Recombination non-ideality factor at reversed bias for drain	-	Default to its source value
ISBJT	BJT injection saturation current	A/m ²	1e-6
IDBJT	BJT injection saturation current	A/m ²	1e-6
ISDIF	Body to source/drain injection saturation current	A/m ²	1e-7
IDDIF	Body to source/drain injection saturation current	A/m ²	1e-7
ISREC	Recombination in depletion saturation current	A/m ²	1e-5
IDREC	Recombination in depletion saturation current	A/m ²	1e-5
ISTUN	Reverse tunneling saturation current	A/m ²	0.0
IDTUN	Reverse tunneling saturation current	A/m ²	0.0
LN	Electron/hole diffusion length	m	2e-6
VREC0	Voltage dependent parameter for recombination current for source	V	0
VREC0D	Voltage dependent parameter for recombination current for drain	V	Default to its source value
VTUN0	Voltage dependent parameter for tunneling current for source	V	0
VTUN0D	Voltage dependent parameter for tunneling current for drain	V	Default to its source value
NBJT	Power coefficient of channel length dependency for bipolar current	-	1
LBJT0	Reference channel length for bipolar current	m	0.20e-6
VABJT	Early voltage for bipolar current	V	10
AELY	Channel length dependency of early voltage for bipolar current	V/m	0
AHLI	High level injection parameter for bipolar current for source	-	0

Parameter	Description	Unit	Default
AHLID	High level injection parameter for bipolar current for drain	-	Default to its source value
RBODY	Intrinsic body contact sheet resistance	W / square	0.0
RBSH	Extrinsic body contact sheet resistance	W / square	0.0
RSH	Source drain sheet resistance in ohm per square	W / square	0.0
RHALO	Body halo sheet resistance	W / m	1e15
RSW	Zero bias lightly-doped source resistance per unit width for RDSMOD=1	$\Omega (\mu\text{m})^{W_r}$	50
RDW	Zero bias lightly-doped drain resistance per unit width for RDSMOD=1	$\Omega (\mu\text{m})^{W_r}$	50
RSWMIN	Lightly-doped source resistance per unit width at high V_{gs} and zero V_{bs} for RDSMOD=1	$\Omega (\mu\text{m})^{W_r}$	0
RDWMIN	Lightly-doped source resistance per unit width at high V_{gs} and zero V_{bs} for RDSMOD=1	$\Omega (\mu\text{m})^{W_r}$	0
DVTP0	First parameter for V_{th} shift due to pocket	m	0.0
DVTP1	Second parameter for V_{th} shift due to pocket	V^{-1}	0.0
DVTP2	Third parameter for V_{th} shift due to pocket	V^{-1}	0.0
DVTP3	Forth parameter for V_{th} shift due to pocket	V^{-1}	0.0
DVTP4	Fifth parameter for V_{th} shift due to pocket	V^{-1}	0.0
PDITS	Coefficient for drain-induced V_{th} shifts	V^{-1}	1e-20
PDITSL	Length dependence of drain-induced V_{th} shifts	m^{-1}	0
PDITSD	V_{ds} dependence of drain-induced V_{th} shifts	V^{-1}	0
FPROUT	Effect of pocket implant on rout degradation	$V/m^{0.5}$	0.0
MINV	V_{gsteff} fitting parameter for moderate inversion	-	0.0

BSIMSOI4 Model — Gate-to-body Tunneling Parameters

Parameter	Description	Unit	Default
IGMOD	Gate current model selector	-	0

Parameter	Description	Unit	Default
TOXQM	Oxide thickness for I_{gb} calculation	m	Tox
NTOX	Power term of gate current	-	1
TOXREF	Target oxide thickness	m	2.5e-9
EBG	Effective bandgap in gate current calculation	V	1.2
ALPHAGB1	First V_{ox} dependent parameter for gate current in inversion	1/V	0.35
BETAGB1	Second V_{ox} dependent parameter for gate current in inversion	$1/V^2$	0.03
VGB1	Third V_{ox} dependent parameter for gate current in inversion	V	300
VEVB	V_{aux} parameter for valence band electron tunneling	-	0.075
ALPHAGB2	First V_{ox} dependent parameter for gate current in accumulation	1/V	0.43
BETAGB2	Second V_{ox} dependent parameter for gate current in accumulation	$1/V^2$	0.05
VGB2	Third V_{ox} dependent parameter for gate current in accumulation	V	17
VECB	V_{aux} parameter for conduction band electron tunneling	-	0.026
AIGBCP2	First V_{gp} dependent parameter for gate current in accumulation in AGBCP2 region	$1/V^2$	0.043
BIGBCP2	Second V_{gp} dependent parameter for gate current in accumulation in AGBCP2 region	$1/V^2$	0.0054
CIGBCP2	Third V_{gp} dependent parameter for gate current in accumulation in AGBCP2 region	$1/V^2$	0.0075

BSIMSOI4 Model — AC and Capacitance Parameters

Parameter	Description	Unit	Default
XPART	Charge partitioning rate flag	-	-
CGSO	Non LDD region source-gate overlap capacitance per channel length	F/m	Calculated

Parameter	Description	Unit	Default
CGDO	Non LDD region drain-gate overlap capacitance per channel length	F/m	Calculated
CGEO	Gate substrate overlap capacitance per unit channel length	F/m	0.0
CJSWG	Source (gate side) sidewall junction capacitance per unit width (normalized to 100nm T_{si})	F/m ²	1e-10
CJSWGD	Drain (gate side) sidewall junction capacitance per unit width (normalized to 100nm T_{si})	F/m ²	Default to its source value
PBSWG	Source (gate side) sidewall junction capacitance built-in potential	V	0.7
PBSWGD	Drain (gate side) sidewall junction capacitance built-in potential	V	Default to its source value
MJSWG	Source (gate side) sidewall junction capacitance grading coefficient	V	0.5
MJSWGD	Drain (gate side) sidewall junction capacitance grading coefficient	V	Default to its source value
TT	Diffusion capacitance transit time coefficient	Second	1e-12
NDIF	Power coefficient of channel length dependency for diffusion capacitance	-	-1
LDIF0	Channel-length dependency coefficient of diffusion cap.	-	1
VSDFB	Source/drain bottom diffusion capacitance flatband voltage	V	Calculated
VSDTH	Source/drain bottom diffusion capacitance threshold voltage	V	Calculated
CSDMIN	Source/drain bottom diffusion minimum capacitance	V	Calculated
ASD	Source/drain bottom diffusion smoothing parameter	-	0.3
CSDESW	Source/drain sidewall fringing capacitance per unit length	F/m	0.0

Parameter	Description	Unit	Default
CGSL	Light doped source-gate region overlap capacitance	F/m	0.0
CGDL	Light doped drain-gate region overlap capacitance	F/m	0.0
CKAPPA	Coefficient for lightly doped region overlap capacitance fringing field capacitance	F/m	0.6
CF	Gate to source/drain fringing field capacitance	F/m	Calculated
CLC	Constant term for the short channel model	m	0.1x10 ⁻⁷
CLE	Exponential term for the short channel model	None	0.0
DLC	Length offset fitting parameter for gate charge	m	lint
DLCB	Length offset fitting parameter for body charge	m	0
DLBG	Length offset fitting parameter for backgate charge	m	0.0
DWC	Width offset fitting parameter from C-V	m	wint
DELVT	Threshold voltage adjust for C-V	V	0.0
FBODY	Scaling factor for body charge	-	1.0
ACDE	Exponential coefficient for charge thickness in capMod=3 for accumulation and depletion regions	m/V	1.0
MOIN	Coefficient for the gate-bias dependent surface potential.	V ^{1/2}	15.0
CFRCOEFF	Sidewall fringe capacitance coefficient	-	1

BSIMSOI4 Model — Temperature Parameters

Parameter	Description	Unit	Default
TNOM	Temperature at which parameters are expected	⁰ C	27
UTE	Mobility temperature exponent	None	-1.5
KT1	Temperature coefficient for threshold voltage	V	-0.11
KT11	Channel length dependence of the temperature coefficient for threshold voltage	V*m	0.0
KT2	Body-bias coefficient of the Vth temperature effect	None	0.022

Parameter	Description	Unit	Default
UA1	Temperature coefficient for U_a	m/V	4.31e-9
UB1	Temperature coefficient for U_b	(m/V) ²	-7.61e-18
UC1	Temperature coefficient for U_c	1/V	-0.056
AT	Temperature coefficient for saturation velocity	m/sec	3.3e4
TCJSWG	Temperature coefficient of C_{jswgs}	1/K	0
TPBSWG	Temperature coefficient of P_{bswgs}	V/K	0
TCJSWGD	Temperature coefficient of C_{jswgd}	1/K	Default to its source value
TPBSWGD	Temperature coefficient of P_{bswgd}	V/K	Default to its source value
CTH0	Normalized thermal capacity	(W*sec)/m°C	1e-5
PRT	Temperature coefficient for R_{dsw}	$\Omega - \mu m$	0
RTH0	Normalized thermal resistance	m°C/W	0
NTRECF	Temperature coefficient for N_{recf}	-	0
NTRECR	Temperature coefficient for N_{recl}	-	0
XBJT	Power dependence of j_{bjt} on temperature	-	1
XDIFS	Power dependence of j_{difs} on temperature	-	Xbjt
XRECS	Power dependence of j_{recl} on temperature	-	1
XTUNS	Power dependence of j_{tuns} on temperature	-	0
XDIFD	Power dependence of j_{difd} on temperature	-	Xbjt
XRECD	Power dependence of j_{recl} on temperature	-	1
XTUND	Power dependence of j_{tund} on temperature	-	0
WTH0	Minimum width for thermal resistance calculation	m	0
TVBCI	Temperature coefficient for V_{bci}	-	0

BSIMSOI4 Model — Built-in Potential Lowering Model Parameters

Parameter	Description	Unit	Default
SOIMOD	SOI model selector. • 0: BSIMPD • 1: unified model for PD&FD • 2: ideal FD • 3: auto selection by BSIMSO	-	0
VBSA	Offset voltage due to non-idealities	V	0
NOFFFD	Smoothing parameter in FD module	-	1
VOFFFD	Smoothing parameter in FD module	V	0
K1B	First backgate body effect parameter	-	1
K2B	Second backgate body effect parameter for short channel effect	-	0
DK2B	Third backgate body effect parameter for short channel effect	-	0
DVBD0	First short channel effect parameter in FD module	-	0
DVBD1	Second short channel effect parameter in FD module	-	0
MOINFD	Gate bias dependence coefficient of surface potential in FD module	-	1e3
VBS0PD	Upper bound of built-in potential lowering for BSIMPD operation	V	0.0
VBS0FD	Lowering bound of built-in potential lowering for ideal FD operation	V	0.5

BSIMSOI4 Model — RF Model Parameters

Parameter	Description	Unit	Default
RGATEMOD	Gate resistance model selector • rgateMod ◦ 0: No gate resistance ◦ 1: Constant gate resistance ◦ 2: Rii model with variable resistance ◦ 3: Rii model with two nodes	-	0

Parameter	Description	Unit	Default
RBODYMOD	RbodyMod • 0: Noi • 1:Two-resistor body resistance model	-	0
RSHG	Gate electrode sheet resistance	Ω /square	0.1
XRCRG1	Parameter for distributed channel-resistance effect for intrinsic input resistance	-	12.0
XRCRG2	Parameter to account for the excess channel diffusion resistance for intrinsic input resistance	-	1.0
NGCON	Number of gate contacts	-	1
XGW	Distance from the gate contact to the channel edge	m	0.0
XGL	Offset of the gate length due to variations in patterning	m	0.0
RBSB	Resistance between sbNode and bNode	Ω	50.0
RBDB	Resistance between dbNode and bNode	Ω	50.0
GBMIN	Conductance parallel with RBSB/RBDB	Ω	1e-12

BSIMSOI4 Model — Noise Model Parameters

Parameter	Description	Unit	Default
FNOIMOD	Flicker noise model selector	-	1
TNOIMOD	Thermal noise model selector	-	0
NTNOI	Noise factor for short-channel devices for TNOIMOD=0 or 2	-	1.0
TNOIA	Coefficient of channel length dependence of total channel thermal noise	-	1.5
TNOIB	Channel length dependence parameter for channel thermal noise partitioning	-	3.5
RNOIA	Thermal noise parameter	-	0.577
RNOIB	Thermal noise parameter	-	0.37
W0FLK	Flicker noise width dependence parameter	-	-1

Parameter	Description	Unit	Default
BF	Flicker noise length dependence exponent	-	2.0

BSIMSOI4 Model — Stress Model Parameters

Parameter	Description	Unit	Default
SA	Distance between OD edge to poly from one side	m	0.0
SB	Distance between OD edge to poly from another side	m	0.0
SD	Distance between neighboring fingers	m	0.0
SAREF	Reference distance between OD and edge to poly of one side	m	1e-6
SBREF	Reference distance between OD and edge to poly of another side	m	1e-6
WLOD	Width parameter for stress effect	M	0.0
KU0	Mobility degradation/enhancement	M	0.0
KVSAT	Saturation velocity degradation/enhancement parameter for stress effect	M	0.0
TKU0	Temperature coefficient of KU0	-	0.0
LKU0	Length dependence of KU0	-	0.0
WKU0	width dependence of KU0	-	0.0
PKU0	Cross-term dependence of KU0	-	0.0
LLODKU0	Length parameter for u0 stress effect	-	0.0
WLODKU0	Width parameter for u0 stress effect	-	0.0
KVTH0	Threshold shift parameter for stress effect	Vm	0.0
LKVTH0	Length dependence of KVHT0	-	0.0
WKVTH0	Width dependence of KVHT0	-	0.0
PKVTH0	Cross term dependence of KVHT0	-	0.0
LLODVTH	Length parameter for Vth stress effect	-	0.0
WLODVTH	Width parameter for Vth stress effect	-	0.0
STK2	K2 shift factor related to Vth0 change	M	0.0

Parameter	Description	Unit	Default
LODK2	K2 shift modification factor for stress effect	-	1.0
STETA0	Eta0 shift factor related to vth0 change	M	0.0
LODETA0	Eta0 shift modification factor for stress effect	-	1.0

BSIMSOI4 Model — Parameter Range Limit

Simulation reports either a warning or a fatal error if BSIMSOI4 parameters fall outside predefined ranges. These range limitations prevent (or at least warn of) potential numerical problems.

To control the maximum number of simulation warning messages printing to the output file, use:

```
.OPTION WARNLIMIT=#
```

In the preceding `.OPTION` statement, `#` is the maximum number of warning messages that simulation reports. The default `WARNLIMIT` value is 1. In some cases, simulation checks parameters only if you set the `PARMAMCHK=1` model parameter.

Parameter	Limit
ACDE	< 0.1, > 1.6 warn
B1+WEFF	b1+weff = 0 fatal
CLC	< 0 fatal
DELTA	< 0 fatal
DROUT	< 0 fatal
DSUB	<0 fatal
DVT1	< 0 fatal
DVT1W	< 0 fatal
FPROUT	< 0 fatal
KVSAT	< -1.0, >1.0 warn
LODETA0	<= 0 warn
LODK2	<= 0 warn
LPE0	< -leff fatal

Parameter	Limit
LPEB	< -leff fatal
MOIN	< 5, > 25 warn
MOINFD	< 5 warn
NF	< 1 fatal
NGATE	< 0, > 1e25 fatal
NOFF	< 0.1, > 4.0 warn
NPEAK	<= 0 fatal
PCLM	<=0 fatal
PD	< w warn
PDITS	< 0 fatal
PDITSL	< 0 fatal
PS	< w warn
SAREF	<= 0 fatal
SBREF	<= 0 fatal
TBOX	<= 0 fatal
TOX	tox+delttox <= 0 fatal
TOXM	toxm+delttox <= 0 fatal
U0	<= 0 fatal
VSAT	<= 0 fatal
W0	w0+weff =0 fatal
WLOD	< 0 warn
If paramchk=1 following parameter limit range is added	
A2	< 0.01, > 1 warn
AELY	< 0 warn
AHLI	< 0 warn
ALPHAGB1	< 0 warn

Parameter	Limit
ALPHAGB2	< 0 warn
ASD	< 0 warn
B1	$b1 + w_{eff} < 1e-7$ warn
BELTAVOX	≤ 0 warn
BETA0	< 0 warn
BETA1	< 0 warn
BETA2	< 0 warn
BETAGB1	< 0 warn
BETAGB2	< 0 warn
CDSC	< 0 warn
CDSCD	< 0 warn
CGDO	< 0 warn
CGEO	< 0 warn
CGSO	< 0 warn
CSDESW	< 0 warn
CSDMIN	< 0 warn
CTH0	< 0 warn
DVT0	< 0 warn
DWBC	< 0 warn
EBG	< 0 warn
ETA0	< 0 warn
FBJTII	< 0 warn
IDBJT	< 0 warn
IDDIFF	< 0 warn
IDREC	< 0 warn
IDTUN	< 0 warn

Parameter	Limit
ISBJT	< 0 warn
ISDIFF	< 0 warn
ISREC	< 0 warn
ISTUN	< 0 warn
K1W1	< 0 warn
K1W2	< 0 warn
KETAS	< 0 warn
LEFF	<= 5e-8 warn
LEFFCV	<= 5e-8 warn
LII	< 0 warn
LPE0	< 0 warn
NBJT	< 0 warn
NDIODE	< 0 warn
NDIODED	< 0 warn
NFACTOR	< 0 warn
NGATE	< 1e18 warn
NIGC	<= 0 warn
NPEAK	<= 1e15, >= 1e21 warn
NSUB	<= -1e21, >= 1e21 warn
NTOX	< 0 warn
NTUN	< 0 warn
NTUND	< 0 warn
PDIBL1	< 0 warn
PDIBL2	< 0 warn
PIGCD	<= 0 warn
POXEDGE	<= 0 warn

Parameter	Limit
RBODY	< 0 warn
RBSH	< 0 warn
RDS0	< 0.001 warn
RDSW	< warn
RHALO	< 0 warn
RTH0	< 0 warn
SII1	< 0 warn
SII2	< 0 warn
SIID	< 0 warn
TII	< 0 warn
TOXQM	< 0 warn
TOXREF	< 0 warn
TT	< 0 warn
VECB	< 0 warn
VEVB	< 0 warn
VGB1	< 0 warn
VGB2	< warn
VOXH	< 0 warn
VREC0	< 0 warn
VREC0D	< 0 warn
VSATTEMP	< 1e3 warn
VTUN0	< 0 warn
VTUN0D	< 0 warn
W0	$w0 + w_{eff} < 1e-7$ warn
WEFF	$\leq 1e-7$ warn
WEFFCV	$\leq 1e-7$ warn

Parameter	Limit
WTH0	< 0 warn
ACDE	< 0.1, > 1.6 warn
AGIDL	< 0 warn
AHLID	< 0 warn
BGIDL	< 0 warn
CAPMOD	< 2 warn
CGIDL	< 0 warn
DLBG	< 0 warn
EGIDL	< 0 warn
ESATII	<= 0 warn
MOIN	< 5, > 25 warn
NTRECF	< 0 warn
NTRECR	< 0 warn
RBODY	< 0 warn
RBSH	< 0 warn
TCJSWG	< 0 warn
TCJSWGD	< 0 warn
TPBSWG	< 0 warn
TPBSWGD	< 0 warn
XJ	> tsi warn

BSIMSOI4 Model — Version Updates

This section covers the following topics:

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- [BSIMSOI4 Version 4.5 Update](#)
- [BSIMSOI4 Version 4.4 Update](#)

- [BSIMSOI4 Version 4.3.1 Update](#)
- [BSIMSOI4 Version 4.3 and 4.2 Updates](#)
- [BSIMSOI4 Version 4.1 Update](#)

BSIMSOI4 Version 4.6 Update

The following changes were made for the BSIMSOI4 version 4.6 release:

- Added RDSMOD=2 to avoid additional internal nodes.
- Improved the implementation of source/drain resistance when RSH=0, to speed up the simulation.
- Introduced parameters for better fitting accuracy when using MTRLMOD=0 (body-contacted device).
- Binning of built-in potential lowering (ΔV_{bi}) model parameters.
- Fixed certain bugs.

BSIMSOI4 Version 4.5 Update

The following changes were made for the BSIMSOI4 version 4.5 release:

- Added the correlated thermal noise model (not supported in the PrimeSim Continuum solution).
- Decoupled DC and AC DIBL parameters. Changed some hard-coded material parameters to model parameters.
- Corrected an issue so that when values of NRS and NRD are zero, the source/drain conductance is set to 1.0e3 instead of zero.
- Corrected inconsistency in drain current when self-heating is on.

BSIMSOI4 Version 4.4 Update

The following changes were made for the BSIMSOI4 version 4.4 release:

- Implemented a check on the physical thickness and doping level of the channel
- Enhanced sidewall fringe capacitance formulation.
- Bug fixes for derivative expressions in SOIMOD=1 and 2.

BSIMSOI4 Version 4.3.1 Update

In BSIMSOI4 version 4.3.1 the following updates are added:

- The temperature derivative expressions are greatly improved.
- Bug fixes related to `mtrlMod=1` are incorporated.
- Thermal noise NF issue is fixed in SOI4.3.1.

For a description of this model, see the *BSIMSOI4.3.1 MOSFET Model User's Manual* at <http://www-device.eecs.berkeley.edu/~bsimsoi/get.html>

BSIMSOI4 Version 4.3 and 4.2 Updates

In BSIMSOI4 versions 4.3 and 4.2, the following features were added:

- Some bugs such as charge derivative issues are fixed in version 4.2.
- The charge derivative issues in v4.2 and earlier versions are greatly improved in version 4.3.
- Many expressions and derivatives are greatly improved in version 4.3, such as GISL.
- A model parameter `BSOIUPDATE=0|1` has been added using a flag to control thermal noise nf fix for v. 4.3. Its default value is 0, providing “No fix” for backward compatibility. Setting `BSOIUPDATE=1` turns on the PrimeSim Continuum solution thermal noise nf fix.

BSIMSOI4 Version 4.1 Update

In BSIMSOI4.1, the following features were added:

- A new material model (`mtrlMod`)
- Asymmetric GIDL/GISL model and new GIDL/GISL model (`gidlMod`)
- A new impact-ionization current model
- An improved Coulombic scattering model for high k/metal gate
- An improved body-contact model to characterize the opposite-type gate
- A new ΔV_{bi} model to simplify the parameter extraction
- A new `VgsteffCV` model for C-V, which is similar to `Vgsteff` in I-V
- A new gate current component in body contact region
- An improved DITS model with more flexibility and better fit

For a description of this model, see the *BSIMSOI4.1 MOSFET Model User's Manual* at <https://www-device.eecs.berkeley.edu/~bsimsoi/get.html>.

LEVEL 71: TFT Model

Thin-film transistor (TFT) process technology has been the industry mainstream technology of choice to implement various flat-panel display applications. While there exist many similarities in the electrical terminal characteristics between TFTs and their crystalline silicon counterparts, unique charge trapping/de-trapping and charge carrier transport mechanisms associated with the grains and grain boundaries in the channel region have led to significant technical challenges in TFT device modeling for accurate and scalable SPICE simulations.

The Level 71 TFT model is a unified model. It is used for polycrystalline silicon (p-si), but it is also applicable to amorphous silicon (a-si). The model has been developed based on BSIMSOI, with high accuracy in geometrical, bias, and temperature scaling. It explicitly and accurately considers the following physical effects: non-ideality of the subthreshold swing, quasi-saturation, bias-dependent source/drain resistance, kink in the saturation region, self heating, large leakage current, and many other related device behaviors.

This TFT model is fully supported by Synopsys' parameter extraction tool, Aurora. For detail on usage, contact the Synopsys support teams.

The following sections discuss these topics:

- [TFT Model — General Syntax](#)
- [TFT Model Parameters](#)

TFT Model — General Syntax

Syntax:

```
Mxxx nd ng ns ne [np] [nb] [nT] mname [L=val]
+ [M=val] [AS=val] [PS=val] [NRD=val]
+ [NRS=val] [NRB=val] [CTH0=val]
+ [NSEG=val] [PDBCP=val] [PSBCP=val] [AGBCP=val]
+ [AEBBCP=val] [DELTOX=val] [TNODEOUT]
+ [Off] [FRBODY] [BJToff=val] [IC=Vds, Vgs, Vbs, Ves, Vps]
```

TFT Model Parameters

This section covers the following topics:

- [TFT Model — Argument Descriptions](#)
- [TFT Model — General Parameters](#)
- [TFT Model — Process Parameters](#)

- [TFT Model — DC Parameters](#)
- [TFT Model — AC and Capacitance Parameters](#)
- [TFT Model — Temperature Parameters](#)
- [TFT Model — Bias-Dependent Source \(Rs\) / Drain \(Rd\) Parameters](#)

TFT Model — Argument Descriptions

Argument	Description
<i>Mxxx</i>	TFT element name. Must begin with M, followed by up to 1023 alphanumeric characters.
<i>nd</i>	Drain terminal node name or number.
<i>ng</i>	Front gate node name or number.
<i>ns</i>	Source terminal node name or number.
<i>ne</i>	Back gate (or substrate) node name or number.
<i>np</i>	External body contact node name or number.
<i>nb</i>	Internal body node name or number.
<i>nT</i>	Temperature node name or number.
<i>mname</i>	MOSFET model name reference.
<i>L</i>	TFT channel length in meters. This parameter overrides DEFL in an OPTIONS statement. Default=DEFL with a maximum of 0.1m.
<i>W</i>	MOSFET channel width in meters. This parameter overrides DEFW in an OPTIONS statement. Default=DEFW.
<i>M</i>	Multiplier to simulate multiple TFT in parallel. The M setting affects all channel widths, diode leakages, capacitances, and resistances. Default=1.
<i>AD</i>	Drain diffusion area. Overrides .OPTION DEFAD statement. Default=DEFAD.
<i>AS</i>	Source diffusion area. Overrides .OPTION DEFAS statement. Default=DEFAS.
<i>PD</i>	Drain junction perimeter, including channel edge. Overrides .OPTION DEFPD.
<i>PS</i>	Source junction perimeter including channel edge. Overrides .OPTION DEFPS.
<i>NRD</i>	Number of squares of drain diffusion for the drain series resistance. Overrides .OPTION DEFNRD.
<i>NRS</i>	Number of squares of source diffusion for the source series resistance. Overrides .OPTION DEFNRS.
<i>NRB</i>	Number of squares for the body series resistance.

Argument	Description
FRBODY	Coefficient of the distributed body resistance effects. Default=1.0
RTH0	Thermal resistance per unit width: If you do not specify RTH0, simulation extracts it from the model card.If you specify RTH0, it overrides RTH0 in the model card.
CTH0	Thermal capacitance per unit width: If you do not specify CTH0, simulation extracts it from the model card.If you specify CTH0, it overrides CTH0 in the model card.
NBC	Number of body contact isolation edge.
NSEG	Number of segments for partitioning the channel width.
PDBCP	Parasitic perimeter length for the body contact a the drain side.
PSBCP	Parasitic perimeter length for the body contact at the source side.
AGBCP	Parasitic gate-to-body overlap area for the body contact.
AEBCP	Parasitic body-to-substrate overlap area for the body contact.
VBSUSR	Optional initial Compared value of Vbs that you specify for transient analysis.
DELTOX	Shift in gate oxide thickness (TOX). That is, the difference between the electrical and physical gate oxide/insulator thicknesses.
TNODEOUT	Temperature node flag indicating the use of the T node. ⁴² See footnote below.
OFF	Sets the initial condition of the element to OFF in DC analysis.
BJTOFF	Turning off BJT if equal to 1.
IC	Initial guess in the order (drain, front gate, internal body, back gate, external voltage). (ignores Vps for 4-terminal devices) Use these only if you specify UIC in the .TRAN statement. The .IC statement overrides it.

TFT Model — General Parameters

Parameter	Unit	Default	Description
MOBMOD	-	1	Mobility model selector
NOIMOD	-	1	Flag for the noise model
SHMOD	-	0	Flag for self-heating: • 0=No • 1=Self-heating

42. If you do not set *TNODEOUT*, you can specify 3 nodes for a device to float the body. Specifying 4 nodes implies that the 4th node is the external body contact node with a body resistance between the internal and external terminals. This configuration applies to a distributed body resistance simulation. If you set *TNODEOUT*, simulation interprets the last node as the temperature node. You can specify 4 nodes to float the device. Specifying 5 nodes implies body contact. 6 nodes is a body-contacted case with an accessible internal body node. You can use the temperature node to simulate thermal coupling.

Parameter	Unit	Default	Description
TFTMOD	-	0	Flag for the transistor operation mode: • 0:Partially depleted • 1:Dynamic depleted • 2:Partially depleted

TFT Model — Process Parameters

Parameter	Unit	Default	Description
DTOXCV			Difference between the electrical and physical gate oxide thicknesses, due to the effects of the gate poly depletion and the finite channel charge layer thickness.
NCH	1/cm ³	1e15	Channel doping concentration
NGATE	1/cm ³	0	Poly gate doping concentration
NSUB	1/cm ³	6.0e16	Substrate doping concentration
TOX	m	4.5e-8	Gate oxide thickness
TSI	m	1.0e-8	Silicon film thickness

TFT Model — DC Parameters

Parameter	Unit	Default	Description
A0	-	2.5	Bulk charge effect coefficient for the channel length
A1	1/V	0.68	First non-saturation effect parameter
A2	-	0.18	Second non-saturation effect parameter
AELY	V/m	-0.1	Channel length dependency of the early voltage for the bipolar current
AGIDL	1/W	1e-7	GIDL constant
AGS	1/V	0.15	Gate bias coefficient of Abulk
AHLI	-	0.1	High-level injection parameter for the bipolar current
ALPHA0	m/V	1e-12	First parameter of the impact ionization current
B0	m	4.5e-8	Bulk charge effect coefficient for the channel width
B1	m	1e-7	Bulk charge effect width offset

Parameter	Unit	Default	Description
BETA0	1/V	40	First Vds dependence parameter of the impact ionization current
BETA1	-	5	Second Vds dependence parameter of the impact ionization current
BETA2	V	0	Third Vds dependence parameter of the impact ionization current
BETAG	-	3	Gate induced grain boundary barrier lowering coefficient
BGIDL	V/m	9.5e9	GIDL exponential coefficient
CDSC	F/m ²	-1e-3	Drain/source to the channel coupling capacitance
CDSCB	F/m ²	0	Body-bias sensitivity of cdsc
CDSCD	F/m ²	1e-3	Drain-bias sensitivity of cdsc
CGIDL	-	3.7e-2	Body voltage factor for gidl
CIT	F/m ²	0.0	Interface trap capacitance
DELTA	-	2.3	Effective Vds parameter
DGIDL	-	2.5	Electric field exponent factor for gidl
DROUT	-	0.2	L dependence coefficient of the DIBL correction parameter in Rout
DSUB	-	5	DIBL coefficient exponent
DVT0	-	0	First coefficient of the short-channel effect on Vth
DVT0W	-	0	First coefficient of the narrow width effect on Vth for a small channel length
DVT1	-	1	Second coefficient of the short-channel effect on Vth
DVT1W	-	5.3e6	Second coefficient of the narrow width effect on Vth for a small channel length
DVT2	1/V	-0.032	Body-bias coefficient of the short-channel effect on Vth
DVT2W	1/V	-0.032	Body-bias coefficient of the narrow width effect on Vth for a small channel length
DWB	m/V ^{1/2}	0.0	Coefficient of the substrate body bias dependence of Weff
DWBC	m	0.0	Width offset for the body contact isolation edge

Parameter	Unit	Default	Description
DWG	m/V	2.3e-8	Coefficient of the gate dependence of Weff
ESATI	V/m	1.e7	Saturation channel electric field for the impact ionization current
ETA0	-	0	DIBL coefficient in the subthreshold region
ETAB	1/V	-0.07	Body-bias coefficient for the DIBL effect in the subthreshold region
FBJTII	-	0.0	Fraction of the bipolar current affecting the impact ionization
GGIDL	-	6e-1	Gate voltage coefficient for gidl
ISBJT	A/m ²	1.4e-7	BJT injection saturation current
ISDIF	A/m ²	5e-5	Body to source/drain injection saturation current
ISREC	A/m ²	1.0e-5	Recombination in the depletion saturation current
ISTUN	A/m ²	0.0	Reverse tunneling saturation current
K1	V ^{1/2}	0.35	First-order body effect coefficient
K1W1	m	0	First-order effect width dependent parameter
K1W2	m	0	Second-order effect width dependent parameter
K2	-	-2e-2	Second-order body effect coefficient
K3	-	6e-4	Narrow coefficient
K3B	1/V	0	Body effect coefficient of k3
KB1	-	1	Backgate body charge coefficient
KETA	1/V	-4.6e-2	Body-bias coefficient of the bulk charge effect
KETAS	V	0.0	Surface potential adjustment for the bulk charge effect
KG	-	1	Offset voltage for gate induced grain boundary barrier lowering
LBJT0	m	1.3e-7	Reference channel length for the bipolar current
LII	-	2.8e-6	Channel length dependence parameter for the impact ionization current
LINT	m	0.0	Length offset fitting parameter from I-V without bias

Parameter	Unit	Default	Description
LGB1	-	1e-2	Grain boundary factor inducing effective channel length decrease
LGB2	-	7e-4	Depletion near grain boundary inducing effective channel length decrease
LGB3	-	2e-2	Gate voltage factor for effective channel length
LN	m	1.8-6	Electron/hole diffusion length
NBJT	-	1.3	Power coefficient of the channel length dependency for the bipolar current
NDIO	-	0.5	Diode non-ideality factor
NLN	-	2	Channel length exponent for amplification coefficient of parasitic BJT
NFACTOR	-	20	Subthreshold swing factor
NGIDL	V	-22	GIDL Vds enhancement coefficient
NLX	m	1.5e-6	Lateral non-uniform doping parameter
NRECF0	-	2.0	Recombination non-ideality factor at the forward bias
NRECR0	-	10	Recombination non-ideality factor at the reversed bias
NTUN	-	10.0	Reverse tunneling non-ideality factor
PCLM	-	9e-4	Channel length modulation parameter
PDIBLC1	-	1	Correction parameter for the DIBL effect of the first output resistance
PDIBLC2	-	1.5e-3	Correction parameter for the DIBL effect of the second output resistance
PRWB	1/V	-7.7e-7	0
PRWG	1/V ^{1/2}	-1e-2	Gate-bias effect coefficient of R _{dsw}
PVAG	-	5.6e2	Gate dependence of the Early voltage
RBODY	ohm/m ²	0.0	Intrinsic body contact sheet resistance
RBSH	ohm/m ²	0.0	Extrinsic body contact sheet resistance
RDSW	ohm/m ²	7e4	Parasitic resistance per unit width

Parameter	Unit	Default	Description
RSH	ohm/square	0.0	Source/drain sheet resistance in ohm per square
SII0	1/V	0.37	First Vgs dependence parameter for the impact ionization current
SII1	1/V	0.08	Second Vgs dependence parameter for the impact ionization current
SII2	1/V	0.08	Third Vgs dependence parameter for the impact ionization current
SIID	1/V	5e-3	Vds dependence parameter of the drain saturation voltage for the impact ionization current
TII	-	0	Temperature dependence parameter for the impact ionization current
U0	cm ² /(V-sec)	NMOS-600 PMOS-200	Mobility at Temp=Tnom
UA	m/V	9.6e-9	First-order mobility degradation coefficient
UB	(m/V) ²	1e-21	Second-order mobility degradation coefficient
UC	1/V	-4e-11	Body-effect of the mobility degradation coefficient
UG	m ² /(V-sec)	9e-2	Grain boundary barrier scattering mobility at Temp=Tnom
VABJT	V	10	Early voltage for the bipolar current
VDSATII0	V	0.6	Nominal drain saturation voltage at threshold for the impact ionization current
VECB	v	0.026v	Electron tunneling from the conduction band
VEVB	v	0.075v	Electron tunneling from the valence band
VOFF	v	-0.6	Offset voltage in the subthreshold region for large W and L values
VREC0	V	0.0	Voltage dependent parameter for the recombination current
VSAT	m/sec	5e5	Saturation velocity at Temp=Tnom

Parameter	Unit	Default	Description
VTH0	v	NMOS 0.7 PMOS -0.7	Threshold voltage @ Vbs=0 for a long, wide device
VTUN0	V	0.0	Voltage dependent parameter for the tunneling current
W0	m	2.5e-5	Narrow width parameter
WINT	m	0.0	Width offset fitting parameter from I-V without bias
WR	-	0.8	Width offset from Weff for the Rds calculation

TFT Model — AC and Capacitance Parameters

Parameter	Unit	Default	Description
ACDE	m/V	1.0	Exponential coefficient for the charge thickness in the CapMod=3 for the accumulation and depletion regions
ASD	V	0.3	Smoothing parameter for the source/drain bottom diffusion
CF	F/m	cal.	Fringing field capacitance of the gate-to-source/drain
CGDL	F/m	0.0	Overlap capacitance for the lightly-doped drain-gate region
CGDO	F/m	0	Non LDD region drain-gate overlap capacitance per channel length
CGEO	F/m 0	Gate	Substrate overlap capacitance per unit channel length
CGSL	F/m	0.0	Overlap capacitance for the lightly-doped source-gate region
CGSO	F/m	cal.	Calculated Non LDD region source-gate overlap capacitance per channel length
CJSWG	F/m ²	1.e-10	Source/drain (gate side) sidewall junction capacitance per unit width (normalized to 100nm Tsi)
CKAPPA	F/m	0.6	Coefficient for the fringing field capacitance for the overlap capacitance in the lightly-doped region
CLC	m	0.1e-7	Constant term for the short-channel model
CLE	-	0.0	Exponential term for the short-channel model

Parameter	Unit	Default	Description
CSDSW	F/m	0.0	Fringing capacitance per unit length for the source/drain sidewall
CSDMIN	V	cal.	Minimum capacitance for the source/drain bottom diffusion
DELVT	V	0.0	Threshold voltage adjustment for C-V
DLBG	m	0	Length offset fitting parameter for the backgate charge
DLC	m	lint	Length offset fitting parameter for the gate charge
DLCB	m	lint	Length offset fitting parameter for the body charge
DWC	m	wint	Width offset fitting parameter from C-V
FBODY	-	1.0	Scaling factor for the body charge
LDIF0	-	1	Channel length dependency coefficient of the diffusion cap.
MJSWG	V	0.5	Grading coefficient of the source/drain (gate side) sidewall junction capacitance
MOIN	$V^{1/2}$	15.0	Coefficient for the gate-bias dependent surface potential
NDIF	-	-1	Power coefficient of the channel length dependency for the diffusion capacitance
PBSWG	V	0.7	Built-in potential of the source/drain (gate side) sidewall junction capacitance
TT	second	1ps	Diffusion capacitance transit time coefficient
VSDFB	V	cal.	Flatband voltage for the source/drain bottom diffusion capacitance
VSDTH	V	cal.	Threshold voltage for the source/drain bottom diffusion capacitance
XPART	-	0	Charge partitioning rate flag

TFT Model — Temperature Parameters

Parameter	Unit	Default	Description
AT	m/sec	3.3e4	Temperature coefficient for Ua
CTH0	(W*S)/m°C	0	Normalized thermal capacity

Parameter	Unit	Default	Description
KT1	V	-0.11	Temperature coefficient for the threshold voltage
KT2	-	0.022	Body-bias coefficient of the threshold voltage temperature effect
KTIL	V* μ m	0	Channel length dependence of the temperature coefficient for the threshold voltage
NTRECF	-	0	Temperature coefficient for Nrecf
NTRECR	-	0	Temperature coefficient for Nrecr
PRT	Ω - μ m	0	Temperature coefficient for Rdsw
RTH0	m°C/W	0	Normalized thermal resistance
TCJSWG	1/K	0	Temperature coefficient of Cjswg
TNOM	°C	25	Temperature at which simulation expects parameters
TPBSWG	V/K	0	Temperature coefficient of Pbswg
UA1	m/V	4.31e-9	Temperature coefficient for Ua
UB1	(m/V) ²	-7.61e-18	Temperature coefficient for Ub
UC1	1/V	-0.056	Temperature coefficient for Uc
UTE	-	-1.5	Mobility temperature exponent
XBJT	-	1	Power dependence of jbjt on the temperature
XDIF	-	Same as XBJT	Power dependence of jdif on the temperature
XREC	-	1	Power dependence of jrec on the temperature
XTUN	-	0	Power dependence of jtun on the temperature

TFT Model — Bias-Dependent Source (Rs) / Drain (Rd) Parameters

Parameter	Unit	Default	Description
CRD	Same as crs		Coefficient of Rd due to gate and external drain bias
CRS		1.0	Coefficient of Rs due to gate and external source bias

Parameter	Unit	Default	Description
DELTA VGD	Same as deltavgs		Smoothing parameter for RD dependence on VGD
DELTA VGS	-	1.0e-4	Smoothing parameter for RS dependence on VGS
ERDD	Same as erss		External drain voltage dependent parameter in PRWDD
ERDG1	Same as ersg1		Gate-bias dependent parameter in PRWGD
ERDG2	Same as ersg2		Gate-bias dependent parameter in PRWDD
ERSG1	-	1.0	Gate-bias dependent parameter in PRWGS
ERSG2	-	1.0	Gate-bias dependent parameter in PRWSS
ERSS	-	1.0	External source voltage dependent parameter in PRWSS
PRWBS	$V^{-0.5}$	0.0	Body-bias dependence of LDD resistance in source side
PRWDB	Same as prwbs		Body-bias dependence of LDD resistance in drain side
PRWDD	Same as prwss		External drain-bias and gate-bias dependence of LDD resistance in drain side
PRWGD	Same as prwgs		Gate-bias dependence of LDD resistance in drain side
PRWGS	V^{-1}	1.0	Gate-bias dependence of LDD resistance in source side
PRWSS	V^{-1}	1.0	External source-bias and gate-bias dependence of LDD resistance in source side
RDW	$\text{ohm}(\mu\text{m})^{\text{wr}}$	100.0	Zero bias lightly-doped drain resistance $R_d(V)$ per unit width for RDMOD=1
RDWMIN	$\text{ohm}(\mu\text{m})^{\text{wr}}$	0.0	Lightly-doped drain resistance per unit width at high V_{gs} and zero V_{bs} for RDMOD=1
RDWMIN0	Ohm	0.0	Drain resistance constant component
RSW	$\text{ohm}(\mu\text{m})^{\text{wr}}$	100.0	Zero bias lightly-doped source resistance $R_s(V)$ per unit width for RSMOD=1
RSWMIN	$\text{ohm}(\mu\text{m})^{\text{wr}}$	0.0	Lightly-doped source resistance per unit width at high V_{gs} and zero V_{bs} for RSMOD=1

Parameter	Unit	Default	Description
RSWMIN0	Ohm	0.0	Source resistance constant component
WR	-	0.0	Width dependence exponent of lightly-doped source and drain resistance

LEVEL 72: BSIM-CMG MOSFET Model

The BSIM-CMG model is a compact model to describe the behavior of multiple-gate MOSFET devices. There are two sub-modules for the PrimeSim Continuum solution built-in model:

- SOIMG module for the BSIM-CMG SOI model
- BULKMG module for the BSIM-CMG BULK model

This model captures all important multi-gate transistor behaviors. The solution of Poisson's equation includes volume inversion; hence the subsequent I-V formulation automatically captures the volume inversion effect. Analysis of the electrostatic potential in the body of CMG MOSFETs provides the model equation for short channel effects (SCE). The short channel model also captures the extra electrostatic control from the top gate (tri-gate) or top/bottom gates (omega-gate or all-around-gate). The SCE model predicts the threshold (V_t) roll-off, drain-induced-barrier lowering (DIBL) and subthreshold slope degradation in close agreement with 2-D device simulation results.

Included in the model are other important effects such as: mobility degradation, multiple surface-orientations, velocity saturation, velocity overshoot, series resistance, channel length modulation, quantum mechanical effects, poly depletion, gate tunneling current, gate-induced-drain-leakage, and parasitic capacitance models.

This model supports the advanced MOSFET self-heating simulation solution.

Note:

With version 105, the model name was renamed as BSIMMG; later it was changed back to BSIM-CMG with version 105.03.

See also: [Output Templates for BSIM-CMG \(Level 72\)](#), [BSIM6 \(Level 77\)](#), and [BSIM-IMG \(Level 78\)](#).

The following sections discuss these topics:

- [BSIM-CMG Model — General Syntax](#)
- [BSIM-CMG Model — Version Updates](#)
- [BSIM-CMG Model Parameters \(Version \$\geq\$ 105\)](#)
- [BSIM-CMG Model Parameters \(Version \$<\$ 105\)](#)
- [BSIM-CMG Model \(Version=105.03\) — WPE Model Support](#)
- [BSIM-CMG Model — Deactivating Model Equations](#)
- [BSIM-CMG Model — Searching Models as Function of NFIN and L](#)

- [BSIM-CMG Model — Binning Calculation](#)
- [BSIM-CMG Model — Additional Operating Point Outputs](#)

BSIM-CMG Model — General Syntax

Syntax:

```
Mxxx nd ng ns nb [nt] mname [L=val] [M=val]
.model mname n(p)mos LEVEL=72 VERSION=val ...
```

Where `nt` indicates the temperature node.

Level 72 is the BSIM-CMG model in PrimeSim Continuum. Before BSIM-CMG102, the model parameter `TECHMOD=0` (default) was used for the BSIM-CMG BULK model and `TECHMOD=1` for the BSIM-CMG SOI model. With BSIMCMG102, and going forward, use model parameter `BULKMOD=1` (default) for the BSIM-CMG BULK model and `BULKMOD=0` for the BSIM-CMG SOI model.

For example, to use the BSIM-CMG model in the PrimeSim Continuum solution, specify:

```
M1 drain gate source NCH L=1u
```

For the BSIM-CMG BULK model, specify NCH as:

```
.MODEL NCH NMOS LEVEL=72 VERSION=1.01 TECHMOD=0
.MODEL NCH NMOS LEVEL=72 VERSION=102 BULKMOD=1
```

For the BSIM-CMG SOI model, specify NCH as:

```
.MODEL NCH NMOS LEVEL=72 VERSION=1.01 TECHMOD=1
.MODEL NCH NMOS LEVEL=72 VERSION=102 BULKMOD=0
```

BSIM-CMG Model — Version Updates

This section covers the following topics:

- [BSIM-CMG 111.2.1 Updates](#)
- [BSIM-CMG 111.2.0 Updates](#)
- [BSIM-CMG 111.1.0 Updates](#)
- [BSIM-CMG 111.0.0 Updates](#)
- [BSIM-CMG 110.0.0 for Nanosheet FETs](#)
- [BSIM-CMG 109.0.0/110.0.0 Updates](#)
- [BSIM-CMG 108.0.0 Updates](#)

- [BSIM-CMG 107.0 Updates](#)
- [BSIM-CMG 106.1.0 Updates](#)
- [BSIM-CMG 106 Updates](#)
- [BSIM-CMG 105.04 Updates](#)
- [BSIM-CMG 105.031 Updates](#)
- [BSIM-CMG 105.03 Updates](#)
- [BSIMMG 105 Updates](#)
- [BSIM-CMG 104.1](#)
- [BSIM-CMG 104.0](#)
- [BSIM-CMG 103 Updates](#)

BSIM-CMG 111.2.1 Updates

BSIM-CMG 111.2.1 implements the following features:

- Fix code implementation issue in `junction_cap` macro

BSIM-CMG 111.2.0 Updates

BSIM-CMG 111.2.0 implements the following features:

- Maximum number of bodies allowed in the GAA (NGAA) has been increased from 3 to 6
- Enhance gate to substrate capacitance model for GAA (`GEOMOD=5`)
- Support for cryogenic temperature model: cryogenic models can be enabled by `CRYOMOD` switch, which can have the following three settings
 - `CRYOMOD = 0`: turns off cryogenic models
 - `CRYOMOD = 1`: most physical cryogenic models
 - `CRYOMOD = 2`: cryogenic temperature expressions converge to BSIM-CMG111.1.0 temperature expressions for $T > 210$ K (-63.15 C)
- Improve accuracy of C-V fitting for high V_d and V_g with `CVMOD=1`
- Improve the accuracy of $I-V$ near V_{dsat} at high V_g : S/D velocity saturation model
- Support parasitic substrate GIDL/GISL model: this enhance separates out D to S/S to D and D to E/S to E components of GIDL/GISL for `GEOMOD=2|3|5`. `GIDLMOD = 2` is used to enable this feature

- Add protection for junction capacitance grading coefficient to avoid numerical issues
- Fixed several bugs

BSIM-CMG 111.1.0 Updates

BSIM-CMG 111.1.0 implements the following features:

- Extend the capability of BSIM-CMG model to capture the behavior gate-all-around FETs (GAAFETs) accurately. It introduces:
 - Support GEOMOD = 5 for GAAFETs.
 - A new mobility module(MOBSCMOD=1) captures geometry dependent mobility scaling for GAAFETs.
 - A new quantum confinement module(SUBBANDMOD=1)models geometry dependent subband energies and density of states
 - A new parasitic capacitance module(CGEMOD=3) models the vertically stacked GAAFET geometry.

BSIM-CMG 111.0.0 Updates

BSIM-CMG 111.0.0 implements the following features:

- Improved clamping on Dmob.
- Renamed model parameter Imin to IMIN.
- Improved fitting flexibility of NF50 at low Ids.
- Igate current clamp concern
- Removed DEVTYPE, use -1 for ptype.
- Added binning for DVTP0 and DVTP1.
- Improved NQSMOD=2 convergence.
- L offset parameters, (LINT, DLBIN and XL) made binnable.
- Introduced gate current scaling/adjustment factors- IGB0MULT and IGC0MULT.
- Fixed several bugs.

BSIM-CMG 110.0.0 for Nanosheet FETs

The following are the highlights of this enhancement:

- Set `GEOMOD = 2` to activate the nanosheet module. This module supports multiple nanosheets per fin (limited to a maximum of three).
- To improve accuracy in quantum confinement effects, set `SUBBANDMOD=1`. The model updates capture the effects of subband separation in nanosheet FETs, which results in peaks and valleys in capacitance, transconductance, and other higher derivatives of charge and current.
- A new geometry-dependent parasitic fringe capacitance network for nanosheet FETs is included and is turned on by using `CGEOMOD = 3`.
- Model usage:

Add subversion `bdmcver` to support BDMC NSHEET model for backward compatibility.

```
.model nfet nmos level=72 version=110 bdmcver=0.1 geomod=2 cgeomod=3  
subbandmod=1
```

BSIM-CMG 109.0.0/110.0.0 Updates

BSIM-CMG 109.0.0/110.0.0 implemented the following features:

- A unified model for various fin geometries.
- A better modeling of body bias effects in I-V and C-V by using `BULKMOD=2`.
- Model parameters to support the `ASYMMOD=1`: `PCLM`, `PDIBL2`, `VSAT`, `KSATIV`, `DVTSHIFT`, `CIT`, `U0`, `UA`, `UD`, `UC`, `EU`.
- Added an optional temperature node for self-heating analysis.
- Added BSIM4-like support for VFBS in gate-channel tunneling current.
- Added additional bias/charge sensitivity to lateral non-uniform doping model.
- Included a new DITS term.
- Added `NFINNOM` as an instance parameter with a default value of 1.
- Modified/updated the `IIMOD=2` clamps. Also, the clamp limits model parameters namely `IIMOD2CLAMP1` and `IIMODCLAMP2` are included.
- Implemented a smooth clamp on gate current to prevent large values.
- The gate sidewall term to the bulk should reflect the dimension along `TFIN`, not `Weff0` which covers the other three sides of channel.

- Implemented an expression for thermal resistance taking NFIN and FPITCH dependence into account.
- Thermal noise model update: the model of TNOIMOD =1 in the previous versions of BSIM-CMG are no longer supported. From BSIM-CMG 109.0.0, only TNOIMOD =0 and 1 is supported. While TNOIMOD = 0 is the same as before, TNOIMOD =1 is the same as TNOIMOD =2 in the previous versions of BSIM-CMG. TNOIMOD=1 in BSIM-CMG109.0.0/110.0 is a correlated thermal noise model where both drain and gate noise are implemented as current sources.
- Fixed several bugs.

BSIM-CMG 108.0.0 Updates

BSIM-CMG 108.0.0 implemented the following features:

- TEMPMOD switch introduced to switch between two types of temperature dependence of parameters.
- Thermal noise model TNOIMOD=1 is added from BSIM4 model.
- New parameter ATCV added to model temperature dependence of ATCV.
- Improvement in Overlap Capacitance model.
- RDSMOD=2 is added to avoid additional nodes.
- Thermal Noise Model TNOIMOD = 2 added from BSIM4 model.
- Leff equation update for binning calculations.
- Added option to replace some pre-computed variables with parameters, such as THETASCE, THETADIBL, THETASS, NVTM.
- Add less-than-zero check on Leff1.
- Less restrictive clamps on parameters modeling NFIN dependence of other parameters.
- Parameters added to ASYMMOD=1 model, such as U0, UA, UC, UD, EU, PDIBL2, KSATIV, DVTSHIFT, PCLM, VSAT.
- TYPE added as a parameter to define device type for consistency with other CMC models.
- Fixed several bugs.

BSIM-CMG 107.0 Updates

BSIM-CMG 107.0 implemented the following features:

- Long channel DIBL (also called Drain-Induced Vth Shift or DITS) was added and new parameters `DVTP0` and `DVTP1` were introduced.
- Average charge weighing factor was added through the new parameter `CHARGEWF`.
- Length scaling of `SCE` and `SS` were de-coupled and the new binnable parameter `DVT1SS` was introduced.
- A linear length scaling equation for `PHIG` was added.
- New `NFIN` scaling equations and parameters were added for the following parameters: `PHIG`, `CDSC`, `CDSCD`, `CDSCDR`, `NBODY`, `VSAT`, `VSAT1`, `VSAT1R`, `U0`.
- Temperature dependence on `ETA0` and `ETA0R` was added.
- The default values of `COVS` and `COVD` were at the unrealistic value of 25 pF. The default values were set to zero.
- For `CGEOMOD=1` only, `CGEO1SW=1` now enables the parameters `COVS`, `COVD`, `CGSP`, and `CGDP` to be in F per fin per gate-finger per unit channel width. The default value of `GEO1SW` switch is zero.
- To be consistent with `SOI`, `RTH` and `CTH` equations in the self-heating sub-model were updates.
- The initial guess for the surface potential calculation was improved. The model is now infinitely scalable with respect to `TFIN` and `NBODY` without any clamping on `NBODY`.
- Unused model parameters `NINTNOI` and `PINTNOI` were removed.
- Minor updates in the technical manual including equations 3.332 and 3.334; the equations were updated in manual to be consistent with the code.
- Fixed several bugs.

BSIM-CMG 106.1.0 Updates

BSIM-CMG 106.1.0 implemented the following features:

- Two different expressions are adopted for V_{fbsd} for Poly and Metal Gate, respectively.
- For gate current model, in order to decouple the oxide thickness used in I-V model (through the gate current) and C-V model and facilitate the model tuning, a new parameter, `TOXG`, is introduced and used in the gate tunneling current model. The default value of `TOXG` is equal to `TOXP`.

- For gate-induced source and drain leakage (GIDL/GISL), the model now retains zero current at zero bias and it is smooth with respect to bias. For `BULKMOD=1`, the singularity in the denominator of body-bias factor, $V_{de}^3 / (CGIDL + V_{de}^3)$, was removed and for $V_{de} > 0$ the GIDL is zero. Similar fixes were applied to GISL model.
- For temperature dependence of body effect model, lateral NUD model, and non-saturation effect model, the model parameters `K1`, `K1SAT`, `K0`, `K0SI`, `A1`, and `A2` now have linear temperature dependency. For example, $K1_t = K1_i + K11_i * (T - TNOM)$.
- Modify temperature and length dependence of impact ionization current for `iimod=2`.
- Current (I_s) clamping in accumulation, `Imin` is now a model parameter with the default value of $1.0E-15$.
- Built-in `cosh()` overflow is added.
- Improve gate resistance model.
- `NQSMOD=3` was disabled.
- The definitions of the physical intrinsic capacitances, `CEDI` and `CEEI`, were corrected.
- Threshold voltage definition is added.
- Fixed many bugs.

BSIM-CMG 106 Updates

BSIM-CMG 106 implemented the following features:

- Removed some unused source codes.
- Used built-in hyperbolic functions instead of macros.
- Added more significant digits to definition of electronic charge.
- Fixed some bugs.

BSIM-CMG 105.04 Updates

BSIM-CMG 105.04 implemented the following features:

- GIDL Current Model modified. New parameter `PGIDL/PGISL` introduced and model retains zero current at zero bias.
- Gate Current Model modified:
 - Default values of `BIGBINV`, `AIGBACC`, `BIGBACC`, `NIGBACC` were changed
 - Gate-to-Channel Current, `Igcs/Igcd`. Partition function implemented with parameter `PIGCD` (Binned).

- Modification to the CLM Model:
 - Fixed discontinuity in implementation at $PCLMG=0$ for CLM Model
 - CLM Model for I-V Modified. Parameter VASAT removed. The old equation was known to cause unphysical “wiggles.”
- ETAMOB binning corrected and linear temperature dependence added with new parameter EMOBT.
- Velocity Saturation Model for better I_d , sat and G_m , sat fitting.
- Parameters DELTAVSAT, DVTSHIFT and KT1 are now binned.
- DTEMP handle has been implemented.
- CGEOMOD=0 case, removing QM effects in Fringe Caps.
- Drain to Source Fringe Cap now available for all CGEOMOD.
- Introduced DLCACC parameter for accumulation region capacitance (CAPMOD=1 and BULKMOD=1).
- Velocity Saturation Model for Short Channel C-V introduced. New parameters VSATCV (binned) added.
- CLM Model for C-V modified:
 - M_{clmCV} factor simplified. Parameters VASATCV and PCLMGCV removed.
 - Corrected the wrong implementation of in the charge equations.
- Added relevant equations for Short Channel CV.
- Parameter NFIN has been converted from integer to real to enable optimization.
- Gate to Substrate Overlap cap scalability corrected. New parameter CGBN introduced.
- Temperature Dependence for Subthreshold Swing enhanced with new parameter TSS (binned).
- RDSMOD=1 is enhanced to capture quasi-saturation / current crowding for high voltage devices. New parameters RSDR, RDDR, PRSDR, PRDDR are introduced and new equations are added. Temperature dependence through TRSDR and TRDDR parameters.

- Asymmetric Model - ASYMMOD Switch:
 - To model highly asymmetric devices, seven parameters in the model are identified and their reverse mode equivalents are created, such as CDSCDR, ETA0R, PDIBL1R, PTWGR, VSAT1R, RSDRR, and RDDRR.
 - Additionally, parameter PRWG is split into PRWGS and PRWGD for source and drain side in RDSMOD=1. However for RDSMOD=0, PRWGS takes over and replaces PRWG. PRWGD is not used.
- Junction capacitance equations have been changed. The transition from standard junction capacitance equation to linear cap does not happen at $V_{es} = 0$ / $V_{ed} = 0$ (like BSIM4), but at $V_{es} = 0.9 * V_{bi}$ (built-in voltage of diode). This moves discontinuity in second derivative of capacitance far into forward bias where the device does not usually operate.
- New parameters PHIBE, K1, K1ST, K1SAT are introduced for body Effect in BULKMOD=1.
- Added more significant digits to definition of electronic charge from $1.6e-19$ to $1.60219e-19$ (Coulombs).
- Quantum Mechanical Effects:
 1. Introduced new parameter QMTCENCVA (binned) to replace QMTCENCV for the accumulation region capacitance.
 2. Some bug fixes for variable 'Tcen' calculations in both inversion and accumulation region.
- Introduced Non-saturation effect with new parameters A1 and A2. To be used to improve $I_{d,sat}$ and $G_{m,sat}$ fitting.
- Lateral NUD Model introduced to create IV-CV V_{th} shift.
- Fixed many bugs.

BSIM-CMG 105.031 Updates

BSIM-CMG 105.031 implemented the following features:

- GIDL Current model modified for zero current at zero bias. This update introduced the equation around parameter $CGIDL/CGISL$ (binned) from BSIM4 for BULKMOD=1. For BULKMOD=0, $CGIDL/CGISL$ parameters are not used.
- Gate Current Model ($igcs/igcd$, igs/igd) has been modified. Partition function implemented with parameter PIGCD (Binned).
- Removed CLM effect in Charge model due to faulty implementation.
- Fixed discontinuity in implementation at $PCLMG=0$ for CLM model.

BSIM-CMG 105.03 Updates

BSIM-CMG 105.03 implemented the following features.

- Restored the name BSIM-CMG from its change to BSIMMG (see [BSIMMG 105 Updates](#))
- Added PCLMG, VASAT, and ETAMOB to the list of binnable parameters
- Introduced DLBIN, an 'L' offset parameter for binning.
- Introduced the DVTSHIFT parameter as a handle to create Vth shift.
- This release replaces all older mobility models with a single new mobility model. The new mobility model uses UA, EU, UD, and UCS parameters from MOBMOD=0 of 105.02 release.
- This version adds a New Velocity Saturation (Ion degradation) model. Parameters VSAT1 (binned) and DELTAVSAT replace the old parameter THETASAT. This is a backward *incompatible* change In BSIMMG105 alpha2.
- Changed the RDSMOD=0 implementation for better accuracy.
- Modified the CLM Model to prevent cross-correlation of binning parameters.
- Added an Igb accumulation component for IGBMOD=1.
- Changed the Vds asymmetric function for better higher order derivative predictability.

BSIMMG 105 Updates

BSIMMG v.105 implemented the following features.

- Implemented the new geometry-dependent FinFET source/drain resistance model $R_{GEOMOD}=1$ for the purpose of variability modeling.
- Implemented a new geometry-dependent FinFET outer fringe capacitance model $C_{GEOMOD}=1$ for the purpose of variability modeling.
- Implemented a bias-dependent overlap capacitance model (adapted from BSIM4).
- SOI multi-gate devices ($BULKMOD=0$) now support 4 terminals only.
- Changed the formulation for current degradation due to velocity saturation in the linear region.
- Changed the definitions of AS, AD, PS, PD for consistency with BSIM4: ASEO, ADEO, PSEO, PDEO now refers to the total source/drain area or perimeter rather than the per-finger value. AS is separated into a junction portion only for bulk multi-gate (ASEJ, PSEJ) and a oxide portion (ASEO, PSEO). AD is separated into ADEJ, PDEJ, ADEO and PDEO.

- Implemented Source/drain to substrate capacitance for both `BULKMOD=0, 1`, controlled by parameters `ASEO`, `ADEO`, `PSEO`, `PDEO`, and `CSDESW`.
- The model now distinguishes `toxe` (electrical oxide thickness) and `toxp` (physical oxide thickness). `DEOT` is a new parameter that represents the oxide-equivalent thickness of the inversion layer. For backward compatibility with BSIM-CMG 104, set `DEOT=0`.
- The model now distinguishes `toxe` (electrical oxide thickness) and `toxp` (physical oxide thickness). `DEOT` is a new parameter that represents the oxide-equivalent thickness of the inversion layer. For backward compatibility with BSIM-CMG 104, set `DEOT=0`.
- New I-V and C-V model.
- QM Model changes: `Weff/Toxeff` equations changed and correct version introduced. Introduced model parameters `QMTCENIV` and `QMTCENCV` (quasi-switches) to independently turn on/off `Weff` correction for IV and CV, and `coxeff` correction for CV.
- Accumulation Capacitance added. `CAPMOD=1` turns on additional Surface potential computation required and stamps accumulation charge too. Currently allowed only for `BULKMOD=1` case.
- Added gate sidewall junction capacitances and currents.
- Separated `GMOD` into `IGBMOD` and `IGCMOD`
- Renamed BSIMCMG to BSIMMG.
- Added `Igb` for SOI multi-gate devices (`BULKMOD=0`). (Previously, `Igb` was only calculated for bulk multi-gate devices.)
- Added BSIM4 style NQS model. `NQSMOD=2`.
- Added New Charge segmentation based NQS model. `NQSMOD=3`.
- Fixed many bugs.

BSIM-CMG 104.1

BSIM-CMG 104.1 implemented the following features:

- Asymmetric `DIBL` parameters supported.
- Bulk-Charge is made bias independent through correct implementation.
- `CLMMOD=1` case replaced with newer model.
- Many bug fixes were implemented.

BSIM-CMG 104.0

BSIM-CMG 104 implemented the following features:

- Channel length modulation model `CLMMOD=1` is added based on the BSIM4 formulation.
- Instance parameters `NRS` and `NRD` and associated equations are added based on the BSIM4 formulation.
- The equations associated with parameter `PVAG` are modified to provide more flexibility for output conductance fitting.
- Added a New Impact Ionization Current Model and Model Switch `IIMOD`; BSIM4 based `Iii` model into `IIMOD=1` block and BSIM SOI based `Iii` model into `IIMOD=2` block.
- Improved the Gate Electrode Resistance Model; introduced new parameters `RGEXT` and `RGFIN` and removed parameter `RGELTD`.
- Changed the Internal Resistance formula.
- Implemented many bug fixes.

BSIM-CMG 103 Updates

BSIM-CMG 103 implemented the following features:

- This release supports a Cylindrical gate geometry through `GEOMOD = 3` with an associated short channel scale length and quantum effects model.
- The existing I-V has been enhanced to model Poly-depletion accurately.
- Self-heating is now supported with addition of a Temperature node.
- Junction Capacitance and Junction Current equations were revamped with source-drain asymmetry supported. The asymmetry is now also in `GIDL/GISL` currents.
- Length-dependent equations have been added for a Global Parameter Extraction without binning.
- `SHMOD`, `RGATEMOD`, `NQSMOD` and `RDSMOD` also control the number of internal nodes for faster simulations. Many bug fixes were implemented.

BSIM-CMG Model Parameters (Version >= 105)

Note:

The name BSIMMG is the same as BSIM-CMG and refers to the same model; from version 105.03 the model name (BSIMMG) was reverted to BSIM-CMG, its original name.

This section covers the following topics:

- [BSIM-CMG Model \(Version>=105\) — Instance Parameters](#)
- [BSIM-CMG Model \(Version>=105\) — Control and Process Parameters](#)
- [BSIM-CMG Model \(Version>=105\) — Basic Model Parameters](#)
- [BSIM-CMG Model \(Version>=105\) — Geometry Dependent Parasitic Parameters](#)
- [BSIM-CMG Model \(Version>=105\) — Temperature Dependent and Self-Heating Parameters](#)

BSIM-CMG Model (Version>=105) — Instance Parameters

Note:

Binnable parameters are marked as ^(b). Instance parameters which are also model parameters are marked as ^(m).

Parameter	Unit	Default	Min	Max	Description
ADEJ ^(m)	m ²	0	0	-	Drain junction area (all fingers; for bulk MuGFETs, BULKMOD = 1)
ADEO ^(m)	m ²	0	0	-	Drain to substrate overlap area through oxide (all fingers)
ASEJ ^(m)	m ²	0	0	-	Source junction area (all fingers; for bulk MuGFETs, BULKMOD = 1)
ASEO ^(m)	m ²	0	0	-	Source to substrate overlap area through oxide (all fingers)
CDSP ^(m)	F	0	0	-	Constant drain to source fringe capacitance (for CGEOMOD = 1)
CGDP ^(m)	F	0	0	-	Constant gate to drain fringe capacitance (for CGEOMOD = 1)
CGSP ^(m)	F	0	0	-	Constant gate to source fringe capacitance (for CGEOMOD = 1)
COVS ^(m)	For F/m, see CGEO 1SW	0	0	-	Constant gate to source overlap capacitance (for CGEOMOD = 1)
COVD ^(m)	For F/m, see CGEO 1SW	0	0	-	Constant gate to drain overlap capacitance (for CGEOMOD = 1)

Parameter	Unit	Default	Min	Max	Description
$D^{(m)}$	m	40n	1n	-	Diameter of cylinder (for GEOMOD = 3)
$FPITCH^{(m)}$	m	80n	TFIN	-	Fin pitch
$L^{(m)}$	m	30n	1n	-	Designed gate length
$LRSD^{(m)}$	m	L	0	-	Length of the source/drain
NF	-	1	1	-	Number of fingers
$NFIN^{(m)}$	-	1	>0	-	Number of fins per finger
$NFINNOM^{(m)}$	-	1.0	>0	-	Nominal number of fins per finger
$NGCON^{(m)}$	-	1	1	2	Number of gate contacts
$NRD^{(m)}$	-	0	0	-	Number of drain diffusion squares (for RGEOMOD = 0)
$NRS^{(m)}$	-	0	0	-	Number of source diffusion squares (for RGEOMOD = 0)
$PDEO^{(m)}$	m	0	0	-	Perimeter of drain to substrate overlap region through oxide (all fingers)
$PSEO^{(m)}$	m	0	0	-	Perimeter of source to substrate overlap region through oxide (all fingers)
$PDEJ^{(m)}$	m	0	0	-	Drain junction perimeter (all fingers; for bulk MuGFETs, BULKMOD = 1)
$PSEJ^{(m)}$	m	0	0	-	Source junction perimeter (all fingers; for bulk MuGFETs, BULKMOD = 1)
RDC	Ω	0	-	-	Drain contact resistance for per-finger device
RSC	Ω	0	-	-	Source contact resistance for per-finger device
$TFIN^{(m)}$	m	15n	1n	-	Body (fin) thickness
$IGB0MULT^{(m)}$	-	1	0	-	Gate to body current scale factor
$IGC0MULT^{(m)}$	-	1	0	-	Gate to channel current scale factor
$LCOVS^{(m)}$	F	0	-	-	L-term of COVS
$NCOVS^{(m)}$	F/m	0	-	-	N-term of COVS

Parameter	Unit	Default	Min	Max	Description
PCOVS ^(m)	F	0	-	-	P-term of COVS
LCOVD ^(m)	F	LCOVS	-	-	L-term of COVD
NCOVD ^(m)	F/m	NCOVS	-	-	N-term of COVD
PCOVD ^(m)	F	PCOVS	-	-	P-term of COVD

BSIM-CMG Model (Version>=105) — Control and Process Parameters

Parameter	Unit	Default	Min	Max	Description
TYPE	-	NMOS	PMOS	NMOS	NMOS=1, PMOS=0
VERSION	-	106.1	-	-	Model version number
DEVTYPE	-	NMOS	PMOS	NMOS	NMOS=1, PMOS=0
ASYMMOD	-	0	0	1	- Asymmetric I-V model selector 0 = turn off - Reverse mode parameters ignored, 1 = turn on
BULKMOD	-	0	0	1	Substrate model selector. 0 = multi-gate on SOI substrate, 1 = multi-gate on bulk substrate
TEMPMOD	-	0	0	1	Temperature dependence model selector
TNOIMOD	-	0	0	2	Before BSIM-CMG109.0.0, thermal noise model selector, 0 = charge-based, 1 = holistic, 2 = correlated noise model
TNOIMOD	-	0	0	1	In BSIM-CMG109.0.0/110.0, thermal noise model selector, 0 =charge-based, 1 = correlated noise model
CGEO1SW	-	0	0	1	For CGEOMOD=1 only, GEO1SW=1 enables the parameters COVS, COVD, CGSP, and CGDP to be in F per n, per gate-finger, per unit channel width.
COREMOD	-	0	0	1	Simplified surface potential solution; 0=Turn off, 1= Turn on (lightly-doped or undoped)

Parameter	Unit	Default	Min	Max	Description
GEOMOD	-	1	0	3	Structure selector; 0=double gate, 1=triple gate, 2=quadruple gate, 3=cylindrical gate
MOBMOD	-	0	0	1	Mobility model selector 0 = BSIM-based, 1 = PSP-based
RDSMOD	-	0	0	2	Bias-dependent source/drain resistance model selector (controls s_{int} and d_{int} nodes); 0 = internal, 1 = external
IGCMOD	-	0	0	1	Model selector for I_{gc} , I_{gs} , and I_{gd} ; 1=turn on, 0=turn off
IGBMOD	-	0	0	1	Model selector for I_{gb} ; 1=turn on, 0=turn off
GIDLMOD	-	0	0		GIDL/GISL current switcher; 1=on, 0=off
IIMOD	-	0			Impact ionization model switch; 0 = OFF, 1 =BSIM4 based, 2 = BSIMSOI based
NQSMOD	-	0	0	1	NQS gate resistor and g_i node switcher; 1=on, 0=off
SHMOD	-	0	0	1	Self-heating and T node switcher; 1=on, 0=off
RGATEMOD	-	0		1	Gate electrode resistor and g_e node switcher; 1=on, 0=off
RGEOMOD	-	0	0	1	Bias independent parasitic resistance model selector
CGEOMOD	-	0	0	2	Parasitic capacitance model selector; 1=on, 0=off
CAPMOD	-	0	0	1	accumulation region capacitance model selector; 0=no accumulation capacitance, 1=accumulation capacitance included
BG0SUB	eV	1.12	-	-	Band gap of the channel material at 300.15K
DELTAW	m	0.0	-	-	Reduction of effective width due to shape of fin

Parameter	Unit	Default	Min	Max	Description
DELTAWCV	m	0.0	-	-	CV reduction of effective width due to shape of fin
DLC	m	0.0	-	-	Length reduction parameter for CV (dopant diffusion effect)
DLCACC	m	0.0	-	-	Length reduction parameter for CV in accumulation region (BULKMOD = 1; CAPMOD = 1)
EASUB	eV	4.05	0	-	Electron affinity of the substrate material
EOT	m	1.0n	0.1n	-	SiO_2 equivalent gate dielectric thickness (including inversion layer thickness)
EOTBOX	m	140n	1n	-	SiO_2 equivalent buried oxide thickness
EPSROX	-	3.9	1	0	Relative dielectric constant of the gate insulator
EPSRSUB	-	11.9	1	-	Relative dielectric constant of the channel material
FECH	-	1.0	0	-	End-channel factor, for different orientation/shape; mobility difference between the side channel and the top channel is handled by this parameter.
FECHCV	-	1.0	0	-	CV end-channel factor, for different orientation/shape; (Mobility difference between the side channel and the top channel is handled by this parameter)
HFIN	m	30n	1n	-	Fin height; instance parameter, also
LINT	m	0.0	-	-	Length reduction parameter (dopant diffusion effect)
LL	$m^{(LLN+1)}$	0.0	-	-	Length reduction parameter (dopant diffusion effect)
LLC	$m^{(LLN+1)}$	0.0	-	-	Length reduction parameter for CV (dopant diffusion effect)
LLN	-	1.0	-	-	Length reduction parameter (dopant diffusion effect)

Parameter	Unit	Default	Min	Max	Description
NBODY ^(b)	m ⁻³	1e22	1e18	5e24	Channel doping concentration; model currently supports NBODY < 5 x10 ²⁴ m ⁻³ ; for larger values use BSIM4 or BSIM4SOI
NCOSUB	m ⁻³	2.86e25	-	-	Conduction band density of states at 300.15K
NGATE ^(b)	m ⁻³	0	-	-	Parameter for Poly Gate doping; =0 for metal gate;
NIOSUB	m ⁻³	1.1e16	-	-	Intrinsic carrier concentration of channel at 300.15K
NSD	m ⁻³	2e26	2e25	1e27	S/D doping concentration
PHIG ^(b)	eV	4.61	0	-	Work function of gate
TOXP	m	1.2n	0.1n		Physical oxide thickness
XL	m	0	-	-	L offset for channel length due to mask/etch effect
Imin	A/m ²	1E-15	-	-	Parameter for voltage clamping for inversion region calculation in accumulation.
SH_WARN	-	0	0	1	Warning on self-heating network being disabled, 0 = Warning turned off, 1= Warning turned on
IGCLAMP	-	1	0	1	Igs/Igd clamp selector, default value 1, 0 = Igs/Igd clamps turned off, 1= Igs/Igd clamps turned on
PHIGLT	/m	0	-	-	Coupled NFIN and Length dependence of gate work function
ACH_UFCM	m ²	1	>0	inf	Area of the channel for the unified model
CINS_UFCM	F/m	1	>0	inf	Insulator capacitance for the unified model
W_UFCM	m	1	>0	inf	Effective channel width for the unified model
TFIN_TOP	m	1.50e-08	0	inf	Top body (fin) thickness for trapezoidal triple gate

Parameter	Unit	Default	Min	Max	Description
TFIN_BASE	m	1.50e-08	1.00e-09	inf	Base body (fin) thickness for trapezoidal triple gate

BSIM-CMG Model (Version>=105) — Basic Model Parameters

Note:

Binnable parameters are marked with ^(b).

Name	Unit	Default	Min	Max	Description
$A1^{(b)}$	V^{-2}	0.0	-	-	Non-saturation effect parameter in strong inversion region
$A2^{(b)}$	V^{-1}	0.0	-	-	Non-saturation effect parameter in moderate inversion region
$AGIDL^{(b)}$	Ω^{-1}	6.055e-12	-	-	Pre-exponential coefficient for GIDL
$AGISL^{(b)}$	Ω^{-1}	AIGDL	-	-	Pre-exponential coefficient for GISL
$AIGBACC^{(b)}$	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e-2	-	-	Parameter for Igb in accumulation
$AIGBINV^{(b)}$	$(Fs^2 / g)^{0.5} m^{-1}$	1.11e-2	-	-	Parameter for Igb in inversion
$AIGC^{(b)}$	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e-2	-	-	Parameter for Igc in inversion
$AIGD^{(b)}$	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e2	-	-	Parameter for Igd in inversion
$AIGEN^{(b)}$	$m^{-3} V^{-1}$	0.0	-	-	Parameter for R/G current (Experimental)
$AIGS^{(b)}$	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e2	-	-	Parameter for Igs in inversion
$ALPHA0^{(b)}$	$m \cdot V^{-1}$	0.0	-	-	First parameter of Iii (IIMOD=1)
$ALPHA1^{(b)}$	V^{-1}	0.0	-	-	L scaling parameter of Iii (IIMOD=1)
$ALPHAII^{(b)}$	-	0.0	-	-	Pre-exponential constant for Iii (IIMOD=2)

Name	Unit	Default	Min	Max	Description
BETA0 ^(b)	V ⁻¹	0.0	-	-	Vds dependent parameter of Iii (IIMOD=1)
BETAI0 ^(b)	V ⁻¹	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
BETAI1 ^(b)	-	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
BETAI2 ^(b)	V	0.1	-	-	Vds dependent parameter of Iii (IIMOD=2)
BGIDL ^(b)	V/m	0.3e9	-	-	Exponential coefficient for GIDL
BGISL ^(b)	V/m	BGIDL			Exponential coefficient for GISL
BIGBACC ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igb in accumulation
BIGBINV ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	9.49e-4	-	-	Parameter for Igb in inversion
BIGC ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igc in inversion
BIGS ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igs in inversion
BIGD ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	BIGS	-	-	Parameter for Igd in inversion
BIGEN ^(b)	m ⁻³ V ⁻³	0.0		-	Parameter for R/G current (Experimental)
BVD	V	BVS			Drain diode breakdown voltage
BVS	V	10.0			Source diode breakdown voltage
CDSC ^(b)	F/m ²	7e-3	0.0	-	Coupling capacitance between S/D and channel
CDSCD ^(b)	F/m ²	7e-3	0.0	-	Drain-bias sensitivity of CDSC
CDSCDN1	-	0	-0.08	-	NFIN dependence of CDSCD
CDSCDN2	-	1e5	1e-5	-	NFIN dependence of CDSCD
CDSCDRN1	-	CDSCDN1	-0.08	-	NFIN dependence of CDSCDR

Name	Unit	Default	Min	Max	Description
CDSCDRN2	-	CDSCDN2	1e-5	-	NFIN dependence of CDSCDR
CDSCN1	-	0	-0.08	-	NFIN dependence of CDSC
CDSCN2	-	1e5	1e-5	-	NFIN dependence of CDSC
CFD ^(b)	F/m	CFS	0.0	-	Drain-side outer fringe cap (for CGEOMOD = 0)
CFS ^(b)	F/m	2.5e-11	0.0	-	Source-side outer fringe cap (for parasitic capacitance calculation)
CGBN	F/m	0	0.0	-	Gate-substrate overlap capacitance per unit channel length per finger per fin
CGBO	F/m	0	0.0	-	Gate-substrate overlap capacitance per unit channel length per finger per gate contact
CGDL ^(b)	F/m	CGSL	0.0	-	Overlap capacitance between gate and lightly-doped drain region (for CGEOMOD = 0, 2)
CGDO	F/m	calculated	0.0	-	Non LDD region drain-gate overlap capacitance per unit channel width (for CGEOMOD = 0, 2)
CGIDL ^(b)	V ³	0.2	-	-	Parameter for body bias effect of GIDL
CGSL ^(b)	F/m	0	0.0	-	Overlap capacitance between gate and lightly-doped source region (for CGEOMOD = 0, 2)
CGSO	F/m	calculated	0.0	-	Non LDD region source-gate overlap capacitance per unit channel width (for CGEOMOD = 0, 2)
CHARGEWF	-	0	-1	1	Average channel charge weighting (sampling) factor, • +1: source-side • ,0: middle, - • 1: drain-side

Name	Unit	Default	Min	Max	Description
CIGBACC ^(b)	V ⁻¹	7.5e-2	-	-	Parameter for Igb in accumulation
CIGBINV ^(b)	V ⁻¹	6.00e-3	-	-	Parameter for Igb in inversion (BULKMOD only)
CIGC ^(b)	V ⁻¹	0.075	-	-	Parameter for Igc in inversion
CIGD ^(b)	V ⁻¹	CIGD	-	-	Parameter for Igd in inversion
CIGS ^(b)	V ⁻¹	0.075	-	-	Parameter for Igs in inversion
CIT ^(b)	F/m ²	0.0	-	-	Parameter for interface trap
CJD	F/m ²	CJS	0.0	-	Unit area drain-side junction capacitance at zero bias
CJS	F/m ²	0.0005	0.0	-	Unit area source-side junction capacitance at zero bias
CJSWD	F/m	CJSWS	0.0	-	Unit length sidewall junction capacitance at zero bias (drain-side)
CJSWS	F/m	5.0e-10	0.0	-	Unit length sidewall junction capacitance at zero bias (source-side)
CJSWGD	F/m	CJSWS1	0.0	-	Unit length gate sidewall junction capacitance at zero bias (drain-side)
CJSWGS	F/m	0.0	0.0	-	Unit length gate sidewall junction capacitance at zero bias (source-side)
CKAPPAD ^(b)	V	CKAPPAS	0.02	-	Coefficient of bias-dependent overlap capacitance for the drain side (for CGEOMOD = 0, 2)
CKAPPAS ^(b)	V	0.6	0.02	-	Coefficient of bias-dependent overlap capacitance for source side (for CGEOMOD = 0, 2)
COVD ^(b)	F/m	COVS	0.0	-	Constant gate-to-drain overlap cap (for CGEOMOD = 1)

Name	Unit	Default	Min	Max	Description
COVS ^(b)	F/m	0	0.0	-	Constant gate-to-source overlap cap (for CGEOMOD = 1)
CSDSW	F/m	0	0.0	-	Source/drain sidewall fringing capacitance per unit length
DELTA ^(b) VSAT	-	1.0	0.01	-	Velocity saturation parameter in the linear region
DELVFBACC	V	0.0	-	-	Additional V_{fb} shift required for accumulation region
DLCIGD	m	DLCIGS	-	-	Delta L for Igd model
DLCIGS	m	0	-	-	Delta L for Igs model
DROUT ^(b)		1.06	>0.0	-	L dependence of DIBL effect on Rout
DSUB ^(b)	-	1.06	>0	-	DIBL exponent coefficient
DVT0 ^(b)	-	0.0	0.0	-	SCE coefficient
DVT1 ^(b)	-	0.60	>0	-	SCE exponent coefficient
DVTP0	-	0	-	-	Coefficient for Drain-Induced Vth Shift (DITS)
DVTP1	-	0	-	-	DITS exponent coefficient
DVTSHIFT	V	0.0	0.0	-	Additional Vth shift handle
DVT1SS ^(b)	-	DVT1	>0	-	Subthreshold Swing exponent coefficient
EF	-	1.0	>0.0	2.0	Flicker noise frequency exponent
EGIDL ^(b)	V	0.2	-	-	Band bending parameter for GIDL
EGISL ^(b)	V	EGIDL	-	-	Band bending parameter for GISL
EIGBINV ^(b)	V	1.1	-	-	Parameter for Igb in inversion (BULKMOD only)
EM	V/m	4.1e7	-	-	Flicker noise parameter

Name	Unit	Default	Min	Max	Description
EOTACC	m	EOT	0.1n	-	S_iO_2 equivalent gate dielectric thickness for accumulation region
ESATII ^(b)	V/m	1.0e7	-	-	Saturation channel E-Field for Iii (IIMOD=2)
ETA0 ^(b)	-	0.60	0.0	-	DIBL coefficient
ETAMOB ^(b)	-	2.0	-	-	Effective field parameter
ETAQM	-	0.54	-	-	Body-charge coefficient for QM charge centroid
EU ^(b)	cm/MV	0.3	>0.0	-	Phonon / surface roughness scattering parameter
IJTHSREV	A	0.1	10 I_{sbs}	-	Reverse source diode breakdown limiting current
IJTHDFWD	A	IJTHSFWD	10 I_{sbs}	-	Forward drain diode breakdown limiting current
IJTHSFWD	A	0.1	10 I_{sbs}	-	Forward source diode breakdown limiting current
IJTHDREV	A	IJTHSREV			Reverse drain diode breakdown limiting current
JSD	A/m ²	JSS	0.0	-	Bottom drain junction reverse saturation current density
JSS	A/m ²	1.0e-4	0.0	-	Bottom source junction reverse saturation current density
JSWD	A/m	JSWS	0.0	-	Unit length reverse saturation current for isolation-edge sidewall drain junction
JSWGD	A/m	JSWGS	0.0	-	Unit length reverse saturation current for isolation-edge sidewall drain junction
JSWGS	A/m	0	0.0	-	Unit length reverse saturation current for isolation-edge sidewall source junction
JSWS	A/m	0	0.0	-	Unit length reverse saturation current for isolation-edge sidewall source junction

Name	Unit	Default	Min	Max	Description
JTSD	A/m^2	JTSS	0.0	-	Bottom drain junction trap-assisted saturation current density
JTSS	A/m^2	1.0e-4	0.0	-	Bottom source junction trap-assisted saturation current density
K0(b)	V	0.0	-		Lateral NUD parameter
K0SI(b)	-	1.0	>0	-	Correction factor for strong inversion/gm
K1 ^(b)	$V^{1/2}$	0.0	-	-	Body-effect coefficient for subthreshold region (before BSIM-CMG 109)
K1 ^(b)	$V^{1/2}$	1.0e-6	1e-6	-	Body-effect coefficient for subthreshold region in BSIM-CMG 109/110
K1SAT	$V^{-1/2}$	0.0	0.0	-	Body-effect coefficient for saturation region
K1RSCE ^(b)	$V^{1/2}$	0.0	-	-	Pre-factor for reverse short channel effect
KSATIV ^(b)	-	1.0	-	-	Parameter for long channel Vdsat
LII ^(b)	V -m	0.5e-9	-	-	Channel length dependent parameter of lII (IIMOD=2)
LINTIGEN	m	0.0	-	$L_{eff}/2$	L_{int} offset for R/G current
LINTNOI	m	0.0	-	$L_{eff}/2$	L_{int} offset for flicker noise calculation
LPA	-	1.0	-	-	Mobility L power coefficient
LPE0 ^(b)	m	5e-9	$-L_{eff}$	-	Equivalent length of pocket region zero bias
MEXP ^(b)	-	4	2	-	Smoothing function factor for Vdsat
MJD	-	MJS	-	-	Drain bottom junction capacitance grading coefficient

Name	Unit	Default	Min	Max	Description
MJS	-	0.5	-	-	Source bottom junction capacitance grading coefficient
MJSWGD	-	MJSWGS	-	-	Gate-edge sidewall junction capacitance grading coefficient (drain-side)
MJSWGS	-	MJSWS	-	-	Gate-edge sidewall junction capacitance grading coefficient (source-side)
MJSWS	-	0.33			Isolation-edge sidewall junction capacitance grading coefficient (source-side)
MJSWD	-	MJSWS	-	-	Isolation-edge sidewall junction capacitance grading coefficient (drain-side)
NBODYN1	-	0	-0.08	-	NFIN dependence of NBODY
NBODYN2	-	1e5	1e-5	-	NFIN dependence of NBODY
NIGBACC ^(b)	-	1.0	>0.0	-	Parameter for Igb in accumulation
NIGBINV ^(b)	-	3.0	> 0.0	-	Parameter for Igb in inversion (BULKMOD only)
NIGC ^(b)	-	1.0	>0	-	Parameter for Igc in inversion
NJS	-	1.0	0.0	-	Source junction emission coefficient
NJD	-	NJS	0.0	-	Drain junction emission coefficient
NJTS	-	20	0.0	-	Non-ideality factor for JTSS
NOIA	$\text{eV}^{-1} \text{s}^{1-\text{EF}} \text{m}^{-3}$	6.250e39	-	-	Flicker noise parameter
NOIB	$\text{eV}^{-1} \text{s}^{1-\text{EF}} \text{m}^{-1}$	3.125e24	-	-	Flicker noise parameter
NOIC	$\text{eV}^{-1} \text{s}^{1-\text{EF}} \text{m}$	8.750e7	-	-	Flicker noise parameter
NSEG	-	5	4	10	Number of channel segments for NQSMOD=3
NTGEN ^(b)	-	1.0	>0.0	-	Parameter for R/G current (Experimental)

Name	Unit	Default	Min	Max	Description
NTNOI	-	1.0	0.0	-	Thermal noise parameter
NTOX ^(b)	-	1.0	-	-	Exponent for gate oxide ratio
NVTM	V	nkT/q	-	-	If provided NVTM overrides nkT/q calculated in the model
PBD	V	PBS	0.01	-	Bottom junction built-in potential (drain-side)
PBS	V	1.0	0.01	-	Bottom junction built-in potential (source-side)
PBSWD	V	PBSWS	0.0	-	Isolation-edge sidewall junction built-in potential (drain-side)
PBSWGD	V	PBSWGS	0.01	-	Gate-edge sidewall junction built-in potential (drain-side)
PBSWGS	V	PBSWS	0.01	-	Gate-edge sidewall junction built-in potential (source-side)
PBSWS	V	1.0	0.01	-	Isolation-edge sidewall junction built-in potential (source-side)
PCLM ^(b)	-	0.013	-	-	Channel length modulation (CLM) parameter
PCLMCV ^(b)	-	0.013	>0.0	-	Channel length modulation (CLM) parameter for C-V
PCLMG ^(b)	-	0	-	-	Gate bias dependent parameter for channel length modulation (CLM)
PDIBL1 ^(b)	-	1.30	0.0	-	Parameter for DIBL effect on Rout in forward mode
PDIBL1R ^(b)	-	2e-4	0.0	-	Parameter for DIBL effect on Rout in reverse mode
PDIBL2 ^(b)	-	2e-4	0.0	-	Parameter for DIBL effect on Rout
PGIDL ^(b)	-	1.0	-	-	Exponent of electric field for GIDL
PGISL ^(b)	-	1.0	-	-	Exponent of electric field for GISL

Name	Unit	Default	Min	Max	Description
PHIBE ^(b)	V	0.7	0.2	1.2	Body-effect voltage parameter
PHIGL	eV=m	0	-	-	Length dependence of gate work function
PHIGN1	-	0	-0.08	-	NFIN dependence of PHIG
PHIGN2	-	1e5	1e-5	-	NFIN dependence of PHIG
PHIN ^(b)	V	0.05	-	-	Nonuniform vertical doping effect on surface potential
PIGCD ^(b)	-	1.0	>0.0	-	Vds dependence of I_{gcs} and I_{gcd}
POXEDGE ^(b)	-	1	>0.0	-	Factor for the gate edge T_{ox}
PQM	-	0.66	-	-	Fitting parameter for QM charge centroid (inversion)
PQMACC	-	0.66	-	-	Fitting parameter for QM charge centroid (accumulation)
PRWGS ^(b)	V^{-1}	0.0	0.0	-	Source side quasi-saturation parameter
PRWGD ^(b)	V^{-1}	PRWGS	0.0	-	Drain side quasi-saturation parameter
PSAT ^(b)	-	2.0	2.0	-	Exponent for field for velocity saturation
PTWG ^(b)	v^{-2}	0.0	-	-	Correction factor for velocity saturation in forward mode
PTWGR ^(b)	v^{-2}	0.0	-	-	Correction factor for velocity saturation in reverse mode
PVAG ^(b)	-	1.0	-	-	Vgs dependence on early voltage
QM0	V	1e-3	>0	-	Normalization parameter for QM charge centroid (inversion)
QM0ACC	V	1e-3	>0	-	Normalization parameter for QM charge centroid (accumulation)

Name	Unit	Default	Min	Max	Description
QMFACTOR ^(b)	-	0.0	-	-	Pre-factor for QM V_{th} shift correction
QMTCENCV ^(b)	-	0.0	-	-	Pre-factor/switch for QM effective width and oxide thickness correction for CV
QMTCENCVA ^(b)	-	0.0	-	-	Pre-factor/switch for QM effective width and oxide thickness correction for accumulation region CV
QMTCENIV ^(b)	-	0.0	-	-	Pre-factor/switch for QM effective width correction for IV
RDSWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	-	-	RDSMOD = 0 S/D extension resistance per unit width at high V_{gs}
RDSW ^(b)	$\Omega \cdot \mu_m^{WR}$	40	0.0	-	RDSMOD = 0 zero bias S/D extension resistance per unit width
RDWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	0.0	-	RDSMOD = 1 drain extension resistance per unit width at high V_{gs}
RDW ^(b)	$\Omega \cdot \mu_m^{WR}$	50	0.0	-	RDSMOD = 1 zero bias drain extension resistance per unit width
RSW ^(b)	$\Omega \cdot \mu_m^{WR}$	50	0.0	-	RDSMOD = 1 zero bias source extension resistance per unit width
RSWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	0.0	-	RDSMOD = 1 source extension resistance per unit width at high V_{gs}
RSDR	V^{-PRSDR}	0.0	0.0	-	RDSMOD = 1 source side drift resistance parameter in forward mode
RSDRR	V^{-PRSDR}	RSDR	0.0	-	RDSMOD = 1 drain side drift resistance parameter in reverse mode
PRSDR	-	1.0	0.0	-	RDSMOD = 1 drain side drift resistance parameter in forward mode

Name	Unit	Default	Min	Max	Description
PRDDR	-	PRSDR	-	-	RDSMOD = 1 drain side drift resistance parameter in reverse mode
RGFIN	Ω	1.0e-3	1.0e-3	-	Effective gate electrode resistance per fin per finger
RGEXT	Ω	0.0	0.0	-	Effective gate electrode external resistance (Experimental)
RNOIA	-	0.577	-	-	Thermal noise parameter
RNOIB	-	0.37	-	-	Thermal noise parameter
RSHD	Ω	RSHS	0.0	-	Drain-side sheet resistance
RSHS	Ω	0.0	0.0	-	Source-side sheet resistance
SIIO ^(b)	V^{-1}	0.5	-	-	Vgs dependent parameter of Iii (IIMOD=2)
SI11 ^(b)	-	0.1	-	-	Vgs dependent parameter of Iii (IIMOD=2)
SI12 ^(b)	V	0.0	-	-	Vgs dependent parameter of Iii (IIMOD=2)
SIID ^(b)	V	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
TETA0	1/K	0.0	-	-	Temperature dependence of DIBL coefficient
TETA0R	1/K	0.0	-	-	Temperature dependence of Reverse-mode DIBL coefficient
THETADIBL	-	θ_{DIBL}	-	-	If provided THETADIBL overrides θ_{DIBL} calculated in the model
THETASCE	-	θ_{SCE}	-	-	If provided THETADIBL overrides θ_{SCE} calculated in the model
THETASW	-	θ_{SW}	-	-	If provided THETADIBL overrides θ_{SW} calculated in the model
TNOIA	m^{-1}	1.5	0.0	-	Thermal noise parameter

Name	Unit	Default	Min	Max	Description
TNOIB	m^{-1}	3.5	0.0	-	Thermal noise parameter
TOXREF	m	1.2nm	> 0.0	-	Nominal gate oxide thickness for gate tunneling current
U0 ^(b)	$m^2/V \cdot s$	3e-2	-	-	Low field mobility
U0N1	-	0	-0.08	-	NFIN dependence of U0
U0N2	-	1e5	1e-5	-	NFIN dependence of U0
UA ^(b)	$(cm/MV)^{EU}$	0.3	>0.0	-	Phonon / surface roughness scattering parameter
UCS ^(b)	-	1.0	>0.0	-	Coulombic scattering parameter (Experimental)
UD ^(b)	cm/MV	0.0	>0.0	-	Coulombic scattering parameter (Experimental)
UP ^(b)	μm^{LPA}	0.0	-	-	Mobility L coefficient
VSAT ^(b)	m/s	85000	-	-	Saturation velocity for the saturation region
VSAT1 ^(b)	m/s	VSAT	-	-	Saturation velocity for the linear region in forward mode
VSAT1N1	-	0	-0.08	-	NFIN dependence of VSAT1
VSAT1N2	-	1e5	1e-5	-	NFIN dependence of VSAT1
VSATN1	-	0	-0.08	-	NFIN dependence of VSAT
VSATN2	-	1e5	1e-5	-	NFIN dependence of VSAT
VSAT1R ^(b)	m/s	VSAT1	-	-	Saturation velocity for the linear region in reverse mode
VSAT1RN1	-	VSAT1N1	-0.08	-	NFIN dependence of VSAT1R
VSAT1RN2	-	VSAT1N2	1e-5	-	NFIN dependence of VSAT1R
VSATCV ^(b)	m/s	VSAT	-	-	Saturation velocity for the capacitance model
WR ^(b)	-	1.0	-	-	W dependence parameter of S/D extension resistance

Name	Unit	Default	Min	Max	Description
XJBVD	-	XJBVS	-	-	Fitting parameter for source diode breakdown current
XJBVS	-	1.0	-	-	Fitting parameter for source diode breakdown current
XRCRG1 ^(b)		12.0	0.0 or ≥ 1 0 ⁻³	-	Parameter for non quasi-static gate resistance (NQSMOD=1) and NQSMOD=2
XRCRG2 ^(b)	-	1.0	-	-	Parameter for non quasi-static gate resistance (NQSMOD=1) and NQSMOD=2
ALPHAII0(b)	m.V ⁻¹	0.0	-	-	First parameter of Iii (IIMOD=2)
ALPHAII1(b)	V ⁻¹	0.0	-	-	L scaling parameter of Iii (IIMOD=2)
TOXG	m	TOXP	>0.0	-	Oxide thickness for gate current model.
K1SI(b)	-	K0SI	>0	-	Correction factor for strong inversion used in M _{ob}
PSATCV ^(b)	-	PSAT	2.0	-	Exponent for field for velocity saturation for the capacitance model.
DELTA _V SATC ^(b)	-	DELTA _V SAT	0.01	-	Velocity saturation parameter in the linear region for the capacitance model.
ETA0LT	/m	0	-	-	Coupled NFIN and length dependence of ETA0
ADVTP0	-	0	-	-	Pre-exponential coefficient for DITS
BDVTP0	-	1.00e-07	-	-	Exponential coefficient for DITS (BDVTP0 not equal to 0)
ADVTP1	-	0	-	-	Pre-exponential coefficient for DVTP1

Name	Unit	Default	Min	Max	Description
BDVTP1	-	1.00e-07	-	-	Exponential coefficient for DVTP1 (BDVTP1 not equal to 0)
DVTP2	-	0	-	-	DITS model parameter
K2SI	-	K0SI	-	-	Correction factor for strong inversion used in Mob
K0SISAT	-	0	-	-	Correction factor for strong inversion used in Mnud
K2SISAT	-	K0SISAT	-	-	Correction factor for strong inversion used in Mob
K2SAT	-	0	-	-	Correction factor for K2 in saturation (high Vds)
K2	-	0	-	-	Body effect coefficient for BULKMOD==2
U0LT	/m	0	-	-	Coupled NFIN and length dependence of U0
VFBS	V	0			Flat band voltage for S/D region
VFBSDCV	V	VFBS			Flat band voltage for S/D region for C-V calculations
QMFACTO RCV	-	0	0	inf	Charge dependence taking QM effects into account
ALPHA_U FCM	-	0.5556	0	inf	Mobile charge scaling term taking QM effects into account
IIMOD2CLA MP1	V	0.1	>0	inf	Clamp1 of SII1*Vg term in IIMOD=2 model
IIMOD2CLA MP2	V	0.1	>0	inf	Clamp2 of SII0*Vg term in IIMOD=2 model
IIMOD2CLA MP3	V	0.1	>0	inf	Clamp3 of ratio term in IIMOD=2 model
DVSATCL AMP	m	0.01	0.01	-	Minimum clamp on Dvsat
DMOBCLAMP	m	0.01	0.01	-	Minimum clamp on Dmob

Name	Unit	Default	Min	Max	Description
TNOIK2	-	0.1	-	-	Empirical parameter for sensitivity of RNOIK

BSIM-CMG Model (Version>=105) — Geometry Dependent Parasitic Parameters

Name	Unit	Default	Min	Max	Description
ARSDEND	m ²	0	0	-	Extra raised source/drain cross sectional area at the two ends of the FinFET
ASILIEND	m ²	0	0	-	Extra silicide cross sectional area at the two ends of the FinFET
CGEOA	-	1.0	-	-	Fitting parameter for CGEOMOD=2
CGEOB	m ⁻¹	0	-	-	Fitting parameter for CGEOMOD=2
CGEOC	m ⁻¹	0	-	-	Fitting parameter for CGEOMOD=2
CGEOD	m ⁻¹	0	-	-	Fitting parameter for CGEOMOD=2
CGEOE	-	1.0	-	-	Fitting parameter for CGEOMOD=2
CRATIO	-	0.5	0	1	Ratio of the corner area filled with silicon to the total corner area
DELTAP RSD	m	0.0	FPI TCH	-	Change in silicon/silicide interface length due to non-rectangular epi
EPSRSP	-	3.9	1	-	Relative dielectric constant of the gate sidewall spacer material
HEPI	m	10n	-	-	Height of the raised source/drain on top of the fin
LDG	m	5n	0	-	Lateral diffusion gradient in the fin extension region
LSP	m	0.2(L+XL)	0	-	Thickness of the gate sidewall spacer
NSDE	m ⁻³	2×10^{25}	10^{25}	10^{26}	Active doping concentration at the channel edge
PRSDEND	m	0	0	-	Extra silicon/silicide interface perimeter at the two ends of the FinFET

Name	Unit	Default	Min	Max	Description
RGEOA	-	1.0	-	-	Fitting parameter for RGEOMOD=1
RGEOB	m^{-1}	0	-	-	Fitting parameter for RGEOMOD=1
RGEOC	m^{-1}	0	-	-	Fitting parameter for RGEOMOD=1
RGEOD	m^{-1}	0	-	-	Fitting parameter for RGEOMOD=1
RGEOE	m^{-1}	0	-	-	Fitting parameter for RGEOMOD=1
RHOC	$\Omega\text{-}m^2$	10^{-18}	10^{-9}		Contact resistivity at the silicon/silicide interface
RHOEXT	$\Omega\text{-}m$	RHORSD	0	-	Average resistivity of silicon in the fin extension region
RHORSD	$\Omega\text{-}m$	calculated	0	-	Average resistivity of silicon in the raised source/drain region
SDTERM	-	0	0	1	Indicator of whether the source/drain are terminated with silicide
TGATE	m	30n	0	-	Gate height on top of the hard mask
TMASK	m	30n	0	-	Height of the hard mask on top of the fin
TSILI	m	10n	-	-	Thickness of the silicide on top of the raised source/drain

BSIM-CMG Model (Version>=105) — Temperature Dependent and Self-Heating Parameters

Name	Unit	Default	Min	Max	Description
AT ^(b)	1 / K	0.005	-	-	Saturation velocity temperature coefficient
ATCV ^(b)	1/K	AT	-	-	Saturation velocity temperature coefficient for C-V
CTH0	J/K	1.0e-5	0.0	-	Thermal capacitance for self-heating calculation
EMOBT ^(b)	-	0.0	-	-	Temperature Coefficient of ETAMOB

Name	Unit	Default	Min	Max	Description
IGT ^(b)	-	2.5	-	-	Gate current temperature coefficient
IIT ^(b)	-	-0.5	-	-	Impact ionization temperature coefficient(IIMOD=1)
KT1	V	0.0	-	-	V _{th} temperature coefficient
KT1L	V · m	0.0	-	-	V _{th} temperature coefficient
PRT ^(b)	1 / K	0.001	-	-	Series resistance temperature coefficient
PTWGT ^(b)	1 / K	0.004	-	-	PTWG temperature coefficient
RTH0	K/W	0.01	0.0	-	Thermal resistance for self-heating calculation
TBGASUB	(eV) / (°K)	7.02e-4	-	-	Bandgap temperature coefficient
TBGBSUB	°K	1108.0	-	-	Bandgap temperature coefficient
TCJ	1 / K	0.0	-	-	Temperature coefficient for CJS/CJD
TCJSW	1 / K	0.0	-	-	Temperature coefficient for CJSWS1/CJDWD1
TCJSWG	1 / K	0.0	-	-	Temperature coefficient for CJSWGS/CJSWGD
TGIDL ^(b)	1 / K	-0.003	-	-	GISL/GIDL temperature coefficient
TI ^(b)	-	0.0	-	-	Impact ionization temperature coefficient(IIMOD=2)
TMEXP ^(b)	1 / K	0	-	-	Temperature coefficient for V _{dseff} smoothing
TNOM	°C	27	0.0	-	Temperature at which the model is extracted in Celsius
TPB	1 / K	0.0	-	-	Temperature coefficient for PBS/PBD
TPBSW	1 / K	0.0	-	-	Temperature coefficient for PBSWS/PBDWD
TBSWG	1 / K	0.0	-	-	Temperature coefficient for PBSWGS/PBSWGD

Name	Unit	Default	Min	Max	Description
TRSDR ^(b)	1 / K	0.0	-	-	Source side drift resistance temperature coefficient
TRDDR ^(b)	1 / K	0.0	-	-	Drain side drift resistance temperature coefficient
TSS ^(b)	1 / K	0.0	-	-	Subthreshold swing temperature coefficient
UA1 ^(b)	-	1.032e-3	-	-	Mobility temperature coefficient for UA
UC1 ^(b)	-	0.056e-9	-	-	Mobility temperature coefficient for UC
UCSTE ^(b)	-	-4.775e-3	-	-	Mobility temperature coefficient
UD1 ^(b)	-	0.0	-	-	Mobility (MOBMOD=0) temperature coefficient (Experimental)
UTE ^(b)	-	0.0	-	-	Mobility temperature coefficient
UTL ^(b)	-	-1.5e-3	-	-	Mobility temperature coefficient
WTH0	m	0.0	0.0	-	Width-dependence coefficient for self-heating calculation
XTIS	-	3.0	-	-	Source junction current temperature exponent
XTID	-	XTIS	-	-	Drain junction current temperature exponent
ALPHAII01 ^(b)	m.V ⁻¹ /K	0.0	-	-	Temperature dependence of ALPHAII0
ALPHAII11 ^(b)	V ⁻¹ /K	0.0	-	-	Temperature dependence of ALPHAII1
ALPHA01 ^(b)	m.V ⁻¹ /K	0.0	-	-	Temperature dependence of ALPHA0
ALPHA11 ^(b)	V ⁻¹ /K	0.0	-	-	Temperature dependence of ALPHA1
AIGBINV1 ^(b)	($\frac{E}{s^2/g}$) ^{0.5} m ⁻¹ /K	0.0	-	-	Temperature dependence of AIGBINV
AIGBACC1 ^(b)	($\frac{E}{s^2/g}$) ^{0.5} m ⁻¹ /K	0.0	-	-	Temperature dependence of AIGBACC

Name	Unit	Default	Min	Max	Description
AIGC1 ^(b)	$(\frac{E}{s^2/g})^{0.5} m^{-1}/K$	0.0	-	-	Temperature dependence of AIGC
AIGS1 ^(b)	$(\frac{E}{s^2/g})^{0.5} m^{-1}/K$	0.0	-	-	Temperature dependence of AIGS
AIGD1 ^(b)	$(\frac{E}{s^2/g})^{0.5} m^{-1}/K$	0.0	-	-	Temperature dependence of AIGD
A11 ^(b)	V^{-2}/K	0.0	-	-	Temperature dependence of non-saturation effect parameter for strong inversion region
A21 ^(b)	V^{-1}/K	0.0	-	-	Temperature dependence of non-saturation effect parameter for moderate inversion region
K01 ^(b)	V/K	0.0	-	-	Temperature dependence of K0
K0SI1 ^(b)	$1/K$	0.0	-	-	Temperature dependence of K0SI
K11 ^(b)	$V^{-1/2}/K$	0.0	-	-	Temperature dependence of K1
K1SI1 ^(b)	$1/K$	0.0	-	-	Temperature dependence of K1SI
K1SAT1 ^(b)	$V^{-1/2}/K$	0.0	-	-	Temperature dependence of K1SAT
ASHEXP	-	1	-	-	Exponent to tune RTH dependence of NFINTOTAL
BSHEXP	-	1	-	-	Exponent to tune RTH dependence of NF
K2SI1	-	K0SI1	-	-	Temperature dependence of K2SI
K0SISAT1	-	0	-	-	Temperature dependence of K0SISAT
K2SISAT1	-	K0SISAT1	-	-	Temperature dependence of K2SISAT
K2SAT1	-	0	-	-	Temperature dependence of K2SAT
K21	-	0	-	-	Temperature dependence of K2

BSIM-CMG Model (Version>=105) — Variability Modeling Parameters

Name	Unit	Default	Min	Max	Description
DTEMP	K	0.0	-	-	Device temperature shift handle
DELVTRAND	V	0.0	-	-	Threshold voltage shift handle
U0MULT	-	1.0	-	-	Multiplier to mobility (or more precisely divides D_{mob} , D_{mobs})
IDS0MULT	-	1.0	-	-	Multiplier to source-drain channel current
TFIN ⁽ⁱ⁾	m	15n	1n		Body (fin) thickness
FPITCH ⁽ⁱ⁾	m	80n	TFIN	-	Fin Pitch
XL ^(mod)	m	0	-	-	L offset for channel length due to mask/etch effect
NBODY ^(mod)	m ⁻³	1e22	1e18	5e24	Channel (body) doping concentration
EOT ^(mod)	m	1.0n	0.1n	-	SiO2 equivalent gate dielectric thickness (including inversion layer thickness)
TOXP ^(mod)	m	1.2n	0.1n	-	Physical oxide thickness
RSHS ^(mod)	Ω	0.0	0.0	-	Source-side sheet resistance
RSHD ^(mod)	Ω	RSHS	0.0	-	Drain-side sheet resistance
RHOC ^(mod)	$\Omega \cdot m^2$	1p	10 ⁻¹⁸	10 ⁻⁹	Contact resistivity at the silicon / silicide interface
RHORSD ^(mod)	$\Omega \cdot m$	Calculated	-	-	Average resistivity of silicon in the raised source/drain region
RHOEXT ^(mod)	$\Omega \cdot m$	RHORSD	-	-	Average resistivity of silicon in the fin extension region

BSIM-CMG Model Parameters (Version < 105)

This section covers the following topics:

- [BSIM-CMG Model \(Version<105\) — Instance Parameters](#)
- [BSIM-CMG Model \(Version<105\) — Control and Process Parameters](#)
- [BSIM-CMG Model \(Version<105\) — Basic Parameters](#)
- [BSIM-CMG Model \(Version<105\) — Temperature Dependence and Self-Heating Parameters](#)

BSIM-CMG Model (Version<105) — Instance Parameters

Note:

Binnable parameters are marked as ^(b). Instance parameters which are also model parameters are marked as ^(m).

Parameter	Unit	Default	Description
AD ^(m)	m ²	0	Drain area
AS ^(m)	m ²	0	Source area
CDSP ^(m)	F	0	Constant drain to source fringe capacitance
CGDP ^(m)	F	0	Constant gate to drain fringe capacitance
CGSP ^(m)	F	0	Constant gate to source fringe capacitance
D ^(m)	m	40e-9	Diameter of cylinder
L ^(m)	m	30e-9	Channel length
NF	-	1	Number of fingers
NFIN ^(m)	-	10	Number of fins per finger
NRD	-	0	Number of drain diffusion squares
NRS	-	0	Number of source diffusion squares
PD1 ^(m)	m	0	1st part of the drain perimeter
PD2 ^(m)	m	0	2nd part of the drain perimeter
PS1 ^(m)	m	0	1st part of the source perimeter
PS2 ^(m)	m	0	2nd part of the source perimeter

Parameter	Unit	Default	Description
RDC	Ω	0	Source contact resistance for per-finger device (after v.104)
RSC	Ω	0	Source contact resistance for per-finger device (after v.104)
TFIN ^(m)	m	15e-9	Fin (channel) thickness; currently supports TFIN < 50x ¹⁰⁻⁹ m; for larger values use BSIM4 or BSIM4SOI.

BSIM-CMG Model (Version<105) — Control and Process Parameters

Parameter	Unit	Default	Description
BULKMOD	-	1	BULK/SOI model selector (Version102 and later) In Verilog-A the number of terminals cannot be controlled by a model parameter. If you are running the Verilog-A version of this model, select the bulk or SOI module by running bulkmg.va or soimg.va
CGEOMOD	-	0	Geometry-dependent parasitic capacitance model selector; 1=on, 0=off
CLMMOD	-	0	Model selector for channel length modulation
COREMOD	-	0	Simplified surface potential solution; 0=off, 1= on (lightly-doped or undoped)
GEOMOD	-	1	Structure selector; 0=double gate, 1=triple gate, 2=quadruple gate, 3=cylindrical gate
GIDLMOD	-	0	GIDL/GISL current switch; 1=on, 0=off
IGMOD	-	0	Gate current switch; 1=on, 0=off
IIMOD	-	0	Impact ionization model switch; 0 = OFF, 1 =BSIM4 based, 2 = BSIMSOI based
MOBMOD	-	0	Mobility model selector 0 = BSIM-based, 1 = PSP-based
NQSMOD	-	0	NQS gate resistor switch; 1=on, 0=off
RDSMOD	-	0	Source/drain resistance selector; 0 = internal, 1 =external
RGATEMOD	-	0	Gate electrode resistor switcher; 1=on, 0=off
SHMOD	-	0	Self-heating switch; 1=on, 0=off

Parameter	Unit	Default	Description
DEVTYPE		NMOS	NMOS=1, PMOS=0
BG0SUB	eV	1.12	Band gap of the channel material at 300.15K
DELTAW	m	0.0	Reduction of effective width due to shape of fin
DELTAWCV	m	0.0	CV reduction of effective width due to shape of fin
DLBIN	m	0.0	Length reduction parameter for binning
DLC	m	0.0	Channel length reduction parameter for CV
DLCACC	m	0.0	Length reduction parameter for CV in accumulation region (BULKMOD =1;CAPMOD = 1)
EASUB	eV	4.05	Electron affinity of the substrate material
EOT	m	0.9e-9	Effective gate dielectric thickness relative to SiO2
EPSROX	-	3.9	Relative dielectric constant of the gate insulator
EPSRSUB	-	11.9	Relative dielectric constant of the channel material
FECH	-	1.0	End-channel factor, for different orientation/shape; this parameter handles the mobility difference between the side channel and the top channel
FECHCV	-	1.0	CV end-channel factor, for different orientation/shape
HFIN	m	35e-9	Fin height; instance parameter, also
LINT	m	0.0	Channel length reduction parameter
LL	$m^{(LLN+1)}$	0.0	Channel length reduction parameter
LLC	$m^{(LLN+1)}$	0.0	Channel length reduction parameter for CV
LLN	-	1.0	Channel length reduction parameter
NBODY ^(b)	m^{-3}	1e22	Channel doping concentration; model currently supports $NBODY < 5 \times 10^{24} m^{-3}$; for larger values use BSIM4 or BSIM4SOI
NC0SUB	m^{-3}	2.86e25	Conduction band density of states at 300.15K
NGATE ^(b)	m^{-3}	0	Parameter for Poly Gate doping; =0 for metal gate
NI0SUB	m^{-3}	1.1e16	Intrinsic carrier concentration of channel at 300.15K
PHIG ^(b)	eV	4.61	Work function of gate

Parameter	Unit	Default	Description
XL	m	-5e-9	L offset for channel length due to mask/etch e effect

BSIM-CMG Model (Version<105) — Basic Parameters

Name	Unit	Default	Min	Max	Description
A1(b)	V^{-1}	0.0	-	-	Non-saturation effect parameter in strong inversion region
A2(b)	V^{-2}	0.0	-	1	Non-saturation effect parameter in moderate inversion region
AGIDL ^(b)	Ω^{-1}	6.055	-	-	Pre-exponential coefficient for GIDL
AGISL ^(b)	Ω^{-1}	AIGDL	-	-	Pre-exponential coefficient for GISL
AIGBINV ^(b)	$(Fs^2 / g)^{0.5} m^{-1}$	1.11e-2	-	-	Parameter for Igb in inversion
AIGC ^(b)	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e-2	-	-	Parameter for Igc in inversion
AIGD ^(b)	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e2	-	-	Parameter for Igd in inversion
AIGS ^(b)	$(Fs^2 / g)^{0.5} m^{-1}$	1.36e2	-	-	Parameter for Igs in inversion
ALP ^(b)	-	0.013	-	-	CLMMOD = 0 Channel Length Modulation (CLM) pre-factor
ALPHA0 ^(b)	$m \cdot V^{-1}$	0.0	-	-	First parameter of Iii
ALPHA1 ^(b)	V^{-1}	0.0	-	-	L scaling parameter of Iii
ALPHAII ^(b)	-	0.0	-	-	Pre-exponential constant for Iii (IIMOD=2)
BETA0 ^(b)	V^{-1}	0.0	-	-	Vds dependent parameter of Iii
BETAII0 ^(b)	V^{-1}	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
BETAII1 ^(b)	-	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
BETAII2 ^(b)	V	0.0	-	-	Vds dependent parameter of Iii (IIMOD=2)
BGIDL ^(b)	V/m	0.3e9	-	-	Exponential coefficient for GIDL

Name	Unit	Default	Min	Max	Description
BGSL ^(b)	V/m	BIGDL			Band bending parameter for GISL
BIGBINV ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	9.49e-4	-	-	Parameter for Igb in inversion (BULKMOD only)
BIGC ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igc in inversion
BIGD ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igd in inversion
BIGS ^(b)	$(Fs^2 / g)^{0.5} m^{-1} V^{-1}$	1.71e-3	-	-	Parameter for Igs in inversion
BVD	V	BVS			Drain diode breakdown voltage
BVS	V	10.0			Source diode breakdown voltage
CDSC ^(b)	F/m ²	7e-3	0.0	-	Coupling capacitance between S/D and channel
CDSCD ^(b)	F/m ²	7e-3	0.0	-	Drain-bias sensitivity of CDSC
CDSCDR ^(b)	F/m ²	CDSCD	0.0		Reverse mode drain-bias sensitivity of CDSC
CFD ^(b)	F/m	CFS	0.0	-	Drain-side outer fringe cap (for parasitic capacitance calculation)
CFS ^(b)	F/m	2.5e-11	0.0	-	Source-side outer fringe cap (for parasitic capacitance calculation)
CGBN	F/m	0	0.0	-	Gate-substrate overlap capacitance per unit channel length per finger per fin
CIGBINV ^(b)	V ⁻¹	6.00e-3	-	-	Parameter for Igb in inversion (BULKMOD only)
CIGC ^(b)	V ⁻¹	0.075	-	-	Parameter for Igc in inversion
CIGD ^(b)	V ⁻¹	0.075	-	-	Parameter for Igd in inversion
CIGS ^(b)	V ⁻¹	0.075	-	-	Parameter for Igs in inversion
CIT ^(b)	F/m ²	0	-	-	Parameter for interface trap
CJD	F/m ²	CJS	0.0	-	Unit area drain-side junction capacitance at zero bias
CJS	F/m ²	0.003	0.0	-	Unit area source-side junction capacitance at zero bias

Name	Unit	Default	Min	Max	Description
CJSWD1	F/m	CJSWS1	0.0	-	1st unit length drain-side sidewall junction capacitance at zero bias
CJSWD2	F/m	CJSWS1	0.0	-	2nd unit length drain-side sidewall junction capacitance at zero bias
CJSWS1	F/m	3.0e-10	0.0	-	1st unit length source-side sidewall junction capacitance at zero bias
CJSWS2	F/m	3.0e-10	0.0	-	2nd unit length source-side sidewall junction capacitance at zero bias
COVD ^(b)	F	COVS	0.0	-	Constant gate-to-drain overlap cap (for parasitic capacitance calculation)
COVS ^(b)	F	2.5e-11	0.0	-	Constant gate-to-source overlap cap (for parasitic capacitance calculation)
CS ^(b)	-	0.0	>0.0	-	Coulombic scattering parameter
DLCIGD	m	DLCIGS	-	-	Delta L for Igd model
DLCIGS	m	0	-	-	Delta L for Igs model
DROUT		1.06	>0.0	-	0.56
DSUB ^(b)	-	1.06	>0	-	DIBL exponent coefficient
DVT0 ^(b)	-	0.0	0.0	-	SCE coefficient
DVT1 ^(b)	-	0.60	>0	-	SCE exponent coefficient
EGIDL ^(b)	V	0.2			Band bending parameter for GIDL
EGISL ^(b)	V	EGIDL	-	-	Pre-exponential coefficient for GISL
EIGBINV ^(b)	V	1.1	-	-	Parameter for Igb in inversion (BULKMOD only)
ESATII ^(b)	V/m	1.0e7	-	-	Saturation channel E-Field for Iii (IIMOD=2)
ETA0 ^(b)	-	0.60	0.0	-	DIBL coefficient
ETA0R ^(b)		ETA0	0.0		Reverse mode DIBL coefficient
ETAMOB ^(b)	-	2.0	-	-	Effective field parameter

Name	Unit	Default	Min	Max	Description
EU ^(b)	cm/MV	2.5	>0.0	-	MOBMOD = 0 phonon / surface roughness scattering
IJTHDFWD	A	IJTHS FWD	10 I _{sbs}	-	Forward drain diode breakdown limiting current
IJTHDREV	A	IJTHS REV			Reverse drain diode breakdown limiting current
IJTHSFWD	A	0.1	10 I _{sbs}	-	Forward source diode breakdown limiting current
IJTHSREV	A	0.1	10 I _{sbs}	-	Reverse source diode breakdown limiting current
JSD	A/m ²	JSS	0.0	-	Bottom drain junction reverse saturation current density
JSS	A/m ²	1.0e-4	0.0	-	Bottom source junction reverse saturation current density
JSWD1	A/m	JSWS1	0.0	0.0	Unit length reverse saturation current for 1st sidewall drain junction
JSWD2	A/m	JSWS2	0.0	0.0	Unit length reverse saturation current for 2nd sidewall drain junction
JSWS1	A/m	0	0.0	0.0	Unit length reverse saturation current for 1st sidewall source junction
JSWS2	A/m	JSWS1	0.0	0.0	Unit length reverse saturation current for 2nd sidewall source junction
K1RSCE ^(b)	V ^{1/2}	0.0	-	-	Pre-factor for reverse short channel effect
K0(b)	V	-	0.0	-	Lateral NUD parameter
K0SI(b)	-	1.0	>0	-	Correction factor for strong inversion/gm
K1(b)	V ^{1/2}	1.0	0.0	-	Body-effect coefficient for subthreshold region
K1SAT(b)	V ^{-1/2}	0.0	-	-	Body-effect coefficient for saturation region

Name	Unit	Default	Min	Max	Description
KSATIV ^(b)	-	1.3	-	-	Parameter for reverse short channel effect
LII ^(b)	V -m	0.5e-9	-	-	Channel length dependent parameter of lii (IIMOD=2)
LOVD ^(b)	m	LOVS	0.0	-	Gate-to-drain overlap length (for parasitic capacitance calculation)
LOVS ^(b)	m	1.0e-9	0.0	-	Gate-to-source overlap length (for parasitic capacitance calculation)
LPA	-	1.0	-	-	Mobility L power coefficient
LPE0 ^(b)	m	5e-9	-Leff	-	Equivalent length of pocket region zero bias
MEXP ^(b)	-	4	2	-	Smoothing function factor for Vdsat
MJD	-	MJS	-	-	Drain bottom junction capacitance grading coefficient
MJS	-	0.5	-	-	Source bottom junction capacitance grading coefficient
MJSWD1	-	MJSWS1	-	-	2nd drain sidewall junction capacitance grading coefficient
MJSWD2	-	MJSWS2	-	-	2nd drain sidewall junction capacitance grading coefficient
MJSWS1	-	0.33			1st source sidewall junction capacitance grading coefficient
MJSWS2	-	MJSWS1	-	-	2nd source sidewall junction capacitance grading coefficient
MUE ^(b)	cm/MV	1.2	>0.0	-	MOBMOD=1 Mobility reduction coefficient
NIGBINV ^(b)	-	3.0	> 0.0	-	Parameter for Igb in inversion (BULKMOD only)
NIGC ^(b)	-	1	>0	-	Parameter for Igc in inversion
NJD	-	NJS	0.0	-	Drain junction emission coefficient
NJS	-	1.0	0.0	-	Source junction emission coefficient
NSD ^(b)	m ⁻³	2e26	-	-	S/D doping concentration

Name	Unit	Default	Min	Max	Description
NTOX(b)	-	1.0	-	-	Nominal gate oxide thickness for Gate tunneling current
PBD	V	PBS	0.01	-	Unit area drain-side junction capacitance at zero bias
PBS	V	1.0	0.01	-	Unit area source-side junction capacitance at zero bias
PBSWD1	V	PBSWS1	0.0	-	Built-in potential for 1st drain-side sidewall junction capacitance
PBSWD2	V	PBSWS2	0.01	-	Built-in potential for 2nd drain-side sidewall junction capacitance
PBSWS1	V	1.0	0.01	-	Built-in potential for 1st source-side sidewall junction capacitance
PBSWS2	V	PBSWS1	0.01	-	Built-in potential for 2nd source-side sidewall junction capacitance
PCLM ^(b)	-	0.013	-	-	CLMMOD = 1 Channel Length Modulation parameter
PCLMG	-	0	-	-	CLMMOD = 1 Gate bias dependent parameter for channel Length Modulation (CLM)
PDIBL1(b)	-	1.30	0.0	-	Parameter for DIBL effect on Rout
PDIBL2(b)	-	2e-4	0.0	-	Parameter for DIBL effect on Rout
PDIBL1R(b)	-	PDIBL1	0.0	-	Parameter for DIBL effect on Rout in reverse mode
PGIDL(b)	-	1.0	-	-	Exponent of electric field for GIDL
PGISL(b)	-	1.0	-	-	Exponent of electric field for GISL
PHIBE ^(b)	V	0.7	0.2	1.2	Body-effect voltage parameter
PHIN ^(b)	V	0.05	-	-	Nonuniform vertical doping effect on surface potential
POXEDGE ^(b)	-	1	>0	-	Factor for the gate edge Tox
PRWG ^(b)	V ⁻¹	0.0	0.0	-	Gate bias dependence of S/D extension resistance

Name	Unit	Default	Min	Max	Description
PRWGS ^(b)	V^{-1}	0.0	0.0	-	Source side quasi-saturation parameter
PRWGD ^(b)	V^{-1}	PRWGS	0.0	-	Drain side quasi-saturation parameter
PSAT ^(b)	-	2.0	2.0	-	Exponent for field for velocity saturation
PTWG ^(b)	V^{-2}	0.0	-	-	Correction factor for velocity saturation
PTWGR ^(b)	V^{-2}	PTWG	-	-	Correction factor for velocity saturation in reverse mode
PVAG ^(b)	-	1.0	-	-	V _{gs} dependence on early voltage
QMFAC ^(b)	-	0.0	-	-	Pre-factor for QM correction
QMTCENCVA ^(b)	-	0.0	-	-	Pre-factor/switch for QM effective width and oxide thickness correction for accumulation region CV
RDSW ^(b)	$\Omega \cdot \mu_m^{WR}$	40	0.0	-	RDSMOD = 0 zero bias S/D extension resistance per unit width
RDSWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	-	-	RDSMOD = 0 S/D extension resistance per unit width at high V _{gs}
RDW ^(b)	$\Omega \cdot \mu_m^{WR}$	50	0.0	-	RDSMOD = 1 zero bias drain extension resistance per unit width
RDWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	0.0	-	RDSMOD = 1 drain extension resistance per unit width at high V _{gs}
RSDR	V^{-PRSDR}	RSDR	0.0	-	RDSMOD = 1 source side drift resistance parameter in forward mode
RDDRR	V^{-PRDDR}	RDDR	0.0	-	RDSMOD = 1 source side drift resistance parameter in reverse mode
RDDR	V^{-PRDDR}	RSDR	0.0	-	RDSMOD = 1 drain side drift resistance parameter in forward mode

Name	Unit	Default	Min	Max	Description
RDDRR	V^{-PRDDR}	RDDR	0.0	0.0	RDSMOD = 1 drain side drift resistance parameter in reverse mode
PRSDR	-	1.0	0.0	-	RDSMOD = 1 drain side drift resistance parameter in forward mode
PRDDR	-	PRSDR	0.0	-	RDSMOD = 1 drain side drift resistance parameter in reverse mode
RGEXT	Ω	0.0	0.0	-	Effective gate electrode external resistance (Experimental)
RGFIN	Ω	1.0e-3	1.0e-3	-	Effective gate electrode resistance per n per finger
RSHD	Ω	RSHS	0.0	-	Drain-side sheet resistance
RSHS	Ω	0.0	0.0	-	Source-side sheet resistance
RSW ^(b)	$\Omega \cdot \mu_m^{WR}$	50	0.0	-	RDSMOD = 1 zero bias source extension resistance per unit width
RSWMIN	$\Omega \cdot \mu_m^{WR}$	0.0	0.0	-	RDSMOD = 1 source extension resistance per unit width at high V_{gs}
SIIO ^(b)	V^{-1}	0.0	-	-	V_{gs} dependent parameter of I_{ii} (IIMOD=2)
SI1 ^(b)	-	0.0	-	-	V_{gs} dependent parameter of I_{ii} (IIMOD=2)
SI2 ^(b)	V	0.0	-	-	V_{gs} dependent parameter of I_{ii} (IIMOD=2)
SIID ^(b)	V	0.0	-	-	V_{ds} dependent parameter of I_{ii} (IIMOD=2)
THETAMU	-	1.0	>0.0	-	MOBMOD = 1 mobility reduction exponent
THETASAT ^(b)	V^{-1}	2.0	-	-	Velocity saturation parameter
TOXREF	m	12nm	>0.0	-	Nominal gate oxide thickness for Gate tunneling current
U0 ^(b)	$m^2/V\cdot s$	6e-2	-	-	Low field mobility

Name	Unit	Default	Min	Max	Description
UA ^(b)	(cm/MV) ^{EU}	0.3	>0.0	-	MOBMOD = 0 phonon / surface roughness scattering
UC ^(b)	$\left(1.0e-6 \cdot \frac{cm}{MV^2}\right)^{Rg}$	0.0	>0.0	-	Body effect coefficient for mobility (BULKMOD=1)
UCS ^(b)	-	1.0	>0.0	-	MOBMOD = 0 Coulombic scattering (Experimental)
UD ^(b)	cm/MV	0.0	>0.0	-	MOBMOD = 0 Coulombic scattering (Experimental)
UP ^(b)	μm^{LPA}	0.0	-	-	Mobility L coefficient
VASAT	V	0.2	-	-	CLMMOD = 1 Channel Length Modulation (CLM) parameter
VP	V	0.0	-	-	CLMMOD = 0 CLM log dependence parameter
VSAT ^(b)	m/s	85000	-	-	Saturation velocity
VSATCV(b)	m/s	VSAT	-	-	Saturation velocity for the capacitance model
VSAT1R(b)	m/s	VSAT1	-	-	Saturation velocity for the linear region in reverse mode
WR ^(b)	-	1.0	-	-	W dependence parameter of S/D extension resistance

BSIM-CMG Model (Version<105) — Temperature Dependence and Self-Heating Parameters

Name	Unit	Default	Min	Max	Description
AT ^(b)	1 / (°K)	0.005	-	-	Saturation velocity temperature coefficient
CTH0	J/K	1.0e-5	0.0	-	Thermal capacitance for self-heating calculation
EMOBT(b)	-	0.0	-	-	Temperature Coefficient of ETAMOB
IGT ^(b)	-	2.5	-	-	Gate current temperature coefficient

Name	Unit	Default	Min	Max	Description
IIT ^(b)	-	-0.5	-	-	Impact ionization temperature coefficient(IIMOD=1)
KT1	V	-0.3	-	-	V_{th} temperature coefficient
KT1L	V · m	0.0	-	-	V_{th} temperature coefficient
PRT ^(b)	-	0.001	-	-	Series resistance temperature coefficient
PTWGT ^(b)	1 / (°K)	0.004	-	-	PTWG temperature coefficient
RTH0	K/W	0.01	0.0	-	Thermal resistance for self-heating calculation
STCS	-	0.0	-	-	Mobility (MOBMOD=1) temperature coefficient
STMUE	-	0.0	-	-	Mobility (MOBMOD=1) temperature coefficient
STTHETAMU	-	1.5	-	-	Mobility (MOBMOD=0) temperature coefficient
STTHETASAT	-	1.0	-	-	Saturation velocity temperature coefficient
TBGASUB	(eV) / (°K)	7.02e-4	-	-	Bandgap temperature coefficient
TBGBSUB	°K	1108.0	-	-	Bandgap temperature coefficient
TCJ	1 / (°K)	0.0	-	-	Temperature coefficient for cjs/cjd
TCJSW1	1 / (°K)	0.0	-	-	Temperature coefficient for cjsws1/cjdwd1
TCJSW2	1 / (°K)	0.0	-	-	Temperature coefficient for cjsws2/cjdwd2
TGIDL ^(b)	1 / (°K)	-0.003	-	-	GISL/GIDL temperature coefficient
TII ^(b)	-	0.0	-	-	Impact ionization temperature coefficient(IIMOD=2)
TMEXP ^(b)	1 / (°K)	0	-	-	Temperature coefficient for V_{dseff} smoothing
TNOM	°C	27	0.0	-	Temperature at which the model is extracted (in Celsius)
TPB	1 / (°K)	0.0	-	-	Temperature coefficient for pbs/pbd

Name	Unit	Default	Min	Max	Description
TPBSW1	1 / ($^{\circ}K$)	0.0	-	-	Temperature coefficient for pbsws1/pbdwd1
TPBSW2	1 / ($^{\circ}K$)	0.0	-	-	Temperature coefficient for pbsws2/pbdwd2
TSS(b)	1/K	0.0	-	-	Subthreshold swing temperature coefficient
TRSDR(b)	1/K	0.0			Source side drift resistance temperature coefficient
TRDDR(b)	1/K	TRSDR	-	-	Drain side drift resistance temperature coefficient
UA1 ^(b)	-	-1.1	-	-	Mobility (MOBMOD=0) temperature coefficient
UC1(b)	-	0.056e-9	-	-	Mobility temperature coefficient for UC
UCSTE ^(b)	-	-4.775e-3	-	-	Mobility (MOBMOD=0) temperature coefficient (Experimental)
UD1 ^(b)	-	0.0	-	-	Mobility (MOBMOD=0) temperature coefficient (Experimental)
UTE ^(b)	-	-1.4	-	-	Mobility temperature coefficient
WTH0	m	0.0	0.0	-	Width-dependence coefficient for self-heating calculation
XTID	-	XTIS	-	-	Source junction current temperature exponent
XTIS	-	3.0	-	-	Drain junction current temperature exponent

BSIM-CMG Model (Version=105.03) — WPE Model Support

This section covers the following topics:

- [BSIM-CMG Model \(Version=105.03\) — WPE Model Parameters](#)
- [BSIM-CMG Model \(Version=105.03\) — WPE Instance Parameters](#)

BSIM-CMG Model (Version=105.03) — WPE Model Parameters

Parameter	Default	Min	Max	Bin	Description
WPEMOD	0	0	1	No	Flag for WPE model (WPEMOD=1 to activate this model)
K2WE	0.0	-	-	Yes	Threshold shift factor for well-proximity effect
KU0WE	0.0	-	-	Yes	K2 shift factor for well-proximity effect
KVTH0WE	0.0	-	-	Yes	Threshold shift factor for well-proximity effect
NK2WE	0.0	-	-	Yes	K2 shift factor for well-proximity effect
NKU0WE	0.0	-	-	Yes	Mobility degradation factor for well-proximity effect
NKVTH0WE	0.0	-	-	Yes	Threshold shift factor for well-proximity effect
PK2WE	0.0	-	-	Yes	K2 shift factor for well-proximity effect
PKU0WE	0.0	-	-		Mobility degradation factor for well-proximity effect
PKVTH0WE	0.0	-	-	Yes	Threshold shift factor for well-proximity effect
SCREF	1.0e-6	0.0	-	No	Reference distance to calculate SCA, SCB, and SCC (unit: m)
WEB	0.0	-	-	No	Coefficient for SCB
WEC	0.0	-	-	No	Coefficient for SCC

BSIM-CMG Model (Version=105.03) — WPE Instance Parameters

Parameter	Default	Min	Max	Description
SC	0.0	0.0	-	Distance to a single well edge (unit: m)
SCA	0.0	0.0	-	Integral of the first distribution function for scattered well dopant
SCB	0.0	0.0	-	Integral of the second distribution function for scattered well dopant
SCC	0.0	0.0	-	Integral of the third distribution function for scattered well dopant

BSIM-CMG Model — Deactivating Model Equations

The following are the parameter settings to turn-off BSIM-CMG model equations:

Parameter	Description
CDSC=0 CDSCD=0 CIT=0	Subthreshold slope degradation
ETA0=0	Drain induced barrier lowering
DVT0=0	Vt roll-off (at low drain bias)
K1RSCE=0	Reverse short channel effect
QMFACTOR=0 QMTCENIV=0 QMTCENCV=0	Quantum effect
UA=0 EU=1 (to prevent 0 to the 0'th power) UD=0	Mobility degradation due to vertical field & Coulomb scattering
RGEOMOD=0 RSHS=0 RSHD=0 RDSDMOD=0 RDSWMIN=0 RDSW=0	Source drain resistance
ASYMMOD=0	Asymmetry Model
KSATIV=1000 VSAT=1e9	Drain saturation voltage
NGATE=0	Polysilicon Depletion
PCLM=0	Channel Length Modulation
THETASAT=0	Effect of velocity saturation in the linear region
PDIBL1=0 PDIBL2=0	Output conductance due to DIBL

Parameter	Description
CGEOMOD=0 CFS=0 CFD=0 CGSO=0 CGDO=0 CSDESW=0 CGBO=0 ASEO=0 ADEO=0	Parasitic Capacitance
IIMOD=0	Impact Ionization
IGBMOD=0 IGCMOD=0	Gate Current
GIDLMOD=0	GIDL Current
NQSMOD=0 XRCRG1=0	NQS Gate Resistance
NOIA=0 NOIB=0 NOIC=0	Flicker Noise
NTNOI=0	Thermal Noise
SHMOD=0	Self-heating
RGATEMOD=0	Gate electrode resistance

BSIM-CMG Model — Searching Models as Function of NFIN and L

BSIM-CMG uses `NFIN` and `L` (Length) to determine model bins. The automatic model selection program searches a data file for a MOSFET model where the `NFIN` and `L` are within the range specified in the MOSFET element statement. The simulation then uses this model statement. Note that `NFIN`, `NFINMAX`, and `NFINMIN` are double type numbers for bin selection. The algorithm for automatic model selection based on `NFIN` and `L` is the same as conventional MOSFET.

To search a data file for MOSFET models within a specified range of `NFIN` and `L`:

- Provide a root extension for the model reference name (in the `.MODEL` statement).
- Use the model geometric range parameters (`LMIN`, `LMAX`, `NFINMIN`, and `NFINMAX`). These model parameters define the range of physical length and `NFIN` dimensions to which the MOSFET model applies.

BSIM-CMG Model — Binning Calculation

For a given L , $NFIN$, each model parameter $PARAM_i$ is calculated as a function of $PARAM$, a length-dependent term, $LPARAM$, a number of fins per finger ($NFIN$) dependent term, $NPARAM$, and a product $L \times NFIN$ term, $PPARAM$:

$$PARAM_i = PARAM + \frac{10^{-6}}{L_{eff} + DLBIN} \cdot LPARAM + \frac{1.0}{NFIN} \cdot NPARAM + \frac{10^{-6}}{NFIN(L_{eff} + DLBIN)}$$

For the list of binnable parameters, please refer to [BSIM-CMG Model Parameters \(Version >= 105\)](#).

BSIM-CMG Model — Additional Operating Point Outputs

BSIM-CMG additional operating point outputs are available in BSIM-CMG version 105.03 or later. These additional operating point outputs include total capacitances and intrinsic capacitances. Additional operating point information is printed out when either the option `ext_op` is set to 1 or the environment variable `ext_op` is set to 1 (default is 0).

Name	Descriptions
CGB	Total gate-to-bulk capacitance
CDG	Total drain-to-gate capacitance
CDS	Total drain-to-source capacitance
CDB	Total drain-to-bulk capacitance
CSD	Total source-to-drain capacitance
CSG	Total source-to-gate capacitance
CSB	Total source-to-bulk capacitance
CBD	Total bulk-to-drain capacitance
CBG	Total bulk-to-gate capacitance
CBS	Total bulk-to-source capacitance
CGGBO	Intrinsic gate capacitance
CGDBO	Intrinsic gate-to-drain capacitance
CGSBO	Intrinsic gate-to-source capacitance
CGBBO	Intrinsic gate-to-bulk capacitance

Name	Descriptions
CSGBO	Intrinsic source-to-gate capacitance
CSDBO	Intrinsic source-to-drain capacitance
CSSBO	Intrinsic source capacitance
CSBBO	Intrinsic source-to-bulk capacitance
CDGBO	Intrinsic drain-to-gate capacitance
CDDBO	Intrinsic drain capacitance
CDSBO	Intrinsic drain-to-source capacitance
CDBBO	Intrinsic drain-to-bulk capacitance
CBGBO	Intrinsic bulk-to-gate capacitance
CBDBO	Intrinsic bulk-to-drain capacitance
CBSBO	Intrinsic bulk-to-source capacitance
CBBBO	Intrinsic bulk capacitance

LEVEL 73: HiSIM_HV Model

HiSIM (Hiroshima University STARC IGFET Model) is the first complete surface-potential-based MOSFET model for circuit simulation based on the drift-diffusion theory, which was originally developed by Pao and Sah. The model has been extended for power MOSFETs by considering the resistance effect explicitly, which is named HiSIM_HV.

There are two types of structures commonly used for high voltage applications. One is the asymmetrical laterally diffused structure called LDMOS and the other is originally the symmetrical structure, which we distinguish by referring to it as HVMOS. However, the asymmetrical HVMOS structure is also possible. HiSIM_HV is valid for modeling all these structure types.

The most important features of LDMOS/HVMOS devices, different from the conventional MOSFET, originate from the drift region introduced to achieve the sustainability of high voltages. By varying the length and the dopant concentration of the drift region, various devices with various operating bias conditions are realized for the LDMOS structure:

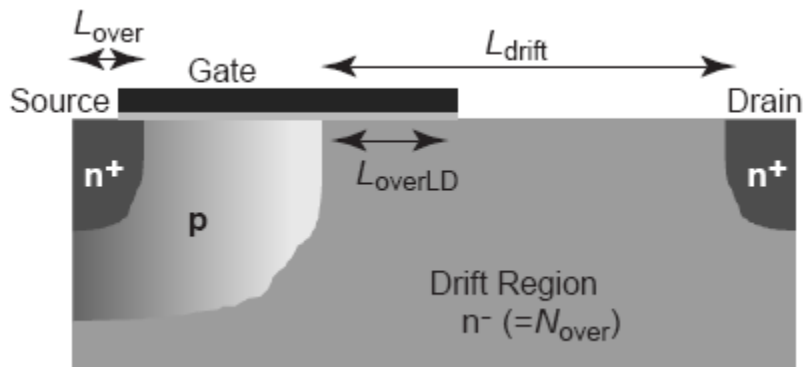
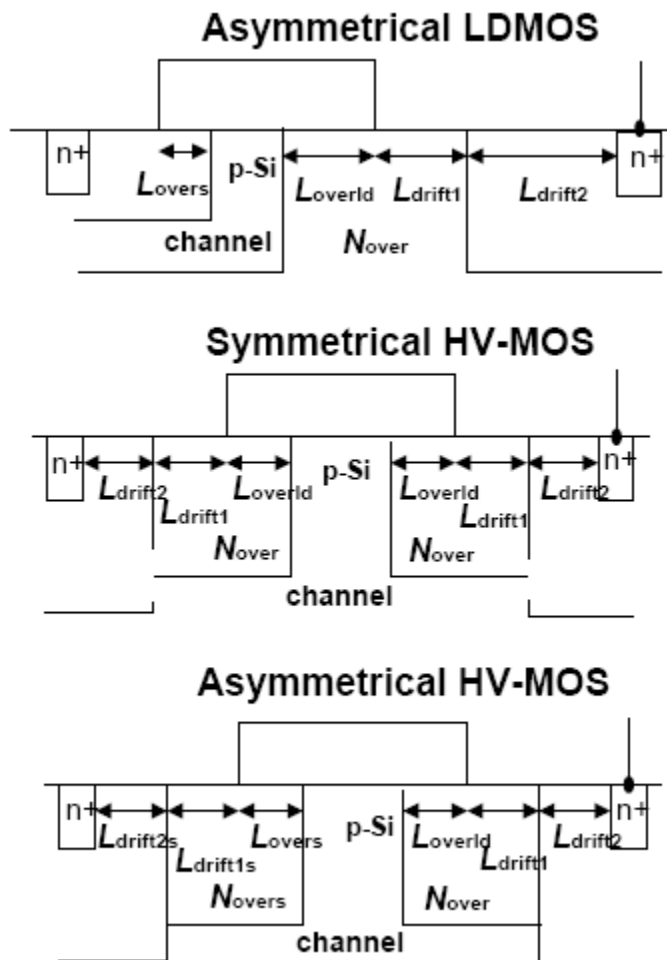
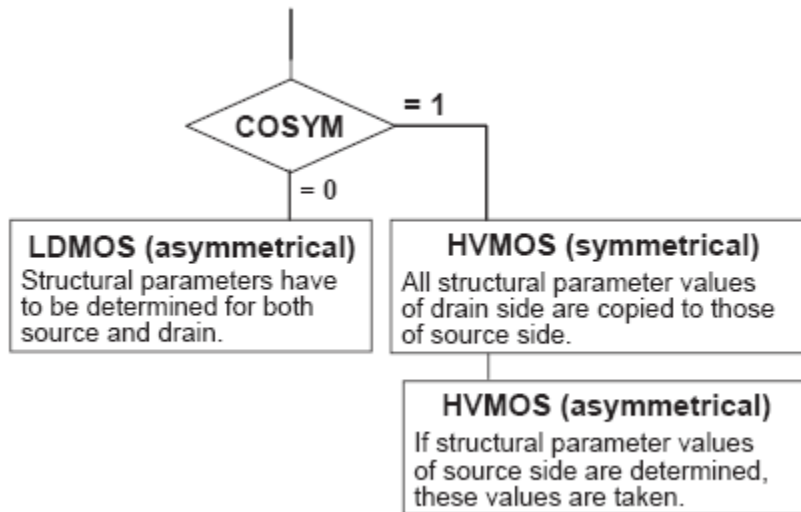


Figure 14 Device parameters in HiSIM_HV



To make the structural definition easy, the flag COSYM is introduced as shown in [Figure 15](#). COSYM=0 refers to the asymmetrical LDMOS, and all structural parameters have to be determined independently. COSYM=1 refers to symmetrical/asymmetrical HVMOS. If parameter values of the source side are given, they are activated. If they are not given, parameter values of the drain side are copied to the source side automatically.

Figure 15 COSYM flag usage: =0 for LDMOS, =1 for HVMOS



[Table 3](#) summarizes the structural parameters to be determined. If the overlap length LOVER is determined instead of LOVERS, then LOVER is taken for LOVERS. Model parameters for resistances at the source side and the drain side are distinguished by RS and RD for the asymmetrical HVMOS.

Table 3 HiSIM HV 1.2.0 model parameters introduced

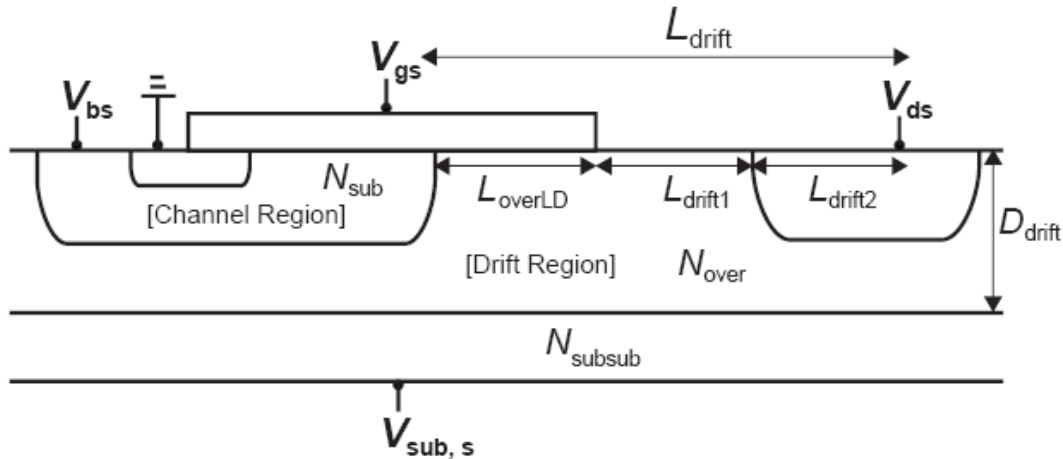
Flag	Structure	Source	Drain
COSYM=0	LDMOS	LOVERS RS	LOVERLD LDRIFT1S LDRIFT2S NOVERS RD
COSYM=1	Symmetrical HVMOS	-	LOVERLD LDRIFT1 LDRIFT2 NOVER RD
COSYM=1	Asymmetrical HVMOS	LOVERS LDRIFT1S LDRIFT2S NOVERS RS	LOVERLD LDRIFT1S LDRIFT2S NOVERS RD

The currently supported version includes the substrate node Vsub as schematically shown in [Figure 16](#), where model parameters DDRIFT and NSUBSUB are newly introduced

for D_{drift} and N_{subsub} , respectively. The node inclusion is done by selecting Flag COSUBNODE=0 as the 5th node.

For details and usage, and current parameters, contact the Synopsys support team.

Figure 16 Schematic of a LDMOS with the substrate node $V_{sub,s}$



Note:

HiSIM-LDMOS source code, and all copyrights, trade secrets or other intellectual property rights in and to the source code in its entirety, is owned by Hiroshima University and STARC.

This section covers the following topics:

- [HiSIM_HV Model — General Syntax](#)
- [HiSIM_HV Model Parameters](#)
- [HiSIM_HV Model Updates](#)
- [HiSIM_HV Model — Template Output](#)

HiSIM_HV Model — General Syntax

The following lists and describes the HiSIM_HV Level 73 general parameters.

```
Mxxx nd ng ns [nb] [nsub] [nth] mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val]
+ [NRS=val] [NRD=val] [XGW=val] [XGL=val]
+ [NF=val] [NGCON=val] [RBPB=val] [RBPD=val]
+ [RBPS=val] [SA=val] [SB=val] [SD=val] [DTEMP=val]
+ [NSUBCDFM=val] [SUBLD1=val] [SUBLD2=val]
+ [LDRIFT1=val] [LDRIFT2=val] [LDSRIFT1S=val] [LDRIFT2S=val]
```

```
+ [LOVER=val] [LOVERLD=val] [LOVERS=val]
+ [COSELFHEAT=val] [CONSUBNODE=val]
```

Notes for Mxxx Card

- *nd*, *ng*, *ns*, *nb*: drain, gate, source, and bulk node, respectively
 - *nsub*: substrate node
 - *nth*: thermal node
 - The square brackets “[]” indicate optional content which may be omitted.
 - If 5 nodes are specified and COSUBNODE=0, the 5th node is *nth*.
 - If 5 nodes are specified and COSUBNODE=1, the 5th node is *nsub*.
 - The following table shows the effects when 6 nodes are specified for a specific version of HiSIM_HV model:
- | Version | Effect |
|-------------------------|--|
| HiSIM_HV 1.2.4 or 2.1.0 | If COSUBNODE=0, the PrimeSim Continuum solution issues an error in parser.
If COSUBNODE=1, the 5th node is <i>nsub</i> and 6th node is <i>nth</i> . |
| Earlier | The 5th node is <i>nsub</i> and the 6th node is <i>nth</i> no matter COSUBNODE is 0 1. |
- COSUBNODE is a new instance parameter and at the same time is recognized as a model parameter for the specification and the recognition of the node order.Originally COSELFHEAT was treated as a model parameter as other flags. It is additionally recognized as an instance parameter.
 - See [HiSIM_HV Model Parameters](#) for model options and descriptions.

HiSIM_HV Model Parameters

This section covers the following topics:

- [HiSIM_HV Model — General Parameters](#)
- [HiSIM_HV Model — Control Option Flags](#)

HiSIM_HV Model — General Parameters

Parameter	Default	Description
L	2.0 μm	Gate length (L_{gate})
W	5.0 μm	Gate width (W_{gate})
Diode		
AD	0.0	Area of drain junction
AS	0.0	Area of source junction
PD	0.0	Perimeter of drain junction
PS	0.0	Perimeter of source junction
Source/Drain Resistance		
NRS	1.0	Number of source squares
NRD	1.0	Number of drain squares
Gate Resistance		
M	1.0	Multiplication factor
NF	1.0	Number of gate fingers
NGCON	1.0	Number of gate contacts
XGL	0.0	Offset of the gate length
XGW	0.0	Distance from the gate contact to the channel edge
Substrate Network		
RBPB	Model parameter value	Substrate Resistance Network
RBPD	Model parameter value	Substrate Resistance Network
RBPS	Model parameter value	Substrate Resistance Network
RBDB		Obsolete
RBSB		Obsolete

Parameter	Default	Description
Length of Diffusion		
SA	0.0	Length of diffusion between gate and STI
SB	0.0	Length of diffusion between gate and STI
SD	0.0	Length of diffusion between gate and gate
Temperature		
DTEMP	0.0	Device temperature change
SHEMAX	500	Limiter for the temperature increase due to the self-heating effect
Design for Manufacturability		
NSUBCDFM	0.0	Substrate impurity concentration
Substrate Current		
SUBLD1	Model parameter value	Substrate current induced in L_{drift}
SUBLD2	Model parameter value	Substrate current induced in L_{drift}
Resistance		
LDRIFT1	0.0	Length of lightly doped drift region
LDRIFT2	$1.0\mu\text{m}$	Length of heavily doped drift region
LDRIFT1S	0.0	Length of lightly doped drift region in source side
LDRIFT2S	$1.0\mu\text{m}$	Length of heavily doped drift region in source side
Overlap		
LOVER	0.0	Length of overlap region in source side for LDMOS
LOVERLD	Model parameter value	Length of overlap region in drain side
LOVERS	Model parameter value	Length of overlap region in source side for HVMOS

Parameter	Default	Description
COSELFHEAT	Model parameter value	Flag to switch on the self-heating effect
COSUBNODE	Model parameter value	Flag for selection of the 5th node

HiSIM_HV Model — Control Option Flags

Description	Flag
Selects asymmetrical (LDMOS) or HVMOS structure	- COSYM = 0: asymmetrical LDMOS (default) - COSYM = 1: symmetrical and asymmetrical HVMOS
Include contact resistances Rs and Rd	- CORSRD= 0: no - CORSRD = 1 & $RS / RD \neq 0$: yes, as internal resistance nodes - CORSRD=2 & $RS / RD \neq 0$: yes, analytical description - CORSRD=3 & $RS / RD \neq 0$: yes, both internal nodes and analytical description (default) - CORSRD = -1 & $RS / RD \neq 0$: yes, as external resistance nodes
Adds overlap charges/capacitances to intrinsic ones	- COADOV = 0: no - COADOV = 1: yes (default)
Selects bias-dependent overlap capacitance model at drain side	- COOVLP = 0: constant overlap capacitance - COOVLP = 1: yes (default) including constant values as option
Selects bias-dependent overlap capacitance model at source side	- COOVLPS = 0: constant overlap capacitance - COOVLPS = 1: yes (default) including constant values as option
Selects method for calculating potential in overlap region	- COQOVSM = 0: analytical equation excluding inversion charge - COQOVSM = 1: iterative solution (default) - COQOVSM = 2: analytical equation including inversion charge
Considers self-heating effect	- COSELFHEAT = 0: no (default) - COSELFHEAT = 1: yes
Calculates substrate current Isub	- COISUB = 0: no (default) - COISUB = 1: yes

Description	Flag
Calculates gate current I_{gate}	• COIIGS = 0: no (default) • COIIGS = 1: yes
Calculates GIDL current I_{GIDL}	• COGIDL = 0: no (default) • COGIDL = 1: yes
Calculates STI leakage current $I_{ds,STI}$	• COISTI = 0: no (default) • COISTI = 1: yes
Invokes non-quasi-static	• CONQS = 0: no (default) • CONQS = 1: yes
Includes gate-contact resistance	• CORG = 0: no (default) • CORG = 1: yes
Invokes substrate resistance network	• CORBNET = 0: no (default) • CORBNET = 1: yes
Calculates 1/f noise	• COFLICK = 0: no (default) • COFLICK = 1: yes
Calculates thermal noise	• COTHRML = 0: no (default) • COTHRML = 1: yes
Calculates induced-gate and cross-correlation noise	• COIGN = 0 & COTHRML = 0: no (default) • COIGN = 1 & COTHRML = 1: yes
Uses previous $_S$ for the iteration	• COPPRV = 0: no • COPPRV = 1: yes (default)
Considers parameter variations for DFM support	• CODFM = 0: no (default) • CODFM = 1: yes
Uses previous I_{ds} is used for calculating source/drain resistance effect ($RS_{and/or}RD \neq 0$): This flag is deactivated.	• COIPRV = 0: no • COIPRV = 1: yes (default)
Selects temperature dependence of models	$R_{d0,temp}$ T T0 T T $R_{dvd,temp}$ T0 T0 T T V_{max} T0 T0 T T0 N_{invd} T0 T0 T T0 • COTEMP=0 (default and backward compatible) • COTEMP=1 • COTEMP=2 • COTEMP=3 where T includes the temperature increase by the self-heating effect and T0 omits it.
Selects the 5th node	• COSUBNODE = 0: the 5th node is the thermal node. • COSUBNODE = 1: the 5th node is the V_{sub} node.

Description	Flag
Selects the Ldrift	• COLDRIFT = 0: Ldrift includes LOVER (default & backward compatible) • COLDRIFT = 1: without LOVER.
Selection for output message whether model parameter is within recommended range	• COERRREP = 0: no message is given. • COERRREP = 1: range check result is given (default).
Selection for the depletion mode model	• CODEP=0: Conventional HVMOS/LDMOS model (default). • CODEP=1: 2.20 depletion mode HVMOS/LDMOS model. • CODEP=2: 2.30 depletion mode HVMOS/LDMOS model.
Improvement of the channel length dependence for Gds	• CODDLT=1: New $V_{ds,eff}$ model (default) • CODDLT=0: Previous $V_{ds,eff}$ model
Hard break down is calculated	• COHBD = 0: no (default). • COHBD = 1 or -1: yes.

HiSIM_HV Model Updates

This section covers the following topics:

HiSIM_HV Version 2.5.1 Updates

The following are the highlights of the HiSIM_HV version 2.5.1 upgrade:

- Added a flag on body-bias clamping for the intrinsic MOSFET model.
 - Added flag: COVBSCL = 1 (Default). COVBSCL = 0 inactivates the bias clamping.
- Added a flag on body-bias clamping for the overlap capacitance model.
 - Added flag: COQOVCL = 1 (Default). COQOVCL = 0 inactivates the bias clamping.
- User-defined maximum and boundary values in common with the two kinds of bias clamping options.
 - Added parameters: VBFWDMX = 0.8 (Default) and VBFWDBND = 0.4 (Default).
- Removal of the special bypass procedure near around the gate threshold voltage for the continuity of I_{ds} .
- Bug-fix for glitches in I_{ds} under very low V_{ds} condition (P_{ds_ini} is updated).
- Local protections of P_{ds} for the CLM model (L_{red}) and the high-field mobility model (μ).

- Bug-fix to avoid FPE in the power function of negative base value (Eyd for the thermal noise model).
- Incorporated the Vds mode into the flicker noise for the correct Pnoise simulation.
- Bug-fix of Vds-mode handling in the simplified overlap charge model (Qgos & Qgod).
- Fully iterative Ps0LD is introduced as a new option of the model flag COQOVSM to solve the discontinuity in overlap capacitance at its flat-band. COQOVSM = 4 activates this model option.
- Bug-fix of Qbu overwritten in the deep PT model: Qbu is localized not to affect AC characteristics.
- Extension of the deep punchthrough (PT) current model: Added a model option with a constant potential value of PS0PT, especially for the negative drain voltage condition.
 - Added parameter: PS0PT = 0 (Default). PS0PT > 0 activates this model option.
- Numerical protection in the deep PT model: Argument of sqrt gets negative because of (Vbipn - Vbscl).
- Incorporated the FD condition into the analytical solution (initial guess) of the surface potential for the overlap charge model.
- Code improvement on the calculation of the overlap charge to avoid possible numerical difficulties around the flatband condition for the gate overlap region.
- Numerical protections for the unclamped bias (-Vxbgmt) case in the overlap charge model.
- Revised the equation of the potential barrier height in the deep PT model. This update eliminates the Vds dependence of the deep PT current under the positive high drain voltage condition.

HiSIM_HV Version 2.5.0 Updates

The following are the highlights of the HiSIM_HV version 2.5.0 upgrade:

- Bug-fix of body-bias clamping for the overlap capacitance model.
- Reactivated the special case near around the gate threshold voltage for the continuity of the intrinsic capacitance.
- Temperature dependence on the diode built-in potentials.

Added parameters: TPBBD=0, TPBBDSW=0, TPBBDSWG=0, TPBBS=0, TPBBSSW=0, TPBBSSWG=0.

- Issue of gm sharpness:
 - Code improvement on the calculation of the overlap charge to avoid possible numerical difficulties around the flatband condition for the gate overlap region. The overlap charge modulates the conductivity of the drift region, and hence gm.
 - Auxiliary adjustment parameters for carrier concentration in the drift region are introduced:

NDRILIM=1, NDRIDLT=1, and NDRIPW=1. A gm roll-off due to drift resistance can be adjusted.
 - Additionally, a sensible use of model parameters for saturation velocity and mobility such as VMAX, RDRVMAX, RDRCAR, and so on, and to refrain from overusing NINVD relax gm sharpness.
- Update of model parameter boundary (RDTEMP1, RDTEMP2, RDVDTEMP1, and RDVDTEMP2).
- Corrections of model parameter units (RD, RS, RDVD, RDTEMP1, RDTEMP2, RDVDTEMP1, RDVDTEMP2, RTHTEMP1, and RTHTEMP2).
- Trench overlap capacitance.

Added parameter: TOXB = TOX (7e-9, Default)
- Improvement of gm and gds in depletion mode.

Improvement of Vbs-dependence for both linear and saturation conditions in depletion mode.

Added flag: COVDSRES=2 (default for CODEP=3; valid for CODEP=3 only)

Added parameters: DEPVLEAK = 1 (valid for CODEP=3 only) DEPPB0 = 0.5 (mobility model for the resistor region compatible with HiSIM2 3.0.0 or later; valid for CODEP=3 only)
- Implementation of diode recovery.

Added flag: CORECOVERY=0 (default)

Added parameters: NDIBOT=1e16 INJ1=1.0 INJ2=10.0 NQS=5e-9 TAU=2e-7 WI=5e-6 DEPNQS=0.0 TAUT=0.0 INJT= 0.0
- Fix on aging model.

Added flag: COFIXAG = 0 (Default). COFIXAG = 1 activates the fixed codes.
- Short-channel effect (Deep punchthrough current).

Added parameters: COPT=0 COPSPT=0 NJUNC=1e20 XJPT=3E-8 MUPT=0.0 VFBPT = 0.0 PSLIMPT=0.0

- Additional Vbs dependence in the mobility for the drift region.

Added parameters: RDRMUEBS1=0.0 RDRMUEBS2=0.0

For detailed information on the newly-added parameters, see the *HiSIM HV 2.5.0 User's Manual* published by the Hiroshima University.

HiSIM_HV Version 2.4.3 Updates

The following are the highlights of the HiSIM_HV version 2.4.3 upgrade:

- Added a flag (COVBSCL) on the activation of forward body-bias clamping for the intrinsic MOSFET model.
 - COVBSCL = 1 (Default).
 - COVBSCL = 0 inactivates the bias clamping.
- Added a flag (COQOVCL) on the activation of forward body-bias clamping for the overlap capacitance model.
 - COQOVCL = 1 (Default).
 - COQOVCL = 0 inactivates the bias clamping.
- Added auxiliary parameters for forward body-bias clamping, each for setting the hard limit (VBFWDMX) of forward bias and the softness of clamping.

Added parameters:

- VBFWDMX = 0.8 (Default)
- VBFWDBND = 0.4 (Default)

The clamping mechanism starts to work when forward bias exceeds VBFWDBND. Both parameters considers a positive value, irrespective of NMOS or PMOS.

- 3rd quadrant drain-to-source current equation was implemented. Added parameters:
 - COPT=0 (must be 1 for activation)
 - XJPT (default:3E-8, depending on device structure)
 - NJUNC (default:1e20, depending on device structure)
 - MUPT (default:0.0; nonzero for activation)
 - VFBPT (default: 0.0)
 - PSLIMPT (default:0.0)
 - PSOPT (default: 0.0)

These may need an adjustment for better fit.

- Code improvement on the calculation of the overlap charge to avoid possible numerical difficulties around the flatband condition for the gate overlap region.
- Removal of the special bypass procedure near around the gate threshold voltage for the continuity of I_{ds} .
- Bug-fix for glitches in I_{ds} under very low V_{ds} condition (P_{ds_ini} is updated).
- Local protections of P_{ds} for the CLM model (L_{red}) and the high-field mobility model (μ).
- Bug-fix to avoid FPE in the power function of negative base value (E_{yd} for the thermal noise model).
- Incorporated the V_{ds} mode into the flicker noise for the correct P_{noise} simulation.
- Bug-fix of V_{ds} mode handling in the simplified overlap capacitance model (Q_{gos} & Q_{god}).
- Incorporated the FD condition into the analytical solution (initial guess) of the surface potential for the overlap capacitance model.

HiSIM_HV Version 2.4.2 Updates

The following are the highlights of the HiSIM_HV version 2.4.2 upgrade:

- **Bug fix:**

Q_{bd_swg} and Q_{bs_swg} bugfix when $CODIO=0$ for backward compatibility.

HiSIM_HV Version 2.4.1 Updates

The following are the highlights of the HiSIM_HV version 2.4.1 upgrade:

- Improvement on G_m at high V_d with back bias dependence. Improvement might cause backward incompatibility for $CODEP=3$.

Introduced geometry-dependence to $NDEPM$. $NDEPML$ and $NDEPMLP$ are added parameters with defaults 0 and 1, respectively.

HiSIM_HV Version 2.4.0 Updates

The following are the highlights of the HiSIM_HV version 2.4.0 upgrade:

- New features:
 - Depletion mode model ($CODEP=3$)
 - Resistance model for source drift region ($CORDRIFT=1$)

- The new flags CORS and CORD are introduced for activation of the newer model for drift region resistance (CORDRIFT=1) on the source and the drain side, respectively.
- Source/Bulk and Drain/Bulk junction diode's internal node connection (CODIO=1)
- Snapback model (COSNP=1)
- NQS effect in overlap charge (CONQSOV=1)
- Gate/drain overlap capacitance for trench-gate structures (COTRENCH=1)
- Bug fixes:
 - Parameter boundaries revised for mitigating possible math errors.
 - Revision of some numerical codes for mitigating loss of accuracy
 - Unphysical Cdd

HiSIM_HV Version 2.3.4 Updates

The following are the highlights of the HiSIM_HV version 2.3.4 upgrade:

Specification changes:

- Added a flag (COVBSCCL) on the activation of forward body-bias clamping for the intrinsic MOSFET model. COVBSCCL = 1 (Default). COVBSCCL = 0 inactivates the bias clamping.
- Added a flag (COQOVCL) on the activation of forward body-bias clamping for the overlap capacitance model. COQOVCL = 1 (Default). COQOVCL = 0 inactivates the bias clamping.
- Added auxiliary parameters for forward body-bias clamping (see 1 and 2 above), each for setting the hard limit (VBFWDMX) of forward bias and the softness of clamping. Added parameters: VBFWDMX = 0.8 (Default) and VBFWDBND = 0.4 (Default). The clamping mechanism starts to work when forward bias exceeds VBFWDBND. Both parameters take a positive value, irrespective of NMOS or PMOS.
- 3rd quadrant drain-to-source current equation was implemented. Added parameters: COPT=0 (must be 1 for activation), XJPT (default:3E-8, depending on device structure), NJUNC (default:1e20, depending on device structure), MUPT (default:0.0; nonzero for activation), VFBPT (default: 0.0), PSLIMPT (default:0.0), PSOPT (default: 0.0). These may need an adjustment for better fit.
- Improvement on the calculation of the overlap charge to avoid possible numerical difficulties around the flatband condition for the gate overlap region.

- Removal of the special bypass procedure near around the gate threshold voltage for the continuity of I_{ds} .
- Bug-fix for glitches in I_{ds} under very low V_{ds} condition
- Local protections for the CLM model and the high-field mobility model.
- Incorporated the V_{ds} mode into the flicker noise for the correct Pnoise simulation.

HiSIM_HV Version 2.3.3 Updates

The following are the highlights of the HiSIM_HV version 2.3.3 upgrade:

Specification changes:

- The lower boundary of the parameter range for LDRIFT2 is changed from open boundary (excludes 0) to closed one (includes 0) to match the document description.
- Implemented a parameter range check for LDRIFT1+LDRIFT2 to protect against zero division in the subsequent calculations in the drift resistance part.

HiSIM_HV Version 2.3.2 Updates

The following are the highlights of the HiSIM_HV version 2.3.2 upgrade:

- Improvements:
 - Computation $Ps0_iniA$ in $HSMHV_{eval}Q_{over}$
 - Small is used in the HiSIM-HV code
 - RD/RS calculations in $CORDRIFT=0$
 - Negative N_{drift} in $HSMHV_eval_rdrift.inc$
 - Optimization of the code in "if(mode > 0)" block
- Bug fixes:
 - Non zero $I_{ds}(1e^{-12})$ at $V_{ds}=0$
 - Discontinuity in Cox versus V_{gb} simulation
 - Cover limitation according to the overlap depth $DDRIFT$
 - Discontinuity in $Cover$ at V_{fb}

HiSIM_HV Version 2.3.1 Updates

The following are the highlights of the HiSIM_HV version 2.3.1 upgrade:

- Introduction of a new temperature clipping method for the self-heating effect which uses COSELFHEAT.
 - COSELFHEAT=0: no self-heating effect (default).
 - COSELFHEAT=1: self-heating effect with the power clipping method for backward compatibility.
 - COSELFHEAT=2: self-heating effect with the temperature clipping method (new).
- Introduction of a new model parameter SHEMAXDLT for self-heating effect, to enable the smoothing adjustments during clipping.

HiSIM_HV Version 2.3.0 Updates

The following are the highlights of the HiSIM_HV version 2.3.0 upgrade:

- Break down model.
- Improvement of the CORDRIFT=1 resistance model with 5th node.
- Improvement of the Qover model with Cgd dependency on Vds.
- Added TK (actual device temperature) and DTSH (increase in temperature due to self-heating) to OP variables.
- Improvement of the depletion mode model with CODEP=2, where CODEP=1 refers the version 2.2.0 model.
- Improvement of the ranges in the parameter declaration.

The checked parameters are: L, W, NRD, NRS, NGCON, XL, XGL, NF, SA, SB, SD, AD, AS, PD, PS, LDRIFT1, LDRIFT2, VMAX, TOX, XLD, DDLTMAX, DDLTSLP, NOVER, NOVERS, SAREF, SBREF, RSH, RSHG, XQY, NSUBC, PARL2, LP, NSUBP, MUEPH1, MUESR1, BB, SLG, NSTI, MUESTI1, MUESTI2, NSUBPSTI1, NSUBPSTI2, LPEXT, TPOLY, GLEAK5, VZADD0, PZADD0, KAPPA, CGDO, CGSO, DLY1, DLY3, TNOM, GBMIN, CTH0, XLDLD, DDRIFT, NSUBSUB, NDEPM, TNDEP, DEPMUE2, DEPVMAX, DEPMUEPH1, DEPBBS, DEPDDL, RDRMUE, RDRVMAX, RDRDJUNC, RDRDL1, RDRDL2, RDRQOVER, QOVADD, QOVJUNC, SHEMAX, GDSLEAK, RDRBB, GMIN, RMIN, ISBREAK, RWELL, NJ, NJSW, MJ, MJSW, MJSWG, PB, PBSW, PBSWG, NJD, NJSWD, MJD, MJSWD, MJSWG, PBD, PBSWD, PBSWGD, NJS, NJSWS, MJS, MJWS, MJWSGS, PBS, PBSWS, PBSWGS, RBPB, RBP, RBPS, LDRIFT1S, LDRIFT2S, RD, RS, FALPH, CGBO, RTH0, and CVDSOVER.

- Added range checking of the model parameter value after calculation of Lgate, Wgate and the binning.

Checked parameters are LLD, WLD, GDLD, MUESTI1, and NSUBPSTI1.

- Cleaning the source code.
- Improvement of the temperature dependence of I_{sub} .
- Bug fixes:
 - FPE in the $1/f$ noise calculation.
 - Negative the effective channel width (W_{eff}) of the drift region and resulting negative capacitance.
 - Discontinuity of the capacitances when QY model is activated.
 - Discontinuity in the 2nd derivative of I_{sub} in CODEP=1.
 - Non-convergence when CODEP=1 and COIIGS=1.
 - Removal of the minimum value checking for NINDVT1 and NINVDT2.
 - Added range checking of the model parameter value after calculation of L_{gate} , W_{gate} , and the temperature dependence.

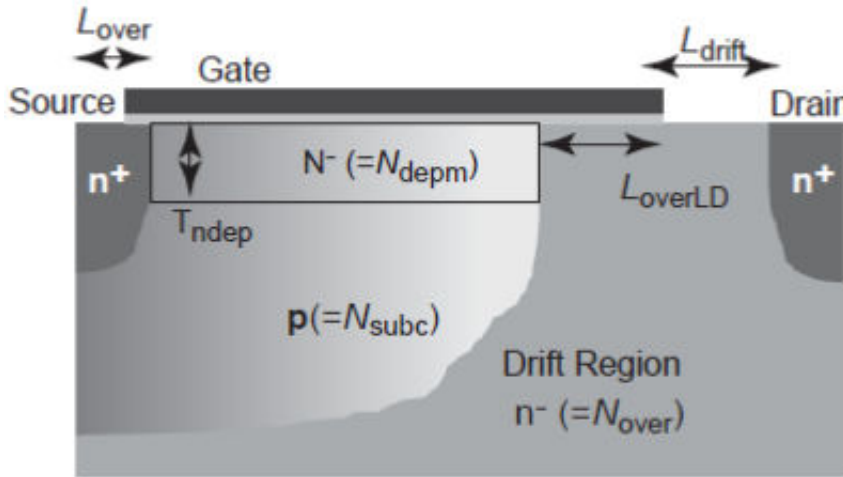
Checked parameters are VMAX , NINVD , NSUBC , NSUBP , MUEPH1 , MUESR0 , RTH0 , RDRMUE , RDRVMAX , RDRBB , QOVADD, and DEPVMAX.

- Removal of the un-initialization issue.
- Removal of the hidden state issues.

HiSIM_HV Version 2.2.0 Updates

The following are the highlights of the HiSIM_HV version 2.2.0 upgrade:

- Model option for depletion mode device:



- Newly added model parameters for the selection of the depletion mode model:

CODEP=1: depletion mode model

CODEP=0: previous model (default)

- Newly added model parameters for the resistor part in the surface layer:

DEPMUE0, DEPMUE1: Coulomb scattering

- Newly added model parameters for the back part in the surface layer:

DEPMUEBACK0, DEPMUEBACK1: Coulomb scattering

- Newly added model parameters both for the resistor part and the back part:

DEPMUEPH0, DEPMUEPH1: Phonon scattering

DEPVAMX, DEPBb: High field mobility

DEPVDSEF1, DEPVDSEF2: Vds dependence

- Newly added model parameters for temperature dependence:

DEPMUETMP: Temperature dependence of the mobility

DEPVTEMP: Temperature dependence of the saturation velocity

Caution:

STI leakage, channel length modulation, thermal noise and induced gate noise are not supported yet in this depletion mode model.

- User-recommended improvement: Modified temperature dependence of drift mobility model.

- Improvement of the channel length dependence for Gds:

`CODDLT=1`: New $V_{ds,eff}$ model (default)

`CODDLT=0`: Previous $V_{ds,eff}$ model

- Changed range of certain parameter values:
 - Maximum value of `SHEMAX` is changed from 600 to 900.
 - Minimum value of `XWDC` is changed from -10nm to -500nm.
 - Minimum value of `VFBOVER` is changed from -1.0 to -1.2.
 - Maximum value of `VFBC` is changed from -0.4 to 0.0.
 - Maximum value of `SC3` is changed from 200nm to 20 μ m.
- Bug fixes and parameter clean-up:
 - Elimination of division by zero problems
 - Modification of comments for some model parameters (`LNOVERS`, `WNOVERS`, `PNOVERS`)
 - Invalidation of `GMIN` parameter
 - Charge and capacitance calculation (noise and fringing capacitance)
 - Addition of source resistance in the noise calculation
 - Removal of `LDRIFT` parameter
 - Undefined variable in VA-Code
 - Not-used variable in VA-Code
 - V_{bs} dependence of the printed output for V_{th}
 - Change of model name in the error messages
 - Changed default value setting of the `LOVERS` parameter. If the `LOVERS` parameter is not specified, the value of the `LOVER` parameter is used.

HiSIM_HV Version 2.1.0 Updates

The following are the highlights of the HiSIM_HV version 2.1.0 upgrade:

- Changed model descriptions:
 - Degradation of the mobility in the drift region with new model parameter `RDRBB`.
 - Additional overlap capacitance with new model parameter `QOVADD`.
 - Improved smoothness of drift resistance at $V_{ds}=0$.
- Changed default/min/max values of model parameters `NOVER`, `NOVERS`, `SCP1`, `SC1`, `FN1`, `NSTI`, `RDRMAX`, `RDRQOVER`, `QME1`, and `QME2` in the source code.
- Changed range check method of model parameter `VFBC`.
- Bug fixes:
 - Range check of `CGSO`, `CGDO`, `CGBO`, `NSUBCDFM`, `SUB2L`, and `SC4`.
 - Calculation of the derivatives.
 - Temperature dependence of the output parameter `VON` for the threshold voltage.
 - Removal of unnecessary code.
 - Removal of the function `FABS` (Verilog-A code only).
 - `NF` dependence of `RD`, `RDVD`, `RD23`, and `RTH`.
 - Removal of unused variables.
 - Calculation of the temperature derivatives.
 - Temperature dependence of the output parameter `Vth`.
 - Error check of the 6th node.
 - Inconsistency between Verilog-A code and C-code.
 - Calculation of the output parameter `Vdsat`.
 - Redundant calculations of the temperature variables.
 - Duplicated output of error message.

HiSIM_HV Version 2.0.1 Updates

The following are the highlights of the HiSIM_HV version 2.0.1 upgrade:

Bug Fixes:

- Missing initialization of `NSUBC`, `NSUBP` and `MUEPH1` in the `hsmhvtemp.c` for repeating simulation.
- Reference voltage (**Vbs -> Vbse**) in the *Qover* model used when model parameter `CVDSOVER != 0`.
- Derivative calculations with respect to temperature (C code only).
- Maximum gate width for the range check when instance parameter `NF != 1`.
- Derivative calculations in the *Qover* model code (`hsmhveval_qover.h`) (C code only).
- Floating point exception in the *QME* model code.
- Floating point exception in the smoothing functions.
- Floating point exception when model parameters `RDVDL << 0` and `RD23L << 0`.
- Floating point exception when model flag/parameters `COSUBNODE != 0`, `COSYM=0` and `NOVER * (NSUBSUB+NOVER) = 0`.
- Floating point exception when model flag `CORDRIFT=1` and model parameter `RDRDJUNC = 0`.
- Derivative calculations when model parameter `COSUBNODE = 1` (C code only).
- Different maximum values of model parameters `RSH` and `RSHG` from User's Manual.
- Inactivated model parameter `RSH` and instance parameter `NRS` when model flag `CORDRIFT = 1`.
- Incorrect clamping of the `Vbs` when `Vbs > 0` (Verilog-A code only).
- Incorrect minimum resistance value `RMIN` (Verilog-A code only).
- Incorrect gate resistance definition when model flag `CORG = 1` and model parameters `RSHG > 0` (Verilog-A code only).

HiSIM_HV 2.0.1 — Junction Diode Model Parameters

Following are the junction diode model parameters:

- `sbdb_dio`: [Table 4](#)
- `t_cap_dio`: [Table 5](#)
- `d_dio`: [Table 6](#)

- `s_dio`: [Table 7](#)
- `g_sbdb_dio`, `g_d_dio`, `g_s_dio`: [Table 8](#)

Table 4 *General Junction Diode Parameters*

Name	Default	Unit	Description
JS0	0.5E-6	A/m ²	Saturation current density
JS0SW	0.0	A/m	Side wall saturation current density
NJ	1.0	-	Emission coefficient
NJSW	1.0		Sidewall emission coefficient
XTI	2.0	-	Junction current temperature exponent coefficient
CJ	5.0E-04	F/m ²	Bottom junction capacitance per unit area at zero bias
CJSW	5.0E-10	F/m	Source/drain sidewall junction capacitance grading coefficient per unit length at zero bias
CJSWG	5.0E-10	F/m	Source/drain gate sidewall junction capacitance per unit length at zero bias
MJ	5.0E-0		Bottom junction capacitance grading coefficient
MJSW	0.33E0		Source/drain sidewall junction capacitance grading coefficient
MJSWG	0.33E0		Source/drain gate sidewall junction capacitance grading coefficient
PB	1.0E0	V	Bottom junction build-in potential
PBSW	1.0E0	V	Source/drain sidewall junction build-in potential
PBSWG	1.0E0	V	Source/drain gate sidewall junction build-in potential
XTI2	0.0E0	-	Temperature coefficient
CISB	0.0E0	-	Reverse bias saturation current
CVB	0.0E0	-	Bias dependence coefficient of cisb
CTEMP	0.0E0	-	Temperature coefficient

Table 4 *General Junction Diode Parameters (Continued)*

Name	Default	Unit	Description
CISBK	0.0E0	A	Reverse bias saturation current
CVBK	CVB	-	Inactivated by CVB
DIVX	0.0E0		[1/V]
VDIFFJ	0.6E-3	V	Threshold voltage for S/D junction diode

Table 5 *Temperature Dependence of Junction Capacitance Parameters*

Name	Default	Unit	Description
TCJBD	0.0		Temperature dependence of cjbd
TCJBS	0.0		Temperature dependence of cjbs
TCJBDSW	0.0		Temperature dependence of cjbdsw
TCJBSSW	0.0		Temperature dependence of cjbsw
TCJBDSWG	0.0		Temperature dependence of cjbdswg
TCJBSSWG	0.0		Temperature dependence of cjbswg

Table 6 *Drain Diode Parameters*

Name	Default	Unit	Description
JS0D	JS0	A/m ²	Saturation current density for drain junction
JS0SWD	JS0SW	A/m	Side wall saturation current density for drain junction
NJD	NJ	-	Emission coefficient for drain junction
NJSWD	NJSW		Sidewall emission coefficient for drain junction
XTID	XTI	-	Junction current temperature exponent coefficient for drain junction
CJD	CJ	F/m ²	Bottom junction capacitance per unit area at zero bias for drain junction

Table 6 Drain Diode Parameters (Continued)

Name	Default	Unit	Description
CJSWD	CJSW	F/m	Sidewall junction capacitance grading coefficient per unit length at zero bias for drain junction
CJSWGD	CJSWG	F/m	Gate sidewall junction capacitance per unit length at zero bias for drain junction
MJD	MJ		Bottom junction capacitance grading coefficient for drain junction
MJSWD	MJSW		Sidewall junction capacitance grading coefficient for drain junction
MJSWGD	MJSWG		Gate sidewall junction capacitance grading coefficient for drain junction
PBD	PB	V	Bottom junction build-in potential for drain junction
PBSWD	PBSW	V	Sidewall junction build-in potential for drain junction
PBSWGD	PBSWG	V	Gate sidewall junction build-in potential for drain junction
XTI2D	XTI2	-	Temperature coefficient for drain junction
CISBD	CISB	-	Reverse bias saturation current for drain junction
CVBD	CVB	-	Bias dependence coefficient of cisb for drain junction
CTEMPD	CTEMP	-	Temperature coefficient for drain junction
CISBKD	CISBK	A	Reverse bias saturation current for drain junction
DIVXD	DIVX	1/V	Reverse coefficient coefficient for drain junction
VDIFFJD	VDIFFJ	V	Threshold voltage for junction diode for drain junction

Table 7 Source Diode Parameters

Name	Default	Unit	Description
JS0S	JS0D	A/m ²	Saturation current density for source junction
JS0SWS	JS0SWD	A/m	Side wall saturation current density for source junction
NJS	NJD	-	Emission coefficient for source junction
NJSWS	NJSWD		Sidewall emission coefficient for source junction
XTIS	XTID	-	Junction current temperature exponent coefficient for source junction
CJS	CJD	F/m ²	Bottom junction capacitance per unit area at zero bias for source junction
CJSWS	CJSWD	F/m	Sidewall junction capacitance grading coefficient per unit length at zero bias for source junction
CJSWGS	CJSWGD	F/m	Gate sidewall junction capacitance per unit length at zero bias for source junction
MJS	MJD		Bottom junction capacitance grading coefficient for source junction
MJSWS	MJSWD		Sidewall junction capacitance grading coefficient for source junction
MJSWGS	MJSWGD		Gate sidewall junction capacitance grading coefficient for source junction
PBS	PBD	V	Bottom junction build-in potential for source junction
PBSWS	PBSWD	V	Sidewall junction build-in potential for source junction
PBSWGS	PBSWGD	V	Gate sidewall junction build-in potential for source junction
XTI2S	XTI2D	-	Temperature coefficient for source junction
CISBS	CISBD	-	Reverse bias saturation current for source junction
CVBS	CVBD	-	Bias dependence coefficient of cisb for source junction

Table 7 Source Diode Parameters (Continued)

Name	Default	Unit	Description
CTEMPS	CTEMPD	-	Temperature coefficient for source junction
CISBKS	CISBKD	A	Reverse bias saturation current for source junction
DIVXS	DIVXD	1/V	Reverse coefficient coefficient for source junction
VDIFFJS	VDIFFJD	V	Threshold voltage for junction diode for source junction

Table 8 Geometry Dependence of Diode Parameters

Name	Default	Unit	Description
LJS0	0.0		Length dependence of js0
LJS0SW	0.0		Length dependence of js0sw
LNJ	0.0		Length dependence of nj
LCISBK	0.0		Length dependence of cisbk
LVDIFFJ	0.0		Length dependence of vdiffj
LJS0D	LJS0		Length dependence of js0d
LJS0SWD	LJS0SW		Length dependence of js0swd
LNJD	LNJ		Length dependence of njd
LCISBKD	LCISBK		Length dependence of cisbkd
LVDIFFJD	LVDIFFJ		Length dependence of vdiffjd
LJS0S	LJS0D		Length dependence of js0s
LJS0SWS	LJS0SWD		Length dependence of js0sws
LNJS	LNJD		Length dependence of njs
LCISBKS	LCISBKD		Length dependence of cisbks
LVDIFFJS	LVDIFFJD		Length dependence of vdiffjs
WJS0	0.0		Width dependence of js0

Table 8 *Geometry Dependence of Diode Parameters (Continued)*

Name	Default	Unit	Description
WJS0SW	0.0		Width dependence of js0sw
WNJ	0.0		Width dependence of nj
WCISBK	0.0		Width dependence of cisbk
WVDIFFJ	0.0		Width dependence of vdiffj
WJS0D	WJS0		Wength dependence of js0d
WJS0SWD	WJS0SW		Wength dependence of js0swd
WNJD	WNJ		Wength dependence of njd
WCISBKD	WCISBK		Wength dependence of cisbkd
WVDIFFJD	WVDIFFJ		Wength dependence of vdiffjd
WJS0S	WJS0D		Wength dependence of js0s
WJS0SWS	WJS0SWD		Wength dependence of js0sws
WNJS	WNJD		Wength dependence of njs
WCISBKs	WCISBKD		Wength dependence of cisbks
WVDIFFJS	WVDIFFJD		Wength dependence of vdiffjs
PJS0	0.0		Cross-term dependence of js0
PJS0SW	0.0		Cross-term dependence of js0sw
PNJ	0.0		Cross-term dependence of nj
PCISBK	0.0		Cross-term dependence of cisbk
PVDIFFJ	0.0		Cross-term dependence of vdiffj
PJS0D	PJS0		Cross-term dependence of js0d
PJS0SWD	PJS0SW		Cross-term dependence of js0swd
PNJD	PNJ		Cross-term dependence of njd
PCISBKD	PCISBK		Cross-term dependence of cisbkd
PVDIFFJD	PVDIFFJ		Cross-term dependence of vdiffjd
PJS0S	PJS0D		Cross-term dependence of js0s

Table 8 Geometry Dependence of Diode Parameters (Continued)

Name	Default	Unit	Description
PJS0SWS	PJS0SWD		Cross-term dependence of js0sws
PNJS	PNJD		Cross-term dependence of njs
PCISBKS	PCISBKD		Cross-term dependence of cisbks
PVDIFFJS	PVDIFFJD		Cross-term dependence of vdiffjs

HiSIM_HV Version 2.0.0 Updates

The following are the highlights of the HiSIM_HV version 2.0.0 upgrade:

- Improvements of fitting capability (Refer to the *HiSIM HV 2.0.0 User's Manual* for detailed description).
 - Length dependence of `Rdrift` with new model parameters `RDRVMAXL`, `RDRVMAXLP`, `RDRMUEL`, and `RDRMUELP`.
 - Addition of overlap charge to the carrier concentration of the drift region with new model parameter `RDRQOVER`.
 - Modification of `Rdrift` for improved small `Vds` fitting with introduction of new model parameters `RDRLOWVD1` and `RDRLOWVD2`.
 - Increase of upper limitation for `Rdrift` from 50000 Ohm to 500000 Ohm.
- Bug Fixes:
 - Floating-point exception for large drain voltage when `COQOVSM=2`.
 - Incorrect recognition of the 5th terminal connected to ground (0) when `COSUBNODE=0`.
 - Removal the unnecessary program statements.

Note:

The HiSIM_HV 2.0.0 model does not currently support Template output.

HiSIM_HV Version 1.2.4 Updates

The following are the highlights of the HiSIM_HV version 1.2.4 upgrade:

Bug fixes:

- Incorrectly Referencing `Total W` Instead of `W per Finger`.
- Derivative calculations with respect to temperature (C code only).
- Convergence and `Flg_subNode` setting.

HiSIM_HV Version 1.2.3 Updates

The following are the highlights of the HiSIM_HV version 1.2.3 upgrade:

Bug fixes:

- Missing initialization of `NSUBC`, `NSUBP`, and `MUEPH1` in the `hsmhvtemp.c` for repeating simulation.
- Inaccurate overlap capacitance when model parameter `CVDSOVER != 0`.
- Reference voltage (`Vbs->Vbse`) in the `Qover` model used when model parameter `CVDSOVER != 0`.
- Derivative calculations with respect to temperature (C code only).
- Maximum gate width for the range check when model parameter `NF != 1`.
- Derivative calculations in the `Qover` model code (`hsmhveval_qover.h`) (C code only).
- Floating point exception in the `QME` model code.
- Floating point exception in the smoothing functions.
- Floating point exception when model parameters `RDVDL < 0` and `RD23 < 0`.
- Floating point exception when model flag/parameters `COSUBNODE != 0`, `COSYM = 0`, and `NOVER * (NSUBSUB + NOVER) = 0`.
- Different default value of some model parameters from User Manual.
- Different minimum value and/or the maximum value of some model parameters from User Manual.
- Incorrect clamping of the `Vbs` when `Vbs > 0` (Verilog-A code only).
- Incorrect minimum resistance value `RMIN` (Verilog-A code only).
- Incorrect gate resistance definition when model flag `CORG = 1` and model parameter `RSHG > 0` (Verilog-A code only).

HiSIM_HV Version 1.2.1 Updates

The following are the highlights of the HiSIM_HV version 1.2.1 upgrade:

- Improvements in the lateral-field-induced charge (Q_y) model.
- Improvements in the overlap capacitance (Q_{over}) model. New options are provided to calculate Q_{over} . These options are selected by the flag `COQOVSM`:
 - `COQOVSM=0`: Q_{over} is calculated with an analytical equation excluding the inversion charge.
 - `COQOVSM=1`: Q_{over} is calculated with an iterative procedure including the inversion charge (Default).
 - `COQOVSM=2`: Q_{over} is calculated with an analytical equation including the inversion charge (HiSIM_HV 1.1.1 original).
- A limiter for the temperature increase due to the self-heating effect is introduced by the model parameter `SHEMAX` (Default=500).
- The Vbs clamping (add temperature dependence to V_{bs_max}) is changed to fix a discontinuity in $C_{dd}-V_{ds}$ at $T=125$.
- Fixed derivative calculation.
- Fixed typo in model ask function for parameter `PB`.
- Fixed calculation of the drift resistance when `COSYM=0` and the `Vsub` node connected to 0.

HiSIM_HV Version 1.1.2 Updates

The following are the highlights of the HiSIM_HV version 1.1.2 upgrade:

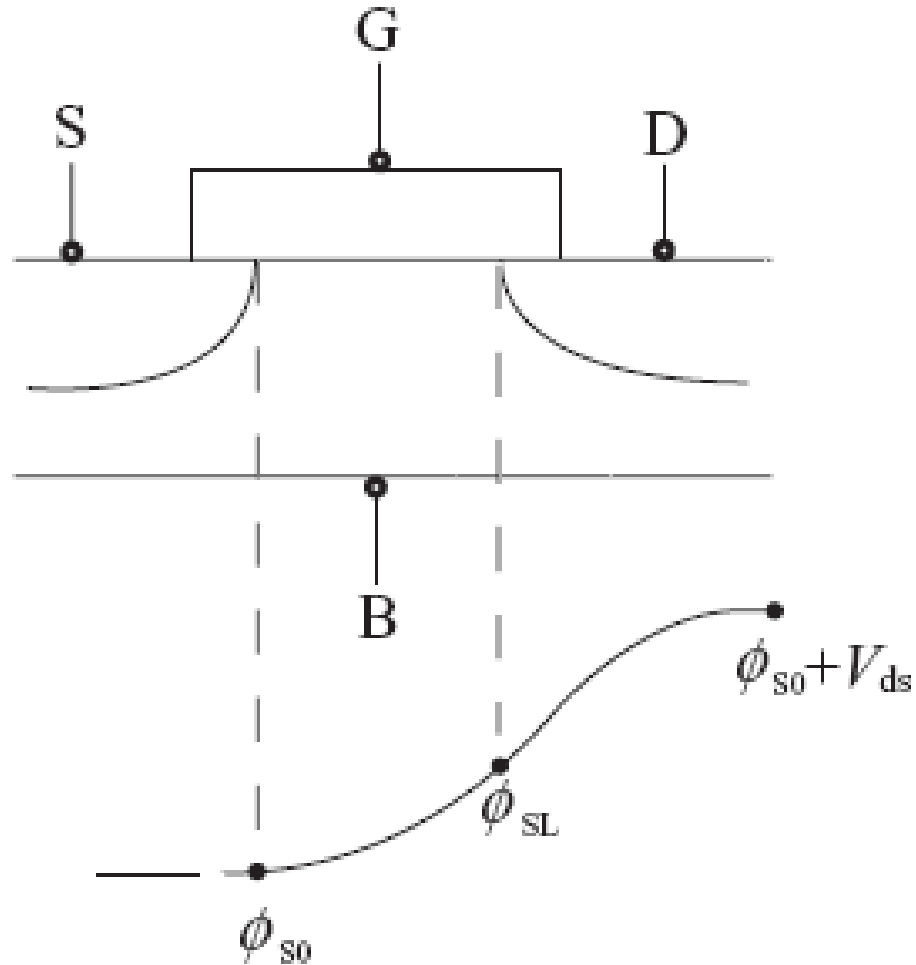
- Improvements in the lateral-field-induced charge (Q_y) model
- Improvements in the overlap capacitance (Q_{over}) model
- New options are provided to calculate Q_{over} . These options are selected by the flag `COQOVSM`:
 - `COQOVSM=0`: Q_{over} is calculated with an analytical equation excluding the inversion charge.
 - `COQOVSM=1`: Q_{over} is calculated with an iterative procedure including the inversion charge (Default)
 - `COQOVSM=2`: Q_{over} is calculated with an analytical equation including the inversion charge (HiSIM_HV 1.1.1 original).

- A limiter for the temperature increase due to the self-heating effect is introduced by the model parameter `SHEMAX` (Default=500).
- The Vbs clamping (add temperature dependence to `Vbs_max`) is changed to fix a discontinuity in `Cdd-Vds` at `T=125`
- Derivative calculation in the `IBPC` model
- Derivative calculation in the `Isub` model: missing `Isub_dVdse`
- Floating-point exception issue in the `Isub` model
- Floating-point exception issue in the `Vdseff` calculation
- Floating-point exception issue for `RD23`
- `Ra` model
- Parameter handling with `COSYM`
- Small resistance cases: introduction of `Res_min`
- Improvement of the drain-side surface potential (`Ps1`) iteration
- `Pds` clippings
- Typo in model ask function for parameter `PB`

HiSIM_HV Version 1.0.1 and 1.0.2 Updates

The PrimeSim Continuum solution supports HiSIM_HV version 1.0.1 and 1.0.2 to address the need for accurate modeling of high-voltage MOSFETs, Hiroshima University and STARC developed the HiSIM-HVMOS model as an extension to the IGFET model. The HVMOS model focuses on symmetrical structure while the HiSIM-LDMOS model is used for asymmetrical structures. According to the *HiSIM_HV User Guide*: the most important advantage of the surface-potential-based modeling is the unified description of device characteristics for all bias conditions. The physical reliability of the drift-diffusion theory has been proved by 2-D device simulations with channel lengths even down to below $0.1\ \mu\text{m}$. To obtain analytical solutions for describing device performances, the charge sheet approximation of the inversion layer with zero thickness has been introduced. Together with the gradual-channel approximation all device characteristics are then described analytically by the channel-surface potentials at the source side (`S0`) and at the drain side (`SL`) (see [Figure 17](#)).

Figure 17 Schematic of the surface potential distribution in the channel



These surface potentials are functions of applied voltages on the four terminals: the gate voltage V_g , the drain voltage V_d , the bulk voltage V_b and the reference potential of the source V_s . The resistance in the contact region causing potential drops also affects the surface potential values. Since the surface potentials are implicit functions of the applied voltages, model-internal iteration procedures are introduced only for calculating ϕ_{S0} , and ϕ_{SL} is calculated in addition to the global iteration of the circuit simulator.

The potential $\phi_s(\Delta L)$ is calculated with ϕ_{S0} , ϕ_{SL} , and V_{ds} together with a fitting parameter.

The most important features of LDMOS/HVMOS devices, different from the conventional MOSFET, originate from the drift region introduced to achieve the sustainability of high voltages. By varying the length as well as the dopant concentration of the drift region,

various devices with various operating bias conditions are realized. In many cases, the drift region acts as the resistance for the current flow and also induces additional charge, which causes the especially unique features of the LDMOS capacitances. Thus, accurate modeling of the drift region is the main task of HiSIM-LDMOS/HV.

For the LDMOS/HVMOS devices the iterative solution is only one possible to model the specific features of this device accurately, because the resistance effect in the drift region is dependent on the bias condition as well as the detailed geometrical LDMOS/HVMOS structure. The basic modeling method is taken over from the HiSIM2 model for advanced MOSFETs, and additional equations for capturing the drift-region effects are included. Since the overlap length is relatively long for LDMOS/HVMOS, accurate surface potential calculation for the overlap region as a function of applied voltages is also necessary for accurate prediction of the high-voltage MOS capacitances.

HiSIM-LDMOS 100 Updates

The following changes were made in the HiSIM-LDMOS 100 model as reported in the SC1 through SC3 release notes provided by Hiroshima University/STARC.

- SC1:
 - Self-heating effect Temperature node is handled as an internal node.
 - The model parameter `RD26` was deleted, and the smoothing parameter was fixed to a constant value.
 - `Qover` Overlap Capacitance: The model description of `Qover` was changed from external bias to internal bias.
 - The charge-partitioning scheme with model parameters `QOVRAT1` and `QOVRAT2` was deactivated.
 - The new model parameter `RD26` was introduced to fit transition characteristics at depletion and/or inversion.
- SC2:
 - The self-heating effect (`COSELFHEAT=1`) was stabilized for the PrimeSim Continuum solution. The temperature node can be treated either as an external or internal node. The NF dependability in the drift resistance model was removed for to improve accuracy.
- SC3:
 - The self-heating effect became applicable for AC analysis Introduction of drift-length (`LDRIFT`) dependence of the drift resistance. Introduction of model parameters `RDTEMP1` and `RDTEMP2` for temperature dependent drift resistance.

Extension to LDMOS and HVMOS: To facilitate the structural definition, the `COSYM` flag was introduced with HiSIM-HV1.0.0. **`COSYM=0`** refers to the asymmetrical LDMOS,

and all drain-side and source-side parameters of the drain side are copied to the source side. **COSYM=1** refers to symmetrical HVMOS, and all parameters have to be determined independently. HiSIM_HV considers the length of the drift region L_{drift} , the overlap length L_{over} , and the impurity concentration of the drift region N_{over} explicitly. In the HVMOS case, the parameter values for the drain side have to be determined, and are copied to the source side automatically. If parameters are not determined, default values are taken.

HiSIM_HV Model — Template Output

The PrimeSim Continuum solution supports parameter template output for all existing versions, HiSIM-HVMOS 1.00, 1.01, 1.02, 1.10, 1.11, 1.12, 1.20, 1.21, 1.23, 2.00, 2.01, and 2.10. See [Output Templates for HiSIM_HV, Level=73](#).

LEVEL 74: MOS Model 20 Model

MOS Model 20 (MM20) is a compact MOSFET model developed by NXP, intended for analog circuit simulation in high-voltage MOS technologies. MOS Model 20 describes the electrical behavior of the region under the thin gate oxide of a high-voltage MOS device, similar to a Lateral Double-diffused MOS (LDMOS) device or an extended-drain MOSFET. It combines the MOSFET-operation of the channel region with that of the drift region under the thin gate oxide in a high-voltage MOS device. As such, MOS Model 20 is aimed as a successor of the combination of MOS Model 9 (MM9) for the channel region in series with MOS Model 31 (MM31) for the drift region under the thin gate oxide, in macro models of various high-voltage MOS devices. MOS Model 20 has especially been developed to improve the convergence behavior during simulation, by having the voltage at the transition from the channel region to the drift region calculated inside the model itself.

The following sections discuss these topics:

- [MOS Model 20 Model — General Syntax](#)
- [MOS Model 20 Model — Instances and Parameters](#)

MOS Model 20 Model — General Syntax

```
Mxxxxx drain gate source bulk mname W=val Wd=val ...
```

MOS Model 20 Model — Instances and Parameters

JUNCAP 200.2 is supported in the PrimeSim Continuum solution MOS Model 20 as an alternative to the MOSFET junction diode model. You can use model parameter, JUNCAPMOD=1, to invoke `Juncap200.2`.

This section covers the following topics:

- [MOS Model 20 — Instance Parameters](#)
- [MOS Model 20 — Geometrical Parameters](#)
- [MOS Model 20 — Electrical Parameters](#)

MOS Model 20 — Instance Parameters

Name	Unit	Default	Description
W	m	2E-5	Drawn width of the channel region
WD	m	2E-5	Drawn width of the drift region

Name	Unit	Default	Description
MULT		1	Number of devices in parallel

MOS Model 20 — Geometrical Parameters

Name	Unit	Default	Description
LEVEL		74	The PrimeSim Continuum solution model level as 74 for MOS Model 20
VERSION		2002.2	VERSION 2002.2 is the default version number to define NXP model level 2002
GEOMOD		1	GEOMOD is used as a switch between geometrical model and electrical model, 0 for geometrical model and 1 for electrical model
JUNCAP MOD		0	JUNCAPMOD as a switch of juncap200.2 model effective, 0 for off and 1 for on
A1CHR		15	Factor of channel weak avalanche current, at reference temperature
A1DRR		15	Factor of drift weak avalanche current, at reference temperature
A2CH	V	73	Exponent of weak avalanche current, related to channel
A2DR	V	73	Exponent of weak avalanche current, related to drift
A3CH		0.8	Factor of the internal drain-source voltage, above which channel weak avalanche occurs
A3DR		0.8	Factor of the internal drain-source voltage, above which drift weak avalanche occurs
ALP		0.002	Factor for channel length modulation
ATH		0	Temperature coefficient of the thermal resistance
BETACCW	A/V^2	7.00E-05	Gain factor of drift region of 1 um wide, at reference temperature
BETW	A/V^2	7.00E-05	Gain factor of a channel region of 1 um wide, at reference temperature
CGDOW	F	0	Gate-to-drain overlap capacitance for a drift region of 1 um wide

Name	Unit	Default	Description
CGSOW	F	0	Gate-to-source overlap capacitance for a drift region of 1 um wide
COXDW	F	7.5E-16	Oxide capacitance for an intrinsic drift region of 1 um wide
COXW	F	7.5E-16	Oxide capacitance for an intrinsic channel region of 1 um wide
CTH	J/K	3.00E-09	Thermal capacitance
DTA	K	0	Temperature offset to the ambient temperature
ETABET		1.6	Temperature scaling exponent for BET
ETABETACC		1.5	Temperature scaling exponent for BETACC
ETARD		1.5	Temperature scaling exponent for RD
ETATHE3		1	Temperature scaling exponent for THE3
ETATHE3D		1	Temperature scaling exponent for THE3D
KODR	$V^{(1/2)}$	1	Body factor of the drift region of an infinitely wide transistor
KOR	$V^{(1/2)}$	1.6	Body factor of the channel region of an infinitely wide transistor
LAMD		0.2	Quotient of the depletion layer thickness to the effective thickness of the drift region at VSB = 0 V
MEXP		2	Smoothing factor for transition from linear to saturation regime
MEXPD		2	Smoothing factor for transition from linear to quasi-saturation regime
MO	V	0	Parameter for the (short-channel) sub-threshold slope
MSDIBL		3	Exponent for the drain-induced barrier lowering dependence on the backgate bias
NFAW	$V^{-1}m^{-4}$	1.4E+25	First coefficient of flicker noise for a channel region of 1 um wide
NFBW	$V^{-1}m^{-2}$	2.00E+08	Second coefficient of flicker noise for a channel region of 1 um wide
NFCW	V^{-1}	0	Third coefficient of flicker noise for a channel region of 1 um wide

Name	Unit	Default	Description
NT	J	1.645E-20	Coefficient of thermal noise, at reference temperature
PHIB	V	0.86	Surface potential at the onset of strong inversion in the channel region, at reference temperature
PHIBD	V	0.78	Surface potential at the onset of strong inversion in the drift region, at reference temperature
RDW	Ohm	4000	On-resistance of a drift region of 1 um wide, at reference temperature
RTH	K/W	300	Thermal resistance
SDIBL	$V^{(-1/2)}$	0.001	Factor for drain-induced barrier lowering
SHMOD		0	SHMOD is used as a switch of self-heating effect, 0 for off and 1 for on
SSF	$V^{(-1/2)}$	1E-12	Factor for static feedback
STA1CH	K^{-1}	0	Temperature scaling coefficient for A1CH
STA1DR	K^{-1}	0	Temperature scaling coefficient for A1DR
STPHIB	V/K	-0.0012	Temperature scaling coefficient for PHIB
STPHIBD	V/K	-0.0012	Temperature scaling coefficient for PHIBD
STVFB	V/K	0	Temperature scaling coefficient for VFB
STVFBD	V/K	0	Temperature scaling coefficient for the flatband voltage of the drift region
SWA1CH		0	Width scaling coefficient for A1CH
SWA1DR		0	Width scaling coefficient for A1DR
SWKO		0	Width scaling coefficient for KO
SWKOD		0	Width scaling coefficient for the body factor of the drift region
SWTHE1		0	Width scaling coefficient for THE1
SWTHE2		0	Width scaling coefficient for THE2
SWTHE3		0	Width scaling coefficient for THE3
SWTHE3D		0	Width scaling coefficient for THE3D

Name	Unit	Default	Description
THE1ACC	V^{-1}	0.02	Mobility reduction coefficient in the drift region due to the vertical electrical field caused by accumulation
THE1R	V^{-1}	0.09	Mobility reduction coefficient of infinitely wide transistor, due to vertical strong-inversion field in a channel region
THE2R	$V^{(-1/2)}$	0.03	Mobility reduction coefficient for $V_{SB} > 0$ of an infinitely wide transistor, due to vertical depletion field in channel region
THE3DR	V^{-1}	0	Mobility reduction coefficient in a channel region of an infinitely wide transistor due to velocity saturation
THE3R	V^{-1}	0.4	Mobility reduction coefficient in a channel region of an infinitely wide transistor due to velocity saturation
TOX	m	3.80E-08	Thickness of the oxide above the channel region
TREF	deg.C	25	Reference temperature
VFB	V	-1	Flatband voltage of the channel region, at reference temperature
VFBD	V	-0.1	Flatband voltage of the drift region, at reference temperature
VP	V	0.05	Characteristic voltage of channel length modulation
WDVAR	m	0	Width offset of the drift region
WVAR	m	0	Width offset of the channel region

MOS Model 20 — Electrical Parameters

Name	Unit	Default	Description
VERSION		2002.2	VERSION 2002.2 is the default version number to define NXP model level 2002
GEOMOD		1	GEOMOD is used as a switch between geometrical model and electrical model, • 0 for geometrical model • 1 for electrical model and
JUNCAPMOD		0	JUNCAPMOD as a switch of juncap200.2 model effective, 0 for off and 1 for on
TYPE		1	TYPE 1: NMOS;TYPE -1: PMOS

Name	Unit	Default	Description
A1CH		15	Factor of channel weak avalanche current, at reference temperature
A1DR		15	Factor of drift weak avalanche current, at reference temperature
A2CH	V	73	Exponent of weak avalanche current, related to channel
A2DR	V	73	Exponent of weak avalanche current, related to drift
A3CH		0.8	Factor of the internal drain-source voltage, above which channel weak avalanche occurs
A3DR		0.8	Factor of the internal drain-source voltage, above which drift weak avalanche occurs
ALP		0.002	Factor for channel length modulation
ATH		0	Temperature coefficient of the thermal resistance
BET	AV^2	7.00E-05	Gain factor of a channel region, at reference temperature
BETACC	AV^2	7.00E-05	Gain factor of drift region of 1 um wide, at reference temperature
CGDO	F	0	Gate-to-drain overlap capacitance for a drift region
CGSO	F	0	Gate-to-source overlap capacitance for a drift region
COX	F	15E-15	Oxide capacitance for an intrinsic channel region
COXD	F	15E-15	Oxide capacitance for an intrinsic drift region
CTH	J/K	3.00E-09	Thermal capacitance
DTA	K	0	Temperature offset to the ambient temperature
ETABET		1.6	Temperature scaling exponent for BET
ETABETACC		1.5	Temperature scaling exponent for BETACC
ETARD		1.5	Temperature scaling exponent for RD
ETATHE3		1	Temperature scaling exponent for THE3
ETATHE3D		1	Temperature scaling exponent for THE3D
KO	$V^{(1/2)}$	1.6	Body factor of the channel region
KOD	$V^{(1/2)}$	1	Body factor of the drift region

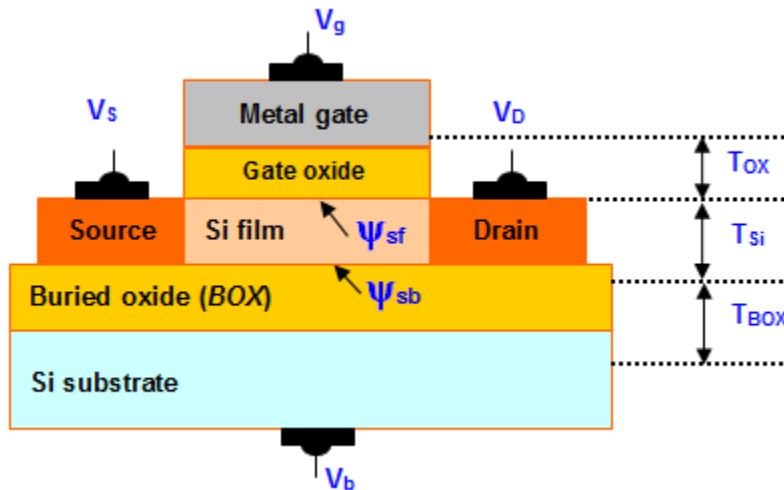
Name	Unit	Default	Description
LAMD		0.2	Quotient of the depletion layer thickness to the effective thickness of the drift region at $V_{SB} = 0$ V
MEXP		2	Smoothing factor for transition from linear to saturation regime
MEXPD		2	Smoothing factor for transition from linear to quasi-saturation regime
MO	V	0	Parameter for the (short-channel) sub-threshold slope
MSDIBL		3	Exponent for the drain-induced barrier lowering dependence on the backgate bias
NFA	$V^{-1m^{-4}}$	7.0E+23	First coefficient of flicker noise for a channel region
NFB	$V^{-1m^{-2}}$	1.0E+07	Second coefficient of flicker noise for a channel region
NFC	V^{-1}	0	Third coefficient of flicker noise for a channel region
NT	J	1.645E-20	Coefficient of thermal noise, at reference temperature
PHIB	V	0.86	Surface potential at the onset of strong inversion in the channel region, at reference temperature
PHIBD	V	0.78	Surface potential at the onset of strong inversion in the drift region, at reference temperature
RD	Ohm	200	On-resistance of a drift region, at reference temperature
RTH	K/W	300	Thermal resistance
SDIBL	$V^{(-1/2)}$	0.001	Factor for drain-induced barrier lowering
SHMOD		0	SHMOD is used as a switch of self-heating effect, 0 for off and 1 for on
SSF	$V^{(-1/2)}$	1E-12	Factor for static feedback
STA1CH	K^{-1}	0	Temperature scaling coefficient for A1CH
STA1DR	K^{-1}	0	Temperature scaling coefficient for A1DR
STPHIB	V/K	-0.0012	Temperature scaling coefficient for PHIB
STPHIBD	V/K	-0.0012	Temperature scaling coefficient for PHIBD
STVFB	V/K	0	Temperature scaling coefficient for VFB

Name	Unit	Default	Description
STVFBD	V/K	0	Temperature scaling coefficient for the flatband voltage of the drift region
THE1	V^{-1}	0.09	Mobility reduction coefficient due to vertical strong-inversion field in a channel region
THE1ACC	V^{-1}	0.02	Mobility reduction coefficient in the drift region due to the vertical electrical field caused by accumulation
THE2	$V^{(-1/2)}$	0.03	Mobility reduction coefficient for $V_{SB} > 0$, due to vertical depletion field in channel region
THE3	V^{-1}	0.4	Mobility reduction coefficient in a channel region due to velocity saturation
THE3D	V^{-1}	0	Mobility reduction coefficient in a channel region due to velocity saturation
TOX	m	3.80E-08	Thickness of the oxide above the channel region
TREF	deg.C	25	Reference temperature
VFB	V	-1	Flatband voltage of the channel region, at reference temperature
VFBD	V	-0.1	Flatband voltage of the drift region, at reference temperature
VP	V	0.05	Characteristic voltage of channel length modulation

LEVEL 76: LETI-UTSOI MOSFET Model

The *UTSOI* [11] model developed by CEA-LETI is a surface-potential based model dedicated to Planar Ultra-Thin SOI MOSFET device. Similarly *PSP* [12] is a surface-potential model dedicated to Planar MOSFET on Bulk Silicon.

The UTSOI model has been designed to be user-friendly for PSP users. The device structure is illustrated by the following figure:



The UTSOI model is designed for transistors with a lightly doped silicon film. The silicon film thickness is typically lower than 10nm.

This model is compatible with the use of thin buried oxide (typically 10nm). However, this model is not designed for double gate transistor where the back Si-SiO₂ interface can be in inversion.

This model supports the advanced MOSFET self-heating simulation solution.

LETI-UTSOI MOSFET Model—General Syntax

The general syntax for including a LETI-UTSOI MOSFET model element in a netlist is:

```
Mxxx nd ng ns nb [nth] mname [L=val] [W=val] [MULT=val]
+ [ASOURCE=val] [ADRAIN=val] [PSOURCE=val] [PDRAIN=val]
+ [SA=val] [SB=val] [SD=val] [NF=val] [DTEMP=val]
+ [DELVTO=val] [FACTUO=val]
```

Note:

Where:

- nd , ng , ns , and nb indicate the drain, gate, source, and bulk node respectively.
- nth is the thermal node.

LETI-UTSOI MOSFET Model Updates

This section covers the following topics:

- [LETI-UTSOI MOSFET Model 102.7 Updates](#)
- [LETI-UTSOI MOSFET Model 102.6 Updates](#)
- [LETI-UTSOI MOSFET Model 102.5 Updates](#)
- [LETI-UTSOI MOSFET Model 102.4 Updates](#)
- [LETI-UTSOI MOSFET Model 2.3.0 Updates](#)
- [LETI-UTSOI MOSFET Model 2.2.0 Updates](#)
- [LETI-UTSOI MOSFET Model 2.1.1 Updates](#)
- [LETI-UTSOI MOSFET Model 2.1.0 Updates](#)
- [LETI-UTSOI MOSFET Model 2.0.0 Updates](#)
- [LETI-UTSOI MOSFET Model 1.1.4 Updates](#)
- [LETI-UTSOI MOSFET Model 1.1.3 Updates](#)
- [LETI-UTSOI MOSFET Model 1.1.2 Updates](#)
- [LETI-UTSOI MOSFET Model 1.1.1 Updates](#)

LETI-UTSOI MOSFET Model 102.7 Updates

The following summarizes the changes from L-UTSOI v102.6 to L-UTSOI v102.7:

- Potential reference shift from the source intrinsic energy level to the source conduction band energy level. This shift allows simulations down to 1K.
- Introduction of the switch SWCRYO to activate low temperature effect model and low temperature accuracy improvement.

- Introduction of low temperature subthreshold slope saturation model. TMIN, ATMIN and BTMIN are the related model parameters.
- Extra global error correction calculation for low temperature simulation.

LETI-UTSOI MOSFET Model 102.6 Updates

The following summarizes the bug fixes and changes from UTSOI model 102.5 to UTSOI model 102.6:

- Introduction of RTA NQS model. KDRIFF, KDIFF, FRACINV and KFRACINV are the related model parameters. Use model parameter SWNQS=1 to turn on NQS effect.

LETI-UTSOI MOSFET Model 102.5 Updates

The following summarizes the bug fixes and changes from UTSOI model 102.4 to UTSOI model 102.5:

- Accuracy and predictability improvements:
 - Introduction of Gate Poly-Depletion model (introduction of SWPDEP flag). NP, NPO and NPL are the related model parameters (in local and global mode).
- Changes in model inputs and outputs:
 - TREF has been added and can be used instead of TR.
- Bug fixes:
 - Improvement of Cgb-Cbg in strong inversion to avoid negative values.
 - Correction of possible division by zero for TMAX parameter.
 - Correction of Flicker noise implementation.
 - Correction of q zero variable conditioning.

LETI-UTSOI MOSFET Model 102.4 Updates

The following summarizes the bug fixes and changes from UTSOI model 2.3.0 to UTSOI model 102.4 :

- Accuracy and predictability improvements:
- Introduction of impact ionization model inspired from PSP model (introduction of SWIMPACT flag). A1, A2, STA2 and A3 are the related model parameters.
- Introduction of extra gate to overlap current prefactor in inversion. FNOVINV, FNOVINVD, GCOVINVFN, and STIGFN are the related model parameters.
- Modification of scaling law for noise model NFA parameter.

LETI-UTSOI MOSFET Model 2.3.0 Updates

The following summarizes the bug fixes and changes from UTSOI model 2.2.0 to UTSOI model 2.3.0:

- Accuracy and predictability improvements:
 - Simplification of the implementation of self-heating.
 - Introduction of a model for the excess noise.
 - Global scale parameters AXL2 and AXLEXP2 are added for more flexibility in AX scaling description.
 - Global scale parameters AGIDLO and DGIDLO are added for more scaling flexibility of the GIDL model.
- Changes in model inputs and outputs:
 - Warnings on clipping of instance, global and local parameters, and channel temperature elevation have been updated.
- Bug fixes:
 - Correction of an undesired W-scaling of Cbs and Cbd capacitances that were introduced in Leti-UTSOI 2.2, for negative values of the narrow channel parameter PNCEW.
 - Correction of a small undesired scaling of the flicker noise in the subthreshold regime, as a function of gate length and back bias.

LETI-UTSOI MOSFET Model 2.2.0 Updates

The following summarizes the bug fixes and changes from UTSOI model 2.1.1 to UTSOI model 2.2.0:

- **Accuracy and predictability improvements**
 - Additional flexibility introduced in the channel length modulation model for more accurate description of current and conductance over applied biases.
 - Model extended to the case of doped thin film transistors, with the introduction of channel doping level local parameter NCH.
 - Channel length dependence over front and back bias in subthreshold and moderate inversion regimes introduced in Leti-UTSOI 2.1. This part of the model is improved for better accuracy, with a physical description of source/drain depletion effect.
 - Mobility model accuracy improved with the introduction of the transverse field dependence of the Coulomb scattering component.

- Series resistance model flexibility increased with the introduction of the source/drain extension component and explicit back bias dependence.
 - Gate-overlap tunneling current description at high drain voltage is improved.
 - Introduction the high longitudinal field dependence of GISL/GIDL currents.
 - Introduction of a physical inner fringe capacitance model, including source/drain depletion effect, for better description of short channel capacitances.
 - Overlap capacitance model flexibility increased to account for the source/drain depletion effect introduced in the DC model.
 - Global scale parameters VFBL2 and VFBLEXP2 added for more flexibility in VFB scaling description.
 - Global scale parameters ALPLEXP2 and ALP1LEXP2 added for more flexibility in ALP and ALP1 scaling descriptions, respectively.
 - Description of the edge effect in gate to substrate parasitic capacitance introduced through the global scale parameter CGBOVO.
 - Geometrical dependences of thermal resistance and thermal capacitance extended to multifinger transistors, by accounting for thermal coupling between fingers.
 - A new strain relaxation model, dedicated to strained-SOI technologies, is introduced besides the existing STI-based stress model.
 - High order derivability of the model around null source-drain voltage is improved.
- **Bug fixes**

LETI-UTSOI MOSFET Model 2.1.1 Updates

The following items summarize the bug fixes and changes from UTSOI Model 2.1.0 to UTSOI Model 2.1.1:

- Improvement of threshold voltage calculation in Operating Point section.
- Modification of smoothing in drain saturation voltage calculation to improve the smoothness of third order derivatives.
- Modification of smoothing in gate current model to improve the smoothness of third order derivatives.

LETI-UTSOI MOSFET Model 2.1.0 Updates

The following summarizes the bug fixes and changes from UTSOI Model 2.0.0 to UTSOI Model 2.1.0:

- The effect of effective channel length dependence on front- and back-gate biases has been introduced in the 2D electrostatic part of the model (subthreshold slope, DIBL) through local parameters `PSCEDL` and `CFDL`, and in overlap capacitance model through parameter `COVDL`.
- The impact of narrow channel effect on front/back gate to interface couplings has been introduced through parameter `PNCE`.
- To improve the description of gate to source/drain overlap tunnelling currents, `GC2OV` and `GC3OV` parameters have been split into `GC2OVINV`, `GC3OVINV`, `GC2OVACC`, and `GC3OVACC`.
- Gate to channel tunnelling current model has been improved for a better description in the subthreshold regime with the introduction of local parameter `NIGINV`.
- Predictability of the 2D electrostatic part of the model has been improved by linking subthreshold slope and DIBL-related parameters (`PSCE` and `CF`) to process parameters (`TSI`, `TOXE`, and `TBOX`) in scaling laws. In addition, introduction of subthreshold slope degradation parameter in model equations has been modified, so that threshold voltage roll-off predictability is also improved.
- Scaling law of `BETN` has been changed for better accuracy. PSP-like scaling law has been adopted.
- Introduction of narrow-short channel effect in VFB scaling law has been modified for better description.
- The impact of 2D electrostatic effect on back-gate depletion has been introduced through calculation of an effective back-gate doping level.
- Warnings about clipping of instance, global/local parameters, and channel temperature elevation have been added. Display is controlled through the `SWCLIPCHK` flag.
- Device temperature node was internal in previous versions and corresponded to 1/100th of channel temperature elevation due to self-heating. This node is now accessible and represents the actual channel temperature elevation. Device instantiation now requires 5 nodes.
- Modification of some parameter clipping: `MULT` min value from 1 to 0, `PSCE` min value from -0.5 to 0 and `AX` max value from 12 to 16. Change of `RTHL`, `RTHW`, and `RTHLW` default values.
- Fix some bugs.

LETI-UTSOI MOSFET Model 2.0.0 Updates

The following summarizes the bug fixes and changes from UTSOI Model 1.1.4 to UTSOI Model 2.0.0 per the CEA-LETI UTSOI release note:

- `FUTSOI2` has been developed with the aim of being able to describe device operations with a strongly inverted back interface. Thus, an original analytical procedure has been developed to calculate the exact values of the surface potentials at front and back interfaces in all operating conditions.
- Besides this surface potential calculation, new drain current and intrinsic charge models have also been developed. It should be noticed that this new model core is valid not only for `FDSOI` technologies, but more generally for all Independent Double Gate device architectures.
- The effect of backplane depletion is introduced in `UTSOI2` and computed through a bulk `MOSFET` like surface potential calculation. This effect can be turned-on or turned-off using the `SWSUBDEP` flag. If the flag is set to 0, the backplane is assumed metallic, as in `UTSOI1`.
- `UTSOI2` offers the possibility to define two different junctions at source and drain sides, as done in PSP. Junction related parameters are thus duplicated, the drain related parameters having the same name as their source side counterparts plus a final "D". This possible junction asymmetry can be turned-on or turned-off using the `SWJUNASYM` flag. When it is turned-off, drain related parameters are ignored.
- It should be noticed that the `UTSOI1` possibility to switch from external to internal series resistance through the `SWRSMOD` flag is no longer included in `UTSOI2`.

LETI-UTSOI MOSFET Model 1.1.4 Updates

The following summarizes the bug fixes and changes from UTSOI Model 1.1.3 to UTSOI Model 1.1.4 per the CEA-LETI UTSOI release note:

- Drain-bulk and source-bulk perimeter capacitances.
- New parameter for high-K gate oxide (`EPSROX`).
- New parameters to describe the channel material (`EPSRSI`, `EG`, `STEG1`, `STEG2`, `NI`, `STNI`).
- Geometrical dependence of `XCOR` and `CS` parameters.
- New implementation of the HF noise model.

- Several bug fixes:
 - A new expression of M_{utmp} variable has been implemented to avoid unexpected transition in the calculation.
 - A new expression of BET_{ith} variable has been implemented.
 - New variables have been added in the HF noise model implementation.

LETI-UTSOI MOSFET Model 1.1.3 Updates

The following summarizes the bug fixes and changes from UTSOI Model 1.1.2 to UTSOI Model 1.1.3 per the CEA-LETI UTSOI release note:

- A switch to turn-off the induced gate noise model.
- Operation point output variables.
- Several bug fixes:
 - A new expression of SP_S_delta0 variable has been implemented in the surface potential calculation.
 - Calculation errors of drain and source has been fixed.
 - New min/max for $GC3CH$, $GC3OV$, $GC3CHO$, $GC3OVO$ parameters have been implemented.
 - A new expression for V_{bx} , G_{mob} , D_d , q_{im} , q_{im1} , $temp$, and $t1$ variables has been implemented.

LETI-UTSOI MOSFET Model 1.1.2 Updates

The following summarizes the bug fixes and changes from UTSOI Model 1.1.1 to UTSOI Model 1.1.2 per the CEA-LETI UTSOI release note:

- Improvement of accurate during the calculation of the surface potential in the overlaps.
- The overlap capacitance and the fringing capacitance contributions are separated (COV parameter is added).
- Maximum NOV and $NOVO$ values are now 10^{21}cm^{-3} .
- Several bug fixes.

LETI-UTSOI MOSFET Model 1.1.1 Updates

This version is the first reference model version as per the CEA-LETI UTSOI release note.

UTSOI MOSFET Model — Global Model Flags and Parameters

Name	Unit	Default	Min.	Max.	Description
TYPE	-	1	-1	1	Channel type parameter, +1=NMOS, -1=PMOS.
TR	°C	21.0	-273.0	-	Reference temperature.
SWSCALE	-	0	0	1	Flag for scaling rules, 0=local parameter set, 1=global parameter set.
VERSION	-	2.0	-	-	Flag for model version.
SWIGATE	-	0	0	1	Flag for gate current, 0=turn off.
SWGIDL	-	0	0	1	Flag for GIDL current, 0=turn off.
SWSHE	-	0	0	1	Flag for self heating effect, 0=turn off.
SWRSMOD	-	0	0	1	Flag for access resistance calculation, 0=includes in mobility model, 1=using internal nodes
SWIGN	-	1	0	1	Flag for induced gate noise, 0=turn off.

UTSOI MOSFET Model — Local Level Parameters (SWSCALE=0)

This section covers the following topics:

- [UTSOI Model — Local Level Instance Parameters](#)
- [UTSOI Model — Local Level Process Parameters](#)
- [UTSOI Model — Local Level Quantum Effect Parameter](#)
- [UTSOI Model — Local Level Interface Coupling Parameter](#)
- [UTSOI Model — Local Level Short Channel Effect Parameter](#)
- [UTSOI Model — Local Level DIBL Parameters](#)
- [UTSOI Model — Local Level Mobility Parameters](#)
- [UTSOI Model — Local Level Series Resistance Parameters](#)
- [UTSOI Model — Local Level Velocity Saturation Parameters](#)
- [UTSOI Model — Local Level Saturation Voltage Parameters](#)

- [UTSOI Model — Local Level Channel Length Modulation Parameters](#)
- [UTSOI Model — Global Level Gate Current Parameters](#)
- [UTSOI Model — Local Level Gate Induced Drain/Source Leakage Current Parameters](#)
- [UTSOI Model — Local Level Charge Model Parameters](#)
- [UTSOI Model — Local Level Self Heating Parameters](#)
- [UTSOI Model — Local Level Noise Model Parameters](#)

UTSOI Model — Local Level Instance Parameters

Name	Unit	Default	Min.	Max.	Description
ASOURCE	m ²	10 ⁻¹²	0	-	Source region area.
ADRAIN	m ²	10 ⁻¹²	0	-	Drain region area.
PSOURCE	m	10 ⁻⁶	0	-	Source region perimeter.
PDRAIN	m	10 ⁻⁶	0	-	Drain region perimeter.
MULT	-	1	1	-	Number of device in parallel.

UTSOI Model — Local Level Process Parameters

Name	Unit	Default	Min.	Max.	Description
VFB	V	0.0	-	-	Flat-band voltage at TR.
STVFB	V/K	5.10 ⁻⁴	-	-	Temperature dependence of VFB.
TOX	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Gate oxide thickness.
EPSROX	-	3.9	1.0	-	Relative permittivity of gate dielectric.
TSI	m	10 ⁻⁸	10 ⁻⁹	10 ⁻⁷	Silicon film thickness.
EPSRSI	-	11.8	11.8	16.5	Relative permittivity of silicon film.
EG	V	1.179	0.6	1.2	Band-gap of silicon film at 300K.
STEG1	V/°C	9,025.10 ⁻⁵	-	-	First temperature coefficient of EG.
STEG2	V/°C ²	3,05.10 ⁻⁷	-	-	Second temperature coefficient of EG.
NI	cm ⁻³	1,45.10 ¹⁰	10 ¹⁰	10 ¹⁴	Intrinsic doping of silicon film at 300K.
STNI	-	1.5	0.5	3.0	Temperature dependence factor of NI.

Name	Unit	Default	Min.	Max.	Description
TBOX	m	10^{-7}	10^{-8}	10^{-6}	Buried oxide thickness.
NSI	cm^{-3}	0.0	-	10^{18}	Lightly silicon film doping, 0=undoped.
NSUB	cm^{-3}	$3 \cdot 10^{18}$	10^{16}	10^{21}	Substrate doping, negative value=N-type, positive value=P-type.
DVFBB	V	0.0	-	-	Offset of back-gate flat-band voltage.
CT	-	0.0	0.0	-	Interface states factor.
TOXOV	m	$2 \cdot 10^{-9}$	10^{-10}	-	Overlap oxide thickness.
NOV	cm^{-3}	0.0	10^{17}	10^{21}	Effective doping of overlap region, 0=no doping effect.
NCH	cm^{-3}	0.0	0.0	10^{19}	Thin film doping (n-type negative value, p-type positive value)

UTSOI Model — Local Level Quantum Effect Parameter

Name	Unit	Default	Min.	Max.	Description
QMC	-	1	0	-	Quantum correction factor.

UTSOI Model — Local Level Interface Coupling Parameter

Name	Unit	Default	Min.	Max.	Description
CIC	-	1.0	0.1	10.0	Substrate bias dependence factor of interface coupling.
NSDDC	cm^{-3}	1.0e^{22}	10^{18}	10^{22}	Source/Drain effective doping level for DC model.
PSCEDLB	-	0.0	0.0	-	Back bias dependence of short channel effect modulation.

UTSOI Model — Local Level Short Channel Effect Parameter

Name	Unit	Default	Min.	Max.	Description
PSCE	-	0.0	0.0	-	SCE-parameter above threshold.

UTSOI Model — Local Level DIBL Parameters

Name	Unit	Default	Min.	Max.	Description
CF	V^{-1}	0.0	0.0	-	DIBL-parameter.
CFB	V^{-1}	0.0	0.0	-	Substrate bias dependence of CF.
CFDLB	-	0.0	0.0	-	Back bias dependence of DIBL modulation.
STCF	-	0.0	-	-	Temperature dependence of CF.

UTSOI Model — Local Level Mobility Parameters

Name	Unit	Default	Min.	Max.	Description
BETN	$m^2/V/s$	$5 \cdot 10^{-2}$	10^{-10}	-	Channel aspect ratio times zero-field mobility
STBET	-	1.0	-	-	Temperature dependence of BETN
MUE	m/V	0.5	0.0	-	Mobility reduction coefficient at TR
STMUE	-	0.0	-	-	Temperature dependence of MUE
THEMU	-	1.5	0.0	-	Mobility reduction exponent at TR.
STTHEMU	-	1.5	-	-	Temperature dependence of THEMU.
CS	-	0.0	0.0	-	Remote Coulomb scattering parameter at TR.
CSB	-	0.0	-	-	Substrate bias dependence of CS
CSBI	-	0.0	0.0	-	Field dependence of Coulomb scattering at back interface
CSFI	-	0.0	0.0	-	Field dependence of Coulomb scattering at front interface
CSTHRB	-	1.0	0.1	-	Remote Coulomb scattering threshold asymmetry parameter
THECS	-	1.5	0.0	-	Remote Coulomb scattering exponent at TR.
STTHECS	-	1.5	-	-	Temperature dependence of THECS.
STCS	-	0.0	-	-	Temperature dependence of CS.

Name	Unit	Default	Min.	Max.	Description
XCOR	V^{-1}	0.0	0.0	-	Non-universality factor.
XCORB	-	1.0	-	-	Asymmetry term of non-universality factor.
STXCOR	-	0.0	-	-	Temperature dependence of XCOR.
FETA	-	1.0	0.0	-	Effective field parameter.

UTSOI Model — Local Level Series Resistance Parameters

Name	Unit	Default	Min.	Max.	Description
RS	W	30.0	0.0	-	Source/Drain series resistance at TR.
RSG	-	0.0	-0.5	-	Gate bias dependence of RS.
RSB	-	0.0	-	-	Back bias dependence of RS.
RSIG	-	0.0	0.0	-	Source/Drain extension resistance coefficient.
THERSG	-	2.0	0.0	-	Gate bias dependence exponent of RS.
STRS	-	1.0	-	-	Temperature dependence of RS.

UTSOI Model — Local Level Velocity Saturation Parameters

Name	Unit	Default	Min.	Max.	Description
THESAT	V^{-1}	0.0	0.0	-	Velocity saturation parameter at TR.
STTHE SAT	-	1.0	-	-	Temperature dependence of THESAT.
THESATB	-	0.0	-0.5	-	Substrate bias dependence of velocity saturation.
THESATG	V^{-1}	0.0	0.0	-	Gate bias dependence of velocity saturation.

UTSOI Model — Local Level Saturation Voltage Parameters

Name	Unit	Default	Min.	Max.	Description
AX	-	10.0	1.0	10.0	Linear/saturation transition factor.

UTSOI Model — Local Level Channel Length Modulation Parameters

Name	Unit	Default	Min.	Max.	Description
ALP	?	0	0	-	CLM pre-factor.
ALP1	V	0	0	-	CLM enhancement factor above threshold.
ALPB	-	0.0	-	-	Back bias dependence of channel length modulation.
VP	V	0.05	10^{-10}	-	CLM logarithm dependence factor.
VPG	-	0.0	0.0	-	Transverse electric field dependence of CLM logarithm factor.

UTSOI Model — Local Level Gate Current Parameters

Name	Unit	Default	Min.	Max.	Description
GCO	-	0.0	-10	10.0	Gate tunneling energy adjustment.
IGINV	A	0.0	0.0	-	Gate to channel current pre-factor in inversion.
IGOVINV	A	0.0	0.0	-	Gate to overlap current pre-factor in inversion.
IGOVACC	A	0.0	0.0	-	Gate to overlap current pre-factor in accumulation.
GC2CH	-	0.375	0.0	10.0	Gate current slope factor for gate to channel current.
GC3CH	-	0.063	-2.0	2.0	Gate current curvature factor for gate to channel current.
GC2OV	-	0.375	0.0	10.0	Gate current slope factor for overlap currents.

Name	Unit	Default	Min.	Max.	Description
GC3OV	-	0.063	-2.0	2.0	Gate current curvature factor for overlap currents.
GCDOV	V^{-1}	0.0	-	-	High drain voltage dependence of overlap gate current.
GCVDOV	V	1.0	-	-	Threshold of high drain voltage effect on overlap gate current.
STIG	-	2.0	-	-	Temperature dependence of all gate currents.
CHIB	V	3.1	1.0	-	Tunneling barrier height.

UTSOI Model — Local Level Gate Induced Drain/Source Leakage Current Parameters

Name	Unit	Default	Min.	Max.	Description
AGIDL	A/V^3	0.0	0.0	-	GIDL pre-factor.
BGIDL	V	41.0	0.0	-	GIDL probability factor at TR.
STBGIDL	V/K	0.0	-	-	Temperature dependence of BGIDL.
CGIDL	-	0.0	-	-	Substrate bias dependence of GIDL.
DGIDL	V^{-1}	0.0	-	-	High longitudinal field parameter of GIDL.
DGIDLD	V^{-1}	0.0	-	-	High longitudinal field parameter of GIDL at drain side.

UTSOI Model — Local Level Charge Model Parameters

Name	Unit	Default	Min.	Max.	Description
COX	F	10^{-14}	0.0	-	Oxide capacitance for intrinsic channel.
CBOX	F/m^2	$5 \cdot 10^{-4}$	0.0	-	Unit area buried oxide capacitance of drain/source region.
CGBOV	F	0.0	0.0	-	Oxide capacitance for gate-substrate overlap.

Name	Unit	Default	Min.	Max.	Description
CFR	F	0.0	0.0	-	Outer fringe capacitance.
COV	F	0.0	0.0	-	Overlap capacitance.
COVDLB	-	0.0	-	-	Overlap capacitance modulation with back bias.
CSDO	F	0.0	0.0	-	Outer drain-source capacitance.
CSDBP	F/m	0.0	0.0	-	Drain/source to substrate perimeter capacitance.
FIF	-	0.0	0.0	-	Inner fringe capacitance prefactor.
FSCEAC	-	0.0	0.0	-	Short channel effect adjustment factor for charge model.
NSDAC	cm ⁻³	10 ²²	10 ¹⁸	10 ²²	Source/Drain effective doping level for AC model.

UTSOI Model — Local Level Self Heating Parameters

Name	Unit	Default	Min.	Max.	Description
RTH	°C/W	1500.0	0.0	-	Substrate thermal resistance
STRTH	-	0.0	0.0	-	Temperature dependence of RTH
CTH	W.s /°C	5.10 ⁻¹⁰	0.0	-	Substrate thermal capacitance

UTSOI Model — Local Level Noise Model Parameters

Name	Unit	Default	Min.	Max.	Description
FNT	-	1.0	0.0	-	Thermal noise coefficient.
NFA	V ⁻¹ /m ⁴	8.10 ²²	0.0	-	First coefficient of flicker noise.
NFB	V ⁻¹ /m ²	3.10 ⁷	0.0	-	Second coefficient of flicker noise.
NFC	V ⁻¹	0.0	0.0	-	Third coefficient of flicker noise.
EF	-	1.0	0.1	-	Frequency coefficient of flicker noise.

UTSOI MOSFET Model — Global Level Parameters (SWSCALE=1)

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UTSOI Model — Global Level Instance Parameters

Name	Unit	Default	Min.	Max.	Description
L	m	10^{-6}	10^{-9}	-	Drawn channel length.
W	m	10^{-6}	10^{-9}	-	Drawn channel width.
SA	m	0.0	-	-	Distance between OD-edge and gate at source side.
SB	m	0.0	-	-	Distance between OD-edge and gate at drain side.

Name	Unit	Default	Min.	Max.	Description
SD	m	0.0	-	-	Distance between neighboring fingers.
ASOURCE	m ²	10 ⁻¹²	0.0	-	Source region area.
ADRAIN	m ²	10 ⁻¹²	0.0	-	Drain region area.
PSOURCE	m	10 ⁻⁶	0.0	-	Source region perimeter.
PDRAIN	m	10 ⁻⁶	0.0	-	Drain region perimeter.
NF	-	1.0	1.0	-	Number of fingers.
MULT	-	1.0	1.0	-	Number of device in parallel.

UTSOI Model — Global Level Geometry Scaling Parameters

Name	Unit	Default	Min.	Max.	Description
LVAR0	m	0.0	-	-	Geometry-independent difference between physical and drawn gate lengths.
LVARL	-	0.0	-	-	Length dependence of LPS.
LVARW	-	0.0	-	-	Width dependence of LPS.
LAP	m	0.0	-	-	Effective channel length reduction per side.
WVAR0	m	0.0	-	-	Geometry-independent difference between physical and drawn field-oxide opening.
WVARL	-	0.0	-	-	Length dependence of WOD.
WVARW	-	0.0	-	-	Width dependence of WOD.
WOT	m	0.0	-	-	Effective reduction of channel width per side.
DLQ	m	0.0	-	-	Effective channel length offset for CV.
DWQ	m	0.0	-	-	Effective channel width offset for CV.

UTSOI Model — Global Level Mechanical Stress Model Parameters

Name	Unit	Default	Min.	Max.	Description
SWSTR ESS	-	1	-	-	Stress model selection flag (1: Previous STI stress model, 2: New channel stress model)
SAREF	m	10^{-6}	10^{-9}	-	Reference distance between OD edge to poly from one side.
SBREF	m	10^{-6}	10^{-9}	-	Reference distance between OD edge to poly from other side.
WLOD	m	0.0	-	-	Width parameter.
KUO	m	0.0	-	-	Mobility degradation/enhancement coefficient.
KVSAT	m	0.0	-1.0	1.0	Saturation velocity degradation/enhancement parameter.
TKUO	-	0.0	-	-	Temperature coefficient of KUO.
LKUO	-	0.0	-	-	Length dependence of KUO.
WKUO	-	0.0	-	-	Width dependence of KUO.
PKUO	-	0.0	-	-	Cross-term dependence of KUO.
LLODKUO	-	0.0	0.0	-	Length parameter for mobility stress effect.
WLODKUO	-	0.0	0.0	-	Width parameter for mobility stress effect.
KVTHO	V.m	0.0	-	-	Threshold shift parameter.
LKVTHO	-	0.0	-	-	Length dependence of KVTHO.
WKVTHO	-	0.0	-	-	Width dependence of KVTHO.
PKVTHO	-	0.0	-	-	Cross-term dependence of KVTHO.
LLODVTH	-	0.0	0.0	-	Length parameter for threshold voltage stress effect.
WLODVTH	-	0.0	0.0	-	Width parameter for threshold voltage stress effect.

Name	Unit	Default	Min.	Max.	Description
STETAO	m	0.0	-	-	ETAO shift factor related to threshold voltage change.
LODETAO	-	1.0		-	ETAO shift modification factor.

Stress Model Parameters for SWSTRESS=1

Name	Unit	Default	Min.	Max.	Description
LOADDETAO	-	1.0	0.0	-	ETAO shift modification factor for stress effect.

Stress Model Parameters for SWSTRESS=2

Name	Unit	Default	Min.	Max.	Description
STRLAMBDA	m	10^{-7}	10^{-9}	10^{-5}	Asymmetry parameter
STRALPHA	-	3.0	0.5	-	Threshold shift parameter
STRDVFB0	V	0.0	-	-	Width dependence of threshold shift parameter
STRWDVFBO	-	0.0	-	-	DIBL variation parameter
STRDCFL	-	0.0	-	-	Mobility degradation/enhancement coefficient
STRRUO	-	0.0	-	-	Temperature dependence of mobility degradation/enhancement coefficient
STRTRUO	-	0.0	-	-	Saturation velocity degradation/enhancement coefficient
STRRVSAT	-	0.0	-	-	Asymmetry parameter

UTSOI Model — Global Level Process Parameters

Name	Unit	Default	Min.	Max.	Description
VFBO	V	0.0	-	-	Geometry-independent flat-band voltage at TR.
VFBL	-	0.0	-	-	Length dependence of VFB.
VFBLEXP	-	1.0	-	-	Exponent describing length dependence of VFB.
VFBL2	-	0.0	0.0	-	Second order length dependence of VFB.
VFBLEXP2	-	2.0	-	-	Exponent of second order length dependence of VFB
VFBW	-	0.0	-	-	Width dependence of VFB.
VFBLW	-	0.0	-	-	Area dependence of VFB.
STVFBO	V/K	5.10^{-4}	-	-	Geometry-independent temperature dependence of VFB.
STVFBL	-	0.0	-	-	Length dependence of STVFB.
STVFBW	-	0.0	-	-	Width dependence of STVFB.
STVFBLW	-	0.0	-	-	Area dependence of STVFB.
TOXO	m	2.10^{-9}	10^{-10}	-	Gate oxide thickness..
EPSROXO	-	3.9	1.0	-	Relative permittivity of gate dielectric.
TSIO	m	10^{-8}	10^{-9}	10^{-7}	Silicon film thickness.
EPSRSIO	-	11.8	11.8	16.5	Relative permittivity of silicon film.
EGO	V	1.179	0.6	1.2	Band-gap of silicon film at 300K.
STEG1O	V/°C	$9,025.10^{-5}$	-	-	First temperature coefficient of EG.
STEG2O	V/°C ²	$3,05.10^{-7}$	-	-	Second temperature coefficient of EG.
NIO	cm ⁻³	$1,45.10^{10}$	10^{10}	10^{14}	Intrinsic doping of silicon film at 300K.
STNIO	-	1.5	0.5	3.0	Temperature dependence factor of NI.

Name	Unit	Default	Min.	Max.	Description
TBOXO	m	10 ⁻⁷	10 ⁻⁸	10 ⁻⁶	Buried oxide thickness.
NCHO	cm ⁻³	0.0	0.0	10 ¹⁹	Thin film doping (n-type: negative value, p-type: positive value)
NSIO	cm ⁻³	0.0	-	-	Lightly silicon film doping, 0=intrinsic doping.
NSUBO	cm ⁻³	-3.10 ¹⁸	10 ¹⁶	10 ²¹	Substrate doping, negative value=N-type, positive value=P-type.
DVFBBO	V	0.0	-	-	Offset of back-gate flat-band voltage.
CTO	-	0.0	0.0	-	Interface states factor.
TOXOVO	m	2.10 ⁻⁹	10 ⁻¹⁰	-	Overlap oxide thickness.
LOV	m	0.0	0.0	-	Length of gate/drain and gate/source overlaps.
NOVO	cm ⁻³	0.0	10 ¹⁷	10 ²¹	Effective doping of overlap region, 0=No doping effect.

UTSOI Model — Global Level Quantum Effect Parameter

Name	Unit	Default	Min.	Max.	Description
QMC	-	1	0	-	Quantum correction factor.

UTSOI Model — Global Level Interface Coupling Parameter

Name	Unit	Default	Min.	Max.	Description
CICO	-	1.0	0.1	10.0	Geometry-independent part of substrate bias dependence factor of interface coupling.
CICL	-	0.0	-	-	Length dependence of CIC.
CICLEXP	-	1.0	-	-	Exponent describing length dependence of CIC.
CICW	-	0.0	-	-	Width dependence of CIC.
CICLW	-	0.0	-	-	Area dependence of CIC.

Name	Unit	Default	Min.	Max.	Description
NSDDCO	cm ⁻³	10 ²²	10 ¹⁸	10 ²²	Source/Drain effective doping level for DC model.
PSCEDLBO	-	0.0	0.0	-	Back bias dependence of short channel effect modulation.

UTSOI Model — Global Level Short Channel Effect Parameters

Name	Unit	Default	Min.	Max.	Description
PSCEL	-	0.0	0.0	-	Length dependence of short channel effect above threshold.
PSCELEXP	-	1.0	-	-	Exponent describing length dependence of PSCE.
PSCEW	-	0.0	0.0	-	Width dependence of PSCE.

UTSOI Model — Global Level DIBL Parameters

Name	Unit	Default	Min.	Max.	Description
CFL	V ⁻¹	0.0	0.0	-	Length dependence of DIBL parameter.
CFLEXP	-	2.0	-	-	Exponent for length dependence of CF.
CFW	-	0.0	-	-	Width dependence of CF.
CFBO	-	0.0	0.0	-	Substrate bias dependence of CF.
CFDLBO	-	0.0	0.0	-	Back bias dependence of DIBL modulation.
STCFO	-	0.0	-	-	Temperature dependence of CF.

UTSOI Model — Global Level Mobility Parameters

Name	Unit	Default	Min.	Max.	Description
UO	m ² /V/s	5.10 ⁻²	-	-	Zero-field mobility at TR.
BETNL	-	0.0	-	-	Second order length dependence of BETN.

Name	Unit	Default	Min.	Max.	Description
BETNLEXP	-	1.0	-	-	Exponent for second order length dependence of BETN.
BETNW	-	0.0	-	-	Second order width dependence of BETN.
STBETO	-	1.0	-	-	Geometry-independent part of temperature dependence of BETN.
STBETL	-	0.0	-	-	Length dependence of STBET.
STBETW	-	0.0	-	-	Width dependence of STBET.
STBETLW	-	0.0	-	-	Area dependence of STBET.
MUEO	m/V	0.5	0.0	-	Mobility reduction coefficient at TR.
STMUEO	-	0.0	-	-	Temperature dependence of MUE.
THEMUO	-	1.5	0.0	-	Mobility reduction exponent at TR.
STTHE MUO	-	1.5	-	-	Temperature dependence of THEMU.
CSO	-	0.0	-	-	Geometry-independent part of remote coulomb scattering parameter at TR.
CSL	-	0.0	-	-	Length dependence of CS.
CSLEXP	-	1.0	-	-	Exponent describing length dependence of CS.
CSW	-	0.0	-	-	Width dependence of CS.
CSLW	-	0.0	-	-	Area dependence of CS.
CSBO	-	0.0	0.0	-	Back bias dependence of CS.
CSBIO	-	0.0	0.0	-	Field dependence of Coulomb scattering at back interface.
CSFIO	-	0.0	0.0	-	Field dependence of Coulomb scattering at front interface.
CSTHRBO	-	1.0	0.1	-	Remote Coulomb scattering threshold asymmetry parameter.
THECSO	-	1.5	0.0	-	Remote coulomb scattering exponent at TR3.

Name	Unit	Default	Min.	Max.	Description
STCSO	-	0.0	-	-	Temperature dependence of CS.
STCSL	-	0.0	-	-	Length dependence of STCS.
STCSW	-	0.0	-	-	Width dependence of STCS.
STCSLW	-	0.0	-	-	Area dependence of STCS.
STTHECSO	-	1.5	-	-	Temperature dependence of THECS.
XCORO	V^{-1}	0.0	-	-	Geometry-independent part non-universality factor.
XCORL	-	0.0	-	-	Length dependence of XCOR.
XCORLEXP	-	1.0	-	-	Exponent describing length dependence of XCOR.
XCORW	-	0.0	-	-	Width dependence of XCOR.
XCORLW	-	0.0	-	-	Area dependence of XCOR.
XCORBO	-	1.0	-	-	Asymmetry term of non-universality factor.
STXCORO	-	0.0	-	-	Temperature dependence of XCOR.
FETAO	-	1.0	0.0	-	Effective field parameter.

UTSOI Model — Global Level Series Resistance Parameters

Name	Unit	Default	Min.	Max.	Description
RSW1	W	30.0	-	-	Source/Drain series resistance for channel width WEN at TR.
RSW2	-	0.0	-	-	Higher-order width scaling of source/drain series resistance.
RSGO	-	0.0	-0.5	-	Gate-bias dependence of RS.
RSIGO	-	0.0	0.0	-	Source/Drain extension resistance coefficient.
RSBO	-	0.0	-	-	Back bias dependence of RS.

Name	Unit	Default	Min.	Max.	Description
THERSGO	-	2.0	-	-	Gate-bias dependence exponent of RS.
STRSO	-	1.0	-	-	Temperature dependence of RS.

UTSOI Model — Global Level Velocity Saturation Parameters

Name	Unit	Default	Min.	Max.	Description
THESATO	V^{-1}	0.0	0.0	-	Geometry-independent Velocity saturation parameter at TR.
THESATL	-	0.0	-	-	Length dependence of THESAT.
THESATLEXP	-	1.0	-	-	Exponent for length dependence of THESAT.
THESATW	-	0.0	-	-	Width dependence of THESAT.
THESATLW	-	0.0	-	-	Area dependence of THESAT.
THESATGO	V^{-1}	0.0	0.0	-	Geometry-independent gate bias dependence of velocity saturation parameter at TR.
STTHESATO	-	1.0	-	-	Geometry-independent of temperature dependence of THESAT.
STTHESATL	-	0.0	-	-	Length dependence of STTHESAT.
STTHESATW	-	0.0	-	-	Width dependence of STTHESAT.
STTHESATLW	-	0.0	-	-	Area dependence of STTHESAT.
THESATBO	V^{-1}	0.0	-0.5	1.0	Substrate bias dependence of velocity saturation.

UTSOI Model — Global Level Saturation Voltage Parameter

Name	Unit	Default	Min.	Max.	Description
AXO	-	10.0	-	-	Geometry-independent of linear/saturation transition factor.
AXL	-	0.0	0.0	-	Length dependence of AX.

Name	Unit	Default	Min.	Max.	Description
AXLEXP	-	1.0	0.0	-	Exponent for length dependence of AX.

UTSOI Model — Global Level Channel Length Modulation Parameters

Name	Unit	Default	Min.	Max.	Description
ALPL1	-	0.0	0.0	-	Length dependence of CLM pre-factor ALP.
ALPLEXP	-	1.0	-	-	Exponent for length dependence of ALP.
ALPL2	-	0.0	-	-	Second order length dependence of ALP.
ALPLEXP2	-	2.0	-	-	Exponent for second order length dependence of ALP.
ALPW	-	0.0	-	-	Width dependence of ALP.
ALP1L1	-	0.0	-	-	Length dependence of CLM enhancement factor above threshold ALP1.
ALP1LEXP	-	0.5	-	-	Exponent for length dependence of ALP1.
ALP1L2	-	0.0	-	-	Second order length dependence of ALP1.
ALP1LEXP2	-	1.5	-	-	Exponent for second order length dependence of ALP1
ALP1W	-	0.0	-	-	Width dependence of ALP1.
ALPBO	-	0.0	-	-	Back bias dependence of channel length modulation.
VPO	V	0.05	10^{-10}	-	CLM logarithm dependence factor.
VPGO	-	0.0	0.0	-	Transverse electric field dependence of CLM logarithm factor.

UTSOI Model — Global Level Gate Current Parameters

Name	Unit	Default	Min.	Max.	Description
GCOO	-	0.0	-10.0	10.0	Gate tunneling energy adjustment
IGINVLW	A	0.0	0.0	-	Gate channel current pre-factor for a channel area of WEN.LEN
IGOVINW	A	0.0	0.0	-	Gate to overlap current pre-factor in inversion for an overlap of WEN.LOV
IGOVACCW	A	0.0	0.0	-	Gate to overlap current pre-factor in accumulation overlap of WEN.LOV
GC2CHO	-	0.375	0.0	10.0	Gate current slope factor for gate to channel current
GC3CHO	-	0.063	-2.0	2.0	Gate current curvature factor for gate to channel current
GC2OVO	-	0.375	0.0	10.0	Gate current slope factor for overlap currents
GC3OVO	-	0.063	-2.0	2.0	Gate current curvature factor for overlap currents
GCDOVL	V ⁻¹	0.0	-	-	High drain voltage dependence of overlap gate current.
GCVDOVO	V	1.0	-	-	Threshold of high drain voltage effect on overlap gate current.
STIGO	-	2.0	-	-	Temperature dependence of all gate currents
CHIBO	V	3.1	1.0	-	Tunneling barrier height

UTSOI Model — Global Level Gate Induced Drain/Source Leakage Current Parameters

Name	Unit	Default	Min.	Max.	Description
AGIDLW	A/V ³	0.0	0.0	-	Width dependence of GIDL pre-factor.

Name	Unit	Default	Min.	Max.	Description
BGIDLO	V	41.0	0.0	-	GIDL probability factor at TR.
STBGIDLO	V/K	0.0	-	-	Temperature dependence of BGIDL.
CGIDLO	-	0.0	-	-	Substrate bias dependence of GIDL.
DGIDLL	V^{-1}	0.0	-	-	High longitudinal field parameter of GIDL.
DGIDLDL	V^{-1}	0.0	-	-	High longitudinal field parameter of GIDL at drain side.

UTSOI Model — Global Level Charge Model Parameters

Name	Unit	Default	Min.	Max.	Description
CGBOVL	F	0.0	-	-	Length dependent gate-substrate overlap capacitance part for a length of LEN.
CFRW	F	0.0	0.0	-	Outer fringe capacitance.
CSDBPO	F/m	0.0	0.0	-	Drain/source to substrate perimeter capacitance.
NSDACO	cm^{-3}	10^{22}	10^{18}	10^{22}	Source/Drain effective doping level for AC model.
FIFW	-	0.0	0.0	-	Inner fringe capacitance prefactor for a width of WEN.
FSCEACO	-	0.0	0.0	-	Short channel effect adjustment factor for charge model.
COVDLBO	-	0.0	-	-	Overlap capacitance modulation with back bias

UTSOI Model — Global Level Self Heating Parameters

Name	Unit	Default	Min.	Max.	Description
RTHO	$^{\circ}C/W$	1500.0	0.0	-	Geometry-independent part of substrate thermal resistance.
RTHL	-	0.0	-	-	Length dependence of RTH.

Name	Unit	Default	Min.	Max.	Description
RTHW	-	0.0	-	-	Width dependence of RTH.
RTHLW	-	0.0	-	-	Area dependence of RTH.
CTHO	W.s/°C	$5 \cdot 10^{-10}$	0.0	-	Geometry-independent part of substrate thermal capacitance.
LAMBTHO	m	10^{-7}	10^{-9}	-	Characteristic length of lateral thermal coupling for multifinger devices.
FTHO	-	0.0	0.0	-	First neighbor thermal coupling factor for multifinger devices.
STRTHO	-	0.0	-	-	Temperature dependence of substrate thermal resistance.

UTSOI Model — Global Level Noise Model Parameters

Name	Unit	Default	Min.	Max.	Description
FNTO	-	1.0	0.0	-	Thermal noise coefficient.
NFALW	V^{-1}/m^4	$8 \cdot 10^{22}$	0.0	-	First coefficient of flicker noise.
NFBLW	V^{-1}/m^2	$3 \cdot 10^7$	0.0	-	Second coefficient of flicker noise.
NFCLW	V^{-1}	0.0	0.0	-	Third coefficient of flicker noise.
EFO	-	1.0	0.1	-	Frequency coefficient of flicker noise.

LEVEL 77: BSIM6 MOSFET Model

The BSIM6 model is a compact model to describe the behavior of bulk planar MOSFET devices.

This model captures all important transistor behaviors. BSIM6, as a charge based physical compact model, is believed to be more accurate for current and charge modeling than the V_{th}-based compact models when devices operation is from the sub-V_{th} to strong inversion region. And so, it is more suitable for low-power, low V_{dd} or RF and analog applications.

BSIM6 is a result of a collaboration and co-development between UC Berkeley and a team at EPFL (École Polytechnique Fédérale de Lausanne) that developed the EKV model in the past. As a result, BSIM6 incorporates a few features from the EKV model, most notably:

- The normalized pinch-off potential.
- The saturation voltage.

Included in the model are other important effects, such as: mobility degradation, velocity saturation, velocity overshoot, series resistance, channel length modulation, quantum mechanical effects, poly depletion, gate tunneling current, gate-induced-drain-leakage, and parasitic capacitance models.

This section covers the following topics:

- [BSIM6 MOSFET Model — General Syntax](#)
- [BSIM6 MOSFET Model — Version Updates](#)
- [BSIM6 MOSFET Model Parameters](#)

BSIM6 MOSFET Model — General Syntax

Syntax:

```
Mxxx nd ng ns nb mname [L=val] [W=val] [M=val]
+ [AD=val] [AS=val] [PD=val] [PS=val]
+ [RGATEMOD=val] [RBODYMOD=val]
+ [GEOMOD=val] [RGEOMOD=val]
+ [NRS=val] [NRD=val] [RBPB=val] [RBPB=val]
+ [RBPS=val] [RBDB=val] [RBSB=val] [NF=val]
+ [MINZ=val] [VFBSDOFF=val] [XGW=val] [NGCON=val]
+ [SA=val] [SB=val] [SC=val] [SD=val]
+ [SCA=val] [SCB=val] [SCC=val]
+ [OFF] [IC=Vds, Vgs, Vbs]
```

BSIM6 MOSFET Model — Version Updates

This section covers the following topic:

- [BSIM-BULK 107.1 Updates](#)
- [BSIM-BULK 107 Updates](#)
- [BSIM-BULK 106.2 Updates](#)
- [BSIM6 Model Version 6.1.1 Updates](#)
- [BSIM6 Model Version 6.1.0 Updates](#)

BSIM-BULK 107.1 Updates

This version includes the following enhancements:

- ABULK Implementation in I-V
- Flexibility in tuning Cgg
- Enhancing the fitting flexibility of Cgd at high Vds
- Drift resistance model Implementation for Id-Vg shows spikes in transient simulation
- Add the flicker noise and white noise for both Rdrift_d and Rdrift_s
- Modeling of drift region with back bias in HV extension
- New model parameters for EDGEFET
- Flat band voltage scaling parameters for I-V
- Improvement to the implementation of body-drain diode
- Gate bias dependence in the depletion model of drift region
- Improved modeling of Id-Vd at high Vg and Vd for HV model
- Change default values of model parameters related to CVMOD = 1
- Improved flexibility in asymmetry mode
- Fix convergence issue due to DVT2EDGE
- Clamping of gate electrode resistance in RGATEMOD=2
- Limit XRCRG1 & XRCRG2 parameters
- Missing multiplication factor in 2nd instance of T5 in Rdss=0
- Removing redundancy in the calculation of Rdss

- Improved smoothing of drift resistance in HV module
- Fix issues raised by VAMPyRE
- Drain-body junction current splitting in RBODYHVMOD
- Protection for junction capacitance grading coefficient (MJX) against 0/0 form
- Smoothing function for psiph
- Add default values of body-bias dependent parameters in HVMOD
- Operating point should include charge Qbdj_ext from external diode
- Clamping Nsat for non-saturation effect
- Limit model parameter MDRIFT
- Allow minr = 0
- Model parameter ABULK added for backward compatibility
- Fix divide by zero error when parameters DMCG / (DMCG+DMCI) are zero
- Limit the layout-dependent parasitic model parameters
- Few other minor changes

BSIM-BULK 107 Updates

This version includes the following enhancements:

- Add new model flag HVMOD for high voltage model
- Add new model flag HVCAP for high voltage capacitance model
- Add new model flag HVCAPS for high voltage capacitance model at source side
- Add new model flag IIMOD for impact ionization in the drift region
- Fix kink due to large PSATB values
- Gm/Id tuning flexibility enhancement
- Flexibility of tuning Cgg in strong inversion for Vds ≠ 0
- Fix negative Cbg issue for CVMOD = 0
- Flexibility of tuning Cgg in accumulation
- Add temperature dependence for the parameter EU
- Changes in default value of the L_H parameter.

- Valid condition for $L \leq L_H$
- Enhance the BSIMBULK flicker noise tuning flexibility
- Evaluating $K2_EDGE$ and $ETA0_EDGE$ independent of STI stress model
- Device type implementation in sub-surface leakage model
- Abnormal flicker noise performance when $EDGEFET=1$
- Aligning BSIMBULK $1/f$ behavior with BSIM4 $1/f$ behavior
- Corrected implementation of sub-surface leakage model
- Enhancing the tuning flexibility in the flicker noise model
- Add flicker noise due to $EDGEFET$
- Fix the impact of non-zero $C0$ on thermal noise
- Fix PSS (periodic state analysis) issue. $Esatnoi$ and Nt parameters are set before they are used
- Implementation of $ABULK$ is updated
- $phib$ and gam are replaced by $phibCV$ and $gamCV$ respectively in $CVMOD=1$
- Add model parameter $NDEPEDGE$ and its binning parameters
- Add new model parameter $minr$ for node collapsing and minimum value of R_{drain}/R_{source}
- Fix incorrect $IGBBACC$ behavior in the accumulation region
- Enhancement for High Voltage modeling
- Rollback $MNUD$ and $MNUD1$ from thermal noise model
- Addressing $IGBACC$ issue
- Implementing $sigvds$ in flicker noise

BSIM-BULK 106.2 Updates

This version includes the following enhancements:

- Module name changed from BSIM6 to BSIMBULK.
- Modified flicker noise model which can also model the flicker noise trends for Halo transistors by making $FNOIMOD = 0/1$.
- Added modified function of $VDSX$ to avoid negative GDS issue.

- Added MNUD model to increase flexibility in IDS - VDS fitting.
- Removed kink around VDS = 0 for GIDLMOD = 1.
- Improved Gamma behavior in the linear Region for TNOIMOD = 1.
- Added model for sub-surface leakage (between drain and source) by flag-SSLMOD.
- Added parameters for EDGEFET to control threshold voltage shift.
- Sign correction in charge calculations.
- Protection on DLCIG and DLCIGD from going negative.
- Removed clamping from UCR.
- Added protection to the following parameters: LP1, LP2, NJS, NJD, XJBVS and XJBVD.
- Added K2EDGE parameter for VTH shift with body bias for EDGEFET.
- Default values of stress parameters for EDGEFET are set the same as the main parameters.
- Used limited log in fringe Capacitance Module.
- Added Clamping in IGD and IGS.
- Deactivated node "T" when RTH0 or SHMOD or both are zero.
- Multiplied with DEV SIGN in the operating point VTH expression.
- Limit (Weff +WTH0) to (Weff +WTH0) > 0 in Self Heating module.
- Added CLM and velocity saturation terms of the charge expressions in manual.
- Modified the name of the parameter "MULUO" to "MULU0".
- SCA, SCB, SCC, and SC are allowed to be real numbers instead of integers.
- DTEMP, MULU0, DELVTO, EDGEFET, and IDS0MULT have modified to both instance and model parameters.
- Few other minor changes.

BSIM6 Model Version 6.1.1 Updates

This version includes the following enhancements:

- Addition of threshold voltage model for operating point information.
- Addition of EDGE FET model (sub-threshold hump module).

- Addition of an external thermal node.
- Addition of a mobility scaling model, similar to BSIM4, which is accessible through MOB-SCALE=1.
- Addition of an asymmetry mode, similar to BSIM-CMG. Accessible through Asymmetry mode=0/1.
- Addition of parameters DELVT0 and MULU0 for process variation.
- Addition of parameters A1 and A2 for better capturing of velocity saturation.
- Addition of parameter K1 for improved body effect model.
- Limitation of mobility parameters to be non-negative.
- Removal of protection on parameters K2 and CDSCB.
- Removal of GMIN.
- Well proximity effect (WPEMOD control flag) similar to BSIM4.
- Multiplication of NF with the effective width in the thermal noise model.
- Removal of clamping of UC.
- Removal of clamping of TNOIA, TNOIB, TNOIC.
- Protection on VTH shift due to DIBL.

BSIM6 Model Version 6.1.0 Updates

The following enhancements were added:

- A self-heating effect model.
- New CVMOD for consistent IV-CV.
- Length (L_{MLT}) and width (W_{MLT}) shrinking parameters.
- BINUNIT parameter to select binning unit.
- New source-drain resistance model with bias-independent external and bias-dependent internal resistance, introduced for RDSMOD=0 (similar to RDSMOD=0 in BSIM4). The RDSMOD=0 model of BSIM6.0.0 (where both bias-dependent and independent parts are internal) is now accessed via RDSMOD=2 in BSIM6.1.0.
- Operating point variable, T_K , which returns total device temperature in degrees of Kelvin.
- Binning equation pattern was modified for robustness by introducing length and width reduction parameters $DLBIN$ and $DWBIN$.

- Calculation of effective perimeter and area of source and drain region from layout when PS, PD, AS, and AD are not given.
- Mobility reduction factor (D_r) due to S/D resistance now considers the effect of velocity saturation (D_{vsat}) and vertical field mobility degradation (D_{mob}).
- L_{eff} and W_{eff} expressions were modified to align with BSIM4.
- Noise names were made similar to noise names in BSIM4.
- Check applied to ensure effective length and width are positive.
- Check applied on $nuEndS$ and $nuEndD$ for smooth operation when any one of them is zero.
- $NFACTOR_t$ is clamped for the lower bound to avoid negative values at low temperatures.
- Body bias dependency of Early voltage due to DIBL ($V_{a;DIBL}$) modified to avoid negative $V_{a;DIBL}$.
- $KVTH0WE$, $K2WE$, $KU0WE$, $KT1$, $KT2$, and $PSATB$ made binnable.
- Effective length and width for binning equations modified.
- $JTSSWGD_t$ and $JTSSWGS_t$ in diode temperature module updated to use W_{effcj} (as in BSIM4) instead of W used in BSIM6.0.0.
- Bug fixes.

BSIM6 MOSFET Model Parameters

The following sections define these parameters:

- [BSIM6 Model — Instance Parameters](#)
- [BSIM6 Model — Control and Process Parameters](#)
- [BSIM6 Model — Basic Parameters](#)
- [BSIM6 Model — Both Model and Instance Parameters](#)
- [BSIM6 Model — High-Speed/RF Parameters](#)
- [BSIM6 Model — Flicker and Thermal Noise Parameters](#)
- [BSIM6 Model — Layout-Dependent Parasitic Parameters](#)
- [BSIM6 Model — Asymmetric Source/Drain Junction Diode Parameters](#)
- [BSIM6 Model — Temperature-Dependent Parameters](#)

- [BSIM6 Model — Stress Effect Parameters](#)
- [BSIM6 Model — Well-Proximity Effect Parameters](#)
- [BSIM6 Model — Edge FET Device and Sub-Surface Leakage Current Parameters](#)

Note:

Binnable parameters are marked as ^(b).

BSIM6 Model — Instance Parameters

Parameter	Unit	Default	Min	Max	Description
AD	m ²	0	0	-	Drain to Substrate Junction Area
AS	m ²	0	0	-	Source to Substrate Junction Area
GEOMOD	-	0	0	10	Same as BSIM4
L	m	10u	1n	-	Designed Gate Length
MINZ	-	0	0	1	Minimize either number of drain or source ends
NF	-	1	1	-	Number of fingers
NGCON	-	1	1	2	Number of gate contacts
NRD	-	0	0	-	Number of drain diffusion squares
NRS	-	0	0	-	Number of source diffusion squares
PD	M	0	0	-	Drain to Substrate Junction Perimeter
PS	M	0	0	-	Source to Substrate Junction Perimeter
RBDB	Ohm	50	1mV	-	Resistance between dbNode and bNode
RBODYMOD	-	0	0	2	Substrate resistance network model selector
RBPB	Ohm	50	1mV	-	Resistance between bNodePrime and bNode
RBPB	Ohm	50	1mV	-	Resistance between bNodePrime and dbNode
RBPS	Ohm	50	1mV	-	Resistance between bNodePrime and sbNode
RBSB	Ohm	50	1mV	-	Resistance between sbNode and bNode
RGATEMOD	-	0	0	2	Gate resistance model selector
RGEOMOD	-	0	0	8	Bias independent parasitic resistance model selector (same as BSIM4)

Parameter	Unit	Default	Min	Max	Description
SA	-	0	0	-	Distance between OD edge from Poly from one side
SB	-	0	0	-	Distance between OD edge from Poly from other side
SC	-	0	0	-	Distance to a single well edge
SCA	-	0	0	-	Integral of the first distribution function for scatted well dopant
SCB	-	0	0	-	Integral of the second distribution function for scattered well dopant
SCC	-	0	0	-	Integral of third distribution function for scattered well dopant
SD	-	0	0	-	Distance between neighboring fingers
VFBSDOFF	-	0	-	-	Source-Drain flat band offset
W	m	10u	-	-	Designed Gate Width (per finger)
XGW	m	0	0	-	Distance from gate contact center to dev edge
EDGEFET	-	0	0	1	Edge FET model flag 0: Disable and 1: Enable If version > 6.11, the default value of EDGEFET is 0. If version <= 6.11, the default value of EDGEFET is 1.
SSLMOD	-	0	0	1	Sub-surface leakage drain current, 0: Turn off and 1: Turn on

BSIM6 Model — Control and Process Parameters

Parameter	Unit	Default	Min	Max	Description
VERSION	-	6.0	-	-	Model version number; VERSION can be 106.2, 6.11, 6.1, or 6.0
ASYMMOD	-	0	0	1	Asymmetry model is ON or OFF
BINUNIT	-	1	0	1	Binning unit selector
COVMOD	-	0	0	1	Bias-independent overlap capacitance:0, Bias-dependent overlap capacitance:1
CVMOD	-	0	0	1	IV-CV: Consistent:0, Dierent:1

Parameter	Unit	Default	Min	Max	Description
DLC ^(b)	m	0	-	-	Length reduction parameter for CV (dopant diffusion effect)
DMCG	-	0	0	-	Distance of mid-contact to gate edge
DMCGT	-	0	0	-	Distance of mid-contact to gate edge in test
DMCI	-	DCMG	0	-	Distance of mid-contact to isolation
DMDG	-	0	0	-	Distance of mid-diffusion to gate edge
DTOX	m	0.0	-	-	Difference between effective dielectric thickness and physical thickness
DWC ^(b)	m	0	-	-	Width reduction parameter for CV (dopant diffusion effect)
EASUB	eV	4.05	-	-	Electron affinity of substrate
EPSROX	-	3.9	1	-	Relative dielectric constant of the gate insulator
EPSRSUB	-	11.9	1	-	Relative dielectric constant of the channel material
GEOMOD	-	0	0	10	Model selector (same as BSIM4)
GIDLMOD	-	0	0	1	Turn-off GIDL model:0, Turn-on GIDL model:1
HVMOD	-	0	0	1	High Voltage series resistance model, 0: Turn Off 1: Turn On
HVCAP	-	0	0	1	High Voltage capacitance model. 0: Turn Off 1: Turn On
HVCAPS	-	0	0	1	High Voltage capacitance model at source side. 0: Turn Off 1: Turn On
IIMOD	-	0	0	1	Flag for impact ionization in the drift region 0: Turn-Off,1: Turn On
LINT ^(b)	m	0	0	-	Length reduction parameter (dopant diffusion effect)
MOBSCALE	-	0	0	1	Two different models for mobility
NDEP ^(b)	m ⁻³	1e24	-	-	channel (body) doping concentration. Global Scaling Parameters - NDEPL1, NDEPLEXP1, NDEPL2, NDEPLEXP2, NDEPW, NDEPWEXP, NDEPWL, NDEPWLEXP

Parameter	Unit	Default	Min	Max	Description
NGATE ^(b)	m ⁻³	5e25	-	-	Parameter for Poly Gate doping. Set NGATE = 0 for metal gates
NIOSUB	m ⁻³	1.1e16	-	-	Intrinsic carrier concentration of channel at 300.15K
NSD ^(b)	m ⁻³	1e26	2e25	1e27	S/D doping concentration
PERMOD	-	1	0	1	Whether PS/PD (when given) includes the gate-edge perimeter: 1: (including the gate-edge perimeter) 0: (not including the gate-edge perimeter)
RBODYMOD	-	0	0	2	Substrate resistance network model selector
RDSMOD	-	0	0	2	0: Bias dependent internal, independent external: 1: External RDS 2: Internal RDS
RGATEMOD	-	0	0	2	Gate resistance model selector
RGEOMOD	-	0	0	8	Bias independent parasitic resistance model selector (same as BSIM4)
SHMOD	-	0	0	1	Turn on Self Heating model: 0, Turn-on : 1
TOXE	m	3.0n	-	-	SiO ₂ equivalent gate dielectric thickness
TOXP	m	=TOXE	-	-	Physical dielectric thickness
TYPE	-	1	-1	1	NMOS=1, PMOS=-1
VFB ^(b)	V	-0.5	-	-	Flat band voltage
WINT ^(b)	m	0	-	-	Width reduction parameter (dopant diffusion effect)
XJ ^(b)	m	1.5e-7	-	-	S/D junction depth
XL	m	0	-	-	L offset for channel length due to mask/etch effect
XW	m	0	-	-	W offset for channel length due to mask/etch effect

BSIM6 Model — Basic Parameters

Name	Unit	Default	Min	Max	Description
ADOS	-	0.0	0.0	-	Quantum mechanical effect pre-factor cum switch in inversion
AGIDL ^(b)	-	0.0	-	-	Pre-exponential coefficient for GIDL. Global Scaling Parameters -AGIDLL, AGIDLW
AGISL ^(b)	-	0.0	-	-	Pre-exponential coefficient for GISL (AGISL < 0 means GISL parameters are the same as GIDL parameters). Global Scaling Parameters -AGISLL, AGISLW
AIGC ^(b)	$(F s^2/g)^{0.5}$	1.36e-2 (NMOS) and 9.8e-3 (PMOS)	-	-	Parameter for I_{gcs} and I_{gcd} . Global Scaling Parameters -AIGCL, AIGCW
AIGD ^(b)	$(F s^2/g)^{0.5}$	1.36e-2 (NMOS) and 9.8e-3 (PMOS)	-	-	Parameter for I_{gd} . Global Scaling Parameters -AIGDL, AIGDW
AIGS ^(b)	$(F s^2/g)^{0.5}$	1.36e-2 (NMOS) and 9.8e-3 (PMOS)	-	-	Parameter for I_{gs} . Global Scaling Parameters -AIGSL, AIGSW
ALPHA0 ^(b)	-	0.0	-	-	First parameter of impact ionization current. Global Scaling Parameters -ALPHA0L, ALPHA0LEXP
BDOS	-	1.0	0.0	-	Charge centroid parameter -slope of CV curve under QME in inversion
BETA0 ^(b)	-	0.0	-	-	First Vds dependent parameter of impact ionization current
BGIDL ^(b)	-	2.3e-9	-	-	Exponential coefficient for GIDL
BGISL ^(b)	-	2.3e-9	-	-	Exponential coefficient for GISL
BIGC ^(b)	$(F s^2/g)^{0.5}$	1.71e-3 (NMOS) and 7.59e-4 (PMOS)	-	-	Parameter for I_{gcs} and I_{gcd}

Name	Unit	Default	Min	Max	Description
BIGD ^(b)	$(F s^2/g)^{0.5}$	1.71e-3 (NMOS) and 7.59e-4 (PMOS)	-	-	Parameter for I_{gd}
BIGS ^(b)	$(F s^2/g)^{0.5}$	1.71e-3 (NMOS) and 7.59e-4 (PMOS)	-	-	Parameter for I_{gs}
BINUNIT	-	1	0	1	Binning unit selector
CDSCB ^(b)	F/m2	7e-03	0.0	-	Body-bias sensitivity of Subthreshold Swing. Global Scaling Parameters -CDSCBL, CDSCBLEXP
CDSCD ^(b)	F/m2	7e-03	0.0	-	Drain-bias sensitivity of Subthreshold Swing. Global Scaling Parameters -CDSCDL, CDSCDLEXP
CF ^(b)	F/m	0.0	0.0	-	Outer fringe cap
CFRCOEFF ^(b)	F/m	1.0	1.0	-	Outer fringe cap coefficient
CGBO	F/m	0.0	0.0	-	Gate-substrate overlap capacitance per unit channel length
CGDL ^(b)	F/m	0.0	0.0	-	Overlap capacitance between gate and lightly-doped drain region
CGDO	F/m	Calculated	0.0	-	Non LDD region drain-gate overlap capacitance per unit channel width
CGIDL ^(b)	-	0.5	-	-	exponential coefficient for GIDL
CGISL ^(b)	-	0.5	-	-	Exponential coefficient for GISL
CGSL ^(b)	F/m	0.0	0.0	-	Overlap capacitance between gate and lightly-doped source region
CGSO	F/m	Calculated	0.0	-	Non LDD region source-gate overlap capacitance per unit channel width

Name	Unit	Default	Min	Max	Description
CIGC ^(b)	$(F s^2/g)^{0.5}$	0.075 (NMOS) and 0.03 (PMOS)	-	-	Parameter for I_{gcs} and I_{gcd}
CIGD ^(b)	$(F s^2/g)^{0.5}$	0.075 (NMOS) and 0.03 (PMOS)	-	-	Parameter for I_{gd}
CIGS ^(b)	$(F s^2/g)^{0.5}$	0.075 (NMOS) and 0.03 (PMOS)	-	-	Parameter for I_{gs}
CIT ^(b)	F/m2	1e-08	0.0	-	Interface trap capacitance
CKAPPAD ^(b)	V	0.6	0.02	-	Coefficient of bias-dependent overlap capacitance for the drain side
CKAPPAS ^(b)	V	0.6	0.02	-	Coefficient of bias-dependent overlap capacitance for the source side
DELTA ^(b)	-	0.125	> 0	0.5	Smoothing factor for V_{ds} to V_{dsat} . Global Scaling Parameters -DELTA, DELTAEXP
DLBIN	-	0.0	-	-	Length reduction parameter for binning
DLCIG ^(b)	m	LINT	-	-	Source/Drain overlap length for I_{gs}
DLCIGD ^(b)	m	DLCIG	-	-	Source/Drain overlap length for I_{gd}
DSUB ^(b)	-	0.375	> 0.0	-	DIBL exponent coefficient
DVTP0 ^(b)	-	1e-10	-	-	Coefficient of drain-induced V_{th} shift for long channel devices with pocket implant
DVTP1 ^(b)	-	0.0	-	-	Coefficient of drain-induced V_{th} shift for long channel devices with pocket implant
DVTP2 ^(b)	-	0.0	-	-	Coefficient of drain-induced V_{th} shift for long channel devices with pocket implant

Name	Unit	Default	Min	Max	Description
DVTP3 ^(b)	-	0.0	-	-	Coefficient of drain-induced Vth shift for long channel devices with pocket implant
DVTP4 ^(b)	-	0.0	-	-	Coefficient of drain-induced Vth shift for long channel devices with pocket implant
DVTP5 ^(b)	-	0.0	-	-	Coefficient of drain-induced Vth shift for long channel devices with pocket implant
DWBIN		0.0	-	-	Width reduction parameter for binning
EGIDL ^(b)	-	0.8	-	-	band bending parameter for GIDL
EGISL ^(b)	-	0.8	-	-	band bending parameter for GISL
ETA0 ^(b)	-	0.08	-	-	DIBL coefficient
ETAB ^(b)	-	-0.07	-	-	Body bias sensitivity to DIBL effect. Global Scaling Parameters ETABLEXP
ETAMOB	-	1.0	-	-	Effective field parameter
ETAQM	-	0.54	0.0	-	Bulk charge coefficient for charge centroid in inversion
EU ^(b)	cm/MV	1.5	> 0.0	-	Phonon / surface roughness scattering. Global Scaling Parameters -EUL, EULEXP, EUW, EUWEXP, EUWL
FPROUT ^(b)	-	0.0	0.0	-	gds degradation factor due to pocket implant. Global Scaling Parameters -FPROUTL, FPROUTLEXP
K2 ^(b)	V	0.0	-	0.0	Vth shift due to nonuniform vertical doping. Global Scaling Parameters -K2L, K2LEXP, K2W, K2WEXP
LLONG	m	10μm	-	-	Length of extracted long channel device
LMLT	-	-	> 0.0		Length of shrinking factor

Name	Unit	Default	Min	Max	Description
NDEPCV ^(b)	m ⁻³	NDEP	-	-	Channel (body) doping concentration for CV. Global Scaling Parameters - NDEPCVL1, NDEPCVLEXP1, NDEPCVL2, NDEPCVLEXP2, NDEPCVW, NDEPCVWEXP, NDEPCVWL, NDEPCVWLEXP
NFACTOR ^(b)		0.0	0.0	-	Subthreshold Swing factor. Global Scaling Parameters -NFACTORL, NFACTORLEXP, NFACTORW, NFACTORWEXP, NFACTORWL
NTOX ^(b)	-	1.0	-	-	Exponent for the gate oxide ratio
PCLM ^(b)	-	0.013	> 0.0	-	Channel Length Modulation (CLM) parameter. Global Scaling Parameters -PCLML, PCLMLEXP
PCLMCV ^(b)	-	PCLM	> 0.0	-	Channel Length Modulation (CLM) parameter for CV. Global Scaling Parameters -PCLMCVL, PCLMCVLEXP
PCLMG	-	0.0	-	-	Gate bias dependent parameter for channel Length Modulation (CLM)
PDIBLC ^(b)	-	2e-4	0.0	-	DIBL effect on Rout. Global Scaling Parameters -PDIBLCL, PDIBLCLEXP
PDIBLCB ^(b)	-	0.0	0.0	-	Body-bias sensitivity on DIBL
PDITS ^(b)	-	0.0	-	-	Drain-induced Vth shift
PDITSD ^(b)	-	0.0	?	-	VDS dependence of Drain-induced Vth shift
PDITSL	-	0.0	?	-	L dependence of Drain-induced Vth shift
PHIN ^(b)	V	0.045	-	-	Vertical nonuniform doping effect on surface potential
PIGCD ^(b)	-	1.0	-	-	Vds dependence of I _{gcs} and I _{gcd} . Global Scaling Parameters -PIGCDL, PIGCDLEXP

Name	Unit	Default	Min	Max	Description
POXEDGE ^(b)	-	1.0	-	-	Factor for the gate oxide thickness in source/drain overlap regions
PRWB ^(b)	V ⁻¹	0.0	0.0	-	body bias dependence of S/D extension resistance. Global Scaling Parameters PRWBL, PRWBLEXP
PRWG ^(b)	V ⁻¹	1.0	0.0	-	gate bias dependence of S/D extension resistance
PSAT	-	1.0	0.25	1.0	Velocity saturation exponent. Global Scaling Parameters -PSATL, PSATLEXP
PSATB	-	0.0	-	-	Velocity saturation exponent for nonzero Vbs
PSATX	-	1.0	0.25	4.0	Fine tuning of PTWG effect
PSCBE1 ^(b)	-	4.24e08	-	-	Substrate current body-effect coefficient
PSCBE2 ^(b)	-	1.0e08	-	-	Substrate current body-effect coefficient
PTWG ^(b)	V ⁻²	0.0	-	-	Correction factor for velocity saturation. Global Scaling Parameters -PTWGL, PTWGLEXP
PVAG ^(b)	-	1.0	-	-	Vgs dependence on early voltage
QM0	-	1e-3	> 0.0	-	Charge centroid parameter -starting point for QME in inversion
RDSW ^(b)	$\Omega - \mu_m^{WR}$	10.0	0.0	-	Zero bias LDD resistance per unit width for RDSMOD=0. Global Scaling Parameters -RDSWL, RDSWLEXP
RDSWMIN ^(b)	$\Omega - \mu_m^{WR}$	0.0	0.0	-	LDD resistance per unit width at high Vgs for RDSMOD = 0
RDW ^(b)	$\Omega - \mu_m^{WR}$	10.0	0.0	-	Zero bias drain extension resistance per unit width. Global Scaling Parameters -RDWL, RDWLEXP

Name	Unit	Default	Min	Max	Description
RDWMIN ^(b)	$\Omega - \mu_m^{WR}$	0.0	0.0	-	Drain extension resistance per unit width at high Vgs
RSH	Ω	0.0	0.0	-	Sheet resistance
RSW ^(b)	$\Omega - \mu_m^{WR}$	10.0	0.0	-	Zero bias source extension resistance per unit width. Global Scaling Parameters -RSWL, RSWLEXP
RSWMIN ^(b)	$\Omega - \mu_m^{WR}$	0.0	0.0	-	Source extension resistance per unit width at high Vgs
TOXREF	m	3.0e-9	-	-	Nominal gate oxide thickness for gate dielectric tunneling current model only
U0 ^(b)	m ² /V - s	67e-02	-	-	Low field mobility. Global Scaling Parameters -U0L, U0LEXP
UA ^(b)	$\frac{cm}{MV}$ (_{EU})	0.001	> 0.0	-	Phonon / surface roughness scattering. Global Scaling Parameters -UAL, UALEXP, UAW, UAWEXP, UAWL
UC ^(b)	-	0.001	> 0.0	-	Body-bias sensitivity on mobility. Global Scaling Parameters -UCL, UCLEXP
UCS ^(b)	-	2.0	1.0	2.0	Coulombic scattering.
UD ^(b)	cm/MV	0.001	> 0.0	-	Coulombic scattering. Global Scaling Parameters -UDL, UDLEXP
VFBCV ^(b)	m ⁻³	VFB	-	-	channel (body) doping concentration for CV. Global Scaling Parameters -VFBCVL, VFBCVLEXP, VFBCVW, VFBCVWEXP, VFBCVWL, VFBCVWLEXP
VFBSDOFF ^(b)	V	0.0	-	-	Flatband Voltage offset parameter
VSAT ^(b)	m/s	1e6	-	-	Saturation velocity. Global Scaling Parameters -VSATL, VSALEXP, VSATW, VSATWEXP

Name	Unit	Default	Min	Max	Description
VSATCV ^(b)	m/s	VSAT	-	-	Saturation velocity for CV. Global Scaling Parameters -VSATCVL, VSACVLEXP, VSATCVW, VSATCVWEXP
WMLT	-	1.0	>0.0		Width of shrinking factor
WR ^(b)	-	1.0	-	-	W dependence parameter of S/D extension resistance
WWIDE	m	10μm	-	-	Width of extracted long channel device

BSIM6 Model — Both Model and Instance Parameters

Name	Unit	Default	Min	Max	Description
DTEMP	K	0.0	-	-	Offset of device temperature
MULU0	m ² /V/s	1.0	-	-	Multiplication factor for low field mobility
DELVTO	V	0	-	-	Zero bias threshold voltage variation
IDSOMULT	-	1.0	-	-	Variability in drain current for miscellaneous reasons

BSIM6 Model — High-Speed/RF Parameters

Name	Unit	Default	Min	Max	Description
GBMIN	Ω	1.0e-12 mho	-	-	Conductance in parallel with each of the five substrate resistances to avoid potential numerical instability due to unreasonably too large a substrate resistance (Warning message is issued if less than 1.0e-20 mho)
RBDB	Ω	50.0 Ω	-	-	Resistance connected between dbNode and bNode (Also an instance parameter).
RBDBX0	Ω	100.0 Ω	-	-	Scaling pre-factor for RBDBX
RDBY0	Ω	100.0 Ω	-	-	Scaling pre-factor for RDBY
RBPB	Ω	50.0 Ω	-	-	Resistance connected between bNodePrime and bNode (Also an instance parameter).

Name	Unit	Default	Min	Max	Description
RBPBX0	Ω	100.0 Ω	-	-	Scaling pre-factor for RBPBX
RBPBXL	-	0.0	-	-	Length scaling parameter for RBPBX
RBPBXNF	-	0.0	-	-	Number of fingers scaling parameter for RBPBX
RBPBXW	-	0.0	-	-	Width scaling parameter for RBPBX
RBPBY0	Ω	100.0 Ω	-	-	Scaling pre-factor for RBPBY
RBPBYL	-	0.0	-	-	Length scaling parameter for RBPBY
RBPBYNF	-	0.0	-	-	Number of fingers scaling parameter for RBPBY
RBPBYW	-	0.0	-	-	Width scaling parameter for RBPBY
RBPD	Ω	50.0 Ω	-	-	Resistance connected between bNodePrime and dbNode (If less than 1.0e-3ohm, reset to 1.0e-3ohm) (Also an instance parameter).
RBPD0	Ω	50.0 Ω	-	-	Scaling pre-factor for RBPD
RPBPD	-	0.0	-	-	Length scaling parameter for RBPD
RPBPDNF	-	0.0	-	-	Number of fingers scaling parameter for RBPD
RPBPDW	-	0.0	-	-	Width scaling parameter for RBPD
RBPS	Ω	50.0 Ω	-	-	Resistance connected between bNodePrime and sbNode (If less than 1.0e-3ohm, reset to 1.0e-3ohm) (Also an instance parameter).
RBPS0	Ω	50.0 Ω	-	-	Scaling pre-factor for RBPS
RBPSL	-	0.0	-	-	Length Scaling parameter for RBPS
RBPSNF	-	0.0	-	-	Number of fingers scaling parameter for RBPS
RBPSW	-	0.0	-	-	Width Scaling parameter for RBPS
RBSB	Ω	50.0 Ω	-	-	Resistance connected between sbNode and bNode (Also an instance parameter).
RBSBX0	Ω	100.0 Ω	-	-	Scaling pre-factor for RBSBX

Name	Unit	Default	Min	Max	Description
RBSBY0	Ω	100.0 Ω	-	-	Scaling pre-factor for RBSBY
RBSDBXL	-	0.0	-	-	Length scaling parameter for RBSBX and RBDBX
RBSDBXNF	-	0.0	-	-	Number of finders scaling parameter for RBSBX and RBDBX
RBSDBXW	-	0.0	-	-	Width scaling parameter for RBSBX and RBDBX
RBSDBYL	-	0.0	-	-	Length scaling parameter for RBSBY and RBDBY
RBSDBYNF	-	0.0	-	-	Number of fingers scaling parameter for RBSBY and RBDBY
RBSDBYW	-	0.0	-	-	Width scaling parameter for RBSBY and RBDBY
XRCRG1 ^(b)	-	12.0	-	-	Parameter for distributed channel-resistance effect for both intrinsic-input resistance and charge-deficit NQS models (Warning message is issued if binned XRCRG1 ≤ 0) distributed channel-resistance effect for both intrinsic-input resistance and charge-deficit NQS models (Warning message is issued if binned XRCRG1 ≤ 0)
XRCRG2 ^(b)	-	1.0	-	-	Parameter to account for the excess channel diffusion resistance for both intrinsic input resistance and charge-deficit NQS models

BSIM6 Model — Flicker and Thermal Noise Parameters

Name	Unit	Default	Min	Max	Description
AF	-	1.0	-	-	Flicker noise exponent
EF	-	1.0	-	-	Flicker noise frequency exponent
EM	-	4.1e7V/m	-	-	Saturation field
KF	-	0:0 ^{A2-EF} s ^{1-EF}	-	-	Flicker noise coefficient
LINTNOI	-	0.0 m	-	-	Length Reduction Parameter Oset

Name	Unit	Default	Min	Max	Description
NOIA	-	NMOS: $6.25e41(\text{eV})^{-1}\text{s}^{1-EF}\text{m}^{-3}$ PMOS: $3.125e26(\text{eV})^{-1}\text{s}^{1-EF}\text{m}^{-3}$	-	-	Flicker noise parameter A
NOIB	-	NMOS: $3.125e26(\text{eV})^{-1}\text{s}^{1-EF}\text{m}^{-1}$ PMOS: $1.5e25(\text{eV})^{-1}\text{s}^{1-EF}\text{m}^{-1}$	-	-	Flicker noise parameter B
NOIC	-	$8.75(\text{eV})^{-1}\text{s}^{1-EF}\text{m}$	-	-	Flicker noise parameter C
NTNOI	-	1.0	-	-	Noise factor for short-channel devices for TNOIMOD=0 only
RNOIA	-	0.577	-	-	Thermal Noise coefficient
RNOIB	-	0.5164	-	-	Thermal Noise coefficient
RNOIC	-	0.395	-	-	Correlation coefficient parameter
TNOIA	-	1.5	-	-	Coefficient of channel-length dependence of total channel thermal noise
TNOIB	-	3.5	-	-	Channel-length dependence parameter for channel $\tau_{\text{RNOIChermal}}$ noise partitioning
TNOIC	-	0	-	-	Length dependent parameter for correlation coefficient

BSIM6 Model — Layout-Dependent Parasitic Parameters

Name	Unit	Default	Min	Max	Description
DMCG	-	0.0m	-	-	Distance from S/D contact center to the gate edge
DMCGT	-	0.0m	-	-	DMCG of test structures
DMCI	-	DMCG	-	-	Distance from S/D contact center to the isolation edge in the channel-length direction
DMDG	-	0.0m	-	-	Same as DMCG but for merged device only
DWJ	-	DWC (in CVmodel)	-	-	Offset of the S/D junction width

Name	Unit	Default	Min	Max	Description
MINZ	-	0 (minimize the drain diffusion fusion number)	-	-	Whether to minimize the number of drain or source diffusions for even-number fingered device (instance parameter only).
NF	-	1	-	-	Number of device fingers (Fatal error if less than one) (instance parameter only).
NGCON	-	1	-	-	Number of gate contacts (Fatal error if less than one; if not equal to 1 or 2, a warning message is issued and reset to 1) (Also an instance parameter).
XGL	-	0.0m	-	-	Offset of the gate length due to variations in patterning
XGW	-	0.0m	-	-	Distance from the gate contact to the channel edge (Also an instance parameter).
XL	-	0.0m	-	-	Channel length offset due to mask/ etch effect
XW	-	0.0m	-	-	Channel width offset due to mask/etch effect

BSIM6 Model — Asymmetric Source/Drain Junction Diode Parameters

Name	Unit	Default	Min	Max	Description
BVD	-	BVS	-	-	Breakdown voltage (If not positive, reset to 10.0V)
BVS	-	10.0 V	-	-	Breakdown voltage (If not positive, reset to 10.0V)
CJD	-	CJS	-	-	Bottom junction capacitance per unit area at zero bias
CJS	-	5.0e-4 F/m2	-	-	Bottom junction capacitance per unit area at zero bias
CJSWD	-	CJSWS	-	-	Isolation-edge sidewall junction capacitance per unit area
CJSWGD	-	CJSWS	-	-	Gate-edge sidewall junction capacitance per unit length
CJSWGS	-	CJSWS	-	-	Gate-edge sidewall junction capacitance per unit length

Name	Unit	Default	Min	Max	Description
CJSWS	-	5.0e-10 F/m	-	-	Isolation-edge sidewall junction capacitance per unit area
IJTHDFWD	-	IJTHSFWD	-	-	Limiting current in forward bias region (If not positive, reset to 0.1A)
IJTHDREV	-	IJTHSREV	-	-	Limiting current in reverse bias region (If not positive, reset to 0.1A)
IJTHSFWD	-	0.1A	-	-	Limiting current in forward bias region (If not positive, reset to 0.1A)
IJTHSREV	-	0.1A	-	-	Limiting current in reverse bias region (If not positive, reset to 0.1A)
JSD	-	JSS	-	-	Bottom junction reverse saturation current density
JSS	-	1.0e-4A/m2	-	-	Bottom junction reverse saturation current density
JSWD	-	JSWS	-	-	Isolation-edge sidewall reverse saturation current density
JSWGD	-	JSWGS	-	-	Gate-edge sidewall reverse saturation current density
JSWGS	-	0.0A/m	-	-	Gate-edge sidewall reverse saturation current density
JSWS	-	0.0A/m	-	-	Isolation-edge sidewall reverse saturation current density
JTSD	-	JTSS	-	-	Bottom trap-assisted saturation current density
JTSS	-	0.0A/m	-	-	Bottom trap-assisted saturation current density
JTSSWD	-	JTSSWS	-	-	STI sidewall trap-assisted saturation current density
JTSSWGD	-	JTSSWGS	-	-	Gate-edge sidewall trap-assisted saturation current density
JTSSWGS	-	0.0A/m	-	-	Gate-edge sidewall trap-assisted saturation current density
JTSSWS	-	0.0A/m2	-	-	STI sidewall trap-assisted saturation current density

Name	Unit	Default	Min	Max	Description
JTWEFF	-	0.0	-	-	Trap-assistant tunneling current density width dependence
MJD	-	MJS	-	-	Bottom junction capacitance grating coefficient
MJS	-	0.5	-	-	Bottom junction capacitance grating coefficient
MJSWD	-	MJSWS	-	-	Isolation-edge sidewall junction capacitance grading coefficient
MJSWGD	-	MJSWS	-	-	Gate-edge sidewall junction capacitance grading coefficient
MJSWGS	-	MJSWS	-	-	Gate-edge sidewall junction capacitance grading coefficient
MJSWS	-	0.33	-	-	Isolation-edge sidewall junction capacitance grading coefficient
NJTS	-	20.0	-	-	Non-ideality factor for JTSS and JTSD
NJTSD	-	NJTS	-	-	Non-ideality factor for JTSS and JTSD
NJTSSW	-	20.0	-	-	Non-ideality factor for JTSSWS and JTSSWD
NJTSSWD	-	NJTSSW	-	-	Non-ideality factor for JTSSWS and JTSSWD
NJTSSWG	-	20.0	-	-	Non-ideality factor for JTSSWGS and JTSSWGD
NJTSSWGD	-	NJTSSWG	-	-	Non-ideality factor for JTSSWGS and JTSSWGD
PBD	-	PBS	-	-	Bottom junction built-in potential
PBS	-	1.0V	-	-	Bottom junction built-in potential
PBSWD	-	PBSWS	-	-	Isolation-edge sidewall junction built-in potential
PBSWGD	-	PBSWGS	-	-	Gate-edge sidewall junction built-in potential
PBSWGS	-	1.0V	-	-	Gate-edge sidewall junction built-in potential

Name	Unit	Default	Min	Max	Description
PBSWS	-	1.0V	-	-	Isolation-edge sidewall junction built-in potential
TNJTS	-	0.0	-	-	Temperature coefficient for NJTS and NJTSD
TNJTSD	-	TNJTS	-	-	Temperature coefficient for NJTS and NJTSD
TNJTSSW	-	0.0	-	-	Temperature coefficient for NJTSSW and NJTSSWD
TNJTSSWD	-	TNJTSSW	-	-	Temperature coefficient for NJTSSW and NJTSSWD
TNJTSSWG	-	0.0	-	-	Temperature coefficient for NJTSSWG and NJTSSWG
TNJTSSWGD	-	TNJTSSWG	-	-	Temperature coefficient for NJTSSWG and NJTSSWG
VTSD	-	VTSS	-	-	Bottom trap-assisted voltage dependent parameter
VTSS	-	10.0 V	-	-	Bottom trap-assisted voltage dependent parameter
VTSSWD	-	VTSSWS	-	-	STI sidewall trap-assisted voltage dependent parameter
VTSSWGD	-	VTSSWS	-	-	Gate-edge sidewall trap-assisted voltage dependent parameter
VTSSWGS	-	10.0 V	-	-	Gate-edge sidewall trap-assisted voltage dependent parameter
VTSSWS	-	10.0 V	-	-	STI sidewall trap-assisted voltage dependent parameter
XJBVD	-	XJBVS	-	-	Fitting parameter for diode breakdown
XJBVS	-	1.0	-	-	Fitting parameter for diode breakdown
XTSD	-	0.02	-	-	Power dependence of JTSS, JTSD on temperature
XTSS	-	0.02	-	-	Power dependence of JTSS, JTSD on temperature
XTSSWD	-	0.02	-	-	Power dependence of JTSSWS, XTSSWD on temperature

Name	Unit	Default	Min	Max	Description
XTSSWGD	-	0.02	-	-	Power dependence of JTSSWGS, JTSSWGD on temperature
XTSSWGS	-	0.02	-	-	Power dependence of JTSSWGS, JTSSWGD on temperature
XTSSWS	-	0.02	-	-	Power dependence of JTSSWS, JTSSWD on temperature

BSIM6 Model — Temperature-Dependent Parameters

Name	Unit	Default	Min	Max	Description
AT ^(b)	m/s	3.3e4 m/s	-	-	Temperature coefficient for saturation velocity
CTH0	-	1.0e-5	-	-	Thermal capacitance for self-heating calculation
DTEMP	-	0	-	-	Variability handle for temperature
KT1	V	-0.11V	-	-	Temperature coefficient for threshold voltage
KT1EXP	-	1.0	-	-	Temperature exponent for threshold voltage
KT1L ^(b)	Vm	0.0Vm	-	-	Channel length dependence of the temperature coefficient for threshold voltage
KT2 ^(b)	-	0.022	-	-	Body-bias coefficient of V_{th} temperature effect
NJD	-	NJS	-	-	Emission coefficients of junction for source and drain junctions, respectively
NJS	-	1.0	-	-	Emission coefficients of junction for source and drain junctions, respectively
PRT ^(b)	-	0.0 ohm-m	-	-	Temperature coefficient for R_{dsw}
PTWGT ^(b)	-	0.0	-	-	Temperature coefficient for PTWG
RTH0	-	0.0	-	-	Thermal resistance for self-heating calculation

Name	Unit	Default	Min	Max	Description
TCJ	1/°K	0.0	-	-	Temperature coefficient of CJ
TCJSW	1/°K	0.0	-	-	Temperature coefficient of CJSW
TCJSWG ^(b)	1/°K	0.0	-	-	Temperature coefficient of CJSWG
TETA0 ^(b)	-	0.0	-	-	Temperature coefficient of ETA0
TNFACTOR ^(b)	-	0.0	-	-	Temperature coefficient of NFACTOR
TNOM	°C	27°C	-	-	Temperature at which the model is extracted
TPB	V/K	0.0	-	-	Temperature coefficient of PB
TPBSW	V/K	0.0	-	-	Temperature coefficient of PBSW
TPBSWG	V/K	0.0	-	-	Temperature coefficient of PBSWG
TVFBSDOFF	1/°K	0.0	-	-	Temperature coefficient of VFBSDOFF
UA1 ^(b)	-	1.0e-9m/V	-	-	Temperature coefficient for UA
UC1 ^(b)	-	0.056V-1 for MOBMOD=1; 0.056e-9m/V ² for MOBMOD=0 and 2	-	-	Temperature coefficient for UC
UCSTE ^(b)	-	-4.775e-3	-	-	Temperature coefficient of coulombic mobility
UD1 ^(b)	-	0:0 (1/m) ²	-	-	Temperature coefficient for UD
UTE ^(b)	-	-1.5	-	-	Mobility temperature exponent
WTH0	-	0.0	-	-	Width-dependence coefficient for self-heating calculation
XTID	-	XTIS	-	-	Junction current temperature exponents for source and drain junctions, respectively
XTIS	-	3.0	-	-	Junction current temperature exponents for source and drain junctions, respectively

BSIM6 Model — Stress Effect Parameters

Name	Unit	Default	Min	Max	Description
KU0	-	0.0m	-	-	Mobility degradation/enhancement coefficient for stress effect
KVSAT	-	0.0m	-	-	Saturation velocity degradation/enhancement parameter for stress effect ($1 \leq kvsat \leq 1$)
KVTH0	-	0.0 Vm	-	-	Threshold shift parameter for stress effect
LKU0	-	1e-6m	-	-	Length dependence of ku0
LKVTH0	-	0.0	-	-	Length dependence of kvth0
LLODKU0	-	0.0	-	-	Length parameter for u0 stress effect (>0)
LLODVTH	-	0.0	-	-	Length parameter for Vth stress effect (>0)
LODETA0	-	1.0	-	-	eta0 shift modification factor for stress effect (>0)
LODK2	-	0.0	-	-	K2 shift modification factor for stress effect (>0)
PKU0	-	0.0	-	-	Cross-term dependence of ku0
PKVTH0	-	0.0	-	-	Cross-term dependence of kvth0
SA	-	0.0	-	-	Distance between OD edge to Poly from one side (If not given or (≤ 0), stress effect is turned-off) (An instance parameter).
SAref	-	1e-6m	-	-	Reference distance between OD and edge to poly of one side (>0.0)
SB	-	0.0	-	-	Distance between OD edge to Poly from other side (If not given or (≤ 0), stress effect is turned-off) (An instance parameter).
SBref	-	1e-6m	-	-	Reference distance between OD and edge to poly of the other side (>0.0)
SD	-	0.0	-	-	Distance between neighboring fingers (For NF>1: If not given or (≤ 0), stress effect is turned-off) (An instance parameter).

Name	Unit	Default	Min	Max	Description
STETA0	-	0.0m	-	-	eta0 shift factor related to Vth0 change
STK2	-	0.0m	-	-	K2 shift factor related to Vth0 change
TKU0	-	0.0	-	-	Temperature coefficient of KU0
WKU0	-	0.0	-	-	Width dependence of ku0
WKVTH0	-	0.0	-	-	Width dependence of kvth0
WLOD	-	0.0m	-	-	Width parameter for stress effect
WLODKU0	-	0.0	-	-	Width parameter for u0 stress effect (>0)
WLODVTH	-	0.0	-	-	Width parameter for Vth stress effect (>0)

BSIM6 Model — Well-Proximity Effect Parameters

Name	Unit	Default	Min	Max	Description
K2WE ^(b)	-	0.0	-	-	K2 shift factor for well proximity effect
KU0WE ^(b)	-	0.0	-	-	Mobility degradation factor for well proximity effect
KVTH0WE ^(b)	-	0.0	-	-	Threshold shift factor for well proximity effect
SC	-	0.0m	-	-	Distance to a single well edge (If not given or =< 0:0, turn-off WPE) (An instance parameter).
SCA	-	0.0	-	-	Integral of the first distribution function for scattered well dopant (If not given, calculated) (An instance parameter).
SCB	-	0.0	-	-	Integral of the second distribution function for scattered well dopant (If not given, calculated) (An instance parameter).
SCREF	-	1e-6m	-	-	Reference distance to calculate SCA, SCB and SCC (<0)
SCC	-	0.0	-	-	Integral of the third distribution function for scattered well dopant (If not given, calculated) (An instance parameter).
WEB	-	0.0	-	-	Coefficient for SCB (>0.0)

Name	Unit	Default	Min	Max	Description
WEC	-	0.0	-	-	Coefficient for SCC (>0.0)

BSIM6 Model — Edge FET Device and Sub-Surface Leakage Current Parameters

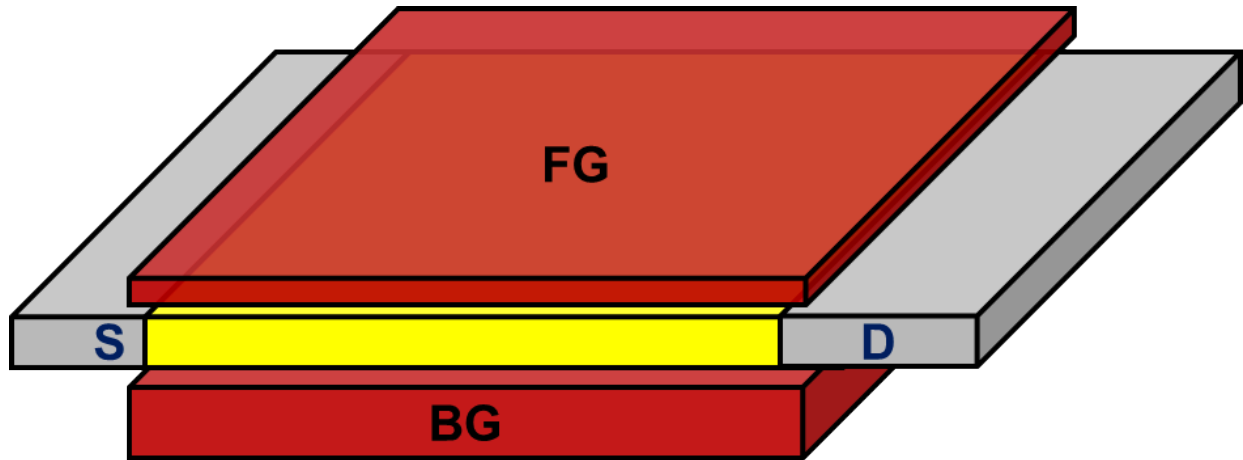
Name	Unit	Default	Min	Max	Description
CDSCBEDGE ^(b)	F/m ²	CDSCB	-	-	CDSCB for Edge FET
CDSCDEGE ^(b)	F/m ²	CDSCD	-	-	CDSCD for Edge FET
CITEDGE ^(b)	F/m ²	CIT	-	-	CIT for Edge FET
DGAMMA	-	0	-	-	Difference in body-bias coefficient between Edge-FET and Main-FET
DGAMMAL	-	0	-	-	L dependence parameter for DGAMMA
DGAMMALEXP	-	1.0	-	-	Exponent of L dependence parameter for DGAMMA
DVT0EDGE	-	2.2	-	-	First coefficient of SCE effect on Vth for Edge FET
DVT1EDGE	-	0.53	-	-	Second coefficient of SCE effect on Vth for Edge FET
DVT2EDGE	-	0.0	-	-	Body-bias coefficient for SCE effect for Edge FET
DVTEDGE	-	0.0	-	-	Vth shift for Edge FET
EDGEFET	-	1	-	-	Edge FET model flag 0: Disable and 1: Enable If version > 6.11, the default value of EDGEFET is 0. If version <= 6.11, the default value of EDGEFET is 1.
ETA0EDGE ^(b)	F/m ²	ETA0	-	-	DIBL parameter for Edge FET
ETABEDGE ^(b)	F/m ²	ETAB	-	-	ETAB for Edge FET
K2EDGE	V	K2	-	-	Vth shift due to body bias
KT1EDGE ^(b)	V	KT1	-	-	Temperature dependence parameter of threshold voltage for edge FET

Name	Unit	Default	Min	Max	Description
KT1EXPEDGE	-	KT1EXP	-	-	Temperature dependence parameter of threshold voltage for edge device
KT1LEDGE ^(b)	F/m ²	KT1L	-	-	Temperature dependence parameter of threshold voltage for edge FET
KT2EDGE ^(b)	V	KT2	-	-	Temperature dependence parameter of threshold voltage for edge FET
NFACTOREDGE ^(b)	-	NFACTOR	-	-	NFACTOR for Edge FET
SSL0	A/m	400	-	-	
SSL1	1/m	3.36e ⁸	-	-	
SSL2	-	0.185	-	-	
SSL3	V	0.3	-	-	
SSL4	1/V	1.4	-	-	
SSLEXP1	-	0.490	-	-	
SSLEXP2	-	1.42	-	-	
SSLMOD	-	1	-	-	Sub-surface leakage drain current, 0: Turn off and 1: Turn on
TETA0EDGE	-	TETA0	-	-	Temperature dependence parameter of DIBL parameter for edge FET
TNFACTORE DGE ^(b)	-	TNFACTOR	-	-	Temperature dependence parameter of sub-threshold slope factor for edge
WEDGE	m	10e-09	1e-09	-	Edge FET width

Level 78: BSIM-IMG Model

BSIM-IMG, the independent multi-gate model described in this section, has been developed to model the electrical characteristics of independent double-gate device structures. It allows front- and back-gate to have different physical configurations such as work function, dielectric thickness, and dielectric constants.

Figure 18 Planar Double-Gate SOI MOSFET



BSIM-IMG models the independent double-gate structure as a four terminal device, containing the source(S), drain(D), front-gate(FG), and back-gate(BG) terminals. The two gates are allowed to have different work functions ($\Delta\psi_1$, $\Delta\psi_2$) and dielectric thicknesses (T_{ox1} , T_{ox2}), as well as different voltage biases.

Physical surface-potential-based formulations are derived in both intrinsic and extrinsic models of BSIM-IMG. Surface potentials and integrated charge densities at the source and drain ends are obtained by solving the Poisson's equation in a fully-depleted lightly-doped body with efficient analytical approximations. Owing to the physics-based surface potential equation, the model captures volume inversion effects and shows excellent scalability as compared to 2D-derived simulation.

To meet the requirements of future devices, parameters for non-silicon channel devices and high-K gate insulators are incorporated into the BSIM-IMG model.

The back-gate of a planar double-gate SOI FET is often used for tuning device threshold voltage (V_{th}). The effect of back-gate on the threshold voltage is also addressed by the model.

Other important effects such as, short-channel effects, mobility degradation, velocity saturation, velocity overshoot, series resistance, channel length modulation, quantum mechanical effects, gate tunneling current, gate-induced-drain-leakage, and parasitic capacitance are also incorporated in the model.

The model is continuous and symmetric at $V_{ds}=0$. This physics-based model is scalable and predictive over a wide range of device parameters.

For the full description of BSIM-IMG including equations and model parameters, refer to the BSIM-IMG technical manual from [the official BSIM-IMG website](#).

This model supports the advanced MOSFET self-heating simulation solution.

This section covers the following topics:

- [BSIM-IMG Model — General Syntax](#)
- [BSIM-IMG Model — Version Updates](#)
- [BSIM-IMG Model Parameters](#)
- [BSIM-IMG Model — Operating Point Information](#)
- [BSIM-IMG Model — Output Templates](#)

BSIM-IMG Model — General Syntax

Syntax:

```
Mxxx nd nfg ns nbg mname [L=val] [M=val]  
.model mname n(p)mos LEVEL=78 VERSION=val ...
```

Level 78 is the BSIM-IMG model in the PrimeSim Continuum solution.

For example, to use the BSIM-IMG model in the PrimeSim Continuum solution, specify:

```
M1 drain fgate source bgate NCH L=1u  
.MODEL NCH NMOS LEVEL=78 VERSION=102.7
```

BSIM-IMG Model — Version Updates

- [BSIM-IMG 103.0.0 Updates](#)
- [BSIM-IMG 102.9.6 Updates](#)
- [BSIM-IMG 102.9.4 Updates](#)
- [BSIM-IMG 102.9.2 Updates](#)
- [BSIM-IMG 102.9.1 Updates](#)
- [BSIM-IMG 102.9.0 Updates](#)
- [BSIM-IMG 102.8 Updates](#)
- [BSIM-IMG 102.7 Updates](#)
- [BSIM-IMG 102.6.1 Updates](#)
- [BSIM-IMG 102.6](#)

BSIM-IMG 103.0.0 Updates

BSIM-IMG version 103.0.0 implements the following features:

- 1/f noise model improvement in subthreshold region
- Set XL as model parameter
- A few bug-fixes

BSIM-IMG 102.9.6 Updates

BSIM-IMG version 102.9.6 implements the following features:

- Potential divide-by-zero errors in gdpr and gspr computations.
- Redundant parameter range checking.
- Unnecessary range check for `RSWMIN_i`, `RDWMIN_i` and `RDSWMIN_i`.
- Modification of gdpr and gspr to include `ids0mult_r_factor`.
- Modification of TNOM < 0 warning statement.
- Changing TNOM macro from MPRco to MPRoo.
- Code implementation correction for RDSMOD=0 and 2.
- Vampyre warning: Bias-dependent '==': if (Rdss==0.0).
- Incorrect TNOIMOD=0 expression.
- Missing impact ionization current in the self-heating power.
- Incorrect gth and cth expressions.
- Incorrect macro for minr.
- Avoiding divide-by-zero error in UDS_t and UDD_t computation.
- Add induced gate noise in thermal noise.
- Add IDS0MULT for local layout effects.
- Cryogenic model implementation.
- Defining EU_t=EU_i for CRYOMOD=0.
- Aging defect screening model.

BSIM-IMG 102.9.4 Updates

BSIM-IMG version 102.9.4 implements the following features:

- Construct lgate model backward compatible with BSIM-IMG102.8.0 by default (IGFLAG=0)
- Add CONVMOD, with default =1 to turn on fixes for non-convergence related bugs in v102.9.2
- Add bias independent contact resistances (RSC and RDC) and their temperature dependent parameters (TRS and TRD) in all RDSMODs
- Add checks for negative value of RSC0 and RDC0
- Fix Rsourcegeo/Rdraingeo multiplication with W
- Temp dependence of bias-independent S/D resistance in RDSMOD=0
- Hidden state warnings for RSC0 and RDC0 when CONVMOD=0
- Correct clamping condition of RSC0_t and RDC0_t for RDSMOD!=0
- Remove bias dependence in RSC0 and RDC0
- Fixed small value of rdstemp causes convergence
- Add warning when DevTemp>TMAXK
- Fix discontinuity in temperature derivative of Mnud
- Check QSREF_i for zero
- Construct 1/f noise model backward compatible with BSIM-IMG102.9.2 by default (FNMOD=0) and improve 1/f noise model subthreshold region.
- Check NOIA for zero when FNMOD=1
- Add sign of modulation in flicker noise model

BSIM-IMG 102.9.2 Updates

BSIM-IMG version 102.9.2 implements the following features:

- Self-heating temperature clamping
- Binning equations for QME effect
- DVTP formulation in line with BSIM4
- A few bug-fixes

BSIM-IMG 102.9.1 Updates

BSIM-IMG version 102.9.1 implements the following features:

- Removed redundant `dvth_rsce` calculation

BSIM-IMG 102.9.0 Updates

BSIM-IMG version 102.9.0 implements the following features:

- Added `Vbg` dependence in gate current module.
- Added external thermal node.
- Improved `gds` fitting in low gate overdrive.
- Added numerical protection for the `sqrt(phist)` term.
- Removed unnecessary range warning messages for `EF` and `NTNOI`.

BSIM-IMG 102.8 Updates

BSIM-IMG version 102.8 implements the following features:

- Added back-gate dependence of `GIDL/GISL` currents.
- Additional binnable parameters are available.
- Adjusted the range of several model parameters.
- Add `NQSMOD` (intrinsic-input impedance).
- Removed the first order `NR` solution in surface potential calculation.

BSIM-IMG 102.7 Updates

BSIM-IMG version 102.7 implements the following features:

- `Igbs` and `Igbd` swapping is corrected.
- `NF` scaling is applied to `Qsbg`, `Qdbg`, and `qfg`.
- Minimum clamping is applied to `Dmob`, `Dmobs`, and `Dvsat`.
- `Vth` is added to the output templates.

BSIM-IMG 102.6.1 Updates

BSIM-IMG version 102.6.1 implements the following features:

- `Qsbg/Qdbg` PFET sign is updated.
- `NF` scaling is applied to the noise model.

- Thermal noise Vds sensitivity is updated.
- Minimum clamp is updated in Rgate for better convergence.
- RDSMOD=2 is introduced to enable the source/drain node collapse.

BSIM-IMG 102.6

The first officially-released BSIM-IMG version is 102.6. For details, refer to BSIM-IMG 102.6 Technical Manual from [the official BSIM-IMG website](#).

BSIM-IMG Model Parameters

Note:

In the following tables, binnable parameters are marked as ^(b). Instance parameters which are also model parameters are marked as ^(m).

This section covers the following topics:

- [BSIM-IMG Model — Instance Parameters](#)
- [BSIM-IMG Model — Control and Process Parameters](#)
- [BSIM-IMG Model — Basic Model Parameters](#)
- [BSIM-IMG Model — Temperature Dependence and Self-Heating Model Parameters](#)

BSIM-IMG Model — Instance Parameters

Parameter	Unit	Default	Min	Max	Description
$L^{(m)}$	m	30n	1n	-	Designed gate length
W	m	1e-6	1n	-	Designed gate width
NF	-	1	1	-	Number of fingers
$AS^{(m)}$	m^2	0	0	-	Source to substrate overlap area through oxide (all fingers)
$AD^{(m)}$	m^2	0	0	-	Drain to substrate overlap area through oxide (all fingers)
$PS^{(m)}$	m	0	0	-	Source to substrate overlap perimeter through oxide (all fingers)
$PD^{(m)}$	m	0	0	-	Drain to substrate overlap perimeter through oxide (all fingers)

Parameter	Unit	Default	Min	Max	Description
NRS ^(m)	-	0	0	-	Number of source diffusion squares (for RGEOMOD=0)
NRD ^(m)	-	0	0	-	Number of drain diffusion squares (for RGEOMOD=0)
XL	m	0	-	-	L offset for channel length due to mask/etch effect
DTEMP	°C	0	-	-	Variability handle for temperature
DELVTRAND	V	0	-	-	Variability in threshold voltage
U0MULT	-	1	-	-	Variability in carrier mobility
RDC	Ω	0	-	-	Drain contact resistance for per-finger device
RSC	Ω	0	-	-	Source contact resistance for per-finger device

BSIM-IMG Model — Control and Process Parameters

Parameter	Unit	Default	Min	Max	Description
VERSION	-	102.91	-	-	Model version number
TYPE	-	NMOS	PMOS	NMOS	NMOS=1, PMOS=-1
WELLTYPE	-	0	0	1	N-well=1, p-well=-1
CHARGEMOD	-	0	0	1	Select the inversion charge density model for the body; 0=simplified, 1=more accurate
RDSMOD	-	0	0	2	Bias-dependent source/drain resistance model selector (controls si and di nodes); 0=internal, 1=external, 2=bias and geometry dependent
IGCMOD	-	0	0	1	Model selector for Igd, Igs, and Igd; 1=turn on, 0=turn off
IGBMOD	-	0	0	1	Model selector for Igb; 1=turn on, 0=turn off

Parameter	Unit	Default	Min	Max	Description
GIDLMOD	-	0	0	1	GIDL/GISL current switcher; 1=turn on, 0=turn off
SHMOD	-	0	0	1	Self-heating mode switch; 1=turn on, 0=turn off
RGATEMOD	-	0	0	1	Gate resistance model selector; 1=turn on, 0=turn off
NFMODE	-	0	-	-	Number of finger selector; 1=W taken as single finger width, 0=W taken as total width like BSIM4
NQSMOD	-	0	0	1	NQS gate resistor and node switcher; 1=on, 0=off
CHARGEWF	-	-	-	-	Average channel charge weighting factor; 1=source-side, 0=middle, -1=drain-side
XW	m	0	-	-	W offset for channel length due to mask/etch effect
LINT	m	0	-	-	Length reduction parameter (dopant diffusion effect)
LL	$m^{(LLN+1)}$	0	-	-	Length reduction parameter (dopant diffusion effect)
LW	m	0	-	-	Length scaling parameter
LWL	m	0	-	-	Length scaling parameter
LLN	-	1	-	-	Length reduction parameter (dopant diffusion effect)
LWN	m	1	-	-	Length scaling parameter
WINT ^(b)	m	0	-	-	Width reduction parameter (dopant diffusion effect)
WL ^(b)	$m^{(WLN+1)}$	0	-	-	Width reduction parameter (dopant diffusion effect)
WW ^(b)	m	0	-	-	Width scaling parameter
WWL ^(b)	m	0	-	-	Width scaling parameter
WLN ^(b)	-	1	-	-	Width reduction parameter (dopant diffusion effect)

Parameter	Unit	Default	Min	Max	Description
WWN ^(b)	m	1	-	-	Width scaling parameter
DLC	m	0	-	-	Length reduction parameter (dopant diffusion effect)
LLC	m	0	-	-	Length scaling parameter
LWC	m	0	-	-	Length scaling parameter
LWLC	m	0	-	-	Length scaling parameter
DWC	m	0	-	-	Width reduction parameter (dopant diffusion effect)
WLC	m	0	-	-	Width scaling parameter
WWC	m	0	-	-	Width scaling parameter
WWLC	m	0	-	-	Width scaling parameter
EOT1	m	1.0n	0.1n	-	SiO2 equivalent front-gate dielectric thickness (including inversion layer thickness)
EOT2	m	140n	0.1n	-	SiO2 equivalent back-gate dielectric thickness (including inversion layer thickness)
EOT1P	m	EOT1	0.1n	-	Physical front-gate dielectric thickness for CV
DTOX1	m	0.0	-	-	Difference between effective dielectric thickness and physical thickness
TSI	m	8n	1n	-	Body thickness
NBODY ^(b)	m ⁻³	1e22	1e18	5e24	Channel doping concentration
NBG	m ⁻³	5e23	0	-	Well or back-gate doping level, zero for metal back-gate
EASUB	eV	4.05	0	-	Electron affinity of the substrate material
NI0SUB	m ⁻³	1.1e16	-	-	Intrinsic carrier concentration of channel at 300.15K
BG0SUB	eV	1.12	0	-	Bandgap of the channel material at 300.15K

Parameter	Unit	Default	Min	Max	Description
NC0SUB	m^{-3}	2.86e25	-	-	Conduction band density of states 300.15K
PHIG1 ^(b)	V	4.61	0	-	Work function of the front-gate
PHIG2 ^(b)	V	EASUB+BG 0SUB for n-well; EASUB for p-well	0	-	Well or back-gate work function, eV, it is modified according to NBG later in the code if 1) the back-gate is NOT metallic and 2) its value is not provided by user
EPSRSUB	-	11.9	0	-	Relative dielectric constant of the substrate material
EPSROX1	-	3.9	0	-	Relative dielectric constant of the front gate insulator material
NSD ^(b)	m^{-3}	2e26	2e25	1e27	S/D doping concentration

BSIM-IMG Model — Basic Model Parameters

Parameter	Unit	Default	Min	Max	Description
CIT ^(b)	F/m^2	0.0	-	-	Parameter for interface trap
CDSC ^(b)	F/m^2	0.14	0.0	-	Coupling capacitance between S/D and channel
CDSCD ^(b)	$\text{F}/\text{m}^2\text{V}$	0.14	0.0	-	Drain-bias sensitivity of CDSC
CBGCBG ^(b)	$\text{F}/\text{m}^2\text{V}$	0.1	0.0	-	Back-gate bias sensitivity of coupling capacitance to channel
CBGCBG0 ^(b)	$\text{F}/\text{m}^2\text{V}$	0.0	0.0	-	Back-gate bias sensitivity of SS for long channel
CBGCBG0P ^(b)	$\text{F}/\text{m}^2\text{V}^2$	0.0	0.0	-	Sub-linear back-gate bias sensitivity of SS
CBGCBGP ^(b)	$\text{F}/\text{m}^2\text{V}^2$	0.0	0.0	-	Non-linear back-gate bias sensitivity of SS
CBGCBGD ^(b)	$\text{F}/\text{m}^2\text{V}^2$	0.0	0.0	-	Back-gate bias sensitivity of CDSCD
DVT0 ^(b)	-	19.20	0.0	-	SCE coefficient
DVT1 ^(b)	-	0.45	0.0	-	SCE exponent coefficient

Parameter	Unit	Default	Min	Max	Description
PHIN ^(b)	V	0.045	-	-	Non-uniform vertical doping effect
ETA0 ^(b)	-	2.00	0.0	-	DIBL coefficient
ETA1 ^(b)	-	0.00	0.0	-	DIBL coefficient for low gate overdrive
ETAB ^(b)	1/V	0.0	0.0	-	DIBL coefficient - back-gate bias dependence
DSUB ^(b)	-	0.375	0.0	-	DIBL exponent coefficient
K1RSCE ^(b)	$V^{1/2}$	-0.32	-	-	Prefactor for reverse short channel effect
LPE0 ^(b)	m	8.2e-9	-L _{eff}	-	Equivalent length of pocket region at zero bias
DSC0 ^(b)	m	0.0	-	-	Parameter for short channel effect at moderate L and high drain bias
DSC1 ^(b)	m	1n	-	-	Parameter for short channel effect at moderate L and high drain bias
ASCL ^(b)	-	0.0	-	-	Parameter for back-gate dependent scale length
BSCL ^(b)	1/V	0.0	-	-	Parameter for back-gate dependent scale length
VSAT ^(b)	m/s	85000	-	-	Saturation velocity in the saturation region
AVSAT ^(b)	-	0	-	-	Saturation velocity in the saturation region for short channel devices
BVSAT ^(b)	-	100e-9	-	-	Saturation velocity coefficient in the saturation region for short channel devices
VSATB ^(b)	1/V	0.0	0.0	-	Saturation velocity parameter for Back-gate bias dependence on mobility at high V _{ds}

Parameter	Unit	Default	Min	Max	Description
AVSATB ^(b)	-	0.0	0.0	-	Saturation velocity parameter for Back-gate bias dependence on mobility at high Vds for short channel devices
BVSATB ^(b)	-	0.0	0.0	-	Saturation velocity parameter for Back-gate bias dependence on mobility at high Vds for short channel devices
VSAT1 ^(b)	m/s	VSAT	-	-	Saturation velocity in the linear region
AVSAT1 ^(b)	-	AVSAT	-	-	Saturation velocity in the linear region for short channel devices
BVSAT1 ^(b)	-	BVSAT	-	-	Saturation velocity coefficient in the linear region for short channel devices
VSATCV ^(b)	m/s	VSAT	-	-	Saturation velocity for the saturation region for C-V
AVSATCV ^(b)	m/s	AVSAT	-	-	Saturation velocity in the saturation region for short channel C-V
BVSATCV ^(b)	m/s	BVSAT	-	-	Saturation velocity coefficient in the saturation region for short channel CV
DELTAVSAT ^(b)	m/s	1	0.01	-	Velocity saturation parameter
KSATIV ^(b)	-	1.0	-	-	Parameter for strong inversion regime for long channel Vdsat
KSUBIV ^(b)	-	1.0	-	-	Parameter for weak inversion regime for long channel Vdsat
MEXP ^(b)	-	4	2	-	Smoothing function factor for Vdsat
AMEXP ^(b)	-	0	0	-	Smoothing function factor for Vdsat in short channel devices
BMEXP ^(b)	-	1	-	-	Smoothing function coefficient for Vdsat in short channel devices
PTWG ^(b)	1/V ²	0.0	-	-	Correction factor for velocity saturation

Parameter	Unit	Default	Min	Max	Description
APTWG ^(b)	-	0.0	-	-	Correction factor for velocity saturation in short channel devices
BPTWG ^(b)	-	100.0e-9	-	-	Coefficient for correction factor for velocity saturation in short channel devices
PTWGB ^(b)	$1/V^3$	0.0	-	-	Parameter for back-gate bias sensitivity in PTWGB
APTWGB ^(b)	-	0.0	-	-	Parameter for back-gate bias sensitivity in PTWGB for short-channel devices
BPTWGB ^(b)	-	100.0e-9	-	-	Parameter for back-gate bias sensitivity in PTWGB for short-channel devices
PTWGB2 ^(b)	$1/V^3$	0.0	-	-	Parameter for back-gate bias sensitivity in PTWGB
APTWGB2 ^(b)	-	0.0	-	-	Parameter for back-gate bias sensitivity in PTWGB for short-channel devices
BPTWGB2 ^(b)	-	100.0e-9	-	-	Parameter for back-gate bias sensitivity in PTWGB for short-channel devices
U0 ^(b)	$m^2/V.s$	3e-2	-	-	Low field mobility
ETAMOD	-	2.0	-	-	Effective field parameter
UP ^(b)	μm^{LPA}	0.0	-	-	Mobility L coefficient
LPA	-	1.0	-	-	Mobility L power coefficient
UA ^(b)	$(cm/MV)^{EA}$	0.3	0.0	-	Phonon/surface roughness scattering
AUA ^(b)	-	0	-	-	Phonon/surface roughness scattering for short-channel devices
BUA ^(b)	-	100.0e-9	-	-	Phonon/surface roughness scattering for short-channel devices
EU ^(b)	cm/MV	2.5	0.0	-	Phonon/surface roughness scattering

Parameter	Unit	Default	Min	Max	Description
AEU ^(b)	-	0	-	-	Phonon/surface roughness scattering for short-channel devices
BEU ^(b)	-	100.0e-9	-	-	Phonon/surface roughness scattering for short-channel devices
UD ^(b)	cm/MV	0.0	0.0	-	Columbic scattering (experimental)
AUD ^(b)	-	0.0	-	-	Columbic scattering (experimental) for short-channel devices
BUD ^(b)	-	50.0e-9	-	-	Columbic scattering (experimental) for short-channel devices
UDB ^(b)	cm/MV	0.0	0.0	-	Back bias sensitivity on columbic scattering (experimental)
AUDB ^(b)	-	0.0	-	-	Back bias sensitivity on columbic scattering (Experimental) for short-channel devices
BUDB ^(b)	-	50.0e-9	-	-	Back bias sensitivity on columbic scattering (Experimental) for short channel devices
UC ^(b)	(cm/MV) ^{EU} .1/V	0.0	0.0	-	Parameter for back-gate bias dependence on mobility at low Vds
AUC ^(b)	-	0.0	-	-	Parameter for back-gate bias dependence on mobility at low Vds for short-channel devices
BUC ^(b)	-	100.0e-9	-	-	Parameter for back-gate bias dependence on mobility at low Vds for short-channel devices
UCS ^(b)	-	1.0	0.0	-	Columbic scattering (experimental)
PCLM ^(b)	-	0.013	0.0	-	Channel length modulation (CLM) parameter

Parameter	Unit	Default	Min	Max	Description
APCLM ^(b)	-	0	-	-	Channel length modulation (CLM) parameter for short-channel devices
BPCLM ^(b)	-	100.0e-9	-	-	Channel length modulation (CLM) parameter for short-channel devices
PCLMG	-	0	-	-	Gate bias dependent parameter for short-channel devices
PCLMCV ^(b)	-	0.013	0.0	-	Channel length modulation (CLM) parameter for C-V
RDSWMIN	$\Omega \cdot \mu\text{m}^{WR}$	0.0	0.0	-	RDSMOD=0 S/D extension resistance per unit width at high Vgs
RDSW ^(b)	$\Omega \cdot \mu\text{m}^{WR}$	100	0.0	-	RDSMOD=0 zero bias S/D extension resistance per unit width at high Vgs
ARDSW ^(b)	-	0	-	-	RDSMOD=0 zero bias S/D extension resistance per unit width at high Vgs for short-channel devices
BRDSW ^(b)	-	100.0e-9	-	-	RDSMOD=0 zero bias S/D extension resistance per unit width at high Vgs for short-channel devices
RSWMIN	$\Omega \cdot \mu\text{m}^{WR}$	0.0	0.0	-	RDSMOD=1 source extension resistance per unit width at high Vgs
RSW ^(b)	$\Omega \cdot \mu\text{m}^{WR}$	50	0.0	1	RDSMOD=1 zero bias source extension resistance per unit width at high Vgs
ARSW ^(b)	-	0	-	-	RDSMOD=1 zero bias source extension resistance per unit width at high Vgs for short-channel devices
BRSW ^(b)	-	100.0e-9	-	-	RDSMOD=1 zero bias source extension resistance per unit width at high Vgs for short-channel devices

Parameter	Unit	Default	Min	Max	Description
RDWMIN	$\Omega \cdot \mu\text{m}^{WR}$	RSWMIN	0.0	1	RDSMOD=1 drain extension resistance per unit width at high Vgs
RDW ^(b)	$\Omega \cdot \mu\text{m}^{WR}$	50	0.0	-	RDSMOD=1 zero bias drain extension resistance per unit width at high Vgs
ARDW ^(b)	-	0	-	-	RDSMOD=1 zero bias drain extension resistance per unit width at high Vgs for short-channel devices
BRDW ^(b)	-	100.0e-9	-	-	RDSMOD=1 zero bias drain extension resistance per unit width at high Vgs for short-channel devices
PRWG ^(b)	1/V	0.0	0.0	-	Front-gate bias dependence of S/D extension resistance
PRWB ^(b)	1/V	0.0	-	-	Back-gate bias dependence of S/D extension resistance
WR ^(b)	-	1.0	-	-	W dependence parameter of S/D extension resistance
RSHS	Ω	0.0	0.0	-	Source-side sheet resistance
RSHD	Ω	RSHS	0.0	-	Drain-side sheet resistance
XGW	m	0	-	-	Distance from gate contact center to device edge
XGL	m	0	-	-	Variation in Ldrawn
NGCON	-	1	1	2	Number of gate contacts
RSHG	Ω	0.1	-	-	Gate sheet resistance
XRCRG1	-	12.0	-	-	Parameter for NQS gate resistance (NQSMOD=1)
XRCRG2	-	1.0	-	-	Parameter for NQS gate resistance (NQSMOD=1)
PDIBL1 ^(b)	-	1.30	0.0	-	Parameter for DIBL effect on Rout
PDIBL2 ^(b)	-	2e-4	0.0	-	Parameter for DIBL effect on Rout

Parameter	Unit	Default	Min	Max	Description
DROUT ^(b)	-	1.06	0.0	-	L dependence of DIBL effect on Rout
PVAG ^(b)	-	1.0	-	-	Vgs dependence on early voltage
AIGBINV ^(b)	$(Fs^2/g)^{0.5}/m$	1.11e-2	-	-	Parameter for Igb in inversion
BIGBINV ^(b)	$(Fs^2/g)^{0.5}/m.V$	9.49e-4	-	-	Parameter for Igb in inversion
CIGBINV ^(b)	1/V	6.00e-3	-	-	Parameter for Igb in inversion
EIGBINV ^(b)	V	1.1	-	-	Parameter for Igb in inversion
NIGBINV ^(b)	-	3.0	0.0	-	Parameter for Igb in inversion
AIGBACC ^(b)	$(Fs^2/g)^{0.5}/m$	1.36e-2	-	-	Parameter for Igb in accumulation
BIGBACC ^(b)	$(Fs^2/g)^{0.5}/m.V$	1.71e-3	-	-	Parameter for Igb in accumulation
CIGBACC ^(b)	1/V	7.5e-2	-	-	Parameter for Igb in accumulation
NIGBACC ^(b)	-	1.0	0.0	-	Parameter for Igb in accumulation
TOXP ^(b)	m	-	-	-	Physical oxide thickness
AIGC ^(b)	$(Fs^2/g)^{0.5}/m$	1.36e-2	-	-	Parameter for Igc in inversion
BIGC ^(b)	$(Fs^2/g)^{0.5}/m.V$	1.71e-3	-	-	Parameter for Igc in inversion
CIGC ^(b)	1/V	7.5e-2	-	-	Parameter for Igc in inversion
DIGC ^(b)	-	1.0	-	-	Parameter for Igc in inversion
PIGCD ^(b)	1/V	1.0	0.0	-	Vds dependence of Igcs and Igcd
DLCIGS	m	0.0	-	-	Delta L for Igs model
DLCIGD	m	DLCIGS	-	-	Delta L for Igd model
AIGD ^(b)	$(Fs^2/g)^{0.5}/m$	AIGS	-	-	Parameter for Igd in inversion
BIGD ^(b)	$(Fs^2/g)^{0.5}/m.V$	BIGS	-	-	Parameter for Igd in inversion

Parameter	Unit	Default	Min	Max	Description
CIGD ^(b)	1/V	CIGS	-	-	Parameter for Igd in inversion
DIGD ^(b)	-	DIGS	-	-	Parameter for Igd in inversion
AIGS ^(b)	$(F_s^2/g)^{0.5}/m$	1.36e-2	-	-	Parameter for Igs in inversion
BIGS ^(b)	$(F_s^2/g)^{0.5}/m.V$	1.71e-3	-	-	Parameter for Igs in inversion
CIGS ^(b)	1/V	0.075	0.0	-	Parameter for Igs in inversion
DIGS ^(b)	-	1.0	-	-	Parameter for Igs in inversion
POXEDGE ^(b)	-	1	0.0	-	Factor for the gate edge Tox
TOXREF	m	1.2nm	0.0	-	Nominal gate oxide thickness for gate tunneling current
NTOX ^(b)	-	1.0	-	-	Exponent for gate oxide ratio
AGISL ^(b)	Ω^{-1}	AGIDL	-	-	Pre-exponential coefficient for GISL
BGISL ^(b)	V/m	BGIDL	-	-	Exponential coefficient for GISL
EGISL ^(b)	V	EGIDL	-	-	Band bending parameter for GISL
PGISL ^(b)	-	1.0	-	-	Exponent of electric field for GISL
VBGISL ^(b)	-	1	-	-	Back-gate correction factor for GISL
VBEGISL ^(b)	V	0.5	-	-	Back band bending parameter for GISL
AGIDL ^(b)	Ω^{-1}	6.055e-12	-	-	Pre-exponential coefficient for GIDL
BGIDL ^(b)	V/m	0.3e-9	-	-	Exponential coefficient for GIDL
EGIDL ^(b)	V	0.2	-	-	Band bending parameter for GIDL
PGIDL ^(b)	-	1.0	-	-	Exponent of electric field for GIDL
VBGIDL ^(b)	-	1	-	-	Back-gate correction factor for GIDL

Parameter	Unit	Default	Min	Max	Description
VBEGIDL ^(b)	V	0.5	-	-	Back band bending parameter for GIDL
ALPHA0 ^(b)	m/V	0.0	-	-	First parameter of Iii
ALPHA1 ^(b)	1/V	0.0	-	-	L scaling parameter of Iii
BETA0	1/V	0.0	-	-	Vds dependent parameter of Iii
LOVS	m	0.0	-	-	Overlap length for g/s overlap (for capacitance calculation)
LOVD	m	LOVS	-	-	Overlap length for g/d overlap (for capacitance calculation)
CFS	F/m	0.0	-	-	Outer fringe cap
CFD	F/m	0.0	-	-	Outer fringe cap
CGSL ^(b)	F/m	0.0	0.0	-	Overlap capacitance between gate and lightly-doped source region
CGDL ^(b)	F/m	CGSL	0.0	-	Overlap capacitance between gate and lightly-doped drain region
CKAPPAS ^(b)	V	0.6	0.02	-	Coefficient of bias-dependent overlap capacitance for source side
CKAPPAD ^(b)	V	CKAPPAS	0.02	-	Coefficient of bias-dependent overlap capacitance for drain side
PCOVBS0	V	0	-	-	Back-gate dependent overlap capacitance clamping shift voltage for the source side
PCOVBD0	V	PCOVBS0	-	-	Back-gate dependent overlap capacitance clamping shift voltage for the drain side
PCOVBS1	-	0	-	-	Parameter of back-gate dependent overlap capacitance for the source side
PCOVBD1	-	PCOVBS1	-	-	Parameter of back-gate dependent overlap capacitance for the drain side

Parameter	Unit	Default	Min	Max	Description
CSDBGSW	F/m	0.0	-	-	Prefactor for bias-dependent inner fringe capacitance model
KBG0PW ^(b)	-	1.0	-	-	Parameter for p-type substrate factor
KBG1PW ^(b)	-	0	-	-	Parameter for length dependence of p-type substrate factor
KBG2PW ^(b)	-	-1	-	-	Parameter for length dependence of p-type substrate factor
DBGPW ^(b)	-	0.12	-	-	Parameter for length dependence of p-type substrate factor
BPFACITORPW ^(b)	-	0.0	0.0	1.0	Back-plane (BP) effect for p-type substrate, 1 means no BP
VKNEE1PW ^(b)	V	0.0	-	-	Back gate voltage at which the p-type back gate depletion below the BOX starts
VKNEE2PW ^(b)	V	1.0	0.0	-	Maximum potential drop below the BOX for p-type substrate
KBG0NW ^(b)	-	1.0	-	-	Parameter for n-type substrate factor
KBG1NW ^(b)	-	0	-	-	Parameter for length dependence of n-type substrate factor
KBG2NW ^(b)	-	-1	-	-	Parameter for length dependence of n-type substrate factor
DBGNW ^(b)	-	0.12	-	-	Parameter for length dependence of n-type substrate factor
BPFACITORNW ^(b)	-	0.0	0.0	1.0	Back-plane (BP) effect for n-type substrate, 1 means no BP
VKNEE1NW ^(b)	V	0.0	-	-	Back gate voltage at which the n-type back gate depletion below the BOX starts

Parameter	Unit	Default	Min	Max	Description
VKNEE2NW ^(b)	V	1.0	0.0	-	Maximum potential drop below the BOX for n-type substrate
EF	-	1.0	0.0	2.0	Flicker noise frequency exponent
LINTNOI	m	0.0	-	$L_{eff}/2$	L_{int} offset for flicker noise calculation
EM	V/m	4.1e7	-	-	Flicker noise parameter
NOIA	$eV^{-1}s^{-1}m^{-3}$	6.250e39	-	-	Flicker noise parameter
NOIB	$eV^{-1}s^{-1}m^{-1}$	3.125e24	-	-	Flicker noise parameter
NOIC	$eV^{-1}s^{-1}m^{-EF}$	8.750e7	-	-	Flicker noise parameter
NTNOI	-	1.0	0.0	-	Thermal noise parameter
QMTCENCV ^(b)	-	0.0	-	-	Prefactor/switch for QM effective width and oxide thickness correction for CV
ETAQM	-	0.54	-	-	Body-gate coefficient for QM charge centroid
QM0	-	1e-3	0.0	-	Normalization parameter for QM charge centroid (inversion)
PQM	-	0.66	-	-	Fitting parameter for QM charge centroid (inversion)

BSIM-IMG Model — Temperature Dependence and Self-Heating Model Parameters

Parameter	Unit	Default	Min	Max	Description
TNOM	C	27	-273.15	-	Temperature at which the model is extracted (in Celcius)
TBGASUB	eV/K	7.02e-4	-	-	Bandgap temperature coefficient
TBGBSUB	K	1108.0	-	-	Bandgap temperature coefficient
KT1 ^(b)	V	0.0	-	-	Vth temperature coefficient

Parameter	Unit	Default	Min	Max	Description
KT1L	V.m	0.0	-	-	Vth temperature coefficient
KT2 ^(b)	-	0.0	-	-	Vth temperature coefficient
KT2L	m	0.0	-	-	Length dependent parameter for Vth temperature Vbg coefficient
UTE ^(b)	-	0.0	-	-	Mobility temperature coefficient
UTL ^(b)	-	-1.5e03	-	-	Mobility temperature coefficient
UA1 ^(b)	-	1.032e-3	-	-	Mobility temperature coefficient for UA
UC1 ^(b)	-	0.0	-	-	Mobility temperature coefficient for UC
UD1 ^(b)	-	0.0	-	-	Mobility temperature coefficient
UCSTE ^(b)	-	-4.775e-3	-	-	Mobility temperature coefficient
AT ^(b)	1/K	-0.00156	-	-	Saturation velocity temperature coefficient
ATL ^(b)	m	0.0	-	-	Length scaling parameter for AT
TMEXP ^(b)	-	0	-	-	Temperature coefficient for smoothing function factor for VSAT
ATB ^(b)	1/K	0.0	-	-	Back bias sensitivity parameter for saturation velocity temperature coefficient
ATBL ^(b)	m	0.0	-	-	Length scaling parameter for ATB
K0 ^(b)	V	0.0	-	-	Lateral NUD voltage parameter
K01 ^(b)	V/K	0.0	-	-	Temperature dependence of K0
K0SI ^(b)	-	1.0	-	-	Correction factor for strong inversion, used in Mund
K0SI1 ^(b)	1/K	0.0	-	-	Temperature dependence of K0SI
PTWGT ^(b)	1/k	0.004	-	-	PTWG temperature coefficient
PRT ^(b)	1/K	0.001	-	-	Series resistance temperature coefficient

Parameter	Unit	Default	Min	Max	Description
TETA0 ^(b)	-	0.0	-	-	Temperature dependence for DIBL effect
IIT ^(b)	-	-0.5	-	-	Impact ionization temperature coefficient
TGIDL ^(b)	1/K	-0.003	-	-	GIDL temperature coefficient
TGISL ^(b)	1/K	-0.003	-	-	GISL temperature coefficient
IGT ^(b)	-	2.5	-	-	Gate current temperature coefficient
RTH0	$\Omega \cdot m \cdot K/W$	0.01	0.0	-	Thermal resistance for self-heating calculation
CTH0	W.s/m/K	1.0e-5	0.0	-	Thermal resistance for self-heating calculation
WTH0	m	0.0	0.0	-	Width-dependence coefficient for self-heating calculation

BSIM-IMG Model — Operating Point Information

BSIM-IMG model supports standard MOSFET DC operating point outputs (see [MOSFET DC Operating Point Output on page 917](#)).

BSIM-IMG Model — Output Templates

See [Output Templates for BSIM-CMG \(Level 72\)](#), [BSIM6 \(Level 77\)](#), and [BSIM-IMG \(Level 78\)](#).

LEVEL 79: EKV3 Model

The EKV3 is an analytical compact MOSFET model that relies on the physics of MOSFET, namely the charge sheet theory, to describe its behavior. EKV3 is a representative of the *charge-based* MOS transistor compact models.

The EKV3 model supports three versions of internal circuitry. Each version covers the needs of certain cases. These versions are listed in the following table.

Mode Name	Description	Internal Nodes
DC	External series resistors	2

Mode Name	Description	Internal Nodes
RF	External series resistors, gate resistor, and full 6 substrate resistor network	
NQS	External series resistors, gate resistor, and full 10 substrate resistor network; minimal channel segmentation	

The following sections discuss these topics:

- [EKV3 Model—PrimeSim Continuum Support](#)
- [EKV3 Model—General Syntax](#)
- [EKV3 Model—Features](#)
- [EKV3 Model—Parameters](#)

EKV3 Model—PrimeSim Continuum Support

Based on different switch-control flags, the PrimeSim Continuum solution supports the following three version of internal circuitry:

- DC: switch-control flag RBODYMOD = 0, RGATEMODE =0, and NQSMOD = 0
- RF: switch-control flag RBODYMOD = 1, RGATEMODE =1, and NQSMOD = 0
- NQS: switch-control flag NQSMOD = 1

EKV3 Model—General Syntax

General syntax:

```
Mxxx nd ng ns nb mname [W=val] [L=val]
.model mname NMOS|PMOS LEVEL=79
```

Level 79 is the EKV3 model in the PrimeSim Continuum solution.

An example of using the EKV3 model in the PrimeSim Continuum solution:

```
M1 drain gate source bulk NCH L=1u
.MODEL NCH NMOS LEVEL=79
```

EKV3 Model—Features

The EKV3 model has the following main features:

- Physical modeling of charges, including accumulation region and polysilicon depletion and quantum mechanical effects
- Bias-dependent overlap charge model
- Inner fringing charge/capacitance
- Non-uniform vertical doping
- NQS (AC, noise)
- RF model-external subcircuit
- Mobility (reduction due to vertical field effect)
- Impact ionization current
- Gate currents
- Longitudinal field effect
- Reverse short channel effect
- Inverse narrow width effect
- Drain induced barrier lowering
- Source and drain charge sharing
- Edge conduction
- Geometrical effects, width scaling
- Noise (flicker/thermal)
- Temperature effects

EKV3 Model—Parameters

The following sections define these parameters:

- [EKV3 Instance Parameters](#)
- [EKV3 Model Parameters](#)
- [EKV3 Model—Operating Point Information](#)

EKV3 Instance Parameters

Parameter	Unit	Default	Min	Max	Description
L	m	10.0e-06	1.0e-09	-	Gate length
W	m	10.0e-06	1.0e-09	-	Total gate width
NF	-	1	1	-	Number of fingers
M	-	1	1	-	Multiplicity factor
AD	m ²	0.0	0.0	1	Area of drain active area
AS	m ²	0.0	0.0	1	Area of source active area
PD	m	0.0	0.0	-	Perimeter of drain active area
PS	m	0.0	0.0	-	Perimeter of source active area
SA	m	0.0	0.0	-	Distance of first gate finger from STI (one side)
SB	m	0.0	0.0	-	Distance of last gate finger from STI (other side)
SD	m	0.0	0.0	-	Distance between neighboring gate fingers

EKV3 Model Parameters

This section discusses the following topics:

- [Switch Parameters](#)
- [Matching Parameters](#)
- [Oxide, Substrate, and Gate Doping Parameters](#)
- [Quantum Effect Parameters](#)
- [Vertical Field Mobility Effect Parameters](#)
- [Coulomb Scattering Parameters](#)
- [Extrinsic Model Parameters](#)
- [Geometrical Parameters](#)
- [Velocity Saturation and CLM-related Parameters](#)
- [Charge Sharing Effect Parameters](#)

- [Drain Induced Barrier Lowering Parameters](#)
- [Inverse Narrow Channel Effect Parameters](#)
- [DITS Parameters](#)
- [Impact Ionization Current Parameters](#)
- [Gate Current Parameters](#)
- [GIDL-GISL Parameters](#)
- [Flicker Noise Parameters](#)
- [Edge Device Parameters](#)
- [STI Stress Effect Parameters](#)
- [Temperature Parameters](#)
- [Overlap Capacitance Parameters](#)
- [Fringing Capacitance Parameters](#)
- [Typical SPICE Model Parameters](#)
- [Parameters for Non-geometrical Approach](#)
- [Temperature Scaling Parameters](#)
- [Source-side Junction Diode Parameters](#)
- [Both Side Junction Diode Parameters](#)
- [Drain-side Junction Diode Parameters](#)
- [Gate and Substrate Resistances Parameters](#)

Switch Parameters

Parameter	Unit	Default	Min	Max	Description
SIGN	-	1	-1	1	1 for NMOS, -1 for PMOS
RBODYMOD	-	0	0	1	Substrate resistance network model selector
RGATEMOD	-	0	0	1	Gate resistance model selector
NQSMOD	-	0	0	1	NQS model selector

Parameter	Unit	Default	Min	Max	Description
TG	-	-1	-1	1	Doping type of gate: -1 for opposite than bulk, 1 for same with bulk, 0 for metal gate; no polysilicon depletion effect
TNOM	°C	27.0	-273.15	-	Nominal temperature
SCALE	-	1.0	0.0	-	Scaling factor for all dimensions
XL	m	0.0	-	-	Optical offset for gate length
XW	m	0.0	-	-	Optical offset for gate width
TH_NOI	-	1.0	0	1	Thermal noise flag (on/off); includes short channel effects but no NQS noise.
NQS_NOI	-	0.0	0	1	NQS noise flag (on/off); includes thermal noise without short channel effects.
INFO_LEVEL	-	0.0	0	-	Flag for information printout; currently only the value of 1 is supported.
QOFF	-	0.0	0	1	Flag for turning off the dynamic behavior of the model (for debugging purposes)

Matching Parameters

Parameter	Unit	Default	Min	Max	Description
AVTO	-	0	-	-	Matching parameter for threshold voltage
AGAMMA	-	0	-	-	Matching parameter for body effect coefficient
AKP	-	0	-	-	Matching parameter for mobility

Oxide, Substrate, and Gate Doping Parameters

Parameter	Unit	Default	Min	Max	Description
COX	F/m ²	0.012	0.0	-	Oxide capacitance per unit area

Parameter	Unit	Default	Min	Max	Description
XJ	m	20.0e-09	0.0	-	Depth of active areas
VTO	V	0.3	-	-	Threshold voltage
PHIF	V	0.45	0.0	-	Bulk Fermi potential
GAMMA	$\sqrt{V}$	0.3	0.0	-	Body effect coefficient
GAMMAG	$\sqrt{V}$	4.1	0.0	-	Body effect coefficient for gate
N0	-	1.0	0.0	-	Long channel slope factor fine tuning
VBI	V	0.0	0.0	-	Built-in voltage drop

Quantum Effect Parameters

Parameter	Unit	Default	Min	Max	Description
AQMA	$V^{1/3}F^{-2/3}$	0.5	0.0	-	Quantum effect coefficient in accumulation
AQMI	$V^{1/3}F^{-2/3}$	0.4	0.0	-	Quantum effect coefficient in inversion
ETAQM	-	0.75	0.0	-	Quantum effect: weight of inversion charge in effective vertical field calculation

Vertical Field Mobility Effect Parameters

Parameter	Unit	Default	Min	Max	Description
KP	F/(V*s)	500.0e-06	0.0	-	Mobility multiplied with COX
E0	V/m	1.0e+10	0.0	-	First order coefficient for mobility reduction due to vertical field
E1	V^2/m^2	3.1e+08	0.0	-	Second order coefficient for mobility reduction due to vertical field
ETA	-	0.5	0.0	-	Weight of inversion charge into calculation of vertical field

Coulomb Scattering Parameters

Parameter	Unit	Default	Min	Max	Description
THC	-	0.0	0.0	-	Coulomb scattering factor
ZC	-	1.0e-06	0.0	-	Coulomb scattering coefficient for normalized inversion charge

Extrinsic Model Parameters

Parameter	Unit	Default	Min	Max	Description
LA	m	1.0	0.0	-	First critical length for KP length scaling
LB	m	1.0	0.0	-	Second critical length for KP length scaling
KA	-	0.0	-	-	Factor for KP length scaling for LA
KB	-	0.0	-	-	Factor for KP length scaling for LB
WKP1	m	1.0e-06	-	-	Width parameter for mobility profile versus width
WKP2	-	0.0	-	-	Amplitude parameter for mobility profile versus width
WKP3	-	1.0	-	-	Span parameter for mobility profile versus width
AVT	-	0.0	-	-	Factor for long and wide channel VTO correction
LVT	m	1.0	0.0	-	Length for long channel VTO correction
WVT	m	1.0	0.0	-	Width for wide channel VTO correction
AGAM	-	0.0	-	-	Factor for long and wide channel GAMMA correction
LGAM	m	1.0	0.0	-	Length for long channel GAMMA correction
WGAM	m	1.0	0.0	-	Width for wide channel GAMMA correction

Parameter	Unit	Default	Min	Max	Description
NFVTA	-	0.0	0.0	-	Number of fingers factor for VTO dependence on NF
NFVTB	-	10000.0	-	-	Amplitude parameter for NFVTA
LWR	m ²	0.0	-	-	Length scaling of WR
LQWR	m	0.0	-	-	Length scaling of QWR
LNWR	m	0.0	-	-	Length scaling of NWR
LDPHIEDGE	m	0.0	-	-	Length scaling of DPHIEDGE
WLR	m ²	0.0	-	-	Width scaling of LR
WQLR	m	0.0	-	-	Width scaling of QLR
WNLR	m	0.0	-	-	Width scaling of NLR
WUCRIT	m	0.0	-	-	Width scaling of UCRIT
WLAMBDA	m	0.0	-	-	Width scaling of LAMBDA
WETAD	m	0.0	-	-	Width scaling of ETAD
WE0	m	0.0	-	-	Width scaling of E0
WE1	m	0.0	-	-	Width scaling of E1
WRLX	m	0.0	-	-	Width scaling of RLX
WUCEX	m	0.0	-	-	Width scaling of UCX
WDPHIEDGE	m	0.0	-	-	Width scaling of DPHIEDGE
WLDPHIEDGE	m ²	0.0	-	-	Area scaling of DPHIEDGE
WLDGAMMAEDGE	m ²	0.0	-	-	Area scaling of DGAMMAEDGE

Geometrical Parameters

Parameter	Unit	Default	Min	Max	Description
DL	m	-10.0e-09	-	-	Difference between effective and drawn gate length

Parameter	Unit	Default	Min	Max	Description
DLC	m	0.0	-	-	Fine tuning difference of effective gate length between current and capacitance behavior
DW	m	-10.0e-09	-	-	Difference between effective and drawn gate width
DWC	m	0.0	-	-	Fine tuning difference of effective gate width between current and capacitance behavior
WDL	m ²	0.0	-	-	Width scaling for narrow devices of L_{eff}
LDW	m ²	0.0	-	-	Length scaling for short devices of W_{eff}
LL	m	0.0	-	-	Base for exponential dependence of L_{eff}
LLN	-	1.0	-	-	Exponent for exponential dependence of L_{eff}

Velocity Saturation and CLM-related Parameters

Parameter	Unit	Default	Min	Max	Description
UCRIT	V/m	5.0e+06	0.0	-	Critical longitudinal field of carriers for VSAT
DELTA	-	2.0	1	2	Order of VSAT model (variable order model 1~2)
LAMBDA	-	0.5	0.0	-	Length modulation coefficient
ACLM	-	0.83	0.0	-	Channel length modulation factor

Charge Sharing Effect Parameters

Parameter	Unit	Default	Min	Max	Description
LETA	-	500.0e-03	0.0	-	CHSH coefficient
LETA0	1/m	0.0	-	-	Length independent CHSH coefficient

Parameter	Unit	Default	Min	Max	Description
LETA2	m	0.0	-	-	Second order length scaling CHSH coefficient
WETA	-	200.0e-03	-	-	Narrow channel CHSH coefficient
NCS	-	1.0	0.0	-	Slope factor dependence on CHSH

Drain Induced Barrier Lowering Parameters

Parameter	Unit	Default	Min	Max	Description
ETAD	-	1.0	0.0	-	DIBL coefficient
SIGMAD	-	1.0	0.0	-	Body effect DIBL coefficient

Inverse Narrow Channel Effect Parameters

Parameter	Unit	Default	Min	Max	Description
WR	m	90.0e-09	0.0	-	Width scaling coefficient for INWE
QWR	Vm ² /F	0.3e-03	-	-	Threshold voltage coefficient of INWE
NWR	m ² /F	5.0e-03	-	-	Body effect coefficient of INWE

DITS Parameters

Parameter	Unit	Default	Min	Max	Description
FPROUT	$\frac{1}{\sqrt{m}}$	1.0e06	0.0	-	Output resistance factor for DITS effect
PDITS	-	0.0	0.0	-	DITS parameter
PDITSL	1/m	0.0	0.0	-	Length scaling factor for DITS effect
PDITSD	1/V	1.0	0.0	-	DITS dependence on drain bias
DDITS	-	0.3	0.0	-	Smoothing parameter for DITS effect

Impact Ionization Current Parameters

Parameter	Unit	Default	Min	Max	Description
IBA	1/m	0.0e+06	0.0	-	IDB coefficient
IBB	V/m	300.0e+06	0.0	-	IDB exponential factor
IBN	-	1.0	0.0	-	IDB factor of VSAT

Gate Current Parameters

Parameter	Unit	Default	Min	Max	Description
XB	V	3.1	0.0	-	Si-SiO ₂ tunneling barrier height
EB	V/m	29.0e+09	0.0	-	Characteristic electrical field for gate current
KG	A/V ²	0.0e-06	0.0	-	Gate current parameter
LOVIG	m	20.0e-09	0.0	-	Overlap length for gate current

GIDL-GISL Parameters

Parameter	Unit	Default	Min	Max	Description
AGIDL	A/V	0.0	0.0	-	Pre-exponential coefficient for GIDL
BGIDL	V/m	2.3e+09	0.0	-	Exponential coefficient for GIDL
CGIDL	V ³	0.5	0.0	-	Body effect parameter for GIDL
EGIDL	V	0.8	0.0	-	Fitting parameter for band bending for GIDL

Flicker Noise Parameters

Parameter	Unit	Default	Min	Max	Description
LFNOI	-	0.0	0.0	-	Flicker noise model selection
KF	-	0.0	0.0	-	Flicker noise parameter
AF	-	1.0	0.0	-	Frequency exponent for flicker noise
EF	-	2.0	0.0	-	Flicker noise transconductance exponent

Parameter	Unit	Default	Min	Max	Description
NT	-	1.0e+14	0.0	-	Number of trapped carriers
ALPHAC	-	10.0e+4	0.0	-	Coulomb scattering coefficient
ALPHAH	-	1.0e-20	0.0	-	Hooge parameter for mobility fluctuation noise mode
KGFN	-	0.0	0.0	-	Gate flicker noise factor
ECN	-	1.0e+4	0.0	-	Critical field value
SDR	-	1.0e-6	0.0	-	Series resistance noise parameter

Edge Device Parameters

Parameter	Unit	Default	Min	Max	Description
WEDGE	V	0.0	0.0	-	Width of edge conduction area
DGAMMAE DGE	V	0.0	-	-	Difference of Fermi potential of edge conduction area with respect to the main part of the channel
DPHIEDGE	$\sqrt{V}$	0.0	-	-	Difference of body effect coefficient between edge conduction area and the main channel

STI Stress Effect Parameters

Parameter	Unit	Default	Min	Max	Description
SAREF	m	0.0	0.0	-	Reference distance from STI, for SA
SBREF	m	0.0	0.0	-	Reference distance from STI, for SB
WLOD	m	0.0	-	-	Distance between the edge device and the STI
KKP	-	0.0	-	-	KP dependence on STISE
LKKP	-	0.0	-	-	Length scaling of KP dependence on STISE

Parameter	Unit	Default	Min	Max	Description
WKKP	-	0.0	-	-	Width scaling of KP dependence on STISE
PKKP	-	0.0	-	-	Area scaling of KP dependence on STISE
TKKP	-	0.0	-	-	Temperature scaling of KP dependence on STISE
LLODKKP	-	1.0	-	-	Length exponent of KP dependence on STISE
WLODKKP	-	1.0	-	-	Width exponent of KP dependence on STISE
KVTO	-	0.0	-	-	VTO dependence on STISE
LKVTO	-	0.0	-	-	Length scaling of VTO dependence on STISE
WKVTO	-	0.0	-	-	Width scaling of VTO dependence on STISE
PKVTO	-	0.0	-	-	Area scaling of VTO dependence on STISE
LLODKVTO	-	1.0	-	-	Length exponent of VTO dependence on STISE
WLODKVTO	-	1.0	-	-	Width exponent of VTO dependence on STISE
KGAMMA	-	0.0	-	-	GAMMA dependence on STISE
LODKGAMMA	-	1.0	-	-	Exponent of GAMMA dependence on STISE
KETAD	-	0.0	-	-	ETAD (DIBL effect) dependence on STISE
LODKETAD	-	1.0	-	-	Exponent of ETAD dependence on STISE
KUCRIT	-	0.0	-	-	VSAT dependence on STISE

Temperature Parameters

Parameter	Unit	Default	Min	Max	Description
TCV	V/ °C	600.0e-06	-	-	Linear temperature dependence of VTO
BEX	-	-1.5	-	-	Exponential temperature dependence of KP
TE0EX	-	0.5	-	-	Exponential temperature dependence of E0
TE1EX	-	0.5	-	-	Exponential temperature dependence of E1
TETA	1/ °C	-0.9e-03	-	-	Linear temperature dependence of ETA
UCEX	-	1.5	-	-	Exponential temperature dependence of UCRIT
TLAMBDA	-	0.0	-	-	Linear temperature dependence of LAMBDA
IBBT	-	800.0e-06	-	-	Linear temperature dependence of IBB
TCVL	m * V/ °C	0.0	-	-	Length dependence of TCV
TCVW	m * V/ °C	0.0	-	-	Width dependence of TCV
TCVWL	m ² * V/ °C	0.0	-	-	Area dependence of TCV

Overlap Capacitance Parameters

Parameter	Unit	Default	Min	Max	Description
GAMMAOV	$\sqrt{m}$	1.6	0.0	-	Body effect coefficient of the overlap area
GAMMAGOV	$\sqrt{m}$	10.0	0.0	-	Body effect coefficient of the gate for the overlap area
VFBOV	V	0.0	-	-	Flat-band voltage of the overlap area
LOV	m	20.0e-09	0.0	-	Length of the overlap area
VOV	-	1.0	0.0	1	Bias coefficient for overlap capacitance

Parameter	Unit	Default	Min	Max	Description
CGSO	F/m	0.0	0.0	-	Gate to source outer fringing capacitance
CGDO	F/m	0.0	0.0	-	Gate to drain outer fringing capacitance
CGBO	F/m	0.0	0.0	-	Gate to bulk outer fringing capacitance

Fringing Capacitance Parameters

Parameter	Unit	Default	Min	Max	Description
KJF	C/m	0.0	0.0	-	Fringing capacitance factor
CJF	1/V	0.0	0.0	-	Fringing capacitance bias factor
VFR	V	0.0	0.0	-	Built-in correction for fringing capacitance
DFR	-	1.0e-03	0.0	-	Smooth factor of fringing capacitance model

Typical SPICE Model Parameters

Parameter	Unit	Default	Min	Max	Description
HDIF	m	0.0e-06	0.0	-	Half length of active area
RSH	Ω	0.0	0.0	-	Square resistance of active area
LDIF	m	0.0	0.0	-	Length of the lightly doped area (LDA)
RS	Ω	0.0	0.0	-	LDA source series resistance
RD	Ω	0.0	0.0	-	LDA drain series resistance

Parameters for Non-geometrical Approach

Parameter	Unit	Default	Min	Max	Description
RLX	Ω / m	-1.0	-	-	Series resistance (symmetric model)

Parameter	Unit	Default	Min	Max	Description
RSX	Ω / m	-1.0	-	-	Source series resistance (asymmetric model)
RDX	Ω / m	-1.0	-	-	Drain series resistance (asymmetric model)

Temperature Scaling Parameters

Parameter	Unit	Default	Min	Max	Description
TR	$\Omega / (^{\circ}C)$	0.0	-	-	First order temperature coefficient of resistors
TR2	$\Omega / (^{\circ}C)^2$	0.0	-	-	Second order temperature coefficient of resistors

Source-side Junction Diode Parameters

Parameter	Unit	Default	Min	Max	Description
NJS	-	1.0	-	-	Slope factor for parasitic diodes (source)
XJBVS	-	0.0	-	-	Breakdown effect coefficient (source)
BVS	V	10.0	-	-	Breakdown voltage (source)
JSS	A/m^2	0.0e-09	-	-	Area component of diode current (source)
JSSWS	A/m	0.0e-12	-	-	Perimeter component of diode current (source)
JSSWGS	A/m	0.0e-12	-	-	Gate side component of diode current (source)
JTSS	A/m^2	0.0e-09	-	-	Area component of trap-assisted current (source)
JTSSWS	A/m	0.0e-12	-	-	Perimeter component of trap-assisted current (source)
JTSSWGS	A/m	0.0e-12	-	-	Gate side component of trap-assisted current (source)
NJTSS	-	1.0	-	-	Area slope factor of trap-assisted current (source)

Parameter	Unit	Default	Min	Max	Description
NJTSSWS	-	1.0	-	-	Perimeter slope factor of trap-assisted current (source)
NJTSSWGS	-	1.0	-	-	Gate side slope factor of trap-assisted current (source)
VTSS	V	0.0	-	-	Area voltage factor of trap-assisted current (source)
VTSSWS	V	0.0	-	-	Perimeter voltage factor of trap-assisted current (source)
VTSSWGS	V	0.0	-	-	Gate side voltage factor of trap-assisted current (source)
CJS	F/m ²	0.0e-06	-	-	Area component of diode capacitance (source)
CJSWS	F/m	0.0e-09	-	-	Perimeter component of diode capacitance (source)
CJSWGS	F/m	0.0e-09	-	-	Gate side component of diode capacitance (source)
PBS	V	0.800	-	-	Area parameter of diode capacitance (source)
PBSWS	V	0.600	-	-	Perimeter parameter of diode capacitance (source)
PBSWGS	V	0.600	-	-	Gate side parameter of diode capacitance (source)
MJS	-	0.900	-	-	Area exponent of diode capacitance (source)
MJSWS	-	0.700	-	-	Perimeter exponent of diode capacitance (source)
MJSWGS	-	0.700	-	-	Gate side exponent of diode capacitance (source)
XTIS	-	3.0	-	-	Temperature dependence of diode (source)
XTSS	-	0.0	-	-	Temperature dependence of area component of trap-assisted current (source)

Parameter	Unit	Default	Min	Max	Description
XTSSWS	-	0.0	-	-	Temperature dependence of perimeter component of trap-assisted current (source)
XTSSWGS	-	0.0	-	-	Temperature dependence of gate side component of trap-assisted current (source)
TNJTSS	-	0.0	-	-	Temperature dependence of NJTSS
TNJTSSWS	-	0.0	-	-	Temperature dependence of NJTSSWS
TNJTSSWGS	-	0.0	-	-	Temperature dependence of NJTSSWGS

Both Side Junction Diode Parameters

Parameter	Unit	Default	Min	Max	Description
GMIN	A/V	0.0	0.0	-	Minimum conductance of diode (source and drain)
TCJ	1/°C	0.0	-	-	Temperature dependence of CJS and CJD
TCJSW	1/°C	0.0	-	-	Temperature dependence of CJSWS and CJSWD
TCJSWG	1/°C	0.0	-	-	Temperature dependence of CJSWGS and CJSWGD
TPB	V/°C	0.0	-	-	Temperature dependence of PBS and PBD
TPBSW	V/°C	0.0	-	-	Temperature dependence of PBSWS and PBSWD
TPBSWG	V/°C	0.0	-	-	Temperature dependence of PBSWGS and PBSWGD

Drain-side Junction Diode Parameters

Parameter	Unit	Default	Min	Max	Description
NJD	-	1.0	-	-	Slope factor for drain diode

Parameter	Unit	Default	Min	Max	Description
XJBVD	-	0.0	-	-	Breakdown effect coefficient of drain diode
BVD	V	10.0	-	-	Breakdown voltage of drain diode
JSD	A/m ²	0.0e-09	-	-	Area component of drain diode current
JSSWD	A/m	0.0e-12	-	-	Perimeter component of drain diode current
JSSWGD	A/m	0.0e-12	-	-	Gate side component of drain diode current
JTSD	A/m ²	0.0e-09	-	-	Area component of trap-assisted current (drain)
JTSSWD	A/m	0.0e-12	-	-	Perimeter component of trap-assisted current (drain)
JTSSWGD	A/m	0.0e-12	-	-	Gate side component of trap-assisted current (drain)
NJTSD	-	1.0	-	-	Area slope factor of trap-assisted current (drain)
NJTSSWD	-	1.0	-	-	Perimeter slope factor of trap-assisted current (drain)
NJTSSWGD	-	1.0	-	-	Gate side slope factor of trap-assisted current (drain)
VTSD	V	0.0	-	-	Area voltage factor of trap-assisted current (drain)
VTSSWD	V	0.0	-	-	Perimeter voltage factor of trap-assisted current (drain)
VTSSWGD	V	0.0	-	-	Gate side voltage factor of trap-assisted current (drain)
CJD	F/m ²	0.0e-06	-	-	Area component of drain diode capacitance
CJSWD	F/m	0.0e-09	-	-	Perimeter component of drain diode capacitance
CJSWGD	F/m	0.0e-09	-	-	Gate side component of drain diode capacitance

Parameter	Unit	Default	Min	Max	Description
PBD	V	0.800	-	-	Area parameter of drain diode capacitance
PBSWD	V	0.600	-	-	Perimeter parameter of drain diode capacitance
PBSWGD	V	0.600	-	-	Gate side parameter of drain diode capacitance
MJD	-	0.900	-	-	Area exponent of drain diode capacitance
MJSWD	-	0.700	-	-	Perimeter exponent of drain diode capacitance
MJSWGD	-	0.700	-	-	Gate side exponent of drain diode capacitance
XTID	-	3.0	-	-	Temperature dependence of drain diode
XTSD	-	0.0	-	-	Temperature dependence of area component of trap-assisted current
XTSSWS	-	0.0	-	-	Temperature dependence of perimeter component of trap-assisted drain diode current
XTSSWGD	-	0.0	-	-	Temperature dependence of gate side component of trap-assisted drain diode current
TNJTSD	-	0.0	-	-	Temperature dependence of NJTSD
TNJTSSWD	-	0.0	-	-	Temperature dependence of NJTSSWD
TNJTSSWGD	-	0.0	-	-	Temperature dependence of NJTSSWGD

Gate and Substrate Resistances Parameters

Parameter	Unit	Default	Min	Max	Description
RGSH	Ω	3.0	0.0	-	Gate square resistance

Parameter	Unit	Default	Min	Max	Description
GC	-	1.0	1.0	2.0	Gate contacts (single sided = 1, double sided = 2)
KRGL1	1/m ²	0.0	0.0	-	Length scaling of gate resistance
RDSBSH	Ω	1.0e+03	0.0	-	Drain to source substrate sheet resistance
RBWSH	Ω / m	3.0e-03	0.0	-	Inner-bulk to external bulk resistance
RBN	Ω	0.0	0.0	-	Inner-bulk to external bulk resistance per finger (for RINGTYPE=1)
RSBWSH	Ω / m	1.0e-03	0.0	-	Inner-bulk source side to external bulk resistance
RSBN	Ω	0.0	0.0	-	Inner-bulk source side to external bulk resistance per finger (for RINGTYPE=1)
RDBWSH	Ω / m	1.0e-03	0.0	-	Inner-bulk drain side to external bulk resistance
RDBN	Ω	0.0	0.0	-	Inner-bulk drain side to external bulk resistance per finger (for RINGTYPE=1)
RINGTYPE	-	1.0	0.0	-	Type of guard ring (bulk contacts) (three sides/horse shoe: 1, two sides or symmetric: 2)

EKV3 Model—Operating Point Information

The EKV3 model supports general MOSFET DC operating point outputs (see the following table).

Quantity	Definition
I _{ds}	D-S drain-to-source current
I _{bs}	B-S bulk-to-source current
I _{bd}	B-D bulk-to-drain current
V _{gs}	G-S gate-source voltage

Quantity	Definition
vds	D-S drain-source voltage
vbs	B-S bulk-source voltage
vth	Threshold voltage
vdsat	Saturation voltage
vod	Overdrive voltage
vdmargin	Drain voltage margin
beta	Beta value
gam eff	Gamma effective
gm	DC gate transconductance
gds	D-S drain-source conductance
gmb	B-S bulk-source conductance
cdtot	Total drain capacitance
cgtot	Total gate capacitance
cstot	Total source capacitance
cbtot	Total bulk capacitance
cgs	Total gate-to-source capacitance
cgd	Total gate-to-drain capacitance

The EKV3 model also supports its own DC operating point outputs (see the following table).

Quantity	Definition
igs	Gate-to-source current
igd	Gate-to-drain current
idsi	Intrinsic current
idsedgen	Edge effect current
idex	Extrinsic current, drain side

Quantity	Definition
igex	Extrinsic current, gate side
isex	Extrinsic current, source side
ibex	Extrinsic current, bulk side
ic	inversion coefficient
avi	Intrinsic gain
gmgid	Gm/Ids
n	Slope factor
vm	Early voltage Ids/gds
sid_flicker	Flicker noise PSD SdID @ 1Hz A ² /Hz
svg_flicker	Flicker noise PSD SdVG @ 1Hz V ² /Hz
sid_thermal	Thermal noise PSD SdID A ² /Hz
svg_thermal	Thermal noise PSD SdVG V ² /Hz

LEVEL 80: HiSIM_SOI Model

HiSIM SOI is valid for all possible SOI-MOSFET structures, released by Compact Model Coalition (CMC).

- [Using HiSIM_SOI Model](#)
- [Model Revisions](#)
- [HiSIM_SOI Model — General Syntax](#)
- [HiSIM_SOI Model Parameters and Equations](#)
- [HiSIM_SOI Instance Parameters](#)

Using HiSIM_SOI Model

To simulate using the HiSIM_SOI mode:

- Set `LEVEL = 80` to identify the model as the HiSIMSOI model.

Model Revisions

- [HiSIM_SOI 1.5 Updates](#)
- [HiSIM_SOI 1.4 Updates](#)
- [HiSIM_SOI 1.3](#)

HiSIM_SOI 1.5 Updates

- Support for body resistance network to facilitate RF modeling. Model parameter CORBNET is used to control this network.
- Added mode parameter COSUBSCALE to scaling for Substrate (body) Current.
- Support for floating body effect in body tie model.

HiSIM_SOI 1.4 Updates

- Channel length offset parameter for capacitance (XLDC) is introduced.
- Pocket impurity concentration NSUBP is set forcefully to NSUBS when NSUBP is less than NSUBS. This takes into effect when VERSION > 1.30. (Default: 1.40).
- Pb2 (2øB) refers to Nsub, SOI impurity concentration with various effects included, such as pocket implantation and STI-induced variation, when VERSION > 1.30 (default: 1.40). When VERSION is less than or equal to 1.30, it refers to NSUBS.

- # HiSIM_SOI 1.3

HISIM_SOI Model — General Syntax

```

graph TD
    Start(( )) --> D6{6th node definition}
    D6 -- yes --> C6{COBCNODE}
    D6 -- no --> D5{5th node definition}
    C6 -- =0 --> E1[Error Exit]
    C6 -- =1 --> BTM[Body Tie Model]
    D5 -- yes --> C5{COBCNODE}
    D5 -- no --> C7{COBCNODE}
    C5 -- =1 --> BTM
    C5 -- =0 --> FBM[Floating Body Model]
    C7 -- =1 --> E2[Error Exit]
    C7 -- =0 --> FBM
    BTM --> End1(( ))
    FBM --> End2(( ))
  
```

```
Mxxx nd ng ns ne [bc] [n6] mname [L=val] [W=val]
+ [M=val] [AD=val] [AS=val] [PD=val] [PS=val]
+ [NRD=val] [NRS=val]
```

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HISIM_SOI Model Parameters and Equations

For model parameters and equations, refer to the HISIM_SOI model manual released by CMC.

HISIM_SOI Instance Parameters

Table 9 Instance Parameters

Parameter	Unit	Default	Minimum	Maximum	Description
L	m	5.0E-6	0	inf	Gate length (Lgate)
W	m	5.0E-6	0	inf	Gate width (Wgate)
AD	m ²	0.0	0	inf	Area of drain junction
AS	m ²	0.0	0	inf	Area of source junction
PD	m	0.0	0	inf	Perimeter of drain junction
PS	m	0.0	0	inf	Perimeter of source junction
NGCON	-	1.0	0	inf	Number of gate contacts
XGW	m	0E0	-	-	Distance from gate contact to channel edge
XGL	m	0E0	0	L	Offset of gate length
NF	-	1.0	0	inf	Number of gate fingers
SA	m	0	0	inf	Length of diffusion between gate and STI
SB	m	0	0	inf	Length of diffusion between gate and STI
SD	m	0	0	inf	Length of diffusion between gate and gate
PDBCP	m	0.0	-	-	Parasitic perimeter length for body contact at drain
PSBCP	m	0.0	-	-	Parasitic perimeter length for body contact at source
LOD	m	10.0E-6	-	-	Diffusion length between gate and STI edge
TEMP	C	0.0	-	-	Device temperature

Table 9 Instance Parameters (Continued)

Parameter	Unit	Default	Minimum	Maximum	Description
DTEMP	C	0	-	-	Device temperature change
NBT	-	1.0	-	-	Number of body contact
LBT	m	0.0	-	-	Length of gate over body-contact well
WBTP	m	0.0	-	-	Distance of gate protrusion over body-contact well
WBTN	m	0.0	-	-	Distance of gate protrusion over body-contact well
ABTN	m ²	0.0	-	-	Area of N ⁺ poly area of body contact
ABTP	m ²	0.0	-	-	Area of P ⁺ poly area of body contact
NRS	-	1.0	0	inf	Number of squares in source
NRD	-	1.0	0	inf	Number of squares in drain
LDRIFT	m	1.0E-6	0	inf	Parameter for drift region length (drain)
LDRIFTS	m	1.0E-6	0	inf	Parameter for drift region length (source)
RBSB	Ohm	50	-	-	body resistance (source side) (CORBNET=1)
RBDB	Ohm	50	-	-	body resistance (drain side) (CORBNET=1)

LEVEL 81: ASM-HEMT Model

ASM-HEMT is an advanced physics-based GaN high electron mobility transistors (HEMT) model for high power, high voltage, and high frequency applications.

- [Using the ASM-HEMT Model](#)
- [Model Revisions](#)
- [ASM-HEMT Model — General Syntax](#)
- [ASM-HEMT Model – Version Updates](#)

- [ASM-HEMT Model Parameters and Equations](#)
- [ASM-HEMT Instance Parameters](#)

Using the ASM-HEMT Model

To simulate using the ASM-HEMT mode in the PrimeSim Continuum simulators:

Set LEVEL = 81 to identify the model as the ASM-HEMT model.

Model Revisions

Version=101.0: The first supported version.

ASM-HEMT Model — General Syntax

Syntax:

```
Mxxx nd ng ns nb [dt] mname [L=val] [W=val] [NF=val] [M=val]
+ [DFP1=val] [LFP1=val] [DFP2=val] [LFP2=val] [DFP3=val] [LFP3=val]
+ [DFP4=val] [LFP4=val] [NGCON=val]
.model mname nmos|pmos level=81
```

ASM-HEMT Model – Version Updates

- [ASM-HEMT 101.4 Updates](#)

ASM-HEMT 101.4 Updates

ASM-HEMT version 101.4 has the following features:

- Updated substrate voltage dependence on the pinch-off voltage. Two new model parameters `tepi` and `asub` added to tune the substrate voltage dependence of pinch-off voltage.
- Added new saturation formulation for access region.
- Fixed nine bugs.

ASM-HEMT 101.3.0 Updates

ASM-HEMT version 101.3.0 implements the following features:

- Support for fast field-plate model. New model formulations for modeling the field-plate behavior in a computationally efficient manner is introduced. These can be selected by the model selector FASTFPMOD.
- Support for new gate- and drain-lag model. A new trapmod allows independent tuning of gate- and drain-lag is introduced.

ASM-HEMT 101.2.0 Updates

ASM-HEMT version 101.2.0 implements the following features:

- Drain-source breakdown effect model added.
- Added temperature dependence to the reverse gate current model.
- Added body node to the device schematic in the model code.
- Support for LF/RF gate resistance model – `rgatemod=2` formulations have been added to model the frequency dependence of gate resistance.
- Added body terminal to the device schematic drawn the model code.
- Support for Gate/drain diode avalanche breakdown model – New formulations added as `GATEMOD=2`

ASM-HEMT 101.1.0 Updates

ASM-HEMT version 101.1.0 implements the following features:

- Include scalable parasitic/off state capacitance model: The off-state capacitances are now updated to scale with device dimensions.
- Kink in capacitance simulation with source-side field plate model: Source side field plate model *sigvdsfp1s* was not correctly included in the model code.
- Non-linear resistance model fix: Access region resistance model updated to avoid any negative number in the power function used in its model formulations.
- Reverse gate leakage model: A new reverse gate leakage model (`gatemod=2`) has been added to the code.
- Negative Cgd possibility for certain parameter combination: Minimum value of Cgd is clamped to zero in the revised code.

- `Gatemod=2` temperature equation and default values: Temperature scaling equations and default values have been updated for correct results.
- Update flicker noise formulation based on Ref. G. Coram et al. DOI: 10.1109/TCAD.2020.2966444.

ASM-HEMT Model Parameters and Equations

For the ASM-HEMT model parameters, see the ASM-HEMT manual released by CMC.

ASM-HEMT Instance Parameters

Parameter	Unit	Default	Minimum	Maximum	Description
L	m	2.50E-07	2.00E-08	inf	Channel length
W	m	2.00E-04	2.00E-08	inf	Channel Width
NF	-	1	1	inf	Number of fingers
DFP1	m	5.00E-08	1.00E-10	inf	Distance of FP1 from 2-DEG charge
LFP1	m	1.00E-06	0	inf	Length of FP1
DFP2	m	1.00E-07	1.00E-10	inf	Distance of FP2 from 2-DEG charge
LFP2	m	1.00E-06	0	inf	Length of FP2
DFP3	m	1.50E-07	1.00E-10	inf	Distance of FP3 from 2-DEG charge
LFP3	m	1.00E-06	0	inf	Length of FP3
DFP4	m	2.00E-07	1.00E-10	inf	Distance of FP4 from 2-DEG charge
DTEMP	C	0	-	-	Temperature variability parameter
LFP4	m	1.00E-06	0	inf	Length of FP4
NGCON	-	1	1	2	Number of gate contacts

LEVEL 82: MVSG Model

MIT Virtual Source GaNFET (MVSG) is a GaN high electron mobility transistor (HEMT) model typically used in RF and power conversion applications. Typical device structure of GaN HEMT is shown in Figure 20. There are five distinct regions of the device, which require modeling attention. The MVSG model captures carrier transport in these different regions of the device with the help of sub-circuit model shown in Figure 21.

Figure 20 Cross-Sectional Schematic of GaN HEMTs on Si

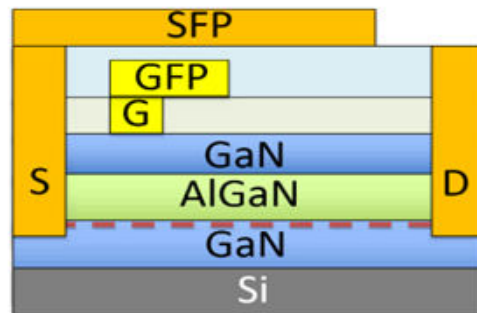
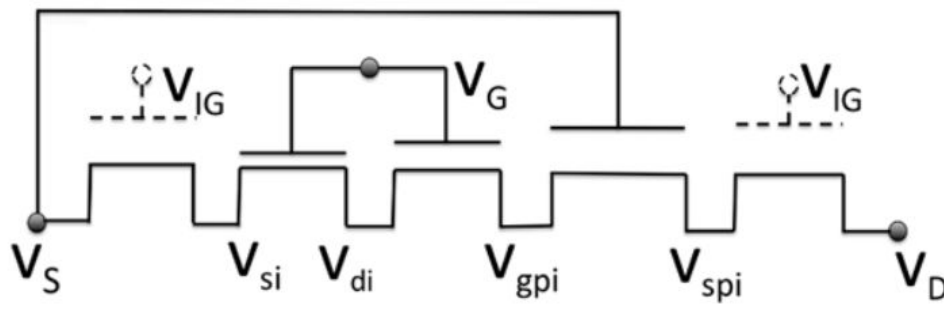


Figure 21 Equivalent circuit for the MVSG Model



- [Using the MVSG Model](#)
- [Model Revisions](#)
- [MVSG Model — General Syntax](#)
- [MVSG Model Parameters and Equations](#)
- [MVSG Instance Parameters](#)

Using the MVSG Model

To simulate using the MVSG mode in the PrimeSim Continuum simulators:

Set LEVEL = 82 to identify the model as the MVSG model.

Model Revisions

Version=1.1: The first supported version.

MVSG V3.2.0 Updates

MVSG V3.2.0 has the following update:

- Added p-GaN module.
 - Parameters related to p-GaN junction diode and junction capacitor.
 - New node `gi2p`: there are nodes `G`, `gi1`, `gi2`, `gi2p` from top to bottom.
 - Local variables related to floating p-gan behavior: `idsh`, `idshsat`, `idshrec`, `csh`.
 - `idsh` calculation, `csh` calculation and contribution.
 - Collapse of `gi2` and `gi2p`, if p-gan calculations are off `flagpgan=0`.
 - Added externally accessible thermal node `dt`.
 - Any connection to `gi2` in V3.1.0 is now connected to `gi2p`, except field plate gate connections (for both source and gate field plates).
 - All field plate voltage definitions remain to `gi2`, so they connect to outside of the p-GaN layer.
-

MVSG Model — General Syntax

Syntax:

```
Mxxx nd ng ns nb mname [L=val] [W=val] [NGF=val] [M=val]  
.model mname nmos|pmos level=82
```

MVSG Model Parameters and Equations

For the MVSG model parameters, see the MVSG manual released by CMC.

MVSG Instance Parameters

Parameter	Unit	Default	Minimum	Maximum	Description
L	m	2.50E-07	-	-	Effective Gate Length
W	m	1.80E-04	-	-	Width per Finger
NGF	-	2	1	inf	Number of fingers
DTEMP	K	0	-	-	Device temperature offset from ambient

Level 83: HISIM-HMG Model

HISIM-HMG is released by Hiroshima University.

- [Model Revision](#)
- [Using HISIM-HMG Model in PrimeSim HSPICE](#)
- [HISIM-HMG Model Parameters](#)
- [HISIM-MG Instance Parameters](#)

Model Revision

Version 2.0 is the supported version.

Using HISIM-HMG Model in PrimeSim HSPICE

To simulate using the HISIM-HMG mode:

Set LEVEL = 83 to identify the model as the HISIM-HMG model.

HISIM-HMG Model Parameters

For model parameters and equations, refer to the HISIM-HMG model manual released by Hiroshima University.

HISIM-MG Instance Parameters

For instance parameters and equations, refer to the HISIM-HMG model manual released by Hiroshima University.

LEVEL 84: HiSIM-SOTB

HiSIM-SOTB is a descendant of HiSIM_SOI valid for ultra-thin SOI and BOX layers applicable even for the double gate structure as a specific structure of SOTB-MOSFET. HiSIM_SOI itself is already valid for any possible SOTB-MOSFET structures, including different types of charges induced within the device. In addition to the HiSIM_SOI treatment, HiSIM-SOTB considers charges induced at both sides of BOX explicitly within the Poisson equation.

Figure 22 Definitions of nodes and device parameters in HiSIM-SOTB

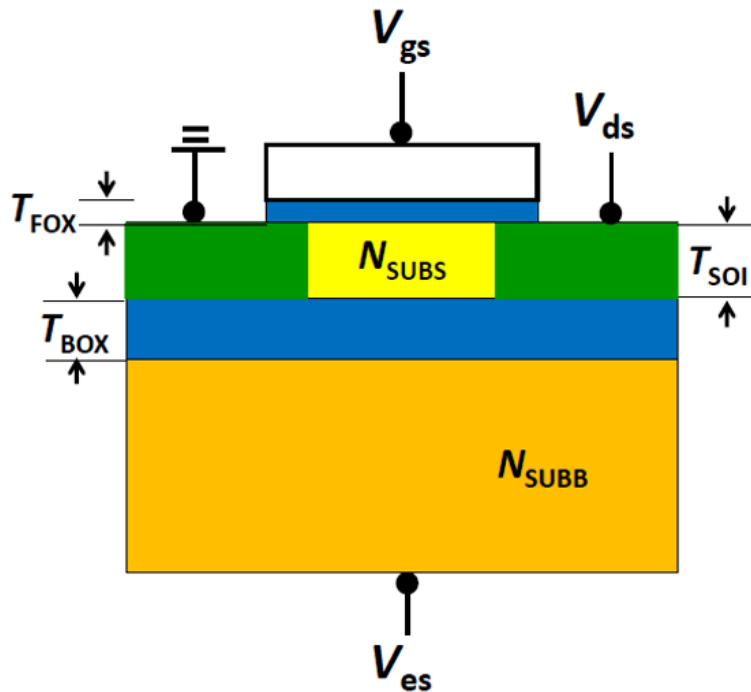
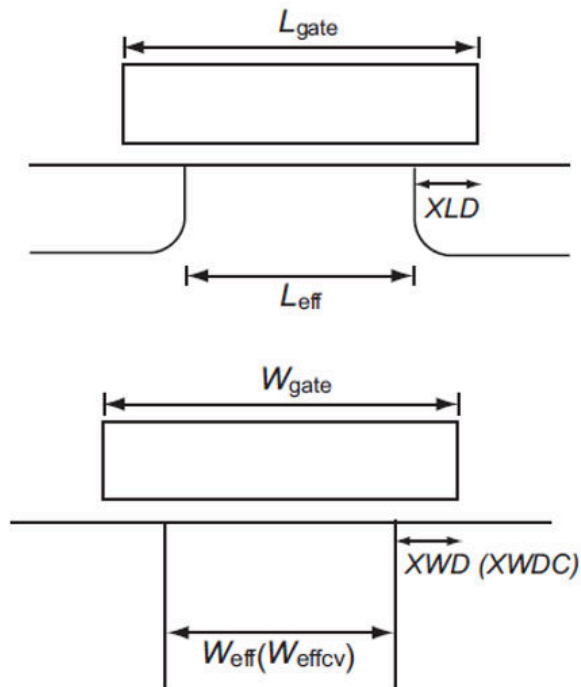


Figure 23 Cross section of HiSIM-SOTB



- [Using the HiSIM-SOTB Model](#)
- [Model Revisions](#)
- [HiSIM-SOTB Model — General Syntax](#)
- [HiSIM-SOTB Model Parameters and Equations](#)
- [HiSIM-SOTB Instance Parameters](#)

Using the HiSIM-SOTB Model

To simulate using the HiSIM-SOTB mode in the PrimeSim Continuum simulators:

Set LEVEL = 84 to identify the model as the HiSIM-SOTB model.

Model Revisions

Version=1.2: The first supported version.

HiSIM-SOTB Model — General Syntax

Syntax:

```
Mxxx nd ng ns nb mname [L=val] [W=val] [NGF=val] [M=val]
.model mname nmos|pmos level=84
```

HiSIM-SOTB Model Parameters and Equations

For the HiSIM-SOTB model parameters, see the HiSIM-SOTB manual released by CMC.

HiSIM-SOTB Instance Parameters

Parameter	Unit	Default	Minimum	Maximum	Description
L	m	5.00E-6	0	-	Gate length (Lgate)
W	m	5.00E-6	0	-	Gate width (Wgate)
XGL	m	0	-L	L	Offset of gate length
XGW	m	0	0	-	Distance from gate contact to channel edge.
M	-	1	0	-	Multiplication factor
NF	-	1	0	-	Number of gate fingers
NGCON	-	1	0	-	Number of gate contacts
SA	m	0	0	-	Length of diffusion between gate and STI
SB	m	0	0	-	Length of diffusion between gate and STI
SD	m	0	0	-	Length of diffusion between gate and gate
SCA	-	0	-	-	Layout characterization factor for WPE
SCB	-	0	-	-	Layout characterization factor for WPE
SCC	-	0	-	-	Layout characterization factor for WPE
TEMP	C	27	-	-	Device temperature

Parameter	Unit	Default	Minimum	Maximum	Description
DTEMP	C	0	-	-	Device temperature change

LEVEL 85: ASM ESD Model

Electrostatic discharge (ESD) diodes are the most popular component for protection of core electronics against any potentially harmful ESD events. Advance SPICE model for ESD diodes (ASM ESD) is a single comprehensive model with ability to capture all the important effects for ESD simulations.

Using the ASM-ESD model

To simulate using the ASM ESD mode in the HSPICE simulator:

Set LEVEL = 85 to identify the model as the ASM ESD model.

Model Revisions

Version=101.0.0: The first supported

ASM ESD Model General Syntax

Syntax:

```
Mxxx c b e dt mname [L=val] [N=val] [DTEMP=val]
.model mname nmos|pmos level=85
```

ASM-ESD Model Parameters and Equations

For the ASM ESD model parameters, see the ASM-ESD manual released by CMC.

ASM-ESD Instance Parameters

Parameter	Unit	Default	Minimum	Maximum	Description
L	m	10.0E-6	20e-9	inf	Finger length
N		1	1.0	inf	Number of fingers
DTEMP	deg C	0.0	-inf	inf	Temperature variability parameter

BSIMx and BSIMx SOI — Supported Instance Parameters

The following lists the instance parameters that are currently supported by the PrimeSim Continuum solution in these four model types:

Instance Parameter	BSIM3	BSIM4	BSIM3 SOI	BSIMSOI4
ACNQSMOD	yes	yes	no	no
AD	yes	yes	yes	yes
AS	yes	yes	yes	yes
CTH0	yes	no	yes	yes
DELDVT0	no	yes	no	no
DELK1	yes	yes	no	yes
DELK2	no	yes	yes	no
DELNFCT	yes	yes	no	yes
DELLPE0	no	yes	no	no
DELRSH	no	yes	no	no
DELRSHG	no	yes	no	no
DELTOX	yes	yes	yes	yes
DELVTO	yes	yes	yes	yes
DELXJ	no	yes	no	no
DTEMP	yes	yes	yes	yes
GEO	yes	yes	no	no
GEOMOD	no	yes	no	no
IC	yes	yes	yes	yes
L	yes	yes	yes	yes
MIN	no	yes	no	no
MULID0	no	yes	yes	yes
MULNGATE	no	yes	no	no
MULU0	yes	yes	yes	yes

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Instance Parameter	BSIM3	BSIM4	BSIM3 SOI	BSIMSOI4
SA6	yes	yes	no	no
SA7	yes	yes	no	no
SA8	yes	yes	no	no
SA9	yes	yes	no	no
SB	yes	yes	no	yes
SB1	yes	yes	no	no
SB10	yes	yes	no	no
SB2	yes	yes	no	no
SB3	yes	yes	no	no
SB4	yes	yes	no	no
SB5	yes	yes	no	no
SB6	yes	yes	no	no
SB7	yes	yes	no	no
SB8	yes	yes	no	no
SB9	yes	yes	no	no
SC	3.22 or later	yes	no	no
SCA	3.22 or later	yes	no	no
SCB	3.22 or later	yes	no	no
SCC	3.22 or later	yes	no	no
SD	yes	yes	no	yes
SOIQ0	no	no	yes	yes
STIMOD	yes	yes	no	no
SW1	yes	yes	no	no
SW10	yes	yes	no	no
SW2	yes	yes	no	no
SW3	yes	yes	no	no

Instance Parameter	BSIM3	BSIM4	BSIM3 SOI	BSIMSOI4
SW4	yes	yes	no	no
SW5	yes	yes	no	no
SW6	yes	yes	no	no
SW7	yes	yes	no	no
SW8	yes	yes	no	no
SW9	yes	yes	no	no
TNODEOUT	yes	no	yes	yes
TRNQSMOD	no	yes	no	no
W	yes	yes	yes	yes
XGW	no	yes	no	no

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- [3] Frohman-Bentchkowski, D. and Grove, A. S. "On the Effect of Mobility Variation on MOS Device Characteristics," *Proc. IEEE*, 56, 1968.
- [4] Fargher, H. E. and Mole, P. J. The Implementation Of A 3 Terminal SOSFET Model In SPICE For Circuit Simulation. GEC VLSI Research Laboratory, MOS1 Division.
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- [9] Duster, J.S., Jeng, M.C., Ko, P. K. and Hu, C. *User's Guide for the BSIM2 Parameter Extraction Program and the SPICE3 with BSIM Implementation*. Industrial Liaison Program, Software Distribution Office, University of California, Berkeley, May 1990.
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- [12] *PSP: An Advanced Surface-Potential-Based MOSFET Model for Circuit Simulation*, G. Gildenblat, X. Li, W. Wu, H. Wang, A. Jha, R. van Langevelde, G. D. J. Smit, A. J. Scholten and D. B. M. Klaassen, IEEE Trans. on Electron Devices, Vol. 53, No. 9, Sept. 2006.

3

MOSFET Capacitance Models

Discusses the use of the available capacitance models.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

This section covers the following topics:

- [MOS Gate Capacitance Models](#)
- [Selecting Capacitor Models](#)
- [Transcapacitance](#)
- [Operating Point Capacitance Printout](#)
- [Element Template Printout](#)
- [Calculating Gate Capacitance](#)
- [MOSFET Gate Capacitance — Model Parameters](#)
- [Specifying XQC and XPART for CAPOP=4, 9, 11, 12, 13](#)
- [Overlap Capacitance Equations](#)
- [CAPOP=0 – SPICE Meyer Gate Capacitances](#)
- [CAPOP=1 – Modified Meyer Gate Capacitances](#)
- [CAPOP=2 – Parameterized Modified Meyer Capacitance](#)
- [CAPOP=3 – Gate Capacitances \(Simpson Integration\)](#)
- [CAPOP=4 – Charge Conservation Capacitance Model](#)
- [CAPOP=5 — No Gate Capacitance](#)
- [CAPOP=6 — AMI Gate Capacitance Model](#)
- [CAPOP=13 — BSIM1-based Charge-Conserving Gate Capacitance Model](#)

- [CAPOP=39 — BSIM2 Charge-Conserving Gate Capacitance Model](#)
- [Calculating Effective Length and Width for AC Gate Capacitance](#)

MOS Gate Capacitance Models

You can use capacitance model parameters with all MOSFET model statements.

Three fixed-capacitance parameters (`CGDO`, `CGSO`, and `CGBO`) represent gate-to-drain, gate-to-source, and gate-to-bulk overlap capacitances to model charge storage, use fixed and nonlinear gate capacitances and junction capacitances. The algorithm used for calculating nonlinear, voltage-dependent MOS gate capacitance depends on the value of the `CAPOP` model parameter.

To model MOS gate capacitances as a nonlinear function of terminal voltages, use Meyer's piecewise linear model for all MOS levels. The charge conservation model is also available for MOSFET model Levels 2 through 7, 13, and 27. For LEVEL 1, you must specify the `TOX` model parameter to invoke the Meyer model. The next three sections describe the Meyer, Modified Meyer, and Charge Conservation MOS Gate Capacitance models.

Some of the charge conserving models (Ward-Dutton or BSIM) can cause "timestep too small" errors if you do not specify other nodal capacitances.

Selecting Capacitor Models

When you select a gate capacitance model, you can choose various combinations of capacitor models and DC models. You can incrementally update older DC models with new capacitance equations, without having to move to a new DC model. You can use the `CAPOP` model parameter to select the gate capacitance and validate the effects of different capacitance models.

The `CAPOP` capacitance model selection parameter selects the capacitor models to use for modeling the MOS gate capacitance:

- gate-to-drain capacitance
- gate-to- source capacitance
- gate-to-bulk capacitance.

You can use `CAPOP` to select several versions of the Meyer and charge conservation model.

Some capacitor models are tied to specific DC models (DC model level in parentheses below). Other models are designated as general; any DC model can use them.

Parameter	Description
CAPOP=0	SPICE original Meyer gate-capacitance model (general)
CAPOP=1	Modified Meyer gate-capacitance model (general)
CAPOP=2	Parameterized Modified Meyer gate-capacitance model (general default)
CAPOP=3	Parameterized Modified Meyer gate-capacitance model with Simpson integration (general)
CAPOP=4	Charge conservation capacitance model (analytic), Levels 2, 3, 6, 7, 13, 28, and 39 only
CAPOP=5	No capacitor model
CAPOP=6	AMI capacitor model (LEVEL 5)
CAPOP=9	Charge conservation model (LEVEL 3)
CAPOP=11	Ward-Dutton model (specialized, LEVEL 2)
CAPOP=12	Ward-Dutton model (specialized, LEVEL 3)
CAPOP=13	Generic BSIM Charge-Conserving Gate Capacitance model (default for Levels 13, 28, 39)
CAPOP=39	BSIM2 Charge-Conserving Gate Capacitance model (LEVEL 39)

CAPOP=4 selects the recommended charge-conserving model from among CAPOP=11, 12, or 13 for the specified DC model.

MOS Level	Default CAPOP	CAPOP=4 selects
2	2	11
3	2	12
13, 28, 39	13	13
Other levels	2	11

The proprietary models (Levels 5, 17, 21, 22, 25, 31, 33), the SOS model (LEVEL 27), and models higher than 49 have their own built-in capacitance routines.

Transcapacitance

If a capacitor has two terminals (1 and 2) with charges named Q1 and Q2 on the two terminals that sum to zero (for example, $Q1 = -Q2$), then the charge is a function of the voltage difference between the terminals ($V12 = V1 - V2$). One quantity ($C = dQ1/dV12$) completely describes the small-signal characteristics of the device.

If a capacitor has four terminals, the sum of the charges on the four terminals must equal zero ($Q1 + Q2 + Q3 + Q4 = 0$). They can depend only on voltage differences, but they are otherwise arbitrary functions. Because three independent charges ($Q1, Q2, Q3$) are functions of three independent voltages ($V14, V24, V34$), you must specify nine derivatives to describe the small-signal characteristics.

You can consider the four charges separately as functions of the four terminal voltages, $Q1(V1, V2, V3, V4), \dots, Q4(V1, V2, V3, V4)$. The derivatives form a four-by-four matrix, dQ_i/dV_j , $i=1,4, j=1,4$. Simulation directly interprets this matrix as AC measurements.

If you apply an AC voltage signal to the j terminal and you ground the other terminals to AC, and if you measure AC current into the i terminal, then the current is the imaginary constant times $2\pi \times \text{frequency} \times dQ_i/dV_j$.

- Because the charges add up to zero, each column of this matrix must add up to zero.
- Because the charges can depend only on voltage differences, each row must add up to zero.

In general, the matrix is not symmetrical:

dQ_i/dV_j need not equal dQ_j/dV_i

This is not an expected event because it does not occur for the two-terminal case. For two terminals, because the rows and columns must add up to zero, $dQ1/dV2$ must equal $dQ2/dV1$.

$$\frac{dQ1}{dV1} + \frac{dQ2}{dV1} = 0, \frac{dQ1}{dV2} + \frac{dQ2}{dV2} = 0$$

For three or more terminals, this relation does not generally hold.

The terminal input capacitances are the diagonal matrix entries:

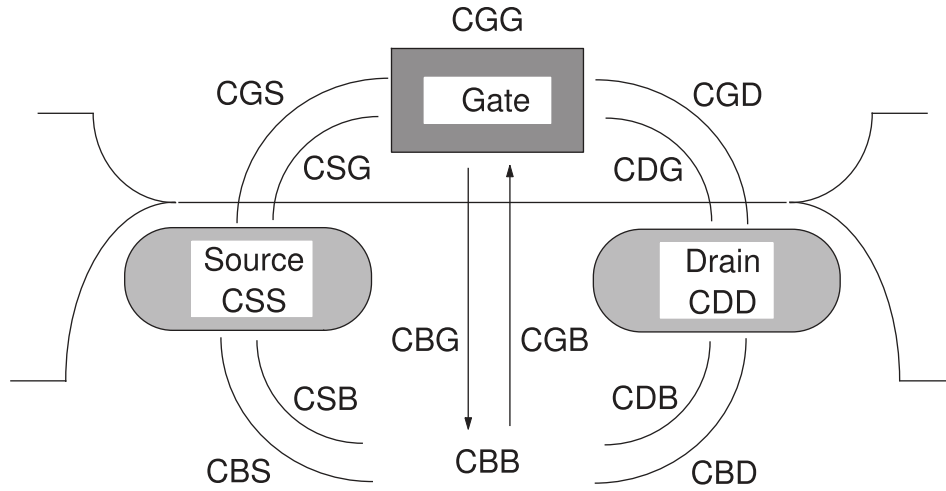
$$C_{ii} = dQ_i/dV_i \quad i=1,4$$

The transcapacitances are the negative of off-diagonal entries:

$$C_{ij} = -dQ_i/dV_j \quad i \text{ not equal to } j$$

All of the C values are normally positive.

Figure 24 MOS Capacitances



In Figure 24, C_{ij} determines the current transferred out of the i node from a voltage change on the j node. The arrows, representing direction of influence, point from node j to node i .

A MOS device with terminals named D, G, S, and B provides:

$$CGG = \frac{dQ_g}{dV_G}, CGD = -\frac{dQ_g}{dV_D}, CDG = -\frac{dQ_D}{dV_G}$$

- CGG represents input capacitance: a change in gate voltage requires a current equal to $CGG \times dV_G/dt$ into the gate terminal.
- CGD represents Miller feedback: a change in drain voltage creates a current equal to $CGG \times dV_G/dt$ out of the gate terminal.
- CDG represents Miller feedthrough, capacitive current out of the drain due to a change in gate voltage.

To show how CGD might not equal CDG , the following example is a simplified model with no bulk charge with a gate charge as a function of V_{GS} only, and with the 50/50 channel charge partitioned into Q_S and Q_D :

$$Q_G = Q(v_{gs}), Q_S = -0.5 \cdot Q(v_{gs}), Q_D = -0.5 \cdot Q(v_{gs}), Q_B = 0$$

Consequently:

$$CGD = -\frac{dQ_G}{dV_D} = 0, CDG = -\frac{dQ_D}{dV_G} = 0.5 \cdot \frac{dQ}{dv_{gs}}$$

Therefore, this model has Miller feed through, but no feedback.

Operating Point Capacitance Printout

The operating point printout reports six capacitances:

Capacitance	Value
cdtot	dQD/dVD
cgtot	dQG/dVG
cstot	dQS/dVS
cbtot	dQB/dVB
cgs	$-dQG/dVS$
cgd	$-dQG/dVD$

These capacitances include gate-drain, gate-source, and gate-bulk overlap capacitance, and drain-bulk and source-bulk diode capacitance. Drain and source refer to node 1 and 3 of the MOS element (physical instead of electrical).

The following table shows the printout quantities for the Meyer models, where charges such as QD are not well defined.

Capacitance	Value
cdtot	$cgd+cdb$
cgtot	$cgs+cgd+cgb$
cstot	$cgs+csb$
cbtot	$cgb+csb+cdb$
cgs	cgs
cgd	cgd

Element Template Printout

The MOS element template printouts for gate capacitance are LX18 to LX23 and LX32 to LX34. From these nine capacitances, you can construct the complete four-by-four matrix of transcapacitances. The nine LX printouts are:

$$\begin{aligned} \text{LX18 (m)} &= dQG/dVGB = CGGBO \\ \text{LX19 (m)} &= dQG/dVDB = CGDBO \end{aligned}$$

$LX20(m) = dQG/dVSB = CGSBO$
 $LX21(m) = dQB/dVGB = CBGBO$
 $LX22(m) = dQB/dVDB = CBDDBO$
 $LX23(m) = dQB/dVSB = CBSBO$
 $LX32(m) = dQD/dVG = CDGBO$
 $LX33(m) = dQD/dVD = CDDBO$
 $LX34(m) = dQD/dVS = CDSBO$

These capacitances include gate-drain, gate-source, and gate-bulk overlap capacitance, and drain-bulk and source-bulk diode capacitance. Drain and source refer to node 1 and 3 of the MOS element (physical instead of electrical).

For an NMOS device with source and bulk grounded:

- LX18 is the input capacitance.
- LX33 is the output capacitance.
- LX19 is the Miller feedback capacitance (gate current induced by voltage signal on the drain).
- LX32 is the Miller feedthrough capacitance (drain current induced by the voltage signal on the gate).

A device operating with node 3 as electrical drain — for example an NMOS device with node 3 at higher voltage than node 1 — is in *reverse mode*.

The LX values are physical, but you can translate them into electrical definitions by interchanging D and S:

$CGG(reverse) = CGG = LX18$
 $CDD(reverse) = CSS = dQS/dVS = d(-QG-QB-QD)/dVS = -LX20-LX23-LX34$
 $CGD(reverse) = CGS = -LX20$
 $CDG(reverse) = CSG = -dQS/dVG = d(QG+QB+QD)/dVG = LX18+LX21+LX32$

For Meyer models, QD and other charges are not well defined. The formulas (such as $LX18 = CGG$, $LX19 = -CGD$) are still true, but the transcapacitances are symmetrical; for example, $CGD = CDG$. In terms of the six independent Meyer capacitances (cgd, cgs, cgb, cdb, csb, and cds), the LX printouts are:

$LX18(m) = CGS + CGD + CGB$
 $LX19(m) = LX32(m) = -CGD$
 $LX20(m) = -CGS$
 $LX21(m) = -CGB \quad LX22(m) = -CDB$
 $LX23(m) = -CSB$
 $LX33(m) = CGD + CDB + CDS$
 $LX34(m) = -CDS$

Calculating Gate Capacitance

The following input file example shows a gate capacitance calculation in detail for a BSIM model. τ_{OX} is chosen so that: $\frac{eox}{tox} = 1e-3F / m^2$

In this example, V_{fb0} , ϕ , and k_1 are chosen so that $v_{th}=1v$. The AC sweep is chosen so that the last point is: $2 \cdot \pi \cdot freq = 1e6s^{-1}$

This section covers the following topics:

- [Gate Capacitance — Input File](#)
- [Gate Capacitance — Calculations](#)
- [Gate Capacitance — Results](#)
- [Gate Capacitance – Plotting Gate Capacitances](#)
- [Gate Capacitance — Control Options](#)
- [Gate Capacitance — Scaling](#)

Gate Capacitance — Input File

This example is based on demonstration netlist `calcap.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
$
m d g 0 b nch l=0.8u w=100u ad=200e-12 as=200e-12
vd d 0 5
vg g 0 5 ac 1
vb b 0 0
.ac dec 1 1.59155e4 1.59155e5
.print CGG=lx18(m) CDD=lx33(m) CGD=par('-lx19(m)')
+ CDG=par('-lx32(m)')
.print ig_imag=ii2(m) id_imag=ii1(m)
.model nch nmos level=13 update=2
+ xqc=0.6 toxm=345.315 vfb0=-1 phi0=1 k1=1.0 muz=600
+ mus=650 acm=2
+ xl=0 ld=0.1u meto=0.1u cj=0.5e-4 mj=0 cjsw=0
.alter
vd d 0 5 ac 1
vg g 0 5
.end
```

Gate Capacitance — Calculations

$$L_{eff} = 0.6u$$

$$\frac{eox}{tox} = 1e-3F / m^2$$

$$Cap = \frac{Leff \cdot Weff \cdot eox}{tox} = 60e-15F$$

BSIM equations for internal capacitance in saturation with xqc=0.4:

$$body = 1 + 0.5 \cdot \left(1 - \frac{1}{(1.744 + 0.8364 \cdot (PHI0 + vsb))} \right) \cdot \frac{K1}{\sqrt{(PHI0 + vsb)}}$$

$$1 + 0.5 \cdot \left(1 - \frac{1}{(1.744 + 0.8364)} \right) = 1.3062$$

$$cgg = Cap \cdot \left(1 - \frac{1}{(3 \cdot body)} \right) = Cap \cdot 0.7448 = 44.69F$$

$$cgd = 0$$

$$cdg = \left(\frac{4}{15} \right) \cdot Cap = 16F$$

$$cdd = 0$$

$$\text{Gate-drain overlap} = (ld + meto) \cdot Weff \cdot \frac{eox}{tox} = 20e-15F$$

Adding the overlaps:

$$cgg = 44.69F + 2 \cdot 20F = 84.69F$$

$$cgd = 20F$$

$$cdg = 36F$$

$$cdd = 20F$$

$$\text{Drain-bulk diode cap} = cj \cdot ad = (0.5e-4) \cdot (200e-12) = 10f$$

Adding the diodes:

$$cgg = 84.69F$$

$$cgd = 20F$$

$$cdg = 36F$$

$$cdd = 30F$$

Gate Capacitance — Results

```
subckt
element 0:m
model 0:nch
cdtot 30.0000f
cgtot 84.6886f
cstot 65.9999f
cbtot 43.4213f
cgs 61.2673f
cgd 20.0000f
```

The calculation and simulation results match.

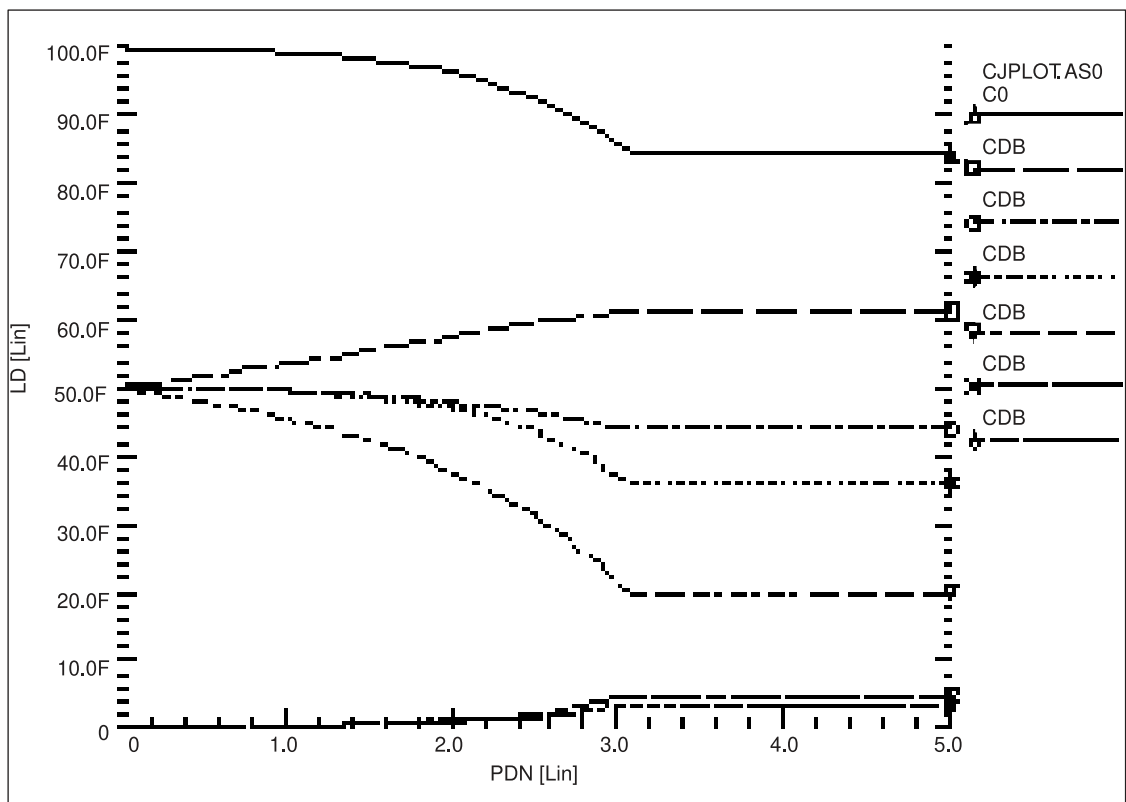
Gate Capacitance – Plotting Gate Capacitances

The following input file shows how to plot gate capacitances as a function of bias. Set `.OPTION DCCAP` to turn on capacitance calculations for a DC sweep. The model used is the same as for the previous gate capacitance calculations.

The following example is based on demonstration netlist `gatecap.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
$ gate capacitance plots
.option dccap=1 post
m d g 0 b nch l=0.8u w=100u ad=200e-12 as=200e-12
vd d 0 0
vg g 0 5
vb b 0 0
.dc vd 0 5 .1
.print vds=v(d) CGG=lx18(m)
+ CGD=par('-lx19(m)') CDG=par('-lx32(m)')
+ CGS=par('-lx20(m)') CSG=par('lx18(m)+lx21(m)+lx32(m)')
+ CGB=par('lx18(m)+lx19(m)+lx20(m)') CBG=par('-lx21(m)')
.model nch nmos
+ level=13 update=2 xqc=0.6 toxm=345.315
+ vfb0=-1 phi0=1 k1=1.0 muz=600 mus=650
+ acm=2 xl=0 ld=0.1u meto=0.1u
+
.end
```

Figure 25 Gate Capacitance



Gate Capacitance — Control Options

The `SCALM`, `CVTOL`, `DCSTEP`, and `DCCAP` control options affect the `CAPOP` models.

- `SCALM` scales the model parameters (ignored in Levels 49 and higher).
- `CVTOL` controls the error tolerance for convergence for the `CAPOP=3` model (see [CAPOP=3 – Gate Capacitances \(Simpson Integration\)](#)).
- `DCSTEP` models capacitances with a conductance during DC analysis.
- `DCCAP` calculates capacitances in DC analysis.

Gate Capacitance — Scaling

.OPTION SCALM scales the CGBO, CGDO, CGSO, COX, LD, and WD parameters according to fixed rules, which are a function of the parameter's units. If the model parameter's units are in meters, simulation multiplies the parameter by SCALM. For example:

- The LD parameter uses units in meters; to obtain its scaled value, simulation multiplies the value of LD by SCALM.
- If the units are in meters squared, simulation multiplies the parameter by $SCALM^2$.
- If the units are in reciprocal meters, the parameter's value is divided by SCALM. For example, because CGBO is in farads/meter, the value of CGBO is divided by SCALM.
- If the units are in reciprocal meters squared, then the parameter is divided by $SCALM^2$.

For the scaling equations specific to each CAPOP level, see the individual CAPOP subsections.

MOSFET Gate Capacitance — Model Parameters

This section covers the following topics:

- [Gate Capacitance — Basic Parameters](#)
- [Gate Capacitance — Gate Overlap Capacitance Parameters](#)
- [Gate Capacitance — Meyer Capacitance Parameters CAPOP=0, 1, 2](#)
- [Gate Capacitance — Charge Conservation Parameters \(CAPOP=4\)](#)

Gate Capacitance — Basic Parameters

Name (Alias)	Units	Default	Description
CAPOP	-	2.0	Capacitance model selector.
COX (CO)	F/m2	3.453e-4	Oxide capacitance. If you do not specify COX, simulation calculates it from TOX. The default corresponds to the TOX default of 1e-7: $COX_{scaled} = COX/SCALM^2$
TOX	m	1e-7	Oxide thickness, calculated from COX (if you specify COX). The program uses the default if you do not specify COX. For TOX>1, simulation assumes that the unit is Angstroms. A level-dependent default can override it. See specific MOSFET levels in this manual.

Gate Capacitance — Gate Overlap Capacitance Parameters

Name (Alias)	Units	Default	Description
CGBO (CGB)	F/m	0.0	Gate-bulk overlap capacitance per meter channel length. If you set WD and TOX, but you do not set CGBO, then simulation calculates CGBO. $CGBO_{scaled} = CGBO/SCALM$
CGDO (CGD, C2)	F/m	0.0	Gate-drain overlap capacitance per meter channel width. If you set LD or METO and TOX, but you do not set CGDO, then simulation calculates CGDO. $CGDO_{scaled} = CGDO/SCALM$
CGSO (CGS, C1)	F/m	0.0	Gate-source overlap capacitance per meter channel width. If you set LD or METO and TOX, but you do not set CGSO, then simulation calculates CGSO. $CGSO_{scaled} = CGSO/SCALM$
LD (LATD, DLAT)	m		Lateral diffusion into channel from source and drain diffusion. If you do not specify either LD or XJ, then the LD default=0.0. If you specify XJ but you do not specify LD, then simulation calculates LD from XJ. $LD_{default} = 0.75 \cdot XJ$ for all levels except LEVEL 4 for which $LD_{default} = 0.75$. $LD_{scaled} = LD \cdot SCALM$ LEVEL 4: $LD_{scaled} = LD \cdot XJ \cdot SCALM$
METO	m	0.0	Fringing field factor for gate-to-source and gate-to-drain overlap capacitance calculation. $METO_{scaled} = METO \cdot SCALM$
WD	m	0.0	Lateral diffusion into channel from bulk along width. $WD_{scaled} = WD \cdot SCALM$

Gate Capacitance — Meyer Capacitance Parameters CAPOP=0, 1, 2

Name (Alias)	Units	Default	Description
CF1	V	0.0	Modified MEYER control for transition of cgs from depletion to weak inversion for CGSO (for CAPOP=2 only)
CF2	V	0.1	Modified MEYER control for transition of cgs from weak to strong inversion region (for CAPOP=2 only)

Name (Alias)	Units	Default	Description
CF3		1.0	Modified MEYER control for the cgs and cgd transition from the saturation region to the linear region as a function of vds (for CAPOP=2 only)
CF4		50.0	Modified MEYER control for the contour of the cgb and cgs smoothing factors
CF5		0.667	Modified MEYER control for the capacitance multiplier for cgs in the saturation region
CF6		500.0	Modified MEYER control for contour of cgd smoothing factor
CGBEX		0.5	cgb exponent (for CAPOP=1 only)

Gate Capacitance — Charge Conservation Parameters (CAPOP=4)

Name (Alias)	Units	Default	Description
XQC		0.5	Coefficient of channel charge share attributed to drain; its range is 0.0 to 0.5. This parameter applies only to CAPOP=4 and some of its level-dependent aliases.

Specifying XQC and XPART for CAPOP=4, 9, 11, 12, 13

Parameter rules for the gate capacitance charge sharing coefficient (X_{QC} & X_{PART}) in the saturation region:

- If you do not specify either X_{PART} or X_{QC} , then simulation uses the 0/100 model.
- If you specify both X_{PART} and X_{QC} , then X_{PART} overrides X_{QC} .
- If you specify X_{PART} , but you do not specify X_{QC} , then:
 - $X_{PART}=0 \rightarrow 40/60$
 - $X_{PART}=0.4 \rightarrow 40/60$
 - $X_{PART}=0.5 \rightarrow 50/50$
 - $X_{PART}=1 \rightarrow 0/100$

- $X_{PART} = \text{any other value less than } 1 \rightarrow 40/60$
- $X_{PART} > 1 \rightarrow 0/100$ If X_{QC} is specified:
- If you specify X_{QC} but you do not specify X_{PART} , then:
 - $X_{QC} = 0 \rightarrow 0/100$
 - $X_{QC} = 0.4 \rightarrow 40/60$
 - $X_{QC} = 0.5 \rightarrow 50/50$
 - $X_{QC} = 1 \rightarrow 0/100$
 - $X_{QC} = \text{any other value less than } 1 \rightarrow 40/60$
 - $X_{QC} > 1 \rightarrow 0/100$

The only difference is the treatment of the 0 parameter value.

After you specify X_{PART}/X_{QC} , the gate capacitance ramps from 50/50 at $V_{ds}=0$ volt (linear region) to the value (with V_{ds} sweep) in the saturation region in X_{PART}/X_{QC} . Ramping the charge-sharing coefficient ensures smooth gate capacitance characteristics.

Overlap Capacitance Equations

The overlap capacitors are common to all models. You can input them explicitly or the program can calculate them. Either way, these overlap capacitors must be added into the respective voltage-variable capacitors before integration, and before the DC operating point reports the combined parallel capacitance.

Gate-to-Bulk Overlap Capacitance

If you specify $CGBO$, then: $CGBO_{eff} = M \cdot L_{eff} \cdot CGBO_{scaled}$

Otherwise: $CGBO_{eff} = 2 \cdot WD_{scaled} \cdot L_{eff} \cdot COX_{scaled} \cdot M$

Gate-to-Source Overlap Capacitance

If you specify $CGSO$, then: $CGSO_{eff} = W_{eff} \cdot CGSO_{scaled}$

Otherwise: $CGSO_{eff} = W_{eff} \cdot (LD_{scaled} + METO_{scaled}) \cdot COX_{scaled}$

Gate-to-Drain Overlap Capacitance

If you specify $CGDO$, then: $CGDO_{eff} = W_{eff} \cdot CGDO_{scaled}$

Otherwise: $CGDO_{eff} = W_{eff} \cdot (LD_{scaled} + METO_{scaled}) \cdot COX_{scaled}$

Simulation calculates the L_{eff} value for each model differently, and saves this value in the corresponding model section. The W_{eff} calculation is not the same as the W_{eff} value in the LEVEL 1, 2, 3, 6, 7 and 13 models.

$W_{eff} = M \cdot (W_{scaled} \cdot WMLT + XW_{scaled})$

The $2 \cdot WD_{scaled}$ factor is not subtracted.

CAPOP=0 – SPICE Meyer Gate Capacitances

Definition: $cap = COX_{scaled} \cdot W_{eff} \cdot L_{eff}$

This section covers the following topics:

- [Gate-Bulk Capacitance \(cgb\)](#)
- [Gate-Source Capacitance \(cgs\)](#)
- [Gate-Drain Capacitance \(cgd\)](#)

Gate-Bulk Capacitance (cgb)

Accumulation, $v_{gs} \leq v_{th} - PH1$: $cgb = cap$

Depletion, $v_{gs} < v_{th}$: $cgb = cap \cdot \frac{v_{th} - v_{gs}}{PH1}$

Strong Inversion, $v_{gs} \geq v_{th}$: $cgb = 0$

Gate-Source Capacitance (cgs)

Accumulation: $v_{gs} \leq v_{th} - \frac{PHI}{2}$

$cgs = 0$

Depletion, $v_{gs} \leq v_{th}$: $cgs = CF5 \cdot cap + \frac{cap \cdot (v_{gs} - v_{th})}{0.75 \cdot PHI}$

Strong Inversion Saturation Region: $v_{gs} > v_{th}$ and $v_{ds} \geq v_{dsat}$:

$cgs = CF5 \cdot cap$

Strong Inversion Linear Region: $v_{gs} > v_{th}$ and $v_{ds} < v_{dsat}$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{dsat} - v_{ds}}{2 \cdot (v_{dsat} + v_{sb}) - v_{ds} - v_{sb}} \right]^2 \right\}$$

Gate-Drain Capacitance (cgd)

The gate-drain capacitance has value only in the linear region.

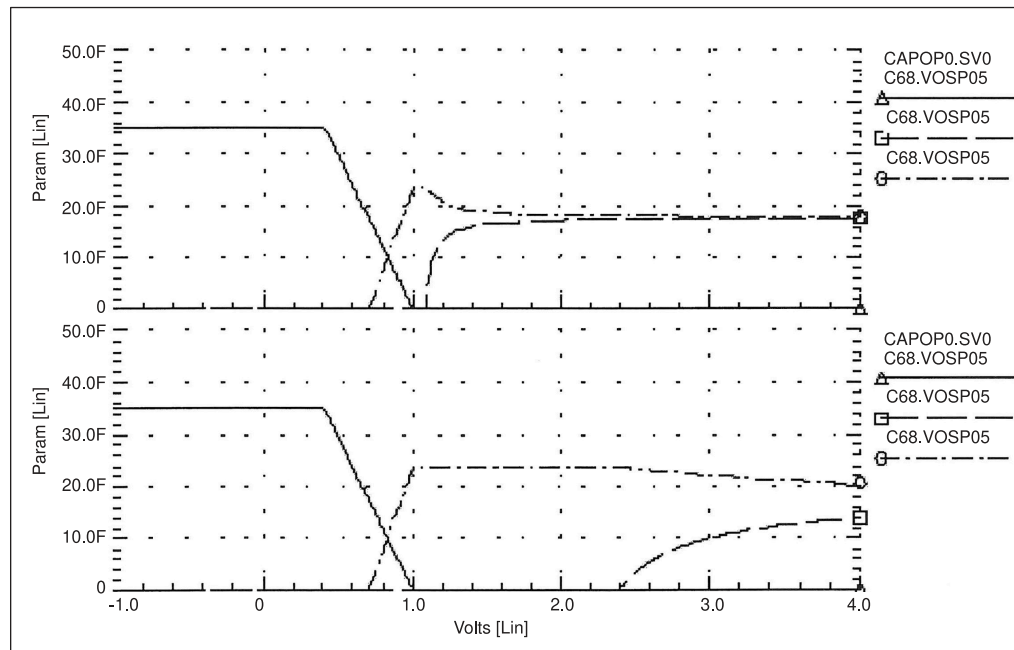
Strong Inversion Linear Region: $v_{gs} > v_{th}$ and $v_{ds} < v_{dsat}$

$$c_{gd} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{dsat} + v_{sb}}{2 \cdot (v_{dsat} + v_{sb}) - v_{ds} - v_{sb}} \right]^2 \right\}$$

The following example is based on demonstration netlist `capop0.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
*file capop0.sp---capop=0 capacitances
*
*this file is used to create spice meyer gate c-v plots
**
*(capop=0) for low vds and high vds
*
.options acct=2 post=2 dccap=1 nomod
.dc vg1 -1 4 .01
.print dc cgb_vdsp05=par('-lx21(m1)') cgd_vdsp05=par('-lx19(m1)')
+ cgs_vdsp05=par('-lx20(m1)')
.print dc cgb_vdsp8=par('-lx21(m2)') cgd_vdsp8=par('-lx19(m2)')
+ cgs_vdsp8=par('-lx20(m2)')
*****
m1 d1 g1 0 0 mn l=5e-6 w=20e-6 $ create capacitances for vds=0.05
m2 d2 g1 0 0 mn l=5e-6 w=20e-6 $ create capacitances for vds=0.80
*****
vd1 d1 0 dc 0.05
vd2 d2 0 dc 0.80
vg1 g1 0 dc 0.0
*
*****
*
.model mn nmos ( level = 2
+ vto = 1.0 gamma = 1.40 nsub = 7.20e15
+ uo = 817 ucrit = 3.04e4 phi=.6
+ uexp = 0.102 neff = 1.74 vmax = 4.59e5
+ tox = 9.77e-8 cj = 0 cjsw = 0 js = 0
+ capop=0 )
.end
```

Figure 26 CAPOP=0 Capacitances



CAPOP=1 – Modified Meyer Gate Capacitances

Define: $cap = COX_{scaled} \cdot W_{eff} \cdot L_{eff}$

In the following equations, G^- , G^+ , D^- , and D^+ are smooth factors. You cannot change the values of these parameters.

This section covers the following topics:

- [Gate-to-Bulk Overlap Capacitance](#)
- [Gate-to-Source Overlap Capacitance](#)
- [Gate-to-Drain Overlap Capacitance](#)

Gate-Bulk Capacitance (cgb)

Accumulation, $v_{gs} \leq v_{fb} - v_{sb}$:

$$cgb = cap$$

Depletion, $v_{gs} \leq v_{th}$:

$$c_{gb} = \frac{cap}{\left[1 + 4 \cdot \frac{v_{gs} + v_{sb} - v_{fb}}{GAMMA^2} \right]^{CGBEX}}$$

Strong Inversion, $v_{gs} > v_{th}$:

$$c_{gb} = \frac{G^+ \cdot cap}{\left[1 + 4 \cdot \frac{GAMMA(v_{sb} + PHI)^2 + v_{sb} + PHI}{GAMMA^2} \right]^{CGBEX}}$$

These equations replace GAMMA with effective γ for model levels higher than 4.

Gate-Source Capacitance (cgs)

Low vds ($v_{ds} < 0.1$)

Accumulation, $v_{gs} \leq v_{th}$ $c_{gs} = CF5 \cdot cap \cdot G^- \cdot D^-$

Weak Inversion, $v_{gs} < v_{th} + 0.1$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ \frac{v_{gs} - v_{th}}{0.1} \cdot \left[1 - \left(\frac{0.1 - v_{ds}}{0.2 - v_{ds}} \right)^2 \cdot D^- \right] + D^- \right\}$$

Strong Inversion, $v_{gs} \geq v_{th} + 0.1$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{gs} - v_{th} - v_{ds}}{2 \cdot (v_{gs} - v_{th}) - v_{ds}} \right]^2 \right\}$$

High vds ($v_{ds} \geq 0.1$)

Accumulation, $v_{gs} \leq v_{th}$ $c_{gs} = CF5 \cdot cap \cdot G^-$

Saturation Region, $v_{gs} < v_{th} + v_{ds}$ $c_{gs} = CF5 \cdot cap$

Linear Region, $v_{gs} \geq v_{th} + v_{ds}$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{gs} - v_{th} - v_{ds}}{2 \cdot (v_{gs} - v_{th}) - v_{ds}} \right]^2 \right\}$$

Gate-Drain Capacitance (cgd)

Low vds ($v_{ds} < 0.1$)

Accumulation, $v_{gs} \leq v_{th}$ $c_{gd} = CF5 \cdot cap \cdot G^- \cdot D^+$

Weak Inversion, $v_{gs} < v_{th} + 0.1$:

$$c_{gd} = CF5 \cdot cap \cdot \left\{ D^+ + \frac{v_{gs} - v_{th}}{0.1} \cdot \max \left[0, 1 - \left(\frac{0.1}{0.2 - v_{ds}} \right)^2 \cdot D^+ \right] \right\}$$

Strong Inversion, $v_{gs} \geq v_{th} + 0.1$:

$$c_{gd} = CF5 \cdot cap \cdot \max \left\{ D^+, 1 - \left[\frac{v_{gs} - v_{th}}{2 \cdot (v_{gs} - v_{th}) - v_{ds}} \right]^2 \right\}$$

High vds ($v_{ds} < 0.1$)

Accumulation, $v_{gs} \leq v_{th}$: $c_{gd} = CF5 \cdot cap \cdot G^- \cdot D^+$

Saturation Region, $v_{gs} < v_{th} + v_{ds}$: $c_{gd} = CF5 \cdot cap \cdot D^+$

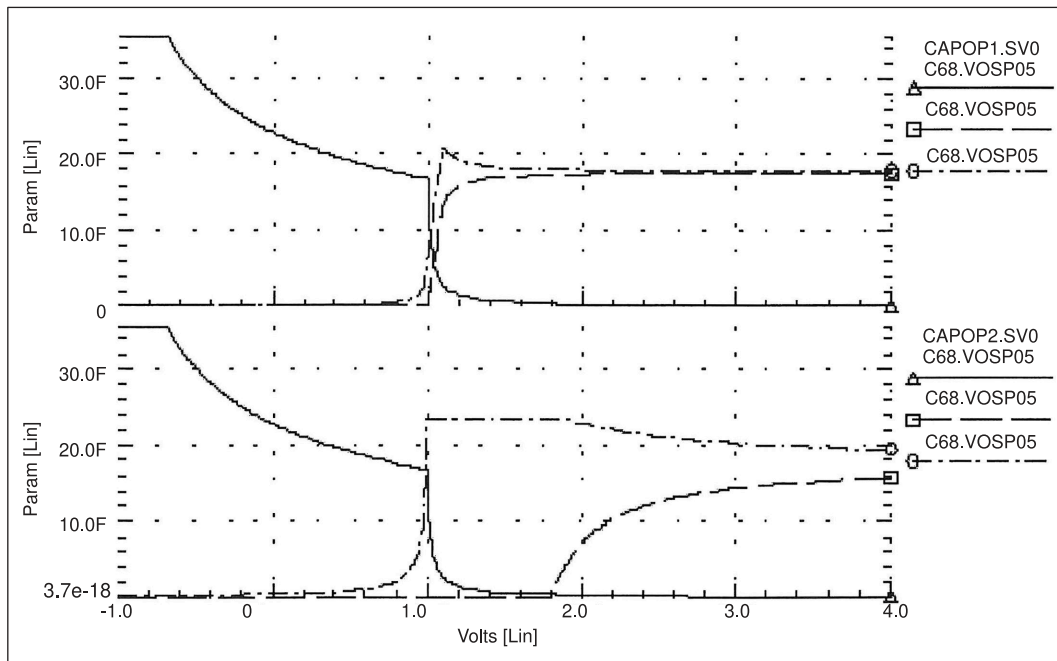
Strong Inversion, $v_{gs} \geq v_{th} + v_{ds}$:

$$c_{gd} = CF5 \cdot cap \cdot \max \left\{ D^+, 1 - \left[\frac{v_{gs} - v_{th}}{2 \cdot (v_{gs} - v_{th}) - v_{ds}} \right]^2 \right\}$$

The following example is based on demonstration netlist `capop1.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
*file capop1.sp---capop1 capacitances
*
*this file creates the modified meyer gate c-v plots
*(capop=1) for low vds and high vds.
*
.options acct=2 post=2 dccap=1 nomod
.dc vg1 -1 4 .01
.print dc cgb_vdsp05=par('-lx21(m1)') cgd_vdsp05=par('-lx19(m1)')
+ cgs_vdsp05=par('-lx20(m1)')
.print dc cgb_vdsp8=par('-lx21(m2)') cgd_vdsp8=par('-lx19(m2)')
+ cgs_vdsp8=par('-lx20(m2)')
*****
m1 d1 g1 0 0 mn l=5e-6 w=20e-6 $creates capacitances for vds=0.05
m2 d2 g1 0 0 mn l=5e-6 w=20e-6 $creates capacitances for vds=0.80
*****
vd1 d1 0 dc 0.05
vd2 d2 0 dc 0.80
vg1 g1 0 dc 0.0
*
*****
*
.model mn nmos ( level = 2
+ vto = 1.0 gamma = 1.40 nsub = 7.20e15
+ tox = 9.77e-8 uo = 817 ucrit = 3.04e4
+ uexp = 0.102 neff = 1.74 vmax = 4.59e5
+ phi = 0.6 cj = 0 cjsw = 0 js = 0
+ capop=1 )
.end
```

Figure 27 CAPOP=1 Capacitances



CAPOP=2 – Parameterized Modified Meyer Capacitance

The CAPOP=2 Meyer capacitance model is the more general form of Meyer capacitance. The CAPOP=1 Meyer capacitance model is the special case of CAPOP=2 if CF1=0, CF2=0.1, and CF3=1.

In the following equations, G^- , G^+ , D^- , and D^+ are smooth factors. You cannot change the values of these parameters.

Definition: $cap = COXscaled \cdot Weff \cdot Leff$

This section covers the following topics:

- [Gate-Bulk Capacitance \(cgb\)](#)
- [Gate-Source Capacitance \(cgs\)](#)
- [Gate-Drain Capacitance \(cgd\)](#)

Gate-Bulk Capacitance (cgb)

Accumulation, $v_{gs} \leq v_{fb} - v_{sb}$: $cgb = cap$

$$\text{Depletion, } v_{gs} \leq v_{th}: c_{gb} = \frac{cap}{\left(1 + 4 \cdot \frac{v_{gs} + v_{sb} - v_{fb}}{GAMMA^2}\right)^{1/2}}$$

$$\text{Inversion, } v_{gs} > v_{th}: c_{gb} = \frac{G^+ \cdot cap}{\left[1 + 4 \cdot \frac{GAMMA(PHI + v_{sb})^{1/2} + PHI + v_{sb}}{GAMMA^2}\right]^{1/2}}$$

These equations replace GAMMA with effective γ for model levels higher than 4.

Gate-Source Capacitance (cgs)

Low vds (vds < 0.1)

Accumulation, $v_{gs} < v_{th} - CF1$: $c_{gs} = CF5 \cdot cap \cdot G^- \cdot D^-$

Depletion, $v_{gs} \leq v_{th} + CF2 - CF1$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ \frac{v_{gs} - v_{th} + CF1}{CF2} \cdot \left[1 - \left(CF2 - \frac{v_{ds}}{2 \cdot CF2 - v_{ds}} \right)^2 \cdot D^- \right] + D^- \right\}$$

Strong Inversion, $v_{gs} > v_{th} + \max(CF2 - CF1, CF3 \cdot v_{ds})$, UPDATE=0

Strong Inversion, $v_{gs} > v_{th} + CF2 - CF1$, UPDATE=1:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{gs} - v_{th} + CF1 - v_{ds}}{2 \cdot (v_{gs} - v_{th} + CF1) - v_{ds}} \right]^2 \right\}$$

High vds

(vds ≥ 0.1)

Accumulation, $v_{gs} < v_{th} - CF1$: $c_{gs} = CF5 \cdot cap \cdot G^- \cdot D^+$, $CF1 \neq 0$

$c_{gs} = CF5 \cdot cap \cdot G^-$, $CF1 = 0$

Weak Inversion, $v_{gs} < v_{th} + CF2 - CF1$, $CF1 \neq 0$:

$$c_{gs} = CF5 \cdot cap \cdot \max\left(\frac{v_{gs} - v_{th} + CF1}{CF2}, D^+\right)$$

Saturation Region, $v_{gs} < v_{th} + CF3 \cdot v_{ds}$: $c_{gs} = CF5 \cdot cap$

Linear Region, $v_{gs} > v_{th} + CF3 \cdot v_{ds}$:

$$c_{gs} = CF5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{gs} - v_{th} - v_{ds}}{2 \cdot (v_{gs} - v_{th}) - v_{ds}} \right]^2 \right\}, \text{UPDATE}=0, CF1=0$$

$$c_{gs} = CG5 \cdot cap \cdot \left\{ 1 - \left[\frac{v_{gs} - v_{th} - CF3 \cdot v_{ds}}{2 \cdot (v_{gs} - v_{th}) - CF3 \cdot v_{ds}} \right]^2 \right\}, \text{UPDATE}=1$$

Gate-Drain Capacitance (c_{gd})

Low v_{ds}, (v_{ds} < 0.1)

Accumulation, v_{gs} ≤ v_{th} – CF1: $c_{gd} = CF5 \cdot cap \cdot G^- \cdot D^-$

Weak Inversion, v_{gs} < v_{th} + CF2 – CF1:

$$c_{gd} = CF5 \cdot cap \cdot \left\{ D^- + \frac{v_{gs}-v_{th}+CF1}{CF2} \cdot \max \left[0, 1 - \left(\frac{CF2}{2 \cdot CF2 - v_{ds}} \right)^2 \cdot D^- \right] \right\}$$

Strong Inversion, v_{gs} ≥ v_{th} + CF2 – CF1:

$$c_{gd} = CF5 \cdot cap \cdot \max \left\{ D^-, 1 - \left[\frac{v_{gs}-v_{th}+CF1}{2 \cdot (v_{gs}-v_{th}+CF1) - v_{ds}} \right]^2 \right\}$$

High v_{ds} (v_{ds} > 0.1)

Accumulation, v_{gs} ≤ v_{th} – CF1: $c_{gd} = CF5 \cdot cap \cdot G^- \cdot DD^+$

Saturation Region, v_{gs} ≤ v_{th} + CF3 · v_{ds}: $c_{gd} = CF5 \cdot cap \cdot DD^+$

DD⁺ is a function of CF3, if UPDATE=1.

Linear Region, v_{gs} > v_{th} + CF3 · v_{ds}:

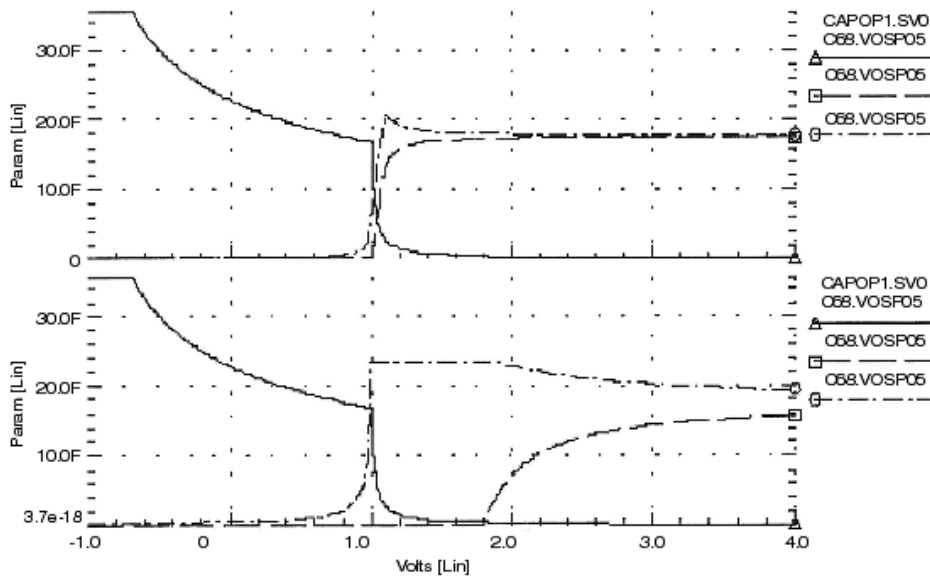
$$c_{gd} = CF5 \cdot cap \cdot \max \left\{ DD^+, 1 - \left[\frac{v_{gs}-v_{th}}{2 \cdot (v_{gs}-v_{th}) - CF3 \cdot v_{ds}} \right]^2 \right\}$$

The following example is based on demonstration netlist capop2.sp, which is available in directory \$installdir/demo/hspice/mos:

```
*file capop2.sp capop=2 capacitances
*
*this file creates parameterized modified gate capacitances
*(capop=2) for low and high vds.
*
.options acct=2 post=2 dccap=1 nomod
.dc vgl -1 4 .01
.print dc cgb_vdsp05=par('-lx21(m1)') cgd_vdsp05=par('-lx19(m1)')
+ cgs_vdsp05=par('-lx20(m1)')
.print dc cgb_vdsp8=par('-lx21(m2)') cgd_vdsp8=par('-lx19(m2)')
+ cgs_vdsp8=par('-lx20(m2)')
*****
m1 d1 g1 0 0 mn l=5e-6 w=20e-6 $creates capacitances for vds=0.05
m2 d2 g1 0 0 mn l=5e-6 w=20e-6 $creates capacitances for vds=0.80
*****
vd1 d1 0 dc 0.05
vd2 d2 0 dc 0.80
vgl g1 0 dc 0.0
*
```

```
*****
*
.model mn nmos ( level = 2
+ vto = 1.0 gamma = 1.40 nsub = 7.20e15
+ tox = 9.77e-8 uo = 817 ucrit = 3.04e4
+ uexp = 0.102 neff = 1.74 phi = 0.6
+ vmax = 4.59e5 cj = 0 cjsw = 0 js = 0
+ capop=2 cf1=0.15 cf2=.2 cf3=.8 cf5=.666)
.end
```

Figure 28 CAPOP=2 Capacitances



CAPOP=3 – Gate Capacitances (Simpson Integration)

The CAPOP=3 model uses the same set of equations and parameters as the CAPOP=2 model. Simulation obtains the charges using Simpson numeric integration instead of the box integration found in the CAPOP=1, 2, and 6 models.

Gate capacitances are not constant values with respect to voltages. The incremental capacitance best describes the capacitance values:

$$C(v) = \frac{d q(v)}{d v}$$

In the preceding equation, $q(v)$ is the charge on the capacitor, and v is the voltage across the capacitor.

The formula for calculating the differential is difficult to derive. Furthermore, the voltage is required as the accumulated capacitance over time. The timewise formula is:

$$i(t) = \frac{d q(v)}{d t} = C(v) \cdot \frac{d v(t)}{d t}$$

The charge is: $q(v) = \int_0^v C(v) d v$

To calculate the current: $i(t) = \frac{d q(v)}{d t} = \left(\frac{d}{d t} \right) \int_0^v C(v) d v$

For small intervals: $I(n+1) = \frac{d q(v)}{d t} = \frac{1}{t(n+1)-t(n)} \int_{V(n)}^{V(n+1)} C(v) d v$

In SPICE, the following equation approximates the integral:

$$I(n+1) = \left(\frac{V(n+1)-V(n)}{t(n+1)-t(n)} \right) \cdot \left(\frac{C[V(n+1)] + C[V(n)]}{2} \right)$$

This last formula is the trapezoidal rule for integration over two points. The charge is approximated as the average capacitance times the change in voltage. If the capacitance is nonlinear, this approximation can be in error. To accurately estimate the charge, use Simpson's numerical integration rule. This method provides charge conservation control.

To use this model parameter:

- Set the CAPOP model parameter to 3 and use the existing CAPOP=2 model parameters.
- Modify the .OPTION RELV (relative voltage tolerance), .OPTION RELMOS (relative current tolerance for MOSFETs), and .OPTION CVTOL (capacitor voltage tolerance) settings.

The default of 0.5 is a good nominal value for CVTOL. The CVTOL option uses the following equation to set the number of integration steps:

$$n = \frac{|V(n+1)-V(n)|}{CVTOL}$$

Use a large value for CVTOL to decrease the number of integration steps for the n to n+1 time interval; this yields slightly less accurate integration results. Using a small CVTOL value increases the computational load, sometimes severely.

CAPOP=4 – Charge Conservation Capacitance Model

The charge conservation method (See "A Charge-Oriented Model for MOS Transistor," Ward, Donald E. and Robert W. Dutton) is not implemented correctly into the SPICE2G.6

program. There are errors in the derivative of charges, especially in LEVEL 3 models. Also, the channel charge partition is not continuous from the linear region to the saturation region.

In the Synopsys MOSFET device models, these problems are corrected. If you specify the CAPOP=4 model parameter, then simulation uses the level-dependent recommended charge conservation model. The xQC model parameter selects ratio of channel charge partitioning between drain/source.

For example, if you set xQC=.4, then in the saturation region, 40% of the channel charge is associated with the drain and the remaining 60% is associated with the source. In the linear region, the ratio is 50/50. Simulation uses an empirical equation to make a smooth transition from 50/50 (linear region) to 40/60 (saturation region).

The capacitance coefficients are the derivative of gate, bulk, drain, and source charges, and are continuous. LEVEL 2, 3, 4, 6, 7, and 13 models include a charge-conservation capacitance model. To invoke this model, set CAPOP=4.

The following example compares only the CAPOP=4 charge conservation capacitance and the CAPOP=9 improved charge conservation capacitance for the LEVEL 3 model. The CGS and CGD capacitances for CAPOP=4 model (SPICE2G.6) show discontinuity at the boundary between the saturation and linear regions. The CAPOP=9 model does not have discontinuity. The modified Meyer capacitances (CAPOP=2) are also provided for comparison. The shape of CGS and CGD capacitances resulting from CAPOP=9 are much closer to those of CAPOP=2.

Example 1:

This example is based on demonstration netlist mcap3.sp, which is available in directory \$installdir/demo/hspice/mos:

```
FILE MCAP3.SP CHARGE CONSERVATION MOSFET CAPS., CAPOP=4,9 LEVEL=3
* CGGB = LX18 (M) DERIVATIVE OF QG WITH RESPECT TO VGB.
* CGDB = LX19 (M) DERIVATIVE OF QG WITH RESPECT TO VDB.
* CGSB = LX20 (M) DERIVATIVE OF QG WITH RESPECT TO VSB.
* CBGB = LX21 (M) DERIVATIVE OF QB WITH RESPECT TO VGB.
* CBDB = LX22 (M) DERIVATIVE OF QB WITH RESPECT TO VDB.
* CBSB = LX23 (M) DERIVATIVE OF QB WITH RESPECT TO VSB.
* CDGB = LX32 (M) DERIVATIVE OF QD WITH RESPECT TO VGB.
* CDDB = LX33 (M) DERIVATIVE OF QD WITH RESPECT TO VDB.
* CDSB = LX34 (M) DERIVATIVE OF QD WITH RESPECT TO VSB.
* SIX NONRECIPROCAL CAPACITANCES (CGB,CBG,CGS,CSG,CGD,AND CDG)
* ARE DERIVED FROM THE ABOVE CAPACITANCE FACTORS.
.OPTION DCCAP=1 POST NOMOD
.PARAM XQC=0.4 CAPOP=4
.DC VGG -2 5 .02
.print CGB=PAR('LX18 (M)+LX19 (M)+LX20 (M) ')
+ CBG=PAR(' -LX21 (M) ')
+ CGS=PAR(' -LX20 (M) ')
+ CSG=PAR(' LX18 (M)+LX21 (M)+LX32 (M) ')
```

Chapter 3: MOSFET Capacitance Models

CAPOP=4 – Charge Conservation Capacitance Model

```

+ CGD=PAR('LX19(M)')
+ CDG=PAR('LX32(M)')
.print
+ CG=par('LX14(M)')
VDD D 0 2.5
VGG G 0 0
VBB B 0 -1
M D G 0 B MOS W=10U L=5U
.MODEL MOS NMOS LEVEL=3 COX=1E-4 VTO=.3 CAPOP=CAPOP
+ UO=1000 GAMMA=.5 PHI=.5 XQC=XQC
+ THETA=0.06 VMAX=1.9E5 ETA=0.3 DELTA=0.05 KAPPA=0.5 XJ=.3U
+ CGSO=0 CGDO=0 CGBO=0 CJ=0 JS=0 IS=0
*
.ALTER
.PARAM CAPOP=9
.END

```

Figure 29 CAPOP=4, 9 Capacitances for LEVEL 3 Model

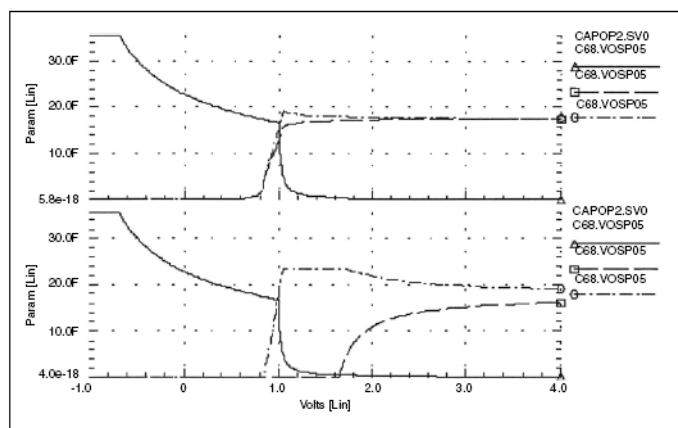
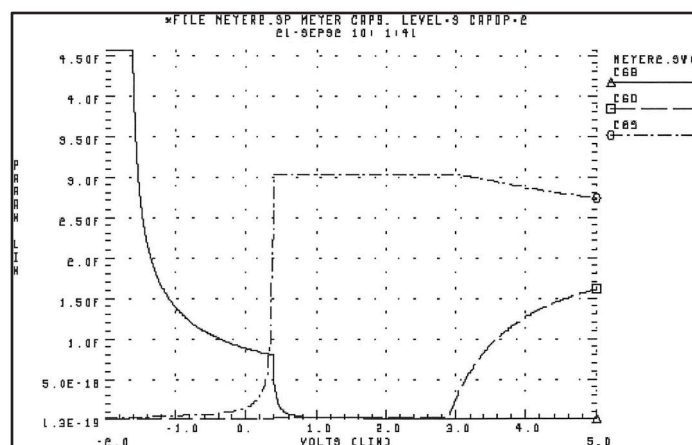


Figure 30 CAPOP=2 Capacitances for LEVEL 3 Model



The example below tests the charge conservation capacitance model (Yang, P., B.D. Epler, and P.K. Chatterjee 'An Investigation of the Charge Conservation Problem') and compares the Meyer and charge conservation models. As the graph in [Figure 32](#) shows, the charge conservation model returns more accurate results.

Example 2:

This example is based on demonstration netlist `chrgpump.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
chrgpump.sp: charge conservation test for charge pump circuit
*test circuit of a mosfet capacitor and a linear capacitor
.options post acct list nomod $ method=gear
+ reltol=1e-3 abstol=1e-6 chgtol=1e-14 $ delmax=0.1ns
.param capop=2
.op
.tran 2ns 470ns sweep capop poi 2 2,9
.ic v(s)=1
*
vin g 0 pulse 0 5 15ns 5ns 5ns 50ns 100ns
vbb 0 b pulse 0 5 0ns 5ns 5ns 50ns 100ns
vdd d d- pulse 0 5 25ns 5ns 5ns 50ns 100ns
*
rc d- s 10k
c2 s 0 10p
m1 d g s b mm w=3.5u l=5.5u
+ad=100p as=100p pd=50u ps=50u nrd=1 nrs=1
*
.model mm nmos level=3 vto=0.7 kp=50e-6 gamma=0.96
+phi=0.5763 tox=50e-9 nsub=1.0e16 ld=0.5e-6
+vmax=268139 theta=0.05 eta=1 kappa=0.5 cj=1e-4
+cjsw=0.05e-9 rsh=20 js=1e-8 pb=0.7
+cgd=0 cgs=0 is=0 js=0
+capop=capop
*
.probe tran vout=v(s)
.probe tran vd=v(d) vg=v(g) vb=v(b)
.print tran v(s) v(d) v(g) v(b)
.end
```

Figure 31 Charge Pump Circuit

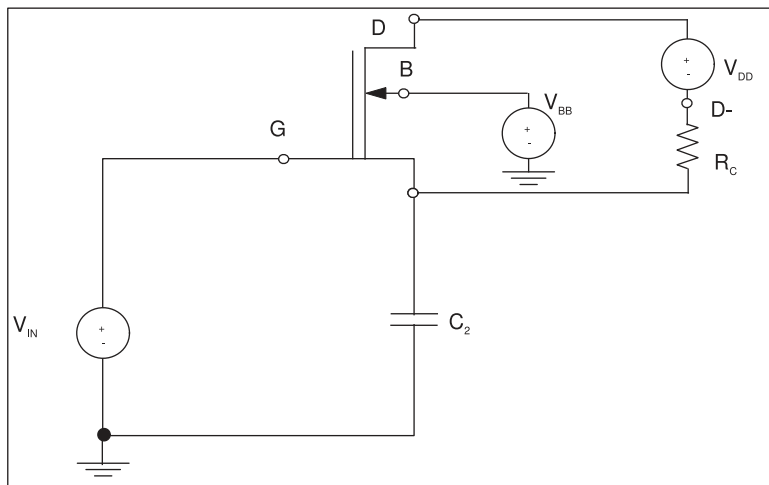
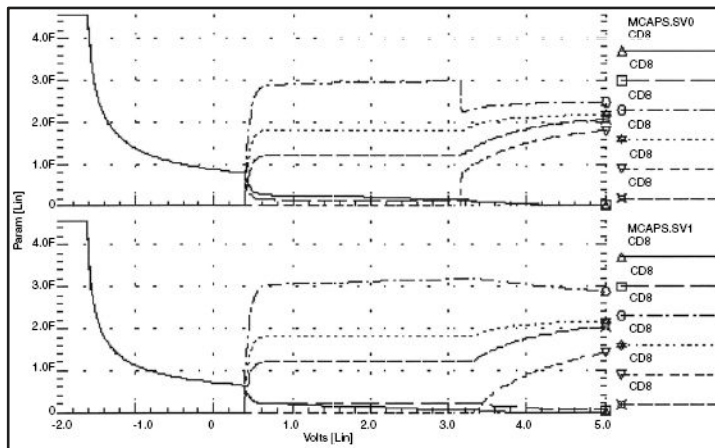


Figure 32 Charge Conservation Test: CAPOP=2 or 9



The following example applies a pulse through a constant capacitance to the gate of a MOS transistor. Ideally, if the model conserves charge, then the voltage at node 20 should become zero when the input pulse becomes zero. Consequently, the model that provides voltage closer to zero for node 20 conserves the charge better. The results of the CAPOP=4 model are better than the CAPOP=2 model.

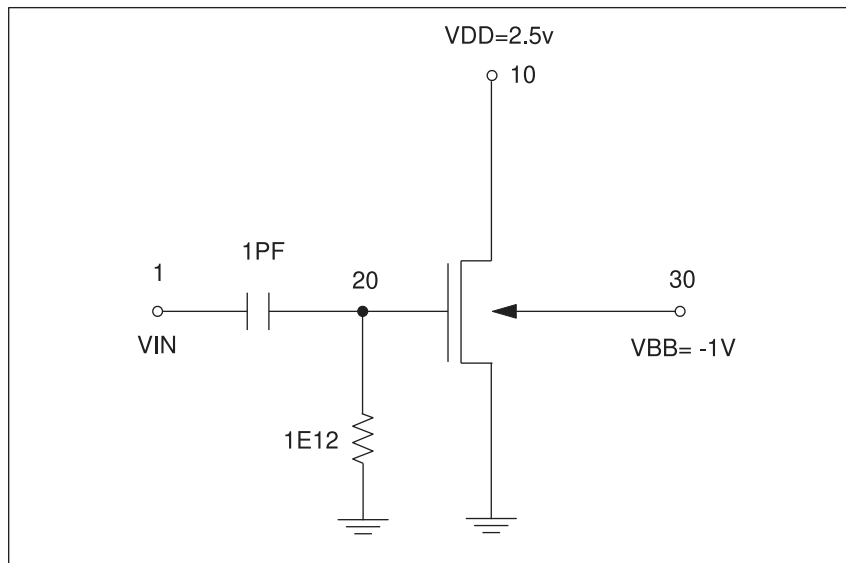
This example compares charge conservation models in SPICE2G.6 and Synopsys device models. The results indicate that the Synopsys device models are more accurate.

Example 3:

This example is based on demonstration netlist `mcap2_a.sp`, which is available in directory `$installdir/demo/hspice/mos`:

```
FILE MCAP2_A.SP
.OPTION SPICE NOMOD DELMAX=.25N POST=2
.PARAM CAPOP=4
.TRAN 1NS 40NS SWEEP CAPOP POI 2 4 2
.PRINT TRAN V(1) V(20)
VIN 1 0 PULSE (0V, 5V, 0NS, 5NS, 5NS, 5NS, 20NS)
CIN 1 20 1PF
RLEAK 20 0 1E+12
VDD 10 0 1.3
VBB 30 0 -1
M 10 20 0 30 MOS W=10U L=5U
.MODEL MOS NMOS LEVEL=2 TOX=250E-10 VTO=.3
+ UO=1000 LAMBDA=1E-3 GAMMA=.5 PHI=.5 XQC=.5
+ THETA=0.067 VMAX=1.956E5 XJ=.3U
+ CGSO=0 CGDO=0 CGBO=0
+ CJ=0 JS=0 IS=0
+ CAPOP=CAPOP
.END
```

Figure 33 Charge Conservation Test Circuit



CAPOP=5 — No Gate Capacitance

If `CAPOP=5` for no capacitors, then simulation does not calculate gate capacitance.

CAPOP=6 — AMI Gate Capacitance Model

Define: $vgst = vgs - \frac{(vth + vfb)}{2}$, $cox = \frac{\epsilon_{ox}}{TOX \cdot 1e-10} \cdot Weff \cdot Leff$

The following equations calculate the cgs gate capacitance in the different regions.

$0.5 \cdot (vth + vfb) > vgs$

$$cgs = 0$$

$0.5 \cdot (vth + vfb) < vgs < vth$

$$\text{For } vgst < vds: cgs = \frac{4}{3} \cdot \frac{cox \cdot vgst}{vth - vfb}$$

$$\text{For } vgst > vds: cgs = arg \cdot \frac{4}{3} \cdot \frac{cox \cdot vgst}{vth - vfb}$$

$vgs > vth$

$$\text{For } vgst < vds: cgs = \frac{2}{3} \cdot cox$$

$$\text{For } vgst > vds: cgs = arg \cdot \frac{2}{3} \cdot cox, arg = vgst \cdot \frac{(3 \cdot vgst - 2 \cdot vds)}{(2 \cdot vgst - vds)^2}$$

The following equations calculate the cgd gate capacitance in the different regions.

$vgs < vth$

$$cgd = 0$$

$vgs > vth$ and $vgst < vds$

$$cgd = 0$$

$vgs > vth$ and $vgst > vds$

$$cgd = arg \cdot \frac{2}{3} \cdot cox, arg = (3 \cdot vgst - vds) \cdot \frac{(vgst - vds)}{(2 \cdot vgst - vds)^2}$$

The following equation combines the cgb gate capacitance with the calculation of both oxide capacitance and depletion capacitance:

$$cgb = \frac{cgbx \cdot cd}{cgbx + cd}$$

Simulation calculates the oxide capacitance ($cgbx$) as:

$$cgbx = cox - cgs - cgd$$

Depletion capacitance (cd) is voltage-dependent:

$$cd = \frac{\epsilon_{si}}{wd} \cdot W_{eff} \cdot L_{eff}, \quad wd = \left(\frac{2 \cdot \epsilon_{si} \cdot vc}{q \cdot NSUB} \right)^{1/2}$$

vc = The effective voltage from channel to substrate (bulk)

The following equations show vc under various conditions:

$$vgs + vsb < vfb$$

$$vc = 0$$

$$vgs + vsb > vfb$$

$$vc = vgs + vsb - vfb$$

$$vgst > 0, vgs < vth, vgst < vds$$

$$vc = \frac{1}{2} \cdot (vth - vfb) + \frac{3}{2} \cdot vgst + vsb$$

$$vgst > 0, vgs < vth, vgst > vds$$

$$vc = \frac{1}{2} \cdot (vth - vfb) + vgst + \frac{1}{2} \cdot vds + vsb$$

$$vgs > vth, vgst < vds$$

$$vc = vth - vfb + \frac{1}{2} \cdot vgst + vsb$$

$$v_{gs} > v_{th}, v_{gst} > v_{ds}$$

$$v_c = v_{th} - v_{fb} + \frac{1}{2} \cdot v_{ds} + v_{sb}$$

CAPOP=13 — BSIM1-based Charge-Conserving Gate Capacitance Model

See [LEVEL 13: BSIM Model](#).

CAPOP=39 — BSIM2 Charge-Conserving Gate Capacitance Model

See [LEVEL 13: BSIM Model](#).

Calculating Effective Length and Width for AC Gate Capacitance

For some MOS processes and parameter extraction method. AC analysis might need different L_{eff} and W_{eff} values than for DC analysis. For AC gate capacitance calculations, substitute the $LDAC$ and $WDAC$ model parameters for LD and WD in the L_{eff} and W_{eff} calculations. You can use LD and WD in L_{eff} and W_{eff} calculations for DC current.

To use $LDAC$ and $WDAC$, enter XL , LD , $LDAC$, XW , WD , and $WDAC$ in the `.MODEL` statement. The model uses the following equations for DC current calculations.

$$L_{eff} = L + XL - 2 \cdot LD$$

$$W_{eff} = W + XW - 2 \cdot WD$$

The model parameters also use the following equations to calculate the AC gate capacitance:

$$L_{eff} = L + XL - 2 \cdot LDAC$$

$$W_{eff} = W + XW - 2 \cdot WDAC$$

The noise calculations use the DC W_{eff} and L_{eff} values.

Use $LDAC$ and $WDAC$ with the standard XL , LD , XW , and WD parameters. Do not use $LDAC$ and $WDAC$ with other parameters, such as DLO and $DW0$.

4

MOSFET Diode Models

Discusses the use of the available MOSFET diode models.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

This section covers the following topics:

- [Selecting MOSFET Diode Models](#)
- [Enhancing Convergence](#)
- [MOSFET Diode Model Parameters](#)
- [Using an ACM=0 MOS Diode](#)
- [Using an ACM=1 MOS Diode](#)
- [Using an ACM=2 MOS Diode](#)
- [Using an ACM=3 MOS Diode](#)
- [MOS Diode Equations](#)

Selecting MOSFET Diode Models

You can use the Area Calculation Method (ACM) parameter to precisely control bulk-to-source and bulk-to-drain diodes within MOSFET models. Use the `ACM` model parameter to select one of three different modeling schemes for the MOSFET bulk diodes. This section discusses the model parameters and model equations used for the different MOSFET diode models.

To select a MOSFET diode model, set the `ACM` parameter within the MOSFET model statements.

- If `ACM=0`, the pn bulk junctions of the MOSFET are modeled in the SPICE style.
- The `ACM=1` diode model is the original ASPEC model.

- The `ACM=2` model parameter specifies the improved diode model, which is based on a model similar to the ASPEC MOSFET diode model.
- The `ACM=3` diode model is a further improvement that deals with capacitances of shared sources and drains, and gate edge source/drain-to-bulk periphery capacitance.
- If you do not set the `ACM` model parameter, the diode model defaults to the `ACM=0` model.
- If `ACM=0` and `ACM=1` models, you cannot specify `HDIF`. In the `ACM=0` model, you cannot specify `LDIF`. The `ACM=1` model does not use the `AD`, `AS`, `PD`, and `PS` geometric element parameters.

Enhancing Convergence

- The `GMIN` option creates a parallel conductance across the bulk diodes and drain-source for transient analysis.
- The `GMINDC` option creates a parallel conductance across the bulk diodes and drain-source for DC analysis.

These options enhance the convergence properties of the diode model, especially when the model has a high off resistance. Use the `RSH`, `RS`, and `RD` parameters to prevent over-driving the diode in either a DC or transient forward bias condition. These parameters also enhance the convergence properties of the diode model.

MOSFET Diode Model Parameters

This section covers the following topics:

- [MOSFET Diode Model — DC Parameters](#)
- [MOSFET Diode Model — Capacitance Parameters](#)
- [MOSFET Diode Model — Drain and Source Resistance Parameters](#)
- [MOSFET Diode Model — MOS Geometry Parameters](#)

MOSFET Diode Model — DC Parameters

Name (Alias)	Units	Default	Description
ACM		0	Area calculation method

Name (Alias)	Units	Default	Description
JS	amp/m ²		Bulk junction saturation current: JSscaled=JS/SCALM2 – for ACM=1 unit is amp/m and JSscaled=JS/SCALM.
JSW	amp/m	0	Sidewall bulk junction saturation current: JSWscaled=JSW/SCALM.
IS	amp	1e-14	Bulk junction saturation current. For the ASPEC=1 option, default=0.
N		1	Emission coefficient.
NDS		1	Reverse bias slope coefficient.
VNDS	V	-1	Reverse diode current transition point.

MOSFET Diode Model — Capacitance Parameters

Name (Alias)	Units	Default	Description
CBD	F	0	Zero bias bulk-drain junction capacitance. Used only when CJ and CJSW are 0.
CBS	F	0	Zero bias bulk-source junction capacitance. Use only when CJ and CJSW are 0.
CJ (CDB, CSB, CJA)	F/m ² μF/m ²	579.11 μF/m ²	Zero-bias bulk junction capacitance: - CJscaled = CJ/SCALM2 —for ACM=1 the unit is F/m. - CJscaled = CJ/SCALM. Default for the ASPEC=0 option is: $CJ = \left(\frac{\epsilon_{Si} \cdot q \cdot NSUB}{2 \cdot PB} \right)^{\frac{1}{2}}$
CJGATE	F/m	CSJW	Zero-bias gate-edge sidewall bulk junction capacitance (ACM=3 only).CJGATEscaled=CJGATE/SCALM - Default = CJSW for the PrimeSim™ Continuum solution releases later than H9007D. - Default = 0 for the PrimeSim™ Continuum solution releases H9007D and earlier, or if you do not specify CJSW.
CJSW (CJP)	F/m	0	Zero-bias sidewall bulk junction capacitance. CJSWscaled = CJSW/SCALM. Default = 0.

Name (Alias)	Units	Default	Description
FC		0.5	Forward-bias depletion capacitance coefficient (not used).
MJ (EXA, EXJ, EXS, EXD)		0.5	Bulk junction grading coefficient.
MJSW (EXP)		0.33	Bulk sidewall junction grading coefficient.
NSUB (DNB, NB)	1/cm ³	1.0e15	Substrate doping.
PB (PHA, PHS, PHD)	V	0.8	Bulk junction contact potential.
PHP	V	PB	Bulk sidewall junction contact potential.
TT	s	0	Transit time

MOSFET Diode Model — Drain and Source Resistance Parameters

Name (Alias)	Units	Default	Description
LRD	ohm/m	0	Drain resistance length sensitivity. Use this parameter with automatic model selection, WRD, and PRD to factor a model for the device size.
LRS	ohm/m	0	Source resistance length sensitivity. Use this parameter with automatic model selection, WRS, and PRS to factor a model for the device size.
PRD	ohm/m ²	0	Drain resistance product (area) sensitivity (used with LRD).
PRS	ohm/m ²	0	Source resistance product (area) sensitivity (used with LRS).
RD	ohm/sq	0.0	Drain ohmic resistance. This parameter is usually the sheet resistance of a lightly- doped region for ACM≥1.
RDC	ohm	0.0	Additional drain resistance due to contact resistance.
RS	ohm/sq	0.0	Source ohmic resistance. This parameter is usually the sheet resistance of a lightly-doped region for ACM ≥ 1.
RSC	ohm	0.0	Source resistance due to contact resistance.
RSH (RL)	ohm/sq	0.0	Drain and source diffusion sheet resistance.

Name (Alias)	Units	Default	Description
WRD	ohm/m	0	Drain resistance width sensitivity (used with LRD).
WRS	ohm/m	0	Source resistance width sensitivity (used with LRS).

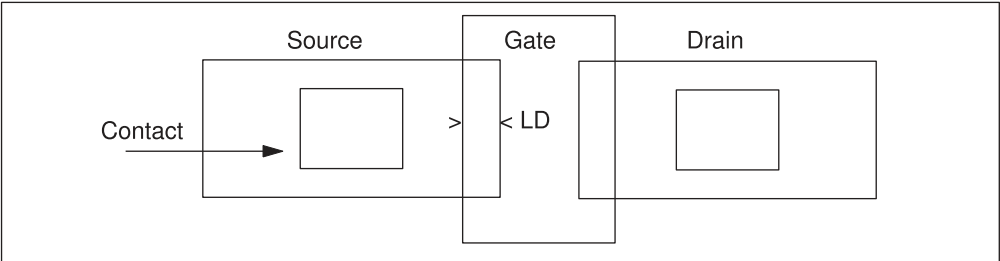
MOSFET Diode Model — MOS Geometry Parameters

Name (Alias)	Units	Default	Description
HDIF	m	0	Length of heavily-doped diffusion, from contact to lightly-doped region (ACM=2, 3 only): $HDIF_{scaled} = HDIF \cdot SCALM$
LD (DLAT,LATD)	m		Lateral diffusion into the channel from the source and drain diffusion. If you do not specify LD and XJ, LD default=0.0. If you specify LD, but you do not specify XJ, then simulation calculates LD from XJ. Default=0.75 · XJ. For LEVEL 4 only, lateral diffusion is derived from LD · XJ. $LD_{scaled} = LD \cdot SCALM$
LDIF	m	0	Length of lightly-doped diffusion adjacent to the gate (ACM=1, 2): $LDIF_{scaled} = LDIF \cdot SCALM$
WMLT		1	Width diffusion layer shrink reduction factor.
XJ	m	0	Metallurgical junction depth: $XJ_{scaled} = XJ \cdot SCALM$
XW (WDEL, DW)	m	0	Accounts for masking and etching effects: $XW_{scaled} = XW \cdot SCALM$

Using an ACM=0 MOS Diode

Figure 34 shows the parameter value settings for a MOSFET diode, designed with a MOSFET that has a channel length of 3 μm and a channel width of 10 μm.

Figure 34 *ACM=0 MOS Diode*



For example, a transistor might include:

$L_D = .5\text{mm}$ $W = 10\text{mm}$ $L = 3\text{mm}$

Parameter	Description
AD	area of drain (about 80 pm^2)
AS	area of source (about 80 pm^2)
CJ	4e-4 F/ m^2
CJSW	1e-10 F/m
JS	1e-8 A/ m^2
JSW	1e-13 A/m
NRD	number of squares for drain resistance
NRS	number of squares for source resistance
PD	sidewall of drain (about 36 μm)
PS	sidewall of source (about 36 μm)

This section covers the following topics:

- [Calculating Effective Areas and Peripheries](#)
- [Calculating Effective Saturation Current](#)
- [Calculating Effective Drain and Source Resistances](#)

Calculating Effective Areas and Peripheries

For $ACM=0$, simulation calculates the effective areas and peripheries as follows:

$$A_{eff} = M \cdot AD \cdot WMLT^2 \cdot SCALE^2$$

$$ASeff = M \cdot AS \cdot WMLT^2 \cdot SCALE^2$$

$$PDef f = M \cdot PD \cdot WMLT \cdot SCALE$$

$$PSeff = M \cdot PS \cdot WMLT \cdot SCALE$$

Calculating Effective Saturation Current

For $ACM=0$, simulation calculates the MOS diode effective saturation currents as follows:

Source Diode Saturation Current

Define: $val = JSscaled \cdot ASeff + JSWscaled \cdot PSeff$

If $val > 0$, then $isbs = val$

Otherwise, $isbd = M \cdot IS$

Drain Diode Saturation Current

Define: $val = JSscaled \cdot ADef f + JSWscaled \cdot PDef f$

If $val > 0$, then $isbd = val$

Otherwise, $isbd = M \cdot IS$

Calculating Effective Drain and Source Resistances

For $ACM=0$, simulation calculates the effective drain and source resistances as follows:

Source Resistance

Define: $val = NRS \cdot RSH$

If $val > 0$, then $RSeff = \frac{val + RSC}{M}$

Otherwise, $RSeff = \frac{RS + RSC}{M}$

Drain Resistance

Define: $val = NRD \cdot RSH$

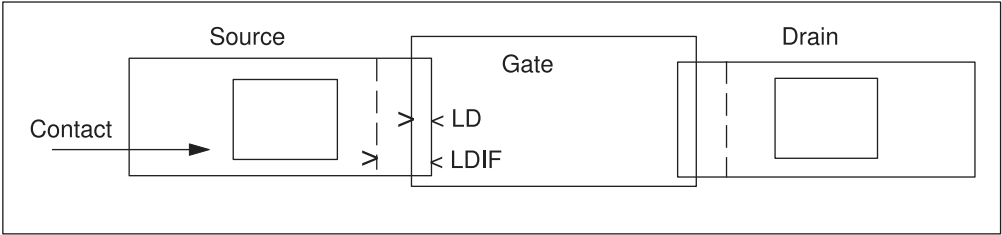
If $val > 0$, then $RDef f = \frac{val + RDC}{M}$

Otherwise, $RDef f = \frac{RD + RDC}{M}$

Using an ACM=1 MOS Diode

If you specify the `ACM=1` model parameter, simulation uses ASPEC-style diodes, and does not use the `AD`, `PD`, `AS`, and `PS` parameters. The `JS` and `CJ` units differ from SPICE-style diodes (`ACM=0`).

Figure 35 *ACM=1 MOS Diode*



Example:

The following table lists parameter value settings for a transistor with the following parameter values:

- `LD=0.5 μm`
- `W=10 μm`
- `L=3 μm`
- `LDIF=0.5 μm`

Parameter	Description
CJ	1e-10 F/m of gate width Note the change from F/m ² (in ACM=0) to F/m.
CJSW	2e-10 F/m of gate width
JS	1e-14 A/m of gate width Note the change from A/m ² (in ACM=0) to A/m
JSW	1e-13 A/m of gate width
NRD	number of squares for drain resistance
NRS	number of squares for source resistance

This section covers the following topics:

- [Calculating Effective Areas and Peripheries](#)
- [Calculating Effective Saturation Current](#)
- [Calculating Effective Drain and Source Resistances](#)

Calculating Effective Areas and Peripheries

For $ACM=1$, simulation calculates the effective areas and peripheries as follows:

$$A_{Def} = W_{eff} \cdot W_{MLT}$$

$$A_{Sef} = W_{eff} \cdot W_{MLT}$$

$$P_{Dff} = W_{eff}$$

$$P_{Sef} = W_{eff}$$

The following equation calculates the W_{eff} value used in the preceding equations:

$$W_{eff} = M \cdot (W_{scaled} \cdot W_{MLT} + XW_{scaled})$$

Note:

The W_{eff} value is not the same as the w_{eff} value in the LEVEL 1, 2, 3, 6, and 13 models. The $2 \cdot WD_{scaled}$ term is not subtracted.

Calculating Effective Saturation Current

For $ACM=1$, the MOS diode effective saturation currents are calculated as follows:

Source Diode Saturation Current

Define: $val = JS_{scaled} \cdot A_{Sef} + JSW_{scaled} \cdot P_{Sef}$

If $val > 0$, then $isbs = val$

Otherwise, $isbs = M \cdot IS$

Drain Diode Saturation Current

Define: $val = JS_{scaled} \cdot A_{Def} + JSW_{scaled} \cdot P_{Def}$

If $val > 0$, then $isbd = val$

Otherwise, $isbd = M \cdot IS$

Calculating Effective Drain and Source Resistances

For $ACM=1$, simulation calculates the effective drain and source resistances as follows.

Source Resistance

For $UPDATE=0$:

$$R_{Seff} = \frac{LD_{scaled} + LDIF_{scaled}}{W_{eff}} \cdot RS + \frac{NRS \cdot RSH + RSC}{M}$$

If $UPDATE \geq 1$, $LDIF=0$, and you specify the `ASPEC` option, then:

$$R_{Seff} = \frac{1}{M} \cdot (RS + NRS \cdot RSH + RSC)$$

Drain Resistance

For $UPDATE=0$:

$$R_{Def} = \frac{LD_{scaled} + LDIF_{scaled}}{W_{eff}} \cdot RD + \frac{NRD \cdot RSH + RDC}{M}$$

If $UPDATE \geq 1$, $LDIF=0$, and you specify the `ASPEC` option, then:

$$R_{Def} = \frac{1}{M} \cdot (RD + NRD \cdot RSH + RDC)$$

See [LEVEL 6: Lattin-Jenkins-Grove \(ASPEC\) Model on page 120](#) and [LEVEL 7: Lattin-Jenkins-Grove \(SPICE\) Model on page 146](#) for more possibilities.

Using an ACM=2 MOS Diode

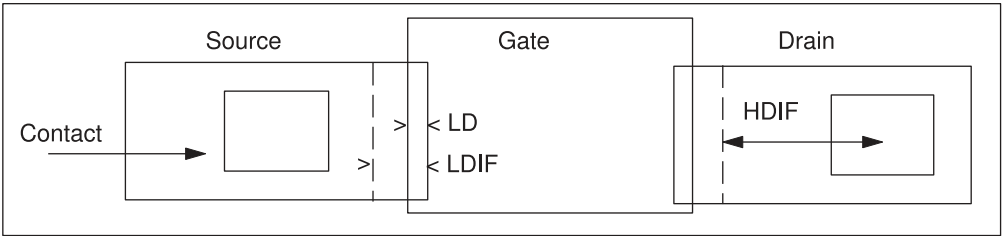
If you set the `ACM=2` model parameter, simulation uses the PrimeSim Continuum solution-style MOS diodes. You can use a fold-back calculation scheme similar to the `ASPEC` method, retaining full model-parameter compatibility with the `SPICE` procedure. This method also supports both lightly-doped and heavily-doped diffusions (the `LD`, `LDIF`, and `HDIF` parameters set the diffusion type). This model preserves the `JS`, `JSW`, `CJ`, and `CJSW` units (used in `SPICE`) for full compatibility.

`ACM=2` automatically generates more-reasonable diode parameter values than those for `ACM=1`. You can generate the `ACM=2` geometry in either of two ways:

- `AD`, `AS`, `PD`, and `PS` element parameters in the element statement generate parasitics. These parameters do not have default option values.
- To suppress the diode, set `IS=0`, `AD=0`, and `AS=0`.

If you set $AS=0$ in the element and $IS=0$ in the model, simulation suppresses the source diode. Use this setting for shared contacts.

Figure 36 *ACM=2 MOS Diode*



Example

For a transistor with $LD=0.07\mu\text{m}$, $W=10\mu\text{m}$, $L=2\mu\text{m}$, $LDIF=1\mu\text{m}$, and $HDIF=4\mu\text{m}$. [Table 10](#) shows typical MOSFET diode parameter values.

Table 10 *ACM=2 MOS Diode Parameters*

Parameter	Description
AD	Area of drain. Default option value for AD is not applicable.
AS	Area of source. Default option value for AS is not applicable.
CJ	$1\text{e-}4\text{ F/m}^2$
CJSW	$1\text{e-}10\text{ F/m}$
JS	$1\text{e-}4\text{ A/m}^2$
JSW	$1\text{e-}10\text{ A/m}$
HDIF	Length of heavily-doped diffusion contact-to-gate (about $2\mu\text{m}$). $HDIF_{\text{eff}}=HDIF \cdot WMLT \cdot SCALM$
LDIF+LD	Length of lightly-doped diffusion (about $0.4\mu\text{m}$).
NRD	Number of squares drain resistance. Default value for NRD does not apply.
NRS	Number of squares source resistance. Default for NRS does not apply.
PD	Periphery of drain, including gate width for ACM=2. No default.
PS	Periphery of source, including gate width for ACM=2. No default.
RD	Resistance (ohm/square) of lightly-doped drain diffusion (about 2000).
RS	Resistance (ohm/square) of lightly-doped source diffusion (about 2000).

Table 10 ACM=2 MOS Diode Parameters (Continued)

Parameter	Description
RSH	Diffusion sheet resistance (about 35).

Calculating Effective Areas and Peripheries

For $ACM=2$, simulation calculates the effective areas and peripheries as follows:

- If you do not specify AD , then $A_{Def} = 2 \cdot HDIF_{eff} \cdot W_{eff}$
Otherwise, $A_{Def} = M \cdot AD \cdot WMLT^2 \cdot SCALE^2$
- If you do not specify AS , then $A_{Sef} = 2 \cdot HDIF_{scaled} \cdot W_{eff}$
Otherwise, $A_{Sef} = M \cdot AS \cdot WMLT^2 \cdot SCALE^2$
- If you do not specify PD , then $P_{Def} = M \cdot (4 \cdot HDIF_{eff} + 2 \cdot W_{eff})$
Otherwise, $P_{Def} = M \cdot PD \cdot WMLT \cdot SCALE$
- If you do not specify PS , then $P_{Sef} = M \cdot (4 \cdot HDIF_{eff} + 2 \cdot W_{eff})$
Otherwise, $P_{Sef} = M \cdot PS \cdot WMLT \cdot SCALE$

The following equations calculate values used in the preceding equation:

$$W_{eff} = W_{scaled} \cdot WMLT + XW_{scaled}$$

$$HDIF_{eff} = HDIF_{scaled}$$

$$HDIF_{scaled} = HDIF \cdot SCALM \cdot WMLT$$

The W_{eff} value is not the same as the W_{eff} value in the LEVEL 1, 2, 3, and 6 models. The $2 \cdot \bar{A}W_{Dscaled}$ term is not subtracted.

Calculating Effective Saturation Currents

For $ACM=2$, simulation calculates the MOS diode effective saturation currents as follows.

Source Diode Saturation Current

Define: $val = JS_{scaled} \cdot A_{Sef} + JSW_{scaled} \cdot P_{Sef}$

If $val > 0$, then $isbs = val$

Otherwise, $isbs = M \cdot IS$

Drain Diode Saturation Current

Define: $val = JSscaled \cdot ADef f + JSWscaled \cdot PDef f$

If $val > 0$, then $isbd = val$

otherwise: $isbd = M \cdot IS$.

Calculating Effective Drain and Source Resistances

For $ACM=2$, simulation calculates the effective drain and source resistances as follows.

Source Resistance

If you specify NRS , then:

$$RSeff = \frac{LDscaled + LDIFscaled}{Weff} \cdot RS + \left(\frac{NRS \cdot RSH + RSC}{M} \right)$$

Otherwise:

$$RSeff = \frac{RSC}{M} + \frac{HDIFeff \cdot RSH + (LDscaled + LDIFscaled) \cdot RS}{Weff}$$

Drain Resistance

If you specify NRD , then:

$$RDef f = \frac{LDscaled + LDIFscaled}{Weff} \cdot RD + \left(\frac{NRD \cdot RSH + RDC}{M} \right)$$

otherwise:

$$RDef f = \frac{RDC}{M} + \frac{HDIFeff \cdot RSH + (LDscaled + LDIFscaled) \cdot RD}{Weff}$$

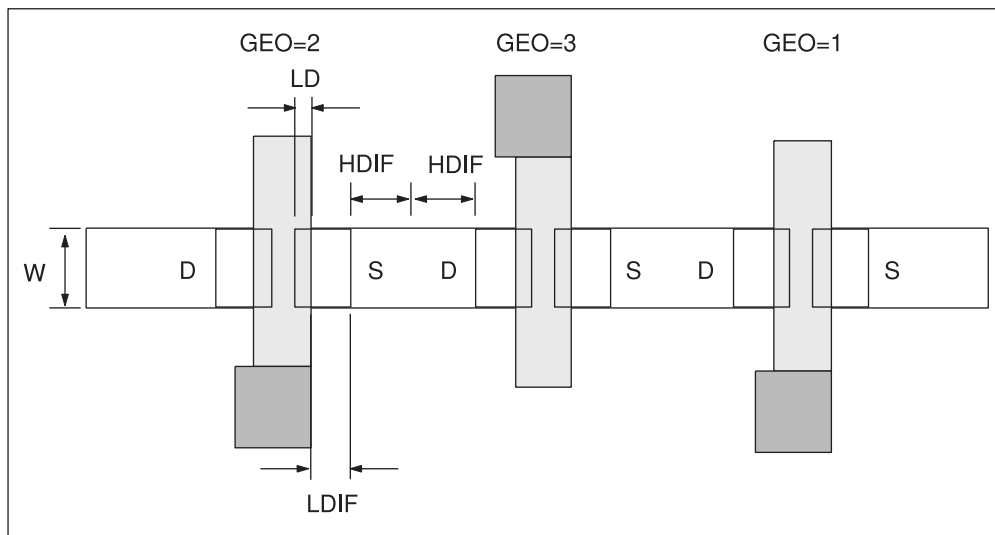
Using an ACM=3 MOS Diode

Use $ACM=3$ to properly model MOS diodes of stacked devices. You can also use the $CJGATE$ parameter to model the drain and source periphery capacitances separately, along the gate edge. Therefore, the PD and PS calculations do not include the gate periphery length. $CJGATE$ defaults to the $CJSW$ value, which in turn defaults to 0.

The AD , AS , PD , and PS calculations depend on the device layout as determined by the value of the GEO element parameter. You can specify the following GEO values in the MOS element description:

- $GEO=0$: other devices do not share the drain and source of the device (default).
- $GEO=1$: another device shares the drain.
- $GEO=2$: another device shares the source.
- $GEO=3$: another device shares the drain and source.

Figure 37 Stacked Devices and Corresponding GEO Values



Calculating Effective Areas and Peripheries

ACM=3 calculates the effective areas and peripheries based on the GEO value.

- If you do not specify AD , then:
 - For $GEO=0$ or 2 , $A_{Def} = 2 \cdot HDIF_{eff} \cdot W_{eff}$
 - For $GEO=1$ or 3 , $A_{Def} = HDIF_{eff} \cdot W_{eff}$
 - Otherwise, $A_{Def} = M \cdot AD \cdot WMLT^2 \cdot SCALE^2$

- If you do not specify AS , then:
 - For $GEO=0$ or 1 , $AS_{eff} = 2 \cdot HDIF_{eff} \cdot W_{eff}$
 - For $GEO=2$ or 3 , $AS_{eff} = HDIF_{eff} \cdot W_{eff}$
 - Otherwise, $AS_{eff} = M \cdot AS \cdot WMLT^2 \cdot SCALE^2$
- If you do not specify PD , then:
 - For $GEO=0$ or 2 , $PDef_{eff} = 4 \cdot HDIF_{eff} + W_{eff}$
 - For $GEO=1$ or 3 , $PDef_{eff} = 2 \cdot HDIF_{eff}$
 - Otherwise, $PDef_{eff} = M \cdot PD \cdot WMLT \cdot SCALE$
- If you do not specify PS , then:
 - For $GEO=0$ or 1 , $PS_{eff} = 4 \cdot HDIF_{eff} + W_{eff}$
 - For $GEO=2$ or 3 , $PS_{eff} = 4 \cdot HDIF_{eff}$
 - Otherwise, $PS_{eff} = M \cdot PS \cdot WMLT \cdot SCALE$

Simulation calculates W_{eff} and $HDIF_{eff}$ as follows:

$$W_{eff} = M \cdot (W_{scaled} \cdot WMLT + XW_{scaled})$$

$$HDIF_{eff} = HDIF_{scaled} \cdot WMLT$$

Note:

The W_{eff} value is not the same as the W_{eff} value in the LEVEL 1, 2, 3, and 6 models. The $2 \cdot WD_{scaled}$ term is not subtracted.

Effective Saturation Current Calculations

The $ACM=3$ model calculates the MOS diode effective saturation currents the same as $ACM=2$.

Effective Drain and Source Resistances

The $ACM=3$ model calculates the effective drain and source resistances the same as $ACM=2$.

MOS Diode Equations

This section describes MOS diode equations.

DC Current

- Simulation parallels the drain and source MOS diodes with G_{MINDC} conductance in the DC analysis.
- Simulation parallels the drain and source MOS diodes with G_{MIN} conductance in the transient analysis.

The total DC current is the sum of the diode current and the conductance current. The diode current is calculated as follows.

Drain and Source Diodes Forward Biased

- $v_{bs} > 0$: $i_{bs} = i_{sbs} \cdot (e^{v_{bs}/(N \cdot v_t)} - 1)$
- $v_{bd} > 0$: $i_{bd} = i_{sbd} \cdot (e^{v_{bd}/(N \cdot v_t)} - 1)$

Drain and Source Diodes Reverse Biased

- For $0 > v_{bs} > V_{NDS}$: $i_{bs} = g_{sbs} \cdot v_{bs}$
- For $v_{bs} < V_{NDS}$: $i_{bs} = g_{sbs} \cdot V_{NDS} + \left(\frac{g_{sbs}}{NDS} \right) \cdot (v_{bs} - V_{NDS})$
- For $0 > v_{bd} > V_{NDS}$: $i_{bd} = g_{sbd} \cdot v_{bd}$
- For $v_{bd} < V_{NDS}$: $i_{bd} = g_{sbd} \cdot V_{NDS} + \left(\frac{g_{sbd}}{NDS} \right) \cdot (v_{bd} - V_{NDS})$

The following equations calculate values used in the preceding equations:

$$|g_{sbs}| = |i_{sbs}| \text{ and } |g_{sbd}| = |i_{sbd}|$$

Using MOS Diode Capacitance Equations

Each MOS diode capacitance is the sum of diffusion and depletion capacitance. Simulation evaluates the diffusion capacitance in terms of the small signal conductance of the diode and a TT model parameter, representing the transit time of the diode. The depletion capacitance depends on which ACM you choose.

To calculate bias-dependent depletion capacitance, define $C0BS$, $C0BD$, $C0BS_SW$, and $C0BD_SW$ intermediate quantities. These depend on geometric parameters, such as AS_{eff} and PS_{eff} , calculated under various ACM specifications.

For $ACM=3$, the $C0BS_SW$ and $C0BD_SW$ intermediate quantities include an extra term to account for $CJGATE$.

ACM=2 includes the CJGATE parameter for backward compatibility. Therefore, the default behavior of CJGATE makes the C0BS_SW and C0BD_SW intermediate quantities the same as for previous versions. The default patterns are:

- If you do not specify CJSW or CJGATE, both default to zero.
- If you do not specify CJGATE, it defaults to CJSW, which defaults to zero.
- If you specify CJGATE, but you do not specify CJSW, then CJSW defaults to zero.

Simulation calculates the C0BS, C0BS_SW, C0BD, and C0BD_SW intermediate quantities as follows.

```
C0BS = CJscaled*ASeff
C0BD = CJscaled*ADeff
```

- If (ACM=0 or 1), then:

```
C0BS_SW = CJSWscaled*PSeff
C0BD_SW = CJSWscaled*PDeff
```

- If (ACM=2) and (PSeff, < Weff), then:

```
C0BS_SW = CJGATEscaled*PSeff
```

- If (ACM=2) and (PSeff, > Weff), then:

```
C0BS_SW = CJSWscaled*(PSeff-Weff) + CJGATEscaled*Weff
```

- If (ACM=2) and (PDeff < Weff), then:

```
C0BD_SW = CJGATEscaled*PDeff
```

- If (ACM=2) and (PDeff > Weff), then:

```
C0BD_SW = CJSWscaled*(PDeff-Weff) + CJGATEscaled*Weff
```

- If (ACM=3), then:

```
C0BS_SW = CJSWscaled*PSeff + CJGATEscaled*Weff
C0BD_SW = CJSWscaled*PDeff + CJGATEscaled*Weff
```

Source Diode Capacitance

- If (C0BS + C0BS_SW) > 0 and vbs < 0, then:

$$capbs = TT \cdot \frac{\partial ibs}{\partial vbs} + C0BS \cdot \left(1 - \frac{vbs}{PB}\right)^{-MJ} \\ + C0BS_SW \cdot \left(1 - \frac{vbs}{PHP}\right)^{-MJSW}$$

If $(C0BS + C0BS_SW) > 0$ and $vbs > 0$, then:

$$capbs = TT \cdot \frac{\partial ibs}{\partial vbs} + C0BS \cdot \left(1 + MJ \cdot \frac{vbs}{PB}\right) \\ + C0BS_SW \cdot \left(1 + MJSW \cdot \frac{vbs}{PHP}\right)$$

Otherwise, if $(C0BS + C0BS_SW) \leq 0$, then:

$$\text{For } vbs < 0: capbs = TT \cdot \frac{\partial ibs}{\partial vbs} + M \cdot CBS \cdot \left(1 - \frac{vbs}{PB}\right)^{-MJ}$$

$$\text{For } vbs > 0: capbs = TT \cdot \frac{\partial ibs}{\partial vbs} + M \cdot CBS \cdot \left(1 + MJ \cdot \frac{vbs}{PB}\right)$$

Drain Diode Capacitance

If $(C0BD + C0BD_SW) > 0$, then:

$$\text{For } vbd < 0: capbd = TT \cdot \frac{\partial ibd}{\partial vbd} + C0BD \cdot \left(1 - \frac{vbd}{PB}\right)^{-MJ}$$

$$\text{For } vbd > 0: capbd = TT \cdot \frac{\partial ibd}{\partial vbd} + C0BD \cdot \left(1 + MJ \cdot \frac{vbd}{PB}\right)$$

Otherwise, if $(AD_{eff} \cdot CJ_{scaled} + PD_{eff} \cdot CJSW_{scaled}) \leq 0$, then:

$$\text{For } vbd < 0: capbd = TT \cdot \frac{\partial ibd}{\partial vbd} + M \cdot CBD \cdot \left(1 - \frac{vbd}{PB}\right)^{-MJ}$$

$$\text{For } vbd > 0: capbd = TT \cdot \frac{\partial ibd}{\partial vbd} + M \cdot CBD \cdot \left(1 + MJ \cdot \frac{vbd}{PB}\right)$$

5

CMC MOS Varactor Model

Discusses the use of the CMC MOSFET resistor varactor models.

This section covers the following topics:

- [LEVEL 7: CMC Varactor Model](#)
- [CMC Varactor Model Updates](#)
- [CMC Varactor Model Parameters](#)

LEVEL 7: CMC Varactor Model

The MOS Varactor compact model is based in part on the PSP MOSFET model and is intended for analog and RF design. It includes dynamic inversion, finite poly doping, quantum mechanics, tunneling currents, and parasitics to model advanced MOS technologies.

The CMC MOS Varactor model is Level 7 in the Synopsys models. To use this model, specify:

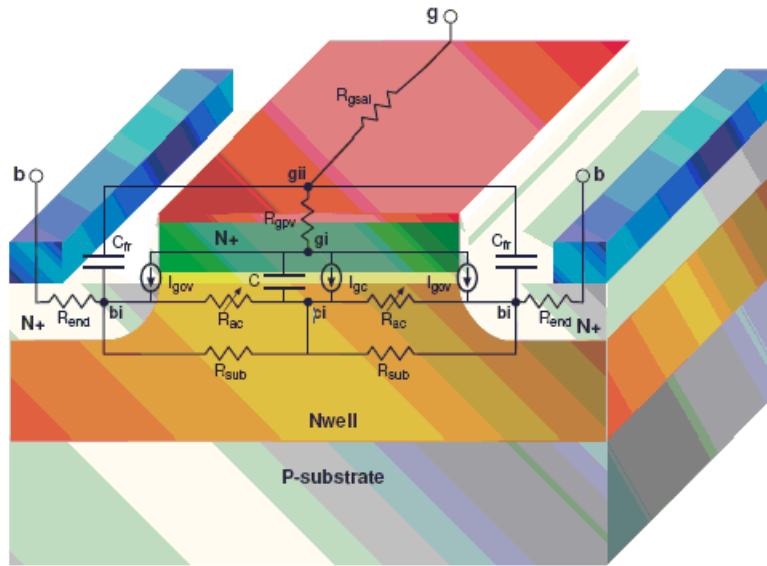
```
Cxxx g [bi] b cmname w=1u l=1u
.MODEL cmname c LEVEL=7
```

General Syntax for CMC MOS Varactor Model, Version 1.2, Version 1.1, Version 1.0, Revision 0.8

The general syntax for a CMC MOS Varactor Model element in a netlist is:

```
Cxxx g [bi] b cname [L=val] [W=val]
+ [M=val] [M_SEG=val] [NGCON=val] [DTA=val]
```

Figure 38 Cross-section of MOS varactor with equivalent circuit model overlapped



In Figure 38, **g**, **bi** and **b** are the external terminals while **gii**, **gi** and **ci** are the internal nodes

The following section describes the PSP-based varactor model parameter set.

CMC Varactor Model Updates

This section covers the following topics:

- [CMC Varactor Model — Version 1.3 Updates](#)
- [CMC Varactor Model — Version 1.2 Updates](#)
- [CMC Varactor Model — Version 1.2 Parameters](#)

CMC Varactor Model — Version 1.3 Updates

The following updates were made in CMC Varactor Version 1.3:

- Extended the range for parameters TOXO and NSUBO.
- Replaced $V(b_cf)$ with $V(b_ov)$, for I_{gov} calculation.
- Replaced $V(b_cf)$ with $V(b_ci)$, for I_{gc} calculation.

CMC Varactor Model — Version 1.2 Updates

The following updates were made in CMC Varactor Version 1.2:

- Added parameter EPSROXO for relative dielectric permittivity.
- Revised gate current computations to avoid numerical problems and to use GC3HVO for HVB current.
- Added parameter RACNOISE to select noise equations for Rac.

CMC Varactor Model — Version 1.2 Parameters

Name	Units	Default	Min.	Max.	Description
EPSROXO		3.9	1.0	N/A	Relative permittivity
RACNOISE		1	0	2	Rac noise model selector: 0: Turn off 1: Use bias-independent conductance 2: Use bias-dependent conductance.

CMC Varactor Model Parameters

This section covers the following topics:

- [CMC Varactor Model — Instance Parameters](#)
- [CMC Varactor Model — Special Parameters](#)
- [CMC Varactor Model — Basic Parameters](#)

CMC Varactor Model — Instance Parameters

Name	Units	Default	Min.	Max.	Description
L	m	10^{-6}	0	-	Design length of a varactor
W	m	10^{-6}	0	-	Design width of a varactor
m		1	0	-	Multiplicity factor
M_SEG		1	1	-	Number of gate segments
NGCON		1	1	2	Number of gate contacts

Name	Units	Default	Min.	Max.	Description
DTA	°C	0	-	-	Local temperature offset with respect to ambient circuit temperature

CMC Varactor Model — Special Parameters

Name	Units	Default	Min.	Max.	Description
version			N/A	N/A	Model version
TMIN	°C	-100	-250	27	Minimum ambient temperature
TMAX	°C	500	27	1000	Maximum ambient temperature

CMC Varactor Model — Basic Parameters

Name	Units	Default	Min.	Max.	Description
CFRL	F/m	0	0	-	Fringing capacitance in length direction
CFRW	F/m	0	0-		Fringing capacitance in width direction
DLQ	m	0	-	-	Length delta for capacitor size
DNSUBO		0	0	100	Doping profile slope parameter
DWQ	m	0	-	-	Width delta for capacitor size
DWR	m	0	-	-	Width delta for substrate resistance calculation
FETA		1.0	0	-	Effective field parameter
EPSROXO	-	3.9	1.0	-	Relative permittivity
LMAX	m	$9.9 \cdot 10^{-8}$	0	-	Maximum allowed drawn length
LMIN	m	10^{-8}	0	-	Minimum allowed drawn length
MNSUBO		1	1	10	Maximum change in absolute doping, limited to 1 order of magnitude up
NSLPO		0.1	0.1	1	Doping profile smoothing parameter

Name	Units	Default	Min.	Max.	Description
NSUBO	m^{-3}	$3 \cdot 10^{23}$	10^{22}	10^{25}	Substrate doping level
REND	$\Omega \cdot m$	10^{-4}	0	-	End resistance (extrinsic well resistance plus vertical contact resistance to well) per width
RPV	$\Omega \cdot m^2$	0	0	-	Vertical resistance down through gate
RSHG	Ω / sq	1	0	-	Gate sheet resistance
RSHS	Ω / sq	1000	0	-	Substrate sheet resistance
STREND		0	-	-	Temperature dependence of R_{end}
STRPV		0	-	-	Temperature dependence of R_{pv}
STRSHG		0	-	-	Temperature dependence of R_{shg}
STRSHS		0	-	-	Temperature dependence of R_{shs}
STUAC		0	-	-	Temperature dependence of U_{ac}
STVFB	V/K	0	-	-	Temperature dependence of V_{fb}
TOXO	m	$2 \cdot 10^{-9}$	$5 \cdot 10^{-10}$	$2 \cdot 10^{-8}$	Oxide thickness
TR	°C	21	-250	1000	Nominal (reference) temperature
UAC	$m^2 / V / s$	$5 \cdot 10^{-2}$	0	-	Accumulation layer zero-bias mobility
UACRED	V^{-1}	0	0	-	Accumulation layer mobility degradation factor
VFBO	V	-1	-	-	Flat-band voltage ⁴³
VNSUBO		0	-5	5	Doping profile corner voltage parameter
WMAX	m	$9.9 \cdot 10^{-8}$	0	-	Maximum allowed drawn width
WMIN	m	10^{-8}	0	-	Minimum allowed drawn width
Switch or switch-like parameters					
SWRES		1	0	1	Switch to control series resistance: 0: exclude 1: include

43. In PSP, VFBO corresponds to NMOS and is negative of the actual value for PMOS. For varactor, you specify the actual value.

Name	Units	Default	Min.	Max.	Description
NPO	m ⁻³	10 ²⁷	10 ²⁴	10 ²⁷	Polysilicon doping level
QMC		1	0	-	Quantum mechanical correction factor
SWIGATE		0	0	1	Flag for gate current: 0: turn off 1: turn on
TAU	s	0.1	0	10	Time constant for inversion charge recombination/generation
RACNOISE		1	0	2	Rac noise model selector: 0: turn off 1: use bias-independent conductance 2: use bias-dependent conductance
TYPE		-1	-1	1	Substrate doping type: -1: n-type +1:p-type
TYPEP		-1	-1	1	Polysilicon doping type: -1: ntype +1! p-type
Additional model parameters for modeling gate tunneling currents					
CHIBO	V	3.1	1.0	-	Tunneling barrier height for electrons
CHIBPO	V	4.5	1.0	-	Tunneling barrier height for holes
GC2EVO		0.375	0	0	EVb gate current slope factor
GC2HVO		0.375	0	0	HVB gate current slope factor
GC2O		0.375	0	0	ECB gate current slope factor
GC3EVO		0.063	-10	10	EVb gate current curvature factor
GC3HVO		0.063	-10	10	HVB gate current curvature factor
GC3O		0.063	-10	10	ECB gate current curvature factor
GCOEVO		0	-10	10	EVb gate tunneling energy adjustment
GCOHVO		0	-10	10	HVB gate tunneling energy adjustment
GCOO		0	-10	10	ECB gate tunneling energy adjustment

Name	Units	Default	Min.	Max.	Description
IGCEVLW	A	0	0	-	EVb gate channel current prefactor for $1\mu m^2$ channel area
IGCHVLW	A	0	0	-	HVB gate channel current prefactor for $1\mu m^2$ channel area
IGINVLW	A	0	0	-	ECB gate channel current prefactor for $1\mu m^2$ channel area
IGMAX	A	10^{-5}	0	-	Maximum gate current
IGOVEVW	A	0	0	-	EVb gate overlap current pre-factor for $1\mu m$ wide gate overlap region
IGOVHVV	A	0	0	-	HVB gate overlap current prefactor for $1\mu m$ wide gate overlap region
IGOVW	A	0	0	-	ECB gate overlap current pre-factor for $1\mu m$ wide gate overlap region
LOV	m	0	0	-	Overlap length
NOVO	m^{-3}	$5 \cdot 10^{25}$	10^{22}	10^{26}	Effective doping level of overlap regions

6

MOSFET Noise Models

Discusses the use of the available MOSFET noise model parameters.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

This section covers the following topics:

- [MOSFET Model — Noise Parameters](#)
- [MOSFET Model — Noise Equations](#)

MOSFET Model — Noise Parameters

This section describes noise model parameters.

Name (Alias)	Units	Default	Description
AF		1.0	Flicker noise exponent.
KF		0.0	Flicker noise coefficient. Reasonable values for KF are in the range 1e-19 to 1e-25 V ² F.
NLEV		2.0	Noise equation selector. Values are 1, 2, or 3.
RD	V ² /Hz		Output thermal noise due to drain resistor.
RS	V ² /Hz		Output thermal noise due to source resistor.

Name (Alias)	Units	Default	Description
RX			Transfers the function of thermal noise to the output. This is not noise, but is a transfer coefficient, which reflects the contribution of thermal noise to the output. For example: $V(\text{output}) = I(\text{local}) * r_x(\text{from local to output})$ Where $V(\text{output})$ is the noise voltage at the output port, $I(\text{local})$ is the local noise current in the specific noise element. It is clear that r_x should have an unit of impedance, therefore we call it transimpedance. By summarizing all the contributions (power) from each independent noisy element, we can get the total noise contribution(power) at the output port.
ID	V^2/Hz		Output channel thermal noise: $ID = RX^2 \cdot P$ (channel thermal noise) ² .
FN	V^2/Hz		Output flicker noise: $FN = RX^2 \cdot P$ (flicker noise) ² .
IFEX			Noise due to floating body
LGS			Shot noise due to I_{gs}
LGD			Shot noise due to I_{gd}
LGF			Shot noise due to I_{gf}
TOT	V^2/Hz		Total output noise: $TOT = RD + RS + ID + FN + IFEX + RG + IGS + IGD + IGF$.

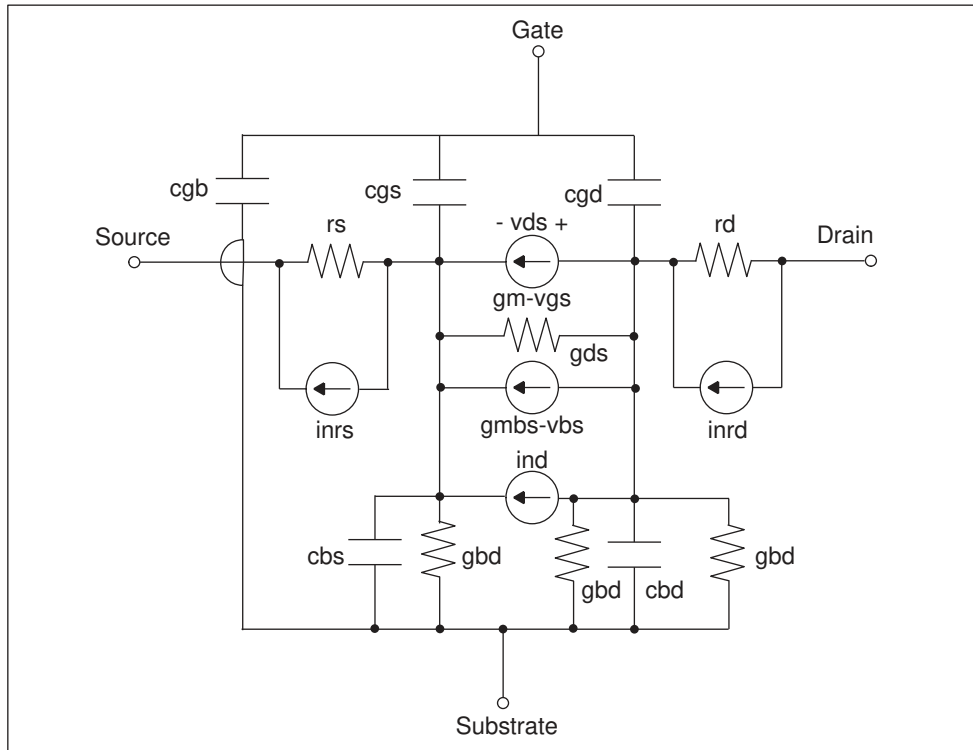
For different MOS models, the total noise equation maybe different; it always equals the sum of the noise types listed. RD, RS, ID, FN are the basic noise types (all MOSFET models contain these).

MOSFET Model — Noise Equations

The MOSFET model noise equations have a selector parameter, `NLEV`, that selects either the original SPICE flicker noise or an equation proposed by Gray and Meyer.

You can model thermal noise generation in the drain and source resistors as two sources, `inrd` and `inrs` (units $\text{amp}/(\text{Hz})^{1/2}$) as shown in [Figure 39](#).

Figure 39 Equivalent Circuit, MOSFET AC Noise Analysis



The following equations calculate the values of these sources:

$$inrs = \left(\frac{4kt}{rs} \right)^{\frac{1}{2}}, inrd = \left(\frac{4kt}{rd} \right)^{\frac{1}{2}}$$

You can model the channel thermal noise and the flicker noise as the ind current source, which the following equation defines:

$$ind^2 = (channelthermalnoise)^2 + (flickernoise)^2$$

If the `NLEV` model parameter is less than 3, then:

$$channelthermalnoise = \left(\frac{8kT \cdot gm}{3} \right)^{\frac{1}{2}}$$

The preceding formula, used in both saturation and linear regions, can lead to wrong results in the linear region. For example, at $V_{DS}=0$, channel thermal noise becomes zero because $gm=0$. This calculation is physically impossible. If you set the `NLEV` model parameter to 3, simulation uses a different equation, which is valid in both linear and saturation regions. (See *Operation and Modeling of the MOS Transistor* by Yanis P. Tsivids, published by McGraw-Hill, 1987, p. 340.)

For NLEV=3

$$\text{channelthermalnoise} = \left(\frac{8kt}{3} \cdot \beta \cdot (v_{gs} - v_{th}) \cdot \frac{1+a+a^2}{1+a} \cdot GDSNOI \right)^{1/2}$$

The following equations calculate the a value used in the preceding equation:

$$a = 1 - \frac{v_{ds}}{v_{dsat}} \text{ Linearregion}$$

$$a = 0 \text{ Saturationregion}$$

Use the A_F and K_F parameters in the small-signal AC noise analysis to determine the equivalent flicker noise current generator, which connects the drain to the source.

NLEV=0 (SPICE):

$$\text{flickernoise} = \left(\frac{KF \cdot Ids^{AF}}{COX \cdot Leff^2 \cdot f} \right)^{\frac{1}{2}}$$

For NLEV=1

$Leff^2$ in the above equation is replaced by $W_{eff} \cdot Leff$.

For NLEV=2, 3

$$\text{flickernoise} = \left(\frac{KF \cdot gm^2}{COX \cdot W_{eff} \cdot Leff \cdot f^{AF}} \right)^{\frac{1}{2}}$$

A

Technology Summary for the PrimeSim Continuum Solution MOSFET Models

Describes the technology used in all the PrimeSim Continuum solution MOSFET models.

For information on MOSFET model parameters, see MOSFET Model Parameters in *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

This section covers the following topics:

- [MOSFET Equivalent Circuits](#)
- [MOSFET DC Operating Point Output](#)
- [Common Threshold Voltage Equations](#)
- [MOSFET Impact Ionization](#)
- [Temperature Parameters and Equations](#)

MOSFET Equivalent Circuits

This section covers the following topics:

- [Equation Variables](#)
- [Using MOSFET Current Convention](#)
- [Using MOSFET Equivalent Circuits](#)

Equation Variables

This section lists the equation variables and constants.

Variable/Quantity	Definition
cbd:cbd	Bulk-to-drain transcapacitance
cbs:cbs	Bulk-to-source transcapacitance

Variable/Quantity	Definition
cbg:cbg	Gate-to-bulk transcapacitance
cgd:cgd	Gate-to-drain transcapacitance
cgs:cgs	Gate-to-source transcapacitance
cds:csd	Drain-to-source transcapacitance
f	Frequency
gbd	Bulk-to-drain dynamic conductance
gbs	Bulk-to-source dynamic conductance
gds	Drain-to-source dynamic conductance (controlled by vds)
gdb	Drain-to-bulk impact ionization conductance
gm	Drain-to-source dynamic transconductance (controlled by vgs)
gmbs	Drain-to-source dynamic bulk transconductance (controlled by vsb)
ibd	Bulk-to-drain DC current
ibs	Bulk-to-source DC current
ids	Drain-to-source DC current
idb	Drain-to-bulk impact ionization current
ind	Drain-to-source equivalent noise circuit
inrd	Drain resistor equivalent noise circuit
inrs	Source resistor equivalent noise circuit
rd	Drain resistance
rs	Source resistance
vsb	Source-to-bulk voltage
vds	Drain-to-source voltage
vgs	Gate-to-source voltage
dt	t-tnom
esi	1.0359e-10F/m dielectric constant of silicon

Variable/Quantity	Definition
k	1.38062e-23 (Boltzmann's constant)
q	1.60212e-19 (electron charge)
t	New temperature of model or element in °K
tnom	tnom = TNOM + 273.15. This variable represents the nominal temperature of parameter measurements in °K (user input in °C).
vt	$k \cdot t/q$
vt(tnom)	$k \cdot tnom/q$

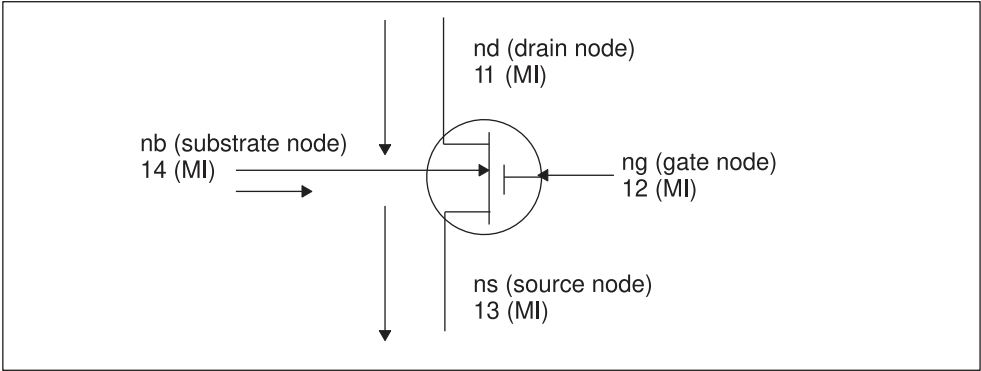
Using MOSFET Current Convention

Figure 40 shows the assumed direction of current flow through a MOS transistor. To print the drain current, use either I(M1) or I1(M1) syntax.

- I2 produces the gate current.
- I3 produces the source current.
- I4 produces the substrate current.

References to bulk are the same as references to the substrate.

Figure 40 MOSFET Current Convention, N-channel



Using MOSFET Equivalent Circuits

Simulators use three equivalent circuits to analyze MOSFETs:

- DC
- Transient
- AC and noise-equivalent circuits

The components of these circuits form the basis for all element and model equations. The equivalent circuit for DC sweep is the same as the one used for transient analysis, but excludes capacitances. [Figure 41](#) through [Figure 43](#) display the MOSFET equivalent circuits.

The fundamental component in the equivalent circuit is the DC drain-to-source current (i_{ds}). Noise and AC analyses do not use the actual i_{ds} current. Instead, the model uses the partial derivatives of i_{ds} with respect to the v_{gs} , v_{ds} , and v_{bs} terminal voltages.

The names for these partial derivatives are as follows.

Transconductance: $gm = \frac{\partial(i_{ds})}{\partial(v_{gs})}$

Conductance: $gds = \frac{\partial(i_{ds})}{\partial(v_{ds})}$

Bulk Transconductance: $gmbs = \frac{\partial(i_{ds})}{\partial(v_{bs})}$

The i_{ds} equation describes the basic DC effects of the MOSFET. Simulation considers the effects of gate capacitance, and of source and drain diodes, separately from the DC i_{ds} equations. Simulation also evaluates the impact ionization equations separately from the DC i_{ds} equation, even though the ionization effects are added to i_{ds} .

Figure 41 Equivalent Circuit, MOSFET Transient Analysis

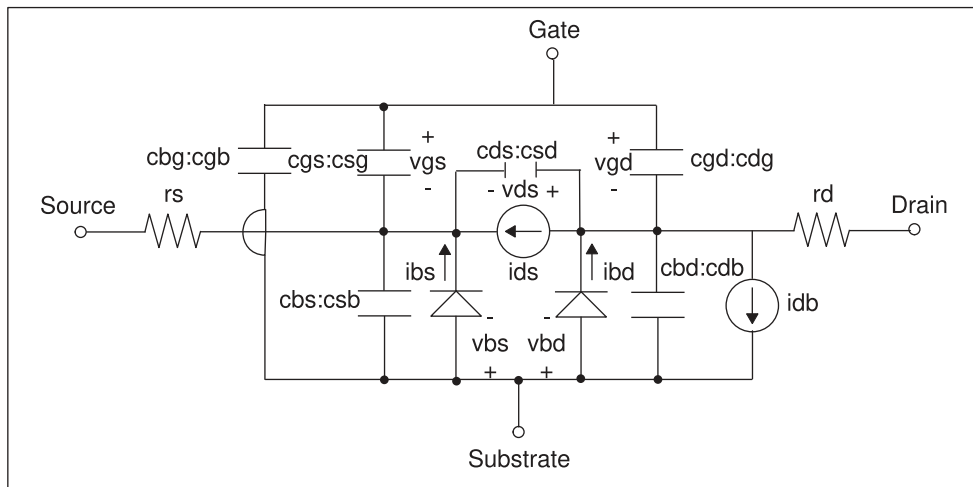


Figure 42 Equivalent Circuit, MOSFET AC Analysis

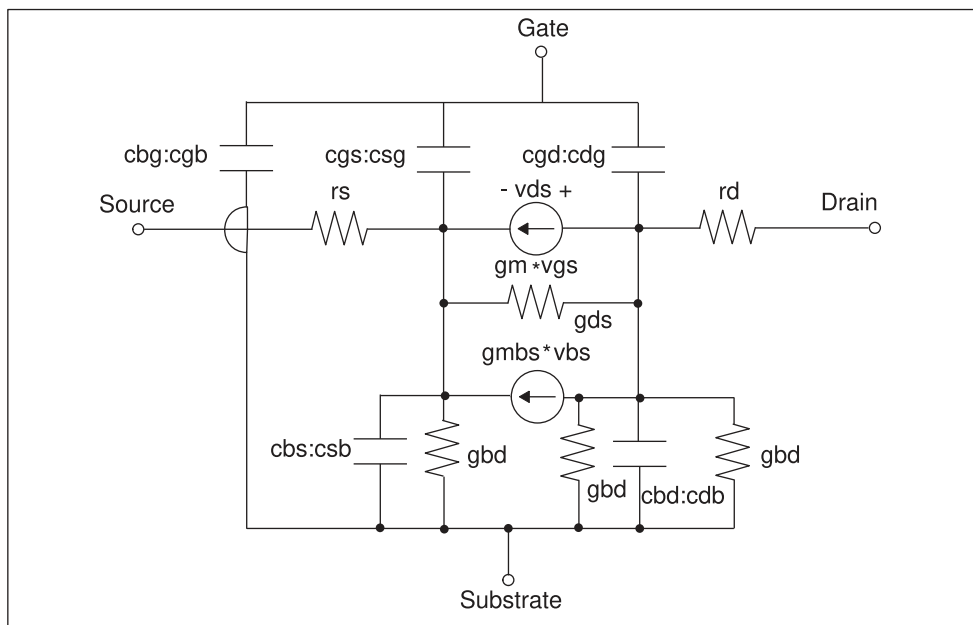
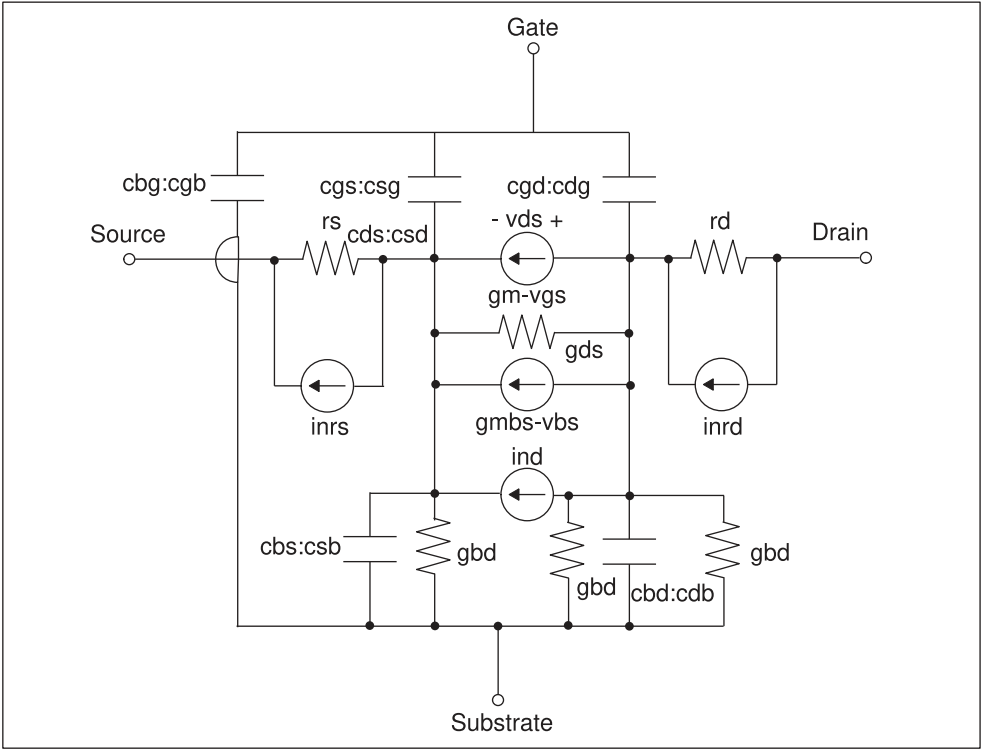


Figure 43 *Equivalent Circuit, MOSFET AC Noise Analysis*



MOSFET DC Operating Point Output

Table 11 *General MOSFET DC Operating Point Output*

Quantities	Definitions
id	Channel current
ibs	B-S bulk-to-source current
ibd	B-D bulk-to-drain current
vgs	G-S gate-source voltage
vds	D-S drain-source voltage
vbs	B-S bulk-source voltage
vth	Threshold voltage
vdsat	Saturation voltage

Table 11 General MOSFET DC Operating Point Output (Continued)

Quantities	Definitions
beta	Beta value
gam eff	Gamma effective
gm	DC gate transconductance
gds	D-S drain-source conductance
gmb	B-S bulk-source conductance
cdtot	Total drain capacitance
cgtot	Total gate capacitance
cstot	Total source capacitance
cbtot	Total bulk capacitance (total floating body capacitance for SOI MOSFET)
cgs	Total gate- to-source capacitance
cgd	Total gate- to-drain capacitance

The PrimeSim Continuum solution also supports extended DC operating point output using `.OPTION EXT_OP=1` or by setting the environment variable `ext_op` to 1.

The extended DC OP output supports MOSFET model Level=49 (BSIM3), Level 54 (BSIM4), Level 69 (PSP), and Level 76 (UTSOI):

Name	Unit	Formulation	Description
Total currents			
Ide	A	-	Total DC drain current
Ige	A	-	Total DC gate current
Ise	A	-	Total DC source current
Ibe	A	-	Total DC bulk current
Internal currents			
Ids	A	-	DC Drain current, excluding avalanche, tunnel, GISL, GIDL, and junction currents
idb	A	-	DC drain-bulk current

Name	Unit	Formulation	Description
isb	A	-	DC source-bulk current
Applied voltages			
V_{ds}	V	-	Drain-Source DC voltage
V_{sb}	V	-	Source-Bulk DC voltage
V_{gs}	V	-	Gate-Source DC voltage
Transconductances			
Gm	A/V	$\delta (i_{de})/\delta (V_{gs})$	DC transconductance
Gmb	A/V	$\delta (i_{de})/\delta (V_{sb})$	DC bulk transconductance
Gds	A/V	$\delta (i_{de})/\delta (V_{ds})$	DC output conductance
Transcapacitors			
Cgs	F	See Transcapacitance	AC Gate-Source capacitance, including overlap capacitances
Cgd	F	See Transcapacitance	AC Gate-Drain capacitance, including overlap capacitances
Csb	F	See Transcapacitance	AC Source-Bulk capacitance
Cgb	F	See Transcapacitance	AC Gate-Bulk capacitance
Cdb	F	See Transcapacitance	AC Drain-Bulk capacitance
Cds	F	See Transcapacitance	AC Drain-Source capacitances
Cdg	F	See Transcapacitance	AC Drain-Gate capacitance
Csg	F	See Transcapacitance	AC Source-Gate capacitance
Csd	F	See Transcapacitance	AC Source-Drain capacitance
Cbd	F	See Transcapacitance	AC Bulk-Drain capacitance

Name	Unit	Formulation	Description
Cbg	F	See Transcapacitance	AC Bulk-Gate capacitance
Cbs	F	See Transcapacitance	AC Bulk-Source capacitance
Capacitors			
Cgg	F	See Capacitance	AC Gate capacitance. Including overlap capacitances
Cdd	F	See Capacitance	AC Drain capacitance
Css	F	See Capacitance	AC Source capacitance
Cbb	F	See Capacitance	AC Bulk capacitance
Derived parameters			
Vth	V	-	Threshold voltage including back bias and drain bias effects
Vgt	V	$V_{gs} - V_{th}$	Effective gate drive voltage including back bias and drain bias effects
Vdss	V	-	Drain saturation voltage at actual bias
Vsat_marg	V	$V_{ds} - V_{dss}$	Vds margin
Self_gain	-	G_m/G_{ds}	Transistor self gain
Rout	Ω	$1/g_{ds}$	AC output resistor
Beff	$A/(V \cdot V)$	$2 \cdot \text{abs}(j\omega_{ide}) / (v_{gt} \cdot v_{gt})$	Gain factor
Fug	Hz	$G_m / [2 \cdot \pi \cdot c_{gg}]$	Unity gain frequency at actual bias
Rgate	Ω	See Rgate	MOS gate resistance
Gmoverid	$1/V$	G_m/I_{ds}	Gm over Id
Vearly	V	$\text{abs}(i_{de})/g_{ds}$	Equivalent early voltage

Common Threshold Voltage Equations

The parameters described in this section works for MOSFET model Level 1, 2, 3, 13, 28, 47, and 50.

This section covers the following topics:

- [Common Threshold Voltage Parameters](#)
- [Calculating PHI, GAMMA, and VTO](#)

Common Threshold Voltage Parameters

The parameters described in this section works for MOSFET model Level 1, 2, 3, 13, 28, 47, and 50.

Name (Alias)	Units	Default	Description
DELVTO	V	0.0	Zero-bias threshold voltage shift.
GAMMA	$V^{1/2}$	0.527625	Body effect factor. If you do not set GAMMA, simulation calculates it from NSUB.
NGATE	$1/\text{cm}^3$	-	Polysilicon gate doping, used for analytical models only. Undoped polysilicon is represented by a small value. If NGATE $\leq$ 0.0, it is set to $1\text{e}+18$.
NSS	$1/\text{cm}^2$	1.0	Surface state density.
NSUB (DNB, NB)	$1/\text{cm}^3$	$1\text{e}15$	Substrate doping.
PHI	V	0.576036	Surface potential. NSUB default= $1\text{e}15$.
TPG (TPS)	-	1.0	Type of gate material for analytical models. LEVEL 4 TPG default=0. The TPG value can be: • TPG = 0 al-gate. • TPG = 1 same as source-drain diffusion. • TPG = -1 gate type opposite to source-drain diffusion.
VTO (VT)	V	-	Zero-bias threshold voltage.

Calculating PHI, GAMMA, and VTO

Use the PHI , GAMMA , and VTO model parameters to calculate threshold voltages. If you do not specify these parameters, simulation calculates them as follows, except for the LEVEL 5 model.

If you do not specify PHI , then: $\text{PHI} = 2 \cdot v_t \cdot \ln\left(\frac{\text{NSUB}}{n_i}\right)$

If you do not specify GAMMA , then:
$$\text{GAMMA} = \frac{(2 \cdot q \cdot \epsilon_{si} \cdot \text{NSUB})^{1/2}}{\text{COX}}$$

The following equations determine the energy gap (eg) and intrinsic carrier concentration used in the above equations:

$$\text{eg} = 1.16 - 7.02e-4 \cdot \frac{t_{nom}^2}{t_{nom} + 1108}$$

$$n_i = 1.45e+10 \cdot \left(\frac{t_{nom}}{300} \right)^{3/2} \cdot e^{\left[\frac{q \cdot \text{eg}}{2k} \left(\frac{1}{300} - \frac{1}{t_{nom}} \right) \right]} (1/\text{cm}^3)$$

In the preceding equation, $t_{nom} = \text{TNOM} + 273.15$.

If you do not specify VTO , then for Al-Gate ($\text{TPG}=0$), the following equation determines the Φ_{ms} work function:
$$\Phi_{ms} = -\frac{\text{eg}}{2} \cdot \text{type} \cdot \frac{\text{PHI}}{2} - 0.05$$

In the preceding equation, type is +1 for n-channel or -1 for p-channel.

For Poly-Gate ($\text{TPG}=\pm 1$), the following equations determine the work function.

If you do not specify the NGATE model parameter, then:

$$\Phi_{ms} = \text{type} \cdot \left(-\text{TPG} \cdot \frac{\text{eg}}{2} - \frac{\text{PHI}}{2} \right)$$

If you specify NGATE , then:
$$\Phi_{ms} = \text{type} \cdot \left[-\text{TPG} \cdot v_t \cdot \ln \left(\frac{\text{NGATE}}{n_i} \right) - \frac{\text{PHI}}{2} \right]$$

If you do not specify the VTO model parameter, then the following equation determines the VTO voltage:
$$\text{VTO} = \text{vfb} + \text{type} \cdot (\text{GAMMA} \cdot \text{PHI}^{1/2} + \text{PHI})$$

In the preceding equation, $\text{vfb} = \Phi_{ms} - \frac{q \cdot \text{NSS}}{\text{COX}} + \text{DELVTO}$.

If you specify VTO , then:
$$\text{VTO} = \text{VTO} + \text{DELVTO}$$
.

MOSFET Impact Ionization

The parameters described in this section work for MOSFET model Level 1, 2, 3, 13, 28, 47, and 50.

Impact ionization current is available for all MOSFET levels. ALPHA , VCR , and IIRAT are the controlling parameters. IIRAT sets the fraction of the impact ionization current sent to the source.

$$\begin{aligned} I_{ds} &= I_{ds_normal} + \text{IIRAT} \cdot I_{\text{impact}} \\ I_{db} &= I_{db_diode} + (1 - \text{IIRAT}) \cdot I_{\text{impact}} \end{aligned}$$

IIRAT defaults to zero, which sends all impact ionization current to bulk. Leave **IIRAT** at its default value unless data is available for both drain and bulk current.

Name (Alias)	Units	Default	Description
ALPHA	1/V	0.0	Impact ionization current coefficient
LALPHA	μm/V	0.0	ALPHA length sensitivity
WALPHA	μm/V	0.0	ALPHA width sensitivity
VCR	V	0.0	Critical voltage
LVCR	μm · V	0.0	VCR length sensitivity
WVCR	μm · V	0.0	VCR width sensitivity
IIRAT		0.0	Portion of impact ionization current sent to the source

This section covers the following topics:

- [Calculating the Impact Ionization Equations](#)
- [Calculating Effective Output Conductance](#)
- [Cascode Example](#)
- [Cascode Circuit](#)

Calculating the Impact Ionization Equations

The following equations calculate the I_{impact} current due to the impact ionization effect:

$$I_{\text{impact}} = I_{\text{ds}} \cdot \text{ALPHA}_{\text{eff}} \cdot (v_{\text{ds}} - v_{\text{dsat}}) \cdot e^{\frac{-V_{\text{CReff}}}{v_{\text{ds}} - v_{\text{dsat}}}}$$

The following equations calculate values used in the preceding equation:

$$\text{ALPHA}_{\text{eff}} = \text{ALPHA} + \text{LALPHA} \cdot 1e-6 \cdot \left(\frac{1}{L_{\text{eff}}} - \frac{1}{L_{\text{REffeff}}} \right)$$

$$\text{WALPHA} \cdot 1e-6 \cdot \left(\frac{1}{W_{\text{eff}}} - \frac{1}{W_{\text{REffeff}}} \right)$$

$$V_{\text{CReff}} = V_{\text{CR}} + \text{LVCR} \cdot 1e-6 \cdot \left(\frac{1}{L_{\text{eff}}} - \frac{1}{L_{\text{REffeff}}} \right)$$

$$\text{WVCR} \cdot 1e-6 \cdot \left(\frac{1}{W_{\text{eff}}} - \frac{1}{W_{\text{REffeff}}} \right)$$

The following equations calculate the $LREF_{eff}$ and $WREF_{eff}$ values used in the preceding equations:

$$LREF_{eff} = LREF + XLREF - 2 \cdot LD \quad WREF_{eff} = WREF + XWREF - 2 \cdot WD$$

Calculating Effective Output Conductance

You can use the element template to directly output gds. For example:

```
.PRINT I (M1) gds=LX8 (M1)
```

If you use impact ionization current, gds is the derivative of I_{ds} only, rather than the total drain current, which is $I_{ds} + I_{db}$.

The complete drain output conductance is:

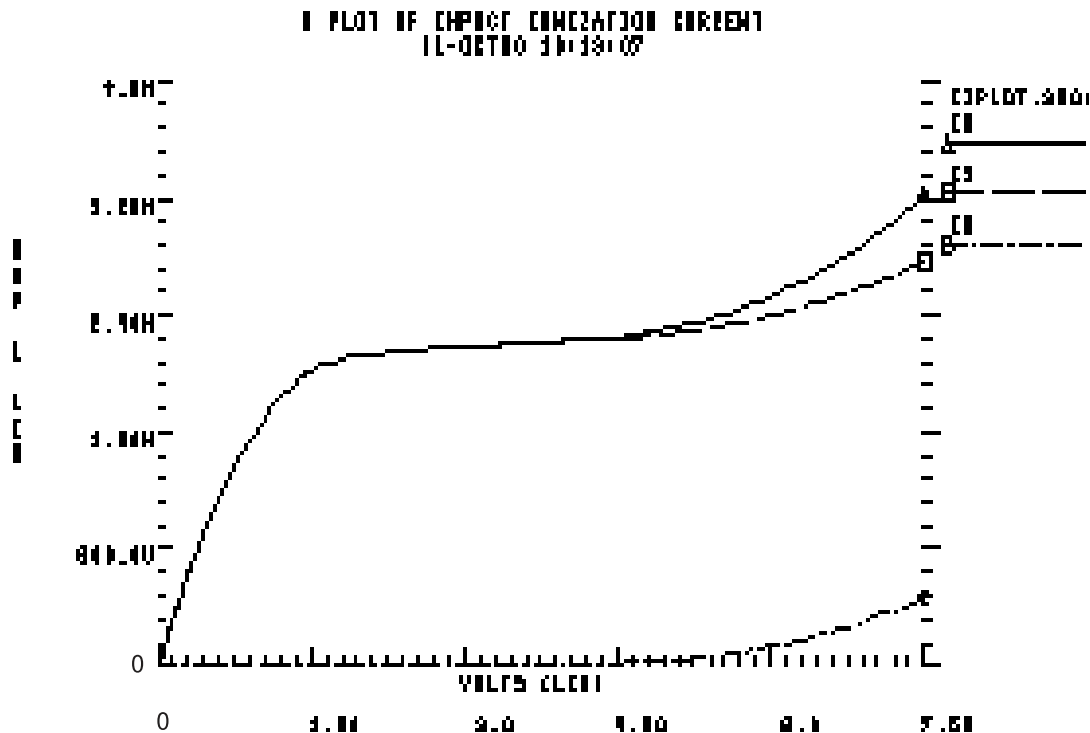
$$g_{dd} = \frac{\partial I_d}{\partial V_d} = \frac{\partial I_{ds}}{\partial V_{ds}} + \frac{\partial I_{db}}{\partial V_{db}} = \frac{\partial I_{ds}}{\partial V_{ds}} + \frac{\partial I_{bd}}{\partial V_{bd}} = g_{ds} + g_{bd}$$

$$G_{dd} = LX8 + LX10$$

For example, to print the drain output resistance of the M1 device:

```
.PRINT rout=PAR('1.0/(LX8 (M1) + LX10 (M1))')
```

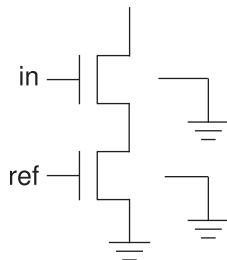

Figure 44 Drain, Source, and Bulk Currents for $v_{gs}=3$ with $IIRAT=0.5$



Cascoding Example

Drain-to-bulk impact ionization current limits the use of cascoding to increase output impedance. The following cascode example shows the effect of changing $IIRAT$. If $IIRAT$ is less than 1.0, the drain-to-bulk current lowers the output impedance of the cascode stage.

Figure 45 Low-frequency AC Analysis Measuring Output Impedance



Cascode Circuit

Example

iirat	gout_ac	rout
0.0	8.86E-6	113 K
0.5	4.30E-6	233 K
1.0	5.31E-8	18.8 Meg

This example is based on demonstration netlist `cascode.sp`, which is available in directory `$<installdir>/demo/hspice/mos`:

```
$ cascode test
.option post
.param pvds=5.0 pvref=1.4 pvin=3.0
vdd dd 0 pvds ac 1
$ current monitor vd
vd dd d 0
vin in 0 pvin
vref ref 0 pvref
x1 d in ref cascode
.macro cascode out in ref
m1 out in 1 0 n L=1u W=10u
mref 1 ref 0 0 n L=1u W=10u
.eom
.param xiirat=0
.ac dec 2 100k 1x sweep xiirat poi 3 0, 0.5, 1.0
.print ir(vd)
.measure gout_ac avg ir(vd)
.model n nmos level=3
+ tox=200 vto=0.8 gamma=0.7 uo=600 kappa=0.05
+ alpha=1 vcr=15 iirat=xiirat
.end
```

Temperature Parameters and Equations

This section covers the following topics:

- [Temperature Parameters](#)
- [Temperature Equations](#)

Temperature Parameters

The following temperature parameters apply to all MOSFET model levels and the associated bulk-to-drain and bulk-to-source MOSFET diode within the MOSFET model. The `TLEV` and `TLEVc` parameters select the temperature equations used to calculate the temperature effects on the model parameters.

Name (Alias)	Units	Default	Description
BEX		-1.5	Low field mobility, UO, temperature exponent.
CTA	1/°K	0.0	Junction capacitance (CJ) temperature coefficient. If TLEVC=1, CTA overrides the default temperature compensation.
CTP	1/°K	0.0	Junction sidewall capacitance (CJSW) temperature coefficient. If TLEVC=1, CTP overrides the default temperature compensation.
EG	eV		Energy gap for PN junction diode for TLEV=0 or 1 with default =1.11; for TLEV=2, default=1.16 1.17 – silicon 0.69 – Schottky barrier diode 0.67 – germanium 1.52 – gallium arsenide
F1EX		0	Bulk junction bottom grading coefficient
GAP1	eV/°K	7.02e-4	First bandgap correction factor (from Sze, alpha term). 7.02e-4 – silicon 4.73e-4 – silicon 4.56e-4 – germanium 5.41e-4 – gallium arsenide
GAP2	°K	1108	Second bandgap correction factor (from Sze, beta term). 1108 – silicon 636 – silicon 210 – germanium 204 – gallium arsenide
LAMEX	1/°K	0	LAMBDA temperature coefficient.
N		1.0	Emission coefficient.
MJ		0.5	Bulk junction bottom grading coefficient.
MJSW		0.33	Bulk junction sidewall grading coefficient.
PTA	V/°K	0.0	Junction potential (PB) temperature coefficient. If you set TLEVC to 1 or 2, PTA overrides the default temperature compensation.
PTC	V/°K	0.0	Fermi potential (PHI) temperature coefficient. If you set TLEVC to 1 or 2, PTC overrides the default temperature compensation.
PTP	V/°K	0.0	Junction potential (PHP) temperature coefficient. If TLEVC=1 or 2, PTP overrides the default temperature compensation.
TCV	V/°K	0.0	Threshold voltage temperature coefficient. Typical values are +1mV for n-channel and -1mV for p-channel.

Name (Alias)	Units	Default	Description
TLEV		0.0	Temperature equation level selector. Set TLEV=1 for ASPEC style. Default is SPICE style. If you invoke the ASPEC option, the program sets TLEV for ASPEC.
TLEVC		0.0	Temperature equation level selector for junction capacitances and potentials. Interacts with TLEV. Set TLEVC=1 for ASPEC style. Default is SPICE style. If you invoke the ASPEC option, the program sets TLEVC for ASPEC.
TRD	1/°K	0.0	Temperature coefficient for drain resistor.
TRS	1/°K	0.0	Temperature coefficient for source resistor.
XTI		0.0	Saturation current temperature exponent. Use XTI=3 for silicon diffused junction. Set XTI=2 for Schottky barrier diode.

MOS Temperature Coefficient Sensitivity Parameters

Model levels 13 (BSIM1), 39 (BSIM2), and 28 (METAMOS) include length and width sensitivity parameters as shown in [Table 12](#). Use these parameters with the Automatic Model Selector to enable more accurate modeling for various device sizes. The default value of each sensitivity parameter is zero to ensure backward compatibility.

Table 12 MOS Temperature Coefficient Sensitivity Parameters

Parameter	Description	Sensitivity Parameters		
		Length	Width	Product
BEX	Low field mobility, UO, temperature exponent	LBEX	WBEX	PBEX
FEX	Velocity saturation temperature exponent	LFEX	WFEX	PFEX
TCV	Threshold voltage temperature coefficient	LTCV	WTCV	PTCV
TRS	Temperature coefficient for source resistor	LTRS	WTRS	PTRS
TRD	Temperature coefficient for drain resistor	LTRD	WTRD	PTRD

Temperature Equations

This section covers the following topics:

- [Energy Gap Temperature Equations](#)
- [Saturation Current Temperature Equations](#)
- [MOS Diode Capacitance Temperature Equations](#)
- [Surface Potential Temperature Equations](#)
- [Threshold Voltage Temperature Equations](#)
- [Mobility Temperature Equations](#)
- [Channel Length Modulation Temperature Equation](#)
- [Calculating Diode Resistance Temperature Equations](#)

Energy Gap Temperature Equations

These equations set the energy gap for temperature compensation:

For $TLEV = 0$ or 1 :

$$egnom = 1.16 - 7.02e-4 \cdot \frac{tnom^2}{tnom + 1108.0}$$

$$eg(t) = 1.16 - 7.02e-4 \cdot \frac{t^2}{t + 1108.0}$$

For $TLEV = 2$:

$$egnom = EG-GAP1 \cdot \frac{tnom^2}{tnom + GAP2}$$

$$eg(t) = EG-GAP1 \cdot \frac{t^2}{t + GAP2}$$

Saturation Current Temperature Equations

$$isbd(t) = isbd(tnom) \cdot e^{\frac{facln}{N}}, isbs(t) = isbs(tnom) \cdot e^{\frac{facln}{N}}$$

The following equation calculates the $facln$ value used in the preceding equations:

$$facln = \frac{egnom}{vt(tnom)} - \frac{eg(t)}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

[MOSFET Diode Models](#) defines the $isbd$ and $isbs$ values.

MOS Diode Capacitance Temperature Equations

$TLEVC$ selects the temperature equation level for MOS diode capacitance.

For $TLEVC=0$:

$$PB(t) = PB \cdot \left(\frac{t}{tnom} \right)^{-vt(t)} \cdot \left[3 \cdot \ln \left(\frac{t}{tnom} \right) + \frac{egnom}{vt(tnom)} \cdot \frac{eg(t)}{vt(t)} \right]$$

$$PHP(t) = PHP \cdot \left(\frac{t}{tnom} \right)^{-vt(t)} \cdot \left[3 \cdot \ln \left(\frac{t}{tnom} \right) + \frac{egnom}{vt(tnom)} \cdot \frac{eg(t)}{vt(t)} \right]$$

$$CBD(t) = CBD \cdot \left[1 + MJ \cdot \left(400u \cdot \Delta t \cdot \frac{PB(t)}{PB} + 1 \right) \right]$$

$$CBS(t) = CBS \cdot \left[1 + MJ \cdot \left(400u \cdot \Delta t \cdot \frac{PB(t)}{PB} + 1 \right) \right]$$

$$CJ(t) = CJ \cdot \left[1 + MJ \cdot \left(400u \cdot \Delta t \cdot \frac{PB(t)}{PB} + 1 \right) \right]$$

$$CJSW(t) = CJSW \cdot \left[1 + MJSW \cdot \left(400u \cdot \Delta t \cdot \frac{PHP(t)}{PHP} + 1 \right) \right]$$

For $TLEVC=1$:

$$PB(t) = PB \cdot PTA \cdot \Delta t, PHP(t) = PHP \cdot PTP \cdot \Delta t$$

$$CBD(t) = CBD \cdot (1 + CTA \cdot \Delta t), CBS(t) = CBS \cdot (1 + CTA \cdot \Delta t)$$

$$CJ = CJ \cdot (1 + CTA \cdot \Delta t), CJSW = CJSW \cdot (1 + CTP \cdot \Delta t)$$

For $TLEVC=2$:

$$PB(t) = PB \cdot PTA \cdot \Delta t, PHP(t) = PHP \cdot PTP \cdot \Delta t$$

$$CBD(t) = CBD \cdot \left(\frac{PB}{PB(t)} \right)^{MJ}, CBS(t) = CBS \cdot \left(\frac{PB}{PB(t)} \right)^{MJ}$$

$$CJ(t) = CJ \cdot \left(\frac{PB}{PB(t)} \right)^{MJ}, CJSW(t) = CJSW \cdot \left(\frac{PHP}{PHP(t)} \right)^{MJSW}$$

For $TLEVC=3$:

$$PB(t) = PB + dpbdt \cdot \Delta t, PHP(t) = PHP + dphpdt \cdot \Delta t$$

$$CBD(t) = CBD \cdot \left(1 - 0.5 \cdot dpbdt \cdot \frac{\Delta t}{PB} \right)$$

$$CBS(t) = CBS \cdot \left(1 - 0.5 \cdot dpbdt \cdot \frac{\Delta t}{PB} \right)$$

$$CJ(t) = CJ \cdot \left(1 - 0.5 \cdot dpbdt \cdot \frac{\Delta t}{PB}\right)$$

$$CJSW(t) = CJSW \cdot \left(1 - 0.5 \cdot dphpdt \cdot \frac{\Delta t}{PHP}\right)$$

If $T_{LEV} = 3$ and $T_{LEV} = 0$ or 1, then:

$$dpbdt = -\frac{\left[egn_{om} + 3 \cdot vt(t_{nom}) + (1.16 - egn_{om}) \cdot \left(2 - \frac{t_{nom}}{t_{nom} + 1108}\right) - PB\right]}{t_{nom}}$$

$$dphpdt = -\frac{\left[egn_{om} + 3 \cdot vt(t_{nom}) + (1.16 - egn_{om}) \cdot \left(2 - \frac{t_{nom}}{t_{nom} + 1108}\right) - PHP\right]}{t_{nom}}$$

For $T_{LEV} = 2$:

$$dpbdt = -\frac{\left[egn_{om} + 3 \cdot vt(t_{nom}) + (EG - egn_{om}) \cdot \left(2 - \frac{t_{nom}}{t_{nom} + GAP2}\right) - PB\right]}{t_{nom}}$$

$$dphpdt = -\frac{\left[egn_{om} + 3 \cdot vt(t_{nom}) + (EG - egn_{om}) \cdot \left(2 - \frac{t_{nom}}{t_{nom} + GAP2}\right) - PHP\right]}{t_{nom}}$$

Surface Potential Temperature Equations

For $T_{LEV} = 0$:

$$PHI(t) = PHI \cdot \left(\frac{t}{t_{nom}}\right) - vt(t) \cdot \left[3 \cdot \ln\left(\frac{t}{t_{nom}}\right) + \frac{egn_{om}}{vt(t_{nom})} \cdot \frac{eg(t)}{vt(t)}\right]$$

For $T_{LEV} = 1$:

$$PHI(t) = PHI - PTC \cdot \Delta t$$

If you do not specify the PHI parameter, simulation calculates it as:

$$PHI(t) = 2 \cdot vt(t) \cdot \ln\left(\frac{NSUB}{ni}\right)$$

The intrinsic carrier concentration, ni , must be temperature updated, and it is calculated from the silicon bandgap at room temperature.

$$ni = 145e16 \cdot \left(\frac{t}{t_{nom}}\right)^{\frac{3}{2}} \cdot \exp\left[EG \cdot \left(\frac{t}{t_{nom}} - 1\right) \cdot \left(\frac{1}{2 \cdot vt(t)}\right)\right]$$

For $T_{LEV} = 2$:

$$PHI(t) = PHI - PTC \cdot \Delta t$$

For $T_{LEV} = 3$:

$$PHI(t) = PHI + dphidt \cdot \Delta t$$

If $T_{LEV} = 3$ and $T_{LEV} = 0$ or 1, then:

$$d\phi/dt = -\frac{[e_{gnom} + 3 \cdot v_t(t_{nom}) + (1.16 - e_{gnom}) \cdot (2 - \frac{t_{nom}}{t_{nom} + 1108}) - PHI]}{t_{nom}}$$

For $T_{LEV} = 2$:

$$d\phi/dt = -\frac{[e_{gnom} + 3 \cdot v_t(t_{nom}) + (EG - e_{gnom}) \cdot (2 - \frac{t_{nom}}{t_{nom} + GAP2}) - PHI]}{t_{nom}}$$

Threshold Voltage Temperature Equations

The threshold temperature equations are:

For $T_{LEV} = 0$:

$$v_{bi}(t) = v_{bi}(t_{nom}) + \frac{PHI(t) - PHI}{2} + \frac{e_{gnom} - e_g(t)}{2}$$

$$V_{TO}(t) = v_{bi}(t) + GAMMA \cdot (PHI(t))^{\frac{1}{2}}$$

For $T_{LEV} = 1$:

$$V_{TO}(t) = V_{TO} - TCV \cdot \Delta t$$

$$v_{bi}(t) = V_{TO}(t) - GAMMA \cdot (PHI(t))^{\frac{1}{2}}$$

For $T_{LEV} = 2$:

$$V_{TO}(t) = V_{TO} + \left(1 + \frac{GAMMA}{2 \cdot PHI^{\frac{1}{2}}}\right) \cdot d\phi/dt \cdot \Delta t$$

$$v_{bi}(t) = V_{TO}(t) - GAMMA \cdot (PHI(t))^{\frac{1}{2}}$$

Mobility Temperature Equations

The MOS mobility temperature equations are:

$$U_O(t) = U_O \cdot \left(\frac{t}{t_{nom}}\right)^{BEX}$$

$$K_P(t) = K_P \cdot \left(\frac{t}{t_{nom}}\right)^{BEX}$$

$$F_1(t) = F_1 \cdot \left(\frac{t}{t_{nom}}\right)^{F1EX}$$

Channel Length Modulation Temperature Equation

If you specify the `LAMEX` model parameter, then the temperature modifies the `LAMBDA` value.

$$LAMBDA(t) = LAMBDA \cdot (1 + LAMEX \cdot \Delta t)$$

Calculating Diode Resistance Temperature Equations

The following equations are examples of the effective drain and source resistance:

$$RD(t) = RS \cdot (1 + TRD \cdot \Delta t), RS(t) = RS \cdot (1 + TRS \cdot \Delta t)$$